Étale Coverings and the Fundamental Group

(SGA 1)

English translation

Contents

VII: does not exist

CHAPTER I

Étale morphisms

To simplify the exposition, we assume that all preschemes under consideration are locally noetherian, at least after no. I.2.

I.1. Notions of differential calculus

Let X be a prescheme over Y, and let $\Delta_{X/Y}$, or simply Δ, be the diagonal morphism $X \to X \times_Y X$. It is an immersion, hence a closed immersion of X into an open subset V of $X \times_Y X$. Let $\mathcal{I}_X$ be the ideal of the closed subprescheme corresponding to the diagonal in V (N.B. if one wishes to proceed intrinsically, without assuming that X is separated over Y—a hypothesis that would be farcical—one should consider the set-theoretic inverse image of $\mathcal{O}_{X\times X}$ in X, and denote by $\mathcal{I}_X$ the augmentation ideal in the latter…). The sheaf $\mathcal{I}_X/\mathcal{I}_X^2$ may be regarded as a quasi-coherent sheaf on X; we denote it by $\Omega^1_{X/Y}$. It is of finite type if $X \to Y$ is of finite type. It behaves well under base change $Y' \to Y$. We also introduce the sheaves $\mathcal{O}_{X\times_Y X}/\mathcal{I}_X^{n+1} = \mathcal{P}^n_{X/Y}$; these are sheaves of *rings* on X, making X into a prescheme that may be denoted by $\Delta^n_{X/Y}$ and called the *n-th infinitesimal neighborhood of X/Y*. The resulting sorites is entirely trivial, although rather long[1]; it would be prudent to speak of it only when there is something useful to say about it, in connection with smooth morphisms.

I.2. Quasi-finite morphisms

Proposition I.2.1. *Let $A \to B$ be a local homomorphism (N.B. the rings are now noetherian), and let $\mathfrak{m}$ be the maximal ideal of A. The following conditions are equivalent:*

(i) *$B/\mathfrak{m}B$ is finite-dimensional over $k = A/\mathfrak{m}$.*

(ii) *$\mathfrak{m}B$ is an ideal of definition, and $B/\mathfrak{r}(B) = \kappa(B)$ is an extension of $k = \kappa(A)$.*

(iii) *The completion $\widehat{B}$ is finite over the completion $\widehat{A}$ of A.*

One then saysthat B *is quasi-finite* over A. A morphism $f\colon X \to Y$ is said to be quasi-finite at x (or the Y-prescheme f is said to be quasi-finite at x) if $\mathcal{O}_x$ is quasi-finite over $\mathcal{O}_{f(x)}$. This is also equivalent to saying that x is *isolated in its fiber* $f^{-1}(f(x))$.A morphism is said to be quasi-finite if it is so at every point.[2]

Corollary I.2.2. *If A is complete, quasi-finite is equivalent to finite.*

One could give the usual chain of implications (i), (ii), (iii), (iv), (v) for quasi-finite morphisms, but this does not seem indispensable here.

I.3. Unramified or net morphisms

Proposition I.3.1. *Let $f\colon X \to Y$ be a morphism of finite type, let $x \in X$, and put $y = f(x)$. The following conditions are equivalent:*

(i) *$\mathcal{O}_x/\mathfrak{m}_y\mathcal{O}_x$ is a finite separable extension of $\kappa(y)$.*

(ii) *$\Omega^1_{X/Y}$ vanishes at x.*

(iii) *The diagonal morphism $\Delta_{X/Y}$ is an open immersion in a neighborhood of x.*

[1]Cf. EGA IV 16.3.

[2]In EGA II 6.2.3 one assumes in addition that f is of finite type.

For the implication (i)⇒(ii), Nakayama's lemma immediately reduces us to the case $Y = \mathrm{Spec}(k)$, $X = \mathrm{Spec}(k')$, where the assertion is well known and, moreover, immediate from the definition of separability. The implication (ii)⇒(iii) follows from a pleasant and easy characterization of open immersions using Krull. For (iii)⇒(i), one is again reduced to the case $Y = \mathrm{Spec}(k)$, with the diagonal morphism an open immersion everywhere. One must then prove that X is finite, with a separable coordinate ring over k; for this one reduces to the case where k is algebraically closed. But then every closed point of X is isolated (it is precisely the inverse image of the diagonal under the morphism $X \to X \times_k X$ defined by x), and hence X is finite. One may therefore suppose that X consists of a single point, with ring A. Thus $A \otimes_k A \to A$ is an isomorphism, whence $A = k$, as required.

Definition I.3.2. (a) *One then says that f is* net*, or equivalently* unramified*, at x, or that X is net, or unramified, at x over Y.*

(b) *Let $A \to B$ be a local homomorphism. It is said to be* net*, or* unramified*, and B is said to be a net, or unramified, local algebra over A, if $B/\mathfrak{r}(A)B$ is a finite separable extension of $A/\mathfrak{r}(A)$; that is, if $\mathfrak{r}(A)B = \mathfrak{r}(B)$ and $\kappa(B)$ is a separable extension of $\kappa(A)$.*[3]

Remarks. The factthat B is unramified over A can already be checked on the completions of A and B. Unramified implies quasi-finite.

Corollary I.3.3. *The set of points at which f is unramified is open.*

Corollary I.3.4. *Let X' and X be two preschemes of finite type over Y, and let $g\colon X' \to X$ be a Y-morphism. If X is unramified over Y, the graph morphism*

$$\Gamma_g\colon X' \longrightarrow X' \times_Y X$$

is an open immersion.

Indeed, it is the inverse image of the diagonal morphism $X \to X \times_Y X$ under

$$g \times_Y \mathrm{id}_X\colon X' \times_Y X \longrightarrow X \times_Y X.$$

One may also introduce the annihilator ideal $\mathfrak{d}_{X/Y}$ of $\Omega^1_{X/Y}$, called the *different ideal* of X/Y. It defines a closed subprescheme of X whose underlying set is the set of points at which X/Y is ramified, that is, not unramified.

Proposition I.3.5. (i) *An immersion is unramified.*

(ii) *The composite of two unramified morphisms is unramified.*

(iii) *A base change of an unramified morphism is unramified.*

This may be seen equally well from criterion (ii) or criterion (iii) (the latter seems more amusing to me). One can of course also be more precise by giving pointwise statements; this is more general only in appearance (except in the setting of Definition I.3.2(b)), and tedious. As usual, one obtains the following corollaries.

Corollaries I.3.6. (iv) *The Cartesian product of two unramified morphisms is unramified.*

(v) *If gf is unramified, then f is unramified.*

(vi) *If f is unramified, then f_{red} is unramified.*

Proposition I.3.7. *Let $A \to B$ be a local homomorphism. Suppose that the residue-field extension $\kappa(B)/\kappa(A)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is unramified if and only if $\widehat{B}$, as an $\widehat{A}$-algebra, is a quotient of $\widehat{A}$.*

[3] Cf. the regrets in Exposé III, no. 1.2.

Remarks.

- When the residue-field extension is not assumed to be trivial, one may reduce to that case by making a suitable finite flat extension over A that kills the extension in question.
- Take the example in which A is the local ring of an ordinary double point of a curve and B is the local ring at a point of the normalization. Then $A \subset B$, and B is unramified over$A$ with trivial residue-field extension; the map $\widehat{A} \to \widehat{B}$ is surjective but *not injective*. We shall therefore strengthen the notion of unramifiedness.

I.4. Étale morphisms. Étale coverings

We shall take for granted whatever is needed concerning flat morphisms; these facts will be proved later, if necessary[4].

Definition I.4.1. (a) *Let $f\colon X \to Y$ be a morphism of finite type. One says that f is* étale *at x if f is flat at x and net (equivalently, unramified) at x. One says that f is étale if it is so at every point. One says that X is étale at x over Y, or that it is a Y-prescheme étale at x, and so on.*

(b) *Let $f\colon A \to B$ be a local homomorphism. One says that f is étale, or that B is étale over A, if B is flat and unramified over A.*[5]

Proposition I.4.2. *For B/A to be étale, it is necessary and sufficient that $\widehat{B}/\widehat{A}$ be étale.*

Indeed, this holds separately for "unramified" and for "flat."

Corollary I.4.3. *Let $f\colon X \to Y$ be of finite type, and let $x \in X$. Whether f is étale at x depends only on the local homomorphism $\mathcal{O}_{f(x)} \to \mathcal{O}_x$, and even only on the corresponding homomorphism between the completions.*

Corollary I.4.4. *Suppose that the residue-field extension $\kappa(A) \to \kappa(B)$ is trivial, or that $\kappa(A)$ is algebraically closed. Then B/A is étale if and only if $\widehat{A} \to \widehat{B}$ is an isomorphism.*

This follows by combining flatness with Proposition I.3.7.

Proposition I.4.5. *Let $f\colon X \to Y$ be a morphism of finite type. Then the set of points at which it is étale is open.*

Indeed, this holds separately for "unramified" and for "flat."

This proposition shows that one may dispense with "pointwise" statements when studying morphisms of finite type that are étale at some point.

Proposition I.4.6. (i) *An open immersion is étale.*

(ii) *The composite of two étale morphisms is étale.*

(iii) *A base change of an étale morphism is étale.*

Indeed, (i) is trivial, and for (ii) and (iii) it is enough to note that the assertions hold for "unramified" and for "flat." To tell the truth, there are also corresponding statements for local homomorphisms (without a finiteness condition); in any event, they will have to appear in the *multiplodoque*, beginning with the unramified case.

Corollary I.4.7. *The Cartesian product of two étale morphisms is likewise étale.*

Corollary I.4.8. *Let X and X' be of finite type over Y, and let $g\colon X \to X'$ be a Y-morphism. If X' is unramified over Y and X is étale over Y, then g is étale.*

[4] Cf. Exposé IV.

[5] Cf. the regrets in Exposé III, no. 1.2.

Indeed, g is the composite of the graph morphism

$$\Gamma_g \colon X \longrightarrow X \times_Y X',$$

which is an open immersion by Corollary I.3.4, and the projection $X \times_Y X' \to X'$, which is étale because it is obtained from the étale morphism $X \to Y$ by the base change $X' \to Y$.

Definition I.4.9. *An* étale covering *(respectively, a* net covering*, also called an unramified covering) of Y is a Y-scheme X that is finite over Y and étale (respectively, unramified) over Y.*

The first condition means that X is defined by a coherent sheaf of algebras $\mathcal{B}$ on Y. The second then means that $\mathcal{B}$ is locally free over Y (with no corresponding local-freeness requirement in the net case), and, moreover, that for every $y \in Y$ the fiber

$$\mathcal{B}(y) = \mathcal{B}_y \otimes_{\mathcal{O}_y} \kappa(y)$$

is a separable algebra over $\kappa(y)$, that is, a finite product of finite separable field extensions of $\kappa(y)$.

Proposition I.4.10. *Let X be a flat covering of Y of degree n (the definition of this term deserved to appear in Definition I.4.9), defined by a coherent locally free sheaf of algebras $\mathcal{B}$. In the usual way one defines the trace homomorphism $\mathcal{B} \to \mathcal{A}$, which is a homomorphism of $\mathcal{A}$-modules, where $\mathcal{A} = \mathcal{O}_Y$. For X to be étale, it is necessary and sufficient that the corresponding trace pairing $(x, y) \longmapsto \mathrm{Tr}_{\mathcal{B}/\mathcal{A}}(xy)$ induce an isomorphism $\mathcal{B} \xrightarrow{\sim} \mathcal{B}^\vee$, or, equivalently, that the* discriminant section

$$d_{X/Y} = d_{\mathcal{B}/\mathcal{A}} \in \Gamma\Big(Y, \bigwedge^n \mathcal{B}^\vee \otimes_{\mathcal{A}} \bigwedge^n \mathcal{B}^\vee\Big)$$

be invertible, or, finally, that the discriminant ideal defined by this section be the unit ideal.

Indeed, one reduces to the case $Y = \mathrm{Spec}(k)$; the assertion is then the well-known separability criterion, and becomes immediate after passing to an algebraic closure of k.

Remark. There will be a less trivial statement below, when one does not assume a priori that X is flat over Y, but instead imposes a normality hypothesis.

I.5. The fundamental property of étale morphisms

Theorem I.5.1. *Let $f \colon X \to Y$ be a morphism of finite type. For f to be an open immersion, it is necessary and sufficient that it be* étale and radicial*.*

Recall that radicial means injective, with radicial residue-field extensions; one may also recall that this means that the morphism remains injective after every base change. Necessity is trivial; it remains to prove sufficiency. We shall give two different proofs, the first shorter and the second more elementary.

1) A flat morphism is open, so (replacing Y by $f(X)$) we may suppose that f is a *homeomorphism* onto Y. After every base change, f remains flat, radicial, and surjective, hence a homeomorphism and therefore closed. Thus f is *proper*. Hence f is *finite* (reference: Chevalley's theorem) and is defined by a coherent sheaf $\mathcal{B}$ of algebras. The sheaf $\mathcal{B}$ is locally free and, by the hypothesis, has rank 1 everywhere. Thus $X = Y$, as required.
2) One may suppose Y and X *affine*. Moreover, one easily reduces to proving the following: if $Y = \mathrm{Spec}(A)$, with A local, and if $f^{-1}(y)$ is nonempty, where y is the closed point of Y, then $X = Y$. Indeed, this will imply that every $y \in f(X)$ has an open neighborhood U such that $X|U = U$. We shall have $X = \mathrm{Spec}(B)$, and we want to prove $A = B$. For this, it is enough to prove the analogous assertion after replacing A by $\widehat{A}$ and B by $B \otimes_A \widehat{A}$, taking into account that $\widehat{A}$ is faithfully flat over A. We may therefore suppose that A is *complete*. Let x be

the point above y. By Corollary I.2.2, $\mathcal{O}_x$ is finite over A, and hence, being flat and radicial over A, is A itself. Thus

$$X = Y \amalg X'$$

as a disjoint union. Since X is radicial over Y, X' is empty. This completes the proof.

Corollary I.5.2. *Let $f\colon X \to Y$ be both a* closed immersion *and* étale*. If X is connected, f is an isomorphism from X onto a connected component of Y.*

Indeed, f is also an open immersion. It follows that:

Corollary I.5.3. *Let X be an unramified Y-scheme, with Y connected. Then every section of X over Y is an isomorphism from Y onto a connected component of X. There is therefore a one-to-one correspondence between the set of such sections and the set of connected components X_i of X for which the projection $X_i \to Y$ is an isomorphism (or, equivalently by Theorem I.5.1, is surjective, étale, and radicial).In particular, a section is determined by its value at one point.*

Only the first assertion requires proof. By Corollary I.5.2, it is enough to observe that a section is a closed immersion (because X is separated over Y) and is étale by Corollary I.4.8.

Corollary I.5.4. *Let X and Y be preschemes over S, with X unramified and separated over S and Y connected. Let $f, g\colon Y \to X$ be two S-morphisms, and let $y \in Y$. Suppose that $f(y) = g(y) = x$ and that the homomorphisms of residue fields $\kappa(x) \to \kappa(y)$ induced by f and g are identical ("f and g coincide geometrically at y"). Then f and g are identical.*

This follows from Corollary I.5.3 by reducing to the case $Y = S$, replacing X by $X \times_S Y$.

Here is a particularly important variant of Corollary I.5.3.

Theorem I.5.5. *Let S be a prescheme, let X and Y be two S-preschemes, and let S_0 be a closed subprescheme of S having the same underlying space as S. Put*

$$X_0 = X \times_S S_0 \qquad \text{and} \qquad Y_0 = Y \times_S S_0,$$

the "restrictions" of X and Y to S_0. Suppose that X is étale over S. Then the natural map

$$\mathrm{Hom}_S(Y, X) \longrightarrow \mathrm{Hom}_{S_0}(Y_0, X_0)$$

is bijective*.*

We are again reduced to the case $Y = S$; the assertion then follows from the "topological" description of the sections of X/Y given in Corollary I.5.3.

Scholium. This result includes both a *uniqueness* assertion and an *existence* assertion for *morphisms*. It may also be expressed, when X and Y are both taken to be étale over S, by saying that the functor

$$X \longmapsto X_0$$

from the category of étale S-schemes to the category of étale S_0-schemes is *fully faithful*; that is, it establishes an *equivalence* of the first category with a *full subcategory* of the second. We shall see below that it is actually an equivalence between the first and the second (which will be an *existence theorem for étale S-schemes*).

The following form of Theorem I.5.5, apparently more general, is often convenient.

Corollary I.5.6 (“Extension theorem for liftings”). *Consider a commutative diagram of morphisms*

$$\begin{array}{ccc} X & \longleftarrow & Y_0 \\ \downarrow & & \downarrow \\ S & \longleftarrow & Y \end{array}$$

in which $X \to S$ is étale and $Y_0 \to Y$ is a bijective closed immersion. Then there is a unique morphism $Y \to X$ that makes the two corresponding triangles commute.

Indeed, replacing S by Y and X by $X \times_S Y$, we are reduced to the case $Y = S$; this is then the special case of Theorem I.5.5 with $Y = S$.

We should also note the following immediate consequence of Theorem I.5.1 (which we did not state as Corollary 1, so as not to interrupt the line of ideas developed after Theorem I.5.1):

Proposition I.5.7. *Let X and X' be two preschemes of finite type and flat over Y, and let $g\colon X \to X'$ be a Y-morphism. For g to be an open immersion (respectively, an isomorphism), it is necessary and sufficient that, for every $y \in Y$, the induced morphism on the fibers*

$$g \otimes_Y \kappa(y)\colon X \otimes_Y \kappa(y) \longrightarrow X' \otimes_Y \kappa(y)$$

be so.

It is enough to prove sufficiency. Since the assertion holds for surjectivity, we are reduced to the case of an open immersion. By Theorem I.5.1, one must verify that g is *radicial*, which is immediate, and that it is *étale*, which follows from Corollary I.5.9 below.

Corollary I.5.8. *(This should be moved to no. I.3.) Let X and X' be two preschemes over Y, let $g\colon X \to X'$ be a Y-morphism, let $x \in X$, and let y be its image in Y. For g to be quasi-finite (respectively, unramified) at x, it is necessary and sufficient that the same be true of $g \otimes_Y \kappa(y)$.*

Indeed, the two algebras over $\kappa(g(x))$ that one must examine to verify that the morphism is quasi-finite (respectively, unramified) at x are the same for g and for $g \otimes_Y \kappa(y)$.

Corollary I.5.9. *With the notation of Corollary I.5.8, suppose that X and X' are flat and of finite type over Y. For g to be flat (respectively, étale) at x, it is necessary and sufficient that $g \otimes_Y \kappa(y)$ be so.*

For “flat,” the statement is included only as a reminder: it is one of the fundamental criteria for flatness.[6] For “étale,” the assertion follows from this, taking Corollary I.5.8 into account.

I.6. Application to étale extensions of complete local rings

This number is a special case of results on formal preschemes that will have to appear in the *multiplodoque*. Nevertheless, here one can get by more cheaply, that is, without the explicit local determination of étale morphisms in no. I.7 (using the Main Theorem). This is perhaps a sufficient reason to keep the present number (even in the *multiplodoque*) in this place.

Theorem I.6.1. *Let A be a complete local ring (noetherian, of course), with residue field k. For every A-algebra B, put*

$$R(B) = B \otimes_A k,$$

[6] Cf. Exposé IV, no. 5.9.

regarded as a k-algebra; it thus depends functorially on B. Then R defines an equivalence *between the category of A-algebras* finite and étale over A *and the category of algebras* of finite rank and separable *over k.*

First, the functor in question is fully faithful, as follows from the more general fact:

COROLLARY I.6.2. *Let B and B' be two A-algebras finite over A. If B is étale over A, then the canonical map*

$$\mathrm{Hom}_{A\text{-}alg}(B, B') \longrightarrow \mathrm{Hom}_{k\text{-}alg}(R(B), R(B'))$$

is bijective.

We are reduced to the case in which A is artinian (by replacing A with $A/\mathfrak{m}^n$); this is then a special case of Theorem I.5.5.

It remains to prove that for every finite separable k-algebra L (why not say "étale"? It is shorter), there is an algebra B, étale over A, such that $R(B)$ is isomorphic to L. We may suppose that L is a separable extension of k; as such, it has a generator x, that is, it is isomorphic to an algebra $k[t]/Fk[t]$, where $F \in k[t]$ is a monic polynomial. Lift F to a monic polynomial $F_1 \in A[t]$, and take

$$B = A[t]/F_1A[t].$$

I.7. Local construction of unramified and étale morphisms

PROPOSITION I.7.1. *Let A be a noetherian ring, B a finite A-algebra, u a generator of B over A, and $F \in A[t]$ such that $F(u) = 0$ (we do not assume that F is monic). Put $u' = F'(u)$, where F' is the derivative of F. Let $\mathfrak{q}$ be a prime ideal of B that does not contain u', and let $\mathfrak{p}$ be its contraction to A. Then $B_{\mathfrak{q}}$ is unramified over $A_{\mathfrak{p}}$.*

In other words, put $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and $X_{u'} = \mathrm{Spec}(B_{u'})$. Then $X_{u'}$ is unramified over Y. The assertion follows from the following, more precise one.

COROLLARY I.7.2. *The different ideal of B/A contains $u'B$, and is equal to it if the natural homomorphism*

$$A[t]/FA[t] \longrightarrow B,$$

which sends t to u, is an isomorphism.

Let J be the kernel of the homomorphism $C = A[t] \to B$. This kernel contains $FA[t]$, and is equal to it in the second case considered in Corollary I.7.2. Since the homomorphism is surjective, $\Omega^1_{B/A}$ is identified with the quotient of $\Omega^1_{C/A}$ by the submodule generated by $J\Omega^1_{C/A}$ and $d(J)$ (the definition of the homomorphism d, and the computation of Ω^1 for a polynomial algebra, should have been made explicit in no. I.1). Identifying $\Omega^1_{C/A}$ with C by means of the basis dt, one obtains

$$B/(B \cdot J'),$$

where J' is the set of images in B of the derivatives of the polynomials $G \in J$; thus the different ideal is generated by J' (and it is enough to take polynomials G that generate J). Since $F \in J$, respectively since F is a generator of J, the proof is complete. (N.B. Corollary I.7.2 should be a proposition and Proposition I.7.1 a corollary.) We obtain:

COROLLARY I.7.3. *Under the hypotheses of Proposition I.7.1, suppose that F is monic and that $A[t]/FA[t] \to B$ is an isomorphism. Then $B_{\mathfrak{q}}$ is étale over $A_{\mathfrak{p}}$ if and only if $\mathfrak{q}$ does not contain u'.*

Indeed, since B is flat over A, being étale is equivalent to being unramified, and Corollary I.7.2 applies.

COROLLARY I.7.4. *Under the hypotheses of Corollary I.7.3, B is étale over A if and only if u' is invertible, or, equivalently, if and only if the ideal generated by F and F' in $A[t]$ is the unit ideal.*

The last criterion follows from the first and Nakayama's lemma (in B).

A monic polynomial $F \in A[t]$ having the property stated in Corollary I.7.4 is called a *separable polynomial.* If F is not monic, one should at least require the coefficient of its leading term to be invertible; when A is a field, this recovers the usual definition.

COROLLARY I.7.5. *Let B be a* finite *algebra over the* local *ring A. Suppose that $K(A)$ is infinite or that B is local. Let n be the rank of $L = B \otimes_A K(A)$ over $K(A) = k$. Then B is unramified (respectively, étale) over A if and only if B is isomorphic to a quotient of (respectively, isomorphic to)*

$$A[t]/FA[t],$$

where F is a monic separable polynomial that may be assumed (respectively, is necessarily) to have degree n.

Only necessity has to be proved. Suppose that B is unramified over A, so that L is separable over k. The hypothesis then implies that L/k has a generator ξ, and hence the ξ^i, $0 \leq i < n$, form a basis of L over k. Let $u \in B$ lift ξ. By Nakayama's lemma, the u^i, $0 \leq i < n$, generate the A-module B (respectively, form a basis of it). In particular, one can find a monic polynomial $F \in A[t]$ such that $F(u) = 0$, and B is isomorphic to a quotient of (respectively, isomorphic to) $A[t]/FA[t]$. Finally, by Corollary I.7.4 applied to L/k, F and F' generate $A[t]$ modulo $\mathfrak{m}A[t]$. Hence, by Nakayama's lemma in $A[t]/FA[t]$, F and F' generate $A[t]$.

THEOREM I.7.6. *Let A be a local ring, and let $A \to \mathcal{O}$ be a local homomorphism such that $\mathcal{O}$ is isomorphic to a localization of an algebra of finite type over A. Suppose that $\mathcal{O}$ is unramified over A. Then there exist an A-algebra B integral over A, a maximal ideal $\mathfrak{n}$ of B, a generator u of B over A, and a monic polynomial $F \in A[t]$, such that*

$$F'(u) \notin \mathfrak{n} \qquad \textit{and} \qquad \mathcal{O} \simeq B_{\mathfrak{n}}$$

as A-algebras. If $\mathcal{O}$ is étale over A, one may take $B = A[t]/FA[t]$.

(Of course, the conditions displayed here are also sufficient...)

Let us first point out the pleasant corollaries.

COROLLARY I.7.7. *The algebra $\mathcal{O}$ is unramified over A if and only if it is isomorphic to a quotient of an algebra of the same form that is étale over A.*

Indeed, take $\mathcal{O}' = B'_{\mathfrak{n}'}$, where $B' = A[t]/FA[t]$ and $\mathfrak{n}'$ is the inverse image of $\mathfrak{n}$ in B'.

COROLLARY I.7.8. *Let $f\colon X \to Y$ be a morphism of finite type and let $x \in X$. Then f is unramified at x if and only if there is an open neighborhood U of x such that $f|_U$ factors as*

$$U \longrightarrow X' \longrightarrow Y,$$

where the first arrow is a closed immersion and the second is an étale morphism.

This is a straightforward translation of Corollary I.7.7.

Let us show how the statement of Theorem I.7.6 follows from the main assertion. By Corollary I.7.7, there is a surjective homomorphism $\mathcal{O}' \to \mathcal{O}$, where $\mathcal{O}'$ has the desired form.If $\mathcal{O}$ is étale over A, then both $\mathcal{O}'$ and $\mathcal{O}$ are étale over A; the morphism $\mathcal{O}' \to \mathcal{O}$ is therefore étale by Corollary I.4.8, and hence is an isomorphism.

Proof of Theorem I.7.6. This repeats a proof from the Chevalley seminar. By the *Main Theorem*, one has $\mathcal{O} = B_{\mathfrak{n}}$, where B is a finite algebra over A and $\mathfrak{n}$ is a maximal ideal of B. Then $B/\mathfrak{n} = K(\mathcal{O})$ is a separable, hence monogenic, extension of k. If $\mathfrak{n}_i$, $1 \leq i \leq r$, are the maximal ideals of B distinct from $\mathfrak{n}$, there is therefore an element $u \in B$ that belongs to every $\mathfrak{n}_i$ and whose image in $B/\mathfrak{n}$ generates that algebra. Moreover,

$$B/\mathfrak{n} = B_{\mathfrak{n}}/\mathfrak{n}B_{\mathfrak{n}} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}},$$

where $\mathfrak{m}$ is the maximal ideal of A. Let us admit for the moment the following lemma.

LEMMA I.7.9. *Let A be a local ring, B a finite A-algebra, and $\mathfrak{n}$ a maximal ideal of B. Let $u \in B$ be an element whose image in $B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$ generates that algebra over $k = A/\mathfrak{m}$, and which belongs to every maximal ideal of B distinct from $\mathfrak{n}$. Put*

$$B' = A[u], \qquad \mathfrak{n}' = \mathfrak{n} \cap B'.$$

Then the canonical homomorphism

$$B'_{\mathfrak{n}'} \longrightarrow B_{\mathfrak{n}}$$

is an isomorphism.

LEMMA I.7.10. *(This should have appeared as a corollary to Proposition I.7.1, before Corollary I.7.5, which it implies.) Let B be a finite A-algebra generated by an element u, and let $\mathfrak{n}$ be a maximal ideal of B such that $B_{\mathfrak{n}}$ is unramified over A. Then there is a monic polynomial $F \in A[t]$ such that*

$$F(u) = 0 \qquad \textit{and} \qquad F'(u) \notin \mathfrak{n}.$$

Indeed, let n be the rank of the k-algebra $L = B \otimes_A k$. By Nakayama's lemma there is a monic polynomial $F \in A[t]$ of degree n such that $F(u) = 0$. Let f be the reduction of F modulo $\mathfrak{m}$. Then L is isomorphic, as a k-algebra, to $k[t]/fk[t]$. Hence, by Corollary I.7.3, $f'(\xi)$ does not belong to the maximal ideal of L corresponding to $\mathfrak{n}$, where ξ denotes the image of t in L, that is, the image of u in L. Since $f'(\xi)$ is the image of $F'(u)$, the proof is complete.

Theorem I.7.6 now follows by combining Lemmas I.7.9 and I.7.10. It remains to prove Lemma I.7.9. Put $S' = B' \setminus \mathfrak{n}'$, so that $B'S'^{-1} = B'_{\mathfrak{n}'}$. We have the diagram

$$\begin{array}{ccccc} B & \longrightarrow & BS'^{-1} & \longrightarrow & BS^{-1} = B_{\mathfrak{n}} \\ \uparrow & & \uparrow & & \\ B' & \longrightarrow & B'S'^{-1} = B'_{\mathfrak{n}'} & & \end{array}$$

where, similarly, $S = B \setminus \mathfrak{n}$, so that $BS^{-1} = B_{\mathfrak{n}}$. There is therefore a natural homomorphism

$$BS'^{-1} \longrightarrow BS^{-1} = B_{\mathfrak{n}}.$$

We prove that it is an isomorphism. Equivalently, the elements of S must be invertible in BS'^{-1}; equivalently, every maximal ideal $\mathfrak{p}$ of the latter ring must be disjoint from S, that is, must induce $\mathfrak{n}$ on B. Since BS'^{-1} is finite over $B'S'^{-1} = B'_{\mathfrak{n}'}$, the ideal $\mathfrak{p}$ contracts to the unique maximal ideal $\mathfrak{n}'B'_{\mathfrak{n}'}$ of $B'_{\mathfrak{n}'}$,and hence to the maximal ideal $\mathfrak{n}'$ of B'. Since B is finite over B', the ideal $\mathfrak{q}$ of B induced by $\mathfrak{p}$, lying over $\mathfrak{n}'$, is necessarily maximal. It does not contain u, and is therefore equal to $\mathfrak{n}$. Here we used the fact that u belongs to every maximal ideal of B distinct from $\mathfrak{n}$.

It remains to prove that $BS'^{-1} = B'S'^{-1}$. Since the former is finite over the latter, Nakayama's lemma reduces us to proving equality modulo $\mathfrak{n}'BS'^{-1}$, and it is enough, a fortiori, to prove equality modulo $\mathfrak{m}BS'^{-1}$. But

$$BS'^{-1}/\mathfrak{m}BS'^{-1} = B_{\mathfrak{n}}/\mathfrak{m}B_{\mathfrak{n}}$$

is generated over k by u (here we use the other property of u). Thus the image of B', and a fortiori of $B'S'^{-1}$, in this quotient is the whole ring, since it is a subring containing k and the image of u.

Remark *. It should be possible to state Theorem I.7.6 for an algebra $\mathcal{O}$ that is only semilocal, so as to subsume Corollary I.7.5 as well. One would assume that $\mathcal{O}/\mathfrak{m}\mathcal{O}$ is a *monogenic* k-algebra. One could then find $u \in B$ whose image in $B/\mathfrak{m}B$ is a generator and which belongs to every maximal ideal of B not arising from $\mathcal{O}$. Lemmas I.7.9 and I.7.10 should adapt without difficulty. More generally, ...

I.8. Infinitesimal lifting of étale schemes. Application to formal schemes

PROPOSITION I.8.1. *Let Y be a prescheme, let Y_0 be a subprescheme, let X_0 be an étale Y_0-scheme, and let x be a point of X_0. Then there exist an étale Y-scheme X, a neighborhood U_0 of x in X_0, and a Y_0-isomorphism*

$$U_0 \xrightarrow{\sim} X \times_Y Y_0.$$

Indeed, let y be the image of x in Y_0. Applying Theorem I.7.6 to the étale local homomorphism $A_0 \to B_0$ of the local rings at y and x in Y_0 and X_0, respectively, one obtains an isomorphism

$$B_0 = (C_0)_{\mathfrak{n}_0}, \qquad C_0 = A_0[t]/F_0A_0[t],$$

where F_0 is monic and $\mathfrak{n}_0$ is a maximal ideal of C_0 that does not contain the class of $F_0'(t)$ in C_0. Let A be the local ring at y in Y, and let $F \in A[t]$ be a monic polynomial whose image under the surjective homomorphism $A \to A_0$ is F_0 (one lifts the coefficients of F_0). Finally, put

$$C = A[t]/FA[t],$$

and let $\mathfrak{n}$ be the inverse image of $\mathfrak{n}_0$ under the natural epimorphism

$$C \longrightarrow C \otimes_A A_0 = C_0.$$

Put

$$B = C_{\mathfrak{n}}.$$

It follows immediately from the construction and Proposition I.7.1 that B is étale over A, and there is an isomorphism

$$B \otimes_A A_0 \xrightarrow{\sim} B_0.$$

We know from EGA, Chapter I, that there exist a Y-scheme X of finite type and a point $z \in X$ above y such that $\mathcal{O}_z$ is A-isomorphic to B.Since the latter is étale over $A = \mathcal{O}_y$, by taking X small enough we may assume that X is étale over Y. Put $X_0' = X \times_Y Y_0$. The local ring at z in X_0' is then identified with

$$\mathcal{O}_z \otimes_A A_0 = B \otimes_A A_0,$$

and is consequently isomorphic to B_0. This isomorphism is induced by an isomorphism from a neighborhood U_0 of x in X_0 onto a neighborhood of z in X_0' (loc. cit.); by taking X small enough, we may assume that the latter neighborhood is all of X_0'. This proves the proposition.

COROLLARY I.8.2. *There is an analogous statement for étale* coverings*, provided the residue field $\kappa(y)$ is infinite.*

The proof is the same, with Corollary I.7.5 in place of Theorem I.7.6.

THEOREM I.8.3. *The functor considered in Theorem I.5.5 is an* equivalence *of* categories.

By Theorem I.5.5, it remains to show that every étale S_0-scheme X_0 is isomorphic to an S_0-scheme $X \times_S S_0$, where X is an étale S-scheme. The underlying topological space of X must necessarily be identical with that of X_0, with X_0 moreover identified with a closed subprescheme of X. The problem is therefore equivalent to the following one: on the topological space $|X_0|$ underlying X_0, find a sheaf of algebras $\mathcal{O}_X$ over $f_0^*(\mathcal{O}_S)$ (where $f_0\colon X_0 \to S_0$ is here regarded as a continuous map of the underlying spaces) that makes $|X_0|$ into an étale S-prescheme X, together with a homomorphism of algebras

$$\mathcal{O}_X \longrightarrow \mathcal{O}_{X_0}$$

compatible with the homomorphism

$$f_0^*(\mathcal{O}_S) \longrightarrow f_0^*(\mathcal{O}_{S_0})$$

on the scalar sheaves, and inducing an isomorphism

$$\mathcal{O}_X \otimes_{f_0^*(\mathcal{O}_S)} f_0^*(\mathcal{O}_{S_0}) \xrightarrow{\sim} \mathcal{O}_{X_0}.$$

Then X will be an étale S-prescheme whose reduction is X_0, and hence will be separated over S, since X_0 is separated over S_0; thus X answers the question. Moreover, if (U_i) is an open covering of X_0, and if a solution to the problem has been found on each U_i, it follows from the uniqueness theorem I.5.5 that these solutions glue (that is, the sheaves of algebras that define them, equipped with their augmentation homomorphisms, glue). One checks at once that the ringed space over S constructed in this way is an étale S-prescheme X, equipped with an isomorphism

$$X_0 \xrightarrow{\sim} X \times_S S_0.$$

It is therefore enough to find a solution locally, and this is provided by Proposition I.8.1.

Corollary I.8.4. *Let S be a locally noetherian formal prescheme, equipped with an ideal of definition $\mathcal{J}$,and let*

$$S_0 = (|S|, \mathcal{O}_S/\mathcal{J})$$

be the corresponding ordinary prescheme. Then the functor

$$\mathfrak{X} \longmapsto \mathfrak{X} \times_S S_0$$

from the category of étale coverings of S to the category of étale coverings of S_0 is an equivalence of categories.

Of course, an étale covering of a *formal* prescheme S means a covering of S, that is, a formal prescheme over S defined by a coherent sheaf of algebras $\mathcal{B}$, such that $\mathcal{B}$ is *locally free* and the residual fibers

$$\mathcal{B}_s \otimes_{\mathcal{O}_s} \kappa(s)$$

of $\mathcal{B}$ are *separable* algebras over $\kappa(s)$. If S_n denotes the ordinary prescheme

$$S_n = (|S|, \mathcal{O}_S/\mathcal{J}^{n+1}),$$

then giving a coherent sheaf of algebras $\mathcal{B}$ on S is equivalent to giving a sequence of coherent sheaves of algebras $\mathcal{B}_n$ on the S_n, equipped with a transitive system of homomorphisms

$$\mathcal{B}_m \longrightarrow \mathcal{B}_n \qquad (m \geq n)$$

that define isomorphisms

$$\mathcal{B}_m \otimes_{\mathcal{O}_{S_m}} \mathcal{O}_{S_n} \xrightarrow{\sim} \mathcal{B}_n.$$

It is immediate that $\mathcal{B}$ is locally free if and only if all the $\mathcal{B}_n$ on the S_n are locally free, and that the separability condition holds if and only if it holds for $\mathcal{B}_0$, or, equivalently, for all the $\mathcal{B}_n$. Thus $\mathcal{B}$ is étale over S if and only if the $\mathcal{B}_n$ are étale over the S_n. In view of this, Corollary I.8.4 follows at once from Theorem I.8.3.

Remark *. It was not necessary in Corollary I.8.4 to restrict oneself to the case of *coverings*. This is, however, the only case used for the moment.

I.9. Permanence properties

Let $A \to B$ be a local étale homomorphism. We examine here several cases in which a given property of A implies the same property for B, or conversely. A number of such assertions already follow from the simple fact that B is *quasi-finite* and *flat* over A, and we shall confine ourselves to "recalling" a few of them: *A and B have the same Krull dimension and the same depth* (Serre's "cohomological codimension", in the terminology still in use). It follows, for example, that *A is Cohen–Macaulay if and only if B is.* Moreover, for every prime ideal $\mathfrak{q}$ of B whose contraction to A is $\mathfrak{p}$, the ring $B_{\mathfrak{q}}$ is again quasi-finite and flat over $A_{\mathfrak{p}}$, provided that B is a localization of an A-algebra of finite type. This follows from the fact that, for a morphism of finite type, the locus where it is quasi-finite, respectively flat, is open. Furthermore, *every* prime ideal $\mathfrak{p}$ of A is the contraction of a prime ideal $\mathfrak{q}$ of B, since B is *faithfully* flat over A. It follows, for example, that *$\mathfrak{p}$ and $\mathfrak{q}$ have the same height*, and also that *A has no embedded prime ideal if and only if B has none.*

We shall therefore restrict ourselves to assertions that are more specific to étale morphisms.

Proposition I.9.1. *Let $A \to B$ be a local étale homomorphism. Then A is regular if and only if B is regular.*

Indeed, let k be the residue field of A, and let L be that of B. Since B is flat over A and $L = B \otimes_A k$, that is, $\mathfrak{n} = \mathfrak{m}B$, where $\mathfrak{m}$ and $\mathfrak{n}$ are the maximal ideals of A and B, respectively, the $\mathfrak{m}$-adic filtration on B is identical to its $\mathfrak{n}$-adic filtration, and hence

$$\mathrm{gr}^*(B) = \mathrm{gr}^*(A) \otimes_k L.$$

It follows that $\mathrm{gr}^*(B)$ is a polynomial algebra over L if and only if $\mathrm{gr}^*(A)$ is a polynomial algebra over k. This proves the proposition. (N.B. We did not use the fact that L/k is separable.)

Corollary I.9.2. *Let $f\colon X \to Y$ be an étale morphism. If Y is regular, then X is regular; the converse holds if f is surjective.*

Proposition I.9.2. *Let $f\colon X \to Y$ be an étale morphism. If Y is reduced, then so is X; the converse holds if f is surjective.*

This is equivalent to the following corollary.

Corollary I.9.3. *Let $f\colon A \to B$ be a local étale homomorphism, with B isomorphic to a localization of an A-algebra of finite type. Then A is reduced if and only if B is reduced.*

If B is reduced, then A is reduced, since $A \to B$ is injective (because B is faithfully flat over A). Conversely, suppose that A is reduced, and let $\mathfrak{p}_i$ be the minimal prime ideals of A. By hypothesis, the natural map $A \to \prod_i A/\mathfrak{p}_i$ is injective. Tensoring with the flat A-module B shows that $B \to \prod_i B/\mathfrak{p}_i B$ is injective, so it is enough to prove that the rings $B/\mathfrak{p}_i B$ are reduced. Since $B/\mathfrak{p}_i B$ is étale over $A/\mathfrak{p}_i$, we are reduced to the case where A is an integral domain.

Let K be its field of fractions. Since $A \to K$ is injective, so is $B \to B \otimes_A K$, because B is flat over A. It therefore remains to prove that the latter ring is reduced. Now B, being a localization of an A-algebra of finite type, is the local ring at a point x of a scheme $X = \mathrm{Spec}(C)$ of finite type and étale over $Y = \mathrm{Spec}(A)$. Consequently $B \otimes_A K$ is a localization, with respect to a suitable multiplicatively closed set, of the ring $C \otimes_A K$ of $X \otimes_A K$. Since $X \otimes_A K$ is étale over K, its ring is a finite product of fields that are finite separable extensions of K; the same is therefore true of $B \otimes_A K$. This proves the corollary.

Corollary I.9.4. *Let $f\colon A \to B$ be a local étale homomorphism, and suppose that A is analytically reduced (that is, the completion $\widehat{A}$ of A has no nilpotent elements). Then B is analytically reduced, and a fortiori reduced.*

Indeed, $\widehat{B}$is *finite* and étale over $\widehat{A}$, so Corollary I.9.3 applies.

Theorem I.9.5. *Let $f\colon A \to B$ be a local homomorphism, with B isomorphic to a localization of an A-algebra of finite type. Then:*

(i) *If f is étale, then A is normal if and only if B is normal.*
(ii) *If A is normal, then f is étale if and only if f is injective and unramified (and in that case B is normal by* (i)*).*

We shall give two different proofs of (i). The first uses certain properties of quasi-finite flat morphisms (recalled at the beginning of this number) without using Theorem I.7.6 (and hence the Main Theorem); in the second proof the roles are reversed. Finally, for (ii), it seems that the Main Theorem is needed in any case.

First proof. We use the following necessary and sufficient condition for a noetherian local ring A of nonzero dimension to be normal.

Serre's criterion. (i) *For every prime ideal $\mathfrak{p}$ of A of height* 1*, $A_{\mathfrak{p}}$ is normal (or, equivalently, regular).*
(ii) *For every prime ideal $\mathfrak{p}$ of A of height ≥ 2, one has* $\operatorname{depth} A_{\mathfrak{p}} \geq 2$.[7]

We shall admit this criterion here; it is supposed to appear in the section on flat morphisms. Its principal advantage is that it does not assume a priori that A is reduced, nor a fortiori that it is an integral domain. Here we may already assume $\dim A = \dim B \neq 0$.

By the facts recalled at the beginning of this number, the prime ideals $\mathfrak{p}$ of A of height 1 (respectively, height ≥ 2) are exactly the contractions to A of the prime ideals $\mathfrak{q}$ of B of height 1 (respectively, height ≥ 2). Finally, if $\mathfrak{p}$ and $\mathfrak{q}$ correspond, then $B_{\mathfrak{q}}$ is étale over $A_{\mathfrak{p}}$, so it has the same depth as $A_{\mathfrak{p}}$, and it is regular if and only if $A_{\mathfrak{p}}$ is regular (Proposition I.9.1). Applying Serre's criterion, we find that A is normal if and only if B is normal.

Second proof. Suppose that B is normal, and let L be its field of fractions and K that of A (the ring A is an integral domain because B is). We saw in the proof of Corollary I.9.3 that $B \otimes_A K$ is a finite product of fields. Since it is contained in L, it is a field, and since it contains B, it is L. An element of K that is integral over A is integral over B, so it belongs to B, since B is normal. It therefore belongs to A, because $B \cap K = A$ (as follows from the faithful flatness of B over A).

Now suppose that A is normal; we prove that B is normal. By Theorem I.7.6, we have $B = B'_{\mathfrak{n}}$, where

$$B' = A[t]/FA[t],$$

with F and $\mathfrak{n}$ as in Theorem I.7.6. Thus $L = B \otimes_A K$ is a localization of

$$B' \otimes_A K = K[t]/FK[t]$$

and is a product of fields, each a finite separable extension of K. This product, as happens whenever an artinian ring—here $B'_K = B' \otimes_A K$—is localized at a multiplicatively closed set, is a direct factor of B'_K. It therefore corresponds to a factorization $F = F_1F_2$ in $K[t]$, and the generator of L corresponding to t is already annihilated by F_1.

But *because A is normal, the polynomials F_i belong to $A[t]$* (when they are taken to be monic). The map $B \to L = B \otimes_A K$ is injective, since $A \to K$ is injective and B is flat over A. Consequently, we already have $F_1(u) = 0$, where u is the class of t in L. If F has been chosen of minimal degree, it follows that $F_2 = 1$. (N.B. one has

$$F'(u) = F_1'(u)F_2(u) + F_1(u)F_2'(u) = F_1'(u)F_2(u),$$

[7] Cf. EGA IV 5.8.6.

because $F_1(u) = 0$; hence $F_1'(u) \neq 0$, since $F'(u) \neq 0$.)

Thus

$$(*) \qquad L = B \otimes_A K = K[t]/FK[t].$$

Consequently, F is separable over K, though of course not necessarily separable over A.(N.B. for the moment, we have essentially shown only that in Theorem I.7.6 one may choose F and $\mathfrak{n}$ in such a way that, with the notation used here, the map

$$B' \longrightarrow B'_{\mathfrak{n}} = B$$

is *injective*. We used the normality of A for this; I do not know whether the assertion remains true without the normality hypothesis.)

We now recall the well-known lemma extracted from Serre's course of last year.

Lemma I.9.6. *Let K be a ring, let $F \in K[t]$ be a monic separable polynomial, let $L = K[t]/FK[t]$, and let u be the class of t in L (so that $F'(u)$ is a unit of L). Then, with $n = \deg F$,*

$$\begin{aligned} \operatorname{Tr}_{L/K}\left(\frac{u^i}{F'(u)}\right) &= 0 && \text{if } 0 \leq i < n-1, \\ \operatorname{Tr}_{L/K}\left(\frac{u^{n-1}}{F'(u)}\right) &= 1. \end{aligned}$$

Corollary I.9.7. *The determinant of the matrix*

$$\left(\operatorname{Tr}_{L/K}\left(\frac{u^j u^i}{F'(u)}\right)\right)_{0 \leq i,j \leq n-1}$$

is equal to $(-1)^{n(n-1)/2}$, and hence is a unit in every subring A of K.

Corollary I.9.8. *Let A be a subring of K, and suppose that $F \in A[t]$.Let V be the A-submodule of L generated by the elements u^i, $0 \leq i \leq n-1$, and let V' be the A-submodule of L consisting of the elements $x \in L$ such that*

$$\operatorname{Tr}_{L/K}(xy) \in A$$

for every $y \in V$ (that is, for $y = u^i$, $0 \leq i \leq n-1$). Then V' has as an A-basis the elements

$$\frac{u^i}{F'(u)}, \qquad 0 \leq i \leq n-1.$$

Corollary I.9.9. *Suppose that K is the field of fractions of a normal integral domain A, and that the coefficients of F belong to A. With the notation of Corollary I.9.8, the module V' then contains the integral closure A' of A in L. Consequently,*

$$A' \subseteq \frac{A[u]}{F'(u)} = F'(u)^{-1}A[u] \subseteq A[u][F'(u)^{-1}].$$

Apply the last corollary to the situation obtained in the proof. Since $F'(u)$ is a unit in B, and B contains $A[u]$, the ring B contains A'. By the Main Theorem (or directly from $B = A[u]_{\mathfrak{n}}$), B is a localization of A'. Since A' is normal, so is B.

Proof of (ii). We proceed as in the preceding proof to show that, in Theorem I.7.6, F can be chosen so that (*) still holds. The only a priori obstacle is that, since B is no longer assumed flat over A, we can no longer assert that $B \to L$ is injective; hence, a priori, the argument applies only to the image B_1 of B under that homomorphism. It follows at once that B_1 is flat over A (being a localization of an A-algebra that is free as an A-module). By Corollary I.4.8, the morphism $B \to B_1$ is étale, hence an isomorphism, which completes the proof.

(From the editorial point of view, the last two proofs should be interchanged, and the formal calculations of the lemma and its corollaries should be put in a separate number.)

Corollary I.9.10. *Let $f\colon X \to Y$ be an étale morphism. If Y is normal, then X is normal; the converse holds if f is surjective.*

Corollary I.9.11. *Let $f\colon X \to Y$ be a dominant morphism, with Y normal and X connected. If f is unramified, then f is étale; hence X is normal and therefore (being connected) irreducible.*

Let U be the set of points at which f is étale. It is open, and it suffices to show that it is also closed and nonempty. The set U contains the inverse image of the generic point of Y (for an algebra over a field, unramified is equivalent to étale); this inverse image is nonempty because X dominates Y. If x lies in the closure of U, then it lies in the closure of an irreducible component U_i of U, hence on an irreducible component $X_i \overset{v}{=} \overline{U_i}$of X that meets U, and consequently dominates Y (for every irreducible component of U, being flat over Y, dominates Y). Let $y = f(x)$ be the image of x in Y. Then the homomorphism $\mathcal{O}_y \to \mathcal{O}_x$ is *injective* (since $\mathcal{O}_y$ is an integral domain). Since $\mathcal{O}_y$ is normal and $\mathcal{O}_y \to \mathcal{O}_x$ is unramified, Theorem I.9.5(ii) shows that this local homomorphism is étale. Thus $x \in U$, so U is closed. Since U is nonempty and X is connected, $U = X$, which proves the corollary.

Corollary I.9.12. *Let $f\colon X \to Y$ be a dominant morphism of finite type, with Y normal and X irreducible. Then the set of points at which f is étale coincides with the complement of the support of $\Omega^1_{X/Y}$, that is, with the complement of the closed subprescheme of X defined by the different ideal $\mathfrak{d}_{X/Y}$.*

(This is the “less trivial” statement alluded to in the remark in no. I.4.)

Remark *. One should beware of assuming that a connected étale covering of an irreducible scheme is itself irreducible when the base is not assumed to be normal. This question will be studied in no. I.11.

I.10. Étale coverings of a normal scheme

Proposition I.10.1. *Let X be a prescheme that is étale and separated over a connected normal prescheme Y with function field K. Then the connected components X_i of X are integral, their function fields K_i are finite separable extensions of K, and each X_i is identified with a nonempty open subset of the normalization of X in K_i (and hence X is identified with a dense open subset of the normalization of Y in $R(X) = L = \prod K_i$).*

By Corollary I.9.10, X is normal; a fortiori its local rings are integral domains, so the connected components of X are irreducible. Since X_i is normal, finite, and dominant over Y, it follows from a special case (an almost trivial one, moreover) of the Main Theorem that X_i is an open subset of the normalization of X in the field K_i of X.

Corollary I.10.2. *Under the hypotheses of Proposition I.10.1, X is finite over Y (i.e. an étale covering of Y) if and only if X is isomorphic to the normalization Y' of Y in $L = R(X)$ (the ring of rational functions on X).*

Indeed, we know that this normalization is finite over Y (Y being normal and R/K separable); conversely, if X is finite over Y, then it is finite over Y', so its image in Y' is closed; on the other hand, that image is dense.

An algebra L of finite rank over K will be said to be *unramified over X* (or simply unramified over K, if X is understood) if L is a separable algebra over K, that is, a direct product of separable extensions K_i, and if the normalization Y' of Y in L (the disjoint union of the normalizations of Y in the K_i) is unramified (that is, étale by Corollary I.9.11) over Y. Thus:

Corollary I.10.3. *For every X finite over Y such that every irreducible component of X dominates Y, let $R(X)$ denote the ring of rational functions on X (the product of the* *local rings at the generic points of the irreducible components of* X*), so that* $X \longmapsto R(X)$ *is a functor with values in algebras of finite rank over* $K = R(Y)$*. This functor establishes an equivalence between the category of connected étale coverings of* Y *and the category of extensions* L *of* K *unramified over* Y*.*

The inverse functor is the normalization functor.

Suppose that Y is affine, say $Y = \mathrm{Spec}(A)$, where A is normal with field of fractions K. Let L be a finite K-algebra that is a product of fields. By definition, the normalization Y' of Y in L is then isomorphic to $\mathrm{Spec}(A')$, where A' is the normalization of A in L. To say that L is unramified over Y means that A' is unramified over A, or equivalently that it is étale over A. If A is local, this is equivalent to saying that the local rings $A'_{\mathfrak{n}}$, where $\mathfrak{n}$ ranges over the finite set of maximal ideals of A', that is, its prime ideals lying over the maximal ideal $\mathfrak{m}$ of A, are unramified (that is, étale) over the local ring A. Finally, let us also note that the discriminant criterion (Proposition I.4.10) can be applied in this situation. More generally, a variant of that criterion should be stated as follows, without a preliminary flatness condition when X dominates Y, with Y nevertheless assumed locally integral: the maps $A \longrightarrow B$ and $B \longrightarrow B \otimes_A K = L$ are injective—then $\mathrm{Tr}_{L/K}$ is defined—and $\mathrm{Tr}_{L/K}(xy)$ induces a *fundamental bilinear form* $B \times B \longrightarrow A$; that is, there exist elements $x_i \in B$ ($1 \le i \le n$, where $n = \mathrm{rank}_K L$) such that $\mathrm{Tr}_{L/K}(x_i x_j) \in A$ for all i, j, and $\det\Big(\big(\mathrm{Tr}_{L/K}(x_i x_j)\big)_{1\le i,j\le n}\Big)$ is invertible in A.

The sorites (I.4.6) immediately implies the sorites of unramifiedness in the classical setting:

Proposition I.10.4. *Let* Y *be a normal integral prescheme with function field* K*.* (i) K *is unramified over* Y*.* (ii) *If* L *is an extension of* K *unramified over* Y*, if* Y' *is a normal prescheme with function field* L *that dominates* Y *(for example, the normalization of* Y *in* L*), and if* M *is an extension of* L *unramified over* Y'*, then* M/K *is unramified over* X *(*transitivity *of unramifiedness).* (iii) *Let* Y' *be a normal integral prescheme dominating* Y*, with function field extension* K'/K*; if* L *is an extension of* K *unramified over* Y*, then* $L \otimes_K K'$ *is an extension of* K' *unramified over* Y' *(the* translation *property).*

Moreover:

Corollary I.10.5. *Under the hypotheses of Proposition I.10.4*(iii)*, if* $Y = \mathrm{Spec}(A)$ *and* $Y' = \mathrm{Spec}(A')$*, then* $\bar{A}'$*,the coordinate ring of the normalization of* Y' *in* $L' = L \otimes_K K'$*, is identified with* $\bar{A} \otimes_A A'$*, where* $\bar{A}$ *is the normalization of* A *in* L*.*

Usually, people (who are reluctant to consider nonintegral rings, even when they are direct products of fields) state the translation property in the following (weaker) form:

Corollary I.10.6. *Under the hypotheses of Proposition I.10.4*(iii)*, let* L_1 *be a* composite extension *of* L/K *(unramified over* Y*) and of* K'/K*. Then* L_1/K' *is unramified over* Y'*. If* $Y = \mathrm{Spec}(A)$ *and* $Y' = \mathrm{Spec}(A')$*, one also has*

$$\overline{A'} = A[\overline{A}, A'].$$

That is, the ring $\overline{A'}$*, the normalization of* A' *in* L_1*, is the* A*-algebra generated by* A' *and by the normalization* $\overline{A}$ *of* A *in* L*.*

This last assertion is false without the unramifiedness hypothesis, even for composite extensions of number fields . . .

To conclude this section, we shall give the interpretation of the notion of an étale covering that corresponds to its intuitive picture: there should be the "maximum number" of points above the point $y \in Y$ under consideration, and in particular there should not be "several points merged together" above y. To prove the results in this direction in the desired generality, we shall admit Proposition I.10.7 below (whose proof will appear in the *multiplodoque*, Chapter IV, no. 15, and uses Chevalley's technique of constructible sets and a little descent theory ...).

A morphism of finite type $f\colon X \to Y$ is called *universally open* if, for every base extension $Y' \to Y$ (with Y' locally noetherian), the morphism

$$f'\colon X' = X \times_Y Y' \longrightarrow Y'$$

is open, that is, carries open sets to open sets. It is enough to consider the case in which Y' is of finite type over Y (and even the case in which Y' is of the form $Y[t_1, \dots, t_r]$, where the t_i are indeterminates).

A universally open morphism is *a fortiori* open (the converse is false). On the other hand, if f is open and X and Y are irreducible, then all the components of all the fibers of f have the same dimension, namely the dimension of the generic fiber $f^{-1}(z)$, where z is the generic point of Y. Finally, if Y is normal, this last condition already implies that f is universally open (Chevalley's theorem). It follows, for example, that if $f\colon X \to Y$ is quasi-finite, with Y normal and irreducible, then f is universally open (equivalently, open) if and only if every irreducible component of X dominates Y. Recall also that a flat morphism of finite type is open, and hence universally open. With these preliminaries established, let us "recall" the following proposition.

PROPOSITION I.10.7. *Let $f\colon X \to Y$ be a quasi-finite, separated, universally open morphism. For every $y \in Y$, let $n(y)$ be the "geometric number of points of the fiber $f^{-1}(y)$", equal to the sum of the separable degrees of the residue-field extensions $\kappa(x)/\kappa(y)$, for $x \in f^{-1}(y)$. Then the function $y \longmapsto n(y)$ on Y is upper semicontinuous. If it is constant in a neighborhood of y (that is, if $n(y) = n(z_i)$, where the z_i are the generic points of the irreducible components of Y that contain y), then there must exist a neighborhood U of y such that $X|_U$ is finite over U.* [8]

COROLLARY I.10.8. *If $y \longmapsto n(y)$ is constant and Y is geometrically unibranch,*[9] *then the irreducible components of X are disjoint.*

PROPOSITION I.10.9. *Let $f\colon X \to Y$ be an* étale*, separated morphism. With the notation of Proposition I.10.7, the function $y \longmapsto n(y)$ is upper semicontinuous. For it to be constant in a neighborhood of y (that is, for $n(y) = n(z_i)$, where the z_i are the generic points of the irreducible components of Y that contain y), it is necessary and sufficient that there exist an open neighborhood U of y such that $X|_U$ is finite over U, that is, is an* étale covering *of U.*

COROLLARY I.10.10. *For a separated étale morphism $f\colon X \to Y$, with Y connected, to be finite (that is, to make X an* étale covering *of Y), it is necessary and sufficient that all the fibers of f have the same geometric number of points.*

In Proposition I.10.7 and its corollary there was no normality hypothesis on Y. If such a hypothesis is imposed, one obtains the stronger statement below (which is most often taken as the definition of the unramifiedness of a covering).

THEOREM I.10.11. *Let $f\colon X \to Y$ be a* quasi-finite*, separated morphism. Suppose that Y is* irreducible*, that every component of X dominates Y, and that X is reduced (that is, $\mathcal{O}_X$ has no nilpotent elements). Let n be the degree of X over Y (the sum of the degrees, over the field K of Y, of the fields K_i of the irreducible components X_i of X). Let y be a normal point of Y. Then the geometric number $n(y)$ of points of X above y is at most n, and equality holds if and only if there exists an open neighborhood U of y such that $X|_U$ is an* étale covering *of U.*

[8] Cf. EGA IV, 15.5.1.

[9] For the definition, see no. I.11 below.

The implication from the existence of such an étale neighborhood to $n(y) = n$ is trivial. Let us prove the converse.[10] Let z be the generic point of Y. We have

$$n(z) = \text{the sum of the separable degrees of the } K_i/K \leq n,$$

and Proposition I.10.7 gives $n(y) \leq n(z) \leq n$. Equality implies that $X|_U$ is finite over U for a suitable neighborhood U of y. We may therefore suppose that X is finite over Y and that the function $n(y')$ on Y is constant. By Corollary I.10.8, X is then the disjoint union of its irreducible components; to prove that it is unramified at y, we are reduced to the case in which X is irreducible, hence integral. Finally, we may suppose that $Y = \mathrm{Spec}(\mathcal{O}_y)$. The theorem then reduces to the following classical statement.

Corollary I.10.12. *Let A be a normal local ring (noetherian, as always) with fraction field K; let L be a finite extension of K, of degree n and separable degree n_s; and let B be a subring of L, finite over A, with fraction field L. Let $\mathfrak{m}$ be the maximal ideal of A, and let n' be the separable degree of $B/\mathfrak{m}B$ over $A/\mathfrak{m}A = k$ (that is, the sum of the separable degrees of the residue-field extensions of this ring). Then $n' \leq n_s$, and* a fortiori *$n' \leq n$. Equality in the latter inequality holds if and only if B is unramified (equivalently, étale) over A.*

It remains only to show that $n' = n$ implies that B is étale over A. Recall the proof when k is infinite. It is enough to show that $R = B/\mathfrak{m}B$ is separable over k. If it were not, a familiar lemma would imply that there is an element $a \in R$ whose minimal polynomial over k has degree $> n'$. This element is the image of an element $x \in B$, whose minimal polynomial over K (as an element of L) has degree at most n. Moreover, the coefficients of this last polynomial lie in A, since A is normal. Reduction modulo $\mathfrak{m}$ therefore gives a monic polynomial $F \in k[t]$, of degree at most $n = n'$, such that $F(a) = 0$, a contradiction.

In the general case (where k may be finite), return to geometric language and put $Y = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$. Consider $Y' = \mathrm{Spec}(A[t])$, which is faithfully flat over Y, and the generic point y' of the fiber $\mathrm{Spec}(k[t])$ of Y' over y. Then X is unramified over Y at y if and only if

$$X' = X \times_Y Y' = \mathrm{Spec}(B[t])$$

is unramified at y' over Y', as one sees immediately. By the choice of y', its residue field is $k(t)$, hence infinite. Since y' is a normal point of Y', this reduces the assertion to the preceding case.

I.11. Some complements

We have already said that a connected étale covering of an integral scheme need not be integral. Here are two examples of this fact.

a) Let C be an algebraic curve with an ordinary double point x, let C' be its normalization, and let a and b be the two points of C' above x. Let C'_i $(i = 1, 2)$ be two copies of C', and let a_i and b_i be the points of C'_i corresponding respectively to a and b. In the disjoint union $C'_1 \amalg C'_2$, identify a_1 with b_2, and a_2 with b_1. (The reader is left to make the identification process precise; it will be explained in Chapter VI of the *multiplodoque*, but, for curves over an algebraically closed field, it is treated in Serre's book on algebraic curves.) The result is a *connected* and *reducible* curve C'', which is an étale covering of C of degree 2. More generally, the reader will verify that the connected "Galois" étale coverings C'' of C for which the inverse image $C'' \times_C C'$ is a *trivial* covering of C' (that is, isomorphic to a disjoint union of some number of copies of C') are "cyclic" of degree n, and that, for every integer $n > 0$, one can construct a connected cyclic étale covering of degree n. In the language of the fundamental group, which will be developed later, this

[10]The French text names these two directions in the opposite order; the argument itself starts from $n(y) = n$ and proves the existence of the étale neighborhood, as translated here.

means that the quotient of $\pi_1(C)$ by the closed normal subgroup generated by the image of

$$\pi_1(C') \longrightarrow \pi_1(C)$$

(the homomorphism induced by the projection) is isomorphic to the profinite completion of $\mathbb{Z}$. More precisely, one should be able to show that the fundamental group of C is isomorphic to the topological free product of the fundamental group of C' with the profinite completion of $\mathbb{Z}$. Notice that questions of this kind gave rise to "descent theory" for schemes.

b) Let A be a complete local integral domain. Its normalization A' is known to be finite over A (Nagata); it is therefore a complete semilocal ring, and hence local because it is an integral domain. Suppose that the residue-field extension L/k thereby defined is not radicial (in the contrary case, A will be called geometrically unibranch; see below). This is the case, for example, for the ring

$$\mathbb{R}[[s,t]]/(s^2+t^2)\mathbb{R}[[s,t]],$$

where $\mathbb{R}$ is the field of real numbers. Let k' be a finite Galois extension of k such that $L \otimes_k k'$ splits, and let B be a finite étale algebra over A corresponding to the residue-field extension k' (recall that B is essentially unique). Then

$$B' = A' \otimes_A B$$

over B has residue algebra $L \otimes_k k'$, which is not local. Thus B' is not a local ring and, being complete, has zero divisors. But B' is contained in the total ring of fractions of B: indeed, it is free over A', hence torsion-free over A', hence torsion-free over A, and is consequently contained in

$$B' \otimes_A K = B'_{(K)} = A'_{(K)} \otimes_K B_{(K)} = B_{(K)},$$

since $A'_{(K)} = K$. It follows that B is not an integral domain. In the case of the ring

$$\mathbb{R}[s,t]/(s^2+t^2)\mathbb{R}[s,t],$$

taking $k'/k = \mathbb{C}/\mathbb{R}$, the ring B obtained is the local ring, at their point of intersection, of two intersecting lines in the plane.

Notice also that, if there exists a connected étale covering X of an integral Y that is not irreducible, then every irreducible component of X gives an example of an unramified covering X' of Y, dominating Y, that is not étale over Y. Thus, in example a), C' is unramified over C without being étale at the two points a and b. This can also be seen directly by inspecting the completions of the local rings at x and a: from the "formal" point of view, C' at a is identified with a closed subscheme of C at x, namely one of the two "branches" of C passing through x.

In a) and b), the failure of the conclusions of Proposition I.9.5(i) and (ii) is seen to be directly connected with the fact that a point of Y "splits" into *distinct* points of the normalization. In b), the fact that the residue-field extension is not radicial is to be interpreted geometrically in this way. More precisely, we shall say that a local integral domain A is *geometrically unibranch* if its normalization has only one maximal ideal and the corresponding residue-field extension is radicial. A point y of an integral prescheme is said to be geometrically unibranch if its local ring is so. Examples are a normal point, an ordinary cusp of a curve, and so on. It seems that, if Y has a point that is not unibranch, there always exists a connected, nonirreducible étale covering of Y; this is at least what we have shown in case b), when Y is the spectrum of a complete local ring. On the other hand, one can show that *if every point of Y is geometrically unibranch, then every connected unramified Y-prescheme dominating Y is étale* and irreducible. The proof repeats that of Proposition I.9.5, using the following generalization of Theorem I.8.3, which will be proved later by means of descent theory:[11]

Let $Y' \to Y$ be a finite, radicial, surjective morphism (that is, what might be called a "universal homeomorphism"). Consider the functor

$$X \longmapsto X \times_Y Y' = X'$$

from Y-preschemes to Y'-preschemes. This functor induces an equivalence between the category of étale Y-schemes and the category of étale Y'-schemes. For example, this result may be applied when Y' is the normalization of Y, with Y assumed unibranch (and Y' finite over Y, as is true in all cases encountered in practice), or when a Y'' is "sandwiched" between Y and its normalization; in the latter case the normalization need no longer be finite over Y.

[11]Cf. IX 4.10. For a more direct proof, cf. EGA IV, 18.10.3, using a variant of Proposition I.9.5 for geometrically unibranch local rings.

CHAPTER II

Smooth morphisms: generalities and differential properties

References to Exposé I are indicated by I. Recall that all rings are noetherian and all preschemes locally noetherian.

II.1. Generalities

Let Y be a prescheme, and let $t_1, \dots, t_n$ be indeterminates. We write

$$Y[t_1, \dots, t_n] = Y \otimes_{\mathbb{Z}} \mathbb{Z}[t_1, \dots, t_n]. \tag{II.1.1}$$

Thus $Y[t_1, \dots, t_n]$ is a Y-scheme, affine over Y, defined by the quasi-coherent sheaf of algebras $\mathcal{O}_Y[t_1, \dots, t_n]$. Giving a section of this prescheme over Y is therefore equivalent to giving n sections of $\mathcal{O}_Y$, which correspond to the images of the t_i under the associated homomorphism. If Y' lies over Y, then

$$Y[t_1, \dots, t_n] \times_Y Y' = Y'[t_1, \dots, t_n]. \tag{II.1.2}$$

Consequently, giving a Y-morphism from Y' to $Y[t_1, \dots, t_n]$ is equivalent to giving n sections of $\mathcal{O}_{Y'}$. We also have

$$(Y[t_1, \dots, t_n])[t_{n+1}, \dots, t_m] = Y[t_1, \dots, t_m], \tag{II.1.3}$$

by the analogous formula for polynomial rings over $\mathbb{Z}$. Formula (II.1.2) shows that $Y[t_1, \dots, t_n]$ varies functorially with Y.

The Y-scheme $Y[t_1, \dots, t_n]$ is of finite type and flat over Y.

Definition II.1.1. *Let $f\colon X \to Y$ be a morphism, making X a Y-prescheme. We say that f is* smooth[1] *at $x \in X$, or that X is* smooth over Y at *x, if there exist an integer $n \geq 0$, an open neighborhood U of x, and an étale Y-morphism from U to $Y[t_1, \dots, t_n]$. We say that f is* smooth *(respectively, that X is* smooth over *Y) if it is smooth at every point of X.*

An algebra B over a ring A is said to be smooth at a prime ideal $\mathfrak{p}$ of B if $\operatorname{Spec}(B)$ is smooth over $\operatorname{Spec}(A)$ at $\mathfrak{p}$; B is said to be smooth over A if it is smooth over A at every prime ideal $\mathfrak{p}$ of B. Finally, a local homomorphism $A \to B$ of local rings is said to be smooth (or B is said to be smooth over A) [2] *if B is a localization of an algebra B_1 of finite type and smooth over A.*

The notion that X is smooth over Y is local on X and on Y. If X is smooth over Y, then it is locally of finite type over Y.

Proposition II.1.1. *The set of points x of X at which f is smooth is open.*

This is immediate from the definition.

Corollary II.1.2. *If B is smooth over A at $\mathfrak{p}$, then it is smooth over A at every prime ideal $\mathfrak{q}$ of B contained in $\mathfrak{p}$.*

Proposition II.1.1 also implies that the last two definitions in Definition II.1.1 agree wherever both apply.

[1] Former terminology: f is *simple* at x, or x is a *simple* point for f. This terminology caused confusion in various contexts (simple algebras, simple groups) and had to be abandoned.

[2] It is preferable in this case, as in EGA IV, 18.6.1, to say that B is *essentially smooth* over A.

PROPOSITION II.1.3. (i) *An étale morphism, and in particular an open immersion or an identity morphism, is smooth.*

(ii) *A base extension of a smooth morphism is smooth.*

(iii) *The composite of two smooth morphisms is smooth.*

Assertion (i) is immediate from the definition. More precisely, we have the following.

COROLLARY II.1.4.

$$\textit{étale} = \textit{quasi-finite} + \textit{smooth}.$$

Assertion (ii) follows at once from the analogous fact for étale morphisms (I I.4.6) and for the projections $Y[t_1, \ldots, t_n] \to Y$; see (II.1.2). Assertion (iii) follows formally because it holds separately for étale morphisms (I I.4.6) and for projections of the form $Y[t_1, \ldots, t_n] \to Y$, by (II.1.3), together with the two facts just cited for (ii). Indeed, suppose that Y is smooth over Z and that X is smooth over Y; we prove that X is smooth over Z. We may suppose that Y is étale over $Z[t_1, \ldots, t_n]$, and that X is étale over $Y[s_1, \ldots, s_m]$. The first hypothesis implies that $Y[s_1, \ldots, s_m]$ is étale over

$$(Z[t_1, \ldots, t_n])[s_1, \ldots, s_m] = Z[t_1, \ldots, t_n, s_1, \ldots, s_m].$$

It follows that X is étale over $Z[t_1, \ldots, t_n, s_1, \ldots, s_m]$, as required.

Remark II.1.5. The integer n occurring in Definition II.1.1 is indeed determined: it is immediately seen to be the dimension of the local ring of x in its fiber $f^{-1}(f(x))$. It is called the *relative dimension* of X over Y. Relative dimension is additive under composition of morphisms.

II.2. Some criteria for smoothness of a morphism

THEOREM II.2.1. *Let $f\colon X \to Y$ be a morphism locally of finite type, let $x \in X$, and put $y = f(x)$. Then f is smooth at x if and only if* (a) *f is flat at x, and* (b) *the fiber $f^{-1}(y)$ is smooth over $\kappa(y)$ at x.*

Since the composite of two flat morphisms is flat, and $Y[t_1, \ldots, t_n] \to Y$ is flat, every smooth morphism is flat. Together with Proposition II.1.3(ii), this proves the necessity of the two conditions. Conversely, suppose that (a) and (b) hold. Let V be an affine neighborhood of y, with ring A, and let U be an affine neighborhood of x above V, with ring B. After shrinking U, condition (b) allows us to suppose that there is an étale $\kappa(y)$-morphism

$$g\colon U|_{f^{-1}(y)} \longrightarrow \operatorname{Spec} k[t_1, \ldots, t_n], \qquad k = \kappa(y),$$

defined by n sections g_i of the structure sheaf of $U|_{f^{-1}(y)}$. We may plainly suppose that the g_i, which a priori are elements of

$$B \otimes_A k = BS^{-1}, \qquad S = A - \mathfrak{p},$$

where $\mathfrak{p}$ is the prime ideal of A corresponding to y, come from sections of the structure sheaf of U. Thus g is induced by a morphism, still denoted by g,

$$g\colon U \longrightarrow Y[t_1, \ldots, t_n],$$

after multiplying all the g_i, if necessary, by the same nonzero element of k. Now U is flat over Y by (a), as is $Y[t_1, \ldots, t_n]$; moreover, g induces an étale morphism between the fibers over y. Therefore g is étale at x, by I I.5.8, as required.

COROLLARY II.2.2. *Let S be a prescheme, and let $f\colon X \to Y$ be an S-morphism of finite type, where Y is of finite type and flat over S. Let $x \in X$, and let s be its image in S. Then f is smooth at x if and only if X is flat (or: smooth) over S at x, and the induced morphism on the fibers over s,*

$$f_s\colon X_s \longrightarrow Y_s,$$

is smooth at x.

Only sufficiency requires proof; it follows from the criterion in Theorem II.2.1, together with the flatness criterion I I.5.9.

To state the next result, "recall" that a morphism $f\colon X \to Y$ locally of finite type is said to be *equidimensional* at $x \in X$ if, on writing $y = f(x)$, there is an open neighborhood U of x, every component of which dominates a component of Y, such that, for every $y' \in Y$, all the irreducible components of $f^{-1}(y') \cap U$ have one and the same dimension, independent of y'. In this condition it is enough to take for y' the generic points of the irreducible components of Y passing through y, *and* the point y itself. If, for example, X and Y are integral and f is dominant, the condition says that the components of $f^{-1}(y)$ passing through x have the "correct" dimension, that is, the dimension of the generic fiber. Recall that their dimension is always at least that of the generic fiber.

Suppose that f is equidimensional at x, that the dimension of its fiber at x is n, and that

$$g\colon U \longrightarrow Y' = Y[t_1, \dots, t_n]$$

is a Y-morphism from a neighborhood U of x, inducing on the fibers over y a morphism that is quasi-finite at x. Equivalently, suppose that g itself is quasi-finite at x. One then shows that every irreducible component of U passing through x dominates an irreducible component of Y'. Moreover, by the "normalization lemma", such a g always exists. Conversely, if there is a quasi-finite Y-morphism g from an open neighborhood U of x to a Y-scheme of the form $Y' = Y[t_1, \dots, t_n]$, such that every component of U passing through x dominates a component of Y', then f is equidimensional at x. With this understood, we have:

Proposition II.2.3. *Let $f\colon X \to Y$ be a morphism locally of finite type, let $x \in X$, and put $y = f(x)$. Suppose that $\mathcal{O}_y$ is normal. Then f is smooth at x if and only if f is equidimensional at x and the fiber $f^{-1}(y)$ is smooth over $\kappa(y)$ at x.*

It follows at once from the definition that a smooth morphism is equidimensional. Notice, however, that a flat morphism of finite type need not be equidimensional at x, even if its fiber at x is irreducible. Let us prove the converse. Since $f^{-1}(y)$ is smooth over $\kappa(y)$ at x, we may suppose, after replacing X by a suitable neighborhood of x if necessary, that there is a Y-morphism

$$g\colon X \longrightarrow Y[t_1, \dots, t_n] = Y'$$

which induces an étale morphism on the fibers over y, and is therefore quasi-finite at x. Thus g is unramified, and, since f is equidimensional at x, the irreducible components of X passing through x each dominate an irreducible component of Y'. Consequently, the homomorphism

$$\mathcal{O}_{y'} \longrightarrow \mathcal{O}_x, \qquad y' = g(x),$$

induced by g is injective. This homomorphism is also unramified, and $\mathcal{O}_{y'}$ is normal because it is a localization of $\mathcal{O}_y[t_1, \dots, t_n]$, which is normal since $\mathcal{O}_y$ is normal. Hence $\mathcal{O}_{y'} \to \mathcal{O}_x$ is étale, by I I.9.5(ii).

Remarks II.2.4. The preceding statement remains valid if the hypothesis that $\mathcal{O}_y$ is normal is replaced by the weaker hypothesis that Y is *geometrically unibranch* at y; see I I.11. Indeed, I I.9.5 remains valid under this hypothesis.

Let us also take this opportunity to point out that, if the residue field of a local integral domain A is algebraically closed, then analytic irreducibility—that is, the condition that $\widehat{A}$ be an integral domain—implies geometric unibranchedness. The converse also holds in every category of "good rings", more precisely in a category of rings stable under the usual operations and in which the completion of a normal local ring is normal. By Zariski's "theorem of analytic normality", this condition is satisfied in the category of affine algebras and their localizations.[3]

[3]Cf. EGA IV, 7.8.

Finally, "recall" in the present context the following result of Hironaka,[4] which can sometimes be used to ensure that $f^{-1}(y)$ is a reduced scheme, that is, that it coincides with what many algebraic geometers would loosely regard as the fiber of f above y without multiplicities, namely $f^{-1}(y)_{\mathrm{red}}$.

PROPOSITION II.2.5. *Let $f\colon X \to Y$ be a dominant morphism of finite type between reduced preschemes, and let $y \in Y$ be a point such that $\mathcal{O}_y$ is regular. Suppose that every component of $f^{-1}(y)$ has multiplicity* 1 *(in the sense defined below), and that $f^{-1}(y)_{\mathrm{red}}$ is normal. Then $f^{-1}(y)$ is reduced, hence normal; X is normal at every point of $f^{-1}(y)$; and X is flat over Y at every point of $f^{-1}(y)$.*

A component Z of $f^{-1}(y)$ is said to have *multiplicity* 1 if, with x denoting the generic point of Z, the following conditions hold:

(i) $\dim \mathcal{O}_x = \dim \mathcal{O}_y$, that is, Z is not an "excess component" or, equivalently, is not of "excessively large dimension";

(ii) the maximal ideal of $\mathcal{O}_x$ is generated by the maximal ideal of $\mathcal{O}_y$, which a priori, by the choice of x, generates an ideal of definition of $\mathcal{O}_x$.

Taking Proposition II.2.3, or Theorem II.2.1, into account, we obtain:

COROLLARY II.2.6. *Let $f\colon X \to Y$ be a dominant morphism of finite type between reduced preschemes, and let $y \in Y$ be a point such that $\mathcal{O}_y$ is regular. Then f is smooth at all points of X above y if and only if the components of $f^{-1}(y)$ have multiplicity* 1 *and $f^{-1}(y)_{\mathrm{red}}$ is smooth over $\kappa(y)$.*

In the past, this situation was considered especially when Y was the spectrum of a discrete valuation ring A, and was commonly described in such terms as: "if the reduction of X with respect to the given valuation is nice".... Moreover, X then denoted a closed subscheme (if one may say so) of some $\mathbb{P}^n_K$, where K is the fraction field of A; and, for want of adequate language, the more intrinsic role of an object "defined over A" rather than merely over K was scarcely apparent.

II.3. Permanence properties

PROPOSITION II.3.1. *Let $f\colon X \to Y$ be a morphism, let $x \in X$, and put $y = f(x)$. Suppose that f is smooth at x. Then $\mathcal{O}_x$ is reduced (respectively, regular; respectively, normal) if and only if $\mathcal{O}_y$ has the corresponding property.*

This statement is known when X is of the form $Y[t_1, \ldots, t_n]$, and it was proved in I, no. I.9, for an étale morphism. The general case follows at once from Definition II.1.1.

We shall not detail here the other permanence properties, which already follow either from flatness alone or from the fact that X is locally quasi-finite and flat over a Y-prescheme of the form $Y[t_1, \ldots, t_n]$, that is, as we shall say, from the fact that X is Cohen–Macaulay over Y. We point out only that this latter fact implies

$$\begin{aligned} \dim \mathcal{O}_x &= \dim \mathcal{O}_y + n - d, \\ \operatorname{depth} \mathcal{O}_x &= \operatorname{depth} \mathcal{O}_y + n - d, \end{aligned} \tag{II.3.1}$$

where n is the dimension of the fiber of f at x, and d is the transcendence degree of $\kappa(x)$ over $\kappa(y)$. Thus, on putting

$$\operatorname{codepth} = \dim - \operatorname{depth},$$

we obtain[5]

$$\operatorname{codepth} \mathcal{O}_x = \operatorname{codepth} \mathcal{O}_y. \tag{II.3.2}$$

It follows, for example, that $\mathcal{O}_x$ is Cohen–Macaulay (respectively, has no embedded components) if and only if the same is true of $\mathcal{O}_y$.

[4] Cf. EGA IV, 5.12.10.

[5] For these formulas, cf. EGA IV, 6.1 and 6.3.

II.4. Differential properties of smooth morphisms

For simplicity, we shall for the most part restrict ourselves to first-order differential calculus, giving only brief indications for higher orders, where the results are just as simple.

For the definition of the sheaf $\Omega^1_{X/Y}$ of first differentials of a Y-prescheme X, see I, no. I.1. Suppose that X and Y are S-preschemes and that the morphism $f\colon X \to Y$ is an S-morphism. Then f defines a homomorphism of sheaves of modules, compatible with f,

$$f^*\colon \Omega^1_{Y/S} \longrightarrow \Omega^1_{X/S}. \tag{II.4.1}$$

In other words, $\Omega^1_{X/S}$ is *contravariant* in the S-prescheme X. Equivalently, (II.4.1) gives a homomorphism of modules on X,

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}, \tag{4.1bis}$$

also denoted by f^*, for lack of a better notation. It belongs to the canonical exact sequence

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0. \tag{II.4.2}$$

All these homomorphisms are defined by the requirements that they be local, which reduces their definition to the affine case, and that they commute with the differential operators d. The exactness of (II.4.2) is classical and immediate. In the affine case, for a homomorphism $B \to C$ of A-algebras, it becomes the exact sequence

$$\Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{C/A} \longrightarrow \Omega^1_{C/B} \longrightarrow 0. \tag{4.2bis}$$

Lemma II.4.1. *Let $f\colon X \to Y$ be a morphism of S-preschemes. If f is unramified (respectively, étale), then*

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is surjective (respectively, an isomorphism). The converse in the unramified case is true if f is assumed locally of finite type.

The unramified assertion follows from the exact sequence (II.4.2) and I I.3.1; it may also be seen directly as part of the étale case. For a direct proof of the latter, consider the diagram

$$\begin{array}{ccccc} X & \xrightarrow{\Delta_{X/Y}} & X \times_Y X & \longrightarrow & X \times_S X \\ & & \downarrow & & \downarrow \\ & & Y & \xrightarrow{\Delta_{Y/S}} & Y \times_S Y. \end{array}$$

Here $X \times_Y X$ is the fiber product of Y and $X \times_S X$ over $Y \times_S Y$. Since f is unramified, $X \to X \times_Y X$ is an open immersion. Hence the conormal sheaf of the composite immersion $\Delta_{X/S}$ is isomorphic to the inverse image on X of the conormal sheaf of $X \times_Y X \to X \times_S X$. In the étale case, $X \to Y$ is flat, so $X \times_S X \to Y \times_S Y$ is flat. The conormal sheaf of $X \times_Y X \to X \times_S X$ is therefore the inverse image of the conormal sheaf of $Y \to Y \times_S Y$, that is, the inverse image of $\Omega^1_{Y/S}$. The assertion follows.

Lemma II.4.2. *Let $X = Y[t_1, \dots, t_n]$, where Y is an S-prescheme. Then the sequence of canonical homomorphisms*

$$0 \longrightarrow f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact, and $\Omega^1_{X/Y}$ is free with basis $\mathrm{d}_{X/Y}t_1, \dots, \mathrm{d}_{X/Y}t_n$.

The verification, which is purely affine, is immediate. Notice that the exactness at the right was already known from (II.4.2).

Combining these two statements with Definition II.1.1, we obtain:

Theorem II.4.3. *Let $f\colon X \to Y$ be a smooth morphism of S-preschemes. Then:*

(i) *the sequence of canonical homomorphisms*

$$0 \longrightarrow f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S} \longrightarrow \Omega^1_{X/Y} \longrightarrow 0$$

is exact;

(ii) $\Omega^1_{X/Y}$ *is locally free, and its rank* n *at* x *is equal to the relative dimension of* f *at* x.

Corollary II.4.4. *The homomorphism*

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is injective, and its image in $\Omega^1_{X/S}$ *is locally a direct summand.*

Let $u\colon F \to G$ be a homomorphism of modules on the prescheme X. It is said to be *universally injective* at $x \in X$ if the homomorphism $F_x \to G_x$ of $\mathcal{O}_x$-modules is injective and remains so after tensoring with every $\mathcal{O}_x$-algebra, or, equivalently, with every $\mathcal{O}_x$-module. It is enough, for example, that there exist an open neighborhood U of x on which u identifies $F|_U$ with a direct summand of $G|_U$. This condition is also necessary when F and G are free of finite type in a neighborhood of x. More precisely, in that case the following conditions are equivalent:

(i) u is injective at x, and $\operatorname{Coker}(u)$ is free at x;

(ii) there exists an open neighborhood U of x on which u identifies $F|_U$ with a direct summand of $G|_U$;

(iii) u is universally injective at x;

(iv) the homomorphism on the fibers induced by u,

$$F_x \otimes_{\mathcal{O}_x} \kappa(x) \longrightarrow G_x \otimes_{\mathcal{O}_x} \kappa(x),$$

is injective;

(v) the transpose homomorphism $G^\vee \to F^\vee$ is surjective at x, or, equivalently, in a neighborhood of x.

The implications form a circle: (iv) implies (v) by Nakayama's lemma, while (v) implies (i) because a locally free quotient sheaf is necessarily a direct summand. Geometrically, the situation means that u corresponds to an isomorphism of the vector bundle whose sheaf of sections is F onto a vector subbundle of the analogous vector bundle defined by G. It is, of course, not enough for this that $F \to G$ merely be injective.

Corollary II.4.5. *Let* $f\colon X \to Y$ *be a morphism of* S*-preschemes, locally of finite type. Let* $x \in X$, *put* $y = f(x)$, *and let* s *be the common image of* x *and* y *in* S. *Suppose that* Y *is smooth over* S *at* y. *Then the following conditions are equivalent:*

(i) f *is smooth at* x;

(ii) X *is smooth over* S *at* x, *and*

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is universally injective at x; *that is, it is injective at* x, *and its cokernel* $\Omega^1_{X/Y}$ *is free at* x.

Necessity follows from Proposition II.1.3(iii) and Theorem II.4.3(i), (ii). Let us prove sufficiency. Since the elements $\mathrm{d}g$, $g \in \mathcal{O}_x$, generate $(\Omega^1_{X/Y})_x$, we can choose g_i, $1 \leq i \leq n$, such that the images of the $\mathrm{d}g_i$ form a basis of this module. After shrinking X, we may suppose that the g_i come from sections of $\mathcal{O}_X$, and hence define a Y-morphism

$$g\colon X \longrightarrow Y' = Y[t_1, \ldots, t_n].$$

Using the hypothesis and Lemma II.4.2, one sees immediately that the corresponding homomorphism

$$g^*(\Omega^1_{Y'/S}) \longrightarrow \Omega^1_{X/S}$$

is an isomorphism at x. We are thus reduced to proving the following corollary.

Corollary II.4.6. *Let $f\colon X \to Y$ be a morphism between S-preschemes that are smooth over S. Then f is étale at $x \in X$ if and only if*

$$f^*(\Omega^1_{Y/S}) \longrightarrow \Omega^1_{X/S}$$

is an isomorphism at x.

Necessity is known from Lemma II.4.1, and the same lemma shows that the stated condition implies that f is unramified at x. By Corollary II.2.2, we are reduced to the case $S = \mathrm{Spec}(k)$. Since Y is smooth over k, it is regular, hence in particular normal. By I I.9.5(ii), it remains only to prove that $\mathcal{O}_y \to \mathcal{O}_x$ is injective, or, equivalently, that $\mathcal{O}_y$ and $\mathcal{O}_x$ have the same dimension. These dimensions are respectively the ranks of $\Omega^1_{Y/k}$ and $\Omega^1_{X/k}$ at y and x, and are therefore equal by hypothesis.

Remarks II.4.7. When X and Y are smooth over S, the smoothness criterion in Corollary II.4.5(ii) for $f\colon X \to Y$ can also be stated by saying that, for every $x \in X$, the *tangent* map of f at x, relative to the base S, is surjective. This tangent map is the transpose of the homomorphism between the finite-dimensional $\kappa(x)$-vector spaces obtained as the fibers at x of $f^*(\Omega^1_{Y/S})$ and $\Omega^1_{X/S}$. This is a familiar hypothesis, especially among those who work with analytic spaces. Their usual nonsingularity hypothesis, which says that X and Y are "smooth over $\mathbb{C}$"; see no. II.5, seems to be due only to the fear that singular points of algebraic varieties or analytic spaces still inspire in many geometers.

We record the following special case of Corollary II.4.6.

Corollary II.4.8. *Let X be an S-prescheme, and let*

$$g\colon X \longrightarrow S[t_1, \dots, t_n]$$

be an S-morphism defined by sections g_i, $1 \le i \le n$, of $\mathcal{O}_X$. Let $x \in X$ be a point at which X is smooth over S. Then g is étale at x if and only if the $\mathrm{d}g_i$*, $1 \le i \le n$, form a basis of $\Omega^1_{X/S}$ at x. Equivalently, their images form a basis of the $\kappa(x)$-vector space*

$$\Omega^1_{X/S}(x) = (\Omega^1_{X/S})_x \otimes_{\mathcal{O}_x} \kappa(x).$$

Let X be a prescheme, and let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$. Then $\mathcal{J}/\mathcal{J}^2$ may be regarded as a coherent sheaf on Y; it is the *conormal sheaf* of Y in X. If X is an S-prescheme, there is a canonical exact sequence of quasi-coherent sheaves on Y:

$$\mathcal{J}/\mathcal{J}^2 \xrightarrow{\mathrm{d}} \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow 0. \tag{II.4.3}$$

The right-hand part is simply (II.4.2), with the roles of X and Y interchanged and with $\Omega^1_{Y/X} = 0$. The homomorphism

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is induced, on passing to quotients, by the generally nonlinear map $g \mapsto \mathrm{d}g$. The exactness of (II.4.3) is classical and immediate. In the affine case, for a surjective homomorphism $B \to C$ of A-algebras with kernel J, it becomes

$$J/J^2 \longrightarrow \Omega^1_{B/A} \otimes_B C \longrightarrow \Omega^1_{C/A} \longrightarrow 0, \qquad C = B/J. \tag{4.3bis}$$

This is the exact sequence already used implicitly in the proof of I I.7.2.

Proposition II.4.9. *Let X be an S-prescheme, let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in X$. Let g_i, $1 \le i \le n$, be sections of $\mathcal{O}_X$ defining an S-morphism*

$$g\colon X \longrightarrow S[t_1, \dots, t_n] = X',$$

and let p be an integer with $0 \le p \le n$. Suppose that X is smooth over S at x*. Then the following conditions are equivalent:*

(i) *There is an open neighborhood X_1 of x in X such that $g|_{X_1}$ is* étale *and $Y_1 = Y \cap X_1$, the trace of Y on X_1, is the* inverse image *of the closed subprescheme*

$$Y' = S[t_{p+1}, \dots, t_n] \quad \textit{of} \quad X' = S[t_1, \dots, t_n].$$

Equivalently, the g_i, $1 \le i \le p$, generate $\mathcal{J}|_{X_1}$. Thus the diagram

$$\begin{array}{ccc} Y_1 & \longrightarrow & X_1 \\ \downarrow & & \downarrow \textit{étale} \\ Y' = S[t_{p+1}, \dots, t_n] & \longrightarrow & X' = S[t_1, \dots, t_n] \end{array}$$

is cartesian.

(ii) *Y is* smooth over S at x*; the germs of the g_i, $1 \le i \le p$, lie in $\mathcal{J}_x$; the elements $\mathrm{d}g_i(x)$, $1 \le i \le n$, form a* basis *of the $\kappa(x)$-vector space $\Omega^1_{X/S}(x)$; and the $\mathrm{d}g'_i(x)$, $p+1 \le i \le n$, form a* basis *of $\Omega^1_{Y/S}(x)$, where g'_i denotes the restriction of g_i to Y and all differentials are relative to S.*

(iii) *The g_i, $1 \le i \le p$, form a* system of generators *of $\mathcal{J}_x$, and the $\mathrm{d}g_i(x)$, $1 \le i \le n$, form a* basis *of $\Omega^1_{X/S}(x)$ over $\kappa(x)$.*

(iv) *Y is* smooth over S at x*; the g_i, $1 \le i \le p$, form a* minimal system of generators *of $\mathcal{J}_x$; and the $\mathrm{d}g'_i(x)$, $p+1 \le i \le n$, form a* basis *of $\Omega^1_{Y/S}(x)$ over $\kappa(x)$.*

Moreover, under these conditions, $\mathcal{J}/\mathcal{J}^2$ is free over Y at x, with the classes of the g_i, $1 \le i \le p$, as a basis at x, and the canonical homomorphism

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is universally injective *at x.*

Remark *. It follows that p is determined by the other conditions: it is the *rank* of the free Y-module $\mathcal{J}/\mathcal{J}^2$ at x, the *minimum number of generators* of $\mathcal{J}_x$ on X, and also the integer characterized by the fact that the relative dimension of Y over S at x is $n-p$.

Proof of Proposition II.4.9. Suppose first that (i) holds. By I I.4.6(iii), Y_1 is étale over Y'; by definition it is therefore smooth over S at x, of relative dimension $n-p$, and the same is true of Y. Corollary II.4.8, applied to g and to its restriction, shows that the $\mathrm{d}g_i$, $1 \le i \le n$, form a basis of $\Omega^1_{X/S}$ at x, and that the $\mathrm{d}g'_i$, $p+1 \le i \le n$, form a basis of $\Omega^1_{Y/S}$ at x. The exact sequence (II.4.3) then shows that the classes of the g_i, $1 \le i \le p$, are linearly independent in the Y-module $\mathcal{J}/\mathcal{J}^2$ at x. Since these g_i generate $\mathcal{J}_x$, their classes modulo $\mathcal{J}_x^2$ form a basis of $\mathcal{J}/\mathcal{J}^2$ at x. Consequently, the g_i, $1 \le i \le p$, form a minimal system of generators of $\mathcal{J}_x$, and the homomorphism

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

in (II.4.3) is universally injective at x: it carries a basis of a module free at x to part of a basis of another such module, both being Y-modules. This proves that (i) implies (ii), (iii), and (iv), as well as the final assertions of Proposition II.4.9.

Condition (iii) implies (i) by Corollary II.4.8.

Condition (ii) also implies (i). Indeed, the requirement that the germs of $g_1, \dots, g_p$ lie in $\mathcal{J}_x$ means that, after replacing X by an open neighborhood of x, the map g induces a morphism $h\colon Y \to Y'$. By Corollary II.4.8, the two basis requirements mean that g and h are étale at x. Let Y'' be the inverse image of Y' under g. Then Y is a closed subprescheme of Y'', and Y'' is étale over Y' at x, by I I.4.6(iii), because g is étale at x. The immersion $Y \to Y''$ is therefore itself étale, by I I.4.8, and hence is an open immersion, by I I.5.8 or I I.5.2. Replacing X once more by a suitable open neighborhood X_1 of x, we obtain (i).

The preceding argument establishes the equivalence of (i), (ii), and (iii), and shows that they imply (iv). It remains to prove that (iv) implies (ii). Since $\Omega^1_{X/S}$ is free at x, this will be immediate once we know that the smoothness of Y over S at x implies that $\mathcal{J}/\mathcal{J}^2$ is free over Y at x, and that

$$\mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is universally injective at x. This assertion is included in the following theorem.

THEOREM II.4.10. *Let X be an S-prescheme smooth over S, let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in X$. The following conditions are equivalent:*

(i) Y *is* smooth over S at x.

(ii) *There exist an open neighborhood X_1 of x in X, an étale S-morphism*

$$g\colon X_1 \longrightarrow X' = S[t_1, \dots, t_n],$$

and an integer p, such that $Y_1 = Y \cap X_1$ is the inverse image *under g of the closed subprescheme*

$$Y' = S[t_{p+1}, \dots, t_n] \quad \textit{of} \quad X' = S[t_1, \dots, t_n].$$

(iii) *There exist generators g_i, $1 \le i \le p$, of $\mathcal{J}_x$ such that the $\mathrm{d}g_i$ form part of a basis of $\Omega^1_{X/S}$ at x, or equivalently such that the $\mathrm{d}g_i(x)$ are linearly independent over $\kappa(x)$ in $\Omega^1_{X/S}(x)$.*

(iv) *The sheaf $\mathcal{J}/\mathcal{J}^2$ is free over Y at x, and the canonical homomorphism*

$$\mathrm{d}\colon \mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$$

is universally injective at x. Equivalently, the sequence of canonical homomorphisms

$$0 \longrightarrow \mathcal{J}/\mathcal{J}^2 \longrightarrow \Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y \longrightarrow \Omega^1_{Y/S} \longrightarrow 0$$

is exact at x, and $\Omega^1_{Y/S}$ is locally free at x.

Proof. We already know from the preceding argument that (ii) implies (i), (iii), and (iv). Let us prove that (i) implies (ii), which will at the same time complete the proof of Proposition II.4.9. By Theorem II.4.3(ii), the last two terms of the exact sequence (II.4.3) are locally free Y-modules at x. The images of the elements $\mathrm{d}g$, $g \in \mathcal{O}_X$, generate $\Omega^1_{X/S} \otimes_{\mathcal{O}_X} \mathcal{O}_Y$ at x, and hence their images generate $\Omega^1_{Y/S}$ there. We can therefore choose g_i, $p+1 \le i \le n$, in $\mathcal{O}_X$ such that the $\mathrm{d}g_i'$ form a basis of $\Omega^1_{Y/S}$. By the exactness of (II.4.3), we can then complete the $\mathrm{d}g_i$, $p+1 \le i \le n$, to a basis of the middle term by elements of the form $\mathrm{d}g_i$, $1 \le i \le p$, with $g_i \in \mathcal{J}_x$. The g_i come from sections of $\mathcal{O}_X$ on a neighborhood of x, which we may take to be X. We are now in condition (ii) of Proposition II.4.9, and we have proved that it implies condition (i) there. This gives condition (ii) of the present theorem.

The implication (iii) $\Rightarrow$ (ii) in the present theorem follows immediately from the implication (iii) $\Rightarrow$ (i) in Proposition II.4.9. Thus (i), (ii), and (iii) are equivalent and imply (iv). Finally, (iv) implies (iii): choose elements $g_i \in \mathcal{J}_x$ whose classes form a basis of $\mathcal{J}_x/\mathcal{J}_x^2$. They generate $\mathcal{J}_x$ by Nakayama's lemma, and universal injectivity says that their differentials form part of a basis.

The preceding proof also gives the following result.

COROLLARY II.4.11. *Let X be an S-prescheme, let Y be a closed subprescheme defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in Y$. Suppose that both X and Y are* smooth over S at x. *Let g_i, $1 \le i \le p$, be sections of $\mathcal{J}$. The following conditions are equivalent:*

(i) *The g_i* generate *$\mathcal{J}_x$, and the $\mathrm{d}g_i(x)$ are* linearly independent *over $\kappa(x)$ in $\Omega^1_{X/S}(x)$.*

(ii) *The classes of the g_i modulo $\mathcal{J}^2$ form a basis of $\mathcal{J}/\mathcal{J}^2$ at x.*

(iii) *The g_i form a minimal system of generators of $\mathcal{J}_x$.*

(iv) *There exist further sections g_i, $p+1 \leq i \leq n$, of $\mathcal{O}_X$ on a neighborhood X_1 of x in X, such that, together with the preceding sections, they define an étale morphism*

$$g\colon X_1 \longrightarrow X' = S[t_1, \dots, t_n]$$

for which $Y_1 = Y \cap X_1$ is the inverse image *of the closed subprescheme*

$$Y' = S[t_{p+1}, \dots, t_n] \quad \textit{of} \quad X' = S[t_1, \dots, t_n].$$

In particular:

Corollary II.4.12. *Let X be an S-prescheme, let F be a section of $\mathcal{O}_X$, let Y be its zero subprescheme, defined by the coherent ideal $F\mathcal{O}_X$, and let $x \in Y$. Suppose that X is smooth over S at x. Then Y is smooth over S at x if and only if either F vanishes in a neighborhood of x, or $\mathrm{d}F(x) \neq 0$, where $\mathrm{d}F(x)$ denotes the image of $\mathrm{d}F$ in the $\kappa(x)$-vector space $\Omega^1_{X/S}(x)$.*

The condition is sufficient by criterion (iii) of Theorem II.4.10. It is necessary because, as $\mathcal{J}$ is generated by one element, $\mathcal{J}/\mathcal{J}^2$ at x must first be free of rank at most 1. If its rank is 0, that is, if $\mathcal{J}/\mathcal{J}^2 = 0$ at x, then $\mathcal{J} = 0$ at x by Nakayama's lemma, so F vanishes in a neighborhood of x. If its rank is 1, then F is a minimal system of generators of $\mathcal{J}$ at x, and the conclusion follows from the equivalence of (i) and (iii) in Corollary II.4.11.

Corollary II.4.13. *Let Y be an S-prescheme locally of finite type, let S' be a flat S-prescheme, put $Y' = Y \times_S S'$, let $x' \in Y'$, and let $x \in Y$ be its canonical image. Then Y is smooth over S at x if and only if Y' is smooth over S' at x'. In particular, if $S' \to S$ is flat and surjective, then Y is smooth over S if and only if Y' is smooth over S'.*

Only sufficiency remains to be proved; necessity was established in Proposition II.1.3(ii). After replacing Y by a suitable neighborhood of x, and Y' by its inverse image, we may suppose that Y is affine and of finite type over the affine prescheme S. Thus Y is isomorphic to a closed subprescheme of

$$X = S[t_1, \dots, t_n].$$

It follows that Y' is a closed subprescheme of $X' = X \times_S S'$. Since X is smooth over S, the prescheme X' is smooth over S', and the smoothness criteria in Theorem II.4.10 apply. Criterion (iv) gives the result immediately.

Remarks II.4.14. Criterion (iii) of Theorem II.4.10 deserves to be called the *Jacobian criterion for smoothness*. It makes it possible, in theory, to determine whether a given S-prescheme Y is smooth over S at a point x of Y, since there always exists a neighborhood of Y isomorphic to a subprescheme of a smooth S-prescheme X, for example $X = S[t_1, \dots, t_n]$. Moreover, it is for $X = S[t_1, \dots, t_n]$, $S = \mathrm{Spec}(()A)$, that the Jacobian criterion is usually stated (of course, in the classical case considered by Zariski, A was a 44
field). We leave it to the reader to give the statement for an ideal J of $A[t_1, \dots, t_n]$ and a prime ideal containing it, to which one is thus led. Let us note that it now seems clear (especially since Nagata succeeded, by nondifferential methods, in generalizing Zariski's theorem that the set of regular points of an algebraic scheme is open) that the Jacobian criterion is of scarcely any interest except in the form in which we give it here (i.e. using exclusively *relative* differentials and not *absolute* differentials, i.e. differentials relative to the ring of absolute constants $\mathbb{Z}$). As is so often the case, the use of differentials is more convenient here than that of derivations. Finally, let us note that if Y is smooth over S at x, of relative dimension n, then there exists an open neighborhood of x in Y isomorphic to a subprescheme of $X = S[t_1, \dots, t_n]$ with $n = m + 1$, as follows from the definition and from (I I.7.6).

Let A be a noetherian ring, let x_i $(1 \leq i \leq n)$ be elements of A, and let J be the ideal generated by the x_i. The x_i are said to form a *regular system of generators* of J if the canonical surjective homomorphism

$$(A/J)[t_1, \dots, t_n] \to \mathrm{gr}^J(A)$$

defined by the x_i (where the second member denotes the graded ring associated with A filtered by the powers of J) is an *isomorphism.* This condition also means that

(i) The canonical surjective homomorphism

$$S_{A/J}(J/J^2) \to \mathrm{gr}^J(A)$$

(where the first member denotes the symmetric algebra of the A/J-module J/J^2) is an isomorphism, and

(ii) J/J^2 is free and has as a basis the classes of the $x_i \mod J^2$.

In this form, we see that if $J \neq A$, the x_i form a *minimal system of generators* of J, and that *every other minimal system of generators* of J *is a regular system of generators* (N.B. "minimal"is taken in the strict sense: minimum number of elements, which is equivalent to the sense: minimal under inclusion only if A is local); on the other hand, if $J = A$, every system of generators of J is regular.

The regularity condition for a system of generators of an ideal is stable under localization by any multiplicatively closed set; on the other hand, it is immediately seen that in order for (x_i) to be a minimal system of generators of J, it already suffices that for every *maximal ideal* $\mathfrak{m}$ *containing* J, the x_i define a regular system of generators of $JA_{\mathfrak{m}}$ in $A_{\mathfrak{m}}$. We are thus reduced to the case where A is a local ring with maximal ideal $\mathfrak{m}$, and where the x_i lie in $\mathfrak{m}$. *Then the x_i form a regular system of generators of J if and only if they form an A-sequence in the sense of Serre*[6], i.e. if for every i such that $1 \leq i \leq n$, x_i is a non-zero-divisor of 0 in $A/(x_1, \dots, x_{i-1})A$. 45

Finally, in the case where A is an algebra over a ring B, and where A/J is isomorphic as a B-algebra to B (so that J is the kernel of a homomorphism of B-algebras $A \to B$), the x_i form a regular system of generators of J if and only if the canonical homomorphism

$$B[[t_1, \dots, t_n]] \to \hat{A}$$

defined by the x_i (where the second member denotes the separated completion $\varprojlim A/J^{n+1}$ of A for the topology defined by the powers of J) is an *isomorphism* (it is in any case *surjective*).

All these facts are well known, and doubtless appear, up to minor differences, in Serre's course on commutative algebra written up by Gabriel. (Where one finds N other characterizations of A-sequences, in the case where A is a local ring).

Let J be an ideal in a noetherian ring A. We shall say that J is a *regular ideal* if for every prime ideal $\mathfrak{p}$ of A, $JA_{\mathfrak{p}}$ admits a regular system of generators. It is plainly enough to verify this for $\mathfrak{p} \supset J$, and one may moreover restrict oneself to maximal $\mathfrak{p}$. More generally, let $\mathcal{J}$ be an Ideal on a locally noetherian prescheme X; one says that $\mathcal{J}$ is a *regular Ideal* if for every $x \in X$, $\mathcal{J}_x$ is an ideal of $\mathcal{O}_x$ which admits a regular system of generators. This is equivalent to the conjunction of the following two conditions:

(a) The canonical surjective homomorphism

$$S_{\mathcal{O}_X/\mathcal{J}}(\mathcal{J}/\mathcal{J}^2) \to \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$$

is an isomorphism, and

(b) The sheaf of $\mathcal{O}_X/\mathcal{J}$-Modules $\mathcal{J}/\mathcal{J}^2$ is locally free.

One also says then that the subprescheme Y of X defined by $\mathcal{J}$ (thus such that $\mathcal{O}_Y$, extended by 0, is isomorphic to $\mathcal{O}_X/\mathcal{J}$) is *regularly immersed* in X, and one defines in the same way (in the evident manner) the notion of a morphism that is 46

[6]We shall now rather say "A-regular sequence", cf. EGA 0_{IV} 15.1.7 and 15.1.11.

a *regular immersion*, (respectively*regular at a point* x), an immersion morphism $Y \to X$ identifying Y (respectivelya suitable neighborhood of x) with a regularly immersed closed subprescheme of an open subset of X. (One must not say: regular subprescheme, for that would mean that the local rings of Y are regular). Finally, sections x_i of $\mathcal{J}$ are called a *regular system of generators* if for every $x \in X$, the corresponding elements of $\mathcal{O}_x$ form a regular system of generators of $\mathcal{J}_x$ (terminology compatible with that introduced for generators of an ideal of a ring). This also means that the canonical surjective homomorphism

$$\mathcal{O}_Y[t_1, \dots, t_n] \to \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$$

defined by the x_i is an isomorphism. If it is known in advance that the Ideal $\mathcal{J}$ is regular, this means simply that at every point x *of* Y, the x_i define a *basis* of $\mathcal{J}/\mathcal{J}^2$ over $\mathcal{O}_{Y,x}$. (N.B. this condition is empty if Y is empty). Thus, in order that $\mathcal{J}$ admit a regular system of generators, it is necessary and sufficient that $\mathcal{J}$ be regular, and that the $\mathcal{O}_Y$-Module $\mathcal{J}/\mathcal{J}^2$ be globally free (and not merely locally free), i.e. that the canonical homomorphism $S_{\mathcal{O}_Y}(\mathcal{J}/\mathcal{J}^2) \to \mathrm{gr}^{\mathcal{J}}(\mathcal{O}_X)$ be surjective, and that the $\mathcal{O}_Y$-Module $\mathcal{J}/\mathcal{J}^2$ be globally free.

An *augmented ring is said to be regular* if the augmentation ideal is regular. Thus, if A is a local ring, regarded as augmented over its residue field k, then A is a regular local ring if and only if it is a regular augmented ring.

(To tell the truth, it seems that it was unnecessary to begin by giving the preliminary chain of reasoning for rings; it is advantageous to begin with sheaves at once. If one wants something in the noetherian case, it is the definition adopted here — a priori less restrictive than that by Serre's A-sequences — which seems preferable for the needs of differential calculus. Of course, to do things properly, one would also have to develop at least part of the theory of smooth morphisms in the nonnoetherian setting[7], probably by starting from the Jacobian criterion, so as to obtain, if possible, all the essential formal 47
properties of smooth morphisms and of ètale morphisms, i.e. smooth and quasi-finite morphisms; only the converses making use of noetherian hypotheses).

After these long terminological preliminaries, a little theorem:

Theorem II.4.15. *Let X be an S-prescheme locally of finite type, let Y be a closed subprescheme of X defined by a coherent sheaf of ideals $\mathcal{J}$, and let $x \in X$. Suppose now that Y is* smooth over S at x, *with no assumption on X. Then the following conditions are equivalent:*

(i) *X is smooth over S at x;*
(ii) *the immersion $i \colon Y \to X$ is regular at x, that is, $\mathcal{J}_x$ is a regular ideal of $\mathcal{O}_x$.*

Corollary II.4.16. *Suppose that Y is smooth over S. Then X is smooth over S in a neighborhood of Y, that is, at every point of Y, if and only if Y is regularly embedded in X, or equivalently the immersion $i \colon Y \to X$ is regular.*

Proof of Theorem II.4.15. Suppose first that (i) holds. Apply criterion (ii) of Theorem II.4.10. Since the resulting morphism $g \colon X_1 \to X'$ is flat, regular immersions are preserved under its inverse image. It is therefore enough to prove that

$$Y' = S[t_{p+1}, \dots, t_n]$$

is regularly embedded in $X' = S[t_1, \dots, t_n]$, which is immediate: $t_1, \dots, t_p$ form a regular system of generators of the ideal defining Y' in X'.

Conversely, suppose that (ii) holds. Let g_i, $1 \le i \le p$, be a regular system of generators of $\mathcal{J}_x$, and choose elements g_i, $p+1 \le i \le n$, of $\mathcal{O}_{X,x}$ whose images g_i' in $\mathcal{O}_{Y,x}$ define an étale morphism

$$Y_1 \longrightarrow Y' = S[t_{p+1}, \dots, t_n]$$

[7]As is said in the foreword, this has now been done, cf. EGA IV 17, 18

from a neighborhood Y_1 of x in Y. The g_i, $1 \leq i \leq n$, come from sections of $\mathcal{O}_X$ on a neighborhood X_1 of x. We may suppose that $X_1 = X$ and $Y_1 = Y$. They define a morphism

$$g\colon X \longrightarrow X' = S[t_1, \dots, t_n],$$

and it remains only to prove that g is étale at x.

After shrinking X, we may suppose that $g_1, \dots, g_p$ form a regular system of generators of $\mathcal{J}$ throughout X. In particular, they generate $\mathcal{J}$, so the closed subprescheme Y of X is the inverse image under g of the closed subprescheme Y' of X'. Put $x' = g(x)$. The fiber of $X \to X'$ over x' is then the same as the fiber of $Y \to Y'$ over x', and is therefore étale over $\kappa(x')$. Thus g is unramified at x; it remains to prove that g is flat there.

Let $\mathcal{J}'_{x'}$ be the ideal of Y' in $\mathcal{O}_{X',x'}$. The associated graded ring of $\mathcal{O}_{X',x'}$, filtered by the powers of $\mathcal{J}'_{x'}$, is free over $\mathcal{O}_{Y',x'}$ in every degree. The associated graded ring of $\mathcal{O}_{X,x}$, filtered by the powers of $\mathcal{J}_x = \mathcal{J}'_{x'}\mathcal{O}_{X,x}$, is canonically isomorphic to the tensor product of the former graded ring with $\mathcal{O}_{Y,x}$; indeed, the two graded rings are polynomial rings in p indeterminates over $\mathcal{O}_{Y',x'}$ and $\mathcal{O}_{Y,x}$, respectively. Finally,

$$\mathcal{O}_{X,x}/\mathcal{J}_x = \mathcal{O}_{Y,x}$$

is flat over

$$\mathcal{O}_{X',x'}/\mathcal{J}'_{x'} = \mathcal{O}_{Y',x'}.$$

A general flatness criterion now applies: for a local homomorphism of noetherian local rings $A' \to A$, with an ideal $J' \neq A'$ such that the associated graded ring is free over A'/J' in every degree, these conditions imply flatness. Hence X is flat over X' at x, as required.

Corollary II.4.17. *Let X be a prescheme locally of finite type over Y, let i be a section of X over Y, let $y \in Y$, and put $x = i(y)$. Let $\mathcal{J}$ be the sheaf of ideals on X defined by the subprescheme $i(Y)$, which we assume closed in order to simplify the statement, a condition automatically satisfied when X is a scheme. The following conditions are equivalent:*

(i) *X is smooth over Y at x;*

(ii) *i is a regular immersion at y;*

(iii) *the completion of the $\mathcal{O}_y$-algebra $\mathcal{O}_x$ for the topology defined by the powers of $\mathcal{J}_x$ is isomorphic to a formal power-series algebra*

$$\mathcal{O}_y[[t_1, \dots, t_n]];$$

(iii bis) *there is an open neighborhood U of y such that the sheaf of $\mathcal{O}_Y$-algebras*

$$\varprojlim_m i^*(\mathcal{O}_X/\mathcal{J}^{m+1})$$

over U is isomorphic to a sheaf of the form $\mathcal{O}_Y[[t_1, \dots, t_n]]$;

(iv) *there exist an open neighborhood U of y, an open neighborhood V of x, and a Y-morphism*

$$g\colon V \longrightarrow U[t_1, \dots, t_n]$$

such that g is étale, i induces a section of V over U, and g carries this section to the zero section of $U[t_1, \dots, t_n]$ over U.

The equivalence of (i) and (ii) is the special case $Y = S$ of Theorem II.4.15. The equivalence of (ii) and (iii), and morally that of (ii) and (iii bis), was noted in the preceding "reminders". The equivalence of (i) and (iv) follows readily from the equivalence of (i) and (ii) in Theorem II.4.10.

Corollary II.4.18. *Let X be a prescheme smooth over S. Then the diagonal morphism*

$$\Delta_{X/S}\colon X \longrightarrow X \times_S X$$

is a regular immersion*; equivalently, X is* differentially smooth *over S.*

Indeed, this is a special case of Corollary II.4.16, since both X and $X \times_S X$ are smooth over S.

Remarks II.4.18. Recall from I I.1 that, if X is a prescheme over S, one introduces on X the quasi-coherent sheaves of algebras

$$\mathcal{P}^n_{X/S} = \mathcal{O}_{X\times_S X}/\mathcal{I}_X^{n+1},$$

where $\mathcal{I}_X$ is the sheaf of ideals defining the diagonal in $X \times_S X$. It is regarded as a sheaf of $\mathcal{O}_X$-algebras by means of the first projection $\mathrm{pr}_1 \colon X \times_S X \to X$. The $\mathcal{P}^n_{X/S}$ form a projective system of algebras on X. Its projective limit, denoted by $\mathcal{P}^\infty_{X/S}$, is the structure sheaf of the formal completion of $X \times_S X$ along the diagonal, where we now suppose that X is locally of finite type over S, so that the $\mathcal{P}^n_{X/S}$ are coherent.

To say that X is differentially smooth over S, that is, that the diagonal $\Delta_{X/S}$ is a regular immersion, is also to say that $\mathcal{P}^\infty_{X/S}$ is regular as a sheaf of algebras augmented toward $\mathcal{O}_X$. Equivalently, $\Omega^1_{X/S}$ is locally free and the canonical surjective homomorphism

$$\mathrm{Sym}_{\mathcal{O}_X}(\Omega^1_{X/S}) \longrightarrow \mathrm{gr}_*(\mathcal{P}^\infty_{X/S})$$

is an isomorphism. Equivalently again, every point of X has an open neighborhood on which the augmented sheaf of algebras $\mathcal{P}^\infty_{X/S}$ is isomorphic to a sheaf $\mathcal{O}_X[[t_1, \ldots, t_n]]$.

Let s be a section of X over S, and let $\mathcal{J}$ be the sheaf of ideals on X that it defines; for simplicity, suppose that $s(S)$ is closed. There are then canonical isomorphisms of augmented $\mathcal{O}_X$-algebras

$$s^*(\mathcal{P}^n_{X/S}) = \mathcal{O}_X/\mathcal{J}^{n+1}, \qquad s^*(\mathcal{P}^\infty_{X/S}) = \varprojlim_n \mathcal{O}_X/\mathcal{J}^{n+1}. \tag{II.4.4}$$

These isomorphisms are functorial under base change in the evident sense and, taking this fact into account, give another characterization of the sheaves of algebras $\mathcal{P}^n_{X/S}$ on S. If, for example, $S = \mathrm{Spec}(k)$, where k is a field, then giving a section s of X over S is equivalent to giving a k-rational point $x \in X$, and the preceding formulas say that there is an isomorphism of k-algebras

$$\mathcal{P}^n_{X/S}(x) = \mathcal{O}_x/\mathfrak{m}_x^{n+1}. \tag{II.4.5}$$

This justifies the name *sheaf of principal parts of order n on X relative to S* for $\mathcal{P}^n_{X/S}$.

Formula (II.4.4) also shows that *if X is differentially smooth over S at every point of $s(S)$, then X is smooth over S at every point of $s(S)$*, by Corollary II.4.17; the converse is also true, by Corollary II.4.18. Together with Corollary II.4.13, this readily implies that, if X is an S-prescheme locally of finite type, then X *is smooth over S if and only if it is flat and differentially smooth over S*. Notice that the flatness hypothesis is essential, as one sees by taking X to be a closed subprescheme of S.

Recall also that $\mathcal{P}^n_{X/S}$ has a *second algebra structure*, obtained from the second projection $\mathrm{pr}_2 \colon X \times_S X \to X$. This structure is obtained from the first by means of the *canonical involution* of the sheaf of rings $\mathcal{P}^n_{X/S}$ induced by the symmetry automorphism of $X \times_S X$. We denote by $d^n_{X/S}$, or simply by d^n, the homomorphism of sheaves of rings

$$d^n_{X/S} \colon \mathcal{O}_X \longrightarrow \mathcal{P}^n_{X/S} \tag{II.4.6}$$

corresponding to this second algebra structure. Under the isomorphism (II.4.4), this homomorphism carries a section f of $\mathcal{O}_X$ to a section $d^n(f)$ of $\mathcal{P}^n_{X/S}$ whose inverse image under a section s of X over S is identified with the canonical image of f in $\Gamma(X, \mathcal{O}_X/\mathcal{J}^{n+1})$. This justifies the name *system of principal parts of order n of f* for $d^n f$, especially when $S = \mathrm{Spec}(k)$, as in (II.4.5).

Finally, the homomorphism (II.4.6) may be regarded as the *universal differential operator of order at most n* on $\mathcal{O}_X$, relative to the constant prescheme S. [8] Here a

[8]For everything concerning this paragraph, see EGA IV, 16.8–16.12.

differential operator of order at most n from $\mathcal{O}_X$ to a module F means a homomorphism of sheaves D that factors as

$$D\colon \mathcal{O}_X \xrightarrow{d^n} \mathcal{P}^n_{X/S} \xrightarrow{u} F,$$

where u is a homomorphism of $\mathcal{O}_X$-modules, uniquely determined by D. This agrees with the intuitive recursive definition: D has order at most n if, for every section g of $\mathcal{O}_X$ on an open set $U \subseteq X$, the map

$$f \longmapsto D(fg) - gD(f)$$

is a differential operator of order at most $n-1$ on U.

It follows that, *if X is differentially smooth over S, the sheaf of rings of differential operators of all orders has the familiar simple structure* from differential calculus on differentiable manifolds. In particular, it has locally a basis as an $\mathcal{O}_X$-module formed by the *divided powers* in mutually commuting operators $\delta/\delta x_i$, $1 \leq i \leq n$. If $\mathcal{O}_S$ is a sheaf of $\mathbb{Q}$-algebras, where $\mathbb{Q}$ is the field of rational numbers, it is enough to take ordinary polynomials in the $\delta/\delta x_i$. In this case, exceptionally, local freeness of $\Omega^1_{X/S}$ already suffices for X to be differentially smooth over S.

Remark II.4.19. The terminology “regular immersion”, “regular ideal”, and so forth, introduced in this section met with fairly strong and general opposition from Chevalley and Serre. The terms “Cohen–Macaulay ideal”, “Macaulay ideal”, and “Macaulayan ideal” were proposed; morally, this would also require the adoption of “Cohen–Macaulay immersion” or “Macaulay immersion”. That terminology, however, conflicts with another already used in forthcoming versions of the *multiplodoque*, where a morphism of finite type is said to be “Cohen–Macaulay” at a point if it is flat there and the local ring of the fiber through that point is Cohen–Macaulay. Until a satisfactory solution is found, we shall retain, with every reservation, the terminology introduced in this section.[9]

II.5. Case of a base field

Proposition II.5.1. *Let k be a field, let X be a prescheme of finite type over k, let $x \in X$, and let n be the dimension of X at x. Let*

$$f\colon X \longrightarrow \operatorname{Spec}(k[t_1, \ldots, t_n]) = Y$$

be the morphism defined by elements $f_i \in \Gamma(X, \mathcal{O}_X)$. The following conditions are equivalent (and imply that X is smooth over k at x, and hence, by Proposition II.3.1, regular at x):

(i) *f is étale at x;*
(ii) *the $\mathrm{d}f_i$ form a basis of $\Omega^1_{X/k}$ at x;*
(iii) *the $\mathrm{d}f_i$ generate $\Omega^1_{X/k}$ at x.*

Since (i) implies that X is smooth over k at x, the implication (i)⇒(ii) is a special case of Corollary II.4.8, and (ii)⇒(iii) is immediate. It remains to prove (iii)⇒(i). If (iii) holds, Lemma II.4.1 shows that f is unramified at x. After replacing X by an open neighborhood of x, it is therefore quasi-finite, and hence dominant by dimension considerations. Since Y is regular, it follows from I I.9.5(ii), or from I I.9.11, that f is étale.

Corollary II.5.2. *Under the hypotheses preceding the equivalent conditions in Proposition II.5.1, suppose that $\kappa(x)$ is a* finite separable *extension of k, and that $f_i \in \mathfrak{m}_x$ for $1 \leq i \leq n$. Then the preceding conditions are also equivalent to:*

(iv) *the f_i form a system of generators of $\mathfrak{m}_x$; equivalently, their classes modulo $\mathfrak{m}_x^2$ form a basis of $\mathfrak{m}_x/\mathfrak{m}_x^2$ over $\kappa(x)$.*

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[9] This is the terminology adopted in EGA 0_{IV}, 15.1.7.

Indeed, (iv) implies (iii) by the exact sequence

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega^1_{\mathcal{O}_x/k} \longrightarrow \Omega^1_{\kappa(x)/k} \longrightarrow 0, \tag{II.5.1}$$

since $\Omega^1_{\kappa(x)/k} = 0$, because $\kappa(x)$ is étale over k. Conversely, (ii) implies (iv): both X and $\operatorname{Spec}(\kappa(x))$ are smooth over k at x, so by Theorem II.4.10(iv) the preceding exact sequence may be preceded by 0.

Corollary II.5.3. *Let x be a point of a prescheme X of finite type over k. If X is smooth over k at x, then $\mathcal{O}_x$ is regular. Conversely, if $\kappa(x)$ is a finite separable extension of k and $\mathcal{O}_x$ is regular, then X is smooth over k at x.*

Indeed, the converse follows from Corollary II.5.2, by taking a regular system (f_i) of generators of $\mathfrak{m}_x$. (Instead of Corollary II.5.2, one may also invoke Theorem II.4.15.) We conclude:

Proposition II.5.4. *Let X be a prescheme of finite type over k. If X is smooth over k, then it is regular. Conversely, if k is perfect and X is regular, then X is smooth over k.*

For the converse, Corollary II.5.3 shows that X is smooth over k at every closed point, hence everywhere, since the locus on which it is smooth is open.

Theorem II.5.5. *Let X be a prescheme of finite type over k, let $x \in X$, let n be the dimension of X at x, and let k' be a perfect field extending k. The following conditions are equivalent:*

(i) *X is smooth over k at x;*
(ii) *$\Omega^1_{X/k}$ is locally free of rank n at x;*
(ii bis) *$\Omega^1_{X/k}$ is generated by n elements at x;*
(iii) *X is differentially smooth over k at x;*
(iv) *there exists an open neighborhood U of x such that $U \otimes_k k'$ is regular; that is, all its local rings are regular.*

Theorem II.4.3(ii) gives (i)$\Rightarrow$(ii), the implication (ii)$\Rightarrow$(ii bis) is immediate, and Proposition II.5.1 gives (ii bis)$\Rightarrow$(i). Since X is flat over k, Corollary II.4.18 gives (i)$\Leftrightarrow$(iii). Condition (i) implies (iv), because smoothness is preserved under base extension and implies regularity. Conversely, if (iv) holds, then Proposition II.5.4 shows that $U \otimes_k k'$ is smooth over k', and hence U is smooth over k by Corollary II.4.13.

Taking x to be the generic point of X, with X assumed irreducible, we obtain:

Corollary II.5.6. *Let K be a local Artinian ring which is a localization of a finitely generated algebra over the field k (for example, K may be a finitely generated extension of k), and let n be the transcendence degree over k of its residue field. The following conditions are equivalent:*

(i) *K is a finite separable extension of a purely transcendental extension $k(t_1, \dots, t_n)$ of k.*
(ii) *$\Omega^1_{K/k}$ is a free K-module of rank n.*
(ii bis) *$\Omega^1_{K/k}$ is a K-module generated by n elements.*
(iii) *The completion O' of $K \otimes_k K$, for the topology defined by the powers of the augmentation ideal of $K \otimes_k K \to K$, is a "regular" augmented K-algebra, that is, isomorphic to a formal power-series algebra over K. (If K is a field, this is equivalent to saying that O' is a regular local ring.)*
(iv) *K is a separable extension of k.*

Indeed, K may always be regarded as the local ring at the generic point of an irreducible scheme X of finite type over k, and the conditions in question are the conditions bearing the same names in Theorem II.5.5, taking for k' in (iv) an algebraically closed extension of k. Only the implication

$$K \text{ separable over } k \quad \Longrightarrow \quad X \text{ smooth over } k \text{ at } x$$

requires proof. By Corollary II.4.13, we are immediately reduced to the case where the base field is k', hence algebraically closed; there is then a k-rational point a of X. By Proposition II.5.4, X is smooth over k at a, and is therefore all the more smooth over k at the generic point x. [10]

When K is a finitely generated extension of k, the equivalence of (i), (ii), (ii bis), and (iv) is well known. Notice, however, that we have used none of these already known equivalences. Of course, Proposition II.5.1 contains as a special case the familiar fact that a sequence of elements x_i, $1 \le i \le n$, is a separating transcendence basis of K over k if and only if the $\mathrm{d}x_i$ form a basis of the K-module $\Omega^1_{K/k}$.

Corollary II.5.7. *Let X be a prescheme of finite type over a field k. Then X is smooth over k if and only if $\Omega^1_{X/k}$ is locally free and the local rings at the generic points of the irreducible components of X are separable extensions of k. (The latter condition is automatically satisfied if k is perfect and X is reduced.)*

We may assume that X is connected. Let n be the rank of $\Omega^1_{X/k}$, assumed locally free. By the hypothesis and Corollary II.5.6, n is also the transcendence degree of the extensions of k defined by the local rings at the generic points of X. Thus all the irreducible components of X have dimension n, and the conclusion follows from Theorem II.5.5.

Care is needed here. If K is a finite extension of k, not necessarily separable, then $\Omega^1_{K/k}$ is a free k-module. Thus, putting $X = \operatorname{Spec}(K)$, the sheaf $\Omega^1_{X/k}$ is locally free and X is reduced, although X need not be smooth over k. Extending scalars to the algebraic closure of k then gives a similar example in which k is algebraically closed, but X is no longer reduced.

Corollary II.5.8. *Let X be a prescheme of finite type over the field k, let $x \in X$, let n be the dimension of X at x, and let $p = \dim \mathcal{O}_x$, that is, the codimension in X of the closure Y of x. Thus $n - p$ is the transcendence degree of $\kappa(x)$ over k. Let f_i, $1 \le i \le n$, be elements of $\mathcal{O}_x$, with $f_i \in \mathfrak{m}_x$ for $1 \le i \le p$. The following conditions are equivalent:*

(i) *The germ at x of the morphism*

$$X \longrightarrow \operatorname{Spec}(k[t_1, \dots, t_n])$$

defined by the f_i is étale at x.

(ii) *The elements f_i, $1 \le i \le p$, generate $\mathfrak{m}_x$, that is, they form a regular system of parameters of $\mathcal{O}_x$, and the classes in $\kappa(x)$ of the f_j, $p+1 \le j \le n$, form a separating transcendence basis. In other words, the $\mathrm{d}\overline{f}_j$, $p+1 \le j \le n$, form a basis of $\Omega^1_{\kappa(x)/k}$, or equivalently generate that module.*

Suppose first that (i) holds. By Corollary II.4.8, the $\mathrm{d}f_i(x)$ form a basis of $\Omega^1_{X/k}(x)$; hence their images $\mathrm{d}\overline{f}_i(x)$ generate $\Omega^1_{\kappa(x)/k}$ over $\kappa(x)$. Since $\overline{f}_i = 0$ for $1 \le i \le p$, it is enough to retain the $\mathrm{d}\overline{f}_i(x)$ with $p+1 \le i \le n$. The transcendence degree of $\kappa(x)$ over k is $n-p$, so Corollary II.5.6, applied to $K = \kappa(x)$, shows that Y is smooth over k at its generic point x, and that the $\mathrm{d}\overline{f}_i(x)$, $p+1 \le i \le n$, form a basis of $\Omega^1_{\kappa(x)/k}$ over $\kappa(x)$. Condition (ii) of Proposition II.4.9 is therefore satisfied, and so is condition (iii); in particular the f_i, $1 \le i \le p$, generate $\mathfrak{m}_x$. Since $\mathcal{O}_x$ has dimension p, they form a regular system of parameters at x. This proves (ii).

Conversely, suppose that (ii) holds. The exact sequence (II.5.1) shows that the $\mathrm{d}f_i(x)$ generate $\Omega^1_{X/k}(x)$, and (i) follows from Proposition II.5.1.

[10] See the Errata at the end of the present Exposé II (p. 38).

Corollary II.5.9. *Let X be a prescheme of finite type over the field k, let $x \in X$, let n be the dimension of X at x, and let $p = \dim \mathcal{O}_x$, that is, the codimension of the closure Y of x in X. Thus $n-p$ is the transcendence degree of $\kappa(x)$ over k. The following conditions are equivalent:*

(i) *$\mathcal{O}_x$ is regular and $\kappa(x)$ is a separable extension of k.*

(ii) *X is smooth over k at x, and the canonical homomorphism*

$$\mathfrak{m}_x/\mathfrak{m}_x^2 \longrightarrow \Omega^1_{\mathcal{O}_x/k} \otimes_{\mathcal{O}_x} \kappa(x) = \Omega^1_{X/k}(x)$$

is injective.

(iii) *There exist $f_i \in \mathcal{O}_x$, $1 \leq i \leq n$, with $f_i \in \mathfrak{m}_x$ for $1 \leq i \leq p$, such that the germ at x of the morphism from X to* $\operatorname{Spec}(k[t_1, \dots, t_n])$ *defined by the f_i is étale at x. Equivalently, by Proposition II.5.1, the* $\mathrm{d}f_i(x)$ *generate $\Omega^1_{X/k}(x)$.*

(iv) *There exist $f_i \in \mathcal{O}_x$, $1 \leq i \leq n$, such that the f_i, $1 \leq i \leq p$, generate $\mathfrak{m}_x$, and the* $\mathrm{d}f_j(x)$*, $p+1 \leq j \leq n$, generate $\Omega^1_{\kappa(x)/k}$ over $\kappa(x)$.*

The equivalence of (iii) and (iv) follows from Corollary II.5.8. By Proposition II.4.9, these conditions are also equivalent to X being smooth over k at x together with condition (ii) of Theorem II.4.10. They are therefore equivalent to X being smooth over k at x together with condition (iv) of that theorem, which is condition (ii) above. Equally, they are equivalent to X being smooth over k at x together with condition (i) of Theorem II.4.10; here the latter simply means that $\kappa(x)$ is separable over k. This implies condition (i) above. To prove the converse it remains only to establish the following corollary.

Corollary II.5.10. *Let x be a point of a prescheme X of finite type over the field k, and suppose that $\kappa(x)$ is separable over k. Then X is smooth over k at x if and only if it is regular at x, that is, if and only if the local ring $\mathcal{O}_x$ is regular.*

Indeed, if X is regular at x, one can plainly find elements $f_i \in \mathcal{O}_x$, $1 \leq i \leq n$, satisfying condition II.5.9(iv).

Errata. In the proof of Corollary II.5.6 above, we used the fact that a nonempty reduced scheme of finite type over an algebraically closed field has at least one regular, hence smooth, point. This fact is usually proved by differential methods, using Zariski's theorem that the set of regular points of X is open. To avoid a circular argument, one must prove the following directly. Let K/k be a finitely generated separable extension, and suppose that $f_i \in K$, $1 \leq i \leq n$, are such that the $\mathrm{d}_{K/k} f_i$ form a basis of $\Omega^1_{K/k}$. Then n is the transcendence degree of K over k; equivalently, the f_i are algebraically independent.

The usual proof uses Mac Lane's criterion; see Bourbaki, *Algèbre*, Chapter V, §9, Theorem 2. Choose a nonzero polynomial $g \in k[t_1, \dots, t_n]$ of minimal degree such that $g(f_1, \dots, f_n) = 0$. Then

$$\sum_i \frac{\partial g}{\partial t_i}(f_1, \dots, f_n)\, \mathrm{d}f_i = 0.$$

Since the $\mathrm{d}f_i$ form a basis of $\Omega^1_{K/k}$, each $\partial g/\partial t_i$ vanishes at $(f_1, \dots, f_n)$, and hence is the zero polynomial by the minimality of g. In characteristic zero this gives $g = 0$. In characteristic $p \neq 0$, one has $g = h(t_1^p, \dots, t_n^p)$. Mac Lane's criterion then shows that $h \in k[t_1, \dots, t_n]$ also vanishes at $(f_1, \dots, f_n)$, and the minimality of g again gives $g = 0$.

CHAPTER III

Smooth morphisms: lifting properties

III.1. Formally smooth homomorphisms

In Exposé II we confined ourselves to homomorphisms of finite type and, consequently, in the case of local homomorphisms $A \to B$ of local rings, to the case where B is isomorphic to a localization of a finitely generated A-algebra. This case is insufficient for various applications, especially in formal or analytic geometry. For example, the formal power-series ring $B = A[[t_1, \ldots, t_n]]$ has, from the viewpoint of formal geometry, the properties of a smooth algebra over A. Likewise, in analytic geometry, the local ring at a point (y, z) of a product $Y \times \mathbb{C}^n$, regarded as an algebra over the local ring at y, has these properties. Indeed, its completion is isomorphic to the algebra of formal power series in n indeterminates over the completion of the base ring $\mathcal{O}_y$. This leads to the following definition.

Definition III.1.1. *Let $u\colon A \to B$ be a local homomorphism of local rings, which, as always, are assumed Noetherian. Suppose that the residue-field extension $\kappa(B)/\kappa(A)$ is finite. We say that u is a* formally smooth homomorphism*, or that the algebra B is* formally smooth over A*, if there exists a finite local $\overline{A}$-algebra A', free over $\overline{A}$, such that the local components of the semilocal ring*

$$B' = \overline{B} \otimes_{\overline{A}} A'$$

are A'-isomorphic to formal power-series algebras over A'. [1]

Here $\overline{A}$ and $\overline{B}$ denote the completions of A and B. Since B' is finite and free over $\overline{B}$, it is indeed a semilocal ring, a direct product of complete local rings. Each of these is again a free $\overline{B}$-module, and therefore has the same dimension as $\overline{B}$, hence as B. It follows that the number of variables t_i in the formal power-series rings of Definition III.1.1 is

$$\dim \overline{B} - \dim \overline{A} = \dim B - \dim A,$$

and in particular is independent of the chosen local component. It is immediately seen to be also the dimension of the ring $B \otimes_A k = B/\mathfrak{m}B$, where $k = A/\mathfrak{m}$ is the residue field of A. We call it the *relative dimension of B over A*.

Remarks III.1.2. Definition III.1.1 plainly depends only on the homomorphism of completions $\overline{A} \to \overline{B}$ induced by $A \to B$, which to some extent justifies the terminology. We here retract the terminology of I I.3.2(b) and I I.4.1(b), which risks being misleading, and prefer to say "formally unramified" and "formally étale" in the cases treated by those definitions. We reserve "unramified" and "étale" for the case where B is a localization of a finitely generated A-algebra. [2]

The reader will verify at once that "formally étale" is equivalent to "formally smooth and quasi-finite". Finally, there is a reasonable definition of "formally smooth" without any preliminary hypothesis on the residue extension $\kappa(B)/\kappa(A)$, assumed finite here. Among other cases, it includes local homomorphisms $A \to B$ for which B is flat over A and $B/\mathfrak{m}B$ is a separable extension of $A/\mathfrak{m} = k$, not necessarily of finite type. For

[1] For a more general and more conceptual definition, motivated by Theorem III.2.1 below, see EGA 0_{IV}, 19.3.1.

[2] Or, better, "essentially unramified" and "essentially étale"; compare EGA IV, 18.6.1.

example, a Cohen p-ring is formally smooth over the ring of p-adic integers. In this general setting, the lifting property for homomorphisms (compare Theorem III.2.1) should be taken as the definition. For the applications we have in mind, the case treated in Definition III.1.1 will suffice. Henceforth, "formally smooth" will tacitly mean "with finite residue extension".

LEMMA III.1.3. *If B is formally smooth over A, then B is flat over A.*

Since flatness is invariant under completion, we may suppose that A and B are complete. Since flatness is also invariant under a flat local, hence faithfully flat, extension of the base ring, Definition III.1.1 reduces us to the case where B is a formal power-series algebra over A. As an A-module, B is then isomorphic to a product of A-modules isomorphic to A, and is therefore A-flat, since the base ring A is Noetherian.

We now place ourselves under the hypotheses of Definition III.1.1. Since the residue extensions of the local components of B' over A' are trivial, $L \otimes_k k'$ is an Artinian k'-algebra whose local components have trivial residue extensions, where L, k, and k' are the residue fields of B, A, and A', respectively. This condition, necessary for the finite free extension A' to satisfy the requirement in Definition III.1.1, is also sufficient, as follows immediately from Proposition III.1.4(i) and Proposition III.1.5 below.

PROPOSITION III.1.4. *Let $A \to B$ be a local homomorphism of local rings with finite residue extension, and let A' be a finite local A-algebra. Then $B' = B \otimes_A A'$ is finite over B, and is therefore a Noetherian semilocal ring.*

(i) *If B is formally smooth over A, then the localizations of B' at its maximal ideals are formally smooth over A'.*
(ii) *The converse holds if A' is free over A.*

We reduce at once to the case in which A and B are complete.

For (i), let A'' be a finite free local extension of A such that the local components of $B'' = B \otimes_A A''$ are formal power-series algebras over A''. Make the scalar extension

$$A'' \longrightarrow A'' \otimes_A A' \longrightarrow A''',$$

where A''' is a local component of $A'' \otimes_A A'$. The local components of

$$B'' \otimes_{A''} A''' = B \otimes_A A'''$$

are then formal power-series algebras over A'''. We also have

$$B \otimes_A A''' = (B \otimes_A A') \otimes_{A'} A''' = B' \otimes_{A'} A'''.$$

Moreover, since A'' is free over A, the algebra $A'' \otimes_A A'$ is free over A'; its direct factor A''' is therefore free over the local ring A'. This proves that the local components of B' are formally smooth over A'.

For (ii), let A'' be a finite free local A'-algebra such that the local components of

$$B' \otimes_{A'} A'' = B \otimes_A A''$$

are formal power-series algebras over A''. Since A' is free over A, the algebra A'' is also free over A, and hence B is formally smooth over A.

PROPOSITION III.1.5. *Let $A \to B$ be a local homomorphism of local rings with* trivial *residue extension. Then B is formally smooth over A if and only if $\overline{B}$ is isomorphic to a formal power-series algebra over $\overline{A}$.*

Only necessity remains to be proved, and we may suppose that A and B are complete. Let $\mathfrak{m}$ and $\mathfrak{n}$ be the maximal ideals of A and B, respectively, and choose elements $t_1, \ldots, t_n \in \mathfrak{n}$ whose classes form a basis of the vector space

$$(\mathfrak{n}/\mathfrak{n}^2)/\operatorname{Im}(\mathfrak{m}/\mathfrak{m}^2) = \mathfrak{n}/(\mathfrak{n}^2 + \mathfrak{m}B).$$

These elements define a local homomorphism of A-algebras

$$B_1 = A[[t_1, \ldots, t_n]] \longrightarrow B.$$

We shall prove that it is an isomorphism. It is enough to prove, for every power $\mathfrak{m}^q$, that reduction modulo $\mathfrak{m}^q$ gives an isomorphism

$$B_1/\mathfrak{m}^q B_1 \longrightarrow B/\mathfrak{m}^q B,$$

since B_1 and B are the projective limits of the corresponding rings modulo $\mathfrak{m}^q$, as q varies. The A-modules B and B_1 are flat. Consequently, their associated graded rings for the $\mathfrak{m}$-adic filtration are obtained by tensoring $B_1/\mathfrak{m}B_1$, respectively $B/\mathfrak{m}B$, over $k = A/\mathfrak{m}$ with $\mathrm{gr}(A)$. We are thus reduced to showing that

$$B_1/\mathfrak{m}B_1 \longrightarrow B/\mathfrak{m}B$$

is an isomorphism. In view of Lemma III.1.3, this reduces us to the case in which A is a field k.

On the other hand, let A' be a finite free local A-algebra such that $B \otimes_A A'$ is a formal power-series algebra over A'. (This algebra is local, since the residue extension of B over A is trivial.) To prove that $B_1 \to B$ is an isomorphism, it is enough to prove that

$$B_1 \otimes_A A' \longrightarrow B \otimes_A A'$$

is an isomorphism. This reduces us to the case in which B is already a formal power-series algebra. This reduction should in fact have been made before reducing to the case of a base field. But then B is a regular local ring with coefficient field k, and it is well known—and follows immediately by considering the associated graded rings for the $\mathfrak{n}_1$-adic and $\mathfrak{n}$-adic filtrations on B_1 and B, where $\mathfrak{n}_1$ is the maximal ideal of B_1—that $B_1 \to B$ is an isomorphism. This completes the proof.

Corollary III.1.6. *If B is formally smooth over A, then there exists a finite local A-algebra A' such that the local components of*

$$\overline{B} \otimes_{\overline{A}} \overline{A'} = \overline{B \otimes_A A'}$$

are isomorphic to formal power-series algebras over $\overline{A'}$.

Indeed, let L/k be the residue extension of B/A, and choose a finite extension k'/k such that the residue extensions of the local components of the k'-algebra $L \otimes_k k'$ are trivial. Let A' be a finite free A-algebra such that $A'/\mathfrak{m}A' = k'$. Such an algebra exists: one may reduce step by step to the case in which k'/k is generated by one element, and then lift to A the coefficients of the minimal polynomial of a generator of k' over k. The algebra A' is local. At each maximal ideal of $B \otimes_A A'$, the residue extension over the residue field k' of A' is trivial, and Proposition III.1.5 completes the proof.

Corollary III.1.7. *Let $A \to B$ be a local homomorphism of local rings. Then B is formally smooth over A if and only if B is flat over A and $B/\mathfrak{m}B$ is formally smooth over $k = A/\mathfrak{m}$.*

After making a suitable finite free local extension A' of A and using Proposition III.1.4(ii), we reduce to the case in which the residue extension of B/A is trivial. Proposition III.1.4(i) and Lemma III.1.3 show that the stated conditions are necessary. For sufficiency, it is enough to observe that, under the present hypotheses, the proof of Proposition III.1.5 shows that B is a formal power-series algebra over A, after replacing A and B by their completions, as we may.

Remark III.1.8. It would not be difficult to develop, for formally smooth homomorphisms, analogues of all the properties of smooth morphisms studied in Exposé II. For differential properties, however, this requires modifying the usual definition of Kähler differentials (cf. I I.1), with completed tensor products replacing ordinary tensor products. We shall merely gesture toward these abysses here; the preceding results suffice for our purposes.

It remains to connect formal smoothness with the notion of smoothness developed in Exposé II, which we have not yet used at all:

Proposition III.1.9. *Let $A \to B$ be a local homomorphism, where B is a localization of an A-algebra of finite type. Then B is smooth over A if and only if it is formally smooth over A.*

Using Corollary III.1.7 and Theorem II.2.1, we reduce to the case in which A is a field k.

Using Proposition III.1.4(ii) and Corollary II.4.13, a suitable extension k'/k reduces us to the case in which the residue extension of B/k is trivial. By Corollary II.5.2, respectively Proposition III.1.5, the algebra B is smooth over k, respectively formally smooth over k, if and only if B is a regular local ring, respectively its completion is a formal power-series algebra over k. These two conditions are well known to be equivalent when the residue extension is trivial.

III.2. The characteristic lifting property of formally smooth homomorphisms

Theorem III.2.1. *Let $A \to B$ be a local homomorphism of local rings inducing a finite residue-field extension. The following conditions are equivalent:*

- (i) *B is formally smooth over A.*
- (ii) *For every local homomorphism $A \to C$, where C is a complete local ring, every ideal $J \subseteq \mathfrak{r}(C)$, and every local A-homomorphism $B \to C/J$, there exists an A-homomorphism $B \to C$, necessarily local, which lifts it.*
- (iii) *For every A-algebra C, not necessarily Noetherian, every nilpotent ideal J of C, and every continuous A-homomorphism $B \to C/J$, that is, one which vanishes on some power of $\mathfrak{r}(B)$, there exists an A-homomorphism $B \to C$ lifting it; this lift is necessarily continuous as well.*
- (iv) *The same lifting assertion as in* (ii) *and* (iii)*, with C restricted to be a local Artinian ring finite over A.*
- (iv bis) *The same as* (iv)*, with the additional requirement $J^2 = 0$.*

Remark. In the remainder of this Exposé we shall use only the implication (iv)⇒(i), or (iv bis)⇒(i). The direct implication (i)⇒(ii) will be proved by another method in the next section when B is a localization of a finitely generated A-algebra. Recall that in the "good" theory of Cohen theorems,[3] property (ii) or (iii) becomes the definition of formally smooth homomorphisms, whereas Definition III.1.1 becomes a characteristic property valid only when the residue extension is finite.

Notice that neither property (ii) nor property (iii) is more general than the other. One could give an equivalent property encompassing both by taking a linearly topologized ring C, separated and complete, a closed topologically nilpotent ideal of C, and a continuous homomorphism $A \to C$, thereby making C a topological A-algebra. We leave this modification to the reader.

Proof of Theorem III.2.1. We shall prove

$$\text{(i)} \Longrightarrow \text{(iii)} \Longrightarrow \text{(ii)}$$

[3] See EGA 0_{IV}, 19.3 and 19.8.

and then (iv) $\Rightarrow$ (i). Since (ii) $\Rightarrow$ (iv) is trivial, and the equivalence of (iv) and (iv bis) follows by immediate induction on an integer n such that $J^n = 0$, this will complete the proof.

(i)$\Rightarrow$(iii). An immediate induction reduces us to the case $J^2 = 0$. Continuity of the given map, together with $J^2 = 0$, implies that some power of the maximal ideal $\mathfrak{m}$ of A acts trivially on C. After quotienting by that power of $\mathfrak{m}$, and observing from Proposition III.1.4(i) that $B/\mathfrak{m}^q B$ remains formally smooth over $A/\mathfrak{m}^q$, we may suppose that A is Artinian. By Lemma III.1.3, B is flat over A, and hence is free as an A-module because A is Artinian. There is therefore an A-linear map

$$w\colon B \longrightarrow C$$

lifting the given homomorphism $u\colon B \to C/J$. Put

$$f(x, y) = w(xy) - w(x)w(y) \qquad (x, y \in B).$$

Then $f(x, y) \in J$, so f is an A-bilinear map from $B \times B$ to J. A lift $v\colon B \to C$ of u is an algebra homomorphism if and only if there is an A-linear map $g\colon B \to J$ such that $v = w + g$ and

$$g(1) = 1 - w(1),$$
$$g(xy) - u(x)g(y) - u(y)g(x) = -f(x, y) \qquad (x, y \in B).$$

Here the action of C/J on J is well defined because $J^2 = 0$.

This is a system of linear equations in $\mathrm{Hom}_A(B, J)$, with right-hand sides in J. It has a solution if and only if the corresponding system in

$$\mathrm{Hom}_A(B, J) \otimes_A A', \qquad J' = J \otimes_A A',$$

has a solution, where A' is any faithfully flat A-algebra. Choose a finite free local A-algebra A' such that $B' = B \otimes_A A'$ is a formal power-series algebra over A'; we may suppose A and B complete in this argument. Since A' is finite free over A,

$$\mathrm{Hom}_A(B, J) \otimes_A A' = \mathrm{Hom}_{A'}(B', J').$$

The resulting system in $\mathrm{Hom}_{A'}(B', J')$ is exactly the one determining the A'-algebra homomorphisms

$$B' \longrightarrow C' = C \otimes_A A'$$

which lift the scalar extension $u'\colon B' \to C'/J'$ of u, by correcting the scalar extension $w'\colon B' \to C'$ of w with an A'-linear map $g'\colon B' \to J'$. Notice that B generates B' as an A'-module.

We are thus reduced to proving (iii) when

$$B = A[[t_1, \ldots, t_n]]$$

is a formal power-series algebra over A. Lift the images of the t_i in C/J arbitrarily to elements $z_i \in C$. Since the z_i mod J are nilpotent, by continuity of $u\colon B \to C/J$, and J itself is nilpotent, the z_i are nilpotent. They therefore define a continuous homomorphism of topological A-algebras from B to the discrete ring C, which plainly lifts u.

(iii)$\Rightarrow$(ii). Let $\mathfrak{n}$ be the maximal ideal of C, and, for each integer $q > 0$, set

$$C_q = C/\mathfrak{n}^q, \qquad J_q = (J + \mathfrak{n}^q)/\mathfrak{n}^q.$$

Thus C_q/J_q is naturally a quotient algebra of C/J. Moreover, the composite homomorphism

$$u_q\colon B \longrightarrow C/J \longrightarrow C_q/J_q$$

is continuous when C_q/J_q is given the discrete topology, and J_q is a nilpotent ideal of C_q. We construct successively A-homomorphisms

$$v_q\colon B \longrightarrow C_q$$

such that (a) v_q lifts u_q, and (b) v_q lifts v_{q-1}. The induction can be carried out as follows. The maps

$$u_q\colon B \longrightarrow C/(J+\mathfrak{n}^q), \qquad v_{q-1}\colon B \longrightarrow C/\mathfrak{n}^{q-1}$$

induce the same homomorphism

$$B \longrightarrow C/((J+\mathfrak{n}^q)+\mathfrak{n}^{q-1}) = C/(J+\mathfrak{n}^{q-1}) = C_{q-1}/J_{q-1},$$

namely u_{q-1}. They therefore determine a homomorphism

$$B \longrightarrow C/J'_q, \qquad J'_q = (J+\mathfrak{n}^q)\cap\mathfrak{n}^{q-1} \supseteq \mathfrak{n}^q,$$

from which both are obtained by reduction. It remains only to lift this homomorphism from B to the quotient of C_q by $J'_q/\mathfrak{n}^q$. This ideal is contained in J_q, hence is nilpotent, so the required lift exists by hypothesis (iii).

The maps v_q now determine a homomorphism from B to the inverse limit C of the rings C_q. Since J is closed, it is the inverse limit of the ideals J_q; consequently this homomorphism lifts u.

(iv)⇒(i). First observe that if condition (iv) holds, then it also holds over A' for every local component of $B\otimes_A A'$, whenever A' is a finite local A-algebra. Choose A' free over A and such that the residue extensions of the local components of $B' = B\otimes_A A'$ over A' are trivial. Proposition III.1.4 (ii) then reduces us to the case in which the residue extension of B over A is trivial. We shall prove the following slightly more precise result.

Corollary III.2.2. *Under the hypotheses of Theorem III.2.1, suppose in addition that the residue extension of B over A is trivial. Then the equivalent conditions of Theorem III.2.1 are also equivalent to each of the following two conditions (with A and B assumed complete in* (v)*):*

(iv ter) *The same assertion as in* (iv)*, but with the finite local Artinian A-algebra C required to have trivial residue extension (and, if desired, with $J^2=0$).*

(v) *There is a local A-homomorphism, where*

$$n = \dim(\mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B)),$$

$$u\colon B \longrightarrow B_1 = A[[t_1,\dots,t_n]],$$

which induces an isomorphism

$$\mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B) \xrightarrow{\sim} \mathfrak{n}_1/(\mathfrak{n}_1^2+\mathfrak{m}B_1),$$

where $\mathfrak{n}$ and $\mathfrak{n}_1$ are the maximal ideals of B and B_1, respectively, and $\mathfrak{m}$ is the maximal ideal of A.

Proof. Since (iv bis) plainly implies (iv ter)—even without using the square-zero proviso—it is enough to prove

$$\text{(iv ter)} \Rightarrow \text{(v)} \Rightarrow \text{(i)}.$$

(iv ter)⇒(v). Choose a basis $a_1,\dots,a_n$ of $\mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B)$. This determines a local homomorphism of A-algebras

$$B \longrightarrow B_1/(\mathfrak{n}_1^2+\mathfrak{m}B_1) = k[t_1,\dots,t_n]/(t_1,\dots,t_n)^2.$$

By (iv ter) it can be lifted successively to homomorphisms of A-algebras from B to $B_1/\mathfrak{n}_1^2, B_1/\mathfrak{n}_1^3,\dots$. Passing to the inverse limit gives a homomorphism $B\to B_1$ having the required property.

(v)⇒(i). In the commutative diagram

$$\begin{array}{ccccccc}
\mathfrak{m}/\mathfrak{m}^2 & \longrightarrow & \mathfrak{n}/\mathfrak{n}^2 & \longrightarrow & \mathfrak{n}/(\mathfrak{n}^2+\mathfrak{m}B) & \longrightarrow & 0\\
\downarrow & & \downarrow & & \downarrow & & \\
\mathfrak{m}/\mathfrak{m}^2 & \longrightarrow & \mathfrak{n}_1/\mathfrak{n}_1^2 & \longrightarrow & \mathfrak{n}_1/(\mathfrak{n}_1^2+\mathfrak{m}B_1) & \longrightarrow & 0.
\end{array}$$

the rows are exact and the two outer vertical arrows are surjective. The middle vertical arrow is therefore surjective as well; since B is complete, it follows that $u\colon B \to B_1$ is surjective. Choose elements $x_i \in B$, $1 \leq i \leq n$, lifting the t_i. They define an A-algebra homomorphism $B_1 \to B$, which is surjective for the same reason as u, and whose composite with u is the identity by construction. Thus $B_1 \to B$ is also injective and hence is an isomorphism. We have therefore also proved the following.

COROLLARY III.2.3. *Under the hypotheses of Corollary III.2.2*(v)*, the map u is necessarily an isomorphism.*

This completes the proof that B is formally smooth over A. At the same time we have recovered Proposition III.1.5, although there is little merit in doing so.

III.3. Local infinitesimal lifting of morphisms into a smooth S-scheme

THEOREM III.3.1. *Let $f\colon X \to Y$ be a morphism locally of finite type. The following conditions are equivalent:*

(i) *f is smooth.*

(ii) *For every prescheme Y' over Y, every closed subprescheme $Y'_0 \subseteq Y'$ having the same underlying space as Y', every Y-morphism $g_0\colon Y'_0 \to X$, and every $z \in Y'_0$, there exist an open neighborhood U of z in Y' and a Y-morphism $g\colon U \to X$ extending $g_0|_{Y'_0 \cap U}$.*

(ii bis) *For Y', Y'_0, and z as in* (ii)*, put*

$$X' = X \times_Y Y', \qquad X'_0 = X \times_Y Y'_0.$$

Every section of X'_0 over Y'_0 extends to a section of X' over an open neighborhood U of z.

(iii) *For every Y-scheme Y' which is the spectrum of a local Artinian ring finite over some $\mathcal{O}_y$, $y \in Y$, every nonempty closed subprescheme $Y'_0 \subseteq Y'$, and every Y-morphism $g_0\colon Y'_0 \to X$, there exists a Y-morphism $g\colon Y' \to X$ extending g_0.*

(iii bis) *For Y' and Y'_0 as in* (iii)*, put*

$$X' = X \times_Y Y', \qquad X'_0 = X \times_Y Y'_0.$$

Every section of X'_0 over Y'_0 extends to a section of X' over Y'.

Proof. The equivalence of (ii) and (ii bis), and that of (iii) and (iii bis), are immediate, as is the implication (ii) $\Rightarrow$ (iii). It remains to prove

$$\text{(i)} \Rightarrow \text{(ii)} \qquad \text{and} \qquad \text{(iii)} \Rightarrow \text{(i)}.$$

(i)$\Rightarrow$(ii). Put $x = g_0(z)$. By replacing X with a suitable open neighborhood of x, and Y' with the prescheme induced on the inverse image of that open set under g_0, we may suppose that X is étale over $Y[t_1, \ldots, t_n]$. The composite Y-morphism

$$Y'_0 \longrightarrow X \longrightarrow Y[t_1, \ldots, t_n]$$

is defined by n sections of $\mathcal{O}_{Y'_0}$. Near z, these extend to sections of $\mathcal{O}_{Y'}$, so after shrinking we may suppose that the displayed morphism extends to a Y-morphism

$$Y' \longrightarrow Y[t_1, \ldots, t_n].$$

By I I.5.6, there is then a unique Y-morphism $g\colon Y' \to X$ lifting this morphism and at the same time extending g_0.

(iii)⇒(i). Since the locus where f is smooth is open, it is enough to show that it contains every $x \in X$ which is closed in its fiber. Put $y = f(x)$. Then $\mathcal{O}_x$ is an $\mathcal{O}_y$-algebra which is a localization of a finitely generated algebra and has finite residue extension. Hypothesis (iii) implies that every homomorphism from $\mathcal{O}_x$ to A/J, where A is a local Artinian algebra finite over $\mathcal{O}_y$ and J is an ideal contained in its radical, lifts to a homomorphism $\mathcal{O}_x \to A$. Here we use the fact that a morphism $\mathrm{Spec}(B) \to X$, with B local and the closed point mapping to x, is determined bijectively by a local homomorphism $\mathcal{O}_x \to B$. Theorem III.2.1 therefore shows that $\mathcal{O}_x$ is formally smooth over $\mathcal{O}_y$, and hence smooth over $\mathcal{O}_y$ by Proposition III.1.9.

Corollary III.3.2. *Let $f\colon X \to Y$ be as in Theorem III.3.1. The following conditions are equivalent:*

(i) *f is étale.*
(ii) *Condition* (ii) *of Theorem III.3.1 holds, with the extension g of g_0 to U unique.*
(iii) *Condition* (iii) *of Theorem III.3.1 holds, with g unique.*

It is enough to observe in the proof of (i) ⇒ (ii) above that uniqueness, when Y'_0 is not identical to Y' near z, is possible only if $n = 0$, a condition already known to be sufficient.

Corollary III.3.3. *Let X be a prescheme locally of finite type over a complete local ring A, let y be the closed point of $Y = \mathrm{Spec}(A)$, and let $x \in f^{-1}(y)$ be rational over $\kappa(y)$. If X is smooth over A at x, then there exists a section s of X over Y passing through x, that is, satisfying $s(y) = x$.*

In particular, if X is smooth over A, then the natural map

$$\Gamma(X/Y) \longrightarrow \Gamma(X \otimes_A k/k)$$

from the sections of X over Y to the set of k-rational points of the fiber $f^{-1}(y) = X \otimes_A k$ is surjective. This fact was especially well known and used when A is a discrete valuation ring and X is proper over A, in fact projective. In that case the sections of X over Y, that is, the A-valued points of X, also identify with the rational sections, namely with the K-valued points of

$$X \otimes_A K = X_K,$$

which is a proper smooth scheme over the fraction field K of A.

III.4. Local infinitesimal lifting of smooth S-schemes

Theorem III.4.1. *Let Y be a locally Noetherian prescheme, let Y_0 be a closed subprescheme with the same underlying space, let X_0 be a smooth Y_0-prescheme, and let $x \in X_0$. There exist an open neighborhood U_0 of x, a prescheme X smooth over Y, and a Y_0-isomorphism*

$$h\colon U_0 \xrightarrow{\sim} X \times_Y Y_0.$$

Moreover, if (U'_0, X', h') is another solution of this problem, then it is isomorphic to the first in a neighborhood of x.

We leave it to the reader to make the last assertion precise. For a fixed U_0, a solution of the problem amounts to giving on U_0 a sheaf of algebras $\mathcal{B}$ over $f_0^{-1}(\mathcal{O}_Y)$, where f_0 is the continuous map underlying the structural morphism $U_0 \to Y_0$, together with a ring homomorphism

$$\mathcal{B} \longrightarrow \mathcal{O}_{U_0}$$

compatible with $f_0^{-1}(\mathcal{O}_Y) \to f_0^{-1}(\mathcal{O}_{Y_0})$, such that:

(a) this homomorphism induces an isomorphism

$$\mathcal{B} \otimes_{f_0^{-1}(\mathcal{O}_Y)} f_0^{-1}(\mathcal{O}_{Y_0}) \xrightarrow{\sim} \mathcal{O}_{U_0};$$

(b) U_0, equipped with $\mathcal{B}$, becomes a smooth Y-prescheme.

In this formulation, the precise meaning of the local uniqueness assertion becomes particularly clear.

Proof. We may first suppose that X_0 is étale over

$$Y_0[t_1, \ldots, t_n] = Y_0'.$$

The latter may be regarded as a closed subprescheme, with the same underlying space, of $Y' = Y[t_1, \ldots, t_n]$. By I I.8.3, there exist a prescheme X étale over Y' and a Y_0'-isomorphism

$$X \times_{Y'} Y_0' \xrightarrow{\sim} X_0.$$

This proves existence. For uniqueness, use the lifting property in Theorem III.3.1(ii) for smooth morphisms, together with the following lemma.

Lemma III.4.2. *Let Y be a prescheme, let Y_0 be a closed subprescheme defined by a locally nilpotent sheaf of ideals $\mathcal{J}$, let X and X' be Y-preschemes, and let $u\colon X \to X'$ be a Y-morphism. Suppose that X is flat over Y. Then u is an isomorphism if and only if*

$$u_0\colon X \times_Y Y_0 \longrightarrow X' \times_Y Y_0$$

is an isomorphism.

This follows easily by passing to the affine case and considering the associated graded objects. Notice also that the analogous statement obtained by replacing "isomorphism" with "closed immersion" is valid without the flatness hypothesis.

Remark III.4.3. It is essential to observe that the local lifting X obtained in Theorem III.4.1 is *not canonical.* In other words, the local isomorphism between two solutions is not unique: in general there are nontrivial Y-automorphisms of X inducing the identity on the closed subprescheme $X_0 = X \times_Y Y_0$. Consequently, in constructing global infinitesimal liftings of smooth preschemes one should expect a cohomological obstruction, which will be made precise below in Section III.6.

III.5. Global infinitesimal lifting of morphisms

Let T be a topological space, let $\mathcal{G}$ be a sheaf of groups on T, and let $\mathcal{P}$ be a sheaf of sets on T on which $\mathcal{G}$ acts, on the right for definiteness. We say that $\mathcal{P}$ is *formally principal homogeneous* under $\mathcal{G}$ if the familiar homomorphism

$$\mathcal{G} \times \mathcal{P} \longrightarrow \mathcal{P} \times \mathcal{P}$$

of sheaves of sets, induced by the action of $\mathcal{G}$ on $\mathcal{P}$, is an *isomorphism.* Equivalently, for every $x \in T$, the stalk $\mathcal{P}_x$ is either empty or a principal homogeneous space under the ordinary group $\mathcal{G}_x$. Equivalently again, for every open subset $U \subseteq T$, the set $\mathcal{P}(U)$ is either empty or a principal homogeneous space under the ordinary group $\mathcal{G}(U)$.

We say that $\mathcal{P}$ is a *principal homogeneous sheaf* under $\mathcal{G}$ if it is formally principal homogeneous and all its stalks $\mathcal{P}_x$ are nonempty; in other words, if every $\mathcal{P}_x$ is a nonempty principal homogeneous space under $\mathcal{G}_x$. [4] Recall that the set of isomorphism classes of principal homogeneous sheaves under $\mathcal{G}$ is identified with the cohomology set $H^1(T, \mathcal{G})$. When $\mathcal{G}$ is commutative, this is the usual cohomology group of T with coefficients in $\mathcal{G}$. Thus every principal homogeneous sheaf $\mathcal{P}$ has a characteristic class

$$c(\mathcal{P}) \in H^1(T, \mathcal{G}).$$

This class vanishes if and only if $\mathcal{P}$ is trivial—that is, isomorphic to $\mathcal{G}$, acting on itself by right translations—or, equivalently, if and only if $\mathcal{P}$ has a section.

[4] It seems preferable to adopt the shorter and more expressive term "*torsor under $\mathcal{G}$*", introduced in the thesis of J. Giraud.

PROPOSITION III.5.1. *Let S be a prescheme, let X and Y be preschemes over S, and let Y_0 be a closed subprescheme of Y, defined by an ideal sheaf $\mathcal{J}$ on Y with $\mathcal{J}^2 = 0$. Let $g_0 : Y_0 \to X$ be an S-morphism, and let $\mathcal{P}(g_0)$ be the sheaf on Y whose sections on an open subset U are the S-morphisms $g: U \to X$ extending $g_0|_{U \cap Y_0}$. Then $\mathcal{P}(g_0)$ is naturally a* formally principal homogeneous sheaf *under the commutative sheaf of groups*

$$\mathcal{G} = \mathcal{H}om_{\mathcal{O}_{Y_0}}(g_0^* \Omega^1_{X/S}, \mathcal{J}).$$

Put $\mathcal{P} = \mathcal{P}(g_0)$. For every open subset U of Y, we must define a map

$$\mathcal{P}(U) \times \mathcal{G}(U) \longrightarrow \mathcal{P}(U)$$

such that: (a) for fixed $g \in \mathcal{P}(U)$, the map $s \mapsto gs$ from $\mathcal{G}(U)$ to $\mathcal{P}(U)$ is bijective; (b) $\mathcal{P}(U)$ becomes a right $\mathcal{G}(U)$-set; and (c) these maps are compatible with restriction to every open subset $V \subseteq U$. Condition (c) will be immediate, so for simplicity we may take $U = Y$. We leave condition (b), which is local in nature, to the reader. It remains, for a given $g \in \mathcal{P}(Y)$, to define a natural bijection from $\mathcal{G}(Y)$ onto $\mathcal{P}(Y)$.

We are therefore given an S-morphism $g: Y \to X$, and seek a canonical bijection

$$\mathrm{Hom}_{\mathcal{O}_{Y_0}}(g_0^* \Omega^1_{X/S}, \mathcal{J}) \xrightarrow{\sim} \mathcal{P}(Y), \tag{*}$$

where $\mathcal{P}(Y)$ is the set of S-morphisms $g' : Y \to X$ that induce on Y_0 the same morphism $g_0 : Y_0 \to X$ as g. Giving such a g' is equivalent to giving an S-morphism $h: Y \to X \times_S X$ such that

$$\mathrm{pr}_1 \circ h = g, \qquad h \circ i = (g_0, g_0),$$

where $\mathrm{pr}_1 : X \times_S X \to X$ is the first projection, $i: Y_0 \to Y$ is the canonical immersion, and $(g_0, g_0) = \Delta_{X/S} g_0 : Y_0 \to X \times_S X$ has components g_0, g_0. This is summarized by the diagram

$$\begin{tikzcd} Y_0 \arrow[r, "h_0=(g_0,g_0)"] \arrow[d, "i"'] & X \times_S X \arrow[d, "\mathrm{pr}_1"] \\ Y \arrow[ur, "h"] \arrow[r, "g"] & X. \end{tikzcd}$$

The morphism h_0 factors through the diagonal immersion $\Delta_{X/S}$. Since Y lies in the first infinitesimal neighborhood of Y_0, that is, since $\mathcal{J}^2 = 0$, every such h necessarily factors, uniquely, through the first infinitesimal neighborhood of the diagonal. As an X-prescheme via pr_1, this neighborhood is identified with the relative spectrum X' of the sheaf of algebras

$$\mathcal{O}_X \oplus \Omega^1_{X/S},$$

where the second summand is a square-zero ideal; the diagonal morphism $X \to X'$ corresponds to the canonical augmentation of this algebra sheaf.

Put

$$Y' = X' \times_X Y, \qquad Y'_0 = Y' \times_Y Y_0 = X' \times_X Y_0.$$

The desired morphisms h are then in bijection with the sections u of Y' over Y that extend a fixed section u_0 of Y'_0 over Y_0. More explicitly, Y' is the relative spectrum over Y of

$$\mathcal{A} = g^*(\mathcal{O}_X \oplus \Omega^1_{X/S}) = \mathcal{O}_Y \oplus g^* \Omega^1_{X/S},$$

and Y'_0 is the relative spectrum over Y_0 of

$$\mathcal{A}_0 = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0} = \mathcal{O}_{Y_0} \oplus g_0^* \Omega^1_{X/S}.$$

The section u_0 is defined by the canonical augmentation $\mathcal{A}_0 \to \mathcal{O}_{Y_0}$. Hence $\mathcal{P}(Y)$ is identified with the algebra homomorphisms $\mathcal{A} \to \mathcal{O}_Y$ inducing that augmentation on $\mathcal{A}_0$.

For brevity put $\mathcal{M} = g^* \Omega^1_{X/S}$ and $\mathcal{M}_0 = \mathcal{M} \otimes_{\mathcal{O}_Y} \mathcal{O}_{Y_0}$. Such an algebra homomorphism is determined by an $\mathcal{O}_Y$-linear homomorphism $\mathcal{M} \to \mathcal{O}_Y$. The condition on $\mathcal{A}_0$ says precisely that its image lies in $\mathcal{J}$; conversely, because $\mathcal{J}^2 = 0$, every $\mathcal{O}_Y$-linear map $\mathcal{M} \to \mathcal{J}$ defines such an algebra homomorphism. We therefore obtain

$$\mathrm{Hom}_{\mathcal{O}_Y}(\mathcal{M}, \mathcal{J}) = \mathrm{Hom}_{\mathcal{O}_{Y_0}}(\mathcal{M}_0, \mathcal{J}),$$

because the action of $\mathcal{O}_Y$ on $\mathcal{J}$ factors through $\mathcal{O}_{Y_0}$. This is precisely the canonical bijection (*).

Taking into account the implication (i) ⇒ (iii) in Theorem III.3.1, we obtain:

Corollary III.5.2. *If X is smooth over S, at least at the points of $g_0(Y_0)$, then $\mathcal{P}$ is in fact a* principal homogeneous sheaf *under the commutative sheaf of groups $\mathcal{G}$. In this case it may also be written*

$$\mathcal{G} = g_0^*(\mathfrak{g}_{X/S}) \otimes_{\mathcal{O}_{Y_0}} \mathcal{J},$$

where $\mathfrak{g}_{X/S} = (\Omega^1_{X/S})^\vee$ is the sheaf on X dual to $\Omega^1_{X/S}$, that is, the tangent sheaf*, or* sheaf of derivations*, of X relative to S. The displayed formula follows from the fact that $\Omega^1_{X/S}$ is then locally free of finite rank.*

In particular, this principal homogeneous sheaf determines a cohomology class in $H^1(Y_0, \mathcal{G})$. Its vanishing is necessary and sufficient for the existence of an S-morphism g extending g_0. If such an extension exists, the set of all possible extensions is a homogeneous space under the group $H^0(Y_0, \mathcal{G})$.

In applications of the methods of formal geometry, the situation is most often the following. Let X and Y be S-preschemes, let $\mathcal{I}$ be a coherent ideal sheaf on S, let S_n denote the closed subprescheme of S defined by $\mathcal{I}^{n+1}$, and put

$$X_n = X \times_S S_n, \qquad Y_n = Y \times_S S_n.$$

Suppose that an S_n-morphism

$$g_n \colon Y_n \longrightarrow X_n$$

is given. Equivalently, this is an S-morphism $Y_n \to X$, or an S_{n+1}-morphism $Y_n \to X_{n+1}$, since any such morphism necessarily induces $Y_n \to X_n$. We seek to extend it to an S_{n+1}-morphism

$$g_{n+1} \colon Y_{n+1} \longrightarrow X_{n+1}.$$

If this process can be continued indefinitely, it yields a morphism $\widehat{Y} \to \widehat{X}$ between the formal preschemes obtained by completing Y and X with respect to the ideals $\mathcal{I}\mathcal{O}_Y$ and $\mathcal{I}\mathcal{O}_X$, respectively.

Apply Proposition III.5.1 with

$$(S, X, Y, Y_0, g_0) = (S_{n+1}, X_{n+1}, Y_{n+1}, Y_n, g_n).$$

The sheaf $\mathcal{G}$ is now the sheaf of homomorphisms from

$$g_n^*\Omega^1_{X_{n+1}/S_{n+1}}$$

to

$$\mathcal{J} = \mathcal{I}^{n+1}\mathcal{O}_Y/\mathcal{I}^{n+2}\mathcal{O}_Y.$$

Since $\mathcal{J}$ is annihilated by $\mathcal{I}\mathcal{O}_Y$, the first sheaf may be replaced by its restriction to $Y_0 = Y \times_S S_0$. Let $h_0 \colon Y_0 \to X_{n+1}$ be the composite $Y_0 \to Y_n \xrightarrow{g_n} X_n \to X_{n+1}$, and let $g_0 \colon Y_0 \to X_0$ be the morphism induced by g_n. Equivalently, h_0 is the composite $Y_0 \xrightarrow{g_0} X_0 \to X_{n+1}$. The restricted sheaf is $h_0^*\Omega^1_{X_{n+1}/S_{n+1}}$. Since the inverse image of $\Omega^1_{X_{n+1}/S_{n+1}}$ on

$$X_0 = X_{n+1} \times_{S_{n+1}} S_0$$

is $\Omega^1_{X_0/S_0}$, we may write

$$\mathcal{G} = \mathcal{H}om_{\mathcal{O}_{Y_0}}\left(g_0^*\Omega^1_{X_0/S_0}, \mathcal{I}^{n+1}\mathcal{O}_Y/\mathcal{I}^{n+2}\mathcal{O}_Y\right).$$

Thus we obtain the following corollary.

COROLLARY III.5.3. *Let $S, X, Y, \mathcal{I}, g_n$ be as above. Let $\mathcal{P}(g_n)$ be the sheaf on Y whose sections over an open set U are the extensions g_{n+1} of g_n to an S_{n+1}-morphism $Y_{n+1} \to X_{n+1}$. Then $\mathcal{P}(g_n)$ is a pseudo-torsor under the sheaf of groups*

$$\mathcal{G} = \mathcal{H}om_{\mathcal{O}_{Y_0}}\left(g_0^*\Omega^1_{X_0/S_0}, \mathrm{gr}^{n+1}_{\mathcal{I}\mathcal{O}_Y}(\mathcal{O}_Y)\right).$$

In particular:

COROLLARY III.5.4. *If, in addition, X is smooth over S, at least at the points of $g_0(Y_0)$, then $\mathcal{P}(g_n)$ is a torsor. It therefore defines an obstruction class in $H^1(Y_0, \mathcal{G})$, whose vanishing is necessary and sufficient for the existence of a global extension g_{n+1} of g_n. If such an extension exists, the set of all global extensions is a principal homogeneous space under $H^0(Y_0, \mathcal{G})$. In this case the sheaf $\mathcal{G}$ may also be written*

$$\mathcal{G} = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes_{\mathcal{O}_{Y_0}} \mathrm{gr}^{n+1}_{\mathcal{I}\mathcal{O}_Y}(\mathcal{O}_Y).$$

Proceeding step by step, we see that if all the $H^1(Y_0, \mathcal{G}_n)$ vanish, where

$$\mathcal{G}_n = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes \mathrm{gr}^{n}_{\mathcal{I}\mathcal{O}_Y}(\mathcal{O}_Y),$$

then, starting with any g_k, one may successively extend it to $g_{k+1}, \ldots$. In particular, if $\mathcal{I}$ is nilpotent, an extension of g_k to Y exists. The required H^1-vanishing holds in particular when Y_0 is affine. Thus:

COROLLARY III.5.5. *In Theorem III.3.1, one obtains another equivalent necessary and sufficient condition by restricting the Y' occurring in* (ii)*, or in* (ii bis)*, to be affine and requiring a global extension g of g_0 over all of Y'.*

Notice that the proof of Theorem III.3.1 could not have given this result directly.

An important case is that in which Y is flat over S. Then

$$\mathrm{gr}^n(\mathcal{O}_Y) = \mathrm{gr}^n(\mathcal{O}_S) \otimes_{\mathcal{O}_{S_0}} \mathcal{O}_{Y_0}.$$

If, in addition, the $\mathrm{gr}^n(\mathcal{O}_S)$ are locally free over S, then

$$\mathcal{G}_n = \mathcal{H}om_{\mathcal{O}_{Y_0}}(g_0^*\Omega^1_{X_0/S_0}, \mathcal{O}_{Y_0}) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}^n(\mathcal{O}_S).$$

Equivalently, if $\Omega^1_{X_0/S_0}$ is also locally free, for example if X is smooth over S,

$$\mathcal{G}_n = g_0^*(\mathfrak{g}_{X_0/S_0}) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}^n(\mathcal{O}_S).$$

For example, suppose that S is affine with coordinate ring A, and that $\mathcal{I}$ is defined by an ideal $I \subseteq A$. Then, for every i,

$$H^i(Y_0, \mathcal{G}_n) = H^i(Y_0, \mathcal{G}_0) \otimes_A \mathrm{gr}^n_I(A).$$

Indeed, the question is local on S_0, and reduces to tensoring with a free module. In this case, the vanishing of $H^1(Y_0, \mathcal{G}_0)$ implies that every obstruction to the successive extensions of g_n vanishes. We obtain:

COROLLARY III.5.6. *Let $(S, X, Y, \mathcal{I}, g_n)$ be as above. Suppose in addition that X is smooth over S, that Y is flat over S, that S is affine, and that*

$$\mathrm{gr}^n(\mathcal{O}_S) = \mathcal{I}^n/\mathcal{I}^{n+1}$$

is locally free for every n. Then the obstruction to constructing g_{n+1} lies in

$$H^1(Y_0, \mathcal{G}_0) \otimes_A \mathrm{gr}^{n+1}_I(A),$$

where A is the ring of S, I is the ideal defining $\mathcal{I}$, and

$$\mathcal{G}_0 = g_0^*(\mathfrak{g}_{X_0/S_0}).$$

If $H^1(Y_0, \mathcal{G}_0) = 0$, then g_n extends to an $\widehat{S}$-morphism

$$\widehat{g}\colon \widehat{Y} \longrightarrow \widehat{X}.$$

The same result plainly remains valid if, instead of ordinary S-preschemes X and Y, one starts with adic formal $\widehat{S}$-preschemes $\mathfrak{X}$ and $\mathfrak{Y}$. For example, it can be used to show that certain formal schemes proper over a complete local ring are in fact algebraic. Proceeding as in Lemma III.4.2, one obtains:

COROLLARY III.5.7. *Under the hypotheses of Corollary III.5.6, if g_0 is an isomorphism, then so is $\widehat{g}$.*

The same statement holds for closed immersions.

PROPOSITION III.5.8. *Let A be a complete local ring with maximal ideal $\mathfrak{m}$ and residue field k. Let $\mathfrak{X}$ and $\mathfrak{Y}$ be two $\mathfrak{m}$-adic formal preschemes over A, flat over A; that is, for every n, X_n and Y_n are flat over $A_n = A/\mathfrak{m}^{n+1}$. Suppose that*

$$X_0 = \mathfrak{X} \otimes_A k$$

is smooth over k and that

$$H^1(X_0, \mathfrak{g}_{X_0/k}) = 0.$$

Then every k-isomorphism $Y_0 \xrightarrow{\sim} X_0$ extends to an A-isomorphism $\mathfrak{Y} \xrightarrow{\sim} \mathfrak{X}$. This extension is unique if, in addition,

$$H^0(X_0, \mathfrak{g}_{X_0/k}) = 0.$$

This gives, in particular, a uniqueness result for a smooth formal prescheme over A reducing to a given prescheme X_0, provided $H^1(X_0, \mathfrak{g}_{X_0/k}) = 0$. Furthermore, if $\mathfrak{X}$ and $\mathfrak{Y}$ arise as the formal completions of ordinary schemes X and Y proper over A, then the existence theorem for sheaves in formal geometry (see Bourbaki Seminar, no. 182, and EGA III, 5.4.1) gives a bijection between the A-isomorphisms $Y \xrightarrow{\sim} X$ and the A-isomorphisms of their formal completions. Hence:

COROLLARY III.5.9. *Proposition III.5.8 remains valid when $\mathfrak{X}$ and $\mathfrak{Y}$ are replaced by ordinary A-schemes X and Y proper over A.*

Finally, suppose that $\mathfrak{X}$ is a formal scheme proper over A, while $\mathfrak{Y} = \widehat{Y}$ for an ordinary scheme Y proper over A. Proposition III.5.8 gives sufficient conditions for an isomorphism $\mathfrak{X} \xrightarrow{\sim} \widehat{Y}$ to exist, and hence for $\mathfrak{X}$ to be algebraic, that is, isomorphic to $\widehat{X}$ for an ordinary scheme X proper over A; this X is then canonically determined. This applies in particular when $X_0 = \mathbb{P}^r_k$, and more generally when X_0 is a Severi–Brauer scheme, that is, becomes isomorphic to projective space over the algebraic closure of k. Every formal scheme proper and flat over A with special fiber $\mathbb{P}^r_k$ is algebraizable; more precisely, it is isomorphic to the $\mathfrak{m}$-adic formal completion of $\mathbb{P}^r_A$. Consequently, by the existence theorem, every ordinary scheme proper over a complete local ring A, with special fiber $\mathbb{P}^r_k$, is isomorphic to $\mathbb{P}^r_A$.

Using descent theory, one can prove that when A is not complete, X becomes isomorphic to $\mathbb{P}^r$ after a finite étale base extension $A \to A'$. In this form the result remains valid when the special fiber is a Severi–Brauer scheme.

III.6. Global infinitesimal lifting of smooth S-schemes

Under the hypotheses of Theorem III.4.1, we ask whether there is a prescheme X smooth over Y such that $X \times_Y Y_0$ is Y_0-isomorphic to X_0, knowing that such a scheme exists locally on X_0. Returning to the step-by-step construction, replace Y by S. Let S_0 be a closed subprescheme of S defined by a sheaf of ideals $\mathcal{I}$, no longer necessarily locally nilpotent, and let S_n be the closed subprescheme of S defined by $\mathcal{I}^{n+1}$. Suppose that a prescheme X_n, smooth over S_n, has been given. We seek an S_{n+1}-prescheme X_{n+1} reducing to X_n, that is, equipped with an isomorphism

$$X_n \xrightarrow{\sim} X_{n+1} \times_{S_{n+1}} S_n,$$

and smooth over S_{n+1}, or equivalently flat over S_{n+1} by II II.2.1.

As observed in Section III.4, such data amount to a sheaf of algebras $\mathcal{B}$ over $f^{-1}(\mathcal{O}_{S_{n+1}})$, where f is the continuous map underlying the structural morphism $X_n \to S_n$, together with an augmentation $\mathcal{B} \to \mathcal{O}_{X_n}$ compatible with

$$f^{-1}(\mathcal{O}_{S_{n+1}}) \longrightarrow f^{-1}(\mathcal{O}_{S_n}),$$

and satisfying the two conditions (a) and (b) already given there. We need only recall that they are local on the underlying topological space of X_n. Theorem III.4.1 gives a solution locally, unique locally up to a nonunique isomorphism. We first make this last point precise.

Proposition III.6.1. *Let X_{n+1} over S_{n+1} reduce to X_n over S_n. The sheaf, on the underlying topological space of X_n, equivalently of X_0, of S_{n+1}-automorphisms of X_{n+1} which induce the identity on X_n is canonically isomorphic, as a sheaf of groups, to*

$$\mathcal{G} = \mathfrak{g}_{X_0/S_0} \otimes_{\mathcal{O}_{S_0}} \operatorname{gr}^{n+1}_{\mathcal{I}}(\mathcal{O}_S).$$

Indeed, by Corollary III.5.4 and Lemma III.4.2, this automorphism sheaf is a torsor under $\mathcal{G}$. It has the distinguished section given by the identity automorphism of X_{n+1}, and hence identifies with $\mathcal{G}$ as a sheaf of sets. Compatibility with the group structures is immediate. It is also a special case of a more general compatibility, omitted here, between the torsor structures of Proposition III.5.1 and Corollary III.5.3 and the composition of morphisms.

In particular, the sheaf on X_0 of germs of automorphisms of X_{n+1}, with the structures just described, is commutative. Thus, if X'_{n+1} is another solution and is isomorphic to X_{n+1} over an open set $U \subseteq X_0$, the isomorphism

$$\mathcal{A}ut(X_{n+1})|_U \xrightarrow{\sim} \mathcal{A}ut(X'_{n+1})|_U$$

obtained by transport of structure through an isomorphism $X_{n+1}|_U \xrightarrow{\sim} X'_{n+1}|_U$ does not depend on the chosen isomorphism. Under the identifications of Proposition III.6.1, it is simply the identity of $\mathcal{G}$.

We deduce:

Corollary III.6.2. *Let X_{n+1} and X'_{n+1} be smooth over S_{n+1} and reduce to X_n. Then the sheaf, on the underlying space of X_0, of S_{n+1}-isomorphisms*

$$X_{n+1} \xrightarrow{\sim} X'_{n+1}$$

which induce the identity on X_n is naturally a torsor under $\mathcal{G}$.

This says precisely that X_{n+1} and X'_{n+1} are locally isomorphic and that the sheaf of germs of automorphisms of either one is $\mathcal{G}$.

By Theorem III.4.1, one can choose an open cover (U_i) of X_n, which may be assumed affine, and for each i a smooth scheme X^i over S_{n+1} reducing to U_i. For simplicity suppose that X_n is separated. The intersections $U_{ij} = U_i \cap U_j$ are then affine. Since $H^1(U_{ij}, \mathcal{G}) = 0$ for the quasi-coherent sheaf $\mathcal{G}$, Corollary III.6.2 shows that $X^i|_{U_{ij}}$ and $X^j|_{U_{ij}}$ are isomorphic. Choose an isomorphism

$$f_{ji} \colon X^i|_{U_{ij}} \xrightarrow{\sim} X^j|_{U_{ij}}.$$

It is determined up to a section of $\mathcal{G}$ over U_{ij}. For every triple of indices put

$$f^{(k)}_{ji} = f_{ji}|_{U_{ijk}}, \qquad U_{ijk} = U_i \cap U_j \cap U_k.$$

If

$$f^{(i)}_{kj} f^{(k)}_{ji} = f^{(j)}_{ki}, \tag{1}$$

then the X^i glue through the f_{ji}, and hence define the desired solution $X = X_{n+1}$. More generally, a solution exists if the f_{ji} can be modified to

$$f'_{ji} = f_{ji} g_{ji}, \qquad g_{ji} \in \Gamma(U_{ij}, \mathcal{G}), \tag{2}$$

so that the f'_{ji} satisfy the preceding cocycle condition. This sufficient condition is also necessary. Indeed, any solution X is locally isomorphic to X^i; choosing isomorphisms

$$f_i \colon X|_{U_i} \xrightarrow{\sim} X^i$$

gives

$$f'_{ji} = (f_j|_{U_{ij}})(f_i|_{U_{ij}})^{-1} \colon X^i|_{U_{ij}} \xrightarrow{\sim} X^j|_{U_{ij}},$$

which satisfy the gluing condition.

Now set

$$f_{ijk} = (f^{(j)}_{ki})^{-1} f^{(i)}_{kj} f^{(k)}_{ji}. \tag{3}$$

This is an automorphism of $X^i|_{U_{ijk}}$, which we identify with a section of $\mathcal{G}$ by Proposition III.6.1. A direct formal calculation, left to the reader, shows that $f = (f_{ijk})$ is a Čech 2-cocycle of the open cover $\mathcal{U} = (U_i)$ with coefficients in $\mathcal{G}$. The same calculation shows that, after the modification (2), the gluing condition (1) for the f'_{ij} is equivalent to

$$f = dg, \tag{4}$$

where $g = (g_{ij})$ is regarded as a Čech 1-cochain of $\mathcal{U}$ with coefficients in $\mathcal{G}$. Thus the necessary and sufficient condition for a solution to exist is the vanishing of the cohomology class in $H^2(\mathcal{U}, \mathcal{G})$ defined by the cocycle (3).

Since $\mathcal{U}$ is an affine open cover of the scheme X_0, $H^2(\mathcal{U}, \mathcal{G})$ identifies with $H^2(X_0, \mathcal{G})$. The resulting class in $H^2(X_0, \mathcal{G})$ is plainly independent of the chosen affine cover. We call it the obstruction class to extending X_n to a scheme X_{n+1} smooth over S_{n+1}.

Suppose this obstruction vanishes. The argument just given shows that every solution $X = X_{n+1}$ is isomorphic to one obtained by gluing with isomorphisms f'_{ji}, which may be assumed of the form (2). The gluing condition is then (4). The set of admissible g is a principal homogeneous space under the group $Z^1(\mathcal{U}, \mathcal{G})$ of Čech 1-cocycles. Moreover, two cochains g and g', with $dg = dg' = f$, define isomorphic solutions if and only if the cocycle $g - g'$ is of the form dh, where $h = (h_i) \in C^0(\mathcal{U}, \mathcal{G})$. We therefore obtain:

THEOREM III.6.3. *Let $(S, \mathcal{I}, X_n)$ be as above, and suppose that X_n is separated.*[5] *There is a canonically defined obstruction class in $H^2(X_0, \mathcal{G})$, where $\mathcal{G}$ is defined in Proposition III.6.1. Its vanishing is necessary and sufficient for the existence of a scheme X_{n+1}, smooth over S_{n+1}, which reduces to X_n. If the obstruction vanishes, the set of isomorphism classes, with the isomorphisms inducing the identity on X_n, of S_{n+1}-preschemes X_{n+1} reducing to X_n is naturally a principal homogeneous space under $H^1(X_0, \mathcal{G})$.*

REMARKS III.6.4. From Proposition III.6.1 onward, the arguments are purely formal and are better expressed in the language of local categories, or even general fibered categories. In that general setting, one obtains an obstruction class to the existence of a global object of a category in which objects exist locally, any two objects are locally isomorphic, and the automorphism group of every object is commutative. As a special case it contains the second connecting homomorphism in an exact sequence of sheaves of groups which need not be commutative, studied for example by Grothendieck in Kansas or Tôhoku. The elementary cocycle calculation given here should therefore be regarded as a makeshift necessitated by the absence of a satisfactory reference.

[5]This hypothesis is in fact unnecessary, and the cocycle calculations above can be avoided. See J. Giraud, *Cohomologie non abélienne*, forthcoming from Springer-Verlag in 1971, and compare Remarks III.6.4.

III.6.5. In Theorem III.6.3, the principal homogeneous space under $H^1(X_0, \mathcal{G})$ generally has no distinguished element. After localizing on S, this is reflected by a torsor on S_0 with structural group $R^1 f_*(\mathcal{G})$, which need not be trivial and thus defines a cohomology class in

$$H^1(S_0, R^1 f_*(\mathcal{G}))$$

which need not vanish. This is the situation when the class $d \in H^2(X_0, \mathcal{G})$, although not zero, vanishes locally over S; equivalently, it defines the zero section of $R^2 f_*(\mathcal{G})$, or the zero element of $H^0(S_0, R^2 f_*(\mathcal{G}))$.

III.6.6. At present almost nothing is known about the general algebraic mechanism of the cohomology classes introduced in this section and the preceding one, and nothing precise can be said even in the simplest special cases, such as abelian schemes over Artinian rings. [6] It is to be hoped that someone will work out this particularly interesting question in detail. It is closely connected, in particular, with the moduli theory of algebraic structures.

Corollary III.6.7. *If $H^2(X_0, \mathcal{G}) = 0$, then an X_{n+1} exists; it is unique up to isomorphism if, in addition, $H^1(X_0, \mathcal{G}) = 0$.*

Proceeding step by step, and recalling that an affine scheme is acyclic for a quasi-coherent sheaf, we obtain:

Corollary III.6.8. *Under the hypotheses of Theorem III.4.1, if X_0 is affine, then there exists a prescheme X, smooth over Y, reducing to X_0; it is unique up to a nonunique isomorphism.*

Notice that the direct proof of Theorem III.4.1 could not have given this result.

Corollary III.6.9. *Under the hypotheses of Theorem III.6.3, suppose that S is affine with ring A, that $\mathcal{I}$ is defined by an ideal $I \subseteq A$, and that*

$$\mathrm{gr}^n_{\mathcal{I}}(\mathcal{O}_S) = \mathcal{I}^n/\mathcal{I}^{n+1}$$

is locally free for every n. Then

$$H^i(X_0, \mathcal{G}) = H^i(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}^{n+1}_I(A), \qquad \mathcal{G}_0 = \mathfrak{g}_{X_0/S_0}.$$

Thus the obstruction class to extending X_n lies in

$$H^2(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}^{n+1}_I(A).$$

If it vanishes, the set of isomorphism classes of solutions is a principal homogeneous space under

$$H^1(X_0, \mathcal{G}_0) \otimes_A \mathrm{gr}^{n+1}_I(A).$$

In particular:

Corollary III.6.10. *Under the hypotheses of Corollary III.6.9, suppose that*

$$H^2(X_0, \mathfrak{g}_{X_0/S_0}) = 0.$$

Then there exists an I-adic formal scheme $\mathfrak{X}$ over the $\mathcal{I}$-adic formal completion $\widehat{S}$ of S, smooth over $\widehat{S}$ in the sense that each $\mathfrak{X}_p$ is smooth over S_p, and reducing to X_n; that is, equipped with an isomorphism

$$X_n \xrightarrow{\sim} \mathfrak{X} \times_S S_n.$$

If, in addition,

$$H^1(X_0, \mathfrak{g}_{X_0/S_0}) = 0,$$

then such a formal scheme $\mathfrak{X}$ is unique up to isomorphism.

[6] It is now known that the obstruction always vanishes in this case. [Added in 2003 (MR): see F. Oort, "Finite group schemes, local moduli for abelian varieties and lifting problems," *Algebraic Geometry, Oslo 1970*, Wolters-Noordhoff, 1972, pp. 223–254.]

Indeed, one constructs $X_{n+1}, X_{n+2}, \ldots$ successively and passes to the limit to obtain $\mathfrak{X}$. The uniqueness assertion was already proved in the preceding paragraph.

III.7. Application to the construction of formal and ordinary schemes smooth over a complete local ring A

The results of the preceding section sometimes prove the existence of an $\mathfrak{m}$-adic formal scheme over such a ring, reducing to a given smooth scheme X_0 over k. We distinguish two cases.

(a) Equal characteristics. This includes, in particular, the case where k has characteristic zero. There is then a coefficient field in A, that is, a subfield $k' \subseteq A$ such that $A \to k$ induces an isomorphism $k' \xrightarrow{\sim} k$. In fact, there is then an ordinary scheme smooth over A reducing to X_0, namely

$$X = X_0 \otimes_k A,$$

where A is regarded as a k-algebra through the section $k \to k' \to A$ defined by k'.

This construction is not natural. Already for the dual-number algebra $A = k[t]/(t^2)$, a different lifting homomorphism $k \to A$, defined in this case by an absolute derivation of k into itself, gives an A-scheme X' which in general is not isomorphic to X when

$$H^1(X_0, \mathfrak{g}_{X_0/k}) \neq 0.$$

It would be interesting, when k has characteristic zero, or is imperfect of characteristic $p > 0$, to determine which smooth A-schemes arise in this way and when two homomorphisms $k \to A$ give isomorphic A-schemes.

Nevertheless, the existence of k' implies that the first obstruction to lifting X_0, which lies in

$$H^2(X_0, \mathfrak{g}_{X_0/k}) \otimes_k \mathfrak{m}/\mathfrak{m}^2,$$

must vanish. After X_0 has been lifted to an X_1 smooth over $A/\mathfrak{m}^2$, however, the new obstruction to constructing X_2 need not vanish. It depends on a variable element in a certain principal homogeneous space under

$$H^1(X_0, \mathcal{G}_0) \otimes \mathfrak{m}/\mathfrak{m}^2$$

and lies in

$$H^2(X_0, \mathcal{G}_0) \otimes \mathfrak{m}^2/\mathfrak{m}^3.$$

The situation deserves a detailed study. [7]

(b) Unequal characteristics. In this case essentially nothing is known, except when by good fortune $H^2(X_0, \mathfrak{g}_{X_0/k}) = 0$, in which case an $\mathfrak{m}$-adic formal scheme smooth over A and reducing to X_0 can be constructed. Even for $A = \mathbb{Z}/p^2\mathbb{Z}$ and X_0 an abelian scheme of dimension 2, it was not known whether X_0 could be lifted to an $X = X_1$ smooth over A. [8] Nor was there an example proved not to arise from an ordinary scheme smooth over A; one expected such examples among projective surfaces. [9]

By Cohen's theorem there is a Cohen p-ring B with residue field k and a homomorphism $B \to A$ inducing the identity on residue fields. The strongest lifting result is therefore obtained by taking A itself to be a Cohen p-ring: if an ordinary or formal solution exists over such a ring, one exists over every complete local ring with residue field k. For a

[7] It is presumably described by the Kodaira–Spencer bracket; see the Cartan Seminar 1960/61, Exposé 4.

[8] This has now been proved; see the note to Section III.6.6.

[9] Such an example was subsequently constructed by J.-P. Serre, *Proc. Nat. Acad. Sci. USA* **47** (1961), no. 1, pp. 108–109, at least in certain dimensions. D. Mumford gave an unpublished example with an algebraic surface.

Cohen p-ring, $\mathfrak{m}/\mathfrak{m}^2$ identifies canonically with k. Hence, for every smooth scheme X_0 over a field k of characteristic $p > 0$, there is a cohomology class in

$$H^2(X_0, \mathfrak{g}_{X_0/k})$$

which is the first obstruction to lifting X_0 to a smooth scheme over a Cohen p-ring. It was not known whether this class could be nonzero.[10]

Even if the X_n reducing to X_0 can be constructed successively, this generally yields only a formal scheme $\mathfrak{X}$, smooth over A, reducing to X_0. When X_0 is proper over k, one must still decide whether $\mathfrak{X}$ is algebraizable, so as to obtain an ordinary scheme proper and smooth over A reducing to X_0. The only known criterion, stated in the Bourbaki Seminar and in EGA III, 4.7.1, is the following. Suppose that $\mathfrak{X}$ is proper over A, and that $\mathcal{L}$ is an invertible sheaf on $\mathfrak{X}$ whose restriction $\mathcal{L}_0$ to X_0 is ample, meaning that some positive tensor power $\mathcal{L}_0^{\otimes n}$ comes from a projective immersion of X_0. Then there exist a scheme X projective over A and an ample invertible sheaf on X whose $\mathfrak{m}$-adic completion is $(\mathfrak{X}, \mathcal{L})$.

This reduces us to extending a locally free sheaf $\mathcal{E}_0$ on X_0, chosen invertible and ample for the present purpose, to a locally free sheaf $\mathcal{E}$ on $\mathfrak{X}$. One constructs locally free sheaves $\mathcal{E}_n$ on the X_n successively. The discussion is entirely analogous to Section III.6; compare Remarks III.6.4. The essential role is played by the sheaf of automorphisms of an $\mathcal{E}_{n+1}$ which induce the identity on $\mathcal{E}_n$. This sheaf identifies with

$$(*) \qquad \mathcal{G} = \mathcal{H}om_{\mathcal{O}_{X_0}}\left(\mathcal{E}_0, \mathcal{E}_0 \otimes \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_X)\right) = \mathcal{E}nd_{\mathcal{O}_{X_0}}(\mathcal{E}_0) \otimes \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_X),$$

which is again a sheaf of commutative groups. We obtain:

Proposition III.7.1. *Let S be a prescheme equipped with a quasi-coherent sheaf of ideals $\mathcal{I}$, let X be a prescheme over S, let S_n be the closed subprescheme of S defined by $\mathcal{I}^{n+1}$, and put $X_n = X \times_S S_n$ for every n. Let $\mathcal{E}_n$ be a locally free sheaf on X_n. The obstruction to extending it to a locally free sheaf $\mathcal{E}_{n+1}$ on X_{n+1} is a canonical class in $H^2(X_0, \mathcal{G})$, where $\mathcal{G}$ is the quasi-coherent sheaf in* (*). *Its vanishing is necessary and sufficient for such an extension to exist. If it vanishes, the set of isomorphism classes, with isomorphisms inducing the identity on $\mathcal{E}_n$, of extensions $\mathcal{E}_{n+1}$ is a principal homogeneous space under $H^1(X_0, \mathcal{G})$.*

This proposition has the usual corollaries. We record only that if X is flat over S, then

$$\mathcal{G} = \mathcal{E}nd_{\mathcal{O}_{X_0}}(\mathcal{E}_0) \otimes_{\mathcal{O}_{S_0}} \mathrm{gr}_{\mathcal{I}}^{n+1}(\mathcal{O}_S).$$

Consequently, if S is affine with ring A and the $\mathcal{I}^n/\mathcal{I}^{n+1}$ are locally free, the condition

$$H^2(X_0, \mathcal{G}_0) = 0, \qquad \mathcal{G}_0 = \mathcal{E}nd_{\mathcal{O}_{X_0}}(\mathcal{E}_0),$$

is sufficient for $\mathcal{E}_{n+1}$ to exist, and hence, step by step, for all successive extensions $\mathcal{E}_m$, $m = n, n+1, \ldots$, to exist.

Returning to the original situation, we obtain:

Proposition III.7.2. *Let A be a complete local ring, and let $\mathfrak{X}$ be a formal scheme proper and flat over A. Suppose that X_0 is projective and*

$$H^2(X_0, \mathcal{O}_{X_0}) = 0.$$

Then there exists a scheme X projective over A whose $\mathfrak{m}$-adic formal completion is isomorphic to $\mathfrak{X}$.

Combining this with Corollary III.6.10, we obtain:

[10] It can be nonzero, as noted in Footnote 9.

THEOREM III.7.3. *Let A be a complete local ring with residue field k, and let X_0 be a smooth projective scheme over k such that*

$$H^2(X_0, \mathfrak{g}_{X_0/k}) = H^2(X_0, \mathcal{O}_{X_0}) = 0.$$

Then there exists a smooth projective scheme X over A reducing to X_0.

More generally, if X_n is smooth over $A_n = A/\mathfrak{m}^{n+1}$ and reduces to X_0, then there exist a scheme X smooth and proper over A and an isomorphism

$$X \otimes_A A_n \xrightarrow{\sim} X_n.$$

COROLLARY III.7.4. *Every smooth proper curve over k arises by reduction from a smooth proper curve over A.*

This result, together with the existence theorem for sheaves in formal geometry, will be the essential tool for studying the fundamental group of X_0 by transcendental methods.

CHAPTER IV

Flat morphisms

We give here chiefly the properties of flatness used in the preceding exposés. A more detailed study will appear in Chapter IV of the forthcoming *Éléments de Géométrie Algébrique*[1], where the following situation is treated systematically: X is locally of finite type over a locally Noetherian prescheme Y, and $\mathcal{F}$ is a coherent sheaf on X, flat over Y; one seeks relations among properties of Y, properties of $\mathcal{F}$, and properties of the coherent sheaves induced by $\mathcal{F}$ on the fibers of $X \to Y$, especially with respect to dimension, cohomological dimension, and depth. In particular, this yields a systematic way of obtaining theorems of Seidenberg or Bertini type for hyperplane sections.

The essential result for applying flatness methods in this setting is the following, to be proved below: if Y is integral, X is of finite type over Y, and $\mathcal{F}$ is coherent on X, then there exists a nonempty open subset $U \subseteq Y$ such that $\mathcal{F}$ is flat over Y at every point of X lying over U.

A second, and perhaps still more important, way in which flatness enters algebraic geometry is through descent theory. See, for example, Grothendieck's two Bourbaki Seminar talks on the subject, and, for a more detailed account, Exposés VIII and IX below. Flatness thus appears to be one of the central technical notions of algebraic geometry.

Recall that Serre introduced the notions of flatness and faithful flatness in GAGA. The material of Sections IV.1 and IV.2 below may also be found in Bourbaki's *Algèbre commutative*, which, as its title indicates, does not restrict itself to commutative base rings.[2]

Unlike the preceding exposés, we do not assume that the rings under consideration are necessarily Noetherian.

IV.1. Elementary facts about flat modules

An A-module M is called *flat*, or A-flat when the ring must be specified, if the functor

$$T_M \colon N \longmapsto M \otimes_A N,$$

which is always right exact, is exact; equivalently, if it takes monomorphisms to monomorphisms. This is equivalent to the vanishing of the first left-derived functor, or of all the left-derived functors:

$$\operatorname{Tor}_1^A(M,N) = 0 \qquad \text{for every } N,$$

or, respectively,

$$\operatorname{Tor}_i^A(M,N) = 0 \qquad \text{for every } i > 0 \text{ and every } N.$$

Since the Tor_i commute with filtered direct limits, it is enough to verify these conditions for finitely generated N. Using a filtration of N with cyclic quotients, it is enough even to require

$$\operatorname{Tor}_1^A(M,N) = 0$$

[1] See EGA IV, 11 and 12.

[2] N. Bourbaki, *Algèbre commutative*, Chapter I (Modules plats), Act. Sci. Ind. 1290, Hermann, Paris, 1961.

when N is cyclic, hence of the form A/I for an ideal $I \subseteq A$. Moreover,

$$\mathrm{Tor}_1^A(M, A/I) = 0 \quad \Longleftrightarrow \quad I \otimes_A M \longrightarrow M = A \otimes_A M \text{ is injective.}$$

This follows from the exact Tor-sequence, since $\mathrm{Tor}_1^A(M, A) = 0$. Thus M is flat if and only if, for every ideal I, the natural homomorphism

$$I \otimes_A M \longrightarrow IM$$

is an isomorphism. It is enough to verify this for finitely generated ideals I; a fortiori it is enough to verify that $M \otimes_A -$ is exact on finitely generated modules.

As for every exact functor T, if $T(N')$ is identified with a subobject of $T(N)$ whenever $N' \subseteq N$, then

$$T(N' \cap N'') = T(N') \cap T(N''), \qquad T(N' + N'') = T(N') + T(N'')$$

for any two subobjects N' and N'' of N.

A direct sum of flat modules is flat, and a direct summand of a flat module is flat. Since A is flat over itself, every free module, and hence every projective module, is flat. The tensor product of two flat modules is flat. If M is flat over A, then $M \otimes_A B$ is flat over B after every base change $A \to B$. These assertions follow from associativity of the tensor product and from the fact that a composite of exact functors is exact. Likewise, if M is flat over B and B is flat over A, then M is flat over A.

The exact Tor-sequence, together with the symmetry of Tor, gives:

Proposition IV.1.1. *Let*

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0$$

be an exact sequence of A-modules, with M'' flat. Then:

(i) *the sequence remains exact after tensoring with any A-module N;*
(ii) *M is flat if and only if M' is flat.*

Thus, from the viewpoint of tensor products, flat modules are as well behaved as free or projective modules. In particular, the exact sequence in Proposition IV.1.1 behaves as well as if it split.

Let S be a multiplicatively closed subset of A. Then $S^{-1}A$ is flat over A, since

$$S^{-1}A \otimes_A N = S^{-1}N$$

is an exact functor of N. If M is A-flat, then $S^{-1}M = S^{-1}A \otimes_A M$ is $S^{-1}A$-flat. Conversely, if $M \to S^{-1}M$ is an isomorphism, equivalently if every $s \in S$ acts bijectively on M, then $S^{-1}A$-flatness of M implies A-flatness by transitivity.

More generally, the study of the flatness over Y of a quasi-coherent sheaf $\mathcal{F}$ on a prescheme $X \to Y$ leads to the following two-ring situation.

Proposition IV.1.2. *Let $A \to B$ be a ring homomorphism, let M be a B-module, and let T be a multiplicatively closed subset of B.*

(i) *If M is A-flat, then $T^{-1}M$ is A-flat. Consequently, it is also $S^{-1}A$-flat for every multiplicatively closed subset $S \subseteq A$ whose image lies in T.*
(ii) *Conversely, suppose that for every maximal ideal $\mathfrak{n}$ of B, $M_{\mathfrak{n}}$ is flat over A, or equivalently over $A_{\mathfrak{m}}$, where $\mathfrak{m}$ is the inverse image of $\mathfrak{n}$ in A. Then M is A-flat.*

Indeed, functorially in the A-module N,

$$T^{-1}M \otimes_A N = T^{-1}(M \otimes_A N).$$

Both sides are naturally isomorphic, by associativity, to

$$T^{-1}B \otimes_B M \otimes_B (N \otimes_A B).$$

Thus exactness of $M \otimes_A N$ in N implies that of $T^{-1}M \otimes_A N$, proving (i). It also proves (ii), since exactness of a sequence of B-modules may be checked after localization at every maximal ideal of B.

PROPOSITION IV.1.3. (i) *Let M be a flat A-module. If $x \in A$ is not a zero divisor in A, then it is not a zero divisor on M. In particular, if A is an integral domain, then M is torsion-free.*

(ii) *Suppose that A is an integral domain and that $A_{\mathfrak{m}}$ is a principal ideal domain for every maximal ideal $\mathfrak{m}$ of A, for example when A is a Dedekind domain or a principal ideal domain. An A-module M is flat if and only if it is torsion-free.*

For (i), multiplication by x on M is obtained by tensoring multiplication by x on A with M. For (ii), Proposition IV.1.2(ii) reduces us to the case where A is a principal ideal domain. If M is torsion-free, then for every ideal $I = (x)$, tensoring the injection $I \to A$ with M remains injective, since x is not a zero divisor on M. This is precisely the flatness criterion above.

IV.2. Faithfully flat modules

A functor F from one category to another is called *faithful* if, for every pair of objects X, Y, the map
$$\mathrm{Hom}(X, Y) \longrightarrow \mathrm{Hom}(F(X), F(Y))$$
is injective. For an additive functor between additive categories, this is equivalent to saying that $F(u) = 0$ implies $u = 0$, and it implies that $F(X) = 0$ only if $X = 0$.

For a functor between abelian categories to be faithful and exact, it is necessary and sufficient that a sequence
$$M' \longrightarrow M \longrightarrow M''$$
be exact if and only if its image
$$F(M') \longrightarrow F(M) \longrightarrow F(M'')$$
is exact. Equivalently, F must be exact and $F(X) = 0$ must imply $X = 0$.

Suppose that a family (M_i) of nonzero objects has the property that every nonzero object has a subobject admitting a quotient isomorphic to some M_i. Then F is faithful and exact if and only if F is exact and $F(M_i) \neq 0$ for every i. In the category of modules over a ring A, one may take the family of modules $A/\mathfrak{m}$, where $\mathfrak{m}$ runs through the maximal ideals of A. Indeed, every nonzero module has a nonzero cyclic submodule, isomorphic to A/I for some proper ideal I, and by Krull's theorem this has a quotient $A/\mathfrak{m}$. Hence:

PROPOSITION IV.2.1. *For an A-module M, the following conditions are equivalent:*

(i) *The functor $M \otimes_A -$ is faithful and exact.*

(i bis) *M is flat, and $M \otimes_A N = 0$ implies $N = 0$.*

(i ter) *M is flat, and*
$$M \otimes_A A/\mathfrak{m} \neq 0$$
for every maximal ideal $\mathfrak{m}$ of A.

(ii) *For every sequence of homomorphisms $N' \to N \to N''$, the sequence obtained by tensoring with M is exact if and only if the original sequence is exact.*

In this case M is called a *faithfully flat A-module.* In particular, if M is faithfully flat, a homomorphism $N \to N'$ is a monomorphism, epimorphism, or isomorphism if and only if the homomorphism obtained by tensoring with M has the same property. A faithfully flat module is faithful, since multiplication by f on M is obtained by tensoring multiplication by f on A with M.

The usual transitivity properties follow as in Section IV.1. The tensor product of two faithfully flat modules is faithfully flat. If M is faithfully flat over A, then $M \otimes_A B$ is faithfully flat over B after every base extension $A \to B$. If B is a faithfully flat A-algebra and M is faithfully flat as a B-module, then M is faithfully flat over A.

COROLLARY IV.2.2. *Let $A \to B$ be a local homomorphism of local rings, and let M be a finitely generated B-module. Then M is faithfully flat over A if and only if it is flat over A and nonzero.*

This follows from criterion (i ter) and Nakayama's lemma. In particular, B is flat over A if and only if it is faithfully flat over A.

PROPOSITION IV.2.3. *Let $A \to B$ be a ring homomorphism, and let M be a B-module which is faithfully flat over A. For every prime ideal $\mathfrak{p}$ of A, there exists a prime ideal $\mathfrak{q}$ of B whose contraction to A is $\mathfrak{p}$.*

After quotienting by $\mathfrak{p}$, we reduce to $\mathfrak{p} = 0$. Localizing at the prime ideal 0, we reduce to the case where A is a field. Since M is faithfully flat over A, it is nonzero, and hence $B \neq 0$. The ring B has a prime ideal, which necessarily contracts to the unique prime ideal of A. Geometrically, the existence on $X = \operatorname{Spec}(B)$ of a quasi-coherent sheaf which is faithfully flat relative to A implies that

$$X \longrightarrow Y = \operatorname{Spec}(A)$$

is surjective.

COROLLARY IV.2.4. *Suppose that M is flat over A, finitely generated over B, and has full support:*

$$\operatorname{Supp} M = \operatorname{Spec}(B).$$

For every prime ideal $\mathfrak{p}$ of A, each prime ideal $\mathfrak{q}$ of B minimal among those containing $\mathfrak{p}B$ contracts to $\mathfrak{p}$.

All the hypotheses are preserved after quotienting, so once again we reduce to $\mathfrak{p} = 0$, with A an integral domain. We are reduced to:

COROLLARY IV.2.5. *Let M be as above. Every minimal prime $\mathfrak{q}$ of B contracts to a minimal prime $\mathfrak{p}$ of A.*

Indeed, after localizing at $\mathfrak{p}$ and $\mathfrak{q}$, it is enough to prove the following. Let A and B be local rings, let $A \to B$ be local, let M be a nonzero B-module flat over A, and suppose that B has dimension zero. Then A has dimension zero. By Corollary IV.2.2 and Proposition IV.2.3, every prime ideal of A is induced by a prime ideal of B, hence by the maximal ideal of B, and is therefore the maximal ideal of A.

Geometrically, Corollary IV.2.5 says that every irreducible component of $X = \operatorname{Spec}(B)$ dominates an irreducible component of $Y = \operatorname{Spec}(A)$, provided there is a finitely generated quasi-coherent sheaf on X, with support X, which is flat relative to Y.

Notice that Corollary IV.2.4 did not require M to be faithfully flat over A. In that case, however, there is no guarantee that a prime ideal containing $\mathfrak{p}B$ exists.

PROPOSITION IV.2.6. *Let $i\colon A \to B$ be a ring homomorphism. The following conditions are equivalent:*

(i) *B is faithfully flat as an A-module.*
(ii) *B is flat over A, and* $\operatorname{Spec}(B) \to \operatorname{Spec}(A)$ *is surjective.*
(ii bis) *B is flat over A, and every maximal ideal of A is the contraction of a prime ideal of B.*
(iii) *i is injective and* $\operatorname{Coker} i$ *is a flat A-module.*
(iv) *The functor*

$$M \longmapsto M_{(B)} = M \otimes_A B$$

is exact, and the canonical natural homomorphism $M \to M_{(B)}$ is injective.

(iv bis) *For every ideal $I \subseteq A$, the map*

$$I \otimes_A B \longrightarrow IB$$

is an isomorphism, and the inverse image of IB in A is I.

The implication (i) $\Rightarrow$ (ii) follows from Proposition IV.2.3; (ii) $\Rightarrow$ (ii bis) is immediate; and (ii bis) $\Rightarrow$ (i) follows from criterion (i ter) of Proposition IV.2.1. Proposition IV.1.1 gives (iii) $\Rightarrow$ (iv), and (iv) $\Rightarrow$ (iv bis) is immediate. Condition (iv bis) implies (i) by the ideal criterion for flatness in Section IV.1 and criterion (i ter) of Proposition IV.2.1. The implication (iv) $\Rightarrow$ (iii) follows from the converse form of Proposition IV.1.1.

Finally, (i) implies (iv) as follows. If N is the kernel of

$$M \longrightarrow M \otimes_A B = T(M),$$

then exactness of T makes the map $N \to T(N)$ zero. Hence $T(N) = N \otimes_A B = 0$, and faithful flatness gives $N = 0$.

IV.3. Relations with completion

Let A be a Noetherian ring, let I be an ideal of A, and let $\widehat{A}$ be the separated completion of A for the I-adic topology. For every A-module M, let $\widehat{M}$ be its I-adic completion. It is an $\widehat{A}$-module, and hence there is a canonical homomorphism

$$M \otimes_A \widehat{A} \longrightarrow \widehat{M}.$$

On finitely generated modules, the functor $M \mapsto \widehat{M}$ is exact. This follows readily from Krull's theorem: if $N \subseteq M$, the I-adic topology of N is the topology induced from M. Since $M \mapsto M \otimes_A \widehat{A}$ is right exact, a finite free presentation $L \to L' \to M$ shows that the displayed natural homomorphism is an isomorphism, completion also being right exact. Consequently $M \mapsto M \otimes_A \widehat{A}$ is exact on finitely generated modules. Hence:

Proposition IV.3.1. *Let A be a Noetherian ring and I an ideal of A. The separated I-adic completion $\widehat{A}$ of A is flat over A.*

Corollary IV.3.2. *The ring $\widehat{A}$ is faithfully flat over A if and only if I is contained in the Jacobson radical of A.*

Apply criterion (i ter) of Proposition IV.2.1.

These statements summarize everything that linear algebra alone says about the relation between A and $\widehat{A}$. Corollary IV.3.2 is used especially when A is a Noetherian local ring and I is contained in its maximal ideal $\mathfrak{m}$, most often with $I = \mathfrak{m}$.

IV.4. Relations with free modules

Proposition IV.4.1. *Let A be a ring, I an ideal of A, and M an A-module. Suppose that one of the following hypotheses holds:*

(a) *I is nilpotent;*
(b) *A is Noetherian, I is contained in the Jacobson radical of A, and M is finitely generated.*

Then M is free over A if and only if $M \otimes_A A/I$ is free over A/I and

$$\mathrm{Tor}_1^A(M, A/I) = 0.$$

Necessity is clear; we prove sufficiency. Choose elements (e_i) of M whose images form a basis of

$$M \otimes_A A/I = M/IM$$

over A/I; the family is finite in case (b). Let L be the free A-module on the same index set. The resulting homomorphism $L \to M$ becomes an isomorphism after tensoring with A/I. If Q is its cokernel, then $Q/IQ = 0$, and Nakayama's lemma, valid under either hypothesis, gives $Q = 0$. Thus $L \to M$ is surjective. Let R be its kernel:

$$0 \longrightarrow R \longrightarrow L \longrightarrow M \longrightarrow 0.$$

Since $\operatorname{Tor}_1^A(M, A/I) = 0$, tensoring with A/I gives an exact sequence

$$0 \longrightarrow R/IR \longrightarrow L/IL \longrightarrow M/IM \longrightarrow 0.$$

Hence $R/IR = 0$, and Nakayama again gives $R = 0$. In case (b), note that R is finitely generated because A is Noetherian.

Corollary IV.4.2. *The condition* $\operatorname{Tor}_1^A(M, A/I) = 0$ *may be replaced by the condition that the canonical surjection*

$$(*) \qquad \operatorname{gr}_I^0(M) \otimes_{A/I} \operatorname{gr}_I(A) \longrightarrow \operatorname{gr}_I(M)$$

is an isomorphism.

The condition plainly holds when M is free. Conversely, suppose that $M \otimes_A A/I$ is free over A/I and that (*) is an isomorphism. Construct $L \to M$ as in the proof above. The hypothesis shows that this homomorphism induces an isomorphism on associated graded modules. Its kernel is therefore contained in $\bigcap_n I^n L$, which is zero: trivially in case (a), and by the standard separation theorem in case (b).

Corollary IV.4.3. *Suppose that A/I is a field. The following conditions on M are equivalent:*

(i) *M is free;*
(ii) *M is projective;*
(iii) *M is flat;*
(iv) $\operatorname{Tor}_1^A(M, A/I) = 0$*;*
(v) *the canonical homomorphism* (*) *is an isomorphism.*

Indeed, $M \otimes_A A/I$ is automatically free in this case. The corollary applies in either of the following situations:

(a) M is an arbitrary module over a local ring A whose maximal ideal I is nilpotent, for example an Artinian local ring;
(b) M is a finitely generated module over a Noetherian local ring A.

For reference, we recall:

Corollary IV.4.4. *Let A be a Noetherian local integral domain with maximal ideal $\mathfrak{m} = I$, residue field $k = A/I$, and fraction field K. Let M be a finitely generated A-module. Conditions* (i)–(v) *above are also equivalent to:*

(vi) *The vector spaces $M \otimes_A K$ and $M \otimes_A k$ have the same dimension; equivalently, the rank of M over A equals the minimum number of generators of the A-module M.*

The proof is immediate. The reader may generalize it to the case where A is only assumed reduced: one must then require the ranks of M at all minimal prime ideals of A to equal the dimension of the vector space $M \otimes_A k$.

IV.5. Local criteria for flatness

Proposition IV.5.1. *Let A be a ring equipped with an ideal I, and let M be an A-module. Suppose that*

$$\operatorname{Tor}_1^A(M, A/I^n) = 0 \qquad \text{for every } n > 0.$$

Then the canonical surjective homomorphism

$$(*) \qquad \operatorname{gr}_I^0(M) \otimes_{A/I} \operatorname{gr}_I(A) \longrightarrow \operatorname{gr}_I(M)$$

is an isomorphism. The converse holds if I is nilpotent.

The hypothesis means that the homomorphisms

$$I^n \otimes_A M \longrightarrow I^n M$$

are isomorphisms. It follows immediately that the homomorphisms

$$(I^n/I^{n+1}) \otimes_A M \longrightarrow I^n M/I^{n+1} M$$

are isomorphisms. Conversely, suppose this last condition holds and that I is nilpotent. We prove that $\mathrm{Tor}_1^A(M, A/I^n) = 0$ for every n. This is true for large n; proceed by descending induction, assuming it for $n+1$. There is a commutative diagram

$$\begin{array}{ccccccccc} & & M \otimes_A I^{n+1} & \longrightarrow & M \otimes_A I^n & \longrightarrow & M \otimes_A (I^n/I^{n+1}) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & MI^{n+1} & \longrightarrow & MI^n & \longrightarrow & MI^n/MI^{n+1} & \longrightarrow & 0, \end{array}$$

whose rows are exact. By hypothesis, the last vertical arrow is an isomorphism, while the induction hypothesis says that the first vertical arrow is an isomorphism. The middle vertical arrow is therefore also an isomorphism, completing the proof.

The following proposition was isolated by Serre during the seminar; it allows substantial simplifications in this section.

Proposition IV.5.2. *Let $A \to B$ be a ring homomorphism and let M be an A-module. The following conditions are equivalent:*

(i) *for every B-module N, $\mathrm{Tor}_1^A(M, N) = 0$;*
(ii) $\mathrm{Tor}_1^A(M, B) = 0$, *and $M_{(B)} = M \otimes_A B$ is flat over B.*

There is a functorial isomorphism

$$M \otimes_A N = (M \otimes_A B) \otimes_B N.$$

Regarding the left-hand side as a functor of M, this expresses it as the composite of the two functors

$$M \longmapsto M \otimes_A B, \qquad P \longmapsto P \otimes_B N.$$

The first carries free A-modules to free B-modules, and hence projective modules to projective modules. The spectral sequence for composite functors therefore gives

$$E^2_{p,q} = \mathrm{Tor}_p^B(\mathrm{Tor}_q^A(M, B), N) \Longrightarrow \mathrm{Tor}_{p+q}^A(M, N),$$

and its low-degree terms give an exact sequence

$$\mathrm{Tor}_1^A(M, B) \otimes_B N \longrightarrow \mathrm{Tor}_1^A(M, N) \longrightarrow \mathrm{Tor}_1^B(M \otimes_A B, N) \longrightarrow 0.$$

If (i) holds, then $\mathrm{Tor}_1^A(M, B) = 0$, and the exact sequence shows that $\mathrm{Tor}_1^B(M \otimes_A B, N) = 0$ for every N. Thus $M \otimes_A B$ is flat over B, proving (ii). Conversely, if (ii) holds, the two terms surrounding $\mathrm{Tor}_1^A(M, N)$ vanish, proving (i).

Corollary IV.5.3. *Suppose that $B = A/I$. Then the preceding conditions are also equivalent to:*

(iii) $\mathrm{Tor}_1^A(M, N) = 0$ *for every A-module N annihilated by a power of I.*

Indeed, condition (i) says this when N is annihilated by I. Condition (iii) follows by applying this assertion successively to the modules $I^n N/I^{n+1} N$.

Corollary IV.5.4. *Under the hypotheses of Corollary IV.5.3, the conditions under consideration imply that the functorial homomorphism*

$$(*) \qquad \mathrm{gr}_I^0(M) \otimes_{A/I} \mathrm{gr}_I(A) \longrightarrow \mathrm{gr}_I(M)$$

is an isomorphism, and that $M \otimes_A A/I$ is flat over A/I.

Apply condition (iii) and Proposition IV.5.1. Using the converse in Proposition IV.5.1 when I is nilpotent gives:

COROLLARY IV.5.5. *Let A be a ring equipped with a nilpotent ideal I, and let M be an A-module. The following conditions are equivalent:*

(i) *M is flat over A;*
(ii) *$M \otimes_A A/I$ is flat over A/I, and $\mathrm{Tor}_1^A(M, A/I) = 0$;*
(iii) *$M \otimes_A A/I$ is flat over A/I, and the canonical homomorphism (*) on associated graded modules is an isomorphism.*

These are, respectively, condition (iii) and condition (ii) above, and the conditions of Corollary IV.5.4.

Now drop the assumption that I is nilpotent. A priori, in Corollary IV.5.5 we retain only the implications

$$\text{(i)} \Rightarrow \text{(ii)} \Rightarrow \text{(iii)}.$$

Since condition (iii) remains stable after quotienting by a power of I, Corollary IV.5.5 shows that it implies:

(iv) *for every $n > 0$, $M \otimes_A A/I^n$ is flat over A/I^n.*

We seek conditions under which (iv) implies (i), that is, under which M is flat over A. It is enough that A be Noetherian and that M satisfy the following finiteness condition: *for every finitely generated A-module N, the module $M \otimes_A N$ is separated for the I-adic topology.* It would be enough to check this when N is a finitely generated ideal of A.

Indeed, suppose these conditions hold, and let $N' \to N$ be an injective homomorphism of finitely generated modules. We prove that

$$M \otimes_A N' \longrightarrow M \otimes_A N$$

is injective. It is enough to show that its kernel is contained in every

$$I^n(M \otimes_A N') = \mathrm{Im}(M \otimes_A I^n N' \longrightarrow M \otimes_A N'),$$

or, equivalently, in every

$$\mathrm{Im}(M \otimes_A V'_n \longrightarrow M \otimes_A N') = \mathrm{Ker}(M \otimes_A N' \longrightarrow M \otimes_A (N'/V'_n)),$$

where (V'_n) runs through a countable fundamental system of neighborhoods of zero in N', equipped with its I-adic topology. By Krull's theorem, the I-adic topology of N' is induced by that of N, so we may take

$$V'_n = N' \cap I^n N.$$

Consider the commutative diagram

$$\begin{array}{ccc} M \otimes_A N' & \longrightarrow & M \otimes_A (N'/V'_n) \\ \downarrow & & \downarrow \\ M \otimes_A N & \longrightarrow & M \otimes_A (N/I^n N). \end{array}$$

Since N'/V'_n and $N/I^n N$ are annihilated by I^n, the right-hand vertical homomorphism is obtained from the *injective* homomorphism

$$N'/V'_n \longrightarrow N/I^n N$$

by tensoring over A/I^n with the *flat* (A/I^n)-module $M \otimes_A A/I^n$. It is therefore injective. Hence the kernel of $M \otimes_A N' \to M \otimes_A N$ is contained in the kernel of

$$M \otimes_A N' \longrightarrow M \otimes_A (N'/V'_n),$$

as required.

The finiteness condition just imposed on M holds in particular if M is a finitely generated module over a Noetherian A-algebra B such that IB is contained in the Jacobson radical of B. Indeed, for every finitely generated A-module N, the module $M \otimes_A N$ is finitely generated over B, and is therefore separated by Krull's theorem for the I-adic topology, which is its (IB)-adic topology. We have proved:

THEOREM IV.5.6. *Let $A \to B$ be a homomorphism of Noetherian rings, let I be an ideal of A such that IB is contained in the Jacobson radical of B, and let M be a finitely generated B-module. The following conditions are equivalent:*

(i) *M is flat over A;*
(ii) *$M \otimes_A A/I$ is flat over A/I, and $\mathrm{Tor}_1^A(M, A/I) = 0$;*
(iii) *$M \otimes_A A/I$ is flat over A/I, and the canonical homomorphism*

$$\mathrm{gr}_I^0(M) \otimes_{A/I} \mathrm{gr}_I(A) \longrightarrow \mathrm{gr}_I(M)$$

is an isomorphism;
(iv) *for every $n > 0$, $M \otimes_A A/I^n$ is flat over A/I^n.*

This result is used chiefly when A and B are Noetherian *local* rings, $A \to B$ is a local homomorphism, and I is an ideal of A contained in its maximal ideal; Theorem IV.5.6 may in fact be reduced immediately to this case. An interesting special case is that in which A/I is a field, that is, I is maximal. The condition that $M \otimes_A A/I$ be flat over A/I is then automatic. Moreover, the rings A/I^n are Artinian local rings, so condition (iv) says that the modules $M \otimes_A A/I^n$ are *free* over A/I^n.

COROLLARY IV.5.7. *Let $A \to B$ be a local homomorphism of Noetherian local rings, and let $u\colon M' \to M$ be a homomorphism of finitely generated B-modules. Suppose that M is flat over A. The following conditions are equivalent:*

(i) *u is injective and $\mathrm{Coker}(u)$ is flat over A;*
(ii) *the homomorphism*

$$u \otimes_A k\colon M' \otimes_A k \longrightarrow M \otimes_A k$$

is injective, where k is the residue field of A.

The implication (i)⇒(ii) follows from Proposition IV.1.1. Conversely, u is first of all injective: it is enough to check this on associated graded modules, where it follows from a commutative square that we leave to the reader. Put $M'' = \mathrm{Coker}(u)$. We then have an exact sequence

$$0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0.$$

The long exact sequence of Tor, together with hypothesis (ii) and $\mathrm{Tor}_1^A(M, k) = 0$, gives $\mathrm{Tor}_1^A(M'', k) = 0$. Theorem IV.5.6 therefore shows that M'' is flat over A.

COROLLARY IV.5.8. *Under the hypotheses of Theorem IV.5.6, let J be an ideal of B containing IB and contained in the Jacobson radical. Let $\widehat{A}$ be the I-adic completion of A, and let $\widehat{B}$ and $\widehat{M}$ be the J-adic completions of B and M, respectively. Then M is flat over A if and only if $\widehat{M}$ is flat over $\widehat{A}$.*

The sufficiency already follows readily from Corollary IV.3.2. Apply criterion (iii) of Theorem IV.5.6 to the situations

$$(A, B, I, M) \qquad \text{and} \qquad (\widehat{A}, \widehat{B}, I\widehat{A}, \widehat{M}).$$

The resulting conditions in the two cases are equivalent by Corollary IV.3.2.

COROLLARY IV.5.9. *Let $A \to B \to C$ be local homomorphisms of Noetherian local rings, and let M be a finitely generated C-module. (The ring C occurs only so that a finiteness condition can be imposed on M.) Suppose that B is flat over A, and let k be the residue field of A. The following conditions are equivalent:*

(i) *M is flat over B;*
(ii) *M is flat over A, and $M \otimes_A k$ is flat over $B \otimes_A k$.*

The implication (i)⇒(ii) is immediate. For the converse, apply criterion (iii) of Theorem IV.5.6 to

$$(B, C, I = \mathfrak{m}B, M),$$

where $\mathfrak{m}$ is the maximal ideal of A. Since

$$M \otimes_B (B/I) = M \otimes_B (B \otimes_A k) = M \otimes_A k,$$

the first condition of that criterion says precisely that $M \otimes_A k$ is flat over $B \otimes_A k$, as required. The second condition is satisfied because M and B are flat over A, by associativity of the tensor product.

Of course, using Corollary IV.5.5 instead of Theorem IV.5.6 gives an analogous statement without Noetherian or finiteness hypotheses, provided instead that the ideal $\mathfrak{m}$ of A is nilpotent. The fact that $\mathfrak{m}$ was taken maximal did not enter the argument; nevertheless, in a sense, the case of maximal $\mathfrak{m}$ is the best possible one.

IV.6. Flat morphisms and open subsets

We first recall some results on constructible sets, also proved in circulating notes from the Dieudonné–Rosenlicht Seminar on schemes. [3]

Following Chevalley, a subset of a topological space is called *constructible* if it is a finite union of locally closed subsets.

Lemma IV.6.1. *Let X be a Noetherian topological space and $Z \subseteq X$. Then Z is constructible if and only if, for every irreducible closed subset $Y \subseteq X$, the intersection $Z \cap Y$ is either not dense in Y or contains a nonempty open subset of Y.*

Using a familiar lemma from commutative algebra, one deduces:

Lemma IV.6.2 (Chevalley). *Let $f\colon X \to Y$ be a morphism of finite type between preschemes, with Y Noetherian. Then $f(X)$ is constructible.*

Lemma IV.6.3. *Let X be a Noetherian topological space in which every irreducible closed subset has a generic point, let $U \subseteq X$ be constructible, and let $x \in X$. Then U is a neighborhood of x if and only if every generalization y of x, that is, every y such that $x \in \overline{\{y\}}$, belongs to U.*

In particular:

Corollary IV.6.4. *Let X be a Noetherian topological space in which every irreducible closed subset has a generic point. A subset $U \subseteq X$ is open if and only if it satisfies both of the following conditions:*

(a) *U contains every generalization of each of its points;*
(b) *if $x \in U$, then $U \cap \overline{\{x\}}$ contains a nonempty open subset of $\overline{\{x\}}$.*

Condition (b) and Lemma IV.6.1 show that U is constructible, and Lemma IV.6.3 then shows that U is a neighborhood of each of its points.

Corollary IV.6.5. *Let $f\colon X \to Y$ be a morphism of finite type between preschemes, with Y locally Noetherian. Let $x \in X$ and $y = f(x)$. Then f maps every neighborhood of x onto a neighborhood of y if and only if, for every generalization y' of y, there is a generalization x' of x such that $f(x') = y'$.*

We may suppose that X and Y are affine, hence Noetherian. The condition is sufficient because it is enough to show that $f(X)$ is a neighborhood of y. The image $f(X)$ is constructible by Lemma IV.6.2, and Lemma IV.6.3 applies.

Conversely, let $Y' = \overline{\{y'\}}$, and let F be the union of the irreducible components of $f^{-1}(Y')$ which do not contain x. Then $X - F$ is an open neighborhood of x, so its image

[3] See EGA 0_{III}, 9, and EGA IV, 1.8 and 1.10.

is a neighborhood of y, and in particular contains y'. Choose $x_1' \in X - F$ with $f(x_1') = y'$. An irreducible component of $f^{-1}(Y')$ containing x_1' must also contain x, for otherwise it would lie in F. Let x' be its generic point. Then x' is a generalization of x, and $f(x')$ is a generalization of $f(x_1') = y'$ lying in Y'; hence $f(x') = y'$.

THEOREM IV.6.6. *Let $f\colon X \to Y$ be a morphism locally of finite type, with Y locally Noetherian, and let F be a coherent sheaf on X, with support X, flat relative to Y. Then f is an open morphism; that is, it maps open sets to open sets.*

It is enough to verify the criterion of Corollary IV.6.5 at each $x \in X$. Generalizations of x correspond to prime ideals of $\mathcal{O}_x$, and generalizations of $y = f(x)$ correspond to prime ideals of $\mathcal{O}_y$. We must therefore show that every prime ideal of $\mathcal{O}_y$ is the contraction of a prime ideal of $\mathcal{O}_x$. The stalk F_x is a nonzero finitely generated $\mathcal{O}_x$-module flat over $\mathcal{O}_y$, hence faithfully flat over $\mathcal{O}_y$ by Corollary IV.2.2. Proposition IV.2.3 now finishes the proof.

Remarks. Since flatness is preserved under base change, the preceding theorem also shows that f is universally open. If Y is integral and X is of finite type over Y, it was not known whether the restriction of f to each irreducible component X_i of X must be open, or even merely equidimensional. [4] Here equidimensional means that all components of all fibers have the same dimension; one knows only that X_i dominates Y. The question is related to the following one. Let $A \to B$ be a local homomorphism of Noetherian local rings, with B flat over A and $\mathfrak{m}B$ an ideal of definition of B, which in particular implies $\dim B = \dim A$. Is it true that

$$\dim B/\mathfrak{p}_i = \dim B$$

for every minimal prime ideal $\mathfrak{p}_i$ of B?

We merely note that the answer to the first question is negative if the flatness hypothesis in Theorem IV.6.6 is replaced by the assumption that f is universally open.

LEMMA IV.6.7. *Let A be a Noetherian integral domain, let B be a finitely generated A-algebra, and let M be a finitely generated B-module. There exists a nonzero $f \in A$ such that M_f is free, and hence flat, over A_f.*

Let K be the fraction field of A. Then $B \otimes_A K$ is a finitely generated K-algebra and $M \otimes_A K$ is a finitely generated module over it. Let n be the dimension of the support of this module; we argue by induction on n. If $n < 0$, that is, if $M \otimes_A K = 0$, choose finitely many generators of M over B. Some nonzero $f \in A$ annihilates all of them, hence annihilates M, so $M_f = 0$.

Suppose $n \geq 0$. The B-module M has a prime filtration whose successive quotients are $B/\mathfrak{p}_i$. Since an extension of free modules is free, we reduce to $M = B/\mathfrak{p}$, or, equivalently, to $M = B$ with B an integral A-algebra. Applying Noether normalization to $B \otimes_A K$, after inverting a suitable nonzero element of A we may suppose that B is integral over

$$C = A[t_1, \ldots, t_n].$$

It is then a finitely generated torsion-free C-module. If m is its rank, there is an exact sequence of C-modules

$$0 \longrightarrow C^m \longrightarrow B \longrightarrow M' \longrightarrow 0$$

with M' a torsion C-module. The Krull dimension of the $C \otimes_A K$-module $M' \otimes_A K$ is strictly smaller than the dimension n of $C \otimes_A K$. By induction, after inverting another suitable nonzero $f \in A$, M' is free over A. The module C^m is also free over A, so the sequence splits over A and B is free over A.

[4] The answer to the second question is affirmative and to the first negative, even when f is étale; see EGA IV, 12.1.1.5, and EGA Errata IV, 33.

LEMMA IV.6.8. *Let A be Noetherian, let B be a finitely generated A-algebra, let M be a finitely generated B-module, and let $\mathfrak{p}$ be a prime ideal of B, with contraction $\mathfrak{q}$ in A. Suppose that $M_{\mathfrak{p}}$ is flat over $A_{\mathfrak{q}}$, equivalently over A. Then there exists $g \in B - \mathfrak{p}$ such that:*

(a) $(M/\mathfrak{q}M)_g$ *is flat over* $A/\mathfrak{q}$;
(b) $\mathrm{Tor}_1^A(M, A/\mathfrak{q})_g = 0$.

Apply Lemma IV.6.7 to

$$(A/\mathfrak{q},\ B/\mathfrak{q}B,\ M/\mathfrak{q}M).$$

There is first an $f \in A - \mathfrak{q}$ such that $(M/\mathfrak{q}M)_f$ is flat over $A/\mathfrak{q}$. Since $M_{\mathfrak{p}}$ is flat over A,

$$\mathrm{Tor}_1^A(M, A/\mathfrak{q})_{\mathfrak{p}} = \mathrm{Tor}_1^A(M_{\mathfrak{p}}, A/\mathfrak{q}) = 0.$$

The left-hand module is finitely generated over B, so there exists $g \in B - \mathfrak{p}$ for which (b) holds. Replacing g by gf, we may arrange that (a) holds as well.

COROLLARY IV.6.9. *With the notation of Lemma IV.6.8, let g be as there. For every prime ideal $\mathfrak{p}'$ of B containing $\mathfrak{p}$ and not containing g, the module $M_{\mathfrak{p}'}$ is flat over A, equivalently over $A_{\mathfrak{q}'}$, where $\mathfrak{q}'$ is the contraction of $\mathfrak{p}'$ to A.*

Apply criterion (ii) of Theorem IV.5.6 to

$$(A,\ B_{\mathfrak{p}'},\ \mathfrak{q},\ M_{\mathfrak{p}'}),$$

using localization of Tor.

THEOREM IV.6.10. *Let $f\colon X \to Y$ be a morphism of finite type, with Y locally Noetherian, and let F be a coherent sheaf on X. The set*

$$U = \{x \in X \mid F_x \text{ is flat over } \mathcal{O}_{f(x)}\}$$

is open.

Proof. We may suppose that X and Y are affine, with rings B and A, and that F is defined by a finitely generated B-module M. Apply Corollary IV.6.4. Condition (a) follows immediately from Proposition IV.1.2(i), and condition (b) is exactly Lemma IV.6.8 and Corollary IV.6.9.

For many purposes the following weaker form of Theorem IV.6.10 suffices. It already follows from Lemma IV.6.7, and therefore requires neither constructible sets nor Theorem IV.5.6.

COROLLARY IV.6.11. *Under the hypotheses of Theorem IV.6.10, suppose that Y is integral. There exists a nonempty open subset $V \subseteq Y$ such that F is flat relative to Y at every point of $f^{-1}(V)$.*

Indeed, the open set U contains the fiber over the generic point of Y, since its local ring is a field. As X is of finite type over Y, U therefore contains an open set of the form $f^{-1}(V)$.

Corollary IV.6.11 also readily implies the following statement when Y is Noetherian but not necessarily integral. There exists a partition of Y into locally closed subsets Y_i such that, after giving Y_i its induced reduced structure, the restriction of F to $X_i = X \times_Y Y_i$ is flat relative to Y_i.

CHAPTER V

The fundamental group: generalities

V.0. Introduction

The present Seminar is the continuation of the 1960 Seminar. We shall refer to the latter by abbreviations such as I I.9.7, meaning: Seminar on Algebraic Geometry, Exposé I, no. 9.7. The numbers of the 1961 exposés will follow those of 1960. We shall refer to the *Éléments de Géométrie Algébrique* of Dieudonné and Grothendieck by abbreviations such as EGA I, 8.7.3.

The present exposé summarizes, with slight additions, the final exposés of 1960, which had not been written up.

As in 1961, we shall as a rule restrict ourselves to locally Noetherian preschemes, although this restriction is often inessential. In Exposé VI we shall assume the theory of faithfully flat descent, summarized in Bourbaki Seminar no. 190. If appropriate, we shall give a more detailed account in a later exposé, [1] once the reader has had occasion to appreciate the usefulness of this technique in the theory of the fundamental group.

V.1. Preschemes with a finite group of operators; quotient preschemes

Let X be a prescheme and let a finite group G act on X by automorphisms, on the right for definiteness. If X is affine with ring A, then G acts on A by automorphisms on the left.

For every prescheme Z, the group G acts on the left on the set $\mathrm{Hom}(X, Z)$. We may therefore consider the set

$$\mathrm{Hom}(X, Z)^G$$

of G-invariant morphisms. It depends functorially on Z, and we may ask whether this functor is representable, that is, isomorphic to a functor $Z \mapsto \mathrm{Hom}(Y, Z)$. This means finding a prescheme Y and a G-invariant morphism

$$p\colon X \longrightarrow Y$$

such that, for every Z, the corresponding map $g \mapsto gp$,

$$\mathrm{Hom}(Y, Z) \longrightarrow \mathrm{Hom}(X, Z)^G,$$

is bijective. The pair (Y, p) is then called a *quotient prescheme* of X by G; it is determined up to unique isomorphism.

Proposition V.1.1. *Let A be a ring on which the finite group G acts on the left, let $B = A^G$ be the subring of invariants, let $X = \mathrm{Spec}(A)$ and $Y = \mathrm{Spec}(B)$, and let $p\colon X \to Y$ be the canonical morphism, which is plainly G-invariant. Then:*

(i) *A is integral over B; equivalently, p is an integral morphism.*
(ii) *The morphism p is surjective, its fibers are the G-orbits, and the topology of Y is the quotient topology of that of X.*
(iii) *Let $x \in X$, let $y = p(x)$, and let G_x be the stabilizer of x. Then $\kappa(x)$ is a quasi-Galois algebraic extension of $\kappa(y)$, and the canonical map*

$$G_x \longrightarrow \mathrm{Gal}(\kappa(x)/\kappa(y))$$

[1] See Exposés VI and VIII.

to the group of $\kappa(y)$-automorphisms of $\kappa(x)$ is surjective.
(iv) *The pair (Y, p) is a quotient prescheme of X by G.*

Statements (i), (ii), and (iii) are well-known in commutative algebra, [2] and are recalled only for reference, except for the assertion about the topology. The latter follows from this general fact, an easy consequence of the Cohen–Seidenberg theorem: an integral morphism is closed, that is, it maps closed sets to closed sets. We note at once:

COROLLARY V.1.2. *Under the preceding hypotheses, the natural homomorphism*
$$\mathcal{O}_Y \longrightarrow p_*(\mathcal{O}_X)^G$$
is an isomorphism.

This follows immediately from the formula
$$(S^{-1}A)^G = S^{-1}(A^G),$$
valid for every multiplicatively closed subset S of $B = A^G$. The formula has a module version, and more generally holds after a flat base change $A \to A'$. Apply it when S is generated by one element $f \in B$.

Assertion (ii) and Corollary V.1.2 readily imply (iv). More generally:

PROPOSITION V.1.3. *Let X be a prescheme with a finite group G of automorphisms, and let $p\colon X \to Y$ be an invariant affine morphism such that*
$$\mathcal{O}_Y \xrightarrow{\sim} p_*(\mathcal{O}_X)^G.$$
Then conclusions (i)*,* (ii)*,* (iii)*, and* (iv) *of Proposition V.1.1 still hold.*

For (i), (ii), and (iii), we may suppose that Y, and hence X, is affine. If B and A are their rings, the hypothesis implies $B = A^G$, so Proposition V.1.1 applies. For (iv), use (ii) together with $\mathcal{O}_Y = p_*(\mathcal{O}_X)^G$.

COROLLARY V.1.4. *Under the hypotheses of Proposition V.1.3, for every open subset $U \subseteq Y$, the prescheme U is a quotient of $X|_U = p^{-1}(U)$ by G.*

Indeed, the morphism $p^{-1}(U) \to U$ induced by p satisfies the same hypotheses as p.

Now suppose that X is a Z-prescheme and that the operations of G are Z-automorphisms. By (iv), Y is then a Z-prescheme. Thus:

COROLLARY V.1.5. *The prescheme X is affine, respectively separated, over Z if and only if Y is. If X is of finite type over Z, then it is finite over Y. If, in addition, Z is locally Noetherian, then Y is of finite type over Z.*

Since X is affine, and hence separated, over Y, it is affine, respectively separated, over Z whenever Y is. Conversely, suppose first that X is affine over Z, and let us prove that Y is. By Corollary V.1.4, we may suppose that Z is affine. We are then reduced to proving that Y is affine when X is affine, and this follows from the explicit description $Y = \operatorname{Spec}(A^G)$ in Proposition V.1.1.

Likewise, since $p\colon X \to Y$ is integral, hence universally closed, and surjective, it follows that Y is separated over Z whenever X is. Indeed, in the diagram
$$\begin{array}{ccc} X \times_Z X & \xrightarrow{p \times_Z p} & Y \times_Z Y \\ \uparrow \Delta_{X/Z} & & \uparrow \Delta_{Y/Z} \\ X & \xrightarrow{p} & Y, \end{array}$$
the morphism $X \times_Z X \to Y \times_Z Y$ is closed. It therefore maps the closed diagonal of X to a closed subset of $Y \times_Z Y$, and this image is precisely the diagonal of Y, because p is surjective.

[2] See N. Bourbaki, *Commutative Algebra*, Chapter V, §§1–2, Theorem 2.

If X is of finite type over Z, it is a fortiori of finite type over Y, and is therefore finite over Y, since it is already integral over Y. Suppose further that Z is locally Noetherian; we prove that Y is of finite type over Z. By Corollary V.1.4, we may suppose that Z is affine. The topological space X is quasi-compact and $p\colon X \to Y$ is surjective, so Y is also quasi-compact and hence is a finite union of affine open subsets. Using Corollary V.1.4 again, we reduce to the case in which Y and X are affine. The ring A of X is then a finitely generated algebra over the Noetherian ring C of Z. It is known that $B = A^G$ is likewise a finitely generated C-algebra. Indeed, A is integral, hence finite, over a C-subalgebra $B' \subseteq B$ of finite type; since B' is Noetherian, B is also finite over B', and hence of finite type over C.

Corollary V.1.6. *The prescheme X is affine, respectively a scheme, if and only if Y is.*

Definition V.1.7. *Let X be a prescheme on which a finite group G acts on the right. The action of G is called* admissible *if there exists a morphism $p\colon X \to Y$ having the properties of Proposition V.1.3. This implies that X/G exists and is isomorphic to Y.*

Proposition V.1.8. *Let X be a prescheme on which the finite group G acts on the right. The action is admissible if and only if X is a union of G-invariant affine open subsets, or equivalently, if and only if every G-orbit in X is contained in an affine open subset.*

The first condition plainly implies the second, and the second in turn implies the first. Indeed, let T be a G-orbit and let U be an affine open subset containing it. The intersection of the translates gU, for $g \in G$, is a G-stable open subset U' containing T and contained in the affine open U. In an affine scheme, every finite subset has a fundamental system of affine open neighborhoods, so T has an affine open neighborhood $V \subseteq U'$. Its translates gV are affine and contained in the separated open subset U'. Their intersection is therefore an affine, G-invariant open subset U'' containing T.

Now the condition in Proposition V.1.8 is necessary: if a quotient $p\colon X \to Y$ as in Proposition V.1.3 exists, take the inverse images of an affine open cover of Y. It is also sufficient. Indeed, Proposition V.1.1 constructs the quotients $Y_i = X_i/G$ of the invariant affine opens X_i. In each Y_i, the image of $X_i \cap X_j$ is an open subset Y_{ij}, identified with X_{ij}/G by Corollary V.1.4. We thereby obtain isomorphisms $Y_{ij} \xrightarrow{\sim} Y_{ji}$, with which the Y_i can be glued to construct Y. Serre instead constructs directly the quotient topological space $Y = X/G$, equips it with the sheaf $p_*(\mathcal{O}_X)^G$, and verifies that Y becomes a prescheme satisfying the conditions of Proposition V.1.3.

Corollary V.1.7. *If the action of G on X is admissible, then the action of every subgroup $H \subseteq G$ is admissible as well; in particular, X/H exists.*

This can also be checked directly in the situation of Proposition V.1.3. We may always suppose that X is affine over some Z and that the elements of G act by Z-automorphisms, for example by taking $Z = Y$. Indeed:

Corollary V.1.8. *Suppose that X is affine over Z and that the operations of G are Z-automorphisms. Then the action of G on X is admissible. If X is defined by a quasi-coherent sheaf of algebras $\mathcal{A}$, then Y is defined by the sheaf of invariant algebras $\mathcal{A}^G$.*

Proposition V.1.9. *Suppose that G acts admissibly on X, that $X/G = Y$ is a Z-prescheme, and let $Z' \to Z$ be a base-change morphism. Put*

$$X' = X \times_Z Z', \qquad Y' = Y \times_Z Z'.$$

The group G acts on X' by transport of structure, and the morphism $p'\colon X' \to Y'$ is invariant. If Z' is flat over Z, then p' again satisfies the hypotheses of Proposition V.1.3;

that is,

$$\mathcal{O}_{Y'} \xrightarrow{\sim} p'_*(\mathcal{O}_{X'})^G.$$

Thus G acts admissibly on X', and

$$(X/G) \times_Z Z' \xrightarrow{\sim} (X \times_Z Z')/G.$$

We may plainly take $Z = Y$, and are then reduced to the case in which Y and Y' are affine. We must show the following. Let B be the invariant subring for the action of G on A, and let B' be a flat B-algebra. Then B' is the invariant subring of $A' = A \otimes_B B'$. This is immediate, since the exact sequence

$$0 \longrightarrow B \xrightarrow{i} A \xrightarrow{j} A^{(G)},$$

where the last term denotes a power of A and $j(x) = (s \cdot x - x)_{s \in G}$, remains exact after tensoring with B'.

The flatness hypothesis is essential. In particular, if Y' is a closed subscheme of Y, even a closed point, and X' is its inverse image in X, then in general Y' is *not* identified with X'/G. We shall see that it is so nevertheless when X is étale over Y.

We conclude with a convenient, if elementary, formalism. Let Y be a prescheme. Since coproducts exist in the category of preschemes, for every set E we may form the sum of a family $(Y_i)_{i \in E}$ of preschemes all equal to Y. We denote this prescheme by $Y \times E$. It is characterized by the formula

$$\text{Hom}(Y \times E, Z) = \text{Hom}\big(E, \text{Hom}(Y, Z)\big). \tag{*}$$

Here the second Hom on the right means, of course, the set of maps from E to the set $\text{Hom}(Y, Z)$. There is a canonical morphism

$$Y \times E \longrightarrow Y,$$

making $Y \times E$ a prescheme over Y. Since fiber products commute with coproducts in the category of preschemes, if Y is a Z-prescheme and $Z' \to Z$ is a base change, then

$$(Y \times E) \times_Z Z' = (Y \times_Z Z') \times E.$$

This formula is especially useful when $Z = Y$. The definition also immediately gives

$$(Y \times E) \times F = Y \times (E \times F) = (Y \times E) \times_Y (Y \times F),$$

the last equality also following from the compatibility with fiber products just noted.

For fixed Y, the construction $E \mapsto Y \times E$ is a functor from sets to preschemes over Y. By the preceding formula it commutes with finite products. Thus, for example, every ordinary group G gives rise to a group prescheme $Y \times G$ over Y, which is finite over Y when G is finite. More generally the functor is "left exact," although we shall not use this here. It plainly commutes with coproducts and is also "right exact," as follows at once from the defining formula (*). In particular, if a finite group G acts on the right on a set E, then it acts on the right on $Y \times E$, and

$$(Y \times E)/G = Y \times (E/G).$$

The quotient on the left in fact satisfies the conditions of Proposition V.1.3, as is immediate.

V.2. Decomposition and inertia groups. The étale case

Let G a finite group acting on the right on the prescheme X. If $x \in X$, the stabilizer $G_d(x)$ of x is called the *decomposition group of x*. This group acts canonically (on the left) on the residue field $\kappa(x)$, and the set of elements of $G_d(x)$ which act trivially is called the *inertia group* of x, denoted $G_i(x)$. 111

Suppose that G acts admissibly on X and that Y is a prescheme over a prescheme Z. Fix $z \in Z$, and an algebraically closed extension Ω of $\kappa(z)$ having transcendence degree greater than that of the $\kappa(x)/\kappa(z)$, where x is a point of X above z. We may regard $\mathrm{Spec}(()\Omega)$ as a Z-scheme, and the Ω-valued points of X correspond to the homomorphisms of $\kappa(z)$-algebras $\kappa(x) \to \Omega$, where x is a point of X above z. Since Ω was chosen sufficiently large, every point x of X above z is the underlying point of an Ω-valued point of X. If $X(\Omega)$ and $Y(\Omega)$ denote, respectively, the sets of Ω-valued points of X and Y, there is a natural map

$$X(\Omega) \to Y(\Omega),$$

and, moreover, G acts on $X(\Omega)$ and the preceding map is G-invariant. With this understood, conclusions (ii) and (iii) of V.1.3 may also be interpreted as follows: *the preceding map is surjective and identifies $Y(\Omega)$ with the quotient $X(\Omega)/G$. Moreover, if x is the underlying point of $a \in X(\Omega)$, then the stabilizer of a in G is precisely the inertia group $G_i(x)$.* All this is in fact true without assuming that Ω is "sufficiently large"; the latter hypothesis serves only to ensure that the inertia group of every point of X above z can be characterized as a "geometric" stabilizer. We immediately deduce, for example:

Proposition V.2.1. *Make a base extension $Z' \to Z$, and hence put $X' = X \times_Z Z'$. Let x' be a point of X' and x its image in X. Then $G_i(x) = G_i(x')$.*

In the preceding considerations, it is enough to take for Ω a sufficiently large extension of $\kappa(z')$ (where z, z' are the images of x, x' in Z, Z', respectively).

Proposition V.2.2. *Under the conditions of V.1.3, suppose that Y is locally Noetherian and that X is finite over Y. Let H be a subgroup of G, put $X' = X/H$ (cf. V.1.7), and let $x \in X$, with image x' in X' and image y in Y.* 112

(i) *If $H \supset G_d(x)$, then the homomorphism $\mathcal{O}_y \to \mathcal{O}_{x'}$ induces an isomorphism on completions.*

(ii) *If $H \supset G_i(x)$, then the homomorphism $\mathcal{O}_y \to \mathcal{O}_{x'}$ is étale, i.e. X' is étale over Y at x'.*

Let $Y_1 = \mathrm{Spec}(()\widehat{\mathcal{O}_y})$, and make the base change $Y_1 \to Y$. We obtain $X_1 = X \times_Y Y_1$, finite over Y_1, on which G acts, with quotient Y_1 by V.1.9. Let y_1 be the unique point of Y_1 above y. Since $\kappa(y) = \kappa(y_1)$, the fiber of X at y is isomorphic to that of X_1 at y_1; hence there is a unique point x_1 of X_1 above x. Moreover, by V.1.9, we have $X_1/H = X_1' = X' \times_Y Y_1$. Let x_1' be the image of x_1 in X_1'. It lies above x', and one checks easily (since X' is of finite type over Y) that the homomorphism $\mathcal{O}_{x'} \to \mathcal{O}_{x_1'}$ induces an isomorphism on completions. We are therefore reduced to the case where Y is the spectrum of a complete local ring B, and hence X is the spectrum of a finite B-algebra A, which is a product of finitely many local rings A_x corresponding to the points x_i of X above Y. If A_0 corresponds to $x = x_0$, then A identifies with the ring $\mathrm{Hom}_{G_d}(G, A_0)$ of functions $f\colon G \to A_0$ such that $f(st) = sf(t)$ for $s \in G_d$, the operations of G on these functions being defined by $(uf)(t) = f(tu)$. Thus, if H is any subgroup of G, then A^H is the ring of functions $f\colon G \to A_0$ such that

$$f(stu) = sf(t) \qquad \text{for } s \in G_d,\ u \in H$$

It is therefore a semilocal ring whose local components correspond to the double cosets $G_d a H$ in G; to the double coset defined by $a \in G$ corresponds (by means of the map $f \longmapsto f(a)$) the subring $A_0^{H(a)}$ of A_0, where $H(a) = G_d \cap aHa^{-1}$. Moreover, the local component of A^H corresponding to the image x' of x is also the one corresponding to the double coset $G_d H$ of the identity element; its local component is therefore $A_0^{G_d \cap H}$. Thus, if $G_d \subset H$, we obtain $A_0^{G_d} = A^G = B$, which proves (i). To prove (ii), we may, by passing to a suitable finite flat extension of A and using V.2.1, reduce to the case where the residue-field extension $\kappa(x)/\kappa(y)$ is trivial. But then $G_i(x) = G_d(x)$, and we are reduced to the preceding case.

Corollary V.2.3. *Under the conditions of V.2.2, suppose that $G_i(x) = (e)$. Then X is étale over Y at x. Consequently, if $G_i(x) = (e)$ for every $x \in X$, then $X \to Y$ is an étale morphism.* 113

There is a partial converse:

Corollary V.2.4. *Suppose that X is connected and that the action of G on X is faithful. The morphism $p\colon X \to Y = X/G$ is étale if and only if the inertia groups of the points of X are reduced to the identity element. If this is so, G identifies with the group of all Y-automorphisms of the Y-scheme X.*

In view of V.2.3, we may suppose that X is étale over Y. But if an $s \in G$ belongs to some $G_i(x)$, it then follows from I I.5.4 that s acts trivially on G, and hence is the identity element since G is faithful; this proves the first assertion. Let u be a Y-automorphism of X, and let $x \in X$. By Proposition V.1.3, there exists an $s \in G$ such that $s(x) = u(x)$ and inducing the same residue homomorphism $\kappa(x)/\kappa(x')$ as u. By loc. cit. , we therefore have $s = u$, which completes the proof.

Remark V.2.5. The hypothesis that G acts faithfully is clearly not superfluous in Corollary V.2.4. The same is true of the hypothesis that X is connected, as is seen, for example, by taking $X = Y \times E$, where E is a finite set, and taking G to be the group of permutations of E: G acts with considerable inertia, yet $(Y \times E)/G = Y \times (E/G) = Y$, and X is étale over Y. Taking for G a group strictly smaller than the symmetric group of E but acting transitively on E, one sees that there will also be Y-automorphisms of X which do not come from G.

The standard example of a group G acting without inertia is that of $Y \times G$, on which G acts through its right-translation action on the factor G: a Y-prescheme X with right operator group G is said to be *trivial* if it is isomorphic to $Y \times G$.

The following considerations are useful for relating preschemes with finite operator groups to the notion of a principal bundle in a category (a relation which, although we shall not need it in the rest of the seminar, is important in other contexts). We fix a base prescheme Y and work in the category of Y-preschemes. If G is a finite group, we shall 114
write $G_Y = Y \times G$ for short; this is thus a finite group scheme over Y (cf. no. 1), and if X is a Y-prescheme, then

$$X \times_Y G_Y = X \times G$$

(same reference). Giving a Y-morphism $X \times_Y G_Y \to X$ is therefore equivalent to giving a Y-morphism $X \times G \to X$, i.e. to giving, for each $g \in G$, a Y-morphism $T_g\colon X \to X$. One checks at once that the data of the T_g define on X the structure of a prescheme with right operator group G (i.e. $T_{gg'} = T_{g'}T_g$, $T_e = \mathrm{id}_x$) if and only if the corresponding Y-morphism $X \times_Y G_Y \to X$ defines on X the structure of a Y-prescheme with operator Y-group scheme (in the general sense of objects with a $\mathcal{C}$-operator group in a category $\mathcal{C}$). Suppose this is so. Recall that X is said to be *formally principal homogeneous* under G_Y[3] if the canonical morphism

$$X \times_Y G_Y \to X \times_Y X$$

whose components are respectively pr_1 and the multiplication morphism $\pi\colon X \times_Y G_Y \to X$, is an isomorphism. In the present case, identifying the first member with $X \times G$, the morphism in question is the one which associates to each $g \in G$ the morphism

$$(\mathrm{id}_X, T_g) = (\mathrm{id}_X \times_Y T_g)\Delta_{X/Y}\colon X \to X \times_Y X$$

Consequently, to say that X is formally principal homogeneous under G_Y also means that $X \times_Y X$ is isomorphic to the disjoint sum of the transforms of the diagonal by the elements (e, g) of $G \times G$ (acting on $X \times_Y X$ in the evident way), where e denotes the

[3] One now says instead that X is a pseudo-torsor under G_Y.

identity element of G. If one does not wish to distinguish left from right, and wants a formula which remains applicable to a product of more than two factors identical to X, the condition may be formulated by saying that the canonical morphism

$$X \times_G (G \times G) \to X \times_Y X$$

obtained by assigning to the pair (g, g') the morphism

$$(T_g, T_{g'}) = (T_g \times_Y T_{g'})\Delta_{X/Y} \colon X \to X \times_Y X$$

and making G act on the left on $G \times G$ by the diagonal homomorphism,

$$s(g, g') = (sg, sg'),$$

is an *isomorphism*.

The notion of a *principal homogeneous space* is obtained from that of a formally principal homogeneous space by adding a further axiom ensuring that the "quotient" of X by G_Y exists and is precisely the right unit object of the category, here Y. This axiom may vary with the context, and is often most conveniently stated explicitly (in the yoga of "descent") by requiring that the object with operators become "trivial", i.e. isomorphic to the product $X \times_Y G_Y$ (in the present case $X \times G$), after a suitable base change of a specified type (in such a way, in practice, as to permit the descent technique; cf. Grothendieck, Technique de descente et théorèmes d'existence en Géométrie Algébrique, Sém. Bourbaki no. 190, pages 26 to 28)[4]. In this vein, we record here the characterization of principal homogeneous bundles with group G (in the sense of loc. cit.): 115

Proposition V.2.6. *Let Y be a locally Noetherian prescheme and let X be a Y-prescheme with a finite group G of operators acting on the right. The following conditions are equivalent:*

(i) *X is finite over Y, $Y = X/G$, and the inertia groups of the points of X are reduced to the identity.*
(ii) *There exists a faithfully flat and quasi-compact base change $Y_1 \to Y$ such that $X_1 = X \times_Y Y_1$ is a trivial Y_1-prescheme with operators, i.e. isomorphic to $Y_1 \times G$.*
(ii bis) *As in (ii), but with $Y_1 \to Y$ finite, étale, and surjective.*
(iii) *X is formally principal homogeneous under G_Y, and is faithfully flat and quasi-compact over Y.*

Proof (i)$\Longrightarrow$(ii bis). Take $Y_1 = X$, noting that $X \to Y$ is indeed finite, étale by V.2.3, and surjective. Let us show that X_1 is then trivial over Y_1, which will follow from the

Corollary V.2.7. *If* (i) *holds and X admits a section over Y, then X is a trivial space with operators.*

Indeed, this section makes it possible to define a G-morphism $X \times G \to X$, which is surjective because G is transitive on the fibers of X, injective because G acts without inertia, and finally a local isomorphism by I I.5.3, since X is étale over Y. It is therefore an isomorphism.

(ii bis) trivially implies (ii), which implies (i), since the ingredients of (i) are "invariant" under faithfully flat quasi-compact base extension (cf. the Bourbaki Seminar cited above for "finite"; for inertia groups apply V.2.1, and for $Y = X/G$ use a converse to V.1.9 in the case of a *faithfully flat* base change, which we had forgotten to state explicitly). 116

In proving (i)$\Longrightarrow$(ii bis), we proved (i)$\Longrightarrow$(iii) along the way. Finally, (iii)$\Longrightarrow$(ii), because the first hypothesis in (iii) means precisely that X becomes trivial after making the base change $Y_1 = X$; hence (ii), since X is faithfully flat and quasi-compact over Y.

Definition V.2.8. *A Y-prescheme X with right operator group G satisfying the equivalent conditions of V.2.6 is called a* principal covering of Y, with Galois group G.

[4]Cf. Exp. VIII for the theory of flat descent.

V.3. Automorphisms and morphisms of étale coverings

Proposition V.3.1. *Let X be separated, étale, and of finite type over a locally noetherian prescheme Y, and let G be a finite group acting on X by Y-automorphisms. Then the action of G is admissible, and the quotient prescheme X/G is étale over Y.*

Although X is not assumed finite over Y, it is quasi-projective over Y, so the existence of X/G follows from V.1.8. Let us first prove:

Corollary V.3.2. *The morphism $X \to X/G$ is étale.*

We may clearly suppose first that G acts transitively on the set of connected components of X, and then, by considering the stabilizer of one connected component, that X itself is connected. Finally, we may suppose that G acts faithfully. But then, just as in V.2.4, one sees that G acts without inertia; hence V.2.3 implies that $X \to X/G$ is étale. The conclusion follows from:

Lemma V.3.3 (a regret concerning Exposé I). *Let $X \to X' \to Y$ be morphisms of finite type, let x be a point of X, and let x' and y be its images. Suppose that Y is locally noetherian. If two of the three morphisms in question are étale at the indicated points, then so is the third.*

It remains only to consider the case in which $X \to X'$ and $X \to Y$ are étale at x, and to prove that $X' \to Y$ is étale at x' (which is the case needed for V.3.1). After making 117
a suitable flat extension of the base Y, we reduce to the case in which the residue-field extension $\kappa(x)/\kappa(y)$ is trivial. Consider the homomorphisms $\mathcal{O}_y \to \mathcal{O}_{x'} \to \mathcal{O}_x$, together with the homomorphisms obtained by passing to completions. The hypothesis says that

$$\widehat{\mathcal{O}}_y \longrightarrow \widehat{\mathcal{O}}_x \quad \text{and} \quad \widehat{\mathcal{O}}_{x'} \longrightarrow \widehat{\mathcal{O}}_x$$

are isomorphisms. It follows at once that $\widehat{\mathcal{O}}_y \to \widehat{\mathcal{O}}_{x'}$ is also an isomorphism, which proves the lemma.

Corollary V.3.4. *If X is finite and étale over Y, then X/G is finite and étale over Y.*

Proposition V.3.5. *Let X and X' be two étale coverings of Y. Then every Y-morphism $f\colon X \to X'$ factors as the composite of a surjective étale morphism $X \to X''$ and the canonical immersion $X'' \to X'$ of a subset X'' of X' that is both open and closed.*

It is known (I I.4.8) that f is étale, and hence an open morphism. On the other hand, since X is finite over Y, the morphism f is closed. Thus $f(X) = X''$ is both open and closed in X', and the proof is complete. (N.B. it would have been enough for X', instead of being an étale covering, to be unramified over Y.)

Corollary V.3.6. *With the preceding notation, $X \to X''$ is a strict epimorphism in the category of preschemes, and $X'' \to X'$ is a monomorphism (indeed, a strict monomorphism) in the category of preschemes.*

By definition, the first assertion means that the sequence of morphisms

$$X \times_{X''} X \underset{\mathrm{pr}_2}{\overset{\mathrm{pr}_1}{\rightrightarrows}} X \longrightarrow X''$$

is exact. This follows from the fact that $X \to X''$ is finite and faithfully flat, as is easily seen (cf. Grothendieck, loc. cit.). The dual assertion for $X'' \to X'$ is still more immediate.

Corollary V.3.6 will be useful in the theory of the fundamental group in the next section. Those who dislike the notion of strict epimorphism may replace Corollary V.3.6 by whatever variant they prefer. Let us merely take this opportunity to point out that a factorization $f = f'f''$, with f'' a strict epimorphism and f' a monomorphism, 118 is necessarily unique up to unique isomorphism (in any category). At the same time, however, there may exist a factorization $f = f_1 f_2$ having the dual properties: f_2 is an epimorphism and f_1 a strict monomorphism (again unique up to unique isomorphism), which is not isomorphic to the preceding one. It is enough, for example, to take the category of topological vector spaces (Hausdorff, if desired) and a morphism $u\colon X \to X'$ such that $u(X)$ is not closed.

Proposition V.3.7. *Let Y be a connected locally noetherian prescheme, let y be a point of Y, and let Ω be an algebraically closed extension of $\kappa(y)$. For every X over Y, denote by $X(\Omega)$ the set of points of X with values in Ω. Let X and X' be étale coverings of Y, and let $u\colon X \to X'$ be a Y-morphism such that the corresponding map*

$$X(\Omega) \longrightarrow X'(\Omega)$$

is a bijection. Then u is an isomorphism.

We reduce immediately to the case in which X' is connected. Since $X \to X'$ is finite and étale, the geometric number of points in a fiber of $X \to X'$ is constant, and it is equal to 1 if and only if the morphism in question is an isomorphism. But this number is also the number of elements in a fiber of $X(\Omega) \to X'(\Omega)$, and the conclusion follows.

V.4. Axiomatic conditions for a Galois theory

Let $\mathcal{C}$ be a category and F a covariant functor from $\mathcal{C}$ to the category of finite sets. Assume that the following conditions are satisfied:

(G 1) $\mathcal{C}$ has a terminal object[5], and the fiber product of two objects over a third object in $\mathcal{C}$ exists (this axiom may also be stated by saying that finite inverse limits exist in $\mathcal{C}$).

(G 2) Finite coproducts exist in $\mathcal{C}$ (and hence so does an initial object $\emptyset_{\mathcal{C}}$, playing the role of the empty set), as does the quotient of an object of $\mathcal{C}$ by a finite group of automorphisms.

(G 3) Let $u\colon X \to Y$ be a morphism in $\mathcal{C}$. Then u factors as a composite $X \xrightarrow{u'} Y' \xrightarrow{u''} Y$, where u' is a *strict* epimorphism and u'' is a monomorphism which is an isomorphism onto a direct summand of Y. 119

(G 4) The functor F is left exact (i.e. it sends a terminal object to a terminal object and commutes with fiber products).

(G 5) F commutes with finite coproducts, sends strict epimorphisms to epimorphisms, and commutes with passage to the quotient by a finite group of automorphisms.

(G 6) Let $u\colon X \to Y$ be a morphism in $\mathcal{C}$ such that $F(u)$ is an isomorphism. Then u is an isomorphism.

Our aim is to construct a topological group π, an inverse limit of finite groups, and an equivalence between the category $\mathcal{C}$ and the category $\mathcal{C}(\pi)$ *of finite sets on which π acts continuously* (i.e. in such a way that the stabilizer of a point is an open subgroup, or, equivalently, that there is a discrete quotient group which already acts on the set in question), the equivalence constructed sending the given functor F to the evident inclusion functor from $\mathcal{C}(\pi)$ into the category of finite sets. We note at once that the category $\mathcal{C}(\pi)$ constructed from a topological group π, together with the preceding inclusion functor, does indeed satisfy conditions (G 1) through (G 6).

We proceed in several steps.

a) *Let $u\colon X \to Y$ be a morphism in $\mathcal{C}$. Then u is a monomorphism if and only if $F(u)$ is a monomorphism.* (This uses (G 1), (G 4), and (G 6).)

Indeed, to say that u is a monomorphism means that the projection $\mathrm{pr}_1 \colon X \times_Y X \to X$ is an isomorphism.

[5]Recall that an object e of $\mathcal{C}$ is called a *terminal object* if, for every X in $\mathcal{C}$, $\mathrm{Hom}(X, e)$ has exactly one element. An *initial object* of $\mathcal{C}$ is defined dually.

b) *Every object X of $\mathcal{C}$ is Artinian.*

Indeed, if $X' \to X'' \to X$ are monomorphisms such that $F(X')$ and $F(X'')$ have the same image in $F(X)$, then by a), $F(X') \to F(X'')$ is an isomorphism, and hence $X' \to X''$ is an isomorphism by (G 6).

c) *The functor F is strictly pro-representable.* (cf. Grothendieck, Technique de descente et théorèmes d'existence en Géométrie Algébrique, II, Séminaire Bourbaki 195, February 1960.)

Indeed, by loc. cit. prop. V.3.1, this follows from b) and (G 4). Thus one can find an inverse system, indexed by a directed ordered set I,

$$P = (P_i)_{i \in I}$$

in $\mathcal{C}$, regarded as a pro-object of $\mathcal{C}$, and a functorial isomorphism 120

$$F(X) = \mathrm{Hom}_{\mathrm{Pro}(\mathcal{C})}(P, X) = \varinjlim_i \mathrm{Hom}_{\mathcal{C}}(P_i, X). \tag{$*$}$$

This isomorphism is realized by an element

$$\varphi \in \varprojlim_i F(P_i) = F(P).$$

One may moreover assume that the transition homomorphisms $\varphi_{ji} \colon P_i \to P_j$ ($i \geq j$) are *epimorphisms*, and that *every epimorphism* $P_i \to P'$ is equivalent to an epimorphism $P_i \to P_j$ for a suitable $j \leq i$ (which determines the inverse system P in an essentially unique way).

An object $X \in \mathcal{C}$ is called *connected* if it is not isomorphic to the coproduct of two objects of $\mathcal{C}$ neither of which is isomorphic to the initial object $\emptyset_{\mathcal{C}}$.

d) *The P_i are connected and are not isomorphic to $\emptyset_{\mathcal{C}}$.*

If X is an initial object, then $F(X) = \emptyset$ by (G 5), applied to the coproduct of an empty family, and the converse follows from (G 6). Thus if X' is an object of $\mathcal{C}$ which is not initial, i.e. such that $F(X') \neq \emptyset$, there is no morphism from X' to X. Consequently, if some P_i is initial, then i is a greatest element of the directed ordered index set I, and formula $(*)$ would say that $F(X) = \mathrm{Hom}(P_i, X)$ is a singleton for every X, which is absurd since $F(\emptyset_{\mathcal{C}}) = \emptyset$. The P_i are therefore not isomorphic to $\emptyset_{\mathcal{C}}$.

Suppose that $P_i = A \amalg B$. By (G 5) this gives $F(P_i) = F(A) \amalg F(B)$. In particular, the element a_i of $F(P_i)$ corresponding under $(*)$ to the identity homomorphism $P_i \to P_i$ belongs to $F(A) \amalg F(B)$, say to $F(A)$. This means that there is a $j \geq i$ such that $\varphi_{ij} \colon P_j \to P_i$ factors as $P_j \to A \to P_i = A \amalg B$, where the second arrow is the canonical morphism. Hence $F(P_j) \to F(P_i)$ factors as $F(P_j) \to F(A) \to F(P_i) = F(A) \amalg F(B)$; since $F(P_j) \to F(P_i)$ is surjective by (G 5), it follows that $F(B) = \emptyset$, and hence that B is isomorphic to $\emptyset_{\mathcal{C}}$. 121

e) *Every morphism $u \colon X \to Y$ in $\mathcal{C}$, with X not isomorphic to $\emptyset_{\mathcal{C}}$ and Y connected, is a strict epimorphism. Every endomorphism of a connected object is an automorphism.*

Consider the factorization of u in (G 3). Since $X \neq \emptyset_{\mathcal{C}}$, it follows from (G 6) that $F(X) \neq \emptyset$, hence $F(Y') \neq \emptyset$, hence $Y' \neq \emptyset_{\mathcal{C}}$. Since Y is connected, Y' is therefore identified with Y, and thus u is a strict epimorphism. Suppose that u is an endomorphism of the connected object X; let us show that it is an automorphism. We may assume that X is not isomorphic to $\emptyset_{\mathcal{C}}$, so by what precedes u is a strict epimorphism. Hence $F(u)$ is an epimorphism by (G 5), and since $F(X)$ is a finite set, $F(u)$ is bijective. Thus u is an automorphism by (G 6).

In particular, *every endomorphism of a P_i is an automorphism.*

f) *The following conditions on a P_i are equivalent*: (i) The natural injective map $\mathrm{Hom}(P_i, P_i) \to \mathrm{Hom}(P, P_i) \simeq F(P_i)$ is also surjective, i.e. for every $u \colon P \to P_i$

there is a $v\colon P_i \to P_i$ such that $u = v\varphi_i$ (where φ_i is the canonical homomorphism $P \to P_i$). (ii) The group $\mathrm{Aut}(P_i)$ acts transitively on $F(P_i)$. (iii) The group $\mathrm{Aut}(P_i)$ acts simply transitively on $F(P_i)$.

Upon identifying $\mathrm{Hom}(P, P_i)$ with $F(P_i)$, the map in (i) is precisely $v \longmapsto F(v)(\varphi_i)$. The equivalence of the three conditions then follows from the fact that $\mathrm{Hom}(P_i, P_i) = \mathrm{Aut}(P_i)$ and that the preceding map is already injective.

A P_i satisfying the equivalent conditions (i), (ii), (iii) of f) is called *Galois.*

g) *For every X in $\mathcal{C}$, there is a Galois P_i such that every $u \in \mathrm{Hom}(P, X)$ factors as $P \xrightarrow{\varphi_i} P_i \longrightarrow X$.*

Let $J = \mathrm{Hom}(P, X) = F(X)$. This is a finite set, so there is a P_j such that every $u\colon P \to X$ factors as $P \xrightarrow{j} P_j \longrightarrow X$, or, equivalently, such that the natural morphism

$$P \to X^J \qquad (J = \mathrm{Hom}(P, X))$$

factors as

$$P \xrightarrow{\varphi_j} P_j \to X^J.$$

By (G 3), the morphism $P_j \to X^J$ factors as the composite of a strict epimorphism and a monomorphism, where the former may be taken to be of the form $\varphi_{ij}\colon P_j \to P_i$. We are thus reduced to proving that P_i is Galois. Let k be an index $\geq j$ such that every morphism $P \to P_i$ factors through $P \xrightarrow{\varphi_k} P_k \longrightarrow P_i$. Note that the natural morphism $P_k \to X^J$ still factors as the composite 122

$$P_k \xrightarrow{\varphi_{ik}} P_i \xrightarrow{U} X^J,$$

where the first arrow is a strict epimorphism by e), and the second is a monomorphism. We wish to show that, for a given morphism $\psi\colon P_k \to P_i$, there is an endomorphism v of P_i such that $\psi = v\varphi_{ik}$. For every $u \in \mathrm{Hom}(P_i, X)$, consider $u\psi \in \mathrm{Hom}(P_k, X)$. It is of the form $u'\varphi_{ik}$ for a uniquely determined $u' \in \mathrm{Hom}(P_i, X)$. The map $u \longmapsto u'$ from J to J thus defined by ψ is injective, since ψ is an epimorphism by e); it is therefore bijective because J is finite. The bijection $u \longmapsto u'$ from J to J thus defines an isomorphism $\alpha\colon X^J \xrightarrow{\sim} X^J$ making the diagram

$$\begin{tikzcd}
P_k \arrow[r, "\varphi_{ik}"] \arrow[d, equal] & P_i \arrow[r, "U"] & X^J \arrow[d, "\alpha", "\simeq"'] \\
P_k \arrow[r, "\psi"] & P_i \arrow[r, "U"] & X^J
\end{tikzcd}$$

commutative. By the uniqueness properties of the factorization of a morphism as a composite of a strict epimorphism and a monomorphism, it follows (since ψ too is a strict epimorphism by e)) that one can find a morphism $v\colon P_i \to P_i$ which makes the diagram commutative, as required.

It follows in particular that *the Galois P_i form a cofinal subsystem of the system* (P_j). Since for a Galois object P_i one has

$$\mathrm{Hom}(P, P_i) = \mathrm{Hom}(P_i, P_i) = \mathrm{Aut}(P_i),$$

passage to the limit therefore gives

$$\mathrm{Hom}(P, P) = \varprojlim_i \mathrm{Hom}(P, P_i) = \varprojlim_i \mathrm{Hom}(P_i, P_i) = \varprojlim_i \mathrm{Aut}(P_i),$$

where the inverse limit is taken over the Galois P_i. Moreover, under the identification $\mathrm{Hom}(P, P_i) = F(P_i)$, and taking account of the fact that F sends epimorphisms to epimorphisms, one sees that the transition homomorphisms in the preceding inverse system are surjective. From all this we conclude: 123

h) *One has*

$$\operatorname{Hom}(P,P)=\operatorname{Aut}(P)=\varprojlim_i F(P_i)=\varprojlim_i \operatorname{Aut}(P_i),$$

where the inverse limit is taken over the Galois P_i.

In particular, $\operatorname{Aut}(P)$ appears as the inverse limit of an inverse system of finite groups (with surjective transition homomorphisms); we endow it with the inverse-limit topology of the discrete topologies. *We denote by π and call the fundamental group* (of $\mathcal{C}$ equipped with F) *the group opposite to* $\operatorname{Aut}(P)$. Thus this group acts *on the right* on P; it is the inverse limit of finite groups π_i acting on the right on the Galois P_i, where π_i is the group opposite to $\operatorname{Aut}(P_i)$.

In view of the functorial isomorphism

$$F(X)=\operatorname{Hom}(P,X)$$

and the definition of π, we therefore see that π acts *on the left* on $F(X)$, and moreover acts continuously by g) (for, with the notation of g), it is in fact π_i that acts on $F(X)$). It is immediate that, for every morphism $u\colon X\to Y$ in $\mathcal{C}$, the morphism $F(u)\colon F(X)\to F(Y)$ is compatible with the actions of π. *Henceforth one may therefore regard F as a covariant functor*

$$F\colon \mathcal{C}\to\mathcal{C}(\pi),$$

where $\mathcal{C}(\pi)$ is the category of finite sets on which π acts continuously on the left.

We now define a functor in the opposite direction:

$$G\colon \mathcal{C}\longleftarrow\mathcal{C}(\pi)$$

by the formula

$$G(E)=P\times_\pi E,$$

where $P\times_\pi E$ is defined as a solution of the universal problem summarized by

$$\operatorname{Hom}_{\mathcal{C}}(P\times_\pi E,X)\xrightarrow{\sim}\operatorname{Hom}_\pi(E,\operatorname{Hom}(P,X))$$

(where, in the right-hand side, $\operatorname{Hom}(P,X)=F(X)$ is regarded as a set on which 124
π acts on the left). We must prove the existence of the object $P\times_\pi E$.

i) *Let Q be an object of $\mathcal{C}$ on which a finite group G acts on the right, and let E be a finite set on which G acts on the left. Then $Q\times_G E$ exists, and the canonical map*

$$F(Q)\times_G E\to F(Q\times_G E)$$

is an isomorphism.

Since finite coproducts exist in $\mathcal{C}$ by (G 2), and F commutes with them by (G 5), one is immediately reduced to the case in which G acts transitively on E: indeed, if the E_j are the orbits of G in E, then

$$Q\times_G E=\coprod_j Q\times_G E_j.$$

Let $a\in E$, and let H be its stabilizer. It follows at once from the definition that $Q\times_G E$ is identified with Q/H. Existence thus follows from (G 2), and the compatibility property for F follows from (G 5).

j) *Let E be an object of $\mathcal{C}(\pi)$, and let P_i be Galois and such that π_i already acts on E. Then $P_i\times_{\pi_i}E$ exists and there is a canonical isomorphism*

$$E\xrightarrow{\sim}F(P_i\times_{\pi_i}E).$$

If $j\geq i$ is such that P_j is Galois, then the canonical homomorphism $P_j\times_{\pi_j}E\to P_i\times_{\pi_i}E$ is an isomorphism.

The first assertion is a special case of i), taking account of the fact that π_i acts simply transitively on $F(P_i)$, which is equipped with the marked point φ_i;

this gives an isomorphism $F(P_i) \times_{\pi_i} E \simeq E$. For the second assertion one may, for example, use (G 6).

For every j, let $\mathcal{C}_j$ be the full subcategory of $\mathcal{C}$ formed by those X for which $\mathrm{Hom}(P_j, X) \to \mathrm{Hom}(P, X) \simeq F(X)$ is bijective. We know from g) that $\mathcal{C}$ *is the filtered union of the* $\mathcal{C}_j$. Thus, for $X \in \mathcal{C}_j$, one has

$$\begin{aligned}\mathrm{Hom}_\pi(E, \mathrm{Hom}(P, X)) \simeq \mathrm{Hom}_\pi(E, \mathrm{Hom}(P_j, X)) \simeq \mathrm{Hom}_{\pi_j}(E, \mathrm{Hom}(P_j, X))\\ \simeq \mathrm{Hom}(P_j \times_{\pi_j} E, X),\end{aligned}$$

and, in view of the last assertion in j), one obtains an isomorphism functorial in 125
the object X of $\mathcal{C}_j$:

$$\mathrm{Hom}_\pi(E, \mathrm{Hom}(P, X)) \simeq \mathrm{Hom}(P_i \times_{\pi_i} E, X).$$

Since this is true for every j, and since these functorial isomorphisms, as j varies, induce one another, we conclude:

k) *Under the conditions of* j), *the composite of the canonical morphisms*

$$E \to \mathrm{Hom}(P_i, P_i \times_{\pi_i} E) \to \mathrm{Hom}(P, P_i \times_{\pi_i} E)$$

makes $P_i \times_{\pi_i} E$ *a solution of the universal problem defining* $P \times_\pi E$, i.e. *the latter exists and there is an isomorphism*

$$P \times_\pi E \xrightarrow{\sim} P_i \times_{\pi_i} E.$$

This completes the construction of the functor $G(E)$. On the other hand, there is a functorial homomorphism

$$\alpha\colon \mathrm{id}_{\mathcal{C}(\pi)} \to FG,$$

that is, a homomorphism functorial in the object E of $\mathcal{C}(\pi)$:

$$\alpha(E)\colon E \to FG(E) = F(P \times_\pi E),$$

namely the composite of the canonical morphisms

$$E \to F(P) \times_\pi E \to F(P \times_\pi E)$$

(where the first comes from the marked point $\varphi \in F(P)$). Combining j) and k), one finds:

l) *The homomorphism* α *is an isomorphism.*

Similarly, one defines a functorial homomorphism

$$\beta\colon GF \to \mathrm{id}_{\mathcal{C}},$$

that is, a homomorphism functorial in the object X of $\mathcal{C}$:

$$\beta(X)\colon P \times_\pi F(X) \to X,$$

as the morphism associated with the π-homomorphism 126

$$F(X) \to \mathrm{Hom}(P, X)$$

inverse to the canonical isomorphism $\mathrm{Hom}(P, X) \xrightarrow{\sim} F(X)$.

m) *The composites*

$$F(X) \xrightarrow{\alpha(F(X))} FGF(X) \xrightarrow{F(\beta(X))} F(X)$$
$$G(E) \xrightarrow{G(\alpha(E))} GFG(E) \xrightarrow{\beta(G(E))} G(E)$$

are the identity isomorphisms.

A trotting donkey.

In view of l), it follows that:

n) *The homomorphism* β *is an isomorphism.*

We have thus obtained the promised result:

Theorem V.4.1. *Let $\mathcal{C}$ be a category satisfying conditions* (G 1), (G 2), *and* (G 3) *at the beginning of this section, and let F be a covariant functor from $\mathcal{C}$ to the category of finite sets, satisfying conditions* (G 4), (G 5), *and* (G 6). *Then the preceding canonical constructions define equivalences of categories $F\colon \mathcal{C} \to \mathcal{C}(\pi)$ and $G\colon \mathcal{C}(\pi) \to \mathcal{C}$ which are quasi-inverse to one another. More precisely, there is a pro-object P of $\mathcal{C}$ and a functorial isomorphism $F(X) \xleftarrow{\sim} \operatorname{Hom}(P, X)$; π is the group opposite to the automorphism group of P, endowed with a suitable topology, so that π acts continuously on the sets* $\operatorname{Hom}(P, X) \simeq F(X)$. *Finally, G is given by $G(E) \simeq P \times_\pi E$.*

Remarks V.4.2. The statement of conditions (G 1) through (G 6) becomes simpler and more pleasing if (G 2) and (G 5) are replaced respectively by:

(G' 2) Finite direct limits exist in $\mathcal{C}$.

(G' 5) The functor F is right exact (i.e. it commutes with finite direct limits).

These conditions appear stronger than (G 2) and (G 5), but it follows at once from the structure theorem V.4.1 that they are implied by (G 1) through (G 6). Note, however, that in the cases of interest to us, checking (G 2) and (G 5) does in fact seem simpler than checking (G' 2) and (G' 5). I do not know whether, in condition (G 3), the requirement that u'' be an isomorphism onto a direct summand of Y could be omitted.

V.5. Galois categories 127

Definition V.5.1. *A* Galois category *is a category $\mathcal{C}$ equivalent to a category $\mathcal{C}(\pi)$, where π is a compact group that is a projective limit of finite groups (i.e. is totally disconnected).*

For the definition of $\mathcal{C}(\pi)$, see the beginning of no. V.4. By Theorem V.4.1, $\mathcal{C}$ is Galois if and only if it satisfies conditions (G 1) through (G 3), and there exists a functor F from $\mathcal{C}$ to the category of finite sets satisfying conditions (G 4) through (G 6), that is, one which is exact *and* conservative*, in general terminology. Such a functor is called a* fundamental functor *of the Galois category $\mathcal{C}$*[6]*; it is pro-representable by a pro-object which we denote by P_F. A pro-object P for which the associated functor F is fundamental is called a* fundamental pro-object*. In this way, the category of fundamental functors on $\mathcal{C}$ is anti-equivalent to the category of fundamental pro-objects. If F and P correspond, the group* $\operatorname{Aut} F$ *is therefore isomorphic to the opposite of the group* $\operatorname{Aut} P$*; hence the group denoted by π in the preceding section is none other than* $\operatorname{Aut} P$. *Recall that in the preceding section, starting from a* given *fundamental functor F, we constructed an equivalence of $\mathcal{C}$ with $\mathcal{C}(\pi)$, where $\pi = \operatorname{Aut}(F)$, which carries F to the canonical functor from $\mathcal{C}(\pi)_x$ to the category of finite sets. In the standard case $\mathcal{C} = \mathcal{C}(\pi)$, with F the canonical functor, the fundamental pro-object associated with F is precisely the projective system of discrete quotients π_i of π.*

It may be useful to describe explicitly the category of pro-objects of $\mathcal{C}(\pi)$. We obtain:

Proposition V.5.2. *The category* Pro-$\mathcal{C}(\pi)$ *is canonically equivalent to the category $\mathcal{C}'(\pi)$ of spaces, equipped with a topological group of operators π, which are compact and totally disconnected.*

The latter category contains $\mathcal{C}(\pi)$ as a full subcategory, corresponding to compact discrete spaces with operators, and projective limits exist in it. There is therefore a canonical functor

$$g\colon \text{Pro-}\mathcal{C}(\pi) \to \mathcal{C}'(\pi),$$

which associates to a projective system $Q = (Q_i)$ the object $X = \varprojlim_i Q_i$ of $\mathcal{C}'(\pi)$. To define a functor in the opposite direction, it is enough to define a contravariant functor

[6] It seems preferable to adopt the more expressive term "fiber functor."

from $\mathcal{C}'(\pi)$ to the category of left-exact functors $\mathcal{C} \to \mathbf{Ens}$. For every $X \in \mathcal{C}'(\pi)$, take

$$h(X)(E) = \operatorname{Hom}(X, E)$$

(where Hom is taken in $\mathcal{C}'(\pi)$). It follows immediately from the definitions that h and g are adjoint, and that hg is canonically isomorphic to the identity functor of Pro-$\mathcal{C}(\pi)$. To prove that g and h are quasi-inverse, it remains to show that every object of $\mathcal{C}'(\pi)$ is isomorphic to an object of the form $g(Q)$, with $Q \in$ Pro-$\mathcal{C}(\pi)$. In other words, *every compact, totally disconnected space X equipped with a topological group of operators π is isomorphic to a projective limit of finite discrete spaces with operators.* 128

As a topological space without operators, X is the projective limit of its finite discrete quotients. We are therefore reduced to showing that among these quotients there is a cofinal system invariant under π. For a given quotient X', it is enough to show that the set of its transforms under the operations of π is finite; one then takes the supremum of these transforms, which is an invariant quotient dominating X'. Equivalently, it is enough to find an open normal subgroup π' of π whose elements act trivially on X'. The quotient X' amounts to a finite partition of X into open subsets X_i. By continuity and compactness of π, there exists a neighborhood V of the identity of π such that $s \in V$ implies $s \cdot X_i \subset X_i$ for every i; thus s acts trivially on X'. The open normal subgroups of π form a fundamental system of neighborhoods of the identity, which completes the proof.

Notice, more simply, that the category $\operatorname{Ind}\mathcal{C}(\pi)$ is canonically equivalent to the category of sets on which π acts continuously. We shall not need this fact here.

Proposition V.5.3. *Let $\mathcal{C}$ be a Galois category, let F be a fundamental functor on $\mathcal{C}$, and let $P = (P_i)$ be the associated pro-object, normalized in the usual way. Let $X \in \mathcal{C}$; then X is connected if and only if π acts transitively on $E = F(X)$.*

We reduce to the standard case $\mathcal{C} = \mathcal{C}(\pi)$, with F the canonical functor, where the assertion is immediate.

Corollary V.5.4. *The following conditions on X are equivalent:* (i) *X is connected and $\not\simeq \emptyset_{\mathcal{C}}$;* (ii) *the group π acts transitively on $E = F(X)$, and $F(X) \neq \emptyset$;* (iii) *X is isomorphic to some P_i.*

The equivalence of (i) and (iii) also follows readily from no. V.4, e). 129

Proposition V.5.5. *Let $Q = (Q_i)_{i \in I}$ be a pro-object of $\mathcal{C}$, normalized in the usual way, and let G be the corresponding functor $G(X) = \operatorname{Hom}(Q, X)$ from $\mathcal{C}$ (to* **Ens***). The following conditions are equivalent:*

(i) *G commutes with finite direct sums;*
(ii) *G commutes with the direct sum of two objects;*
(iii) *the Q_i are connected and $\not\simeq \emptyset_{\mathcal{C}}$;*
(iv) *Q is isomorphic to π/H, where H is a closed subgroup of π;*
(v) *the functor G is isomorphic to the functor $E \longmapsto E^H$, the set of H-invariants, defined by a closed subgroup H of π.*

N.B. In conditions (iv) and (v), a fundamental functor is understood to have been chosen, thereby identifying $\mathcal{C}$ with the category $\mathcal{C}(\pi)$.

Proof. We may suppose that $\mathcal{C} = \mathcal{C}(\pi)$. The implication (i)$\Longrightarrow$(ii) is immediate, and (ii)$\Longrightarrow$(iii) is proved as in property d) of no. V.4. Let us prove (iii)$\Longrightarrow$(iv). The set $\varprojlim_i Q_i$ is nonempty, being the projective limit of nonempty finite sets. Let $a \in \varprojlim_i Q_i$. It defines a homomorphism of spaces with operators

$$\pi \to Q$$

which is *surjective*: for every i, the composite $\pi \to Q \to Q_i$ is surjective, since π acts transitively on Q_i by Proposition V.5.3. If H is the stabilizer of a, we obtain an isomorphism $\pi/H \xrightarrow{\sim} Q$. The implications (iv)$\Longrightarrow$(v) and (v)$\Longrightarrow$(i) are again immediate.

Proposition V.5.6. *Let $\mathcal{C}$ be a Galois category, let P be a fundamental pro-object of $\mathcal{C}$, and let F be the associated fundamental functor. Let $P' = (P'_i)_{i \in I}$ be a pro-object of $\mathcal{C}$ in normal form, and let F' be the associated functor $F'(X) = \operatorname{Hom}(P', X)$ from $\mathcal{C}$ to* **Ens**. *The following conditions are equivalent:*

(i) *$P' \simeq P$, or equivalently $F' \simeq F$;*
(ii) *P' is fundamental, or equivalently F' is fundamental;*
(iii) *F' carries a direct sum of two objects to their direct sum, and $X \not\simeq \emptyset_{\mathcal{C}}$ implies $F'(X) \neq \emptyset$;* 130
(iv) *the connected objects of $\mathcal{C}$ distinct from $\emptyset_{\mathcal{C}}$ are exactly those isomorphic to some P'_i.*

We have trivially (i)$\Longrightarrow$(iii) and (i)$\Longrightarrow$(ii); moreover, (ii)$\Longrightarrow$(iv) by Corollary V.5.4, applied to P' in place of P. Furthermore, either (iii) or (iv), by Proposition V.5.5, implies that P' is of the form π/H, where H is a closed subgroup of π. In case (iii), for every open normal subgroup π' of π there is a π-homomorphism $P' = \pi/H \to \pi/\pi'$. Hence $H \subset \pi'$, so $H = (0)$, and consequently (i) holds. This proves the assertion.

Corollary V.5.7. *Let $\mathcal{C}$ be a Galois category. Its fundamental pro-objects are isomorphic, and its fundamental functors are isomorphic.*

In other words, *the category of fundamental functors is a connected groupoid* Γ, which may be called the *fundamental groupoid* of the Galois category $\mathcal{C}$. If $\mathcal{C} = \mathcal{C}(\pi)$, the automorphism group of an object of the fundamental groupoid is isomorphic to π, this isomorphism being well determined up to an inner automorphism. (Recall that a *groupoid* is a category in which every morphism is an isomorphism, and a *connected* groupoid is a groupoid all of whose objects are isomorphic.) The fundamental pro-objects of $\mathcal{C}$ form a connected groupoid equivalent to the *opposite* of the fundamental groupoid. If F, F' are two fundamental functors, associated with fundamental pro-objects P, P', then $\operatorname{Hom}(F, F') = \operatorname{Isom}(F, F')$ is sometimes denoted by $\pi_{F',F}$ and plays the role of a "set of *homotopy classes of paths* from F to F'". In particular, $\pi_{F,F} = \pi_F$ is precisely the *fundamental group of $\mathcal{C}$ at F* constructed in the preceding section. The pro-object P associated with F plays the role of a *universal covering at F* of the final object $e_{\mathcal{C}}$ of $\mathcal{C}$.

It may be convenient to have a description of $\mathcal{C}$, up to equivalence, in terms of its fundamental groupoid Γ, without choosing a particular object F of the latter. To every object X of $\mathcal{C}$ is associated a functor E_X on the fundamental groupoid, defined by

$$E_X(F) = F(X),$$

with values in **Ens**. Such a functor is known in topology as a "local system" on the groupoid. The set $F(X) = E_X(F)$ may be called the *fiber* of X at F, and E_X the fiber functor associated with X. The functor E_X has the following property: *for every F,* 131
$E_X(F)$ is a finite set on which the topological group $\pi_F = \operatorname{Aut}(F)$ acts continuously. For a given covariant functor ξ from the fundamental groupoid to **Ens**, the preceding condition is equivalent to the same condition for any one fixed F. With this understood:

Proposition V.5.8. *The functor $X \longmapsto E_X$ is an equivalence from the category $\mathcal{C}$ to the category of covariant functors from the fundamental groupoid Γ of $\mathcal{C}$ to* **Ens** *which satisfy the condition highlighted above.*

Indeed, let F_0 be an object of the fundamental groupoid and put $\pi_0 = \operatorname{Aut}(F_0)$. The functor $\xi \longmapsto \xi(F_0)$ is an equivalence from the second category in Proposition V.5.8 to the category $\mathcal{C}(\pi_0)$, as is seen immediately. On the other hand, its composite with the functor $X \longmapsto E_X$ is the natural equivalence $\mathcal{C} \to \mathcal{C}(\pi_0)$. It follows that $X \longmapsto E_X$ itself is an equivalence.

Corollary V.5.9. *The category* Pro-$\mathcal{C}$ *is canonically equivalent to the category of covariant functors ξ from the fundamental groupoid Γ to the category of topological spaces* *satisfying the following condition: for every object F of Γ, $\xi(F)$ is a compact, totally disconnected space equipped with a topological group of operators π_F.*

Here again, it is enough to verify this condition on ξ for a single F. The proof is the same as that of Proposition V.5.8.

Remarks V.5.10. Let $(F_s)_{s\in S}$ be a family of objects of the fundamental groupoid Γ. For $s, s' \in S$, put

$$\mathrm{Hom}(s, s') = \mathrm{Hom}(F_s, F_{s'}).$$

The set S thereby becomes a connected groupoid, and the map $s \longmapsto F_s$ becomes a fully faithful functor $f\colon S \to \Gamma$. Consider the functor $X \longmapsto E_X \circ f$ from $\mathcal{C}$ to the functor category $\mathrm{Hom}(S, \mathbf{Ens})$. This gives a variant of Proposition V.5.8, and of Corollary V.5.9, with Γ replaced by S. The resulting statement reduces to Theorem V.4.1 when S consists of one point, and is precisely Proposition V.5.8 itself when S is the set of objects of Γ.

We shall use Corollary V.5.9 to define a canonical pro-object of $\mathcal{C}$. To this end, consider the functor from Γ to the category of topological spaces and indeed of topological 132
groups:

$$f\colon F \longmapsto \mathrm{Aut}(F) = \pi_F.$$

This functor satisfies the condition in Proposition V.5.8. The space with operators $f(F)$ under π_F is simply π_F itself, regarded as a space with operators under itself by inner automorphisms. Thus f corresponds to a pro-object of $\mathcal{C}$ determined up to unique isomorphism. It is in fact a pro-group in $\mathcal{C}$, called the *fundamental pro-group of* $\mathcal{C}$, and plays the role of a local system of fundamental groups. Thus it is a pro-group Π of $\mathcal{C}$ defined by the existence of an isomorphism, functorial in F,

$$F(\Pi) \simeq \pi_F.$$

If X is any pro-object of $\mathcal{C}$, there is a canonical morphism

$$\Pi \times X \to X$$

which makes X an object with a left group of operators Π in Pro-$\mathcal{C}$. It is enough to observe that, for variable F, there is a canonical map

$$\Pi(F) \times X(F) \to X(F),$$

that is,

$$\mathrm{Aut}(F) \times E_X(F) \to E_X(F), \quad \text{or} \quad \pi_F \times F(X) \to F(X),$$

which is functorial in F. It is also functorial in X, so for every morphism $X \to Y$ of pro-objects the corresponding diagram

$$\begin{tikzcd}
\Pi \times X \arrow[r] \arrow[d] & X \arrow[d] \\
\Pi \times Y \arrow[r] & Y
\end{tikzcd}$$

is commutative.

Remarks V.5.11. One must not confuse a fundamental pro-object P, which has no group structure and is connected, with the fundamental pro-group Π, which is a pro-*group* and is generally not connected. More precisely, Π is connected if and only if π_F, acting on itself by inner automorphisms, acts transitively; that is, if and only if π consists only of the identity, or equivalently if $\mathcal{C}$ is equivalent to the category of finite sets. Another essential difference is that Π is determined up to unique isomorphism, whereas P is determined 133
only up to nonunique isomorphism.

Let E be a finite set and consider the constant functor on the groupoid Γ with value E. By Proposition V.5.8, it defines an object of $\mathcal{C}$, denoted by $E_{\mathcal{C}}$, which may also be interpreted as the direct sum of E copies of the final object $e_{\mathcal{C}}$ of $\mathcal{C}$. We may regard

$E_{\mathcal{C}}$ as a functor of E, from the category of finite sets to $\mathcal{C}$. This functor is *exact*, and hence carries finite groups to $\mathcal{C}$-groups, etc. If an object X of $\mathcal{C}$ carries a right action of a finite group G, we may therefore regard X as an object of $\mathcal{C}$ with a right group of operators, the $\mathcal{C}$-group $G_{\mathcal{C}}$. Extending the general terminology for objects with $\mathcal{C}$-groups of operators, we say that X is *formally principal homogeneous* under G if it is formally principal homogeneous under $G_{\mathcal{C}}$, that is, if the canonical morphism

$$X \times G_{\mathcal{C}} \to X \times X$$

induced by the right action of $G_{\mathcal{C}}$ on X is an isomorphism. We say that X is *principal homogeneous* under G if it is so under $G_{\mathcal{C}}$, that is, if it is formally principal homogeneous and, in addition, the quotient $X/G = X/G_{\mathcal{C}}$ is $e_{\mathcal{C}}$.

Fix a fundamental functor, and hence an equivalence of $\mathcal{C}$ with a category $\mathcal{C}(\pi)$. The object X corresponds to a set $E = F(X)$ on which π acts continuously on the left. Giving a right action of G on X is then equivalent to giving a right action of G on the set E which commutes with the action of π. It follows at once that X is principal homogeneous under G if and only if E is a principal homogeneous space under G, that is, if and only if G acts simply transitively on E. Moreover, X is formally principal homogeneous $if and only if$ E is principal homogeneous *or* empty. Comparing with Proposition V.5.3, we see that if X is principal homogeneous under G *and connected*, then the given homomorphism from G to the group opposite to $\operatorname{Aut}(X)$ is an *isomorphism.* Furthermore, an object X of $\mathcal{C}$ is connected and principal homogeneous under the group opposite to $\operatorname{Aut}(X)$ if and only if, with the notation of no. V.4, it is isomorphic to a *Galois* P_i. In the standard case $\mathcal{C} = \mathcal{C}(\pi)$, this means that X is isomorphic to a quotient of π by a normal subgroup.

Continue to fix a fundamental functor F. The data of an object X principal homo- 134
geneous under a finite group G acting on the right, together with a point $a \in F(X)$, are equivalent to the data of a homomorphism from π to G. Indeed, to such a homomorphism we associate the set $E = G$, letting π act on the left through the given homomorphism $\pi \to G$ and the left translations of G, and letting G act on the right by right translations; the marked point a of E is the identity element of G. By the preceding discussion, this gives, in an essentially unique way, every triple (X, G, a) with the properties considered above, since a pointed principal homogeneous space under a group G is identified with G itself. In this way we obtain a direct geometric interpretation of the functor $G \longmapsto \operatorname{Hom}(\pi, G)$ from the category of finite groups to **Ens**. This functor is pro-representable by π, and its consideration would therefore provide another construction of the group π associated with F.

V.6. Exact functors from one Galois category to another

PROPOSITION V.6.1. *Let $\mathcal{C}$ and $\mathcal{C}'$ be two Galois categories, let $H\colon \mathcal{C} \to \mathcal{C}'$ be a covariant functor, let F' be a fundamental functor on $\mathcal{C}'$, and set $F = F' \circ H$. The following conditions are equivalent:*

(i) *H is* exact*, i.e. both left exact and right exact.*
(ii) *H is left exact, carries finite sums to finite sums, and epimorphisms to epimorphisms (or, equivalently, carries objects not isomorphic to $\emptyset_{\mathcal{C}}$ to objects not isomorphic to $\emptyset_{\mathcal{C}'}$).*
(iii) *F is a fundamental functor on $\mathcal{C}$.*

The implication (i)$\Longrightarrow$(ii) is a general categorical fact. Moreover, the first form of (ii) implies the second. Indeed, if X is an object of $\mathcal{C}$, then $X \not\approx \emptyset_{\mathcal{C}}$ if and only if the morphism $X \to e_{\mathcal{C}}$ is an epimorphism; note also that, since F is assumed left exact, it carries $e_{\mathcal{C}}$ to $e_{\mathcal{C}'}$. The second form of (ii) implies (iii), since F, being left exact and therefore pro-representable, satisfies criterion (iii) of V.5.6. Finally, (iii) implies (i), because F is exact and conservative (i.e. it satisfies axiom (G 6) of no. V.4).

Let Γ now be the fundamental groupoid of $\mathcal{C}$, and Γ' that of $\mathcal{C}'$. Thus, if H is exact, 135 then $F' \longmapsto F' \circ H$ is a functor from the groupoid Γ' to the groupoid Γ, which we denote by ${}^tH$:

$${}^tH(F')(X) = F'(H(X)).$$

Using the notation $F(X) = E_X(F)$ introduced in no. V.6, this may also be written

$$E_{H(X)}(F') = E_X({}^tH(F')).$$

This last formula, together with V.5.8 or V.4.1, shows that the exact functor H is determined up to unique isomorphism once the corresponding functor ${}^tH$ is known. Fix F', and set $F = {}^tH(F')$. Then ${}^tH$ defines a homomorphism from $\pi_{F'} = \mathrm{Aut}(F')$ to $\pi_F = \mathrm{Aut}(F)$:

$${}^tH \colon \pi_{F'} \longrightarrow \pi_F \qquad (F = {}^tH(F') = F' \circ H).$$

The preceding formula also shows, in view of V.5.8, that this homomorphism has the following property: for every finite set E on which π_F acts continuously, the group $\pi_{F'}$ also acts *continuously* through the preceding homomorphism $\pi_{F'} \to \pi_F$. Applying this to the quotients of π_F by its open normal subgroups, one sees that the preceding condition also means that the homomorphism in question is continuous.

Conversely, let F be an object of Γ, let F' be an object of Γ', and let

$$u \colon \pi_{F'} \longrightarrow \pi_F$$

be a continuous homomorphism. It determines a manifestly exact functor from $\mathcal{C}(\pi)$ to $\mathcal{C}(\pi')$, and hence, by V.4.1, an exact functor $H \colon \mathcal{C} \to \mathcal{C}'$ for which ${}^tH \colon \pi_{F'} \to \pi_F$ is precisely u. Instead of starting with a group homomorphism, one may start with a *functor*

$$U \colon \Gamma' \longrightarrow \Gamma$$

such that, for *every* $F' \in \Gamma'$ (or for *one* $F' \in \Gamma'$, which amounts to the same thing), the corresponding homomorphism $\pi_{F'} \to \pi_F$ is continuous. Every such functor is isomorphic to one of the form ${}^tH$, where $H \colon \mathcal{C} \to \mathcal{C}'$ is an exact functor determined up to unique isomorphism. Thus: 136

Corollary V.6.2. *A functor $H \colon \mathcal{C} \to \mathcal{C}'$ between Galois categories is exact if and only if there exist equivalences $\mathcal{C}(\pi) \to \mathcal{C}$ and $\mathcal{C}' \to \mathcal{C}(\pi')$ which carry H to the functor $\mathcal{C}(\pi) \to \mathcal{C}(\pi')$ associated with a homomorphism of topological groups $\pi' \to \pi$.*

Corollary V.6.3. *Let $\mathcal{C}$ and $\mathcal{C}'$ be two Galois categories, with fundamental groupoids Γ and Γ'. Then the category of exact functors from $\mathcal{C}$ to $\mathcal{C}'$ is equivalent to the category of functors $U \colon \Gamma' \to \Gamma$ having the following property: for every F' in Γ' (or for one F' in Γ', which amounts to the same thing), if $F = U(F')$, then the homomorphism*

$$\pi_{F'} = \mathrm{Aut}(F') \longrightarrow \pi_F = \mathrm{Aut}(F)$$

defined by U is continuous.

Consider the fundamental pro-group Π of $\mathcal{C}$. An exact functor H carries it to a pro-group $H(\Pi)$ in $\mathcal{C}'$. We shall define a homomorphism

$$\Pi' \longrightarrow H(\Pi),$$

where Π' is the fundamental pro-group of $\mathcal{C}'$, by requiring that, for every object F' of Γ', the corresponding homomorphism

$$F'(\Pi') = \pi_{F'} \longrightarrow F'(H(\Pi)) = \pi_F \qquad (\text{where } F = F' \circ H = {}^tH(F'))$$

be the natural homomorphism

$$\mathrm{Aut}(F') \longrightarrow \mathrm{Aut}(F' \circ H).$$

Since the latter is functorial in F', it does indeed define, by V.5.8, a homomorphism of pro-objects, and in fact of pro-groups, in $\mathcal{C}'$. This homomorphism is said to be *associated* with the functor H.

Now let H' be a second exact functor, from the Galois category $\mathcal{C}'$ to a Galois category $\mathcal{C}''$. It is immediate that

$$^{t}(H'H) = {}^{t}H\,{}^{t}H'$$

(N.B. this is an identity of functors, not merely a canonical isomorphism). There is an analogous transitivity property for the associated homomorphisms of fundamental pro-groups.

We now interpret the properties of the exact functor H in terms of the corresponding homomorphism 137

$$u\colon \pi_{F'} \longrightarrow \pi_F \qquad \text{(where } F = F' \circ H\text{)}.$$

It is convenient to introduce the notion of a *pointed object* of the Galois category $\mathcal{C}$, equipped with its fundamental functor F. By definition, this is an object X of $\mathcal{C}$ together with an element a of $F(X)$. It may therefore be viewed as a finite set on which π_F acts continuously on the left, together with a point a. Consequently, by V.5.3, the *connected* pointed objects of $\mathcal{C}$ are identified with the open subgroups of π_F. If U and V are two such subgroups, corresponding to connected pointed objects X and Y of $\mathcal{C}$, then a pointed homomorphism from X to Y exists if and only if $U \subset V$, and it is then unique.

Of course, the functor H carries pointed objects to pointed objects, since $F = F' \circ H$. Also note that a closed subgroup of a group such as π_F is the intersection of the open subgroups containing it. Thus, if M and N are two closed subgroups, then $M \subset N$ if and only if every open subgroup containing N also contains M. These remarks easily yield the following results:

Proposition V.6.4. *Let X be a connected pointed object of $\mathcal{C}$, associated with an open subgroup U of π_F. Then U contains $u(\pi_{F'})$ (respectively, the closed normal subgroup generated by $u(\pi_{F'})$) if and only if $H(X)$ admits a pointed section (respectively, is completely decomposed).*

By a *section* of an object X of a Galois category $\mathcal{C}$ we mean, implicitly over the final object, a morphism from the final object $e_{\mathcal{C}}$ to X. Equivalently, it is an element a of $F(X)$ fixed by π_F. If X is pointed, this is called a *pointed section* when it is compatible with the pointed structures on X and $e_{\mathcal{C}}$, i.e. when a is precisely the distinguished element of $F(X)$. Such a section is therefore unique, and exists if and only if the distinguished element of $F(X)$ is fixed by π_F. Finally, an object of a Galois category is said to be *completely decomposed* if it is isomorphic to a sum of final objects, i.e. if π_F acts trivially on $F(X)$. For a pointed X, this condition is clearly stronger than the existence of a pointed section. Proposition V.6.4 follows immediately from the preceding definitions and remarks. 138

Corollary V.6.5. *The homomorphism u is trivial if and only if $H(X)$ is completely decomposed for every object X of $\mathcal{C}$.*

Proposition V.6.6. *Let X' be a connected pointed object of $\mathcal{C}'$, associated with an open subgroup U' of $\pi_{F'}$. Then U' contains* $\operatorname{Ker} u$ *if and only if there exist a connected pointed object X of $\mathcal{C}$ and a pointed homomorphism from the distinguished connected component X'_0 of $H(X)$ to X'. Equivalently, X' is isomorphic, as a pointed object, to a quotient of the distinguished connected component of the inverse image of a pointed object of $\mathcal{C}$. If u is surjective, the preceding condition is also equivalent to the following: X' is isomorphic to an object $H(X)$, where X is a pointed object of $\mathcal{C}$.*

(The *distinguished connected component* of a pointed object X of a Galois category $\mathcal{C}$ is the unique connected pointed subobject of X. By V.5.3, it corresponds to the orbit under π_F of the distinguished point of $F(X)$.) Since the fact that U' contains $\operatorname{Ker} u$ does not depend on the chosen pointing of X' (another pointing replaces U' by a conjugate subgroup), we obtain:

Corollary V.6.7. *The subgroup U' contains* $\operatorname{Ker} u$ *if and only if there exist an object X of $\mathcal{C}$, which may be assumed connected, and a morphism from a connected component of $H(X)$ to X'. If u is surjective, this also says that X' is isomorphic to an object of the form $H(X)$.*

Corollary V.6.8. *The homomorphism u is injective if and only if, for every object X' of $\mathcal{C}'$, there exist an object X of $\mathcal{C}$ and a homomorphism from a connected component of $H(X)$ to X'.*

Proposition V.6.9. *The following conditions are equivalent:*

(i) *The homomorphism $u\colon \pi_{F'} \to \pi_F$ is surjective.*
(ii) *For every connected object X of $\mathcal{C}$, the object $H(X)$ is connected.*
(iii) *The functor H is fully faithful.*

The last condition means that, for any two objects X, Y of $\mathcal{C}$, the natural map

$$\operatorname{Hom}(X, Y) \longrightarrow \operatorname{Hom}(H(X), H(Y))$$

is bijective. 139

Corollary V.6.10. *The homomorphism u is an isomorphism if and only if H is an equivalence of categories, or equivalently if the following two conditions hold:*

a) *for every connected object X of $\mathcal{C}$, the object $H(X)$ is connected;*
b) *every object of $\mathcal{C}'$ is isomorphic to an object of the form $H(X)$.*

Proposition V.6.11. *Let $H\colon \mathcal{C} \to \mathcal{C}'$ and $H'\colon \mathcal{C}' \to \mathcal{C}''$ be exact functors between Galois categories. Let F'' be a fundamental functor on $\mathcal{C}''$, set $F' = F''H'$ and $F = F'H$, and consider the associated homomorphisms*

$$u'\colon \pi_{F''} \longrightarrow \pi_{F'} \qquad \text{and} \qquad u\colon \pi_{F'} \longrightarrow \pi_F.$$

The inclusion $\operatorname{Ker} u \supset \Im u'$*, or equivalently the triviality of uu', holds if and only if $H'(H(X))$ is completely decomposed for every object X of $\mathcal{C}$. The inclusion* $\operatorname{Ker} u \subset \Im u'$ *holds if and only if, for every connected pointed object X' of $\mathcal{C}'$ such that $H'(X')$ admits a pointed section, there exist an object X of $\mathcal{C}$ and a homomorphism from a connected component of $H(X)$ to X'.*

The first assertion follows from the last assertion of V.6.4. The second follows by combining V.6.4 and V.6.6.

Remarks V.6.12. Under the conditions of V.6.8, it is not true in general that X' is isomorphic to an object of the form $H(X)$. One can show that every connected object, and hence every object, of $\mathcal{C}'$ is isomorphic to an object of the form $H(X)$ if and only if u is an isomorphism from $\pi_{F'}$ onto a *direct-factor* subgroup of π_F. In practice, however, one constructs directly a right inverse $\pi_F \to \pi_{F'}$ of u, using a suitable exact functor from $\mathcal{C}'$ to $\mathcal{C}$.

Proposition V.6.13. *Let $\mathcal{C}$ be a Galois category equipped with a fundamental functor F, let S be a connected object of $\mathcal{C}$, and let $\mathcal{C}'$ be the category of objects of $\mathcal{C}$ over S. Then $\mathcal{C}'$ is a Galois category, and the functor*

$$X \longmapsto H(X) = X \times S$$

from $\mathcal{C}$ to $\mathcal{C}'$ is exact. Let $a \in F(S)$, and let F' be the functor from $\mathcal{C}'$ to the category of finite sets defined by

$$F'(X') = \text{the inverse image of } a \text{ under } F(X') \longrightarrow F(S).$$

Then there is an isomorphism $F \cong F' \circ H$, and the corresponding homomorphism

$$u\colon \pi_{F'} \longrightarrow \pi_F$$

is an isomorphism from $\pi_{F'}$ onto the open subgroup U of π_F stabilizing the distinguished 140 *element a of $F(S)$.*

The proof is left to the reader.

V.7. The case of preschemes

Let S be a connected locally noetherian prescheme, and let

$$a\colon \operatorname{Spec}(()\Omega) \longrightarrow S$$

be a geometric point of S, with values in an algebraically closed field Ω. Set

$$\mathcal{C} = \text{the category of étale coverings of } S,$$

and, for an object X of $\mathcal{C}$, i.e. an étale covering X of S, set

$$F(X) = \text{the set of geometric points of } X \text{ lying over } a.$$

Thus F becomes a functor from $\mathcal{C}$ to the category of finite sets. Properties (G 1) through (G 6) hold: (G 1) is included among the standard consequences in I I.4.6; (G 2) follows from V.3.4; (G 3) from V.3.5; (G 4) is immediate from the definition; (G 5) follows from V.3.5 and the beginning of no. V.2; and (G 6) is proved in V.3.7. We may therefore apply the results of no. V.4, V.5, and V.6.

In particular, this defines a pro-object P of $\mathcal{C}$ representing F, called the *universal covering of S at the point a*, and a topological group $\pi = \operatorname{Aut}(F) = \operatorname{Aut}^0 P$, called the *fundamental group of S at a*, and denoted by $\pi_1(S, a)$. The functor F then defines an equivalence between $\mathcal{C}$ and the category of finite sets on which $\pi = \pi_1(S, a)$ acts continuously. This equivalence interprets the usual finite projective- and inductive-limit operations on coverings—products, fiber products, sums, passage to quotients, etc.—in terms of the analogous operations in $\mathcal{C}(\pi)$, i.e. the evident operations on finite sets with a π-action. Since the topological connected components of an étale covering are themselves étale coverings, note also that *an object X of $\mathcal{C}$ is connected in $\mathcal{C}$ if and only if it is topologically connected.* By V.5.3, this says that π_1 acts transitively on $F(X)$. 141

For an object X of $\mathcal{C}$ to be faithfully flat and quasi-compact over S—it is already flat and quasi-compact over S—it is necessary and sufficient that $X \to S$ be surjective, i.e. an epimorphism in $\mathcal{C}$, or equivalently that $X \neq \emptyset$. Criterion V.2.6 (iii) therefore shows that *X is a principal covering of S with group G if and only if it is a principal homogeneous space under G in the category $\mathcal{C}$*, in the sense defined in no. V.5.

If a' is another geometric point of S, corresponding to an algebraically closed field Ω', which may differ from Ω and may even have a different characteristic, then it defines another fiber functor $F' = F_{a'}$ from $\mathcal{C}$ to the category of finite sets. This functor is again exact, and hence is isomorphic to $F = F_a$. Consequently, the fundamental groups $\pi_1(S; a)$ for varying a are isomorphic to one another.

Let $\pi_1(S; a, a')$ denote the set of isomorphisms—or, equivalently, the set of homomorphisms—from F_a to $F_{a'}$ between the associated fiber functors. We thereby obtain a *groupoid* whose objects are the geometric points of S, the fundamental groups being the automorphism groups of the objects of this groupoid. The set $\pi_1(S; a', a)$ may be called the *set of homotopy classes of paths from a to a'*. These classes compose in the evident way.

Finally, one may define a pro-group Π_1^S in $\mathcal{C}$, called the *fundamental pro-group of S* or the *local system of fundamental groups on S*, characterized up to unique isomorphism by the existence of an isomorphism, functorial in the geometric point a of S,

$$F_a(\Pi_1^S) = \pi_1(S; a)$$

(cf. Remark V.5.10). In particular, if s is an ordinary point of S, the fiber of Π_1^S at s is a pro-group over $\kappa(s)$, a projective limit of finite étale groups over $\kappa(s)$. One might call this pro-group the *fundamental group of S at the ordinary point s* and denote it by $\pi_1(S, s)$. By definition, its points with values in an algebraically closed extension Ω of $\kappa(s)$ are the elements of $\pi_1(S; a)$, where a is the geometric point of S defined by that extension.

In particular, by taking S to be the spectrum of a field, every field k has canonically and functorially associated with it a pro-group over k, which one might denote by $\pi_1(k)$. It is a projective limit of finite étale groups over k, and its points in an algebraically closed extension Ω of k identify with the elements of the topological Galois group of $\bar{k}/k$, 142
where $\bar{k}$ is the separable closure of k in Ω (cf. V.8.1). This group $\pi_1(k)$ does not yet seem to have attracted the attention of algebraists.

Now let

$$f\colon S' \longrightarrow S$$

be a morphism from one connected locally noetherian prescheme to another, let a' be a geometric point of S', and let $a = f(a')$ be its direct image in S. The inverse-image functor induces a functor from the category $\mathcal{C}(S)$ of étale coverings of S to the category $\mathcal{C}(S')$ of étale coverings of S':

$$f^\bullet\colon \mathcal{C}(S) \longrightarrow \mathcal{C}(S').$$

There is an isomorphism of functors

$$F_a \cong F_{a'} \circ f^\bullet,$$

so $f^\bullet$ is an *exact* functor to which the results of no. V.6 apply. In particular, there is a canonical homomorphism

$$u = \pi_1(f; a')\colon \pi_1(S', a') \longrightarrow \pi_1(S; a) \qquad (a = f(a')).$$

This homomorphism recovers the inverse-image functor as restriction of the acting groups. Thus the properties of $f^\bullet$ are expressed simply in terms of the properties of the associated group homomorphism, as explained in no. V.6. In particular, if S' is an étale covering of S, then u is an isomorphism from $\pi_1(S', a')$ onto the open subgroup of $\pi_1(S, a)$ defining the pointed connected étale covering S' of S, i.e. the stabilizer U of $a' \in F_a(S')$ in $\pi_1(S; a)$.

To interpret the homomorphisms $\pi_1(f; a')$ as the geometric point a' varies, one must, in accordance with no. V.6, consider a homomorphism

$$\Pi_1(f)\colon \Pi_1^{S'} \longrightarrow f^\bullet(\Pi_1^S)$$

of pro-groups over S', and then take the corresponding homomorphism on geometric fibers.

V.8. The case of a normal base prescheme 143

Proposition V.8.1. *Let S be the spectrum of a field k, and let Ω be an algebraically closed extension of k, defining a geometric point a of S with values in Ω. Let $\bar{k}$ be the separable closure of k in Ω. Then there is a canonical isomorphism from $\pi_1(S, a)$ onto the topological Galois group of $\bar{k}/k$.*

Let k' be the algebraic closure of k in Ω; it defines a geometric point b of S with values in k'. The natural homomorphism of functors $F_b \to F_a$ is plainly an isomorphism, because every k-homomorphism from a finite separable extension of k into Ω necessarily has its image in $\bar{k}$, and hence in k'. The group π' of k-automorphisms of k' acts naturally on F_b, giving a homomorphism

$$\pi' \longrightarrow \mathrm{Aut}(F_b) \xrightarrow{\sim} \mathrm{Aut}(F_a) = \pi_1(S; a).$$

It is well known that the natural homomorphism from π' to the group π of automorphisms of $\bar{k}/k$ is an isomorphism. We thus obtain a canonical homomorphism $\pi \to \pi_1(S; a)$, and it remains to show that it is an isomorphism. It is injective because an element of its kernel is an automorphism of $\bar{k}/k$ inducing the identity on every finite separable subextension, and is therefore trivial. It is surjective because, if X is a *connected* étale covering of S, hence defined by a finite separable *extension* L/k, then π acts transitively on the set of k-homomorphisms from L into k', as is well known.

Proposition V.8.2. *Let S be a connected, locally noetherian, normal prescheme, let $K = \kappa(s)$ be its function field, equal to the residue field at its generic point s, and let Ω be an algebraically closed extension of K, defining a geometric point a' of $S' = \mathrm{Spec}(()K)$ and a geometric point a of S. Then the homomorphism*

$$\pi_1(S'; a') \longrightarrow \pi_1(S; a)$$

is surjective. If the first group is identified with the Galois group of the separable closure $\bar{K}$ of K in Ω (cf. V.8.1), then the kernel of the preceding homomorphism corresponds under Galois theory to the subextension of $\bar{K}/K$ obtained by composing the finite extensions of K in Ω that are unramified over S.

The first assertion says that the inverse image on S' of a connected étale covering X of S is connected, i.e. that X is integral; this is precisely I I.10.1. The kernel of the preceding homomorphism consists of those automorphisms of $\bar{K}/K$ that induce the identity on the sets $F_a(X)$, where the étale covering X of S may be assumed connected. 144
This means exactly that the automorphism induces the identity on the finite subextensions of $\bar{K}/K$ that are unramified over S, proving the last assertion.

Remark. Through this interpretation of the fundamental group of the normal prescheme S in terms of ordinary Galois theory, the definition in this case has long been known.

V.9. The case of disconnected preschemes: multi-Galois categories

Let S be a locally noetherian prescheme, and let $(S_i)_{i \in I}$ be its connected components. The category $\mathcal{C}(S)$ of étale coverings of S is equivalent to the product category of the $\mathcal{C}(S_i)$, which may be interpreted in terms of the fundamental groups of the S_i once a geometric point has been chosen in each S_i. In applying descent theory to étale morphisms, it is sometimes inconvenient to choose a geometric point in every S_i. It is then preferable to use the natural generalization of V.5.8, interpreting $\mathcal{C}(S)$ as a category of functors on the groupoid of geometric points of S, regarded as the sum of the groupoids corresponding to the connected components of S. The functors in question take values in the category of finite sets and satisfy the continuity condition analogous to that used in V.5.8.

In practice, one has a family $(a_t)_{t \in E}$ of geometric points of S such that every connected component S_i of S contains at least one of them. As in V.5.10, one may then replace the groupoid of all geometric points of S by the analogous groupoid whose underlying set is E. These considerations should of course be incorporated into general definitions concerning categories equivalent to product categories of categories of the form $\mathcal{C}(\pi)$, which may be called *multi-Galois categories*. We leave the details to the reader.

CHAPTER VI

Fibered categories and descent

VI.0. Introduction

Contrary to what was announced in the introduction to the preceding exposé, it has proved impossible to carry out descent in the category of preschemes, even in special cases, without first developing with sufficient care the language of descent in general categories.

The notion of "descent" provides the general framework for every process of "gluing" objects, and consequently for gluing categories. The most classical gluing situation begins with a topological space X and an open cover (X_i). Suppose that for every i we are given a fibered space E_i over X_i, and for every pair (i, j) an isomorphism

$$f_{ji} \colon E_i|_{X_{ij}} \xrightarrow{\sim} E_j|_{X_{ij}}, \qquad X_{ij} = X_i \cap X_j,$$

satisfying the familiar transitivity condition, written in abbreviated form as $f_{kj} f_{ji} = f_{ki}$. Then there exists a fibered space E over X, determined up to isomorphism by the existence of isomorphisms

$$f_i \colon E|_{X_i} \xrightarrow{\sim} E_i$$

satisfying $f_{ji} = f_j f_i^{-1}$, with the usual abuse of notation.

Let $X' = \coprod_i X_i$, which is a fibered space over X, that is, is equipped with a continuous map $X' \to X$. The family (E_i) can be described more concisely as a fibered space E' over X'. The family of maps f_{ji} becomes an isomorphism between the two pullbacks E''_1 and E''_2 of E' to

$$X'' = X' \times_X X'$$

by the two canonical projections. The gluing condition can then be written as an identity between isomorphisms of the fibered spaces E'''_1 and E'''_3 over the triple fiber product

$$X''' = X' \times_X X' \times_X X',$$

where E'''_i is the pullback of E' to X''' by the i-th canonical projection.

The construction of E from E' and the gluing isomorphism is a typical descent process. Conversely, starting with a fibered space E over X, one says that E is *locally trivial with fiber* F if there is an open cover (X_i) of X such that every $E|_{X_i}$ is isomorphic to $F \times X_i$. Equivalently, the pullback E' of E to $X' = \coprod_i X_i$ is isomorphic to $X' \times F$.

Thus both the gluing of objects and the localization of a property are connected with the study of certain kinds of base change $X' \to X$. In algebraic geometry, many other kinds of base change, notably faithfully flat morphisms $X' \to X$, must be regarded as providing a localization process for preschemes, or other objects, over X. This kind of localization is used just as much as ordinary topological localization, of which the latter is in fact a special case. The same is true, to a lesser extent, in analytic geometry.

Most proofs, being reduced to verifications, are omitted or only sketched. When useful, we spell out the less evident diagrams arising in a proof.

VI.1. Universes, categories, and equivalences of categories

To avoid certain logical difficulties, we shall use the notion of a *universe*: a set large enough that the usual operations of set theory do not lead outside it. A "universe axiom" guarantees that every object belongs to some universe. For details, see the book in preparation by C. Chevalley and the lecturer. [1] Thus **Set** denotes not the category of all sets, a notion which has no meaning, but the category of sets belonging to a fixed universe, which will not be indicated in the notation. Likewise, **Cat** denotes the category of categories belonging to that universe; by definition, the morphisms from an object X of **Cat** to another object Y are the functors from X to Y.

If $\mathcal{C}$ is a category, we denote by $\mathrm{Ob}(\mathcal{C})$ its *set of objects*, and by $\mathrm{Fl}(\mathcal{C})$ its *set of arrows*, or morphisms. We shall therefore write $X \in \mathrm{Ob}(\mathcal{C})$, and avoid the customary abuse $X \in \mathcal{C}$. If $\mathcal{C}$ and $\mathcal{C}'$ are categories, a *functor* from $\mathcal{C}$ to $\mathcal{C}'$ will always mean what is commonly called a *covariant* functor. Its specification includes its source and target categories $\mathcal{C}$ and $\mathcal{C}'$. The functors from $\mathcal{C}$ to $\mathcal{C}'$ form a set $\mathrm{Hom}(\mathcal{C}, \mathcal{C}')$, which is the set of objects of the functor category $\mathrm{Fun}(\mathcal{C}, \mathcal{C}')$. By definition, a *contravariant functor* from $\mathcal{C}$ to $\mathcal{C}'$ is a functor from the opposite category $\mathcal{C}^\circ$ to $\mathcal{C}'$.

We shall use the notions of projective and inductive limits of a functor $F\colon \mathcal{I} \to \mathcal{C}$, and in particular their most common special cases: Cartesian products and fiber products, the dual notions of coproducts and amalgamated sums, and the usual formal properties of these operations.

For example, projective limits exist in the category **Cat** introduced above, provided the indexing categories $\mathcal{I}$ belong to the chosen universe. The set of objects, respectively the set of arrows, of a projective-limit category $\mathcal{C}$ of categories $\mathcal{C}_i$ is obtained by taking the projective limit of the sets of objects, respectively arrows, of the categories $\mathcal{C}_i$. The best-known case is the product of a family of categories. We shall constantly use below the fiber product of two categories over a third.

For matters concerning categories and functors, pending the book already mentioned, see the reference cited as VI.1. It is necessarily very incomplete, even concerning the generalities sketched in this section.

We take this opportunity to make explicit the notion of equivalence of categories, which is not presented satisfactorily in that reference. A functor

$$F\colon \mathcal{C} \longrightarrow \mathcal{C}'$$

is called *faithful*, respectively *fully faithful*, if for every pair of objects S, T of $\mathcal{C}$, the map

$$\mathrm{Hom}(S,T) \longrightarrow \mathrm{Hom}(F(S), F(T)), \qquad u \longmapsto F(u),$$

is injective, respectively bijective. The functor F is called an *equivalence of categories* if it is fully faithful and every object S' of $\mathcal{C}'$ is isomorphic to an object of the form $F(S)$.

Equivalently, there is a functor

$$G\colon \mathcal{C}' \longrightarrow \mathcal{C}$$

quasi-inverse to F, with GF isomorphic to $\mathrm{id}_{\mathcal{C}}$ and FG isomorphic to $\mathrm{id}_{\mathcal{C}'}$. The data of a functor $G\colon \mathcal{C}' \to \mathcal{C}$ and an isomorphism

$$\varphi\colon FG \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}'}$$

amount to choosing, for every $S' \in \mathrm{Ob}(\mathcal{C}')$, an object $S = G(S')$ of $\mathcal{C}$ and an isomorphism $u = \varphi(S')\colon F(S) \xrightarrow{\sim} S'$, compatibly with morphisms. More precisely, there is a unique functor $\mathcal{C}' \to \mathcal{C}$ having the prescribed map $S' \mapsto G(S')$ on objects and for which $S' \mapsto \varphi(S')$ is a natural isomorphism $FG \to \mathrm{id}_{\mathcal{C}'}$.

Finally, if G is quasi-inverse to F, and if isomorphisms

$$\varphi\colon FG \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}'}, \qquad \psi\colon GF \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}}$$

are chosen, then the two compatibility conditions on φ and ψ stated in VI.1, I.1.2 are equivalent. For every chosen φ, there is a unique ψ for which those conditions hold.

[1] The eventual authors are C. Chevalley and P. Gabriel. The book is due to appear in the year 3000. In the meantime, see also SGA 4, Exposé I.

VI.2. Categories over another category

Let $\mathcal{E}$ be a category belonging to $\mathfrak{U}$, hence an object of **Cat**. We may therefore consider the category

$$\mathbf{Cat}_{/\mathcal{E}}$$

of "objects of **Cat** over $\mathcal{E}$." An object of this category is a functor

$$p\colon \mathcal{F} \longrightarrow \mathcal{E}.$$

The category $\mathcal{F}$, equipped with such a functor, is also called a *category over* $\mathcal{E}$, or an $\mathcal{E}$-*category*. Accordingly, an $\mathcal{E}$-functor from a category $\mathcal{F}$ over $\mathcal{E}$ to a category $\mathcal{G}$ over $\mathcal{E}$ is a functor

$$f\colon \mathcal{F} \longrightarrow \mathcal{G}$$

such that

$$qf = p,$$

where p and q are the projection functors for $\mathcal{F}$ and $\mathcal{G}$, respectively. The set of $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$ is thus in bijection with the set of arrows from $\mathcal{F}$ to $\mathcal{G}$ in $\mathbf{Cat}_{/\mathcal{E}}$. This is not literally an identity, since the datum of f alone does not determine $\mathcal{F}$ and $\mathcal{G}$ as categories over $\mathcal{E}$; nevertheless, as in every category $\mathcal{C}_{/S}$, we shall routinely make the harmless abuse of identifying $\mathcal{E}$-functors in the above sense with arrows in $\mathbf{Cat}_{/\mathcal{E}}$.

We denote by

$$\mathrm{Hom}_{\mathcal{E}}(\mathcal{F}, \mathcal{G})$$

the set of $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$. Of course, a composite of $\mathcal{E}$-functors is an $\mathcal{E}$-functor; by definition, the composition in question corresponds to composition of arrows in $\mathbf{Cat}_{/\mathcal{E}}$.

Now consider two $\mathcal{E}$-functors

$$f, g\colon \mathcal{F} \longrightarrow \mathcal{G}$$

and a natural transformation

$$u\colon f \longrightarrow g.$$

We call u an $\mathcal{E}$-*homomorphism*, or a *homomorphism of* $\mathcal{E}$-*functors*, if, for every $\xi \in \mathrm{Ob}(\mathcal{F})$,

$$q(u(\xi)) = \mathrm{id}_{p(\xi)}\,.$$

In words, put

$$S = p(\xi) = qf(\xi) = qg(\xi) \in \mathrm{Ob}(\mathcal{E}).$$

Then the morphism

$$u(\xi)\colon f(\xi) \longrightarrow g(\xi)$$

in $\mathcal{G}$ is an id_S-morphism. More generally, for every morphism $\alpha\colon T \to S$ in $\mathcal{E}$, and every category $\mathcal{G}$ over $\mathcal{E}$, a morphism v in $\mathcal{G}$ is called an α-*morphism* if $q(v) = \alpha$, where $q\colon \mathcal{G} \to \mathcal{E}$ is the projection functor.

If $h\colon \mathcal{F} \to \mathcal{G}$ is a third $\mathcal{E}$-functor and $v\colon g \to h$ is an $\mathcal{E}$-homomorphism, then vu is again an $\mathcal{E}$-homomorphism. Thus the $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$, together with their $\mathcal{E}$-homomorphisms, form a subcategory of the category $\mathrm{Fun}(\mathcal{F}, \mathcal{G})$ of all functors from $\mathcal{F}$ to $\mathcal{G}$. We call it the *category of* $\mathcal{E}$-*functors from* $\mathcal{F}$ *to* $\mathcal{G}$ and denote it by

$$\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}).$$

It is also the kernel subcategory of the pair of functors

$$\mathrm{Fun}(\mathcal{F}, \mathcal{G}) \underset{S}{\overset{R}{\rightrightarrows}} \mathrm{Fun}(\mathcal{F}, \mathcal{E}),$$

where R is the constant functor defined by the object $p \in \mathrm{Fun}(\mathcal{F}, \mathcal{E})$, while S is the functor $f \mapsto q \circ f$ defined by $q\colon \mathcal{G} \to \mathcal{E}$.

To finish these generalities, it remains to define the natural pairings among the categories $\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ by composition of $\mathcal{E}$-functors. In other words, for three categories $\mathcal{F},\mathcal{G},\mathcal{H}$ over $\mathcal{E}$, we want to define a "composition functor"

$$\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})\times\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{G},\mathcal{H})\longrightarrow\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{H}) \tag{i}$$

which, on objects, induces the composition map $(f,g)\mapsto gf$ for $\mathcal{E}$-functors $f\colon\mathcal{F}\to\mathcal{G}$ and $g\colon\mathcal{G}\to\mathcal{H}$.

For this purpose, recall the canonical functor

$$\mathrm{Fun}(\mathcal{F},\mathcal{G})\times\mathrm{Fun}(\mathcal{G},\mathcal{H})\longrightarrow\mathrm{Fun}(\mathcal{F},\mathcal{H}), \tag{ii}$$

which, on objects, is just the composition map $(f,g)\mapsto gf$. It sends an arrow (u,v), where

$$u\colon f\to f',\qquad v\colon g\to g'$$

are arrows in $\mathrm{Fun}(\mathcal{F},\mathcal{G})$ and $\mathrm{Fun}(\mathcal{G},\mathcal{H})$, respectively, to the arrow

$$v*u\colon gf\longrightarrow g'f'$$

defined by

$$(v*u)(\xi)=v(f'(\xi))\circ g(u(\xi))=g'(u(\xi))\circ v(f(\xi)).$$

It is well known that this does define a natural transformation from gf to $g'f'$, and that, as f,g,u,v vary, it defines the functor in (ii); that is,

$$\mathrm{id}_g*\mathrm{id}_f=\mathrm{id}_{gf}, \tag{I}$$

and

$$(v'*u')\circ(v*u)=(v'\circ v)*(u'\circ u). \tag{II}$$

Recall also the associativity formula for the canonical pairings in (ii). It is expressed both by the associativity $(hg)f=h(gf)$ of composition of functors and by

$$(w*v)*u=w*(v*u) \tag{III}$$

for composition products of natural transformations. Here $u\colon f\to f'$ and $v\colon g\to g'$ are as above, and one is also given a natural transformation $w\colon h\to h'$ between functors $h,h'\colon\mathcal{H}\to\mathcal{K}$.

We now claim that, when $\mathcal{F}$ and $\mathcal{G}$ are $\mathcal{E}$-categories, the canonical composition functor in (ii) induces the functor in (i). Since we already know that the composite of two $\mathcal{E}$-functors is an $\mathcal{E}$-functor, this amounts to saying that if $u\colon f\to f'$ and $v\colon g\to g'$ are homomorphisms of $\mathcal{E}$-functors, then $v*u\colon gf\to g'f'$ is again a homomorphism of $\mathcal{E}$-functors. This follows immediately from the definitions. Because the pairings in (i) are induced by those in (ii), they satisfy the same associativity property, expressed by the formulas $(hg)f=h(gf)$ and $(w*v)*u=w*(v*u)$ for $\mathcal{E}$-functors and homomorphisms of $\mathcal{E}$-functors.

To complete the collection of formulas (I)–(III), recall also

$$v*\mathrm{id}_{\mathcal{F}}=v,\qquad \mathrm{id}_{\mathcal{G}}*u=u. \tag{IV}$$

For brevity, one writes $v*f$ or $u*g$ instead of $v*u$ when u, respectively v, is the identity automorphism of f, respectively g.

It follows from the definition of the pairings in (i) that $\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ is a functor in $\mathcal{F}$ and $\mathcal{G}$, from the product category

$$\mathbf{Cat}_{/\mathcal{E}}^{\mathrm{op}}\times\mathbf{Cat}_{/\mathcal{E}}\quad\text{to}\quad\mathbf{Cat}.$$

Indeed, if $g\colon\mathcal{G}\to\mathcal{G}_1$ is an $\mathcal{E}$-functor, hence an object of $\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{G},\mathcal{G}_1)$, then, on putting $\mathcal{H}=\mathcal{G}_1$ in (i), it gives a functor

$$g_*\colon\ \mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})\longrightarrow\mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G}_1).$$

Similarly, an $\mathcal{E}$-functor $f\colon \mathcal{F}_1 \to \mathcal{F}$ defines a functor

$$f^*\colon\ \mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G}) \longrightarrow \mathrm{Fun}_{\mathcal{E}/-}(\mathcal{F}_1,\mathcal{G}).$$

For short, these functors are also denoted by the symbols $f \mapsto g\circ f$ and $g \mapsto g \circ f$, respectively; strictly speaking, those symbols describe only the corresponding maps on sets of objects. The associativity property established above shows that this does indeed give the announced functor

$$\mathbf{Cat}_{/\mathcal{E}}^{\mathrm{op}} \times \mathbf{Cat}_{/\mathcal{E}} \longrightarrow \mathbf{Cat}.$$

VI.3. Base change in categories over $\mathcal{E}$

Since inverse limits (with respect to categories $\mathcal{I}$ that are elements of $\mathfrak{U}$) exist in $\mathbf{Cat}$, the same is true in $\mathbf{Cat}_{/\mathcal{E}}$; in particular, Cartesian products exist there, and are interpreted as fiber products in $\mathbf{Cat}$. In accordance with the general notation, if $\mathcal{F}$ and $\mathcal{G}$ are categories over $\mathcal{E}$, we denote by

$$\mathcal{F}\times_{\mathcal{E}}\mathcal{G}$$

their product in $\mathbf{Cat}_{/\mathcal{E}}$, that is, their fiber product over $\mathcal{E}$ in $\mathbf{Cat}$, regarded as a category over $\mathcal{E}$. Thus $\mathcal{F}\times_{\mathcal{E}}\mathcal{G}$ is equipped with two $\mathcal{E}$-functors pr_1 and pr_2, which define, for every category $\mathcal{H}$ over $\mathcal{E}$, a bijection 153

$$\mathrm{Hom}_{\mathcal{E}}(\mathcal{H},\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{H},\mathcal{F})\times \mathrm{Hom}_{\mathcal{E}}(\mathcal{H},\mathcal{G}).$$

Moreover, this bijection is obtained from an isomorphism of categories

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{F})\times \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{G})$$

by taking the sets of objects on both sides, where the displayed functor has as its components the functors $h \longmapsto \mathrm{pr}_1\circ h$ and $h\longmapsto \mathrm{pr}_2\circ h$ from the left-hand side to the two factors on the right. We leave it to the reader to verify that this indeed gives an isomorphism (the analogous fact holds, more generally, whenever one has an inverse limit of categories—and not only in the case of a fiber product).

Recall also that (as was stated in no. VI.1):

$$\begin{aligned}\mathrm{Ob}(\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) &= \mathrm{Ob}(\mathcal{F})\times_{\mathrm{Ob}(\mathcal{E})}\mathrm{Ob}(\mathcal{G})\\ \mathrm{Fl}(\mathcal{F}\times_{\mathcal{E}}\mathcal{G}) &= \mathrm{Fl}(\mathcal{F})\times_{\mathrm{Fl}(\mathcal{E})}\mathrm{Fl}(\mathcal{G}),\end{aligned}$$

the composition of arrows being performed componentwise.

In what follows, we consider a functor

$$\lambda\colon \mathcal{E}'\to\mathcal{E}$$

and, for every category $\mathcal{F}$ over $\mathcal{E}$, regard $\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'$ as a category over $\mathcal{E}'$ by means of pr_2; in other words, we interpret the "fiber product"operation as a *"base change"* operation, and call the functor $\lambda\colon \mathcal{E}'\to\mathcal{E}$ a *"base-change functor"*. In accordance with the familiar general facts, this yields a functor, called the *base-change functor* for λ: 154

$$\lambda^*\colon \mathbf{Cat}_{/\mathcal{E}}\to\mathbf{Cat}_{/\mathcal{E}'},$$

(adjoint to the "restriction of base"functor which, to every category $\mathcal{F}'$ over $\mathcal{E}'$ with projection functor p', assigns $\mathcal{F}'$, regarded as a category over $\mathcal{E}$ through the functor $p=\lambda p'$). As is well known in the general case of a base-change functor in a category, the base-change functor "commutes with inverse limits", and in particular "transforms"fiber products over $\mathcal{E}$ into fiber products over $\mathcal{E}'$.

Let $\mathcal{F}$ and $\mathcal{G}$ be two categories over $\mathcal{E}$. We shall define a *canonical isomorphism*:

(i) $\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}',\mathcal{G}') \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}',\mathcal{G})$

where $\mathcal{F}'=\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'$, $\mathcal{G}'=\mathcal{G}\times_{\mathcal{E}}\mathcal{E}'$.

For this, consider the functor

$$\mathrm{pr}_1 \colon \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}' \to \mathcal{G},$$

and define (i) by

$$F \longmapsto \mathrm{pr}_1 \circ F,$$

which a priori denotes a functor

(ii) $$\mathbf{Hom}(\mathcal{F}', \mathcal{G}') \to \mathbf{Hom}(\mathcal{F}', \mathcal{G})$$

It therefore remains only to verify that the latter induces a functor on the subcategories in (i), and that this functor is an isomorphism. That (ii) induces a bijection

$$\mathrm{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$$

is the characteristic property of the base-change functor. It therefore remains to prove 155
that if F, G are $\mathcal{E}'$-functors $\mathcal{F}' \to \mathcal{G}'$, then *the map*

$$u \longmapsto \mathrm{pr}_1 \circ u$$

induces a bijection

$$\mathrm{Hom}_{\mathcal{E}'}(F, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathrm{pr}_1 \circ F, \mathrm{pr}_1 \circ G).$$

The verification of this fact is immediate and is left to the reader.

It follows from this isomorphism (i), and from the end of the preceding subsection, that

$$\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}')$$

may be regarded as a functor in $\mathcal{E}', \mathcal{F}, \mathcal{G}$, from the category $\mathbf{Cat}^{\circ}_{/\mathcal{E}} \times \mathbf{Cat}^{\circ}_{/\mathcal{E}} \times \mathbf{Cat}_{/\mathcal{E}}$ to the category $\mathbf{Cat}$, isomorphic to the functor defined by the expression $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G})$. In particular, for fixed $\mathcal{F}$, $\mathcal{G}$, one obtains a functor in $\mathcal{E}'$, $\mathcal{E}' \to \mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F}', \mathcal{G}') = \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}')$, and in particular the projection $\mathcal{E}$-functor $\lambda \colon \mathcal{E}' \to \mathcal{E}$ defines a morphism i.e. a functor

$$\lambda^*_{\mathcal{F},\mathcal{G}} \colon \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \to \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}', \mathcal{G}')$$

which we shall make explicit. On the sets of objects of the two sides, this is the map

$$f \longmapsto f' = f \times_{\mathcal{E}} \mathcal{E}'$$

which expresses the functorial dependence of $\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$ on the object $\mathcal{F}$ over $\mathcal{E}$. On the other hand, consider two $\mathcal{E}$-functors

$$f, g \colon \mathcal{F} \to \mathcal{G}$$

and a homomorphism of $\mathcal{E}$-functors

$$u \colon f \to g,$$

we shall make explicit the corresponding homomorphism of $\mathcal{E}'$-functors: 156

$$u' \colon f' \to g'.$$

For every

$$\xi' = (\xi, S') \in \mathrm{Ob}(\mathcal{F}')$$

with

$$\xi \in \mathrm{Ob}(\mathcal{F}), \quad S' \in \mathrm{Ob}(\mathcal{E}'), \quad p(\xi) = \lambda(S') = S$$

the morphism

$$u'(\xi') \colon f'(\xi') = (f(\xi), S') \to g'(\xi') = (g(\xi), S') \quad \text{in } \mathcal{G}'$$

is defined by the formula

$$u'(\xi') = (u(\xi), \mathrm{id}_{S'})$$

(which is indeed an S'-morphism in $\mathcal{G}'$, since $q(u(\xi)) = \lambda(\mathrm{id}_{S'}) = \mathrm{id}_S$).

Now consider an arbitrary $\mathcal{E}$-functor

$$\lambda' : \mathcal{E}'' \to \mathcal{E}'$$

and the corresponding functor

$$\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}') \to \mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}''),$$

I claim that this functor is none other than the functor obtained by the preceding procedure, starting from $\mathcal{F}'$ and $\mathcal{G}'$ over $\mathcal{E}'$ and regarding $\mathcal{E}''$ as a category over $\mathcal{E}'$, taking into account the isomorphisms of *"transitivity of base change"*:

$$\mathcal{F}' \times_{\mathcal{E}'} \mathcal{E}'' \xrightarrow{\sim} \mathcal{F}'' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'' \quad \text{and} \quad \mathcal{G}' \times_{\mathcal{E}'} \mathcal{E}'' \xrightarrow{\sim} \mathcal{G}'' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'',$$

which induce a canonical isomorphism

$$\mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F}' \times_{\mathcal{E}'} \mathcal{E}'', \mathcal{G}' \times_{\mathcal{E}'} \mathcal{E}'') \xrightarrow{\sim} \mathbf{Hom}_{\mathcal{E}''/-}(\mathcal{F} \times_{\mathcal{E}} \mathcal{E}'', \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'')$$

The verification of this compatibility is immediate and is left to the reader. 157

The functors just defined are compatible with the pairings defined in the preceding subsection, in the following precise sense: if $\mathcal{F}$, $\mathcal{G}$, and $\mathcal{H}$ are categories over $\mathcal{E}$ and if we set

$$\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}', \quad \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}', \quad \mathcal{H}' = \mathcal{H} \times_{\mathcal{E}} \mathcal{E}',$$

then the following diagram of functors is commutative:

$$\begin{tikzcd}
\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G}) \times \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G},\mathcal{H}) \arrow[r] \arrow[d, "\lambda^*_{\mathcal{F},\mathcal{G}} \times \lambda^*_{\mathcal{G},\mathcal{H}}"'] & \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{H}) \arrow[d, "\lambda^*_{\mathcal{F},\mathcal{H}}"] \\
\mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}',\mathcal{G}') \times \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{G}',\mathcal{H}') \arrow[r] & \mathbf{Hom}_{\mathcal{E}'/-}(\mathcal{F}',\mathcal{H}')
\end{tikzcd}$$

where the horizontal arrows are the composition functors defined in the preceding subsection. This commutativity is expressed by the formulas

$$(gf)' = g'f'$$

for $f \in \mathrm{Hom}_{\mathcal{E}}(\mathcal{F},\mathcal{G})$, $g \in \mathrm{Hom}_{\mathcal{E}}(\mathcal{G},\mathcal{H})$ (a formula that simply expresses the functoriality of base change), and by the formula

$$(v * u)' = v' * u'$$

when $u : f \to f_1$ is an arrow of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ and $v : g \to g_1$ is an arrow of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G},\mathcal{H})$. The verification of this formula follows readily from the definitions.

In what follows, we shall be principally interested in $\mathbf{Hom}_{\mathcal{E}}(\mathcal{F},\mathcal{G})$ (and certain noteworthy subcategories thereof) when $\mathcal{F} = \mathcal{E}$, and for this reason we introduce special notation:

$$\mathbf{\Gamma}(\mathcal{G}/\mathcal{E}) = \mathbf{Hom}_{\mathcal{E}}(\mathcal{E},\mathcal{G}), \quad \Gamma(\mathcal{G}/\mathcal{E}) = \mathrm{Ob}(\mathbf{\Gamma}(\mathcal{G}/\mathcal{E})) = \mathrm{Hom}_{\mathcal{E}}(\mathcal{E},\mathcal{G}).$$

Remarks. When $\mathcal{E}$ is a one-point category, i.e. when $\mathrm{Ob}(\mathcal{E})$ and $\mathrm{Fl}(\mathcal{E})$ are each reduced to a single element, which also means that $\mathcal{E}$ is a final object of the category **Cat**, the datum of a category over $\mathcal{E}$ is equivalent to the datum of a category, plain and simple (since there is a unique functor from $\mathcal{F}$ to $\mathcal{E}$). More precisely, $\mathbf{Cat}_{/\mathcal{E}}$ is then isomorphic to **Cat**. Moreover, the categories $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ are then precisely the categories $\mathbf{Hom}(\mathcal{F},\mathcal{G})$. Recall that the fundamental formula 158

$$\mathrm{Hom}(\mathcal{H}, \mathbf{Hom}(\mathcal{F},\mathcal{G})) \xrightarrow{\sim} \mathrm{Hom}(\mathcal{F} \times \mathcal{H}, \mathcal{G})$$

(an isomorphism functorial in the three arguments occurring therein) allows $\mathbf{Hom}(\mathcal{F},\mathcal{G})$ to be interpreted axiomatically, in terms internal to the category **Cat**, so that the familiar formalism of the categories **Hom** appears as a special case of a formalism valid in categories such as **Cat**, in which "**Hom** objects" (defined by the preceding formula) exist. There is an analogous interpretation of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ when $\mathcal{E}$ is again arbitrary, by means of the formula

$$\mathrm{Hom}(\mathcal{H}, \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{F}\times\mathcal{H},\mathcal{G})$$

(an isomorphism functorial in the three arguments). Thus the formal properties set out in no. VI.2, VI.3 are special cases of more general results, valid in categories in which the objects $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{G})$ (when $\mathcal{F}$ and $\mathcal{G}$ are two objects of the category over a third object $\mathcal{E}$) exist.

VI.4. Fiber categories; equivalence of $\mathcal{E}$-categories

Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let $S \in \mathrm{Ob}(\mathcal{E})$. The *fiber category of $\mathcal{F}$ at S* is the subcategory $\mathcal{F}_S$ of $\mathcal{F}$ that is the inverse image of the one-point subcategory of $\mathcal{E}$ defined by S. Thus the objects of $\mathcal{F}_S$ are the objects ξ of $\mathcal{F}$ such that $p(\xi) = S$, and its morphisms are the morphisms u of $\mathcal{F}$ such that $p(u) = \mathrm{id}_S$, i.e. the S-morphisms in $\mathcal{F}$. 159

Of course, $\mathcal{F}_S$ is canonically isomorphic to the fiber product $\mathcal{F}\times_{\mathcal{E}}\{S\}$, where $\{S\}$ denotes the one-point subcategory of $\mathcal{E}$ defined by S, equipped with its inclusion functor into $\mathcal{E}$. It follows, in view of the transitivity of base change, that if one makes the base change $\lambda\colon \mathcal{E}'\to\mathcal{E}$, then for every $S'\in\mathrm{Ob}(\mathcal{E}')$, *the projection*

$$\mathrm{pr}_1\colon \mathcal{F}' = \mathcal{F}\times_{\mathcal{E}}\mathcal{E}' \longrightarrow \mathcal{F}$$

induces an isomorphism

$$\mathcal{F}'_{S'} \longrightarrow \mathcal{F}_S \qquad (\text{where } S = \lambda(S')).$$

Proposition VI.4.1. *Let $f\colon \mathcal{F}\to\mathcal{G}$ be an $\mathcal{E}$-functor. If f is fully faithful, then for every base change $\mathcal{E}'\to\mathcal{E}$, the corresponding functor*

$$f'\colon \mathcal{F}' = \mathcal{F}\times_{\mathcal{E}}\mathcal{E}' \longrightarrow \mathcal{G}' = \mathcal{G}\times_{\mathcal{E}}\mathcal{E}'$$

is fully faithful.

The verification is immediate. More generally, one can show that every projective limit of fully faithful functors (here f and the identity functors in $\mathcal{E},\mathcal{E}'$) is a fully faithful functor.

The analogue of Proposition VI.4.1 in which where "fully faithful" is replaced by "equivalence of categories" is false, already when $\mathcal{G}=\mathcal{E}$. Nevertheless:

Proposition VI.4.2. *Let $f\colon \mathcal{F}\to\mathcal{G}$ be an $\mathcal{E}$-functor. The following conditions are equivalent:*

(i) *There exist an $\mathcal{E}$-functor $g\colon \mathcal{G}\to\mathcal{F}$ and $\mathcal{E}$-isomorphisms*

$$gf \xrightarrow{\sim} \mathrm{id}_{\mathcal{F}}, \qquad fg \xrightarrow{\sim} \mathrm{id}_{\mathcal{G}}.$$

(ii) *For every category $\mathcal{E}'$ over $\mathcal{E}$, the functor*

$$f' = f\times_{\mathcal{E}}\mathcal{E}'\colon \mathcal{F}' = \mathcal{F}\times_{\mathcal{E}}\mathcal{E}' \longrightarrow \mathcal{G}' = \mathcal{G}\times_{\mathcal{E}}\mathcal{E}'$$

is an equivalence of categories. 160

(iii) *f is an equivalence of categories, and for every $S\in\mathrm{Ob}(\mathcal{E})$, the functor*

$$f_S\colon \mathcal{F}_S \longrightarrow \mathcal{G}_S$$

induced by f is an equivalence of categories.

(iii bis) *f is fully faithful, and for every $S\in\mathrm{Ob}(\mathcal{E})$ and every $\eta\in\mathrm{Ob}(\mathcal{G}_S)$, there exist $\xi\in\mathrm{Ob}(\mathcal{F}_S)$ and an S-isomorphism $u\colon f(\xi)\to\eta$.*

Proof. Condition (i) plainly implies that f is an equivalence of categories, a notion defined by the same condition except that the functor isomorphisms are not required to be $\mathcal{E}$-morphisms. Moreover, the functoriality established in the preceding section shows that condition (i) is preserved after a base change $\mathcal{E}' \to \mathcal{E}$. Thus (i)$\Longrightarrow$(ii). Clearly (ii)$\Longrightarrow$(iii), since it is enough to take $\mathcal{E}' = \mathcal{E}$ and $\mathcal{E}' = \{S\}$. The implication (iii)$\Longrightarrow$(iii bis) is still more immediate, so it remains to prove (iii bis)$\Longrightarrow$(i).

For every $\eta \in \mathrm{Ob}(\mathcal{G})$, choose $g(\eta) \in \mathrm{Ob}(\mathcal{F})$ and an isomorphism

$$u(\eta)\colon f(g(\eta)) \longrightarrow \eta$$

such that $q(u(\eta)) = \mathrm{id}_S$, where $S = q(\eta)$. This is possible by the second condition in (iii bis). As is well known and immediate, the full faithfulness of f implies that g can be made, in a unique way, into a functor from $\mathcal{G}$ to $\mathcal{F}$ so that the $u(\eta)$ define a functorial homomorphism, hence an isomorphism,

$$u\colon fg \xrightarrow{\sim} \mathrm{id}_{\mathcal{G}}.$$

Moreover, by construction, g is an $\mathcal{E}$-functor and u an $\mathcal{E}$-homomorphism. The preceding data then determine a functorial isomorphism

$$v\colon gf \longrightarrow \mathrm{id}_{\mathcal{F}},$$

defined by the condition $f * v = u * f$, and one sees at once that this, too, is an $\mathcal{E}$-morphism. This proves the proposition.

Definition VI.4.3. *When the preceding conditions hold, f is called an* equivalence of categories over $\mathcal{E}$*, or an* $\mathcal{E}$-equivalence.

Corollary VI.4.4. *Suppose that the projection functor $p\colon \mathcal{F} \to \mathcal{E}$ is transportable, i.e. that for every isomorphism $\alpha\colon T \to S$ in $\mathcal{E}$ and every object ξ in $\mathcal{F}_T$, there exists an isomorphism u in $\mathcal{F}$ with source ξ such that $p(u) = \alpha$. Then every $\mathcal{E}$-functor $f\colon \mathcal{F} \to \mathcal{G}$ that is an equivalence of categories is an $\mathcal{E}$-equivalence.*

This follows from criterion (iii bis). 161

Corollary VI.4.5. *Let $f\colon \mathcal{F} \to \mathcal{G}$ be an $\mathcal{E}$-equivalence. Then, for every category $\mathcal{H}$ over $\mathcal{E}$, the corresponding functors*

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{G},\mathcal{H}) \longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F},\mathcal{H}),$$
$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{F}) \longrightarrow \mathbf{Hom}_{\mathcal{E}/-}(\mathcal{H},\mathcal{G})$$

(cf. no. VI.2) are equivalences of categories.

This follows from criterion (i) by the usual argument.

VI.5. Cartesian morphisms, inverse images, Cartesian functors

Let $\mathcal{F}$ be a category over $\mathcal{E}$, with projection functor p.

Definition VI.5.1. *Consider a morphism*

$$\alpha\colon \eta \longrightarrow \xi$$

in $\mathcal{F}$, and let

$$S = p(\xi), \qquad T = p(\eta), \qquad f = p(\alpha).$$

The morphism α is called a Cartesian morphism *if, for every $\eta' \in \mathrm{Ob}(\mathcal{F}_T)$ and every f-morphism $u\colon \eta' \to \xi$, there exists a unique T-morphism $\overline{u}\colon \eta' \to \eta$ such that $u = \alpha \circ \overline{u}$.*

Thus, for every $\eta' \in \mathrm{Ob}(\mathcal{F}_T)$, the map $v \longmapsto \alpha \circ v$,

(i) $$\mathrm{Hom}_T(\eta', \eta) \longrightarrow \mathrm{Hom}_f(\eta', \xi),$$

is bijective. Equivalently, the pair (η, α) *represents the functor in η'* $\mathcal{F}_T^{\circ} \to \mathbf{Ens}$ appearing on the right.

If, for a given morphism $f\colon T \to S$ in $\mathcal{E}$ and a given $\xi \in \mathrm{Ob}(\mathcal{F}_S)$, such a pair (η, α)
exists, i.e. if there is a Cartesian morphism α in $\mathcal{F}$ with target ξ and $p(\alpha) = f$, then η is 162
determined in $\mathcal{F}_T$ up to unique isomorphism. One then says that *the inverse image of ξ under f exists*, and an object η of $\mathcal{F}_T$, together with a Cartesian f-morphism $\alpha\colon \eta \to \xi$, is called *an inverse image of ξ under f*. Often such an inverse image is chosen whenever it exists, with $\mathcal{F}$ fixed. It is then denoted by $f^*_{\mathcal{F}}(\xi)$, or simply $f^*(\xi)$, or $\xi \times_S T$ when these notations cause no confusion. The canonical morphism $\alpha\colon \eta \to \xi$ will be denoted below by $\alpha_f(\xi)$.

If the inverse image of every $\xi \in \mathrm{Ob}(\mathcal{F}_S)$ under f exists, one also says that *the inverse-image functor under f in $\mathcal{F}$ exists*; then $f^*(\xi)$ becomes a covariant functor in ξ, from $\mathcal{F}_S$ to $\mathcal{F}_T$. This follows because the right-hand side of (i) depends covariantly on ξ, or more precisely defines a functor from $\mathcal{F}_T^\circ \times \mathcal{F}_S$ to **Ens**.

This functorial dependence of $f^*(\xi)$ is explicit as follows. Consider Cartesian f-morphisms

$$\alpha\colon \eta \longrightarrow \xi, \qquad \alpha'\colon \eta' \longrightarrow \xi',$$

and an S-morphism $\lambda\colon \xi \to \xi'$. There exists a unique T-morphism $\mu\colon \eta \to \eta'$ such that

$$\alpha'\mu = \lambda\alpha,$$

as follows from the fact that α' is Cartesian.

We also record the following immediate fact. Consider a commutative diagram

$$\begin{tikzcd} \xi \arrow[d, "\lambda"'] & \eta \arrow[l, "\alpha"'] \arrow[d, "\mu"] \\ \xi' & \eta' \arrow[l, "\alpha'"'] \end{tikzcd}$$

in $\mathcal{F}$, where α and α' are f-morphisms, λ is an S-isomorphism, and μ a T-isomorphism. 163
Then *α is Cartesian if and only if α' is Cartesian.*

DEFINITION VI.5.2. *An $\mathcal{E}$-functor $F\colon \mathcal{F} \to \mathcal{G}$ is called a* Cartesian functor *if it carries Cartesian morphisms to Cartesian morphisms. Denote by $\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}, \mathcal{G})$ the full subcategory of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G})$ formed by the Cartesian functors.*

For example, regard $\mathcal{E}$ as a category over itself by means of the identity functor. Every morphism of $\mathcal{E}$ is Cartesian, so a Cartesian functor from $\mathcal{E}$ to $\mathcal{F}$ is a section functor $F\colon \mathcal{E} \to \mathcal{F}$ that carries every morphism of $\mathcal{E}$ to a Cartesian morphism. Such a functor is called a Cartesian section *of $\mathcal{F}$ over $\mathcal{E}$.*

PROPOSITION VI.5.3.
(i) *A functor $F\colon \mathcal{F} \to \mathcal{G}$ that is an $\mathcal{E}$-equivalence is Cartesian.*
(ii) *Let $F, G\colon \mathcal{F} \to \mathcal{G}$ be isomorphic $\mathcal{E}$-functors. If one is Cartesian, then so is the other.*
(iii) *The composite of two Cartesian functors $\mathcal{F} \to \mathcal{G}$ and $\mathcal{G} \to \mathcal{H}$ is Cartesian.*

Assertion (iii) is immediate from the definition, (ii) follows from the remark preceding VI.5.2, and (i) follows readily from the definition and criterion VI.4.2 (iii). More precisely, a morphism α in $\mathcal{F}$ is Cartesian if and only if $F(\alpha)$ is Cartesian.

COROLLARY VI.5.4. *Let $F\colon \mathcal{F} \to \mathcal{G}$ be an $\mathcal{E}$-equivalence. Then, for every category $\mathcal{H}$ over $\mathcal{E}$, the corresponding functors $G \longmapsto G \circ F$ and $G \longmapsto F \circ G$ induce equivalences of categories*

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{H}) \xrightarrow{\approx} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}, \mathcal{H}),$$
$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{H}, \mathcal{F}) \xrightarrow{\approx} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{H}, \mathcal{G}).$$

This follows in the usual way from criterion VI.4.2 (i) and VI.5.3 (i), (ii), and (iii). 164
More precisely, *an $\mathcal{E}$-functor $G\colon \mathcal{G} \to \mathcal{H}$ is Cartesian if and only if $G \circ F$ is Cartesian*; likewise, *an $\mathcal{E}$-functor $G\colon \mathcal{H} \to \mathcal{F}$ is Cartesian if and only if $F \circ G$ is Cartesian.*

It follows from VI.5.3 (iii) that, if one considers the subcategory $\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ of $\mathbf{Cat}_{/\mathcal{E}}$ having the same objects as $\mathbf{Cat}_{/\mathcal{E}}$ and Cartesian functors as its morphisms, then, as in no. VI.2, there are composition pairings

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{G}) \times \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G},\mathcal{H}) \longrightarrow \mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{H})$$

induced by those of no. VI.2. These allow $\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{G})$ to be regarded as a functor in $\mathcal{F},\mathcal{G}$, from $\left(\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}\right)^{\circ} \times \mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ to $\mathbf{Cat}$. We shall chiefly need this remark when $\mathcal{F} = \mathcal{G}$.

Definition VI.5.5. *Let $\mathcal{F}$ be a category over $\mathcal{E}$. Denote by*

$$\varprojlim \mathcal{F}/\mathcal{E}$$

the category of Cartesian $\mathcal{E}$-functors $\mathcal{E} \to \mathcal{F}$, i.e. of Cartesian sections of $\mathcal{F}$ over $\mathcal{E}$.

By what has just been said, $\varprojlim \mathcal{F}/\mathcal{E}$ is a functor in $\mathcal{F}$, from the category $\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ to the category $\mathbf{Cat}$.

We shall see below the relations between this operation $\varprojlim$ and the notion of a projective limit of categories, as well as many examples.

VI.6. Fibered categories and prefibered categories. Products and base change therein

Definition VI.6.1. *A category $\mathcal{F}$ over $\mathcal{E}$ is called a* fibered category *(and the functor $\mathcal{F} \to \mathcal{E}$ is then said to be* fibered*) if it satisfies the following two axioms:*

- $\mathrm{Fib}_{\mathrm{I}}$ *For every morphism $f\colon T \to S$ in $\mathcal{E}$, the inverse-image functor by f in $\mathcal{F}$ exists.* 165
- $\mathrm{Fib}_{\mathrm{II}}$ *The composite of two Cartesian morphisms is Cartesian.*

A category $\mathcal{F}$ over $\mathcal{E}$ satisfying condition $\mathrm{Fib}_{\mathrm{I}}$ *is called a* prefibered category over *$\mathcal{E}$.*

If $\mathcal{F}$ is a fibered category (respectivelyprefibered category) over $\mathcal{E}$, a subcategory $\mathcal{G}$ of $\mathcal{F}$ is called a *fibered subcategory* (respectivelya *prefibered subcategory*) if it is a fibered category (respectivelyprefibered category) over $\mathcal{E}$ and, moreover, the inclusion functor is Cartesian. If, for example, $\mathcal{G}$ is a *full* subcategory of $\mathcal{F}$, one sees that this means that, for every morphism $f\colon T \to S$ in $\mathcal{E}$ and every $\xi \in \mathrm{Ob}(\mathcal{G}_S)$, $f^*_{\mathcal{F}}(\xi)$ is T-isomorphic to an object of $\mathcal{G}_T$. Another interesting case is the following: with $\mathcal{F}$ a fibered category over $\mathcal{E}$, consider the subcategory $\mathcal{G}$ of $\mathcal{F}$ having the same objects and whose morphisms are the *Cartesian* morphisms of $\mathcal{F}$; in particular, the morphisms of $\mathcal{G}_S$ are the isomorphisms of $\mathcal{F}_S$. One sees at once that this is indeed a fibered subcategory of $\mathcal{F}$, since in the bijection

$$\mathrm{Hom}_T(\eta',\eta) \xrightarrow{\sim} \mathrm{Hom}_f(\eta',\xi)$$

associated with a Cartesian f-morphism α in $\mathcal{F}$, the T-isomorphisms on the left correspond to the Cartesian morphisms on the right. By definition, the Cartesian sections $\mathcal{E} \to \mathcal{F}$ correspond bijectively to arbitrary $\mathcal{E}$-functors $\mathcal{E} \to \mathcal{G}$ (but note that the natural functor

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{E},\mathcal{G}) \to \mathbf{Hom}_{\mathrm{cart}}(\mathcal{E},\mathcal{F}) = \varprojlim(\mathcal{F}/\mathcal{E})$$

is faithful, but in general is not fully faithful, i.e. is not an isomorphism).

Remarks. Let $\mathcal{F}$ be a category over $\mathcal{E}$. The following conditions are equivalent: (i) Every morphism of $\mathcal{F}$ is Cartesian; (ii) $\mathcal{F}$ is a fibered category over $\mathcal{E}$, and the $\mathcal{F}_S$ are groupoids (i.e. every morphism in $\mathcal{F}_S$ is an isomorphism). In this case $\mathcal{F}$ is said to be a category *fibered in groupoids* over $\mathcal{E}$. These are the categories encountered especially 166
in "moduli theory". If $\mathcal{E}$ is a groupoid, one shows that conditions (i) and (ii) are also equivalent to the following: (iii) $\mathcal{F}$ is a groupoid, and the projection functor $p\colon \mathcal{F} \to \mathcal{E}$ is transportable (cf. VI.4.4). For example, if $\mathcal{E}$ and $\mathcal{F}$ are groupoids such that $\mathrm{Ob}(\mathcal{E})$ and $\mathrm{Ob}(\mathcal{F})$ are each reduced to one point, so that $\mathcal{E}$ and $\mathcal{F}$ are defined, up to isomorphism, by groups E and F, and the functor $p\colon \mathcal{F} \to \mathcal{E}$ is defined by a group homomorphism

$p\colon F \to E$, then $\mathcal{F}$ is fibered over $\mathcal{E}$ if and only if p is surjective, i.e. if p defines an *extension* of the group E by the group $G = \operatorname{Ker} p$.

Proposition VI.6.2. *Let $F\colon \mathcal{F} \to \mathcal{G}$ be an $\mathcal{E}$-equivalence. Then $\mathcal{F}$ is a fibered (respectively, prefibered) category over $\mathcal{E}$ if and only if $\mathcal{G}$ is.*

This follows readily from the definitions and from the observation made above that a morphism α in $\mathcal{F}$ is Cartesian if and only if $F(\alpha)$ is Cartesian.

Proposition VI.6.3. *Let $\mathcal{F}_1, \mathcal{F}_2$ be two categories over $\mathcal{E}$, and let $\alpha = (\alpha_1, \alpha_2)$ be a morphism in $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$. Then α is Cartesian if and only if both its components are Cartesian.*

Let ξ_i be the target and η_i the source of α_i, and let $f\colon T \to S$ be the morphism of $\mathcal{E}$ for which α_1 and α_2 are f-morphisms. For every $\eta' = (\eta'_1, \eta'_2)$ in $\mathcal{F}_T$, there is a commutative diagram

$$\begin{tikzcd}
\operatorname{Hom}_T(\eta', \eta) \arrow[r] \arrow[d] & \operatorname{Hom}_f(\eta', \xi) \arrow[d] \\
\operatorname{Hom}_T(\eta'_1, \eta_1) \times \operatorname{Hom}_T(\eta'_2, \eta_2) \arrow[r] & \operatorname{Hom}_f(\eta'_1, \xi_1) \times \operatorname{Hom}_f(\eta'_2, \xi_2)
\end{tikzcd}$$

whose vertical arrows are bijections. Thus, if either horizontal arrow is a bijection, then so is the other. This already shows that if α_1 and α_2 are Cartesian, so that the lower horizontal arrow is bijective, then α is Cartesian. Conversely, in the preceding diagram first put $\eta'_2 = \eta_2$, so that $\operatorname{Hom}_T(\eta'_2, \eta_2) \neq \emptyset$; this proves that α_1 is Cartesian. Then put $\eta'_1 = \eta_1$, proving that α_2 is Cartesian.

167

Corollary VI.6.4. *Let $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$, and let $F = (F_1, F_2)$ be an $\mathcal{E}$-functor $\mathcal{G} \to \mathcal{F}$. Then F is Cartesian if and only if F_1 and F_2 are Cartesian. This gives an isomorphism of categories*

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2) \xrightarrow{\sim} \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_1) \times \mathbf{Hom}_{\mathrm{cart}}(\mathcal{G}, \mathcal{F}_2),$$

and, in particular, by taking $\mathcal{G} = \mathcal{E}$, an isomorphism of categories

$$\varprojlim(\mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2/\mathcal{E}) \xrightarrow{\sim} \varprojlim(\mathcal{F}_1/\mathcal{E}) \times \varprojlim(\mathcal{F}_2/\mathcal{E}).$$

Corollary VI.6.5. *Let $\mathcal{F}_1$ and $\mathcal{F}_2$ be two fibered (respectively, prefibered) categories over $\mathcal{E}$. Then their fiber product $\mathcal{F} = \mathcal{F}_1 \times_{\mathcal{E}} \mathcal{F}_2$ is a fibered (respectively, prefibered) category over $\mathcal{E}$.*

These results extend to the fiber product of an arbitrary family of categories over $\mathcal{E}$.

Proposition VI.6.6. *Let $\mathcal{F}$ be a category over $\mathcal{E}$, with projection functor p, and let $\lambda\colon \mathcal{E}' \to \mathcal{E}$ be a functor. Regard*

$$\mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}'$$

as a category over $\mathcal{E}'$ by means of the projection functor $p' = p \times_{\mathcal{E}} \mathrm{id}_{\mathcal{E}'}$. Let α' be a morphism of $\mathcal{F}'$. Then α' is Cartesian if and only if its image α in $\mathcal{F}$ is Cartesian.

The proof is immediate and is left to the reader.

Corollary VI.6.7. *For every Cartesian functor $F\colon \mathcal{F} \to \mathcal{G}$ between categories over $\mathcal{E}$, the functor*

$$F' = F \times_{\mathcal{E}} \mathcal{E}'\colon \mathcal{F}' = \mathcal{F} \times_{\mathcal{E}} \mathcal{E}' \longrightarrow \mathcal{G}' = \mathcal{G} \times_{\mathcal{E}} \mathcal{E}'$$

is Cartesian.

Consequently, the functor

$$\mathbf{Hom}_{\mathcal{E}}(\mathcal{F},\mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathcal{E}'}(\mathcal{F}',\mathcal{G}')$$

considered in no. VI.3 induces a functor

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F},\mathcal{G}) \longrightarrow \mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}',\mathcal{G}').$$

In other words, for fixed $\mathcal{F},\mathcal{G}$, one may regard

$$\mathbf{Hom}_{\mathrm{cart}}(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}',\mathcal{G}\times_{\mathcal{E}}\mathcal{E}')$$

as a functor in $\mathcal{E}'$, from the category $\mathbf{Cat}_{/\mathcal{E}}{}^{\circ}$ to $\mathbf{Cat}$. If $\mathcal{F},\mathcal{G}$ are also allowed to vary, one 168
obtains a functor from

$$\mathbf{Cat}_{/\mathcal{E}}{}^{\circ} \times (\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}})^{\circ} \times \mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$$

to $\mathbf{Cat}$.

Taking into account the isomorphism

$$\mathrm{Hom}_{\mathcal{E}'}(\mathcal{F}',\mathcal{G}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}}(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}',\mathcal{G})$$

considered in no. VI.3, the Cartesian $\mathcal{E}'$-functors from $\mathcal{F}'$ to $\mathcal{G}'$ correspond to the $\mathcal{E}$-functors

$$\mathcal{F}\times_{\mathcal{E}}\mathcal{E}' \longrightarrow \mathcal{G}$$

that carry every morphism whose first projection is a Cartesian morphism of $\mathcal{F}$ to a Cartesian morphism of $\mathcal{G}$. Taking $\mathcal{F}=\mathcal{E}$ and changing notation gives:

Corollary VI.6.8. *$\varprojlim(\mathcal{F}'/\mathcal{E}')$ is isomorphic to the full subcategory of $\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{E}',\mathcal{F})$ formed by those $\mathcal{E}$-functors $\mathcal{E}'\to\mathcal{F}$ that carry arbitrary morphisms to Cartesian morphisms. In particular, if $\mathcal{F}$ is a fibered category and $\widetilde{\mathcal{F}}$ is the subcategory of $\mathcal{F}$ whose morphisms are the Cartesian morphisms of $\mathcal{F}$, then there is a bijection*

$$\mathrm{Ob}\varprojlim(\mathcal{F}'/\mathcal{E}') \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{E}/-}(\mathcal{E}',\widetilde{\mathcal{F}}).$$

This specifies how the expression

$$\varprojlim(\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'/\mathcal{E}')$$

is to be regarded as a functor in $\mathcal{E}'$ and $\mathcal{F}$, from $\mathbf{Cat}_{/\mathcal{E}}{}^{\circ}\times\mathbf{Cat}^{\mathrm{cart}}_{/\mathcal{E}}$ to $\mathbf{Cat}$. We shall later obtain a fuller functorial dependence on $\mathcal{E}'$ when $\mathcal{F}$ is required to be a fibered category.

Corollary VI.6.9. *Let $\mathcal{F}$ be a fibered (respectively, prefibered) category over $\mathcal{E}$. Then $\mathcal{F}'=\mathcal{F}\times_{\mathcal{E}}\mathcal{E}'$ is a fibered (respectively, prefibered) category over $\mathcal{E}'$.*

Proposition VI.6.10. *Let $\mathcal{F},\mathcal{G}$ be prefibered categories over $\mathcal{E}$, and let F be a Cartesian $\mathcal{E}$-functor from $\mathcal{F}$ to $\mathcal{G}$. Then F is faithful (respectively fully faithful, respectively* 169
an $\mathcal{E}$-equivalence) if and only if, for every $S\in\mathrm{Ob}(\mathcal{E})$, the induced functor

$$F_S\colon \mathcal{F}_S \longrightarrow \mathcal{G}_S$$

is faithful (respectively fully faithful, respectively an equivalence).

The proof is immediate from the definitions.

To conclude this section, we give some properties of fibered categories that use axiom $\mathrm{Fib}_{\mathrm{II}}$.

Proposition VI.6.11. *Let $\mathcal{F}$ be a prefibered category over $\mathcal{E}$. Then $\mathcal{F}$ is fibered if and only if it satisfies the following condition:*

$\mathrm{Fib}'_{\mathrm{II}}$*: Let $\alpha\colon\eta\to\xi$ be a Cartesian morphism in $\mathcal{F}$ lying over a morphism $f\colon T\to S$ of $\mathcal{E}$. For every morphism $g\colon U\to T$ in $\mathcal{E}$, and every $\zeta\in\mathrm{Ob}(\mathcal{F}_U)$, the map $u\longmapsto\alpha\circ u$,*

$$\mathrm{Hom}_g(\zeta,\eta) \longrightarrow \mathrm{Hom}_{fg}(\zeta,\xi),$$

is bijective.

In other words, in a category *fibered* over $\mathcal{E}$, Cartesian diagrams are characterized by a property that is a priori stronger than the one in the definition (obtained by taking $g = \mathrm{id}_T$ in the preceding statement).

Corollary VI.6.12. *Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let α be a morphism in $\mathcal{F}$. If α is an isomorphism, then $p(\alpha) = f$ is an isomorphism and α is Cartesian. The converse holds if $\mathcal{F}$ is fibered over $\mathcal{E}$.*

Indeed, if α is an isomorphism, then $f = p(\alpha)$ plainly is as well. For every $\eta' \in \mathrm{Ob}(\mathcal{F}_T)$, the map $u \longmapsto \alpha \circ u$,

$$\mathrm{Hom}(\eta', \eta) \longrightarrow \mathrm{Hom}(\eta', \xi),$$

is bijective. Since f is an isomorphism, an element on the left is a T-morphism if and only if its image on the right is an f-morphism. We therefore obtain a bijection

$$\mathrm{Hom}_T(\eta', \eta) \longrightarrow \mathrm{Hom}_f(\eta', \xi),$$

which proves the first assertion. 170

Conversely, suppose that f is an isomorphism and that α satisfies condition $\mathrm{Fib}'_{\mathrm{II}}$, which, when $\mathcal{F}$ is fibered over $\mathcal{E}$, means that α is Cartesian. Then one sees at once that, for every $\zeta \in \mathrm{Ob}(\mathcal{F})$, the map

$$\mathrm{Hom}(\zeta, \eta) \longrightarrow \mathrm{Hom}(\zeta, \xi), \qquad u \longmapsto \alpha \circ u,$$

is bijective. Thus α is an isomorphism.

Corollary VI.6.13. *Let $\alpha\colon \eta \to \xi$ and $\beta\colon \zeta \to \eta$ be two composable morphisms in the category $\mathcal{F}$ fibered over $\mathcal{E}$. If α is Cartesian, then β is Cartesian if and only if $\alpha\beta$ is Cartesian.*

Use the strengthened form of the definition of Cartesian morphisms given in VI.6.11.

VI.7. Cloven categories over $\mathcal{E}$

Definition VI.7.1. *Let $\mathcal{F}$ be a category over $\mathcal{E}$. A* cleavage *of $\mathcal{F}$ over $\mathcal{E}$ is a function that assigns to every $f \in \mathrm{Fl}(\mathcal{E})$ an inverse-image functor f^* for f in $\mathcal{F}$. The cleavage is called* normalized *if $f = \mathrm{id}_S$ implies $f^* = \mathrm{id}_{\mathcal{F}_S}$. A* cloven category *(respectively a* normalized cloven category*) is a category $\mathcal{F}$ over $\mathcal{E}$ equipped with a cleavage (respectively a normalized cleavage).*

It is clear that $\mathcal{F}$ admits a cleavage if and only if it is prefibered over $\mathcal{E}$, and in that case it admits a normalized cleavage. The set of cleavages on $\mathcal{F}$ is in bijection with the set of subsets $K \subset \mathrm{Fl}(\mathcal{F})$ satisfying the following conditions:

(a) Every $\alpha \in K$ is cartesian.
(b) For every morphism $f\colon T \to S$ in $\mathcal{E}$ and every $\xi \in \mathrm{Ob}(\mathcal{F}_S)$, there is a unique f-morphism in K with target ξ.

The cleavage defined by K is normalized if and only if K also satisfies

(c) The identity morphisms of $\mathcal{F}$ belong to K.

The elements of K will be called the *transport morphisms* for the cleavage in question.

The notion of an isomorphism of cloven categories over $\mathcal{E}$ is clear. More generally, one may define morphisms of cloven $\mathcal{E}$-categories to be the $\mathcal{E}$-functors $\mathcal{F} \to \mathcal{G}$ that send transport morphisms to transport morphisms; in particular, these are cartesian functors. In this way the cloven categories over $\mathcal{E}$ are the objects of a category, the *category of cloven categories over $\mathcal{E}$*. The reader may spell out the existence of products: if a category over $\mathcal{E}$ is the product of categories $\mathcal{F}_i$ over $\mathcal{E}$, each supplied with a cleavage, then their product has the corresponding natural cleavage. We likewise leave the notion of base change for cloven categories to the reader.

We denote by $\alpha_f(\xi)$ the canonical morphism

$$\alpha_f(\xi)\colon f^*(\xi) \longrightarrow \xi.$$

As already noted, it is functorial in ξ; that is, there is a natural transformation

$$\alpha_f\colon i_T f^* \longrightarrow i_S,$$

where, for every $S \in \mathrm{Ob}(\mathcal{E})$, $i_S\colon \mathcal{F}_S \to \mathcal{F}$ denotes the inclusion functor.

Now consider morphisms

$$f\colon T \longrightarrow S, \qquad g\colon U \longrightarrow T$$

in $\mathcal{E}$, and let $\xi \in \mathrm{Ob}(\mathcal{F}_S)$. There is a unique U-morphism

$$c_{f,g}(\xi)\colon g^* f^*(\xi) \longrightarrow (fg)^*(\xi)$$

making the diagram

$$\begin{tikzcd}
f^*(\xi) \arrow[d, "\alpha_f(\xi)"'] & g^* f^*(\xi) \arrow[l, "\alpha_g(f^*(\xi))"'] \arrow[d, "c_{f,g}(\xi)"] \\
\xi & (fg)^*(\xi) \arrow[l, "\alpha_{fg}(\xi)"]
\end{tikzcd}$$

commutative, by the definition of $(fg)^*(\xi)$. As ξ varies, this morphism is functorial; in other words, there is a natural transformation

$$c_{f,g}\colon g^* f^* \longrightarrow (fg)^*$$

between functors $\mathcal{F}_S \to \mathcal{F}_U$.

Proposition VI.7.2. *A cloven category $\mathcal{F}$ over $\mathcal{E}$ is fibered if and only if all the transformations $c_{f,g}$ are isomorphisms.*

Taking f to be an isomorphism and g its inverse, and considering $c_{f,g}$ and $c_{g,f}$, gives the following.

Corollary VI.7.3. *If $\mathcal{F}$ is a cloven category fibered over $\mathcal{E}$, then, for every isomorphism $f\colon T \to S$ in $\mathcal{E}$, the functor*

$$f^*\colon \mathcal{F}_S \longrightarrow \mathcal{F}_T$$

is an equivalence of categories.

Proposition VI.7.4. *Let $\mathcal{F}$ be a cloven category over $\mathcal{E}$. Then*

$$\text{A)} \qquad \begin{cases} c_{f,\mathrm{id}_T}(\xi) = \alpha_{\mathrm{id}_T}(f^*(\xi)), \\ c_{\mathrm{id}_S,f}(\xi) = f^*(\alpha_{\mathrm{id}_S}(\xi)), \end{cases}$$

and

$$\text{B)} \qquad c_{f,gh}(\xi) \circ c_{g,h}(f^*(\xi)) = c_{fg,h}(\xi) \circ h^*(c_{f,g}(\xi)).$$

Here f, g, h are successive morphisms

$$V \longrightarrow U \longrightarrow T \longrightarrow S$$

and ξ is an object of $\mathcal{F}_S$.

For a normalized cleavage, the first two relations take the simpler form

$$\text{(A')} \qquad c_{f,\mathrm{id}_T} = \mathrm{id}_{f^*}, \qquad c_{\mathrm{id}_S,f} = \mathrm{id}_{f^*}\,.$$

The third relation is visualized by the commutative diagram

(D)
$$\begin{tikzcd} h^*g^*f^*(\xi) \arrow[r, "c_{g,h}(f^*(\xi))"] \arrow[d, "h^*(c_{f,g}(\xi))"'] & (gh)^*f^*(\xi) \arrow[d, "c_{f,gh}(\xi)"] \\ h^*(fg)^*(\xi) \arrow[r, "c_{fg,h}(\xi)"'] & (fgh)^*(\xi). \end{tikzcd}$$

For fibered categories, where the $c_{f,g}$ are isomorphisms, this commutativity says intuitively that successive use of isomorphisms of the form $c_{f,g}$ does not lead to "contradictory identifications." The same formula, written without the argument ξ and using the convolution product of natural transformations, is

$$c_{fg,h} \circ (h^* * c_{f,g}) = c_{f,gh} \circ (c_{g,h} * f^*).$$

The first two formulas in VI.7.4 are immediate; we sketch the proof of the third. In addition to square (D), consider the square of natural transformations

(D')
$$\begin{tikzcd} g^*f^*(\xi) \arrow[r, "\alpha_g(f^*(\xi))"] \arrow[d, "c_{f,g}(\xi)"'] & f^*(\xi) \arrow[d, "\alpha_f(\xi)"] \\ (fg)^*(\xi) \arrow[r, "\alpha_{fg}(\xi)"'] & \xi, \end{tikzcd}$$

which is commutative by the definition of $c_{f,g}(\xi)$. Join the vertices of (D) to the corresponding vertices of (D') by the natural transformations

$$\begin{array}{ll} \alpha_h(g^*f^*(\xi)), & \alpha_{gh}(f^*(\xi)), \\ \alpha_h((fg)^*(\xi)), & \alpha_{fgh}(\xi). \end{array}$$

The four lateral faces of the resulting cube are also commutative. For the left face this follows because the left column of (D) is obtained from the left column of (D') by applying h^*, and because α_h is natural. For the other three faces it is simply the definition of the c-operations on the remaining three sides of (D). Thus the five faces other than the top face commute. It follows that the two (fgh)-morphisms

$$h^*g^*f^*(\xi) \longrightarrow (fgh)^*(\xi)$$

defined by (D) have the same composite with $\alpha_{fgh}(\xi)\colon (fgh)^*(\xi) \to \xi$; hence they are equal by the definition of $(fgh)^*$.

For what follows, restrict attention to *normalized* cloven categories. Such a category gives rise to the following data:

(a) A map $S \mapsto \mathcal{F}_S$ from $\mathrm{Ob}(\mathcal{E})$ to **Cat**.
(b) A map $f \mapsto f^*$ assigning to every $f \in \mathrm{Fl}(\mathcal{E})$, with source T and target S, a functor $f^*\colon \mathcal{F}_S \to \mathcal{F}_T$.
(c) A map $(f,g) \mapsto c_{f,g}$ assigning to every composable pair of arrows (f,g) of $\mathcal{E}$ a natural transformation $c_{f,g}\colon g^*f^* \to (fg)^*$.

These data satisfy conditions (A') and (B) above. If we had not restricted to normalized cleavages, one further datum would be needed: a map $S \mapsto \alpha_S$ assigning to every object S of $\mathcal{E}$ a natural transformation

$$\alpha_S\colon (\mathrm{id}_S)^* \longrightarrow \mathrm{id}_{\mathcal{F}_S};$$

condition (A') would then be replaced by condition (A).

We shall now show how the preceding data reconstruct, up to unique isomorphism, the normalized cloven category $\mathcal{F}$ over $\mathcal{E}$.

VI.8. Cloven category defined by a pseudofunctor $\mathcal{E}^\circ \to \mathbf{Cat}$

For brevity, call a *pseudofunctor* from $\mathcal{E}^\circ$ to **Cat**—more precisely, a normalized pseudofunctor—a set of data a), b), c) as above satisfying conditions $A')$ and $B)$. In the preceding section we associated a pseudofunctor $\mathcal{E}^\circ \to \mathbf{Cat}$ with a normalized cloven category over $\mathcal{E}$; we now describe the inverse construction. We leave to the reader the verification of most of the details, as well as the fact that these constructions are indeed inverse to one another. More precisely, one should regard the pseudofunctors $\mathcal{E}^\circ \to \mathbf{Cat}$ as the objects of a new category and show that our constructions give mutually quasi-inverse equivalences between this category and the category of cloven categories over $\mathcal{E}$ defined in the preceding section.

Set

$$\mathcal{F}_\circ = \coprod_{S \in \mathrm{Ob}(\mathcal{E})} \mathrm{Ob}(\mathcal{F}(S)),$$

the disjoint union of the sets $\mathrm{Ob}(\mathcal{F}(S))$. (N.B. Here we write $\mathcal{F}(S)$, rather than $\mathcal{F}_S$, for the value of the given pseudofunctor at the object S of $\mathcal{E}$, in order to avoid notational confusion below.) There is an evident map

$$p_\circ \colon \mathcal{F}_\circ \longrightarrow \mathrm{Ob}(\mathcal{E}).$$

Let

$$\overline{\xi} = (S, \xi), \qquad \overline{\eta} = (T, \eta) \qquad \text{with } \xi \in \mathrm{Ob}(\mathcal{F}(S)),\ \eta \in \mathrm{Ob}(\mathcal{F}(T))$$

be two elements of $\mathcal{F}_\circ$, and let $f \in \mathrm{Hom}(T, S)$. Define

$$h_f(\overline{\eta}, \overline{\xi}) = \mathrm{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)).$$

176

If, in addition, $g \colon U \to T$ is a morphism in $\mathcal{E}$ and $\zeta \in \mathrm{Ob}(\mathcal{F}(U))$, define a map, denoted $(u, v) \longmapsto u \circ v$,

$$h_f(\overline{\eta}, \overline{\xi}) \times h_g(\overline{\zeta}, \overline{\eta}) \longrightarrow h_{fg}(\overline{\zeta}, \overline{\xi}),$$

i.e. a map

$$\mathrm{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)) \times \mathrm{Hom}_{\mathcal{F}(U)}(\zeta, g^*(\eta)) \longrightarrow \mathrm{Hom}_{\mathcal{F}(U)}(\zeta, (fg)^*(\xi)),$$

by the formula

$$u \circ v = c_{f,g}(\xi) \cdot g^*(u) \cdot v.$$

Thus $u \circ v$ is the composite

$$\zeta \xrightarrow{v} g^*(\eta) \xrightarrow{g^*(u)} g^* f^*(\xi) \xrightarrow{c_{f,g}(\xi)} (fg)^*(\xi).$$

Also set

$$h(\overline{\eta}, \overline{\xi}) = \coprod_{f \in \mathrm{Hom}(T,S)} h_f(\overline{\eta}, \overline{\xi}).$$

The preceding pairings then define pairings

$$h(\overline{\eta}, \overline{\xi}) \times h(\overline{\zeta}, \overline{\eta}) \longrightarrow h(\overline{\zeta}, \overline{\xi}),$$

while the definition of $h(\overline{\eta}, \overline{\xi})$ gives an evident map

$$p_{\overline{\eta},\overline{\xi}} \colon h(\overline{\eta}, \overline{\xi}) \longrightarrow \mathrm{Hom}(T, S).$$

One now verifies the following points:

1) Composition between elements of $h(\overline{\eta}, \overline{\xi})$ is *associative.* 177
2) For every $\overline{\xi} = (S, \xi)$ in $\mathcal{F}_\circ$, consider the identity element of

$$h_{\mathrm{id}_S}(\overline{\xi}, \overline{\xi}) = \mathrm{Hom}_{\mathcal{F}(S)}(\mathrm{id}_S^*(\xi), \xi) = \mathrm{Hom}_{\mathcal{F}(S)}(\xi, \xi),$$

and its image in $h(\overline{\xi}, \overline{\xi})$. This element is a left and right *identity* for composition between elements of the $h(\overline{\eta}, \overline{\xi})$.

This already shows that one obtains a category $\mathcal{F}$ by setting

$$\mathrm{Ob}(\mathcal{F}) = \mathcal{F}_\circ, \qquad \mathrm{Fl}(\mathcal{F}) = \coprod_{\overline{\xi},\overline{\eta}\in\mathcal{F}_\circ} h(\overline{\eta},\overline{\xi}).$$

(N.B. One cannot simply take $\mathrm{Fl}(\mathcal{F})$ to be the *union* of the sets $h(\overline{\eta},\overline{\xi})$, since these sets need not be disjoint.) Moreover:

3) The maps

$$p_\circ\colon \mathrm{Ob}(\mathcal{F}) \to \mathrm{Ob}(\mathcal{E}), \qquad p_1 = (p_{\overline{\eta},\overline{\xi}})\colon \mathrm{Fl}(\mathcal{F}) \to \mathrm{Fl}(\mathcal{E})$$

define a *functor* $p\colon \mathcal{F} \to \mathcal{E}$. Thus $\mathcal{F}$ becomes a category over $\mathcal{E}$, and the evident map

$$h_f(\overline{\eta},\overline{\xi}) \longrightarrow \mathrm{Hom}(\overline{\eta},\overline{\xi})$$

induces a *bijection*

$$h_f(\overline{\eta},\overline{\xi}) \xrightarrow{\sim} \mathrm{Hom}_f(\overline{\eta},\overline{\xi}).$$

4) The evident maps

$$\mathrm{Ob}(\mathcal{F}(S)) \longrightarrow \mathcal{F}_\circ = \mathrm{Ob}(\mathcal{F}), \qquad \mathrm{Fl}(\mathcal{F}(S)) \longrightarrow \mathrm{Fl}(\mathcal{F}),$$

where the second is defined by the evident maps

$$\mathrm{Hom}_{\mathcal{F}(S)}(\xi,\xi') = h_{\mathrm{id}_S}(\overline{\xi},\overline{\xi'}) \longrightarrow \mathrm{Hom}(\overline{\xi},\overline{\xi'}),$$

define an *isomorphism*

$$i_S\colon \mathcal{F}(S) \xrightarrow{\sim} \mathcal{F}_S.$$

5) For every object $\overline{\xi} = (S,\xi)$ of $\mathcal{F}$, and every morphism $f\colon T \to S$ of $\mathcal{E}$, consider the 178
element $\overline{\eta} = (T,\eta)$ of $\mathcal{F}_T$, where $\eta = f^*(\xi)$, and the element $\alpha_f(\xi)$ of $\mathrm{Hom}(\overline{\eta},\overline{\xi})$ that is the image of $\mathrm{id}_{f^*(\xi)}$ under the map

$$\mathrm{Hom}_{\mathcal{F}(T)}(f^*(\xi), f^*(\xi)) = h_f(\overline{\eta},\overline{\xi}) \longrightarrow \mathrm{Hom}_f(\overline{\eta},\overline{\xi}).$$

This element is Cartesian, and is the identity of $\overline{\xi}$ when $f = \mathrm{id}_S$. In other words, the collection of the $\alpha_f(\xi)$ defines a *normalized cleavage of $\mathcal{F}$ over $\mathcal{E}$.* Moreover, by construction, the diagram of functors

$$\begin{tikzcd} \mathcal{F}(S) \arrow[r, "f^*"] \arrow[d, "i_S"'] & \mathcal{F}(T) \arrow[d, "i_T"] \\ \mathcal{F}_S \arrow[r, "f^*_{\mathcal{F}}"] & \mathcal{F}_T \end{tikzcd}$$

is commutative, where $f^*_{\mathcal{F}}$ is the inverse-image functor under f relative to the cleavage just defined on $\mathcal{F}$. Finally:

6) Under the isomorphisms i_S, the homomorphisms $c_{f,g}$ supplied with the pseudofunctor are carried to the functorial homomorphisms $c_{f,g}$ associated with the cleavage of $\mathcal{F}$.

We give only the verification of 1), which is, if anything, less immediate than the others. It suffices to prove associativity of composition between elements of sets of the form $h_f(\overline{\eta},\overline{\xi})$. Consider morphisms in $\mathcal{E}$

$$S \xleftarrow{f} T \xleftarrow{g} U \xleftarrow{h} V,$$

objects

$$\xi,\ \eta,\ \zeta,\ \tau$$

in $\mathcal{F}(S), \mathcal{F}(T), \mathcal{F}(U), \mathcal{F}(V)$, respectively, and elements

$$\begin{array}{rcl} u \in h_f(\overline{\eta},\overline{\xi}) & = & \mathrm{Hom}_{\mathcal{F}(T)}(\eta, f^*(\xi)), \\ v \in h_g(\overline{\zeta},\overline{\eta}) & = & \mathrm{Hom}_{\mathcal{F}(U)}(\zeta, g^*(\eta)), \\ w \in h_h(\overline{\tau},\overline{\zeta}) & = & \mathrm{Hom}_{\mathcal{F}(V)}(\tau, h^*(\zeta)). \end{array}$$

We must prove 179

$$(u \circ v) \circ w = u \circ (v \circ w),$$

an equality in $\mathrm{Hom}_{\mathcal{F}(V)}(\tau, (fgh)^*(\xi))$. By the definitions, the two sides are obtained by composing along the upper and lower boundaries, respectively, of the following diagram:

$$\begin{tikzcd}
\tau \arrow[r, "w"] \arrow[drr, "v\circ w"'] & h^*(\zeta) \arrow[r, "h^*(v)"] \arrow[rrrr, bend left, "h^*(u\circ v)"] & h^*g^*(\eta) \arrow[rr, "h^*g^*(u)"] \arrow[d, "c_{g,h}(\eta)"'] & & h^*g^*f^*(\xi) \arrow[r, "h^*(c_{f,g}(\xi))"] \arrow[d, "c_{g,h}(f^*(\xi))"] & h^*(fg)^*(\xi) \arrow[d, "c_{fg,h}(\xi)"] \\
& & (gh)^*(\eta) \arrow[rr, "(gh)^*(u)"] & & (gh)^*f^*(\xi) \arrow[r, "c_{f,gh}(\xi)"] & (fgh)^*(\xi).
\end{tikzcd}$$

The middle square is commutative because $c_{g,h}$ is a functorial homomorphism, and the square on the right is commutative by condition B) for a pseudofunctor. This proves the assertion.

Of course, when the pseudofunctor in question already comes from a normalized cloven category $\mathcal{F}'$ over $\mathcal{E}$, one must still specify the natural isomorphism between $\mathcal{F}'$ and $\mathcal{F}$. We leave the details to the reader.

We also leave to the reader the interpretation, in terms of pseudofunctors, of the inverse image of a cloven category $\mathcal{F}$ over $\mathcal{E}$ under a base-change functor $\mathcal{E}' \to \mathcal{E}$.

VI.9. Example: a cloven category defined by a functor $\mathcal{E}^\circ \to$ Cat; split categories over $\mathcal{E}$

Suppose that a functor

$$\varphi\colon \mathcal{E}^\circ \to \mathbf{Cat}$$

is given. It defines a pseudofunctor by setting

$$\mathcal{F}(S) = \varphi(S), \qquad f^* = \varphi(f), \qquad c_{f,g} = \mathrm{id}_{(fg)^*}.$$

180

The construction of the preceding section therefore gives a cloven category $\mathcal{F}$ over $\mathcal{E}$, called the category associated with the functor φ. A cloven category over $\mathcal{E}$ is isomorphic to a cloven category defined by a functor $\varphi\colon \mathcal{E}^\circ \to \mathbf{Cat}$ if and only if it satisfies

$$(fg)^* = g^*f^*, \qquad c_{f,g} = \mathrm{id}_{(fg)^*}.$$

In terms of the set K of transport morphisms, this simply says that *the composite of two transport morphisms is a transport morphism.* A cleavage of a category $\mathcal{F}$ over $\mathcal{E}$ satisfying this condition is called a *splitting* of $\mathcal{F}$ over $\mathcal{E}$, and a category $\mathcal{F}$ over $\mathcal{E}$ equipped with a splitting is called a *split category over* $\mathcal{E}$. This is a special case of the notion of a cloven category. The category of split categories over $\mathcal{E}$ is therefore equivalent to $\mathbf{Hom}(\mathcal{E}^\circ, \mathbf{Cat})$. Note that a split category over $\mathcal{E}$ is, in particular, a cloven category over $\mathcal{E}$.

If $\mathcal{F}$ is a fibered category over $\mathcal{E}$, a splitting of $\mathcal{F}$ need not exist. Suppose, for example, that $\mathrm{Ob}(\mathcal{E})$ and $\mathrm{Ob}(\mathcal{F})$ each consist of one element and that the endomorphisms of those elements form groups E and F, respectively. The projection functor p is then given by a group homomorphism $p\colon F \to E$, which is surjective because p is fibered. One immediately verifies that the set of cleavages of $\mathcal{F}$ over $\mathcal{E}$ is in bijection with the set of maps

$$s\colon E \longrightarrow F \qquad \text{such that} \qquad ps = \mathrm{id}_E;$$

these are the systems of representatives for the cosets modulo the subgroup $G = \operatorname{Ker} p$ of the surjective homomorphism $p\colon F \to E$. A cleavage is a splitting if and only if s is a group homomorphism. Thus a splitting exists precisely when the extension F of E by G is split; when G is commutative, this is expressed by the vanishing of a certain cohomology class in $H^2(E, G)$, where G is regarded as a group on which E acts.

Suppose, however, that $\mathcal{F}$ is a fibered category over $\mathcal{E}$ whose fiber categories $\mathcal{F}_S$ are 181
rigid, i.e. every object of $\mathcal{F}_S$ has trivial automorphism group. It is then easy to prove

that $\mathcal{F}$ admits a splitting over $\mathcal{E}$. First, the existence of a splitting is unchanged if $\mathcal{F}$ is replaced by an $\mathcal{E}$-equivalent category. We may therefore reduce to the case in which the $\mathcal{F}_S$ are both rigid and *reduced*, meaning that two isomorphic objects in $\mathcal{F}_S$ are equal. If G is a rigid reduced category, every isomorphism between two functors $H \to G$, with H arbitrary, is an identity. Consequently, if $\mathcal{F}$ is fibered over $\mathcal{E}$ and its fiber categories are rigid and reduced, then $\mathcal{F}$ has a *unique* cleavage, which is necessarily a splitting. Thus $\mathcal{F}$ is isomorphic to the category defined by a functor

$$\varphi\colon \mathcal{E}^\circ \longrightarrow \mathbf{Cat}$$

whose $\varphi(S)$ are rigid discrete categories, and the functor φ is determined up to isomorphism.

VI.10. Cofibered categories and bifibered categories

Consider a category $\mathcal{F}$ over $\mathcal{E}$, with projection functor

$$p\colon \mathcal{F} \longrightarrow \mathcal{E}.$$

It defines a category $\mathcal{F}^\circ$ over $\mathcal{E}^\circ$, with projection functor

$$p^\circ\colon \mathcal{F}^\circ \longrightarrow \mathcal{E}^\circ.$$

A morphism $\alpha\colon \eta \to \xi$ in $\mathcal{F}$ is called *cocartesian* if it is Cartesian for $\mathcal{F}^\circ$ over $\mathcal{E}^\circ$. Explicitly, this means that, for every object ξ' of $\mathcal{F}_S$, the map $u \longmapsto u \circ \alpha$,

$$\mathrm{Hom}_S(\xi, \xi') \longrightarrow \mathrm{Hom}_f(\eta, \xi'),$$

is bijective. The pair (ξ, α) is then also called a *direct image* of η under f, in the category $\mathcal{F}$ over $\mathcal{E}$. If such a direct image exists for every η in $\mathcal{F}_T$, the direct-image functor under f is said to exist; once chosen, it is denoted by $f_*^{\mathcal{F}}$, or simply f_*. It is characterized by an 182
isomorphism of bifunctors on $\mathcal{F}_T^\circ \times \mathcal{F}_S$:

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_f(\eta, \xi).$$

If f_* exists, then f^* exists if and only if f_* admits a right adjoint, i.e. if and only if there exist a functor

$$f^*\colon \mathcal{F}_S \longrightarrow \mathcal{F}_T$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_T(\eta, f^*(\xi)).$$

Let $g\colon U \to T$ be another morphism in $\mathcal{E}$, and suppose that the inverse and direct images under f, g, and fg exist. Consider the functorial homomorphisms

$$\begin{aligned} c^{f,g}\colon f_*g_* &\longleftarrow (fg)_*, \\ c_{f,g}\colon g^*f^* &\longrightarrow (fg)^*. \end{aligned}$$

If f_*g_* and g^*f^* are regarded as a pair of adjoint functors, as are $(fg)_*$ and $(fg)^*$, then these two homomorphisms are adjoint to one another. Hence one is an isomorphism if and only if the other is. In particular:

Proposition VI.10.1. *Suppose that the category $\mathcal{F}$ over $\mathcal{E}$ is both prefibered and coprefibered. Then it is fibered if and only if it is cofibered.*

Of course, $\mathcal{F}$ is called coprefibered, respectively cofibered, over $\mathcal{E}$ if $\mathcal{F}^\circ$ is prefibered, respectively fibered, over $\mathcal{E}^\circ$. We call $\mathcal{F}$ *bifibered over $\mathcal{E}$* if it is both fibered and cofibered over $\mathcal{E}$.

VI.11. Miscellaneous examples

a) **Arrow categories of $\mathcal{E}$.** Let $\mathcal{E}$ be a category. Denote by $\mathbf{\Delta^1}$ the category associated with the totally ordered two-element set $[0,1]$; it thus has two objects, 183
0 and 1, and, in addition to the two identity morphisms, an arrow $(0,1)$ with source 0 and target 1. Let

$$\mathbf{Fl}(\mathcal{E}) = \mathbf{Hom}(\mathbf{\Delta^1}, \mathcal{E})$$

which is called the *arrow category of* $\mathcal{E}$. The object 1 of $\mathbf{\Delta^1}$ defines a canonical functor, called the *target functor*,

$$\mathbf{Fl}(\mathcal{E}) \to \mathcal{E}$$

(the functor defined by the object 0 of $\mathbf{\Delta^1}$ is called the *source functor*). For every object S of $\mathcal{E}$, the fiber category $\mathbf{Fl}(\mathcal{E})_S$ is canonically isomorphic to the category $\mathcal{E}_{/S}$ of objects of $\mathcal{E}$ over S.

Consider a morphism $f\colon T \to S$ in $\mathcal{E}$. It corresponds to a canonical functor

$$f_*\colon \mathcal{E}_{/T} = \mathcal{F}_T \to \mathcal{E}_{/S} = \mathcal{F}_S$$

and a functorial isomorphism

$$\mathrm{Hom}_S(f_*(\eta), \xi) \xrightarrow{\sim} \mathrm{Hom}_f(\eta, \xi)$$

which thus makes f_* a direct-image functor for f in $\mathcal{F}$. Moreover, here one has

$$(\mathrm{id}_S)_* = \mathrm{id}_{\mathcal{F}_S}, \quad (fg)_* = f_* g_*, \quad c^{f,g} = \mathrm{id}_{(fg)},$$

i.e. $\mathcal{F}$ is equipped with a cosplitting over $\mathcal{E}$. A fortiori, $\mathcal{F}$ is cofibered over $\mathcal{E}$. Note now that the set of morphisms in $\mathcal{F}$ is in bijection with the set of commutative square diagrams in $\mathcal{E}$:

$$\begin{tikzcd} X \arrow[d, "u"'] & Y \arrow[l, "f'"'] \arrow[d, "v"] \\ S & T \arrow[l, "f"] \end{tikzcd}$$

By definition, the morphism in question is Cartesian if the square is Cartesian 184
in $\mathcal{E}$, i.e. if it makes Y a fiber product of X and T over S. The inverse-image functor f^* therefore exists if and only if, for every object X over S, the fiber product $X \times_S T$ exists. It follows from VI.10.1 that if the product of two objects over a third always exists in $\mathcal{E}$, i.e. if $\mathcal{F}$ is prefibered over $\mathcal{E}$, then $\mathcal{F}$ is in fact fibered over $\mathcal{E}$.

b) **Category of presheaves or sheaves on varying spaces.**

Let $\mathcal{E} = \mathbf{Top}$ be the category of topological spaces. If T is a topological space, we denote by $\mathcal{U}(T)$ the category of open subsets of T, whose morphisms are the inclusion maps. If $\mathcal{C}$ is a category, a functor $\mathcal{U}(T)^\circ \to \mathcal{C}$ is called a *presheaf* on T with values in $\mathcal{C}$, and a *sheaf* if it satisfies a left-exactness condition that we shall not repeat here. The *category $\mathcal{P}(T)$ of presheaves on T with values in $\mathcal{C}$* is, by definition, the category $\mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C})$, and the category $\mathcal{F}(T)$ of sheaves on T with values in $\mathcal{C}$ is the full subcategory whose objects are those objects of $\mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C})$ that are sheaves. If $f\colon T \to S$ is a morphism in $\mathcal{E}$, i.e. a continuous map of topological spaces, the order-preserving map $U \longmapsto f^{-1}(U)$ gives a functor $\mathcal{U}(S) \to \mathcal{U}(T)$, hence a functor

$$f_*\colon \mathbf{Hom}(\mathcal{U}(T)^\circ, \mathcal{C}) \to \mathbf{Hom}(\mathcal{U}(S)^\circ, \mathcal{C}),$$

called the *direct-image functor for presheaves under* f. It is immediate that the direct image of a sheaf is a sheaf. Thus the functor

$$f_*\colon \mathcal{P}(T) \to \mathcal{P}(S)$$

induces a functor, also denoted

$$f_*\colon \mathcal{F}(T) \to \mathcal{F}(S).$$

Moreover, one checks trivially, by associativity of the composition of functors, that for a second continuous map $g\colon U \to T$ one has

$$(gf)_* = g_* f_*, \qquad \text{and likewise} \qquad (\mathrm{id}_S)_* = \mathrm{id}_{\mathcal{P}(S)}.$$

In this way we have obtained a functor

$$S \longmapsto \mathcal{P}(S),$$

respectively

$$S \longmapsto \mathcal{F}(S),$$

from $\mathcal{E}$ to **Cat**. In fact, we are interested in the corresponding functor 185

$$S \longmapsto \mathcal{P}(S)^\circ, \qquad \text{respectively} \qquad S \longmapsto \mathcal{F}(S)^\circ.$$

It defines a cofibered, and indeed cosplit, category over the category of topological spaces, called the *cofibered category of presheaves* (respectively*sheaves*) *with values in* $\mathcal{C}$ (understood: on varying spaces). Spelling out the construction of no. VI.8, one sees that a morphism from a presheaf B on T to a presheaf A on S is a pair (f, u) consisting of a continuous map from T to S and a morphism $u\colon A \to f_*(B)$ in the category $\mathcal{P}(S)$. This description is equally valid for morphisms of sheaves, since $\mathcal{F}$ is a full subcategory of $\mathcal{P}$.

In the most important cases, the categories $\mathcal{P}$ and $\mathcal{F}$ over $\mathcal{E}$ are also fibered categories; that is, for every continuous map, the direct-image functors $\mathcal{P}(T) \to \mathcal{P}(S)$ and $\mathcal{F}(T) \to \mathcal{F}(S)$ have an adjoint functor, then denoted f^* and called the inverse-image functor for presheaves, respectivelythe inverse-image functor for sheaves, under the continuous map f. This functor exists, for example, if $\mathcal{C} = \mathbf{Ens}$. One can show that $f^*\colon \mathcal{P}(S) \to \mathcal{P}(T)$ exists whenever inductive limits (relative to diagrams in the Universe under consideration) exist in $\mathcal{C}$. The question is less easy for $\mathcal{F}$. Indeed, even when $\mathcal{C} = \mathbf{Ens}$, the inverse image of a presheaf that is a sheaf is not in general a sheaf; in other words, the inverse-image functor for sheaves is not isomorphic to the functor induced by the inverse-image functor for presheaves (despite the common notation f^*). Thus $\mathcal{F}$ is a cofibered subcategory of $\mathcal{P}$, but not a fibered subcategory; that is, *the inclusion functor* $\mathcal{F} \to \mathcal{P}$ *is not fibrant.*

The cofibered category $\mathcal{P}$ can be deduced from a more general cofibered category—or rather, fibered category—constructed as follows. For every category $\mathcal{U}$ (in the fixed Universe), put

$$\mathcal{P}(\mathcal{U}) = \mathbf{Hom}(\mathcal{U}, \mathcal{C}),$$

and note that $\mathcal{U} \longmapsto \mathcal{P}(\mathcal{U})$ is naturally a contravariant functor in $\mathcal{U}$, from **Cat** 186
to **Cat**. It therefore defines a split category over $\mathcal{E} = \mathbf{Cat}$, which we denote $\mathbf{Cat}_{//\mathcal{C}}$. The objects of this category are pairs $(\mathcal{U}, p)$ consisting of a category $\mathcal{U}$ and a functor $p\colon \mathcal{U} \to \mathcal{C}$, and a morphism from $(\mathcal{U}, p)$ to $(\mathcal{V}, q)$ is essentially a pair (f, u), where f is a functor $\mathcal{U} \to \mathcal{V}$ and u is a homomorphism of functors $u\colon p \to qf$. We leave it to the reader to spell out the composition of morphisms in $\mathbf{Cat}_{//\mathcal{C}}$. The projection functor

$$\mathcal{F} = \mathbf{Cat}_{//\mathcal{C}} \to \mathcal{E} = \mathbf{Cat}$$

assigns to the pair $(\mathcal{U}, p)$ the object $\mathcal{U}$; the fiber category at $\mathcal{U}$ is the category $\mathbf{Hom}(\mathcal{U}, \mathcal{C})$, up to isomorphism. When inductive limits exist in $\mathcal{C}$, one easily shows that the fibered category $\mathbf{Cat}_{//\mathcal{C}}$ over **Cat** is also cofibered over **Cat**; that is, one can define the notion of the *direct image of a functor* $p\colon \mathcal{U} \to \mathcal{C}$ under a functor $f\colon \mathcal{U} \to \mathcal{V}$. The category of presheaves is obtained from the preceding fibered category by the base change

$$\mathbf{Top}^\circ \to \mathbf{Cat}$$

(the functor $S \longmapsto \mathcal{U}(S)$ defined above). This gives a fibered category over $\mathbf{Top}^\circ$, and on passing to the opposite category one obtains the cofibered category $\mathcal{P}$ of presheaves over $\mathbf{Top}$. The notion of inverse image of a functor corresponds to that of direct image of a presheaf, and the notion of direct image of a functor to that of inverse image of a presheaf.

c) **Objects with operators over an object with operators.**
Let $\mathcal{F}$ be a category over $\mathcal{E}$, and let S be an object of $\mathcal{E}$ on which a group G acts, on the left to fix ideas. This object with operators can be interpreted as corresponding to a functor $\lambda\colon \mathcal{E}' \to \mathcal{E}$ from the category $\mathcal{E}'$ defined by G (having a single object and G as its group of endomorphisms) to the category $\mathcal{E}$. By base change it therefore defines a category $\mathcal{F}'$ over $\mathcal{E}'$, which is fibered, respectivelycofibered, when $\mathcal{F}$ is so over $\mathcal{E}$. A section of $\mathcal{E}'$ over $\mathcal{F}'$ (necessarily 187
Cartesian, since $\mathcal{E}'$ is a groupoid and every isomorphism in $\mathcal{F}'$ is Cartesian by VI.6.12) can also be interpreted as an $\mathcal{E}$-functor $\mathcal{E}' \to \mathcal{F}$ over λ, or as an object with operators ξ in $\mathcal{F}$ "over" the object with operators S.

d) **Pairs of quasi-inverse adjoint functors; autodualities.**
When the base category $\mathcal{E}$ is reduced to two objects a, b and, besides the identity arrows, two mutually inverse isomorphisms $f\colon a \to b$ and $g\colon b \to a$ (that is, $\mathcal{E}$ is a rigid connected groupoid with two objects), a normalized cloven category over $\mathcal{E}$ is essentially the same thing as a system consisting of two categories $\mathcal{F}_a$ and $\mathcal{F}_b$ and a *pair of adjoint functors*

$$G\colon \mathcal{F}_a \to \mathcal{F}_b, \qquad F\colon \mathcal{F}_b \to \mathcal{F}_a,$$

which are equivalences of categories (and hence quasi-inverse to one another). One takes for $\mathcal{F}_a$ and $\mathcal{F}_b$ the fiber categories of $\mathcal{F}$, for F and G the functors f^* and g^*, and the two isomorphisms

$$u\colon FG \xrightarrow{\sim} \mathrm{id}_{\mathcal{F}_a}, \qquad v\colon GF \xrightarrow{\sim} \mathrm{id}_{\mathcal{F}_b}$$

are $c_{g,f}$ and $c_{f,g}$. The two usual compatibility conditions between u and v are nothing other than condition VI.7.4 B) for the composites fgf and gfg. It is easy to show that these conditions suffice to imply that one indeed has a pseudofunctor $\mathcal{E}^\circ \to \mathbf{Cat}$.

An interesting case is that in which

$$\mathcal{F}_b = \mathcal{F}_a^\circ, \qquad G = F^\circ, \qquad v = u^\circ.$$

An *autoduality* in a category $\mathcal{C}$ means the data of a functor $D\colon \mathcal{C} \to \mathcal{C}^\circ$ and an isomorphism $u\colon DD^\circ \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}}$ such that u and the isomorphism $u^\circ\colon D^\circ D \xrightarrow{\sim} \mathrm{id}_{\mathcal{C}^\circ}$ make (D, D°) a pair of adjoint functors (necessarily quasi-inverse to one another). This condition is written

$$D(u(x)) = u(D(x)) qquad \text{for every } x \in \mathrm{Ob}(\mathcal{C}).$$

188

e) **Categories over a discrete category $\mathcal{E}$.** One says that $\mathcal{E}$ is a *discrete category* if every arrow in it is an identity arrow, so that $\mathcal{E}$ is determined up to unique isomorphism by the set $I = \mathrm{Ob}(\mathcal{E})$. The data of a category $\mathcal{F}$ over $\mathcal{E}$ is therefore equivalent (up to unique isomorphism) to the data of a family of categories $\mathcal{F}_i$ $(i \in I)$, the fiber categories. Every category $\mathcal{F}$ over $\mathcal{E}$ is fibered, every $\mathcal{E}$-functor $\mathcal{F} \to \mathcal{G}$ is Cartesian, and there is a canonical isomorphism

$$\mathbf{Hom}_{\mathcal{E}/-}(\mathcal{F}, \mathcal{G}) \xrightarrow{\sim} \prod_i \mathbf{Hom}(\mathcal{F}_i, \mathcal{G}_i).$$

In particular, one obtains

$$\mathbf{\Gamma}(\mathcal{F}/\mathcal{E}) = \varprojlim \mathcal{F}/\mathcal{E} \xrightarrow{\sim} \prod_i \mathcal{F}_i.$$

f) **Suppose that $\mathcal{E}$ has exactly two objects S and T,** and, besides the **identity morphisms, one morphism** $f\colon T \to S$. Then a category $\mathcal{F}$ over $\mathcal{E}$ is determined, up to unique $\mathcal{E}$-isomorphism, by two categories $\mathcal{F}_S$ and $\mathcal{F}_T$ and a bifunctor $H(\eta, \xi)$ on $\mathcal{F}_T^\circ \times \mathcal{F}_S$ with values in **Ens**. Indeed, if $\mathcal{F}$ is a category over $\mathcal{E}$, one associates with it the two fiber categories $\mathcal{F}_S$ and $\mathcal{F}_T$ and the bifunctor $H(\eta, \xi) = \mathrm{Hom}_f(\eta, \xi)$. We leave it to the reader to spell out the construction in the opposite direction. For the category in question to be fibered (or prefibered, which amounts to the same thing), it is necessary and sufficient that the functor H be representable with respect to the argument ξ. For it to be cofibered, it is necessary and sufficient that H be representable with respect to the argument η.

g) Let $\mathcal{F} = \mathcal{C} \times \mathcal{E}$, regarded as a category over $\mathcal{E}$ by means of pr_2. Then $\mathcal{F}$ is fibered and cofibered over $\mathcal{E}$, and is even endowed with a canonical splitting and cosplitting, corresponding to the constant functor on $\mathcal{E}$, respectivelyon $\mathcal{E}^\circ$, with values in **Cat** and value $\mathcal{C}$. One has

$$\mathbf{\Gamma}(\mathcal{F}/\mathcal{E}) \simeq \mathbf{Hom}(\mathcal{E}, \mathcal{C}),$$

and $\varprojlim \mathcal{F}/\mathcal{E}$ corresponds to the full subcategory formed by the functors $F\colon \mathcal{E} \to \mathcal{C}$ 189
that send arbitrary morphisms to isomorphisms.

VI.12. Functors on a cloven category

Let $\mathcal{F}$ be a normalized cloven category over $\mathcal{E}$. For every object S of $\mathcal{E}$, denote by

$$i_S\colon \mathcal{F}_S \longrightarrow \mathcal{F}$$

the inclusion functor. Thus, for every morphism $f\colon T \to S$ in $\mathcal{E}$, there is a functorial homomorphism

$$\alpha_f\colon i_T f^* \longrightarrow i_S,$$

where $f^*\colon \mathcal{F}_S \to \mathcal{F}_T$ is the base-change functor defined by the cleavage. Now let

$$F\colon \mathcal{F} \longrightarrow \mathcal{C}$$

be a functor from $\mathcal{F}$ to a category $\mathcal{C}$. For every $S \in \mathrm{Ob}(\mathcal{E})$, set

$$F_S = F \circ i_S\colon \mathcal{F}_S \longrightarrow \mathcal{C},$$

and, for every $f\colon T \to S$ in $\mathcal{E}$, set

$$\varphi_f = F * \alpha_f\colon F_T f^* \longrightarrow F_S.$$

Thus every functor $F\colon \mathcal{F} \to \mathcal{C}$ determines a family (F_S) of functors $\mathcal{F}_S \to \mathcal{C}$ and a family (φ_f) of functorial homomorphisms $F_T f^* \to F_S$. These families satisfy the following conditions:

$$\text{a)} \qquad \varphi_{\mathrm{id}_S} = \mathrm{id}_{F_S}.$$

For morphisms $f\colon T \to S$ and $g\colon U \to T$ in $\mathcal{E}$, the following square of functorial homomorphisms is commutative:

(b))
$$\begin{tikzcd} F_U g^* f^* \arrow[r, "F_U * c_{f,g}"] \arrow[d, "\varphi_g * f^*"'] & F_U(fg)^* \arrow[d, "\varphi_{fg}"] \\ F_T f^* \arrow[r, "\varphi_f"] & F_S. \end{tikzcd}$$

190

The first relation is immediate, and the second follows by applying the functor F to the commutative diagram

$$\begin{tikzcd} g^*f^*(\xi) \arrow[r, "c_{f,g}(\xi)"] \arrow[d, "\alpha_g(f^*(\xi))"'] & (fg)^*(\xi) \arrow[d, "\alpha_{fg}(\xi)"] \\ f^*(\xi) \arrow[r, "\alpha_f(\xi)"] & \xi \end{tikzcd}$$

for a varying object ξ in $\mathcal{F}_S$.

If $G\colon \mathcal{F} \to \mathcal{C}$ is a second functor, giving rise to functors $G_S\colon \mathcal{F}_S \to \mathcal{C}$ and functorial homomorphisms $\psi_f\colon G_T f^* \to G_S$, and if $u\colon F \to G$ is a functorial homomorphism, then the homomorphisms $u * i_S$ give

$$u_S\colon F_S \longrightarrow G_S.$$

For every morphism $f\colon T \to S$ in $\mathcal{E}$, the squares

(c))
$$\begin{tikzcd} F_T f^* \arrow[r, "\varphi_f"] \arrow[d, "u_T * f^*"'] & F_S \arrow[d, "u_S"] \\ G_T f^* \arrow[r, "\psi_f"] & G_S \end{tikzcd}$$

are commutative.

Proposition VI.12.1. *Let $\mathcal{H}(\mathcal{F},\mathcal{C})$ be the category whose objects are pairs consisting of a family (F_S), indexed by $S \in \mathrm{Ob}(\mathcal{E})$, of functors $\mathcal{F}_S \to \mathcal{C}$, and a family (φ_f), indexed by $f \in \mathrm{Fl}(\mathcal{E})$, of functorial homomorphisms $F_T f^* \to F_S$, satisfying conditions* a) and b). *Its morphisms are the families (u_S), indexed by $S \in \mathrm{Ob}(\mathcal{E})$, of homomorphisms $F_S \to G_S$* 191
satisfying commutativity condition c) *above; composition is defined by composing the homomorphisms of functors $\mathcal{F}_S \to \mathcal{C}$. Then the two constructions described above define an* isomorphism

$$K\colon\ \mathbf{Hom}(\mathcal{F},\mathcal{C}) \xrightarrow{\sim} \mathcal{H}(\mathcal{F},\mathcal{C}).$$

It is immediate that these constructions define a functor from the first category to the second. This functor is fully faithful. For given F, G, the map

$$\mathrm{Hom}(F, G) \longrightarrow \mathrm{Hom}(K(F), K(G))$$

is plainly injective. To prove surjectivity, note that condition c) expresses the functoriality of the maps

$$u(\xi) = u_S(\xi)\colon F(\xi) = F_S(\xi) \longrightarrow G(\xi) = G_S(\xi)$$

for homomorphisms of the form $\alpha_f(\xi)$ in $\mathcal{F}$. Functoriality also holds on every fiber category, i.e. for the morphisms in $\mathcal{F}$ that are T-morphisms, with $T \in \mathrm{Ob}(\mathcal{E})$. It therefore holds for every morphism in $\mathcal{F}$, since every f-morphism, with $f\colon T \to S$ in $\mathcal{E}$, is uniquely the composite of a morphism $\alpha_f(\xi)$ and a T-morphism.

It remains to show that K is bijective on objects. The preceding argument already shows that K is injective on objects. For surjectivity, start with a system $(F_S), (\varphi_f)$ satisfying a) and b), and define a map $\mathrm{Ob}(\mathcal{F}) \to \mathrm{Ob}(\mathcal{C})$ by

$$F(\xi) = F_S(\xi) \qquad \text{for } \xi \in \mathrm{Ob}(\mathcal{F}_S) \subset \mathrm{Ob}(\mathcal{F}),$$

and a map $\mathrm{Fl}(\mathcal{F}) \to \mathrm{Fl}(\mathcal{C})$ by

$$F(\alpha_f(\xi)u') = \varphi_f(\xi)\, F_T(u')$$

for every morphism $f\colon T \to S$ in $\mathcal{E}$, every object ξ of $\mathcal{F}_S$, and every T-morphism u' with target $f^*(\xi)$. These maps define a *functor* $F\colon \mathcal{F} \to \mathcal{C}$. Indeed, $F(\mathrm{id}_\xi) = \mathrm{id}_{F(\xi)}$ is immediate. It remains to prove multiplicativity $F(uv) = F(u)F(v)$ for an f-morphism

$u\colon \eta \to \xi$ and a g-morphism $v\colon \zeta \to \eta$, where $f\colon T \to S$ and $g\colon U \to T$ are morphisms of $\mathcal{E}$. Setting $w = uv$, we have

$$u = \alpha_f(\xi)u', \qquad v = \alpha_g(\eta)v', \qquad w = \alpha_{fg}(\xi)w',$$ 192

with

$$w' = c_{f,g}(\xi)g^*(u')v' \qquad (cf.\ no.\ VI.8).$$

With this notation, we must prove commutativity along the outer boundary of the following diagram:

$$\begin{tikzcd}
F_U(\zeta) \arrow[r, "F_U(v')"] \arrow[dr, "F(v)"'] \arrow[rrr, bend left, "F_U(w')"] & F_U g^*(\eta) \arrow[r, "F_U g^*(u')"] \arrow[d, "\varphi_g(\eta)"] & F_U g^* f^*(\xi) \arrow[r, "F_U(c_{f,g}(\xi))"] \arrow[d, "\varphi_g(f^*(\xi))"] & F_U(fg)^*(\xi) \arrow[d, "\varphi_{fg}(\xi)"] \\
& F_T(\eta) \arrow[r, "F_T(u')"'] \arrow[rr, bend right, "F(u)"'] & F_T f^*(\xi) \arrow[r, "\varphi_f(\xi)"'] & F_S(\xi).
\end{tikzcd}$$

The left triangle is commutative by the definition of $F(v)$. The middle square is commutative because it is obtained from the homomorphism $u'\colon \eta \to f^*(\xi)$ by the functorial homomorphism φ_g, and the square on the right is commutative by condition b). The desired conclusion follows.

Now suppose that $\mathcal{C}$ is itself a normalized cloven category over $\mathcal{E}$, which we shall henceforth call $\mathcal{G}$, and that we are interested in the $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$. Every such functor F induces functors

$$F_S\colon \mathcal{F}_S \longrightarrow \mathcal{G}_S$$

on the fiber categories. Moreover, for every morphism $f\colon T \to S$ in $\mathcal{E}$ and every object ξ of $\mathcal{F}_S$, the f-morphism $F(\alpha_f(\xi))$ factors uniquely through a T-morphism

$$\varphi_f(\xi)\colon F_T(f^*_{\mathcal{F}}(\xi)) \longrightarrow f^*_{\mathcal{G}}(F_S(\xi)).$$

Here the subscript $\mathcal{F}$ or $\mathcal{G}$ indicates the cloven category whose inverse-image functor is being used. We thereby obtain a functorial homomorphism between functors from $\mathcal{F}_S$ to $\mathcal{G}_T$:

$$\varphi_f\colon F_T f^*_{\mathcal{F}} \longrightarrow f^*_{\mathcal{G}} F_S.$$ 193

The two systems (F_S) and (φ_f) satisfy the following conditions:

$$\text{a')} \qquad \varphi_{\mathrm{id}_S} = \mathrm{id}_{F_S}.$$

For morphisms $f\colon T \to S$ and $g\colon U \to T$ in $\mathcal{E}$, the following diagram of functorial homomorphisms is commutative:

(b'))

$$\begin{tikzcd}
F_U g^*_{\mathcal{F}} f^*_{\mathcal{F}} \arrow[r, "F_U * c^{\mathcal{F}}_{f,g}"] \arrow[d, "\varphi_g * f^*_{\mathcal{F}}"'] & F_U(fg)^*_{\mathcal{F}} \arrow[dd, "\varphi_{fg}"] \\
g^*_{\mathcal{G}} F_T f^*_{\mathcal{F}} \arrow[d, "g^*_{\mathcal{G}} * \varphi_f"'] & \\
g^*_{\mathcal{G}} f^*_{\mathcal{G}} F_S \arrow[r, "c^{\mathcal{G}}_{f,g} * F_S"] & (fg)^*_{\mathcal{G}} F_S.
\end{tikzcd}$$

We leave to the reader the verification, as well as the statement and proof of the analogue of Proposition VI.12.1. This analogue shows that the set of $\mathcal{E}$-functors from $\mathcal{F}$ to $\mathcal{G}$ is in bijection with the set of systems $(F_S), (\varphi_f)$ satisfying conditions a') and b') above. Under this correspondence, the Cartesian functors are precisely those for which the homomorphisms φ_f are isomorphisms.

Remark. It is usually preferable, of course, to reason directly with fibered categories without using explicit cleavages. In particular, this avoids having to use a cumbersome interpretation such as the one above for the simple notion of an $\mathcal{E}$-functor or a Cartesian $\mathcal{E}$-functor. It was in order to avoid unbearable heaviness and to obtain more intrinsic statements that we had to abandon the approach taken in [**2**], starting from the notion of a 194
cloven category, called there a "fibered category", which now takes second place behind the notion of a fibered category. It is moreover likely that, contrary to the still predominant usage inherited from older habits of thought, universal problems will eventually prove more conveniently handled by not emphasizing *one* solution chosen once and for all, but by placing all solutions on an equal footing.

Bibliography

[1] A. Grothendieck, *Sur quelques points d'algèbre homologique*, Tôhoku Math. J. **9** (1957), pp. 119–221.
[2] A. Grothendieck, *Technique de descente et Théorèmes d'existence, I*, Séminaire Bourbaki 190, December 1959.

CHAPTER VIII

Faithfully flat descent

VIII.1. Descent of quasi-coherent modules 195

Let **Sch** denote the category of preschemes. Proceeding as in VI VI.11.b, one finds that the category of pairs (X, F) consisting of a prescheme X and a module F on X (with morphisms defined as in loc. cit. by means of the direct image of a module under a morphism of ringed spaces) may be regarded as a fibered category over **Sch**. The base-change functor associated with a morphism $f\colon X \to Y$ in **Sch** is the inverse-image functor for modules under f. Notice that the fiber category at $X \in \mathrm{Ob}(\mathbf{Sch})$ of the preceding fibered category is the category *opposite* to the category of modules on X.

Since the inverse image of a quasi-coherent module is quasi-coherent, the full subcategory of the category of pairs (X, F) formed by those pairs for which F is quasi-coherent is a fibered subcategory of the preceding fibered category. (By contrast, without hypotheses on f, the direct image of a quasi-coherent module is not in general a quasi-coherent module.) We shall simply call this the *fibered category of quasi-coherent modules on preschemes.*

Recall also that a morphism $f\colon X \to Y$ of ringed spaces is called *faithfully flat* if it is *flat* (that is, for every $x \in X$, $\mathcal{O}_{X,x}$ is a flat module over $\mathcal{O}_{Y,f(x)}$, (IV)) and *surjective.* The morphism f is called *quasi-compact* if the inverse image under f of every quasi-compact subset is quasi-compact. When f is a morphism of preschemes, this also means that the inverse image under f of every affine open subset of Y is a *finite* union of affine open subsets of X.

Theorem VIII.1.1. *Let $\mathcal{F}$ be the fibered category of quasi-coherent modules on preschemes. Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes. Then g is a morphism of effective $\mathcal{F}$-descent.* 196

Recall[1] that this means two things.

Corollary VIII.1.2 (Descent of homomorphisms of modules). *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes, let F and G be two quasi-coherent modules on S, let F' and G' be their inverse images on S', and finally let F'' and G'' be their inverse images on $S'' = S' \times_S S'$. Consider the diagram of maps of sets defined by the base-change functors for g, p_1, and p_2, where $p_1, p_2\colon S' \times_S S' \rightrightarrows S'$ are the two projections:*

$$\mathrm{Hom}_S(F, G) \longrightarrow \mathrm{Hom}_{S'}(F', G') \rightrightarrows \mathrm{Hom}_{S''}(F'', G'').$$

This diagram is exact; that is, it identifies the first set bijectively with the equalizer of the two displayed maps from the second set to the third.

In other words, the base-change functor for g, $F \longmapsto F'$, defines a *fully faithful* functor from the category of quasi-coherent modules on S to the category of quasi-coherent modules on S' endowed with descent data relative to g. Moreover:

[1]We shall admit here the general theory of descent set out in detail in the article of J. Giraud cited in the footnote to the Warning, a work that we shall cite below as [**D**]. Cf. also [**2**] for a succinct account.

COROLLARY VIII.1.3 (Descent of modules). *For every quasi-coherent module F' on S', every descent datum on F' relative to g is* effective*; that is, F', together with its descent datum, is isomorphic to the inverse image under g of a quasi-coherent module on S (determined up to unique isomorphism by VIII.1.2).*

In other words, the preceding fully faithful functor is in fact an *equivalence.* In practical terms, giving a quasi-coherent module on S amounts to the same thing as giving a quasi-coherent module on S' endowed with descent data relative to g.

Proof of VIII.1.1. Let T first be an S-prescheme that is S-isomorphic to the disjoint 197
union of a family of open subpreschemes S_i induced from S and covering S. It is then evident that the structural morphism $T \to S$ is a morphism of effective $\mathcal{F}$-descent. This means precisely that giving a quasi-coherent module F on S is equivalent to giving quasi-coherent modules F_i on the S_i, together with gluing isomorphisms

$$\varphi_{ji} \colon F_i|S_i \cap S_j \longrightarrow F_j|S_i \cap S_j$$

satisfying the familiar cocycle condition. By VII, 8, it follows that, to verify that $g\colon S' \to S$ is a morphism of effective $\mathcal{F}$-descent, it suffices to verify this for the morphism

$$g_T \colon T' = T \times_S S' \longrightarrow T$$

obtained from g by the base change $T \to S$. Notice that the hypothesis on $T \to S$ is stable under arbitrary base change, so that $T \to S$ is in fact a *universal* morphism of effective $\mathcal{F}$-descent. Taking the S_i to be affine open subsets covering S, we are therefore reduced to the case where S is affine.

Then S' is a finite union of affine open subsets. Taking their disjoint union as an S-scheme gives an affine S-scheme S_1 and a flat surjective S-morphism $S_1 \to S'$. Thus S_1 is also faithfully flat over S. If, therefore, one proves that a faithfully flat affine morphism is a morphism of effective $\mathcal{F}$-descent, hence a strict universal morphism of $\mathcal{F}$-descent (the hypothesis being stable under base change), it follows in particular that the structural morphism $S_1 \to S$ is a strict universal morphism of $\mathcal{F}$-descent. Since there is an S-morphism $S_1 \to S'$, it then follows from [**D**] that $g\colon S' \to S$ is indeed a strict morphism of $\mathcal{F}$-descent.

We are thus reduced to the case in which g is affine. As we have seen, we may then also suppose that S is affine; hence *we may suppose that S and S' are affine.* In this case, VIII.1.2 is equivalent to the following lemma.

LEMMA VIII.1.4. *Let A be a ring, let A' be a faithfully flat A-algebra, let M and N be two A-modules, let M' and N' be the A'-modules obtained by the change of rings $A \to A'$, and let M'' and N'' be the $A'' = A' \otimes_A A'$-modules obtained by the change of rings $A \to A''$. Then the sequence of maps of sets*

$$\mathrm{Hom}_A(M, N) \longrightarrow \mathrm{Hom}_{A'}(M', N') \rightrightarrows \mathrm{Hom}_{A''}(M'', N'')$$

is exact.

Since the homomorphism $N \to N'$ is injective (because A' is faithfully flat over A), the first arrow is injective. It remains to prove that if an A'-homomorphism $u' \colon M' \to N'$ is compatible with the descent data, then it comes from an A-homomorphism $u\colon M \to N$. This simply means that u' maps the subset M of M' into the subset N of N'. The induced map $u\colon M \to N$ is then automatically A-linear because u' is A'-linear, and one sees similarly that u' is necessarily equal to $u \otimes_A A'$.

Now, if $x \in M$, then $u'(x)$ belongs to the equalizer of the pair of maps $N' \rightrightarrows N''$. Thus, to prove VIII.1.4, we are reduced to the following special case (corresponding to $M = A$):

COROLLARY VIII.1.5. *Let N be an A-module. Then the sequence of maps of sets*

$$N \longrightarrow N' \rightrightarrows N''$$

is exact.

Let A_1 be a faithfully flat A-algebra. To prove exactness of the displayed sequence, it is enough to prove exactness after the base change $A \to A_1$. The resulting sequence is plainly the sequence associated with the A_1-module

$$N_1 = N \otimes_A A_1$$

and the A_1-algebra $A'_1 = A_1 \otimes_A A'$. It is therefore enough to find a faithfully flat A-algebra A_1 for which

$$\mathrm{Spec}(()A'_1) \longrightarrow \mathrm{Spec}(()A_1)$$

is a strict $\mathcal{F}$-descent morphism. We may take $A_1 = A'$: the preceding morphism then has a right inverse, and hence, by [**D**], is an effective descent morphism for every category fibered over **Sch**.

It remains to show that if N' is an A'-module equipped with a descent datum relative to $A \to A'$, i.e. with an isomorphism

$$\varphi \colon N'_1 \xrightarrow{\sim} N'_2$$

between the two modules obtained from N' by the two base changes

$$A' \rightrightarrows A' \otimes_A A',$$

then N', together with its descent datum, is isomorphic to a module of the form $N \otimes_A A'$. 199
In view of VIII.1.5, this statement is readily seen to be equivalent to the following:

Lemma VIII.1.6. *Let N' be an A'-module equipped with a descent datum relative to $A \to A'$, where A' is an A-algebra. Let N be the A-submodule of N' formed by the elements x such that*

$$\varphi(x \otimes_{A'} 1_{A'}) = 1_{A'} \otimes_{A'} x,$$

and consider the canonical homomorphism

$$N \otimes_A A' \longrightarrow N',$$

which is compatible with the descent data. If A' is faithfully flat over A, this homomorphism is an isomorphism.

Let us prove the lemma. Again let A_1 be a faithfully flat A-algebra. To show that the homomorphism in question is an isomorphism, it is enough to show that it becomes one after the base change $A \to A_1$. Using the flatness of A_1 over A, one sees that the resulting homomorphism is precisely the one obtained directly from the $A'_1 = A' \otimes_A A_1$-module $N' \otimes_A A_1$, equipped with the descent datum relative to $A_1 \to A'_1$ obtained canonically by base change from the descent datum on N'. Thus it is enough to choose a faithfully flat A-algebra A_1 such that $\mathrm{Spec}(()A'_1) \to \mathrm{Spec}(()A_1)$ is an effective $\mathcal{F}$-descent morphism. As above, take $A_1 = A'$. This proves VIII.1.6, and hence also VIII.1.1.

Corollary VIII.1.7 (Descent of sections of modules). *Let $g \colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes. For every quasi-coherent module G on S, let G' and G'' be its inverse images on S' and $S'' = S' \times_S S'$, respectively. Then the diagram of homomorphisms of modules on S*

$$G \longrightarrow g_*(G') \rightrightarrows h_*(G'')$$

where $h \colon S'' \to S$ is the structural morphism, is exact.

Indeed, this means that, for every open subset U of S, the corresponding diagram of 200
sections over U is exact. We may plainly assume $U = S$, and the asserted exactness is then the special case of VIII.1.2 obtained by taking $F = \mathcal{O}_S$.

Since the inverse-image functor on modules is right exact, VIII.1.1 formally implies:

Corollary VIII.1.8 (Descent of quotient modules). *With the notation of VIII.1.7, let* $\mathrm{Quot}(F)$*, for every quasi-coherent module* F *on a prescheme, denote the set of quasi-coherent quotient modules of* F*. Then the diagram of maps of sets*

$$\mathrm{Quot}(G) \longrightarrow \mathrm{Quot}(G') \rightrightarrows \mathrm{Quot}(G'')$$

is exact.

(The same statement plainly holds with submodules in place of quotient modules, since the two are in bijection.) Taking $G = \mathcal{O}_S$, we obtain:

Corollary VIII.1.9 (Descent of closed subpreschemes). *For every prescheme* X*, let* $H(X)$ *denote the set of closed subpreschemes of* X*. With this notation, and under the hypotheses of VIII.1.7, the diagram of maps of sets*

$$H(S) \longrightarrow H(S') \rightrightarrows H(S'')$$

is exact.

Theorem VIII.1.1 should be supplemented by the following result:

Proposition VIII.1.10 (Descent of properties of modules). *Let* $g\colon S' \to S$ *be faithfully flat and quasi-compact, and let* F *be a quasi-coherent module on* S*. Then* F *is of finite type, respectively of finite presentation, respectively finite locally free, if and only if its inverse image* F' *on* S' *has the corresponding property.*

Only sufficiency requires proof. We may plainly suppose that S is affine. Replacing 201
S' by a sum of affine open subsets covering S', we reduce to the case in which S' is also affine. The assertion is then equivalent to the following:

Corollary VIII.1.11. *Let* A *be a ring, let* A' *be a faithfully flat* A*-algebra, let* M *be an* A*-module, and let* $M' = M \otimes_A A'$*. Then* M *is of finite type, respectively of finite presentation, respectively finite locally free, if and only if* M' *has the corresponding property.*

Indeed, write $M = \varinjlim_i M_i$, where the M_i are the finite-type submodules of M. Then $M' = \varinjlim_i M_i'$. If M' is of finite type, it equals one of the M_i'; faithful flatness then gives $M = M_i$, so M is of finite type. Consequently, there is an exact sequence

$$0 \longrightarrow R \longrightarrow L \longrightarrow M \longrightarrow 0$$

with L finite free, and hence an exact sequence

$$0 \longrightarrow R' \longrightarrow L' \longrightarrow M' \longrightarrow 0$$

with L' finite free. If M' is finitely presented, then R' is of finite type; by what precedes, R is of finite type, and therefore M is finitely presented. Finally, M is finite locally free precisely when it is finitely presented and flat (cf. IV in the noetherian case; the general case is left to the reader). Each of these properties descends, and so does their conjunction. This completes the proof.

Remark VIII.1.12. Combining VIII.1.1 and VIII.1.10 shows that VIII.1.1 remains valid if the fibered category $\mathcal{F}$ is replaced by the fibered subcategory consisting of quasi-coherent modules of finite type, respectively of finite presentation, respectively finite locally free, respectively locally free of a specified rank n.

VIII.2. Descent of affine preschemes over another 202

Since the inverse-image functor for modules is compatible with tensor products and other tensor operations, Theorem VIII.1.1 implies various variants obtained by considering, in place of a single quasi-coherent module, a quasi-coherent module or a system of quasi-coherent modules endowed with various additional structures expressed by means of tensor operations. For example, giving three quasi-coherent modules F, G, and H on S and a pairing

$$F \otimes G \longrightarrow H$$

is equivalent to giving three quasi-coherent modules F', G', and H' on S', endowed with descent data relative to $g\colon S' \to S$, together with a pairing

$$F' \otimes G' \longrightarrow H'$$

"compatible" with those descent data, in the evident sense of the term. For example, if $F = G = H$, one sees that giving a quasi-coherent module F on S endowed with an algebra law (which for the moment we do not require to satisfy any axiom of associativity, commutativity, or the existence of a unit section) is equivalent to giving the same data on S', together with descent data. Using the results of the preceding section, one sees at once that F satisfies any one of the usual axioms just mentioned if and only if F' does.

For example, giving a quasi-coherent algebra $\mathcal{A}$ on S (by which from now on we mean associative, commutative, and equipped with a unit section) is equivalent to giving a quasi-coherent algebra $\mathcal{A}'$ on S' endowed with descent data relative to $g\colon S' \to S$. Recalling the equivalence between the opposite category of quasi-coherent algebras on S and the category of affine S-preschemes over S (EGA II, §1), one obtains at once:

Theorem VIII.2.1. *Let $\mathcal{F}'$ be the fibered category of affine morphisms of preschemes $f\colon X \to S$, regarded as a fibered subcategory of the fibered category of arrows in the category* 203 **Sch** *of preschemes* (VI VI.11.a). *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism of preschemes. Then g is a morphism of effective $\mathcal{F}'$-descent.*

VIII.3. Descent of set-theoretic properties and finiteness properties of morphisms[2]

Proposition VIII.3.1. *Let $f\colon X \to Y$ be an S-morphism, let $g\colon S' \to S$ be a surjective morphism, and let*

$$f'\colon X' = X \times_S S' \longrightarrow Y' = Y \times_S S'$$

be the morphism obtained from f by base change along $g\colon S' \to S$. The morphism f is surjective (respectivelyradicial) if and only if f' is so.

Notice that f' can also be obtained by the base change $Y' \to Y$, which is itself surjective because it is obtained from the surjective morphism $g\colon S' \to S$. On the other hand, for every $y \in Y$ and every $y' \in Y'$ lying over y, there is an isomorphism

$$X'_{y'} \xleftarrow{\sim} X_y \otimes_{\kappa(y)} \kappa(y'),$$

where X_y denotes the fiber of X at y, and $X'_{y'}$ that of X' at y'. It follows that X_y is nonempty (respectivelyhas at most one point, and that point corresponds to a radicial residue-field extension) if and only if $X'_{y'}$ has the same property. This proves VIII.3.1.

Corollary VIII.3.2. *Under the hypotheses of VIII.3.1, if f' is injective (respectively-bijective), then f is likewise.*

Indeed, if $X'_{y'}$ has at most one point (respectivelyexactly one point), the same is true of X_y. This follows because the morphism $X'_{y'} \to X_y$ is surjective, being obtained from the surjective morphism $\operatorname{Spec}(\kappa(y')) \to \operatorname{Spec}(\kappa(y))$.

[2] For other results of the kind treated in Sections 3 and 4, cf. EGA IV 2.3, 2.6, and 2.7.

Proposition VIII.3.3. *With the notation of VIII.3.1, suppose that* $g\colon S' \to S$ *is surjective and quasi-compact (respectivelyfaithfully flat and quasi-compact). The morphism* f *is quasi-compact (respectivelyof finite type) if and only if* f' *is so.*

Only the sufficiency needs to be proved. We may clearly suppose $S = Y$, since the hypothesis on $g\colon S' \to S$ is preserved for $Y' \to Y$. We may also suppose that Y is affine. Then Y' is quasi-compact, and therefore X' is quasi-compact, since f' is so by hypothesis. Let $(X_i)_{i\in I}$ be a family of affine open subsets of X covering X. The X'_i are open subsets covering X', so a finite subfamily covers X'. Since $X' \to X$ is surjective, the corresponding X_i already cover X. Thus X is quasi-compact, that is, f is quasi-compact.

Now suppose that f' is of finite type, and let us prove that f is of finite type, assuming g faithfully flat. Replacing Y' by the disjoint union of a family of affine open subsets covering it, we may suppose that Y' is affine. Finally, by the preceding argument X is covered by finitely many affine open subsets X_i; it remains to show that they are of finite type over Y, knowing that X'_i is of finite type over Y'. This reduces us to the following result.

Corollary VIII.3.4. *Let* B *be an* A*-algebra, let* A' *be a faithfully flat* A*-algebra, and let* $B' = B \otimes_A A'$ *be the* A'*-algebra obtained from* B *by change of rings. Then* B *is of finite type if and only if* B' *is of finite type.*

Again, only the sufficiency needs proof. We have $B = \varinjlim_i B_i$, where the B_i are the finite-type subalgebras of B. Hence $B' = \varinjlim_i B'_i$, and if B' is of finite type over A', then B' is equal to one of the B'_i. Faithful flatness then gives $B = B_i$, so B is of finite type.

Corollary VIII.3.5. *Suppose again that the base-change morphism* $g\colon S' \to S$ *is faithfully flat and quasi-compact. The morphism* f *is quasi-finite if and only if* f' *is quasi-finite.*

Indeed, the property of being quasi-finite is, by definition, the conjunction of being of finite type and having finite fibers. Each of these properties descends under g: the first by VIII.3.3, and the second by the argument of VIII.3.1, which uses only the surjectivity of g.

Remarks VIII.3.6. Let A be a ring and X an A-prescheme. One sees easily that the following conditions are equivalent: 205

(i) There exist a noetherian ring A_0 (which, if desired, may be assumed to be a finite-type subring of A), an A_0-prescheme X_0 of finite type, a homomorphism $A_0 \to A$, and an A-isomorphism
$$X \xrightarrow{\sim} X_0 \times_{A_0} A.$$

(ii) The diagonal morphism $X \to X \times_{\mathrm{Spec}(A)} X$ is quasi-compact (a vacuous condition if X is separated over A), and X is the union of finitely many affine open subsets X_i whose rings B_i are algebras of finite presentation over A; that is, quotients of polynomial algebras in finitely many indeterminates by ideals of finite type.

If X is itself affine, with ring B, these conditions simply mean that B is an algebra of finite presentation over A.

A morphism $f\colon X \to Y$ is called a *morphism of finite presentation*, and X is also said to be of finite presentation over Y, if Y is the union of affine open subsets Y_i such that $X|Y_i$, as a Y_i-prescheme, satisfies the preceding equivalent conditions. The same is then true of $X|Y'$ for *every* affine open subset Y' of Y. This property is stable under base change, and the composite of two morphisms of finite presentation is again of finite presentation.

With these notions in place, condition (ii), by the same argument as in VIII.1.10, shows that Proposition VIII.3.3 remains valid when "of finite type" is replaced by "of finite presentation."

VIII.4. Descent of topological properties

THEOREM VIII.4.1. *Let $g\colon Y' \to Y$ be a morphism, and let Z be a subset of Y. Suppose that g is flat and that there exists a quasi-compact morphism $f\colon X \to Y$ such that $Z = f(X)$ (N.B. if Y is noetherian, this last condition is implied by "Z being constructible"). Then*

$$g^{-1}(\overline{Z}) = \overline{g^{-1}(Z)}.$$

We may assume Y affine, and then Y' affine. Since Y is affine, X is a finite union 206
of affine open subsets X_i, and, replacing X by the disjoint union of the X_i, we may also assume X affine. Let A, A', B be the rings of Y, Y', X, let $B' = B \otimes_A A'$ be that of $X' = X \times_Y Y'$, let I be the kernel of $A \to B$, and let I' be the kernel of $A' \to B'$. Thus the closed subsets of Y and Y' defined by these ideals are respectively the closure of $Z = f(X)$ and the closure of $Z' = f'(X') = g^{-1}(Z)$. We wish to show that the latter equals $g^{-1}(\overline{Z})$; this follows from $I' = IA'$, which in turn follows from the flatness of A' over A.

COROLLARY VIII.4.2. *Let $g\colon Y' \to Y$ be a flat and quasi-compact morphism, and let Z' be a closed subset of Y' saturated for the set-theoretic equivalence relation defined by g. Then $Z' = g^{-1}(\overline{g(Z')})$.*

Indeed, $Z' = g^{-1}(Z)$, with $Z = g(Z')$. We may then apply VIII.4.1, noting that the condition imposed on Z in VIII.4.1 is indeed satisfied by taking for X the prescheme Z' equipped with the reduced structure induced by Y'. (The fact that g is quasi-compact then ensures that the induced morphism $f\colon Z' \to Y$ is quasi-compact.)

Statement VIII.4.2 also means that *the topology on $g(Y')$ induced by Y is the quotient of that on Y'*. In particular:

COROLLARY VIII.4.3. *Let $g\colon Y' \to Y$ be a faithfully flat and quasi-compact morphism. Then g makes Y a quotient topological space of Y', i.e. for a subset Z of Y, Z is closed (respectivelyopen) if and only if $Z' = g^{-1}(Z)$ is so.*

Recall now that two elements a, b of Y' have the same image in Y if and only if they are of the form $p_1(c), p_2(c)$ for a suitable element c of $Y'' = Y' \times_Y Y'$. It follows that if g is surjective, there is an *exact* diagram of sets

$$\mathcal{P}(Y) \longrightarrow \mathcal{P}(Y') \rightrightarrows \mathcal{P}(Y'') \ ,$$

where, for every set E, $\mathcal{P}(E)$ denotes its power set. With this notation, VIII.4.3 may also be interpreted as follows:

COROLLARY VIII.4.4 (Descent of open respectivelyclosed subsets). *Let $g\colon Y' \to Y$ be* 207
as in VIII.4.3. For every prescheme X, let $\mathrm{Ouv}(X)$ *respectively*$\mathrm{Fer}(X)$ *be the set of its open subsets respectivelythe set of its closed subsets. Then the following are* exact *diagrams of maps of sets (deduced from g and the two projections of $Y'' = Y' \times_Y Y'$):*

$$\mathrm{Ouv}(Y) \longrightarrow \mathrm{Ouv}(Y') \rightrightarrows \mathrm{Ouv}(Y'')$$

$$\mathrm{Fer}(Y) \longrightarrow \mathrm{Fer}(Y') \rightrightarrows \mathrm{Fer}(Y'')$$

The following supplements VIII.4.3:

COROLLARY VIII.4.5. *Let $g\colon Y' \to Y$ be as in VIII.4.3, and let Z be a subset of Y for which there exists a quasi-compact morphism $f\colon X \to Y$ with image Z (for example, Z constructible and Y noetherian). Then Z is a locally closed subset of Y if and only if $Z' = g^{-1}(Z)$ is a locally closed subset of Y'.*

It suffices to prove the "if"direction. Let Y_1 be the closed subprescheme of Y given by the closure of Z equipped with the induced reduced structure, and let $Y_1' = Y_1 \times_Y Y'$ be the closed subprescheme of Y' that is the inverse image of Y_1. Its underlying set is the inverse image $g^{-1}(Y_1) = g^{-1}(\overline{Z})$, and hence equals $\overline{Z'}$ by VIII.4.1. Since Z' is locally closed in Y', it is open in $\overline{Z'}$, and hence open in Y_1'. The latter is faithfully flat and quasi-compact over Y_1, so by VIII.4.3 we conclude that Z is open in Y_1, i.e. in $\overline{Z}$, which means that Z is locally closed.

Corollary VIII.4.6. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism, let $f\colon X \to Y$ be an S-morphism, and let $f'\colon X' \to Y'$ be the S'-morphism obtained from it by base change. Suppose that f' is an open map (respectivelya closed map, respectivelyquasi-compact and a homeomorphism into, respectivelya homeomorphism onto); then f has the same property.*

Since Y' is faithfully flat and quasi-compact over Y, we may assume $Y = S$. Let Q be a subset of X. Then (denoting by h the projection morphism $X' \to X$) one has: 208

$$g^{-1}(f(Q)) = f'(h^{-1}(Q)).$$

If Q is open (respectivelyclosed), the same is true of $h^{-1}(Q)$, and hence also of $f'(h^{-1}(Q))$ if f' is assumed to be an open (respectivelyclosed) map; the same is therefore true of $f(Q)$ by the preceding formula and VIII.4.3. This proves the first two assertions in VIII.4.6. It remains to consider the case in which f' is a homeomorphism into, and then to prove that f is a homeomorphism into. (The case of a homeomorphism onto will then follow from VIII.3.1.) By VIII.3.2 f is injective; it remains to prove that the map $X \to f(X)$ is open. We already know that f is quasi-compact by VIII.3.3. It suffices to prove that, for every closed subset Z of X, one has $Z = f^{-1}(\overline{f(Z)})$. This is equivalent to the analogous formula for the inverse images under the surjective map $h\colon X' \to X$, i.e. to

$$Z' = f'^{-1}(g^{-1}(\overline{f(Z)})),$$

where $Z' = h^{-1}(Z)$. But by VIII.4.1, applied to the subset $f(Z)$ of Y, one has $g^{-1}(\overline{f(Z)}) = \overline{g^{-1}(f(Z))}$, and the formula to be proved is equivalent to

$$Z' = f'^{-1}(\overline{f'(Z')}),$$

which follows from the hypothesis that f' is a homeomorphism into.

N.B. In this last argument, already supposing that f is quasi-compact, we did not use the quasi-compactness of g, but only that g is faithfully flat. Thus it is under this hypothesis that one may descend the property "homeomorphism into", or "homeomorphism onto", or, by the preceding argument, the property "f' is quasi-compact and makes $f'(X')$ a quotient topological space of X'".

We shall say that a morphism $f\colon X \to Y$ of preschemes is *universally open* (respectively*universally closed*, respectively*universally bicontinuous*, etc.) if for every base change $Y' \to Y$, $f'\colon X' \to Y'$ is an open morphism (respectivelyclosed, respectivelya homeomorphism onto its image space). We then deduce from VIII.4.6:

Corollary VIII.4.7. *Under the conditions of VIII.4.6, f is universally open (respectivelyuniversally closed, respectivelya universal homeomorphism into, respectivelya universal homeomorphism) if and only if f' is so.* 209

Corollary VIII.4.8. *Under the conditions of VIII.4.6, f is separated (respectivelyproper) if and only if f' is so.*

To say that f is separated means that the diagonal morphism $X \to X \times_Y X$ is closed, or equivalently universally closed, and the first assertion of VIII.4.8 therefore follows from VIII.4.7. To say that f is proper means that f satisfies the conditions a) f is of finite type, b) f is separated, c) f is universally closed. Condition a) does descend by VIII.3.3, b) does as well by what we have just seen, and finally c) does by VIII.4.7.

Remarks VIII.4.9. Recall that when $g\colon Y' \to Y$ is a flat morphism of finite type, with Y locally noetherian, then g is an open morphism (VI IV.6.6), which is a more precise result than VIII.4.3. Note, however, that if f is a faithfully flat and quasi-compact morphism of noetherian preschemes, then f is not in general an open morphism. For example, let Y be an irreducible scheme whose generic point y is not open (for example, an algebraic curve), and take for Y' the disjoint-union scheme $Y \amalg \operatorname{Spec}(()\kappa(y))$. Then the image of the open subset $\operatorname{Spec}(()\kappa(y))$ under the structure morphism $Y' \to Y$ is not an open subset of Y. The reader will also note that various statements of the present exposé become false if the hypothesis that the faithfully flat morphism under consideration is also quasi-compact is dropped. The typical counterexample is obtained by taking for Y' the disjoint-union scheme of the spectra of the local rings at the points of Y. For example, again taking Y to be an irreducible algebraic curve and Z to be the subset of Y consisting of the generic point, its inverse image in Y' is open although Z is not open.

. Various statements of the present exposé remain valid if the hypothesis that Y' is 210
flat over Y is replaced by the following: there exists a module of finite type F on Y', with support Y', that is flat relative to Y; the hypothesis of faithful flatness is then replaced by the preceding one together with the hypothesis that $Y' \to Y$ is surjective. This applies to the first two assertions in VIII.1.10, to VIII.3.3, VIII.3.5, VIII.4.1, and consequently to all the results of the present section.

VIII.5. Descent of morphisms of preschemes

Proposition VIII.5.1. *Let $g\colon S' \to S$ be a morphism of preschemes.*

a) *Suppose that g is surjective, and that the homomorphism*
$$g^*\colon \mathcal{O}_S \to g_*(\mathcal{O}_{S'})$$
is injective. Then g is an epimorphism in the category of preschemes, and even in the category of ringed spaces.

b) *Suppose that g is surjective and makes S a quotient topological space of S'. Let $S'' = S' \times_S S'$ and let $h\colon S'' \to S$ be the structural morphism; consider the canonical diagram of homomorphisms:*
$$\mathcal{O}_S \longrightarrow g_*(\mathcal{O}_{S'}) \rightrightarrows h_*(\mathcal{O}_{S''})\ ,$$
and suppose that this diagram is exact*. Then g is an effective epimorphism in the category of preschemes (and also in the category of ringed spaces), i.e. the diagram*
$$S \longleftarrow S' \leftleftarrows S''$$
is exact.

Proof. a) We must show that a morphism of ringed spaces $f\colon S \to Z$ is determined once fg is known. Since g is surjective, the underlying set map f_0 of f is known; it remains to determine the homomorphism of sheaves of rings $\mathcal{O}_Z \to \mathcal{O}_S$, or, what amounts to the same thing, the homomorphism of sheaves of rings 211
$$u\colon f_0^{-1}(\mathcal{O}_Z) \to \mathcal{O}_S$$
defined by f. We already know the homomorphism
$$(fg)_0^{-1}(\mathcal{O}_Z) = g_0^{-1}(f_0^{-1}(\mathcal{O}_Z)) \to \mathcal{O}_{S'}$$
defined by fg, or, what amounts to the same thing, we have a homomorphism
$$f_0^{-1}(\mathcal{O}_Z) \to g_{0*}(\mathcal{O}_{S'}) = g_*(\mathcal{O}_{S'}).$$
One immediately sees that the latter is none other than the composite of $g^*\colon \mathcal{O}_S \to g_*(\mathcal{O}_{S'})$ with u, and since g^* is injective, u is known when g^*u is known. [N.B. we have obviously not used the fact that $g\colon S' \to S$ is a morphism of preschemes; the statement would hold

for an arbitrary morphism of ringed spaces; the same remark applies to b), both in the category of ringed spaces and in the category of locally ringed spaces. Note also that if g is a morphism of preschemes, not necessarily surjective, but such that $g^*\colon \mathcal{O}_S \to g_*(\mathcal{O}_{S'})$ is injective, then for two morphisms f_1, f_2 from S to a *scheme* Z such that $f_1 g = f_2 g$, one has $f_1 = f_2$; indeed, if I is the Ideal on S that defines the subprescheme of S where f_1, f_2 coincide (the inverse image of the diagonal subprescheme of $Z \times Z$ under (f_1, f_2)), one sees that I is contained in $\operatorname{Ker}(g^*)$].

b) We must show that for every ringed space Z, the following diagram of maps

$$\operatorname{Hom}(S, Z) \longrightarrow \operatorname{Hom}(S', Z) \rightrightarrows \operatorname{Hom}(S'', Z)$$

is exact, and that the same holds when Z is a locally ringed space and one restricts to morphisms of locally ringed spaces. Since we already know from a) that the first map is injective, it remains to see that if $f'\colon S' \to Z$ is a morphism of ringed spaces such that $f'p_1 = f'p_2$, then f' is of the form fg, where $f\colon S \to Z$ is a morphism of ringed spaces. Since g is surjective, it is then evident that if f' is a morphism of locally ringed spaces, the same will hold for f. 212

It follows from the hypothesis on f' that the underlying set map f'_0 is constant on the fibers of the map g_0; hence, since the latter is surjective, f'_0 factors uniquely as $f'_0 = f_0 g_0$, where $f_0\colon S \to Z$ is a map, necessarily continuous since g_0 identifies S with a quotient topological space of S'. Now consider the homomorphism

$$f_0^{-1}(\mathcal{O}_Z) \to g_*(\mathcal{O}_{S'})$$

deduced from the homomorphism $(f_0 g_0)^{-1}(\mathcal{O}_Z) \to \mathcal{O}_{S'}$ corresponding to f'. The hypothesis $f'p_1 = f'p_2$ is then interpreted as saying that the composites of the preceding homomorphism with the two homomorphisms $g_*(\mathcal{O}_{S'}) \rightrightarrows h_*(\mathcal{O}_{S''})$ are the same; hence, by hypothesis b), it factors through a morphism

$$f_0^{-1}(\mathcal{O}_Z) \to \mathcal{O}_S.$$

The latter defines a morphism of ringed spaces $f\colon S \to Z$, which is the desired morphism.

Theorem VIII.5.2. *Let $\mathcal{F}$ be the fibered category of arrows in the category* **Sch** *of preschemes* (VI VI.11.a)*. Then every faithfully flat and quasi-compact morphism $g\colon S' \to S$ is a morphism of $\mathcal{F}$-descent (or, as one also says, a descent morphism in* **Sch***).*

This means the following: let $S'' = S' \times_S S'$, and for two preschemes X, Y over S, consider their inverse images X', Y' over S' and their inverse images X'', Y'' over S'', giving a diagram of maps

$$\operatorname{Hom}_S(X, Y) \longrightarrow \operatorname{Hom}_{S'}(X', Y') \rightrightarrows \operatorname{Hom}_{S''}(X'', Y'')\ ;$$

with these notations, VIII.5.2 means that this diagram is exact. It should be noted that it is not true in general that g is a morphism of effective descent, i.e. that for every prescheme X' over S', every descent datum on X' relative to $g\colon S' \to S$ is effective. The question of effectivity, often delicate, will be examined in no. VIII.7. 213

We have seen in [**D**] (taking into account that fiber products exist in **Sch**) that statement VIII.5.2 is equivalent to the following:

Corollary VIII.5.3. *A faithfully flat and quasi-compact morphism of preschemes is a universal effective epimorphism.*

Since a faithfully flat and quasi-compact morphism remains so under every base change, we are reduced to proving that it is an effective epimorphism. We then apply the criterion VIII.5.1 b), which gives the desired result, taking VIII.4.3 and VIII.1.7 into account.

Corollary VIII.5.4. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism, let $f\colon X \to Y$ be an S-morphism, and let $f'\colon X' \to Y'$ be the S'-morphism deduced from it by the base change $S' \to S$. Then f is an isomorphism if and only if f' is.*

Indeed, if f' is an isomorphism, it is also an isomorphism for the natural descent structures on X', Y', and since the functor $X \longmapsto X'$ from $\mathbf{Sch}_{/S}$ to the category of preschemes over S' endowed with a descent datum relative to g is fully faithful by VIII.5.2, the desired conclusion follows.

Corollary VIII.5.5. *Under the conditions of VIII.5.4, f is a closed immersion (respectivelyan open immersion, respectivelya quasi-compact immersion) if and only if f' is.*

As usual, we may suppose $Y = S$, and only the "if" direction has to be proved. Note that the fact that X'/Y' is endowed with a descent datum relative to $g\colon Y' \to Y$, and that the structural morphism $f'\colon X' \to Y'$ is an immersion, hence a monomorphism, implies that the two subobjects of Y'' obtained as inverse images of X'/Y' under either projection 214
from S'' to S' are the same. If f' is a closed immersion, it follows by virtue of VIII.1.9 that there exists a closed subprescheme X_1 of Y whose inverse image under $g\colon Y' \to Y$ is X'. Thus, by the uniqueness of the solution to a descent problem relative to a morphism of $\mathcal{F}$-descent, it follows that X_1 is Y-isomorphic to X, so $f\colon X \to Y$ is a closed immersion. We proceed in the same way for an open immersion, using VIII.4.4. Finally, if f' is a quasi-compact immersion, f is quasi-compact by virtue of VIII.3.3, so we may apply the criterion VIII.4.5 to the subset $f(X)$ of Y, which proves that $f(X)$ is locally closed since its inverse image $f'(X')$ in Y' is so. Replacing Y by an open subset in which $f(X)$ is closed, we are reduced to the case where f' is a closed immersion, so f is one by what precedes.

Corollary VIII.5.6. *Under the conditions of VIII.5.4, f is affine if and only if f' is.*

We proceed as in VIII.5.5, using VIII.2.1 (One may also use Serre's cohomological criterion [EGA II 5.2], which proves VIII.5.6 without using descent techniques).

Corollary VIII.5.7. *Under the conditions of VIII.5.4, f is integral (respectivelyfinite, respectivelyfinite and locally free) if and only if f' is.*

Only the "if" direction has to be proved, and as usual we may suppose $Y = S$, with Y affine and Y' affine. Since the hypothesis implies that f' is affine, the same is true of f by VIII.5.6, so X and consequently X' are affine. Let $A, A', B, B' = B \otimes_A A'$ be the rings of Y, Y', X, X'. We have $B = \varinjlim_i B_i$, where B_i ranges over the sub-A-algebras of B that are of finite type over A, whence $B' = \varinjlim_i B'_i$, where the B'_i are finite-type subalgebras of the A'-algebra B'. If B' is integral over A', the B'_i are modules of finite type over A'; since A' is faithfully flat over A, the B_i are modules of finite type over A, i.e. B is integral over A. One sees in the same way that if B' is finite over A', B is finite over A. The same conclusion holds for "locally free of finite type", cf. VIII.1.11.

Corollary VIII.5.8. *Under the conditions of VIII.5.4, suppose f quasi-compact and* 215
let $\mathcal{L}$ be an invertible Module on X, and $\mathcal{L}'$ its inverse image on X'. Then $\mathcal{L}$ is ample (respectivelyvery ample) relative to f if and only if $\mathcal{L}'$ is ample (respectivelyvery ample) relative to f'.

Only the "if" direction has to be proved. The hypothesis on $\mathcal{L}$ implies in any case that f' is separated, hence f is separated by VIII.4.8; and since f is quasi-compact and $g\colon Y' \to Y$ is flat, the computation of direct images by affine coverings shows that there are isomorphisms

$$g^*(f_*(\mathcal{L}^{\otimes n})) \xrightarrow{\sim} f'_*(\mathcal{L}'^{\otimes n})$$

for every integer n, and consequently there is an isomorphism

$$g^*(\mathcal{S}) \xrightarrow{\sim} \mathcal{S}',$$

where $\mathcal{S}$ (respectively$\mathcal{S}'$) denotes the quasi-coherent graded Algebra on Y (respectivelyon Y') that is the direct sum of the $f_*(\mathcal{L}^{\otimes n})$ (respectivelythe $f'_*(\mathcal{L}'^{\otimes n})$) for $n \geq 0$. Note that for every $n \geq 0$, the cokernel of the canonical homomorphism $f'^*(\mathcal{S}'_n) \to \mathcal{L}'^{\otimes n}$ is the inverse image under $X' \to X$ of the cokernel of $f^*(\mathcal{S}_n) \to \mathcal{L}^{\otimes n}$, and hence its support Z'_n is the inverse image of the support Z_n. If $\mathcal{L}'$ is ample, the intersection of the Z'_n is empty; hence, since $X' \to X$ is surjective, the intersection of the Z_n is empty, i.e. there is a canonical morphism

$$j\colon X \to \operatorname{Proj}(\mathcal{S})$$

(EGA II 3). Moreover, the analogous morphism

$$j'\colon X' \to \operatorname{Proj}(\mathcal{S}')$$

is none other than the one deduced from the preceding morphism by the base change $Y' \to Y$ (loc. cit.). This being so, to say that $\mathcal{L}'$ is ample relative to f' means that j' is an immersion, necessarily quasi-compact since f' is quasi-compact. Thus, by virtue of VIII.5.5, j is an immersion, i.e. $\mathcal{L}$ is ample relative to f. — We proceed in an entirely analogous manner in the case of "very ample", restricting the above to $n = 1$, and replacing the 216
consideration of $\operatorname{Proj}(\mathcal{S})$ by that of the projective bundle $\mathcal{P}(\mathcal{S}_1)$ associated with $\mathcal{S}_1$.

Recall (EGA II 5.1.1) that a quasi-compact morphism f is called *quasi-affine* if for every affine open subset U of Y, $f^{-1}(U)$ is a prescheme isomorphic to an open subscheme of an affine scheme. It is shown (loc. cit.) that this is equivalent to saying that $\mathcal{O}_X$ is ample (or also: very ample) relative to f. Thus VIII.5.8 implies:

Corollary VIII.5.9. *Under the conditions of VIII.5.4, and assuming f quasi-compact, f is quasi-affine if and only if f' is.*

Remarks VIII.5.10. Hironaka's example of a nonprojective variety shows that one can have a proper morphism $f\colon X \to Y$ of nonsingular algebraic varieties (with Y projective), such that Y is the union of two open subsets Y_i for which $X_i = X \times_Y Y_i$ is projective over Y_i, while f is not projective. Thus, putting $Y' = Y_1 \amalg Y_2$, Y' is faithfully flat and quasi-compact (and even quasi-finite) over Y, and $f'\colon X' \to Y'$ is projective, but f is not projective. One must therefore take care: in order to apply VIII.5.8, and to deduce from the fact that f' is projective the same conclusion for f, one must already have on X' an invertible Module $\mathcal{L}'$ ample for f', *endowed with a descent datum relative to $X' \to X$* (which allows $\mathcal{L}'$ to be regarded as the inverse image of an invertible Module $\mathcal{L}$ on X, which will then be ample for f by virtue of VIII.5.8). When $g\colon S' \to S$ is finite and locally free, however, see VIII.7.7.

VIII.6. Application to finite and quasi-finite morphisms[3]

We shall prove the following two theorems:

Theorem VIII.6.1. *Let $f\colon X \to Y$ be a morphism that is* proper with finite fibers*, with Y locally Noetherian. Then f is finite.*

Theorem VIII.6.2. *Let $f\colon X \to Y$ be a* quasi-finite *and* separated *morphism, with Y locally Noetherian. Then f is quasi-affine, and a fortiori quasi-projective.*

Remarks VIII.6.3. The theorem VIII.6.1 is well known, and is due to Chevalley in 217
the case of algebraic varieties; a simple proof may also be found in [EGA III 4], using the "theorem on holomorphic functions". The proof given here does not use the latter theorem, but instead uses descent theory; we give it as a bonus to the reader, since one

[3]Cf. EGA IV 18.12 for generalizations to preschemes that are not necessarily locally Noetherian

obtains it “for free”at the same time as the proof of VIII.6.2. Recall also ([EGA III 4] or [**1**]) that the global form of Zariski’s “Main Theorem”, deduced from the “theorem on holomorphic functions”, asserts that if $f\colon X \to Y$ is quasi-finite and *quasi-projective*, with Y Noetherian, then X is Y-isomorphic to an open subprescheme of a Y-prescheme *finite* Z. The conjunction of the “Main Theorem”and of VIII.6.2 may therefore be stated as follows:

Corollary VIII.6.4. *Let $f\colon X \to Y$ be a quasi-finite and separated morphism, with Y Noetherian. Then X is Y-isomorphic to an open subprescheme of a finite Y-prescheme Z.*

Another interesting consequence of VIII.6.2 for descent theory will be given along with VIII.7.9.

Proof of VIII.6.1 and VIII.6.2. We shall use the following fact, whose proof is easy[4]:

Lemma VIII.6.5. *Let X be a prescheme of finite type over a locally Noetherian prescheme Y, and let $y \in Y$. In order that there exist an open neighborhood U of y such that $X|U$ is finite (respectivelyquasi-affine, respectively ...) over U, it is necessary and sufficient that $X \times_Y \operatorname{Spec}(()\mathcal{O}_y)$ be finite (respectivelyquasi-affine, respectively ...) over* $\operatorname{Spec}(()\mathcal{O}_y)$.

Since, on the other hand, the property for $f\colon X \to Y$ of being finite, respectivelyquasi-affine, is local on Y, we are reduced, in order to prove VIII.6.1 and VIII.6.2, to the case where Y is the spectrum of a local ring, and hence in particular is of finite dimension. (N.B. by the *dimension of a prescheme Y* we mean the supremum of the Krull dimensions of its local rings). We proceed by induction on

$$n = \dim(Y),$$

the assertion being trivial for $n < 0$. We may therefore suppose $n \geq 0$, and the assertion 218
proved for dimensions $n' < n$. We may again suppose that Y is the spectrum of a Noetherian local ring A, of dimension n. Note that the hypotheses made in VIII.6.1 and VIII.6.2 are stable under base change (this was already used in the initial reduction), so they will remain true after the base change $\operatorname{Spec}(()\widehat{A}) \to \operatorname{Spec}(()A)$. Since the latter is faithfully flat and quasi-compact, the statements VIII.5.7 and VIII.5.9 reduce us to the case where A is moreover complete. Using then the fact that every Noetherian local ring B over A that is quasi-finite over A is finite over A, and the fact that X is separated over Y and the fiber at y consists of isolated points, we obtain a decomposition

$$X = X' \amalg X''$$

where X' is *finite* over Y, and where the fiber of X'' at y is empty. If X is proper over Y, then so is X'', hence its image in Y is closed, and since it does not contain y, it is empty; thus $X'' = \emptyset$ and hence $X = X'$, which shows that X is finite over Y and proves VIII.6.1 (N.B. the induction hypothesis is unnecessary here). If X is quasi-finite over Y, so is X'', but X'' in fact lies over the open subset $Y - \{y\}$ of Y, *which has dimension* $< n$. By the induction hypothesis, X'' is quasi-affine over $Y - \{y\}$, hence also over Y; the same is plainly true of X', and hence also of their sum X, which proves VIII.6.2.

Remark VIII.6.6. Theorems VIII.6.1 and VIII.6.2 remain valid if Y is no longer assumed locally Noetherian, provided one specifies that f is assumed to be of finite presentation (cf. VIII.3.6). Indeed, we may still suppose Y affine, and then one verifies without difficulty that the situation $f\colon X \to Y$ is deduced, by a base change $Y \to Y_0$, from a situation $f_0\colon X_0 \to Y_0$ satisfying the same hypotheses as f, with Y_0 *Noetherian*. Therefore, by VIII.6.1 respectivelyVIII.6.2, f_0 is finite respectivelyquasi-affine, and hence so is f. This kind of argument is often useful for dispensing with Noetherian hypotheses, (which always end up being troublesome in applications). 219

[4] Cf. EGA IV 8

VIII.7. Criteria for the effectivity of a descent datum

As usual, consider a morphism of preschemes

$$g\colon S' \longrightarrow S$$

and an S'-prescheme X'. In accordance with the general facts in VII, 9[5], giving a descent datum on X' relative to g is equivalent to giving an equivalence pair [**3**]

$$q_1, q_2\colon X'' \rightrightarrows X'$$

such that the structural morphism $X' \to S'$ is compatible with this pair and the equivalence pair

$$p_1, p_2\colon S'' = S' \times_S S' \rightrightarrows S'$$

defined by g, and such that the two squares (or either one of them, which amounts to the same thing by symmetry) extracted from the corresponding diagram

$$\begin{array}{ccc} X' & \leftleftarrows & X'' \\ \downarrow & & \downarrow \\ S' & \leftleftarrows & S'' \end{array}$$

(using either p_1, q_1 or p_2, q_2) are *cartesian.* A solution to the descent problem posed by this descent datum, i.e. an object X over S endowed with an isomorphism $X \times_S S' \xleftarrow{\sim} X'$ compatible with the descent data, is equivalent to a *cartesian* square

$$\begin{array}{ccc} X & \xleftarrow{h} & X' \\ \downarrow & & \downarrow \\ S & \xleftarrow{g} & S' \end{array}$$

satisfying $hq_1 = hq_2$.

Since the class of faithfully flat and quasi-compact morphisms is stable under base 220
change, and since a faithfully flat and quasi-compact morphism is an effective epimorphism by VIII.5.3, the general theory [**D**] gives:

Proposition VIII.7.1. *Suppose that $g\colon S' \to S$ is faithfully flat and quasi-compact. A descent datum on X' relative to g is effective if and only if the equivalence relation $R = (q_1, q_2)$ that it defines is effective (i.e. the quotient X'/R exists and X'' becomes the fiber square of X' over X'/R), and the canonical morphism $X' \to X'/R$ is faithfully flat and quasi-compact.*

Thus the question of the effectivity of a descent datum is a special case of the question of the effectivity of an equivalence graph, and various effectivity criteria given in this section may be obtained in this way. Nevertheless, in the context of descent one has Theorem VIII.2.1, which implies that *if X' is affine over S', every descent datum on X' relative to g is effective.* This statement has no analogue for passage to the quotient by a general flat equivalence graph. All the effectivity criteria given here may also be regarded as consequences of that theorem.

Let U' be a subprescheme of X' (or, more generally, a subobject of X' in the category **Sch**). We say that U' is *stable under the descent datum* on X' if U' can be endowed with a descent datum relative to g for which the immersion $U' \to X'$ is compatible with the descent data. This also means that the inverse images of U' in X'' under q_1 and q_2 are the same (or, as one also says, that U' is *stable under the equivalence relation R*). The descent datum in question on U' is then plainly unique and is called the *induced descent datum* from that on X'. With this terminology:

[5]This section does not exist; see Exposé 212 of the Séminaire Bourbaki (1962).

Proposition VIII.7.2. *Let (X'_i) be a covering of X' by open subsets X'_i stable under the descent datum. The descent datum on X' is effective if and only if the induced descent data on all the X'_i are effective.* 221

This is, for example, an immediate consequence of VIII.7.1; the details of the proof are left to the reader.

Corollary VIII.7.3. *Let (S_i) be an open covering of S, and for every i let S'_i and X'_i be deduced from S' and X' by the base change $S_i \to S$. The descent datum on X' is effective if and only if, for every i, the descent datum on X'_i relative to $g_i\colon S'_i \to S_i$ is effective.*

In practice this criterion always reduces us to the case in which S is affine. When S' is affine as well, as is most often the case in applications, we have:

Corollary VIII.7.4. *Suppose that S and S' are affine. The descent datum on X' is effective if and only if X' is the union of affine open subsets X'_i stable under the descent datum.*

Sufficiency follows from VIII.7.2 and the fact that, if X'_i is affine, then it is affine over S', so that VIII.2.1 applies. For necessity, observe that if X' comes from X, and if X is covered by affine open subsets X_i, then the $X'_i = X_i \times_S S'$ are affine open subsets stable under the descent datum and covering X'.

Corollary VIII.7.5. *Let $g\colon S' \to S$ be a faithfully flat, quasi-compact, and* radicial *morphism. Then g is an* effective descent *morphism, i.e. for every X' over S', every descent datum on X' relative to $g\colon S' \to S$ is effective.*

Indeed, by VIII.7.3 we may suppose that S is affine. Since S' is radicial over S, and hence separated over S, it is separated. Moreover, for every $x' \in X'$, the fiber

$$R(x') = q_2(q_1^{-1}(x'))$$

of the set-theoretic equivalence relation defined by R consists of a single point. Indeed, 222
since g is radicial, the same holds for p_1, p_2, which are deduced from g by the base change $S' \to S$, and hence also for q_1, q_2, which are deduced from the former by the base change $X' \to S''$. Therefore *every open subset of X' is stable* under the descent datum. Cover X' by affine open subsets X'_i. They are affine over S', since S' is separated, so the induced descent datum is effective by VIII.2.1. We conclude by VIII.7.2.

Observe that VIII.7.5 gives the only known instance of an effective descent morphism in the category of preschemes, and it is probably the only instance, even if one restricts to noetherian schemes or to schemes of finite type over a field.

When S is assumed locally noetherian and S' of finite type over S, statement VIII.7.5 is also a special case of the following result (which generalizes Weil's Galois descent and Cartier's inseparable descent):

Corollary VIII.7.6. *Let $g\colon S' \to S$ be a finite locally free morphism (i.e. defined by an algebra on S which, as a module, is finite locally free) and suppose that g is surjective (so g is faithfully flat and quasi-compact, and hence a descent morphism). Let X' be an S'-prescheme endowed with a descent datum. This datum is effective if and only if, for every $x' \in X'$, the fiber $R(x') = q_2(q_1^{-1}(x'))$ is contained in an affine open subset (a condition automatically satisfied if X' is quasi-projective over S').*

The parenthetical observation follows from the fact that, if s is the point of S below x', then $R(x')$ is finite and contained in the fiber X'_s, while, since X' is quasi-projective over S' and S' is finite over S, X' is quasi-projective over S; this implies that a fiber of X' over S is contained in an affine open subset.

Since every finite subset of an affine scheme has a fundamental system of affine neighborhoods, the hypothesis is preserved when we restrict over an affine open subset

of S. By VIII.7.3, this reduces us to the case in which S is affine. By VIII.7.4, it is enough to show that x' is contained in an affine open subset *stable* under the descent 223
datum. Indeed, let U be an affine open subset containing $R(x')$. Then the saturation

$$R(X' - U) = q_2(q_1^{-1}(X' - U))$$

does not meet $R(x')$. Moreover, since q_2 is finite (because g, and hence p_2, is finite), and therefore closed, the right-hand side is a closed subset of X'. Let U' be its complement in X'. Thus U' is a *saturated* open subset, and

$$R(x') \subset U' \subset U,$$

where U is affine, although U' need not be affine a priori. Since the finite subset $R(x')$ of the affine scheme U has a fundamental system of affine neighborhoods of the form U_f, one sees, by replacing f by its restriction to U', that there is a section f of $\mathcal{O}_U$ such that

$$R(x') \subset U'_f, \qquad U'_f \text{ is affine.}$$

Set $U'' = q_1^{-1}(U') = q_2^{-1}(U')$, continue to write q_1, q_2 for the induced morphisms $U'' \to U'$, and consider

$$f' = \mathrm{N}_{q_2}(q_1^*(f)),$$

where N_{q_2} denotes the *norm* relative to the finite locally free morphism $q_2 \colon U'' \to U'$. Compatibility of the formation of the norm with base change readily implies that f' is an *invariant* section:

$$q_1^*(f') = q_2^*(f').$$

It follows that $U'_{f'}$ is a saturated open subset of U'. More precisely, writing $Z(f')$ for the zero set of a section f', the properties of norms give

$$Z(f') = q_2(Z(q_1^*(f))) = q_2(q_1^*(Z(f))) = R(U' - U'_f).$$

Consequently, $U'_{f'} = U' - Z(f')$ is saturated, contains $R(x')$, and is contained in U'_f. Since 224
the latter is affine, $U'_{f'}$ is affine as well (it is equal to $(U'_f)_{f''}$, with $f'' = f'|_{U'_f}$). It is therefore a saturated affine open subset containing $R(x')$, and hence x'. This completes the proof.

The same argument applies whenever one has an equivalence relation (or even merely a pre-equivalence relation, cf. [**3**]) on a prescheme X' which is finite and locally free. Moreover, VIII.7.6 is itself a special case of the analogous result for finite locally free pre-equivalence relations; see loc. cit. The same observation applies to VIII.7.7 below.

Alternatively, once a saturated quasi-affine open subset U' containing x' has been obtained, one may invoke VIII.7.9 and VIII.7.2, thereby avoiding the use of norms.

Observe also that under the hypotheses of VIII.7.6, if the descent datum on X' is effective and X' comes from X over S, then the morphism $X' \to X$ is finite, locally free, and surjective, since it is deduced from g by the base change $X \to S$. It follows (EGA II, 6.6.4) that if X' is quasi-projective over S', and hence over S, then X is quasi-projective over S. A relatively ample invertible sheaf on X is obtained by taking the *norm* of an invertible sheaf on X' that is relatively ample over S, or equivalently over S'. Thus we obtain:

Corollary VIII.7.7. *A finite locally free and surjective morphism $g \colon S' \to S$ is an effective descent morphism for the fibered category of preschemes quasi-projective over other preschemes. In other words, for every X' quasi-projective over S', every descent datum on X' relative to g is effective, and the descended S-prescheme X is quasi-projective over S.*

Proposition VIII.7.8. *Let $g \colon S' \to S$ be faithfully flat and quasi-compact. Then g is an effective descent morphism for the fibered category of preschemes Z quasi-compact over a prescheme T, equipped with an invertible sheaf ample relative to T. In particular, let X'* 225 *be an S'-prescheme endowed with a descent datum relative to $g\colon S' \to S$, and let $\mathcal{L}'$ be an invertible sheaf on X', ample relative to S', likewise endowed with a descent datum relative to the given datum on X' (i.e. endowed with an isomorphism between $q_1^*(\mathcal{L}')$ and $q_2^*(\mathcal{L}')$ satisfying the usual transitivity condition). Then the descent datum on X' is effective, and the invertible sheaf $\mathcal{L}$ on the descended prescheme X, obtained from $\mathcal{L}'$ by descent, is ample relative to S.*

The proof is entirely analogous to that of VIII.5.8. On the quasi-coherent graded algebra $\mathcal{S}'$ over S' defined by $\mathcal{L}'$, there is a descent datum. By VIII.1.1, this allows one to construct a quasi-coherent graded algebra $\mathcal{S}$ over S, and hence an S-prescheme $P = \mathrm{Proj}(\mathcal{S})$, such that $P' = \mathrm{Proj}(\mathcal{S}')$, together with its descent datum, is identified with $P \times_S S'$. By hypothesis, X' is identified with an open subset of P', necessarily stable under the descent datum on P'. The descent datum on X' is therefore effective as well, and the descended prescheme is obtained as an open subset of P. The details are left to the reader. In particular, taking $\mathcal{L}' = \mathcal{O}_{X'}$, we find:

Corollary VIII.7.9. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism, and let X' be a prescheme* quasi-affine *over S'. Then every descent datum on X' relative to g is effective, and the descended prescheme X is quasi-affine over S.*

By VIII.6.2, this result applies in particular when S' is locally noetherian and X' is quasi-finite and separated over S', and more generally when S' is arbitrary and X' is of finite presentation, quasi-finite, and separated over S' (cf. VIII.6.6).

Remarks VIII.7.10. The results given in this section exhaust the currently known effectivity criteria, and probably even all the useful criteria that exist[6]. The following counterexamples support this assertion: 226

(i) If S is the spectrum of a field and S' is the spectrum of a quadratic Galois extension, there exists an X' over S', proper and smooth over S', of dimension 3, endowed with a descent datum which is not effective (Serre).
(ii) There exist an S which is the spectrum of a regular local ring of dimension 3 (if desired, the local ring of an algebraic scheme over a field of prescribed characteristic), a principal covering T of S with group $\mathbb{Z}/2\mathbb{Z}$, and the following property. If t denotes one of the points of T above the closed point s of S, and $S' = T - \{t\}$, there exists a regular X', *projective* over S', endowed with a descent datum relative to $g\colon S' \to S$, which is not effective.

These constructions use Hironaka's example of nonprojective varieties. For (i), it is enough to use the fact that there exists over k a proper smooth scheme X_0 of dimension 3 on which $G = \mathbb{Z}/2\mathbb{Z}$ acts without inertia, and in which there are two rational points a, b, over S, conjugate under G, which are not contained in an affine open subset. Set $X' = X_0 \times_k k'$, and let G act on X' by its actions on both factors. This gives a descent datum on X' relative to $g\colon \mathrm{Spec}(()k') \to \mathrm{Spec}(()k)$. Above a, respectively b, there is exactly one point a', respectively b', with quadratic residue-field extension, and a' and b' are conjugate under G, since $X' \to X_0$ is compatible with the actions of G. The points a' and b' cannot be contained in an affine open subset U'. Indeed, otherwise

$$U = X_0 - \mathrm{Im}(X' - U')$$

would be an open subset of X_0 containing a, b, whose inverse image in X' would be contained in U', and hence would be quasi-affine. It would follow that U is quasi-affine, and thus that a, b have an affine neighborhood in U.

[6]This opinion proved partly mistaken; see, for example, J.P. Murre, Sém. Bourbaki 294 (Appendix), May 1965, and special results, notably those of Néron and Raynaud. For the descent of group schemes, cf. M. Raynaud, thesis (1968).

For (ii), one uses the fact that in Hironaka's example X_0 is obtained as a prescheme proper over a projective k-scheme Y, smooth over k. The morphism $f\colon X_0 \to Y$ is in fact birational, but this is irrelevant. The group G also acts on Y compatibly with its action on X_0. Finally, on setting $S' = Y - f(b)$ and $X' = X_0|S'$, the prescheme X' is projective over S'. The prescheme X_0 is then endowed with a natural descent datum relative to the canonical morphism $Y \to S = Y/G$, by virtue of the action of G on X_0 compatible with 227 its action on Y. This descent datum is not effective, since a, b are not contained in an affine open subset. The induced descent datum on X' relative to $g\colon S' \to S$ is then not effective either, as is readily verified.

Bibliography

[D] J. Giraud, Méthode de la descente, Mémoire no. 2 de la Soc. Math. Fr., 1964.
[1] A. Grothendieck, Séminaire Bourbaki: Géométrie formelle et Géométrie algébrique, May 1959, no. 182.
[2] A. Grothendieck, Séminaire Bourbaki: Technique de descente et Théorèmes d'existence I, December 1959, no. 190.
[3] A. Grothendieck, Séminaire Bourbaki: Technique de descente et Théorèmes d'existence III, February 1961, no. 212.

CHAPTER IX

Descent of étale morphisms. Application to the fundamental group

IX.1. Review of étale morphisms 228

We shall review here the properties of étale morphisms (developed in I) that we shall use, taking this opportunity to remove the superfluous noetherian hypotheses from the theory. The reader should note that even if one is interested only in noetherian schemes, the technique of descent leads one to introduce nonnoetherian schemes (such as $\mathrm{Spec}(()\widehat{A} \otimes_A \widehat{A})$, where A is a noetherian local ring). In order to use the language of fibered categories, it is therefore important to define notions such as étale morphisms, etc., without introducing noetherian restrictions. A reader reluctant to verify or accept that the statements below hold without noetherian hypotheses may simply take them under the noetherian hypotheses of I, provided that the same noetherian hypotheses are inserted into the statements of the following sections, and that Definition IX.1.1 below is used for the nonnoetherian schemes that arise in the arguments.

Definition IX.1.1. *Let $f\colon X \to S$ be a morphism of preschemes and x a point of X. We say that f is* étale at x, *or that X is étale over S at x, if there exist an affine open neighborhood U of $s = f(x)$, an affine open neighborhood V of x above U, an affine* noetherian *scheme U_0, an affine U_0-scheme V_0 which is* étale (I)*, a morphism $U \to U_0$, and a U-isomorphism*

$$V \xrightarrow{\sim} V_0 \times_{U_0} U.$$

When S is locally noetherian, this terminology agrees with that of loc. cit. Likewise, we say that *f is étale*, or that *X is étale over S*, if f is étale at every point x of X. With these definitions, the propositions below reduce without difficulty to the noetherian case, where they are proved in I, no. I.4, I.5, and I.7. For details, the reader may consult EGA IV[1]. 229

Remarks IX.1.2. If f is étale at x, then f is *of finite presentation at x* (VIII VIII.3.5), the local ring of x in the fiber $f^{-1}(s)$ is a *finite separable extension* of $\kappa(s)$, and f is *flat at x*. One can prove the converse, so Definition IX.1.1 is the same as in the case where S is locally noetherian, except that the condition *of finite type at x* must be replaced by *of finite presentation at x*. Since this result is delicate to prove, we have not chosen it here as the definition of an étale morphism; it does not lend itself directly to proving the properties that follow.

First, we have trivially:

Proposition IX.1.3. *If $f\colon X \to S$ is étale, then every morphism $f'\colon X' \to S'$ deduced from it by a base change $S' \to S$ is likewise étale.*

Thus one may say that étale morphisms form a *fibered subcategory* of the category of arrows in **Sch** (cf. VI VI.11 a)). The purpose of the present exposé is to study the exactness properties of this category fibered over **Sch**.

[1] More precisely, EGA IV, 17 and 18.

PROPOSITION IX.1.4. *Let $f\colon X \to S$ be a morphism of preschemes. It is an open immersion if and only if it is étale and radicial.*

Cf. I I.5.1. It follows that if X is étale over S, every section of X over S is an open immersion. Applying IX.1.4 once more gives:

COROLLARY IX.1.5. *Let X be an étale S-prescheme. There is a bijection between the set of sections of X over S and the set of open subsets Γ of X for which the morphism* 230
$\Gamma \to S$ induced by the structural morphism is radicial *and* surjective.

If X is separated over S, then Γ is both open and closed in X, but this is immaterial. After an evident base change, IX.1.5 may be put in the following apparently more general form:

COROLLARY IX.1.6. *Let X and Y be two S-preschemes, with Y étale over S. The map $f \longmapsto \Gamma_f$, which assigns to every S-morphism $f\colon X \to Y$ the underlying subset of the graph of f in $X \times_S Y$, is a bijection from $\mathrm{Hom}_S(X, Y)$ onto the set of open subsets Γ of $X \times_S Y$ such that the morphism $\Gamma \to X$ induced by pr_1 is* radicial *and* surjective.

PROPOSITION IX.1.7. *Let S_0 be the subprescheme of S defined by a quasi-coherent nil ideal, i.e. such that S_0 has the same underlying set as S. Then the functor*

$$X \longmapsto X \times_S S_0$$

from the category of preschemes étale over S to the category of preschemes étale over S_0 is an equivalence of categories.

The fact that this functor is fully faithful is an immediate consequence of IX.1.6. Its essential surjectivity is contained in I I.8.3. Observe that under the preceding equivalence, X is of finite type, i.e. quasi-finite, over S (respectively finite, i.e. an étale covering of S) if and only if X_0 satisfies the analogous condition over S_0. The same observation applies to separatedness. These facts are immediate, and are also contained in IX.2.4 below.

COROLLARY IX.1.8. *Let A be a complete noetherian local ring with residue field k. Then the functor*

$$B \longmapsto B \otimes_A k$$

is an equivalence from the category of finite étale algebras over A to the category of finite étale algebras over k (i.e. finite products of finite separable extensions of k).

PROPOSITION IX.1.9. *A prescheme X is an* étale covering *of S, i.e. finite and étale over S, if and only if X is S-isomorphic to the spectrum of an algebra $\mathcal{A}$ on S which is finite locally free as a module and such that, for every $s \in S$, the algebra $\mathcal{A}_s \otimes_{\mathcal{O}_s} \kappa(s)$ is* 231
separable over $\kappa(s)$, and hence in this case is a finite product of finite separable extensions of $\kappa(s)$.

Finally, the following result is less elementary, being the conjunction of I I.8.4 and the *existence theorem for sheaves in algebraic geometry* (EGA III, 5; cf. also [**1**], Th. 3).

THEOREM IX.1.10. *Let S be the spectrum of a complete noetherian local ring, let X be a proper S-scheme, and let X_0 be the fiber of X over the closed point of S, so that X_0 is a closed subscheme of X. Then the restriction functor*

$$X' \longmapsto X' \times_X X_0$$

is an equivalence from the category of étale coverings of X to the category of étale coverings of X_0.

IX.2. Submersive and universally submersive morphisms

DEFINITION IX.2.1. *A morphism $g\colon S' \to S$ of preschemes is called* submersive *if it is surjective and makes S a quotient topological space of S' (i.e. every subset U of S for which $g^{-1}(U)$ is open is itself open). We say that g is* universally submersive *if, for every morphism $T \to S$, the morphism $g'\colon T' = S' \times_S T \to T$ deduced from g by base change is submersive.*

It is immediate that the composite of two submersive (respectively universally submersive) morphisms is submersive (respectively universally submersive), and that a base change of a universally submersive morphism is universally submersive, as the definition was made precisely for this purpose. If fg is submersive (respectively universally submersive), then f is so.

EXAMPLES IX.2.2. *a) A surjective morphism which is open or closed is submersive. Consequently, a surjective universally closed or universally open morphism is universally submersive. For example,* a proper surjective morphism is universally submersive. *Moreover,* a faithfully flat and quasi-compact morphism is universally submersive *(VIII VIII.4.3). These will be the two most important cases for us.*

The arguments of VIII VIII.4.3 may be applied to a submersive or universally sub- 232
mersive morphism $g\colon S' \to S$. In particular, one obtains:

PROPOSITION IX.2.3. *Suppose that $g\colon S' \to S$ is submersive. Then the following diagram of maps is exact:*

$$\mathrm{Ouv}(S) \longrightarrow \mathrm{Ouv}(S') \rightrightarrows \mathrm{Ouv}(S''),$$

where $S'' = S' \times_S S'$, and where $\mathrm{Ouv}(X)$ denotes the set of open subsets of the prescheme X.

PROPOSITION IX.2.4. *Let $g\colon S' \to S$ be universally submersive, let $f\colon X \to Y$ be an S-morphism, and let $f'\colon X' \to Y'$ be the S'-morphism deduced from f by base change. For f to be open (respectively closed), it is sufficient that f' be so. The morphism f is universally open, respectively universally closed, respectively separated, if and only if f' is so. If, in addition, g is quasi-compact and f is locally of finite type, then f is proper if and only if f' is proper.*

For the last assertion, observe that if f' is proper, and hence quasi-compact, then f is quasi-compact (VIII VIII.3.3), and therefore of finite type since it is locally of finite type. Moreover, by what precedes, it is separated and universally closed, and hence proper.

PROPOSITION IX.2.5. *Let S' be a prescheme of finite type over the spectrum S of a complete noetherian local ring. Suppose that the fiber over the closed point s of S is finite, so that the local rings of S' at the points s' of this fiber are finite over $A = \mathcal{O}_s$. Let S'' be the scheme which is the disjoint sum of the spectra of the rings $\mathcal{O}_{S',s'}$ in question, regarded as a finite S-scheme. Then $g\colon S' \to S$ is universally submersive if and only if the structural morphism $S'' \to S'$ is surjective.*

Since there is a natural S-morphism $S'' \to S'$, and a finite surjective morphism is universally submersive by IX.2.2, the stated condition is sufficient. We therefore show that if $S'' \to S'$ is not surjective, then g is not universally submersive. Let t be a point of S not in the image of S''. There is then an S-scheme T, the spectrum of a discrete valuation ring, whose image in S is $\{s, t\}$. The image of S'' in S' is open, since the 233
morphism $S'' \to S'$ is a local isomorphism; moreover, this image contains S'_s and does not meet S'_t. Its inverse image in $T' = S' \times_S T$ is therefore *open*, and is precisely the inverse image of the closed point of T. This shows that $T' \to T$ is not submersive, and hence that $S' \to S$ is not universally submersive.

Remark IX.2.6. Using the criterion IV IV.6.3 for a constructible subset of a noetherian space to be open, one readily obtains the following valuative criterion for a morphism $g\colon S' \to S$ *of finite type*, with S locally noetherian, to be universally submersive: it is necessary and sufficient that, for every S-scheme T which is the spectrum of a discrete valuation ring, on setting $T' = S' \times_S T$, the inverse image in T' of the closed point of T is not open.

IX.3. Descent of étale morphisms of preschemes

Proposition IX.3.1. *Let $g\colon S' \to S$ be a* surjective *morphism of preschemes, let X and Y be two preschemes over S, and let X' and Y' be their inverse images over S'. If Y is unramified over S, then the canonical map*

$$\mathrm{Hom}_S(X,Y) \longrightarrow \mathrm{Hom}_{S'}(X',Y')$$

is injective.

Indeed, by IX.1.6, an S-morphism $f\colon X \to Y$ is determined by the underlying set of its graph Γ, which is a subset of $Z = X \times_S Y$. Since

$$Z' = Z \times_S S' = X' \times_{S'} Y' \longrightarrow Z$$

is surjective, because $S' \to S$ is surjective, the subset Γ is determined by its inverse image in $X' \times_{S'} Y'$, which is precisely the underlying set of the graph of f'. This proves the assertion.

A subset Γ of Z is the graph of an S-morphism $f\colon X \to Y$ if and only if it is open and the morphism $\Gamma \to X$ induced by pr_1 is radicial and surjective; see IX.1. When the 234
first property holds, the second holds if and only if the inverse image Γ' of Γ in Z' satisfies the same condition (VIII VIII.3.1). Finally, if $Z' \to Z$ is known to be submersive, as is in particular the case when $S' \to S$ is universally submersive, then Γ is open if and only if Γ' is open.

Thus $\mathrm{Hom}_S(X,Y)$ is in bijection with the set of open subsets Γ' of Z' for which the projection $\mathrm{pr}_1\colon \Gamma' \to X'$ is radicial and surjective (i.e. those corresponding to an S'-morphism $f'\colon X' \to Y'$), and which are saturated for the equivalence relation defined by $Z' \to Z$. The latter condition says that their two inverse images in

$$Z'' = Z' \times_Z Z' = Z \times_S S'', \qquad S'' = S' \times_S S',$$

under the two projections are equal. These inverse images are the graphs of the two S''-morphisms $X'' \to Y''$ deduced from f' by base change along the two projections $S'' \to S'$. We have therefore proved:

Proposition IX.3.2. *Let $g\colon S' \to S$ be a* universally submersive *morphism of preschemes, let $S'' = S' \times_S S'$, let X and Y be two S-preschemes, let X' and Y' be their inverse images over S', and let X'' and Y'' be their inverse images over S''. If Y is étale over S, then the following canonical diagram of maps is exact:*

$$\mathrm{Hom}_S(X,Y) \longrightarrow \mathrm{Hom}_{S'}(X',Y') \rightrightarrows \mathrm{Hom}_{S''}(X'',Y'').$$

Taking X and Y étale over S, one obtains the following statement, which moreover recovers IX.3.2, even after restricting to $X = S$, a case to which one may always reduce in IX.3.2 by the base change $X \to S$:

Corollary IX.3.3. *A universally submersive morphism of preschemes is a descent morphism for the fibered category of preschemes étale over other preschemes.*

I do not know whether it is necessarily an *effective* descent morphism for the fibered category in question, even under the additional hypotheses that S is noetherian and g is of finite type, and even if one restricts to étale coverings. In the next section we shall nevertheless give useful effectivity criteria.

Corollary IX.3.4. *Let $g\colon S' \to S$ be a universally submersive morphism whose fibers $g^{-1}(s)$ are* geometrically connected*, i.e. such that for every extension $K/\kappa(s)$, the scheme $g^{-1}(s) \otimes_{\kappa(s)} K$ is connected. Then S' is connected whenever S is connected. The functor, defined by g, from the category of preschemes étale over S to the category of preschemes étale over S' is fully faithful.* 235

A subset of S' which is both open and closed is saturated for the set-theoretic equivalence relation defined by g, since the fibers are connected. It is therefore the inverse image of a subset of S, which is necessarily open and closed because g is submersive. Thus if S is connected, so is S'.

This also implies the following fact: if f and g are two morphisms with universally connected fibers, and f is universally submersive, then the composite fg has universally connected fibers. If S'_1 and S'_2 over S have universally connected fibers, then so does $S'_1 \times_S S'_2$. In particular, under the hypotheses of IX.3.4, S'' has universally connected fibers over S.

Now let X and Y be étale over S, and let u' be an S'-morphism from X' to Y'. We prove that it is compatible with the descent data, which gives the desired conclusion by IX.3.3. Let u''_1 and u''_2 be the two S''-morphisms $X'' \to Y''$ deduced from u'. The subprescheme of S'' on which u''_1 and u''_2 coincide is an induced open subprescheme, closed on each fiber, since it is the inverse image of the diagonal prescheme of Y'' over S'' [2]. It is therefore the inverse image of a subset of S. Since it contains the diagonal in S'', it is all of S''. Thus $u''_1 = u''_2$, as required.

IX.4. Descent of étale preschemes: effectivity criteria

Proposition IX.4.1. *Let $g\colon S' \to S$ be a faithfully flat and quasi-compact morphism. Then g is an effective descent morphism for the fibered category of preschemes that are étale, separated, and of finite type over other preschemes.*

Indeed, it is a descent morphism for the fibered category in question, by IX.3.3 or VIII VIII.5.2, as one prefers. It remains to show that if X' is étale, separated, and of finite type over S', and endowed with a descent datum relative to $g\colon S' \to S$, the latter is effective in the fibered category in question. But one sees easily that if X is a prescheme over S, then it is étale over S if and only if it is étale over S' (by Definition IX.1.1 and loc. 236
cit. VIII.3.6). Thus it is étale, separated, and of finite type over S if and only if X' is so over S', cf. for example IX.2.4. It therefore suffices to ensure the effectivity of the descent datum on X for the fibered category of arrows in **Sch**. But X' is quasi-affine over S' by VIII VIII.6.2 and VIII.6.6. We may then conclude by using VIII VIII.7.9. The reader will moreover note that the proof requires less if one restricts oneself to preschemes that are étale and *finite* over other preschemes, since one may then invoke VIII VIII.2.1 directly.

Corollary IX.4.2. *Let $g\colon S' \to S$ be a universally submersive morphism, let X' be an S'-prescheme that is étale, separated, and of finite type, endowed with a descent datum relative to g, let $S_1 \to S$ be a faithfully flat and quasi-compact morphism, and let S'_1 and X'_1 be deduced from S' and X' by base change, so that $S'_1 \to S_1$ is universally submersive, while X'_1 is étale, separated, and of finite type over S'_1, and is endowed with a descent datum relative to $g_1\colon S'_1 \to S_1$. The descent datum on X' is effective if and only if the descent datum on X'_1 is effective.*

This follows from descent theory in categories [**D**], in view of IX.4.1 and IX.3.3.

One proves similarly:

Corollary IX.4.3. *Let $g\colon S' \to S$ be a universally submersive morphism, let X' be an S'-prescheme that is étale and endowed with a descent datum relative to g, and let (S_i) be a covering of S by open subsets. The descent datum is effective if and only if,*

[2] Observe that the fibers of S' over S are separated!

for every i, the corresponding descent datum on $X'_i = X \times_S S_i$, relative to the morphism $g_i \colon S'_i = S' \times_S S_i \to S_i$, is effective.

This last result leads to the following local effectivity criterion:

Proposition IX.4.4. *Let $g \colon S' \to S$ be a morphism of finite presentation* VIII *VIII.3.6 and universally submersive, let X' be a prescheme that is étale and of finite presentation* 237
over S', endowed with a descent datum relative to g, and finally let a be a point of S. In order that there exist an open neighborhood U of a such that the corresponding descent datum on $X'_U = X' \times_S U$ relative to the morphism

$$g_U \colon S'_U = S' \times_S U \to S_U = U$$

is effective, it is necessary and it is sufficient that the corresponding descent datum on $X'_a = X' \times_S \operatorname{Spec}(()\mathcal{O}_a)$, relative to the morphism

$$g_a \colon S'_a = S' \times_S \operatorname{Spec}(()\mathcal{O}_a) \to S_a = \operatorname{Spec}(()\mathcal{O}_a),$$

be effective.

Necessity being trivial, let us prove sufficiency. We thus have an étale prescheme of finite type X_a over S_a, and an isomorphism

$$(*) \qquad X'_a \xrightarrow{\sim} X_a \times_{S_a} S'_a$$

compatible with the descent data. In accordance with a general elementary limit argument for preschemes defined over a direct limit of rings (here the rings A_f, where A is the ring of an affine open neighborhood of a, and where f ranges over the elements of A that do not belong to the prime ideal corresponding to a), one can find an open neighborhood U of a, an étale prescheme of finite type X_U over $U = S_U$, and an S_a-isomorphism $X_a \xrightarrow{\sim} X_U \times_{S_U} S_a$. Moreover, on taking U sufficiently small, one may then suppose that the isomorphism $(*)$ comes from an isomorphism:

$$X'_U \xrightarrow{\sim} X_U \times_{S_U} S'_U;$$

the latter might not be compatible with the descent data; nevertheless, after shrinking U, it will be compatible with the descent data. This completes the proof.

Corollary IX.4.5. *Under the conditions of IX.4.4, the descent datum on X' is effective if and only if, for every $a \in S$, the corresponding descent datum on X'_a, relative to the morphism $S'_a = S' \times_S \operatorname{Spec}(()\mathcal{O}_a) \to \operatorname{Spec}(()\mathcal{O}_a)$, is effective. When S is locally Noetherian and X' is separated over S', one may in the preceding criterion also replace $\mathcal{O}_a$ by its completion.*

The first assertion follows from IX.4.4 and IX.4.3; the second is then a consequence 238
of IX.4.2. Using IX.4.2 once more and the fact that for every Noetherian local ring A, one can find a complete Noetherian local ring B, and a local homomorphism $A \to B$, such that B is flat over A and $B/\mathfrak{m}B$ is a prescribed extension of the residue field $k = A/\mathfrak{m}$ of A, one obtains:

Corollary IX.4.6. *Under the conditions of IX.4.4, suppose in addition that X' is separated over S', and that S is locally Noetherian. The descent datum on X' is effective if and only if, for every prescheme S_1 over S that is the spectrum of a complete local ring with algebraically closed residue field, the corresponding descent datum on $X'_1 = X' \times_S S_1$, relative to the morphism $g_1 \colon S'_1 \to S_1$, is effective.*

Theorem IX.4.7. *Let $g \colon S' \to S$ be a finite and surjective morphism of finite presentation (the latter hypothesis being a consequence of the others if S is locally Noetherian)*[3]*. Then g is an effective descent morphism for the fibered category of preschemes that are étale, separated, and of finite type over other preschemes.*

[3] One can show that it in fact suffices for g to be an *integral* morphism, by reducing to the case in the text through a limit process in the style of EGA IV 8.

We must show that if X' is étale, separated, and of finite type over S', and endowed with a descent datum relative to g, then this datum is effective. Using IX.4.3, one reduces easily to the case where S is Noetherian. By IX.4.5, we may therefore suppose that S is the spectrum of a Noetherian local ring, and hence in particular that

$$\dim S = n < +\infty.$$

We then argue by induction on $\dim S = n$, the assertion being trivial for $n < 0$. Suppose therefore that $n \geq 0$ and that the theorem has been proved for dimensions $n' < n$. By IX.4.6, we are reduced to the case where S is the spectrum of a complete local ring, so S' is a finite union of spectra of complete local rings. We therefore have

$$X' = X'_1 \sqcup X'_2$$

where X'_1 is *finite* over S', and where X'_2 has no point above any of the closed points of S'. 239
Consider the morphisms

$$q_1, q_2 \colon X'' \rightrightarrows X'$$

corresponding to the descent datum, compatible with $p_1, p_2 \colon S'' \rightrightarrows S'$. One sees at once that

$$X'' = q_i^{-1}(X'_1) \sqcup q_i^{-1}(X'_2) \quad i = 1, 2$$

is the analogous canonical decomposition of X'' over S'', which implies $q_1^{-1}(X'_1) = q_2^{-1}(X'_1)$, and consequently X'_1 and X'_2 are endowed with induced descent data. Now let T be the open subset of S complementary to its closed point; then $T' = S' \times_S T$ is the part of S' complementary to the set of closed points, and X'_2, which lies entirely over T', is endowed with a descent datum relative to the morphism $T' \to T$ induced by g. Since the latter is finite and surjective, and $\dim T < \dim S = n$, this descent datum is effective by the induction hypothesis. Thus one sees that it suffices to prove that the descent datum on X'_1 is effective, so we may now suppose that X' is étale and *finite* over S'. (N.B. the inductive argument is unnecessary if one restricts to étale coverings in the statement IX.4.7). Let S_0 then be the spectrum of the residue field of A, let $S'_0 = S' \times_S S_0$, and define S''_0, S'''_0 similarly from the fibered squares and cubes S'' and S''' of S' over S. By IX.1.8, the morphisms $S_0 \to S$, $S'_0 \to S'$, etc. induce equivalences for the categories of étale coverings of S and S_0, on the one hand, and of S' and S'_0 on the other hand, etc. By the standard consequences of descent theory in categories [**D**], it follows that $g \colon S' \to S$ is an effective descent morphism for the fibered category of étale coverings if and only if the same is true of $g_0 \colon S'_0 \to S_0$. But this is indeed the case, for example as a special case of IX.4.1. This completes the proof.

Corollary IX.4.8. *The conclusion of IX.4.7 remains valid if one assumes only that $S' \to S$ is universally submersive, of finite type, and quasi-finite, provided S is assumed locally Noetherian.*

Indeed, by IX.4.6, we may suppose that S is the spectrum of a complete Noetherian local ring. Then, by IX.2.5, there exists a finite and surjective morphism $S_1 \to S$, and an S-morphism $S_1 \to S'$. Since $S_1 \to S$ is a universal strict descent morphism for the fibered category under consideration, by IX.4.7, and $S' \to S$ is a universal descent morphism for that category, IX.4.8 follows from the general formal consequences [**D**]. 240

Corollary IX.4.9. *Let $g \colon S' \to S$ be a morphism of finite type, surjective and universally open, with S locally Noetherian. Then g is an effective descent morphism for the fibered category of preschemes that are étale, separated, and of finite type over other preschemes.*

Proceeding as in IX.4.7, we reduce to the case where S is the spectrum of a complete Noetherian local ring A. Let A_1 be a finite algebra over A, with spectrum S_1, such that $S_1 \to S$ is finite and *surjective*, and hence a universal effective descent morphism for the fibered category under consideration, by IX.4.7. It then follows from the general theorems [**D**] that g is an effective descent morphism for the fibered category under consideration if and only if the corresponding morphism $g_1 \colon S'_1 = S' \times_S S_1 \to S_1$ is so. Since the latter satisfies the same hypotheses as g, we are reduced to proving IX.4.9 with S_1 in place of S. Taking first for A_1 the direct product of the $A/\mathfrak{p}_i$, for the minimal prime ideals $\mathfrak{p}_i$ of A, we are reduced to the case where A is an *integral domain.* One then shows[4] that there exists an integral subscheme S_1 of S', quasi-finite over S and dominating S, passing through a point of the fiber of S' over the closed point y of S (using the fact that S' is universally open and of finite type over the integral Noetherian local scheme S, and $S'_y \neq \emptyset$). Since A is complete, S_1 is finite over S, and since it dominates S, the morphism $S_1 \to S$ is surjective. Replacing S once more by S_1, we are reduced to the case where S' has a section over S, in which case the statement is trivial.

Theorem IX.4.10. *Let $g \colon S' \to S$ be a finite, radicial, and surjective morphism of finite presentation (the latter condition being superfluous if S is locally Noetherian[5]). Then the inverse-image functor induces an equivalence from the category of preschemes that are étale over S to the category of preschemes that are étale over S'.* 241

Since the diagonal morphisms from S' into $S' \times_S S'$ and $S' \times_S S' \times_S S'$ are surjective immersions, and hence by IX.1.9 induce equivalences from the categories of preschemes that are étale over $S' \times_S S'$ respectively$S' \times_S S' \times_S S'$ to the category of preschemes that are étale over S', it follows from the standard consequences of descent [**D**] that every X' étale over S' is endowed with one and only one descent datum relative to $g \colon S' \to S$. Thus IX.3.3 implies that the inverse-image functor under g, from the category of preschemes that are étale over S to the category of preschemes that are étale over S', is *fully faithful.* It remains to show that it is essentially surjective, i.e. that every X' étale over S' is isomorphic to the inverse image of an X étale over S. Since the question is plainly local on S *and on* X', we may suppose that S, S', X' are affine. But then X' is separated and of finite type over S', and we may apply the effectivity criterion IX.4.7.

Corollary IX.4.11. *The conclusion of IX.4.9 remains valid if the hypothesis on g is replaced by: g is faithfully flat, quasi-compact, and radicial.*

The same proof applies, invoking IX.4.1 instead of IX.4.7.

Note that the proof of IX.4.7 is "elementary"in that it does not use the theorems on finiteness and comparison for proper morphisms (EGA III 3, 4, 5). The same is not true of the following result:

Theorem IX.4.12. *Let $g \colon S' \to S$ be a proper and surjective morphism of finite presentation (the latter hypothesis being a consequence of the first if S is locally Noetherian). Then g is an effective descent morphism for the fibered category of étale coverings of preschemes.*

By IX.3.3 and IX.2.2, we are reduced to proving that for every étale covering X' over S', endowed with a descent datum relative to $g \colon S' \to S$, this descent datum is effective. Using IX.4.3, we reduce easily to the case where S is Noetherian, and using IX.4.6, we 242
may therefore suppose that S is the spectrum of a *complete* Noetherian local ring A. Introduce S'' and S''' as usual, let S_0 be the spectrum of the residue field of A, and let S'_0, S''_0, S'''_0 be deduced from S', S'', S''' by the base change $S_0 \to S$, i.e. the fibers of S', S'', S''' at the closed point of S. By IX.1.10, the morphisms $S_0 \to S$, $S'_0 \to S'$, etc. induce equivalences from the category of étale coverings over the target scheme to the category of étale coverings over the source scheme. Consequently, $g \colon S' \to S$ is a strict

[4]Cf. EGA IV 14.3.13 and 14.5.4.

[5]It even suffices for g to be integral, radicial, and surjective, as follows from an easy reduction to the case in the text, in the style of EGA IV 8, cf. SGA 4 VIII 1.1.

descent morphism for the fibered category of étale coverings of preschemes if and only if $g_0\colon S'_0 \to S_0$ is so, which is indeed the case by IX.4.1. This completes the proof of IX.4.12. (In this argument, from IX.1.10 we needed only the fact that the functor considered in IX.1.10 is *fully faithful*, which does *not* use the existence theorem for coherent sheaves in algebraic geometry.)

IX.5. Translation in terms of the fundamental group

Let

$$g\colon S' \to S$$

be an *effective descent morphism* for the fibered category of *étale coverings* of preschemes, for example a proper, surjective morphism of finite presentation (IX.4.12), or a faithfully flat and quasi-compact morphism. Introducing S'', S''' as usual, and denoting by $\mathcal{C}$, $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$ the categories of étale coverings of S, S', S'', and S''', respectively, we thus have a 2-exact diagram of categories

$$(*) \qquad \mathcal{C} \xrightarrow{p^*} \mathcal{C}' \overset{p_1^*,p_2^*}{\rightrightarrows} \mathcal{C}'' \overset{p_{21}^*,p_{32}^*,p_{31}^*}{\substack{\longrightarrow\\[-1ex]\longrightarrow\\[-1ex]\longrightarrow}} \mathcal{C}'''$$

corresponding to the diagram

$$S \xleftarrow{p} S' \overset{p_1,p_2}{\leftleftarrows} S'' \overset{p_{21},p_{32},p_{31}}{\substack{\longleftarrow\\[-1ex]\longleftarrow\\[-1ex]\longleftarrow}} S'''.$$

Suppose that the preschemes S, S', S'', and S''' are disjoint unions of connected preschemes, 243
as will be the case in particular if they are locally connected preschemes, and a fortiori if they are locally noetherian (for example, if S' is of finite type over the locally noetherian prescheme S). The categories $\mathcal{C}$, $\mathcal{C}'$ ... in $(*)$ are then multigalois categories V V.9, each described by a collection of compact totally disconnected topological groups, namely the fundamental groups of the connected components of the preschemes S, S', S'', and S'''. For simplicity, we suppose that S is connected, and we shall then give a procedure for computing its fundamental group in terms of the fibered category formed from $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$, suitably made explicit by means of the fundamental groups describing these categories. The reader will note that the procedure outlined is in fact valid in the general framework of multigalois categories (which need not arise from given preschemes S, S', S'', and S'''). It is moreover the analogue of the well-known procedure for computing the fundamental group of a topological space S that is a locally finite union of closed subspaces S_i (or an arbitrary union of open subspaces S_i), by means of the fundamental groups of the components of the S_i and of the components of the $S_i \cap S_j$. Of course, the analogous situation for preschemes does fall within the general framework of descent, by introducing the prescheme S' that is the disjoint union of the S_i and the canonical morphism $g\colon S' \to S$.

Set

$$E' = \pi_0(S'), \quad E'' = \pi_0(S''), \quad E''' = \pi_0(S'''),$$

where π_0 denotes the "set of connected components" functor. Since the fiber products of S' over S form a simplicial object of **Sch**, it is transformed by the functor π_0 into a simplicial set of which E', E'', and E''' are the components in dimensions 0, 1, and 2. We shall have to use the simplicial maps

$$q_i = \pi_0(p_i), \quad (i = 1, 2) \quad \text{and} \quad q_{ij} = \pi_0(p_{ij}), \quad (i, j) = (2, 1), (3, 2), (3, 1),$$

displayed in the diagram

$$(1) \qquad E' \overset{q_1,q_2}{\leftleftarrows} E'' \overset{q_{21},q_{32},q_{31}}{\substack{\longleftarrow\\[-1ex]\longleftarrow\\[-1ex]\longleftarrow}} E'''.$$

The elements of E' will be denoted with one prime, as in s', while those of E'' respectivelyE''' 244
will be denoted with $''$ respectivelywith $'''$. The fact that S is connected is expressed by

$\pi_0(K) = 0$, where K is the simplicial set defined by $g\colon S' \to S$, or equivalently by the fact that the equivalence relation on E' generated by the pair of maps (q_1, q_2) is transitive.

We shall choose once and for all an element s'_0 in E', and, for every s' in E', an element $\overline{s'} \in E''$ such that[6]

$$q_1(\overline{s'}) = s'_0, \quad q_2(\overline{s'}) = s',$$

thus exhibiting the connectedness of S. For every $s' \in E'$, choose a geometric point $\underline{s}'$ in the connected component s' of S'; this point will in fact enter through the corresponding fiber functor $F'_{s'}$ on the multigalois category $\mathcal{C}'$. The automorphism group of this functor, i.e. the fundamental group of S' at $\underline{s}'$, will be denoted $\pi_{s'}$. We similarly choose $\underline{s}''$ and $\underline{s}'''$, hence functors $F''_{s''}$ and $F'''_{s'''}$, and thus fundamental groups $\pi_{s''}$ and $\pi_{s'''}$. Hence

$$\pi_{s'} = \pi_1(S', \underline{s}'), \quad \pi_{s''} = \pi_1(S'', \underline{s}''), \quad \pi_{s'''} = \pi_1(S''', \underline{s}''').$$

For every $s'' \in E''$, $p_1(\underline{s}'')$ lies in the same connected component as $\underline{q_1(s'')}$, so there exists an isomorphism of functors $F''_{s''} \circ p_1^* \xrightarrow{\sim} F'_{s'}$ (i.e. a "path class" from $p_1(\underline{s}'')$ to $\underline{q_1(s'')}$). This observation applies likewise to q_2 and to the q_{ij}. Choose all these path classes:

$$F''_{s''} \circ p_i^* \xrightarrow{\sim} F'_{q_i(s'')}, \quad F'''_{s'''} \circ p_{ij}^* \xrightarrow{\sim} F'''_{q_{ij}(s''')},$$

(for $i = 1, 2$ and $(i, j) = (2, 1), (3, 2), (3, 1)$). In particular, this yields group homomorphisms: 245

$$\text{(2)} \qquad q_i^{s''}: \pi_{s''} \to \pi_{q_i(s'')}, \quad q_{ij}^{s'''}: \pi_{s'''} \to \pi_{q_{ij}(s''')},$$

(with the same values of i and (i, j)). Finally, recall that the structure of a cloven category with fibers $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$ also includes isomorphisms of functors:

$$p_{21}^* p_1^* \xrightarrow{\sim} p_{31}^* p_1^*, \quad p_{21}^* p_2^* \xrightarrow{\sim} p_{32}^* p_1^*, \quad p_{31}^* p_2^* \xrightarrow{\sim} p_{32}^* p_2^*,$$

deduced from isomorphisms of the two sides, respectively, with the u_i^* ($i = 1, 2, 3$), where the u_i are the three projections from S''' to S'. When these data are made explicit, one obtains, for every s''', a well-determined element

$$\text{(3)} \qquad a_i^{s'''} \in \pi_{v_i(s''')},$$

(where the v_i, $i = 1, 2, 3$, are the three maps $E''' \to E'$ defined by $v_i = \pi_0(u_i)$), which moreover satisfies the conditions:

$$q_1^{s''_1} q_{21}^{s'''} = \mathrm{int}(a_1^{s'''}) q_1^{s''_2} q_{31}^{s'''} \qquad (s''_1 = q_{21}(s'''),\ s''_2 = q_{31}(s''')),$$

and the two analogous conditions involving a_2 and a_3. The reader will moreover note that the data (1), (2), and (3) allow one to reconstruct, up to an equivalence of fibered categories, the fibered category under consideration with fibers $\mathcal{C}'$, $\mathcal{C}''$, and $\mathcal{C}'''$. They must therefore, in principle, allow one to reconstruct $\mathcal{C}$ up to equivalence, and hence its fundamental group up to isomorphism. In fact, we shall determine the fundamental group at the geometric point $p(\underline{s}'_0)$ of S, i.e. the automorphism group of $F'_{s'_0} \circ p^*$.

Note that the datum of an object X' of $\mathcal{C}'$ is essentially equivalent to the datum of finite sets $X'_{s'}$ ($s' \in E'$) on which the $\pi_{s'}$ act continuously. A gluing datum on such an 246
object then amounts to the datum, for every $s'' \in E''$, of a bijection

$$\varphi_{s''}: X'_{q_1(s'')} \xrightarrow{\sim} X'_{q_2(s'')}$$

[6]Care must be taken here: the element $\overline{s'}$, whose existence is implicitly assumed, does not exist in every case. Consequently, Theorem IX.5.1, as stated, does not apply in every case. It is not difficult, however, by following the ideas of the written text, to modify the statement of this theorem so that it gives a computational method applicable in every case. In particular, the corollaries of the said theorem remain valid as stated.

compatible with the actions of $\pi_{s''}$, acting on the two sides through the homomorphisms $q_i^{s''}: \pi_{s''} \to \pi_{q_i(s'')}$. Taking first the s'' of the form $\overline{s'}$, one sees that such a datum defines bijections

$$\psi_{s'}: X'_{s'_0} = F'_0(X') \to X'_{s'}$$

which allow the $X'_{s'}$ to be identified with the same set $F'_0(X') = X'_{s'_0}$, on which all the groups $\pi_{s'}$ will henceforth act. With this understood, the bijections $\varphi_{s''}$ correspond to bijections

$$g_{s''}: F'_0(X') \xrightarrow{\sim} F'_0(X'),$$

subject, on the one hand, to the commutation relations with $\pi_{s''}$:

a) $$g_{s''} q_1^{s''}(g'') = q_2^{s''}(g'') g_{s''} \qquad (s'' \in E'',\ g'' \in \pi_{s''}),$$

and, on the other hand, to the relations

b) $$g_{\overline{s'}} = g_{\overline{s'_0}} \qquad (s' \in E'),$$

which express the manner in which we identified the $X_{s'}$ with one another. On making explicit the condition for such a gluing datum to be a descent datum, one obtains the relations

c) $$a_3^{s'''} g_{q_{31}(s''')} a_1^{s'''} = g_{q_{32}(s''')} a_2^{s'''} g_{q_{21}(s''')} \qquad (s''' \in E''').$$

This gives an equivalence between the category of objects of $\mathcal{C}'$ equipped with a descent datum and the category of finite sets on which the groups $\pi_{s'}$ act continuously, 247
equipped in addition with bijections $g_{s''}$ satisfying the relations A), B), and C). Let G be the group generated by the groups $\pi_{s'}$ and the new generators $g_{s''}$, subject to the relations A), B), and C), and let π be the inverse-limit group of the quotients of G by the subgroups of finite index whose inverse images in the groups $\pi_{s'}$ are open subgroups. One also says that π is the *group of Galois type generated by the $\pi_{s'}$ and the $g_{s''}$, subject to the relations A), B), and C).* It is immediately seen that the category under consideration is also equivalent to the category of finite sets on which the topological group π acts continuously. This proves the following statement:

Theorem IX.5.1. *Let $g: S' \to S$ be a morphism of preschemes which is an effective descent morphism for the fibered category of étale coverings of preschemes (cf. IX.4.9 and IX.4.12). Suppose that S is connected, and that S', its fiber square S'', and its fiber cube S''' are disjoint unions of connected preschemes (as is the case, for example, if S' is of finite type over a locally noetherian connected S). Choose as above: a geometric point in every connected component of S', S'', and S''', certain path classes, an $s'_0 \in E'$, and, for every $s' \in E'$, an $s'' \in E''$ whose two images in E' are s'_0 and s'. (E', E'', and E''' denote respectively the sets of connected components of S', S'', and S'''.) Then the fundamental group of S at the geometric point that is the image of s'_0 is canonically isomorphic to the group of Galois type generated by the $\pi_{s'} = \pi_1(S', \underline{s}')$ $(s' \in E')$ and generators $g_{s''}$ $(s'' \in E'')$, subject to the above relations A), B), and C), which involve the elements of the groups $\pi_{s''} = \pi_1(S'', \underline{s}'')$, and the elements $a_i^{s'''}$ $(i = 1, 2, 3,\ s''' \in E''')$ introduced above.*

Corollary IX.5.2. *Suppose that S' and S'' have only finitely many connected components, and that the fundamental groups of the connected components of S' are topologically finitely generated. Then the fundamental group of S is topologically finitely generated.*

Thus, we shall prove later that the fundamental group of a normal projective scheme 248
over an algebraically closed field is topologically finitely generated. Using Chow's lemma and the normalization of algebraic schemes, it will follow that the same result holds for every proper scheme over an algebraically closed field.

Corollary IX.5.3. *Suppose that S', S'', and S''' have only finitely many connected components, that the fundamental groups of the connected components of S' are topologically* finitely presented*, and that the fundamental groups of the connected components of S'' are topologically* finitely generated*. Then the fundamental group of S is topologically* finitely presented*.*

Note that IX.4.9 (restricted to étale *coverings*) may be expressed by saying that *a finite radicial surjective morphism of noetherian preschemes induces an isomorphism of fundamental groups*; figuratively speaking, one may therefore say that the fundamental group is a *topological invariant* for preschemes. More generally, using IX.5.1, one can make explicit the effect on the fundamental group of operations on preschemes, such as the "pinching"of a prescheme along a finite set of points, which has a simple topological meaning. For example, one obtains:

Corollary IX.5.4. *Let $g\colon S' \to S$ be a finite morphism of finite presentation, and let T be a discrete subset of S. For every $s \in S$, let $n(s)$ be the "geometric number of points"in the fiber $g^{-1}(s)$ (which may also be described as the separable degree of $g^{-1}(s)$ over $\kappa(s)$, the sum of the separable degrees of its residue extensions). Suppose that $n(s) = 1$ for $s \in S - T$. For every $s \in T$, let K_s be an algebraically closed extension of $\kappa(s)$, let I_s be the set of geometric points of S' with values in K_s (a set with $n(s)$ elements), let I'_s be the complement of a chosen point of I_s, and finally let I' be the union of the I'_s. Suppose that S' is connected. Then the fundamental group of S is isomorphic to the group of Galois type generated by the fundamental group of S' and generators g_i ($i \in I'$), subject to no additional relations.*

The details of the proof are left to the reader; the resulting statement is merely the 249
translation into the language of group theory of the fact that the category C of étale coverings of S is equivalent to the category of étale coverings X' of S', *equipped*, for every $s \in T$, with a transitive system of bijections between the $n(s)$ fibers of X' at the points of $g^{-1}(s)$ with values in K_s. (In this intrinsic form, of course, it is no longer necessary to suppose that S' is connected.)

Example IX.5.5. *It is easy to prove that the rational curve $\mathbb{P}^1_k$ over an algebraically closed field k is simply connected*[7]*. Hence the fundamental group of a complete rational curve having exactly one double point, with n analytic branches, is the free group of Galois type generated by $n - 1$ generators. For example, in the case of an ordinary double point, one obtains the fundamental group $\hat{\mathbb{Z}}$, as announced in (I I.11 a)). By contrast, the presence of a cusp (which is a "geometrically unibranch"point) has no effect on the fundamental group.*

Corollary IX.5.6. *Let $g\colon S' \to S$ be a universally submersive morphism of preschemes with geometrically connected fibers, where S is connected. Then S' is connected, and, on choosing a geometric point s' of S' and denoting its image in S by s, the homomorphism*

$$\pi_1(S', s') \to \pi_1(S, s)$$

is surjective. *If g is an effective descent morphism for the fibered category of étale coverings of preschemes (cf.* IX.4.12*), introduce the geometric point $s'' = \mathrm{diag}(s')$ of $S'' = S' \times_S S'$ and the two homomorphisms*

$$p_{1*}, p_{2*}\colon \pi_1(S'', s'') \to \pi_1(S', s')$$

induced by the two projections. Then $\pi_1(S, s)$ is isomorphic to the cokernel of this pair of homomorphisms in the category of groups of Galois type, i.e. to the quotient of $\pi_1(S', s')$ by the closed normal subgroup generated by the elements of the form $p_{1*}(g'')p_{2*}(g'')^{-1}$, 250
with $g'' \in \pi_1(S'', s'')$.

[7] Cf. Exp. XI XI.1.1.

Indeed, by IX.3.4, the functor $X \longmapsto X \times_S S'$ from étale coverings of S to étale coverings of S' is fully faithful, which is equivalent to the homomorphism on fundamental groups being an epimorphism (V V.6.9). The last assertion follows immediately from the description in IX.5.1.

Remark IX.5.7. It is not known at present whether the fundamental group of a proper scheme over an algebraically closed field k is topologically finitely presented[8]. Using IX.5.3, a well-known hyperplane-section technique, and the resolution of singularities of normal surfaces, one is reduced to the case of a *smooth surface over* k. This at least makes it possible to show, by transcendental methods, that the answer is affirmative in characteristic 0 (without having to assume the triangulability of singular algebraic varieties). In characteristic $p > 0$, the principal difficulty seems to lie in the case of curves, whose fundamental group is known only to be a quotient of the group arising in the classical case (cf. the following exposé), while the kernel by which one takes the quotient is very poorly understood.

Remark IX.5.8. One could make explicit other special cases besides IX.5.4 and IX.5.6 in which IX.5.1 takes a particularly simple form. An interesting case is that in which S is the quotient of S' by a finite group of automorphisms Γ. *Then the category of étale coverings of S' is equivalent to the category of étale coverings X' of S' on which the group Γ acts compatibly with its action on S', in such a way that for every $s' \in S'$ and every $g \in \Gamma_{s'}$* (where $\Gamma_{s'}$ denotes *the inertia group* of s' in Γ), *g acts trivially on the fiber $X'_{s'}$*. If S' is connected, this statement is interpreted as follows. Let C'_0 be the category of étale coverings of S' on which Γ acts compatibly with its action on S' (but without necessarily satisfying the preceding condition on the inertia groups of the points of S'). One sees easily that this is a Galois category (V V.5), and that for every geometric point a' of S', 251
the fiber functor $X' \longmapsto X'_{a'}$ on C'_0 is a fundamental functor. Let $\pi_1(S', \Gamma; a') = G$ be the group of automorphisms of this functor, endowed with its usual topology. There is then a canonical exact sequence

$$e \to \pi_1(S', a') \to G \to \Gamma \to e$$

(a special case of (V V.6.13), where one takes for S the trivial covering $S' \times \Gamma$ of S' defined by Γ, on which Γ is made to act in the evident way). Moreover, for every geometric point b' of S', there is an isomorphism $\pi_1(S', \Gamma; b') \to G = \pi_1(S', \Gamma; a')$, defined up to an inner automorphism arising from $\pi_1(S', a')$; and since $\Gamma_{b'}$ maps in the evident way into the first group, one obtains a homomorphism

$$u_{b'} : \Gamma_{b'} \to G,$$

defined up to an inner automorphism (arising from $\pi_1(S', a')$), whose composite with the canonical homomorphism $G \to \Gamma$ is the canonical inclusion $\Gamma_{b'} \to \Gamma$. With this understood, *the fundamental group* $\pi_1(S, a)$ is *canonically isomorphic to the quotient of* $G = \pi_1(S', \Gamma; a')$ *by the closed normal subgroup generated by the images of the homomorphisms* $\Gamma_{b'} \to G$. In particular, the image of $\pi_1(S', a')$ in $\pi_1(S, a)$ is a normal subgroup, and the corresponding quotient is isomorphic to a quotient of Γ. One can moreover reduce the number of "relations"introduced by considering, for every $g \in \Gamma$, $g \neq e$, the coincidence subprescheme S'_g of the automorphisms id_S and g of S, choosing a geometric point $b'_{g,i}$ in each connected component of S'_g, and then choosing one of the corresponding homomorphisms $\pi_1(S', \Gamma; b'_{g,i}) \to G$, thereby obtaining lifts $\bar{g}_i$ of g in G. It then suffices to take the quotient of G by the closed normal subgroup of G generated by the $\bar{g}_i$.

When a' is fixed by Γ, one sees easily that Γ acts naturally on $\pi_1(S', a')$, and G identifies with the corresponding semidirect product. Identifying Γ with a subgroup of G,

[8] This seems very unlikely in the case of smooth curves of genus $g \geq 2$ in characteristic $p > 0$. On the other hand, when π_1 is replaced by its largest prime-to-p quotient, it appears that well-known techniques give an affirmative answer, even without the properness hypothesis. Cf. forthcoming work of J.P. Murre.

one sees that among the relations introduced above, on taking $b' = a'$, one finds "$g = e$"for 252
$g \in I$. Thus *if* S' *has a geometric point* a' *fixed by* I (i.e. a point s' whose inertia group is Γ), *then* $\pi_1(S, a)$ *is a quotient group of the Galois-type quotient group of* $\pi_1(S', a')$ *obtained by* "making trivial" *the actions of* Γ *on* $\pi_1(S', a')$*; and it is even isomorphic to this latter group if one assumes that for every* $g \in G$*, the inertia locus* S'_g *is connected, and therefore contains the underlying point of* a'. Indeed, this last assertion is contained in the second description given above of the relations to be imposed on G.

This last result applies in particular if one takes for S' the Cartesian power X^n of a connected prescheme over an algebraically closed field, and for Γ the symmetric group $\Gamma = \mathfrak{S}_n$, acting in the usual way, so that S is the symmetric power of degree n of X. Taking for a' a geometric point whose underlying point lies on the diagonal, one is under the preceding conditions, since all the inertia loci S'_g contain the diagonal. Using the fact, proved in the following exposé, that if X is proper and connected over k, the fundamental group of X^n identifies with $\pi_1(X)^n$, one obtains the following amusing result: *If* X *is proper and connected over an algebraically closed field* k*, the fundamental group of its symmetric power of degree* n*,* $n \geq 2$*, is isomorphic to the fundamental group of* X *made abelian.* (I do not know whether the analogous fact in algebraic topology is known; it should be possible to establish it by the same descent method.) Take, for example, the rational curve $X = \mathbb{P}^1_k$; one obtains yet another proof of the fact that $\mathbb{P}^r_k$ is simply connected, using the fact that $\mathbb{P}^1_k$ is so. Now take for X a nonsingular curve over k, and let $n \geq 2g - 1$, so that $\mathrm{Sym}^n(X)$ is fibered over the Jacobian J, with projective spaces as fibers, and hence (as will be seen using the results of the next two exposés) has the same fundamental group as J. One thereby recovers, without dévissage, the well-known fact that *the fundamental group of the Jacobian of* X *is isomorphic to the fundamental group of* X *made abelian.*

IX.6. A fundamental exact sequence. Descent by morphisms with relatively connected fibers 253

THEOREM IX.6.1. *Let* S *be the spectrum of an Artinian ring* A *with residue field* k*, let* $\overline{k}$ *be an algebraic closure of* k*, let* X *be an* S*-prescheme, set*

$$X_0 = X \otimes_A k, \qquad \overline{X}_0 = X \otimes_A \overline{k},$$

and let $\overline{a}$ *be a geometric point of* $\overline{X}_0$*, with image* a *in* X *and image* b *in* S*. Suppose that* X_0 *is quasi-compact and* geometrically connected *over* k*. (If* X *is proper over* S*, this means that* $\mathrm{H}^0(X_0, \mathcal{O}_{X_0})$ *is a* local *Artinian ring whose residue field is* radicial *over* k*.) Then the sequence of canonical homomorphisms*

$$e \longmapsto \pi_1(\overline{X}_0, \overline{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(S, b) \longrightarrow e$$

is exact, and

$$\pi_1(S, b) \xleftarrow{\sim} \pi_1(k, \overline{k}) = \textit{the Galois group of } \overline{k} \textit{ over } k.$$

Since fundamental groups are unchanged on killing nilpotent elements, we may suppose that $A = k$, which already makes the last isomorphism evident. Let k' be the separable closure of k in $\overline{k}$, consider $X' = X \otimes_k k'$, and let a' be the image of $\overline{a}$ in X'. There is a sequence of canonical homomorphisms

$$e \longmapsto \pi_1(\overline{X}_0, \overline{a}) \longrightarrow \pi_1(X', a') \longrightarrow \pi_1(S', b') \longrightarrow e,$$

where $S' = \mathrm{Spec}(()k')$. The diagram

$$\begin{tikzcd}
S & X \arrow[l] & \overline{X}_0 \arrow[l] \\
S' \arrow[u] & X' \arrow[l] \arrow[u] & \overline{X}_0 \arrow[l] \arrow[u, equal]
\end{tikzcd}$$

gives a canonical homomorphism from this sequence to the corresponding sequence for X/k. By IX.4.11, this homomorphism of sequences of groups is an isomorphism. We are therefore reduced to proving that the second sequence is exact, i.e. we may assume that k is *perfect*.

Let k_i run through the finite Galois subextensions of k in $\overline{k}$, set $X_i = X \otimes_k k_i$, and let a_i be the image of $\overline{a}$ in X_i. We leave it to the reader to verify that the natural homomorphism

$$\pi_1(\overline{X}_0, \overline{a}) \xrightarrow{\sim} \varprojlim_i \pi_1(X_i, a_i)$$

is an isomorphism. This simply means that an étale covering of $\overline{X}_0$ comes from an étale 254
covering of some X_i, and that the latter is essentially unique after passing to an X_j with $j \geq i$.

Let π_i be the Galois group of k_i over k, i.e. the group opposite to the group of S-automorphisms of $S_i = \mathrm{Spec}(()k_i)$. Since the functor

$$S' \longmapsto X \times_S S'$$

from étale coverings of S to étale coverings of X is fully faithful by (IX.3.4), it follows that π_i is also isomorphic to the group opposite to the group of X-automorphisms of the connected principal covering X_i of X. Consequently, V V.6.13 gives an exact sequence

$$e \longmapsto \pi_1(X_i, a_i) \longrightarrow \pi_1(X, a) \longrightarrow \pi_i \longrightarrow e.$$

Passing to the inverse limit over i in these exact sequences gives an exact sequence, since we are in the category of groups of Galois type. This is precisely the sequence considered in IX.6.1, and the proof is complete.

The geometric translation of exactness on the right in IX.6.1 is the following:

Corollary IX.6.2. *With the preceding notation, let X' be an étale covering of X, and let $\overline{X'}_0$ be the corresponding étale covering of $\overline{X}_0$. The following conditions are equivalent:*

(i) *There exist an S' étale over S and an X-isomorphism $X' \xrightarrow{\sim} X \times_S S'$. In that case S' is determined up to unique isomorphism by IX.3.4.*

(ii) *The covering $\overline{X'}_0$ is completely split over $\overline{X}_0$.*

If X' is connected, these conditions are also equivalent to:

(ii bis) *The covering $\overline{X'}_0$ has a section over $\overline{X}_0$.*

The last qualification is essential. The equivalence of (i) and (ii) means only that $\pi_1(S, b)$ is the quotient of $\pi_1(X, a)$ by the closed normal subgroup generated by the image 255
of $\pi_1(\overline{X}_0, \overline{a})$, and not by that image itself. Under the preceding hypotheses, we shall say that X' is a *geometrically trivial* covering of X.

Remark IX.6.3. In IX.6.1, one cannot replace $\overline{k}$ by an arbitrary algebraically closed extension of k, even if k is already assumed algebraically closed. In other words, it is not true in general that, if X is a connected algebraic scheme over an algebraically closed field k, its fundamental group is unchanged when k is replaced by an algebraically closed extension. This already fails, for example, in characteristic $p > 0$ for the affine line over k, because of phenomena of *higher ramification* at the point at infinity, which give the fundamental group a *continuous* structure. In the next exposé we shall see, however, that such phenomena cannot occur if X is *proper* over k. We shall also prove by transcendental methods that the same holds when k has characteristic zero.

Corollary IX.6.4. *Suppose that a is localized at a point $x \in X$ rational over k, or more generally with residue field radicial over k. Then the exact sequence IX.6.1 splits.*

We may suppose that $S = \mathrm{Spec}(()k)$. If x is rational over k, it corresponds to a section $S \to X$ of X over S taking b to a. This section defines a homomorphism $\pi_1(S, b) \to \pi_1(X, a)$, which is the required splitting. If $\kappa(x)$ is radicial over k, we reduce to the preceding case by the base extension $\mathrm{Spec}(()\kappa(x)) \to \mathrm{Spec}(()k)$.

THEOREM IX.6.5. *Let $f\colon X \to S$ be a proper surjective morphism of finite presentation with geometrically connected fibers, let X' be a prescheme of finite presentation and proper over X, let s be a point of S, let $F = X_s$ be the fiber of X at s, and let F'_1 be a connected component of the fiber $F' = X'_s$ of X' at s. There exist an open neighborhood X'_1 of F'_1 in X', an S-scheme S'_1 étale over S, and an X-isomorphism*

$$X'_1 \xrightarrow{\sim} S'_1 \times_S X$$

if and only if X' is étale over X at every point of F'_1, and F'_1 is a geometrically trivial covering of F.

The necessity of the condition is trivial, so it remains to prove sufficiency. We reduce readily to the case where S is noetherian. Consider the Stein factorization $X \to T \to S$ of f, where T is the spectrum of the algebra $f_*(\mathcal{O}_X)$ on S. Since the fibers of X over S are geometrically connected and f is surjective, the morphism $T \to S$ is finite, surjective, and radicial. Hence, by (IX.4.10), every T' étale over T is the inverse image of an S' étale over S. This reduces us to proving IX.6.5 after replacing S by T, i.e. to the case where $f_*(\mathcal{O}_X) = \mathcal{O}_S$. Now consider the Stein factorization 256

$$X' \longrightarrow S' \longrightarrow S$$

of the proper morphism $h\colon X' \to S$, where S' is the spectrum of the algebra $h_*(\mathcal{O}_{X'})$. The morphisms $X' \to X$ and $X' \to S'$ define a canonical morphism

$$X' \longrightarrow X \times_S S',$$

and our assertion is contained in the following result.

COROLLARY IX.6.6. *Let $f\colon X \to S$ be a proper morphism of locally noetherian preschemes such that $f_*(\mathcal{O}_X) = \mathcal{O}_S$, and let X' be a prescheme proper over X. Consider the* Stein *factorization*

$$X' \longrightarrow S' \longrightarrow S$$

of $X' \to S$, and the canonical morphism $X' \to X \times_S S'$. Let s be a point of S, and let s' be a point of S' above s, corresponding to a connected component F'_1 of the fiber X'_s. The morphism $X' \to X \times_S S'$ is an isomorphism over an open neighborhood U' of s' which is étale over S if and only if X' is étale over X at the points of F'_1, and F'_1 is a geometrically trivial covering of the fiber $F = X_s$.

Necessity is again trivial, so it remains to prove sufficiency. The conclusion also says that

a) the morphism deduced from $X' \to X \times_S S'$ by the base change $\operatorname{Spec}(()\widehat{\mathcal{O}}_{s'}) \to S'$ is an isomorphism;
b) S' is étale over S at s', i.e. $\widehat{\mathcal{O}}_{s'}$ is étale over $\widehat{\mathcal{O}}_s$.

In this form, the conclusion is invariant under the base change $\operatorname{Spec}(()\widehat{\mathcal{O}}_s) \to S$. Since the hypotheses are also stable under this base change, we may suppose that S is the spectrum of a complete noetherian local ring. We may moreover plainly suppose that X' is connected, which here implies that

$$S' = \operatorname{Spec}(()\mathcal{O}_{s'}), \qquad F' = F'_1.$$

The set of points of X' at which X' is étale over X is open and contains the fiber $X'_{s'} = F'$. Since X' is proper over S, it follows that X' is étale over X. On $F = X_s$, it induces an étale covering isomorphic to 257

$$F \otimes_{\kappa(s)} L,$$

where L is étale over $\kappa(s)$. It follows from IX.1.10 that this covering is isomorphic to one of the form $X \times_S T$, with T étale over S. Here again, it is enough to use the full faithfulness of the functor in IX.1.10; this follows from the fact that a formal isomorphism of coherent sheaves on X comes from an isomorphism of those sheaves.

Thus, if T is defined by the finite A-algebra B, then X' is identified with the spectrum of the algebra $\mathcal{O}_X \otimes_A B$ on X. Since $f_*(\mathcal{O}_X) = \mathcal{O}_S$, it follows at once that $h_*(\mathcal{O}_{X'})$ is defined by B. Consequently, the canonical homomorphism $X' \to X \times_S S'$ is precisely the isomorphism $X' \xrightarrow{\sim} X \times_S T$ under consideration. This completes the proof.

Corollary IX.6.7. *Under the hypotheses of IX.6.5, there exist a prescheme S' étale over S and an X-isomorphism*

$$X' \xrightarrow{\sim} X \times_S S'$$

if and only if X' is étale over X and, for every $s \in S$, X'_s is a geometrically trivial covering of X_s.

Indeed, if this is so, then X' is the union of open subsets X'_i isomorphic to inverse images of S'_i étale over S. It is readily seen that these S'_i glue to an S' étale over S, yielding an isomorphism $X' \xrightarrow{\sim} X \times_S S'$. For example, one may say that the X'_i carry descent data relative to $X \to S$, and that these necessarily glue to a descent datum on all of X' relative to $X \to S$. Since the latter is effective on the X'_i, it follows readily, by a sorites omitted from Section IX.4, that it is effective. We may also state IX.6.7 in the following form:

Corollary IX.6.8. *Let $f\colon X \to S$ be a proper surjective morphism of finite presentation with geometrically connected fibers. Then f is an effective descent morphism for the fibered category of preschemes finite and étale over other preschemes. The functor*

$$S' \longmapsto X \times_S S'$$

induces an equivalence from the category of preschemes finite and étale over S to the category of preschemes finite and étale over X which induce a geometrically trivial covering on every fiber X_s. 258

Remark IX.6.9. Let $f\colon X \to S$ be proper and surjective, with S locally noetherian. Then f factors as a morphism $X \to S'$ satisfying the hypothesis of IX.6.8, followed by a finite surjective morphism $S' \to S$ to which IX.4.7 applies. Thus f is the composite of two morphisms which are *universal effective descent morphisms* for the fibered category of preschemes finite and étale over other preschemes. It follows that f itself is a universal effective descent morphism for the fibered category in question. This recovers IX.4.12 by a different method.

Remark IX.6.10. The conclusion of IX.6.7 no longer holds if the hypothesis that f is proper is replaced by the following: X is of finite type over S and has a section over S. Under this replacement, f is universally submersive and is a descent morphism for the fibered category of preschemes étale over other preschemes, but the conclusion fails even when S is the spectrum of a discrete valuation ring and X' is an étale covering of X.

To see this, start with a Z proper over S whose generic fiber is a nonsingular rational curve and whose special fiber Z_0 consists of two intersecting lines. For example, if t is a uniformizer of the valuation ring A, take the closed subscheme Z of $\mathbb{P}^2_A$ defined by the homogeneous equation

$$x^2 + y^2 + tz^2 = 0$$

in homogeneous coordinates x, y, z. Let X be the complement of the singular point a of Z_0 in the union $Z \cup \mathbb{P}^2_k$. The fibers of X are $\mathbb{P}^1_k$ and $\mathbb{P}^2_k - a$, and are therefore geometrically simply connected (i.e. every étale covering of such a fiber is geometrically trivial).

Nevertheless, arguing as in Section IX.4, one readily constructs étale coverings of X which do not come from étale coverings of S, by gluing trivial coverings of $Z - a$ and $\mathbb{P}^2_k - a$. It is possible, on the other hand, that the conclusion of IX.6.7 remains valid if properness is replaced by the assumption that X is *universally open* and of finite presentation over S [9]. This is at least true if the fibers of X over S are assumed geometrically irreducible, 259

[9] This has now been proved with g merely universally open and surjective; cf. SGA 4, XV 1.15.

and not merely geometrically connected. We mention only that this question may be reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field.

The interpretation of IX.6.7 in terms of the fundamental group is the following:

COROLLARY IX.6.11. *Let $f\colon X \to S$ be a proper surjective morphism of finite presentation with geometrically connected fibers. Suppose that X, and hence S, is connected. Let a be a geometric point of X and b its image in S. For every $s \in S$, choose an algebraic closure $\overline{\kappa(s)}$ of $\kappa(s)$, a geometric point a_s of X_s with values in this extension, and a path class from a_s to a. This gives a homomorphism*

$$\pi_1(\overline{X}_s, a_s) \longrightarrow \pi_1(X, a), \qquad \overline{X}_s = X_s \otimes_{\kappa(s)} \overline{\kappa(s)}.$$

Then the homomorphism $\pi_1(X, a) \to \pi_1(S, b)$ is surjective, and its kernel is the closed normal subgroup of $\pi_1(X, a)$ generated by the images of the groups $\pi_1(\overline{X}_s, a_s)$.

REMARK IX.6.12. Under the hypotheses of IX.6.7, with S noetherian, it is readily seen that the set of points $s \in S$ for which X'_s is geometrically trivial over X_s is constructible. If X' is proper over X, it is even open, as follows from IX.6.6. Consequently, if S is a Jacobson prescheme (for example, of finite type over a field), or if X' is proper over X, 260
it is enough, in verifying the hypotheses of IX.6.7, to consider the *closed* points s of S. Likewise, in IX.6.11, it is then enough to use the groups $\pi_1(\overline{X}_s, a_s)$ for closed points s of S.

Bibliography

[D] J. Giraud, Méthode de la descente, Mémoire no. 2 de la Soc. Math. Fr., 1964.
[1] A. Grothendieck, Géométrie Algébrique et Géométrie Formelle, Séminaire Bourbaki t. 11, 1959, no. 182.

CHAPTER X

Specialization theory of the fundamental group

In the present exposé we restrict ourselves to studying the fundamental group of 261
geometric fibers in a *proper* morphism, i.e. the fundamental group of a varying *proper* algebraic scheme. In a later exposé we shall extend the technique used here to étale coverings *tamely ramified* at infinity. This will give, for example, a solution of the *three-point problem* for Galois coverings of order prime to the characteristic (i.e. a determination of the Galois coverings of the line $\mathbb{P}^1_k$ ramified at no more than three specified points and tamely ramified at those points), together with its evident variants.

X.1. The homotopy exact sequence for a proper and separable morphism

Definition X.1.1. *A prescheme X over a field k is called* separable*, or* separable over k*, if $X \otimes_k K$ is reduced for every extension K of k. If $f\colon X \to Y$ is a morphism of preschemes, we say that f is* separable*, or that X* is separable over Y*, if X is flat over Y and, for every $y \in Y$, the fiber $X \otimes_Y \kappa(y)$ is separable over $\kappa(y)$.*

For a prescheme X over a field k, separability also means that X is *reduced* and that the fields $\kappa(x)$, for x the generic point of an irreducible component of X, are separable extensions of k. Thus, if k is perfect, this is equivalent to saying that X is reduced. Observe that if X is separable over Y, then after every base change $Y' \to Y$, $X' = X \times_Y Y'$ is separable over Y'. Under suitable finiteness hypotheses one can also prove that the composite of two separable morphisms is separable. We shall need this only in the 262
following form: *if X is separable over Y, and X' is étale over X, then X' is separable over Y.* This is an immediate consequence of the definitions and I I.9.2. The hypothesis of a separable morphism will otherwise enter through the following proposition:

Proposition X.1.2. *Let $f\colon X \to Y$ be a proper and separable morphism, with Y locally noetherian, and consider its* Stein *factorization*

$$X \xrightarrow{f'} Y' \to Y,$$

where $f'_*(\mathcal{O}_X) = \mathcal{O}_{Y'}$, and where Y', finite over Y, is isomorphic to the spectrum of the algebra $f_*(\mathcal{O}_X)$. Then Y' is an étale covering *of Y.*

This proposition appears in EGA III, 7[1]. Let us indicate the principle of the proof. One readily reduces to the case where Y is the spectrum of a complete local ring A. After making a suitable finite flat extension of A, corresponding to a suitable residue-field extension, one may suppose that the connected components of the fiber over the closed point y are geometrically connected. This also means that $\mathrm{H}^0(X_y, \mathcal{O}_{X_y})$ decomposes as a product of fields equal to $k = \kappa(y)$. We may then suppose X connected, in which case

$$\mathrm{H}^0(X_y, \mathcal{O}_{X_y}) = k.$$

Thus the homomorphism

$$A \longrightarrow \mathrm{H}^0(X_y, \mathcal{O}_{X_y})$$

[1]Cf. EGA III, 7.8.10 (i).

is *surjective.* A general proposition of Künneth type then shows that $f_*(\mathcal{O}_X)$ is defined by an A-module B which is free over A, and that

$$B/\mathfrak{m}B \longrightarrow \mathrm{H}^0(X_y, \mathcal{O}_{X_y}) = k$$

is bijective. In the present case B is therefore an étale algebra over A, which completes the proof.

THEOREM X.1.3. *Let $f\colon X \to Y$ be a proper and separable morphism, with Y locally noetherian and connected, and suppose that $f_*(\mathcal{O}_X) = \mathcal{O}_Y$. This implies that the fibers of X over Y are geometrically connected, and the converse follows from X.1.2. Let y be a point of Y, let $\overline{\kappa(y)}$ be an algebraic closure of $\kappa(y)$, and put*

$$\overline{X}_y = X_y \otimes_{\kappa(y)} \overline{\kappa(y)}.$$

Finally, let X' be a connected *étale covering of X, and put*

$$\overline{X}'_y = X'_y \otimes_{\kappa(y)} \overline{\kappa(y)}.$$

There exist an étale covering Y' of Y and an X-isomorphism 263

$$X' \xrightarrow{\sim} X \times_Y Y'$$

if and only if $\overline{X}'_y$ has a section over $\overline{X}_y$.

Set $Y' = \mathrm{Spec}(()h_*(\mathcal{O}_{X'}))$, where $h\colon X' \to Y$ is the composite $X' \to X \to Y$. It is enough to prove that the canonical Y-morphism

$$X' \longrightarrow X \times_Y Y'$$

is an *isomorphism*, and that Y' is étale over Y. By X.1.2, we already know that Y' is étale over Y. Consequently $X \times_Y Y'$ is étale over X, and hence $X' \to X \times_Y Y'$ is likewise étale (I I.4.8). Moreover, Y' is connected, as the image of the connected prescheme X', so $X \times_Y Y'$ is connected because X has connected fibers over Y (IX IX.3.4 and V V.6.9 (iii)). Thus, to prove that $X' \to X \times_Y Y'$ is an isomorphism, it is enough to show that its degree at *one* point of $X \times_Y Y'$ equals 1. This follows readily from the hypothesis that $\overline{X}'_y$ has a section over $\overline{X}_y$, either by using IX IX.6.6, or more simply by observing that it is enough to prove the existence of such a point in $X \times_Y Y'$ after the base change $\mathrm{Spec}(()\overline{\kappa(y)}) \to Y$, where it is evident. This completes the proof of X.1.3.

Taking account of IX IX.3.4 and the dictionary V V.6.9 and V V.6.11, we may put X.1.3 in the following equivalent form:

COROLLARY X.1.4. *With the preceding notation for $f\colon X \to Y$ and $\overline{X}_y$, let $\overline{a}$ be a geometric point of $\overline{X}_y$, let a be its image in X, and let b be its image in Y. Then the following sequence of group homomorphisms is* exact*:*

$$\pi_1(\overline{X}_y, \overline{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Y, b) \longrightarrow e.$$

REMARKS X.1.5. Observe that the proof of X.1.3 uses X.1.2 in an essential way, and hence the *first comparison theorem* in algebraic-formal geometry. By contrast, the descent theory of Exposé IX entered only through IX IX.3.4, for which a direct proof is easy in 264
the case of a *proper* morphism $f\colon X \to Y$ such that $f_*(\mathcal{O}_X) = \mathcal{O}_Y$. Indeed, let Y' be étale over Y, and suppose that $X' = X \times_Y Y'$ is the disjoint sum of two nonempty open subsets. We prove that the same is true of Y'. We may write $Y' = \mathrm{Spec}(()\mathcal{A})$, and hence $X' = \mathrm{Spec}(()\mathcal{B})$, with

$$\mathcal{B} = \mathcal{A} \otimes_{\mathcal{O}_Y} \mathcal{O}_X.$$

The decomposition of X' as a disjoint sum corresponds to a decomposition of $\mathcal{B}$ as a product of two nonzero algebras $\mathcal{B}_1$ and $\mathcal{B}_2$. Since $f_*(\mathcal{O}_X) = \mathcal{O}_Y$, it follows readily that $f_*(\mathcal{B}) = \mathcal{A}$. Thus $\mathcal{A}$ is a product of the two algebras $f_*(\mathcal{B}_1)$ and $f_*(\mathcal{B}_2)$, which are likewise nonzero because their unit sections are nonzero. This proves the assertion.

. Continue to suppose that f is proper and separable, but make no further assumption on $f_*(\mathcal{O}_X)$. The latter corresponds to a well-determined étale covering Y' of Y, pointed above b by the image b' of a. Applying X.1.4 to the canonical morphism $X \to Y'$, and assuming f surjective, the exact sequence X.1.4 is replaced by the following sequence, analogous to the homotopy exact sequence of fiber spaces in algebraic topology:

$$\pi_1(\overline{X}_y, \overline{a}) \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Y, b) \longrightarrow \pi_0(\overline{X}_y, \overline{a}) \longrightarrow \pi_0(X, a) \longrightarrow \pi_0(Y, b) \longrightarrow e.$$

Of course, in X.1.4 one cannot in general assert that the homomorphism

$$\pi_1(\overline{X}_y, \overline{a}) \longrightarrow \pi_1(X, a)$$

is injective. In algebraic topology its kernel is the image of $\pi_2(Y, b)$, and in algebraic geometry as well one ought to introduce homotopy groups in every dimension, together with the full homotopy exact sequence for a proper morphism satisfying suitable hypotheses, for example being smooth). At present no result of this kind is available, apart from a reasonable, if not definitive, definition of the higher homotopy groups.

Corollary X.1.6. *Let k be an algebraically closed field, and let X and Y be two connected preschemes over k, with X proper over k and Y locally noetherian. Let a* 265
be a geometric point of X and b a geometric point of Y, both with values in the same algebraically closed extension K of k. Let $c = (a, b)$ be the corresponding geometric point of $X \times_k Y$. Then the homomorphism

$$\pi_1(X \times_k Y, c) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b)$$

induced by the two projections $X \times_k Y \to X$ and $X \times_k Y \to Y$ is an isomorphism.

First suppose that $K = k$. Set $Z = X \times_k Y$, consider the projection $f : Z \to Y$ and the underlying point y of the geometric point b of Y, and apply X.1.4. For this purpose, observe that after replacing X by X_{red}, which does not change the fundamental groups in question, we may assume that X is reduced, and hence separable over k. Thus Z is separable over Y, and plainly has geometrically connected fibers, since X is connected. The geometric fiber of Z at b is canonically isomorphic to $X \otimes_k K = X$.

Moreover, since the composite of the morphisms $X \to Z \to X$ is the identity, the homomorphism $\pi_1(X, a) \to \pi_1(Z, c)$ is injective, and X.1.4 gives an exact sequence

$$e \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(Z, c) \longrightarrow \pi_1(Y, b) \longrightarrow e.$$

There is also the canonical exact sequence

$$e \longrightarrow \pi_1(X, a) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b) \longrightarrow \pi_1(Y, b) \longrightarrow e,$$

in which the displayed homomorphisms are the canonical inclusion and projection. Finally, the identity homomorphisms on the outer terms together with the canonical homomorphism

$$\pi_1(Z, c) \longrightarrow \pi_1(X, a) \times \pi_1(Y, b)$$

on the middle terms define a homomorphism from the first exact sequence to the second. The resulting diagram is plainly commutative. Since the homomorphisms on the outer terms are isomorphisms, so is the homomorphism on the middle terms. This proves X.1.6 in this case.

When K is no longer assumed equal to k, the argument gives only an isomorphism

$$\pi_1(Z, c) \longrightarrow \pi_1(X \otimes_k K, a) \times \pi_1(Y, b),$$

and X.1.6 is then equivalent to the following special case: 266

Corollary X.1.7. *Let X be a proper connected scheme over an algebraically closed field k, let k' be an algebraically closed extension of k, let a' be a geometric point of $X \otimes_k k'$, and let a be its image in X. Then the canonical homomorphism*

$$\pi_1(X \otimes_k k', a') \longrightarrow \pi_1(X, a)$$

is an isomorphism.

Surjectivity of this homomorphism is equivalent to the assertion that, if X' is a connected étale covering of X, then $X' \otimes_k k'$ is connected as well. This follows at once from the fact that k is algebraically closed, and is also a special case of IX IX.3.4. The properness hypothesis on X has not yet been used. Injectivity of the homomorphism in question means the following: *every étale covering of $X \otimes_k k'$ is isomorphic to the inverse image of an étale covering of X.*

By an essentially formal argument, one can find a k-subalgebra A of k', of finite type over k, and an étale covering of $X \otimes_k A$ whose inverse image over $X \otimes_k k'$ is isomorphic to the given covering. Let $Y = \mathrm{Spec}(()A)$. This is an integral k-scheme of finite type, and hence has rational points over k. Apply X.1.6 to the fundamental group of $X \times Y$ at a point (a, b) rational over k. Every connected étale covering of $X \times Y$ is then isomorphic to a quotient of a covering $X' \times Y'$, where X' and Y' are Galois étale coverings of X and Y, with groups G and G', by a subgroup H of $G \times G'$. It follows that the inverse image of this covering of $X \times Y$ over $X \times Y'$ is isomorphic to a covering of the form $X'_1 \times Y'$, where X'_1 is an étale covering of X.

Let L be the function field of Y, equal to the fraction field of A inside k'. The étale covering of $X \otimes_k L$ induced by the given covering of $X \times_k Y$ has the following property: there is a finite separable extension L'/L such that its inverse image over $X \otimes_k L'$ is isomorphic to $X'_1 \otimes_k L'$. Since k' is algebraically closed, we may suppose that L' is contained in k'. This proves that the given étale covering of $X \otimes_k k'$ is isomorphic to $X'_1 \otimes_k k'$, as required.

The explicit description just given for étale coverings of a product $X \times_k Y$ immediately 267
implies:

Corollary X.1.8. *Let k be an algebraically closed field, let X and Y be two locally noetherian preschemes over k, let $Z = X \times_k Y$, and let Z' be an étale covering of Z. For every point $y \in Y$ rational over k, let $i_y \colon \mathrm{Spec}(()k) \to Y$ be the canonically associated morphism, let $j_y = \mathrm{id}_X \times_k i_y \colon X \to Z$, and let X'_y be the étale covering of X obtained from Z' by inverse image under j_y. Suppose that Y is connected, and that X or Y is proper over k. Then all the coverings X'_y of X are isomorphic.*

Figuratively, one may say that *a family of étale coverings of X, parametrized by a connected prescheme Y, is constant if X, or the parameter prescheme Y, is proper over k.*

Remarks X.1.9. Corollaries X.1.6 through X.1.8 are due to Lang–Serre [**2**] in the case of normal algebraic schemes. Their work was the initial motivation for the theory of the fundamental group developed in this Seminar. As those authors observed, these results become false when the properness hypothesis is dropped, at least in characteristic $p > 0$. For example, take $X = \mathrm{Spec}(()k[t])$, the affine line. The coverings of X parametrized by the affine line $Y = \mathrm{Spec}(()k[s])$, defined by the equations

$$x^p - x = st,$$

are étale and pairwise nonisomorphic. This contradicts X.1.8, and a fortiori X.1.6. Likewise, if s is regarded as an element transcendental over k in an algebraically closed extension K of k, one obtains an étale covering X' of $X \otimes_k K$ which does not come from an étale covering of X.

X.2. Application of the existence theorem for sheaves: a semicontinuity theorem for the fundamental groups of the fibers of a proper and separable morphism

268

Theorem X.2.1. *Let Y be the spectrum of a* complete *noetherian local ring A, with residue field k, let X be a proper Y-scheme, let $X_0 = X \otimes_A k$, let a_0 be a geometric* *point of X_0, and let a be the corresponding geometric point of X. Then the canonical homomorphism*

$$\pi_1(X_0, a_0) \longrightarrow \pi_1(X, a)$$

is an isomorphism.

This is merely the translation into the language of the fundamental group of the result recalled in IX IX.1.10. It is here that the existence theorem for sheaves in algebraic-formal geometry enters the theory of the fundamental group in an essential way.

Now introduce an algebraic closure $\overline{k}$ of the residue field k, and the geometric fiber

$$\overline{X}_0 = X_0 \otimes_k \overline{k}.$$

We have the exact sequence (IX IX.6.1)

$$e \to \pi_1(\overline{X}_0, \overline{a}) \to \pi_1(X_0, a_0) \to \pi_1(k, \overline{k}) \to e.$$

On the other hand, there is the isomorphism X.2.1, together with the analogous and more elementary isomorphism

$$\pi_1(k, \overline{k}) \longrightarrow \pi_1(Y, b),$$

where b is the image of a in Y. Thus we obtain:

Corollary X.2.2. *With the preceding notation, suppose that $\overline{X}_0 = X_0 \otimes_k \overline{k}$ is connected, and let $\overline{a}_0$ be a geometric point of $\overline{X}_0$, with image a_0 in X, and let b_0 be its image in Y. Then the following sequence of canonical homomorphisms*

$$e \to \pi_1(\overline{X}_0, \overline{a}) \to \pi_1(X, a_0) \to \pi_1(Y, b_0) \to e$$

is exact.

Compare this sequence with the exact sequence X.1.4, but note that a) no flatness or fiberwise separability hypothesis was required for $X \to Y$; and b) one has the important additional fact that *the homomorphism $\pi_1(\overline{X}_0, \overline{a}_0) \to \pi_1(X, a_0)$ is injective.*

This last fact will allow us to compare the fundamental groups of the other geometric fibers of X over Y with that of $\overline{X}_0$. Let y_1 be an arbitrary point of Y, let X_1 be its fiber, and let $\overline{X}_1$ be its geometric fiber with respect to a chosen algebraically closed extension of $\kappa(y_1)$. Let $\overline{a}_1$ be a geometric point of $\overline{X}_1$, let a_1 be its image in X, and let b_1 be its image in Y. Choose a path class from a_1 to a_0, and hence a path class from b_1 to b_0. This gives a commutative diagram of homomorphisms 269

$$\begin{tikzcd}
 & \pi_1(\overline{X}_1, \overline{a}_1) \arrow[r] & \pi_1(X, a_1) \arrow[r] \arrow[d] & \pi_1(Y, b_1) \arrow[r] \arrow[d] & e \\
e \arrow[r] & \pi_1(\overline{X}_0, \overline{a}_0) \arrow[r] & \pi_1(X, a_0) \arrow[r] & \pi_1(Y, b_0) \arrow[r] & e,
\end{tikzcd}$$

in which the two displayed vertical arrows are isomorphisms. Since the second row is exact, we obtain a canonical homomorphism, which we shall call the *specialization homomorphism for the fundamental group*:

$$\pi_1(\overline{X}_1, \overline{a}_1) \longrightarrow \pi_1(\overline{X}_0, \overline{a}_0).$$

It depends only on the chosen path class from a_1 to a_0, and is therefore *defined up to an inner automorphism* of $\pi_1(X, a_0)$. When the first row above is exact as well, it follows at once that the specialization homomorphism is surjective. Taking X.1.4 into account, we therefore obtain:

Corollary X.2.3. *Under the hypotheses of X.2.1, suppose in addition that $f\colon X \to Y$ is* separable (X.1.1) *and that $\overline{X}_0$ is connected. Then $f_*(\mathcal{O}_X) = \mathcal{O}_Y$ by X.1.2. For every geometric fiber $\overline{X}_1$ of X over Y, endowed with a geometric point $\overline{a}_1$, the specialization homomorphism defined above is* surjective.

This is a *semicontinuity* result for the fundamental group, which does not yet seem to have an analogue in algebraic topology. It may moreover be stated under more general hypotheses:

Corollary X.2.4. *Let $f\colon X \to Y$ be a proper morphism with geometrically connected fibers, with Y locally noetherian. Let y_0, y_1 be two points of Y such that $y_0 \in \overline{\{y_1\}}$, and let $\overline{X}_0, \overline{X}_1$ be the geometric fibers of X corresponding to chosen algebraically closed extensions of $\kappa(y_0)$ and $\kappa(y_1)$. Let $\overline{a}_0$, respectively $\overline{a}_1$, be a geometric point of $\overline{X}_0$, respectively $\overline{X}_1$. Then one can naturally define a specialization homomorphism* 270

$$\pi_1(\overline{X}_1, \overline{a}_1) \longrightarrow \pi_1(\overline{X}_0, \overline{a}_0),$$

well-defined up to inner automorphism. This homomorphism is surjective *if f is separable* (X.1.1).

First, it follows from X.1.7 that X.2.4 is essentially independent of the chosen algebraically closed extensions of the residue fields $\kappa(y_0)$ and $\kappa(y_1)$. We may therefore replace Y by a prescheme Y' over Y having a point y'_0 above y_0, respectively a point y'_1 above y_1. Take for Y' the spectrum of the completion of the local ring of Y at y_0, and apply X.2.3.

Remarks X.2.5. The final conclusion of X.2.4, concerning surjectivity of the specialization homomorphism, and a fortiori the results X.1.3 and X.1.4 from which it follows, becomes false if $f\colon X \to Y$ is no longer assumed separable, even for projective schemes over an algebraically closed field of characteristic 0. We shall see examples below both when f is flat but has a nonseparable fiber, although X and Y are smooth over k, and when the fibers of f are separable but f is not flat, for example when $f\colon X \to Y$ is a birational morphism of normal integral schemes; see XI XI.3. In these examples, the fundamental group of the geometric generic fiber may be trivial while that of a suitable geometric special fiber is not.

Moreover, even if $f\colon X \to Y$ is proper and separable as in X.2.4, the specialization homomorphism often fails to be an isomorphism. For example, it is easy to give cases in 271
which $\overline{X}_1$ is a nonsingular elliptic curve, so its fundamental group is commutative and its ℓ-primary component, for a prime ℓ different from the characteristic, is isomorphic to $\mathbb{Z}_\ell^2$; see XI. Meanwhile $\overline{X}_0$ may consist of two nonsingular rational curves meeting in two points, or of two rational curves tangent at one point, or of a rational curve having a singular point which is a cusp. For the complete classification of degenerate elliptic curves, see the recent work of Kodaira [**1**] and Néron. In these three cases the fundamental group of $\overline{X}_0$ is respectively $\widehat{\mathbb{Z}}$, e, and e, and is therefore *strictly smaller* than that of $\overline{X}_1$.

We shall nevertheless obtain below, when f is *smooth*, a bound on the kernel of the specialization homomorphism which implies in particular that, if $\kappa(y_0)$ has characteristic 0, the specialization homomorphism is an isomorphism. Even for a smooth morphism, however, if $\kappa(y_0)$ has positive characteristic, the specialization homomorphism need not be an isomorphism, as one sees for example when X is an abelian scheme over Y, even of relative dimension 1, cf. XI XI.2. A satisfactory theory of specialization of the fundamental group must take account of the *continuous component* of the *true* fundamental group, corresponding to the classification of principal coverings whose structural groups are infinitesimal groups. One might then expect that the true fundamental groups of the geometric fibers of a smooth proper morphism $f\colon X \to Y$ form a well-behaved local system on X, an inverse limit of finite flat group schemes over X[2]. We shall return later to this point of view. Our present aim, by contrast, is to push as far as possible the phenomena common to the topological and scheme-theoretic theories of the fundamental group.

[2]This extremely attractive conjecture is unfortunately disproved by an unpublished example of M. Artin, already when the fibers of f are algebraic curves of a fixed genus $g \geq 2$.

Now let X_0 be a proper smooth connected curve of genus g over an algebraically closed field k. If k has characteristic zero, its fundamental group may be determined by transcendental methods as follows. We know that X_0 is obtained by base extension from a curve defined over an algebraically closed extension of finite transcendence degree of 272
the prime field $\mathbb{Q}$. By X.1.7, we may suppose that k itself has finite transcendence degree over $\mathbb{Q}$. We may therefore regard k as a subfield of the complex numbers $\mathbb{C}$, and a further application of X.1.7 allows us to suppose $k = \mathbb{C}$. It is then not difficult to verify that the fundamental group of X_0 is isomorphic to the profinite completion of the fundamental group of the associated topological space $\widetilde{X}_0$, a compact oriented surface of genus g, with respect to the topology defined by subgroups of finite index[3]. It is classical that the topological fundamental group is generated by $2g$ generators s_i, t_i, $1 \leq i \leq g$, subject to the single relation

$$(s_1 t_1 s_1^{-1} t_1^{-1}) \cdots (s_g t_g s_g^{-1} t_g^{-1}) = 1.$$

Thus the fundamental group of X_0 has $2g$ *topological* generators s_i, t_i, $1 \leq i \leq g$, satisfying the preceding relation.

Now suppose that k has characteristic $p > 0$. Let A be the ring of Witt vectors constructed from k, and let K be an algebraically closed extension of its fraction field. We saw in III III.7.4 that there exists a scheme X over $Y = \mathrm{Spec}(()A)$, proper and smooth over Y, whose reduction is X_0. Applying X.2.3, we obtain a *surjective* homomorphism

$$\pi_1(X_1) \longrightarrow \pi_1(X_0),$$

where $X_1 = X \otimes_A K$. It is immediate[4] that X_1 is smooth over K, connected by (X.1.2), and of dimension 1, and that its genus equals g, by invariance of the Euler–Poincaré characteristic; see EGA III, 7. Since K has characteristic 0, the preceding result applies. We have therefore proved, by *transcendental methods*:

Theorem X.2.6. *Let X_0 be a smooth proper connected algebraic curve over an algebraically closed field k, and let g be its genus. Then $\pi_1(X_0)$ has a system of $2g$* 273
topological generators subject to the relation displayed above. If k has characteristic 0, $\pi_1(X_0)$ is moreover the free group of Galois type on those generators subject to that relation.

Remarks X.2.7. At present, to the author's knowledge, there is no purely algebraic proof of the preceding result, except in genera 0 and 1. To begin with, it is not at all clear how to distinguish $2g$ elements of $\pi_1(X_0)$ which one might then expect to form a system of topological generators. In this respect, the situation of the rational line with n points removed, and the study of its coverings tamely ramified at those points, is more congenial: the ramification groups at the n points provide n elements of the fundamental group under study, and one proves that they generate it topologically, as we shall see later[5]. Yet even in this particularly concrete case no purely algebraic proof seems to be known. Such a proof would plainly be extremely interesting. The only fact about the fundamental group of a curve that appears to admit a purely algebraic proof, apart from the weak finiteness theorem X.2.11 below, proved algebraically by Lang–Serre [**2**], is the determination of its abelianization by means of the Jacobian, mentioned in the last line of IX IX.5.8.

. The final assertion of X.2.6 no longer holds in characteristic $p > 0$, as one sees already in the case of elliptic curves. As we have already pointed out, we do not know whether the fundamental group of X_0 is topologically finitely presented; this seems wholly unlikely.

[3] This deduction was explained in one of the oral exposés which were not written up.

[4] Cf. EGA IV, 12.2.

[5] Cf. Exposé XII. Even these elements are really determined only up to conjugacy, and one must make a judicious *simultaneous* choice of representatives in their conjugacy classes.

THEOREM X.2.8. *Let k be an algebraically closed field, and let X be a proper connected scheme over k. Then the fundamental group of X is topologically finitely generated.*

We proceed by induction on $n = \dim X$, the assertion being trivial for $n \leq 0$. Thus 274
suppose that $n > 0$, and that the theorem has been proved in dimensions $n' < n$. By Chow's lemma (EGA II 5.6.2), there exists a projective scheme X' over k and a surjective morphism $X' \to X$. We may clearly suppose that X' is reduced and, on passing to its normalization, normal. By descent theory, it suffices to prove that the fundamental groups of the connected components of X' are topologically finitely generated (IX IX.5.2). We are thus reduced to the case in which X is *projective* and *normal.* If $n = 1$, it is enough to apply X.2.6. If $n \geq 2$, consider a projective immersion $X \to \mathbb{P}^r_k$ and a hyperplane section $Y = X \cdot H$ (equipped with the induced reduced structure), such that $Y \neq X$, i.e. $H \not\supset X$. We then have $\dim Y < n$, and, taking the induction hypothesis into account, it suffices to prove that $\pi_1(Y) \to \pi_1(X)$ is *surjective.* More generally:

LEMMA X.2.9. *Let X be a proper prescheme over an algebraically closed field k, and let $g\colon X \to \mathbb{P}^r_k$ be a morphism. Suppose that X is irreducible and normal and that $\dim g(X) \geq 2$. Let H be a hyperplane of $\mathbb{P}^r_k$ and let $Y = X \times_{\mathbb{P}^r_k} H$. Then Y is connected, and the homomorphism $\pi_1(Y) \to \pi_1(X)$ is surjective.*

These assertions in fact follow from the following one:

COROLLARY X.2.10. *Under the preceding conditions, let X' be a connected étale covering of X, and let $Y' = X' \times_X Y = X' \times_{\mathbb{P}^r_k} H$ be the induced covering of Y. Then Y' is connected.*

Since X is normal, X' is normal; hence, being connected, X' is irreducible. Moreover, its image in $\mathbb{P}^r_k$ has dimension ≥ 2. A well-known lemma due to Zariski (called the "*Bertini theorem*") therefore implies that if H'_1 is the generic hyperplane in $\mathbb{P}^r_k$, defined over an extension K of k, then $X' \times_{\mathbb{P}^r} H_1$ is universally irreducible, hence universally connected over K. Zariski's connectedness theorem (EGA III 4) then implies that for *every* hyperplane H (defined over an arbitrary extension of k), $X' \times_{\mathbb{P}^r} H$ is geometrically connected. This completes the proof of X.2.10, and hence of X.2.8.

COROLLARY X.2.11 (Lang-Serre). *Under the conditions of X.2.8, for every finite* 275
group G, the set of isomorphism classes of principal coverings of X with group G is finite.

REMARK X.2.12. Under the conditions of X.2.9, when $\dim g(X) \geq 3$ (at least when g is an immersion and X is regular), we shall prove a more precise result, known in algebraic geometry as the "*Lefschetz theorem*": $\pi_1(Y) \to \pi_1(X)$ *is an isomorphism.*[6] In the classical cases there are analogous statements for homology groups and higher homotopy groups, which sooner or later will have to be encompassed within abstract algebraic geometry. Even for Hodge cohomology $\mathrm{H}^p(X, \Omega^q)$, the question does not seem to have been studied yet; moreover, it is hardly likely that for the latter the Lefschetz theorems remain unchanged in characteristic $p > 0$.

REMARK X.2.13. Let R be a complete discrete valuation ring with algebraically closed residue field κ of characteristic $p > 0$ and fraction field K, and let Y be a proper smooth connected curve of genus g over R. Thus there is a surjective specialization morphism $\mathrm{sp} : \pi_1(Y_{\overline{K}}) \to \pi_1(Y_\kappa)$. We have already noted that if K has characteristic 0, sp is not an isomorphism as soon as $g \geq 1$. Suppose that K has characteristic p, so that R is isomorphic to the formal power series ring $\kappa[[T]]$. In equal characteristic $p > 0$, the fundamental group is not determined by the genus g, as one sees already from elliptic curves, which may be ordinary or supersingular. We mention the recent result of A. Tamagawa (not yet published). If G is a profinite group, we denote by G^{res} the quotient

[6]Cf. the SGA 2 seminar (1962), which followed this one, Theorem X 3.10.

profinite group of G given by the inverse limit of the finite solvable topological quotients of G.

Theorem (A. Tamagawa). *Suppose that $g \geq 2$, that the special fiber Y_κ is definable over a finite field, and that the morphism* $\mathrm{sp}^{\mathrm{res}} : \pi_1(Y_{\overline{K}})^{\mathrm{res}} \to \pi_1(Y_\kappa)^{\mathrm{res}}$ *induced by* sp *upon passing to the quotient is bijective. Then the curve Y is constant over R.*

Note that the Galois group of $\overline{K}/K$ is solvable. The preceding statement may also be expressed as follows: suppose that every finite étale covering $X_K \to Y_K$ of the generic fiber Y_K, Galois with solvable Galois group, has potentially good reduction (that is, extends to a finite étale covering of Y after possibly replacing R by its normalization in a finite extension of K). Then Y is constant over R.

X.3. Application of the purity theorem: continuity theorem for the fundamental groups of the fibers of a proper and smooth morphism

Recall without proof[7] the

Purity theorem X.3.1 (Zariski-Nagata). *Let $f\colon X \to Y$ be a quasi-finite and dominant morphism of integral preschemes, with X normal and Y regular and locally Noetherian, and let Z be the set of points of X where f is not étale, i.e. where f is ramified (these are equivalent,* I I.9.5 (ii)*). If $Z \neq X$, then Z has codimension* 1 *in X at all its points, i.e. for every irreducible component Z' of Z with generic point z, the Krull dimension of $\mathcal{O}_{X,z}$ is equal to* 1*.*

Recall that a prescheme is called *normal* respectively*regular* if its local rings are normal respectivelyregular, and that the relation $Z \neq X$ also means that the finite extension $R(Z)/R(X)$ (where R denotes the field of rational functions) is *separable*. Passing to the generic point z of a component Z' of Z, and localizing at the point y of Y below z, one obtains the equivalent statement: 276

Corollary X.3.2. *Let A be a regular Noetherian local ring, and let $A \to B$ be an injective local homomorphism such that B is normal, a localization of an algebra of finite type over A, and* quasi-finite *over A; suppose in addition that* $\dim A (= \dim B) \geq 2$*, and that for every prime ideal $\mathfrak{p}$ of B distinct from the maximal ideal, B is étale over A at $\mathfrak{p}$, i.e. $B_{\mathfrak{p}}$ is étale over $A_{\mathfrak{q}}$ (where $\mathfrak{q} = A \cap \mathfrak{p}$). Then B is étale over A.*

Moreover, it is not difficult to reduce this last statement to the case where A is a *complete* local ring, and hence where B is *finite* over A. Zariski [**5**] gives a simple proof of this result, valid in the equal-characteristic case; the general case is due to Nagata [**3**], who relies on a delicate result of Chow; the latter has not been verified by any of the participants in the Seminar, and ought to be the subject of a later lecture. We mention here only the very simple proof in the special case where $\dim A = 2$, which is sufficient for the most important application we shall make of it in the present section. Since B is normal, it is a B-module of depth (older terminology: cohomological codimension) ≥ 2, and hence it is an A-module of depth ≥ 2; since A is regular of dimension 2, it follows that B is a *free module* over A[8]. It then follows from I I.4.10 that the set of prime ideals $\mathfrak{q}$ of A at which B is ramified over A is the subset of $\mathrm{Spec}(()A)$ defined by a principal ideal (generated by the discriminant of a basis of B over A), and is therefore empty if it is contained in the closed point of $\mathrm{Spec}(()A)$, which proves X.3.2 when $\dim A = 2$.

We shall chiefly use X.3.1 in the equivalent form:

Corollary X.3.3. *Let X be a regular locally Noetherian prescheme, and let U be an open subset of X complementary to a closed subset Z of X of codimension ≥ 2. Then*

[7]For a proof, cf. SGA 2 X X.3.4.

[8]Cf. EGA 0_{IV} 17.3.4

the functor $X' \longmapsto X' \times_X U$ from the category of étale coverings of X to the category of étale coverings of U is an equivalence of categories; in particular, if a is a geometric point of U, the canonical homomorphism $\pi_1(U,a) \to \pi_1(X,a)$ is an isomorphism. 277

The last assertion is plainly a consequence of the first, and for the latter we may plainly suppose that X is connected, and hence irreducible. The normality of X already implies that the functor $X' \longmapsto X' \times_X U$ from the category of locally free coverings (not necessarily étale) of X' to the category of coverings of U is fully faithful, since the functor $\mathcal{E} \longmapsto \mathcal{E}|U$ from the category of locally free Modules on X to the category of locally free Modules on U is so. It therefore remains to prove that for every *étale* covering U' of U, there exists an étale covering X' of X (necessarily unique by what precedes) such that U' is isomorphic to $X' \times_X U$. We may plainly suppose that U' is connected, and hence irreducible since (U being normal) U' is normal. Let K be the field of rational functions on X, or on U (it is the same), and let K' be that of U'; then U' identifies with the normalization of U in K' I I.10.3. Let X' be the normalization of X in K' (EGA II 6.3); then $X' = X \times_X U \cong U'$, while X' is normal and integral, and the structural morphism $f\colon X' \to X$ is *finite* and dominant (since X is normal and K'/K is a finite separable extension). It is étale on $U' = f^{-1}(U) = X' = f^{-1}(Z)$, and since Z has codimension ≥ 2 in X, $f^{-1}(Z)$ has codimension ≥ 2 in X'. It then follows from X.3.1 that X' is étale over X, which completes the proof.

Now let $f\colon X \to Y$ be a rational map from a regular locally Noetherian prescheme X to a prescheme Y, and suppose that f is defined on an open subset U complementary to a closed subset of codimension ≥ 2. Then X.3.3 yields a functor, defined up to isomorphism, from the category of étale coverings of Y to the category of étale coverings of X, and hence, for every geometric point a of U, with image b in Y, a canonical homomorphism

$$\pi_1(X,a) \to \pi_1(Y,b)$$

(deduced from the canonical homomorphism $\pi_1(U,a) \to \pi_1(Y,b)$ by means of the isomorphism $\pi_1(U,a) \xrightarrow{\sim} \pi_1(X,a)$). When f is a dominant morphism, with X and Y integral with function fields K and L, so that K is an extension of L, and with Y normal, these correspondences may be made more precise in terms of field extensions by noting that for every finite extension L' of L that is unramified over Y, the algebra $K' = L' \otimes_L K$ over K is unramified over X. 278

In particular, these observations show that the fundamental group of connected regular locally Noetherian preschemes, pointed by geometric points located in codimension ≤ 1, is a *functor* when the morphisms in this category are taken to be dominant rational maps defined on the complements of closed subsets of codimension ≥ 2. Recalling, for example, that a rational map from a normal scheme over a field k to a proper scheme over k is defined on the complement of a subset of codimension ≥ 2, one obtains:

Corollary X.3.4 (Birational invariance of the fundamental group). *Let k be a field, let X and Y be two regular schemes proper over k, let $f\colon X \to Y$ be a birational map from X to Y, and let Ω be an algebraically closed extension of the function field K of X, making it possible to define the fundamental group of X and the fundamental group of Y. These groups are then canonically isomorphic.*

This also means that if a finite extension K' of K is unramified over one proper nonsingular "model"X of K, then it is unramified over every other proper nonsingular model.

Remark X.3.5. For other applications of the purity theorem, see the works of Abhyankar presented in [**4**], inspired by the results of Zariski [**6**, chap. VIII], proved by topological methods. The latter are far from having been assimilated by "abstract"algebraic geometry and deserve further effort.

We shall need a few elementary facts from ramification theory. Let V be a discrete 279
valuation ring with fraction field K and residue field k, let L be a Galois extension of K with group G, and let V' be the normalization of V in L, which is a free module of rank $n = [L\colon K]$ over V; let $\mathfrak{m}'$ be a maximal ideal of V', let G_d be the subgroup of G consisting of the elements that leave $\mathfrak{m}'$ invariant, so that G_d acts on the residue extension $k' = V'/\mathfrak{m}'$ of k, and let G_i be the subgroup of elements of G_d that act trivially (recall that G_d and G_i are called, respectively, the *decomposition* and *inertia* subgroups of G). One says that L is *tamely ramified* over V if $n_i = [G_i\colon e]$ is prime to the characteristic p of k (a condition that is always satisfied if k has characteristic 0). It is well known that G_i then embeds canonically into the group k'^*, and is therefore isomorphic to the group of n_i-th roots of unity in k'^*; this implies in particular that G_i *is cyclic.* The standard example of this situation is obtained by setting $L = K[t]/(t^n - u)$, where u is a uniformizer of V and n is an integer prime to p: if K contains the n-th roots of unity, L is a totally ramified Galois extension of K, with Galois group $G = G_i$ isomorphic to $\mathbb{Z}/n\mathbb{Z}$.

Lemma X.3.6 (Abhyankar's lemma). *Let V be a discrete valuation ring with fraction field K, let L and K' be two Galois extensions of K that are* tamely ramified *over V, and let n and m be the orders of the corresponding inertia groups , Let L' be a composite extension of L and K' over K. If m is a multiple of n, then L' is unramified over the localizations of the integral closure V' of V in K'.*

Indeed, let W' be the normalization of V' in L', let $\mathfrak{m}'$ be a maximal ideal of V', and let $\mathfrak{n}'$ be a maximal ideal of W' lying above $\mathfrak{m}'$, let $\mathfrak{n}$ be the maximal ideal it induces on the normalization W of V in L, let $G,\ H,\ M$ be the Galois groups of $L,\ K',\ L'$ over K, and let $G_i,\ H_i,\ L'_i$ be the inertia groups corresponding to the chosen maximal ideals. Then M embeds into the product $G \times H$ and M_i into the product $G_i \times H_i$, in such a way
that the projections $M \to G$ and $M \to H$, $M_i \to G_i$ and $M_i \to H_i$ are surjective (the 280
intermediate-field argument). It already follows, since G_i and H_i are by hypothesis cyclic of orders m and n prime to p, that M_i has order prime to p, and hence is cyclic; and since m is a multiple of n, the elements of $G_i \times H_i$ have trivial m-th power, so M_i has order dividing m, and therefore has order equal to m since $M_i \to H_i$ is surjective. This last homomorphism is therefore also injective. But its kernel is the inertia group of $\mathfrak{n}'$ over $\mathfrak{m}'$, which proves that L' is unramified over K' at $\mathfrak{n}'$. This proves the lemma.

Let us now place ourselves under the conditions of X.2.4, where we have a specialization homomorphism

$$\pi_1(\overline{X}_1, \overline{a}_1) \to \pi_1(\overline{X}_0, \overline{a}_0)$$

that is *surjective*, relative to a proper and separable morphism $f\colon X \to Y$. We wish to determine the kernel of this homomorphism. Proceeding as in the proof of X.2.4, we see that for this question we may always suppose that Y is the spectrum of a *complete discrete valuation ring V with algebraically closed residue field* (since one can always find such a ring and a morphism from its spectrum Y' to Y whose image is $\{y_0, y_1\}$). Then $X_0 = \overline{X}_0$, $\kappa(y_0) = k =$ the residue field of V, and $\kappa(y_1) = K =$ the fraction field of V. Let K_s be the separable closure of K, let $\overline{K}$ be its algebraic closure, and for every subring W of $\overline{K}$ containing V, put $X_W = X \otimes_V W$. In particular, we have

$$X_V = X, \qquad X_K = X_1, \qquad X_{\overline{K}} = \overline{X}_1.$$

Moreover, the canonical morphism $\overline{X}_1 = X_{\overline{K}} \to X_{K_s}$ induces an isomorphism on fundamental groups (IX IX.4.11); hence, in view of the isomorphism X.2.1 $\pi_1(X_0) \to \pi_1(X)$, we are reduced to studying the surjective homomorphism

$$\pi_1(X_{K_s}) \to \pi_1(X)$$

associated with the canonical morphism $X_{K_s} \to X$. Determining the kernel of the latter 281
amounts to solving the following problem: *given a connected principal covering Z_{K_s} of*

X_{K_s} *with group* G (thus associated with a homomorphism from $\pi_1(X_{K_s})$ to G), *determine under what conditions it is isomorphic to the inverse image of a principal covering* Z *of* X *with group* G.

First note that K_s is the increasing filtered union of its finite subextensions K' over K, and consequently Z_{K_s} is isomorphic to the inverse image of a principal covering $Z_{K'}$ of $X_{K'}$ for a suitable K' (note, however, that for fixed K', $Z_{K'}$ is not uniquely determined). To say that Z_{K_s} is isomorphic to the inverse image of a principal covering Z of X means that there exists a finite subextension $K'' \supset K'$ of K_s such that $Z_{K''} = Z_{K'} \otimes_{K'} K''$ is isomorphic to $Z \otimes_V K''$. For a finite subextension K' of K_s, now denote by V' the normalization of V in K'; this is a complete discrete valuation ring with residue field k. Thus the canonical morphism $X_{V'} \to X_V$ induces an isomorphism on the fibers over the closed points of $Y = \mathrm{Spec}(()V)$ and $Y' = \mathrm{Spec}(()V')$, and it then follows from X.2.1, applied to X_V and $X_{V'}$, that the induced homomorphism on fundamental groups $\pi_1(X_{V'}) \to \pi_1(X_V)$ is an isomorphism, or equivalently that every principal covering of $X_{V'}$ is the inverse image of a principal covering of X_V determined up to isomorphism. This therefore implies the

LEMMA X.3.7. *Let* $Z_{K'}$ *be a connected principal covering of* $X_{K'}$ *with group* G*, and let* Z_K *be its inverse image over* X_{K_s}*. This latter covering is isomorphic to the inverse image of a principal covering* Z *of* X *if and only if there exists a finite extension* $K'' \supset K'$ *of* K *contained in* K_s *such that the principal covering* $Z_{K''}$ *of* $X_{K''}$ *is induced by a principal covering of* $X_{V''}$.

Suppose in particular that the $X_{V''}$ are normal (it suffices for this, for example, that X_0 be normal, and a fortiori that X_0 be simple, cf. I I.9.1). Since they are connected, 282
they are therefore irreducible. Let L be the field of rational functions of X and X_K, let L' be that of $X_{V'}$ and $X_{K'}$, and let L'' be that of $X_{V''}$ and $X_{K''}$. Then under the conditions of X.3.7, $Z_{K'}$ defines a finite separable extension R' of L', and $Z_{K''}$ defines the extension $R'' = R' \otimes_{L'} L'' = R' \otimes_{K'} K''$. The condition considered in X.3.7 thus also means that there exists a finite separable extension K'' of K' such that $R'' = R' \otimes_{K'} K''$ is *unramified* over the normal scheme $X_{V''}$ with function field $L'' = L' \otimes_{K'} K''$, and not only over the open subset $X_{K''}$ of $X_{V''}$.

From now on, suppose that $f\colon X \to Y$ is a *smooth* morphism, so the morphisms $X_{V'} \to \mathrm{Spec}(()V')$ are smooth, and hence the schemes $X_{V'}$ are *regular*. Note that the fiber of the closed point of $\mathrm{Spec}(()V')$ in $X_{V'}$ is irreducible and of codimension 1. Let $\mathfrak{o}'$ be its local ring; it is therefore a discrete valuation ring with fraction field L' and residue field isomorphic to the field of rational functions of X_0, and thus of the same characteristic as k. Define $\mathfrak{o}''$ similarly in L''; it is plainly the normalization of $\mathfrak{o}'$ in L''. It then follows from the purity theorem X.3.1 or X.3.3 that R'' is unramified over $X_{V''}$ if and only if R'' is unramified over $\mathfrak{o}''$, the normalization of $\mathfrak{o}'$ in L''.

Now note that if u' is a uniformizer of V', it is also a uniformizer of $\mathfrak{o}'$. Thus, if n is an integer prime to the characteristic p of k, and if we take $K'' = K'[t]/(t^n - u)$, then K'' is a finite Galois extension of K', and L'' is isomorphic to $L'[t]/(t^n - u')$, and hence is tamely ramified over $\mathfrak{o}'$ with inertia group of order n. Now suppose that G *has order prime to* p*;* this implies that R' is tamely ramified over $\mathfrak{o}'$, and choose for n a multiple, prime to p, of the order of the inertia group of R' over $\mathfrak{o}'$ (for example, $n = [G\colon e]$). Applying the ABHYANKAR's lemma X.3.6, we see that the condition considered in X.3.7 is satisfied.

This proves the following theorem:

THEOREM X.3.8. *Let* $f\colon X \to Y$ *be a proper and smooth morphism with geometrically connected fibers, with* Y *locally Noetherian, and let* y_0 *and* y_1 *be two points of* Y *such that* $y_0 \in \overline{\{y_1\}}$*; let* $\overline{X}_0$ *and* $\overline{X}_1$ *be the corresponding geometric fibers, and consider the specialization homomorphism* (X.2.4) $\pi_1(\overline{X}_1) \to \pi_1(\overline{X}_0)$*. This homomorphism is surjective, and every continuous homomorphism from* $\pi_1(\overline{X}_1)$ *to a finite group* G *of order prime to the characteristic* p *of* $\kappa(y_0)$ *comes from a homomorphism from* $\pi_1(\overline{X}_0)$ *to* G. 283

In other words:

Corollary X.3.9. *If $\kappa(y_0)$ has characteristic zero, then the specialization homomorphism is an isomorphism. If $\kappa(y_0)$ has characteristic $p > 0$, then the kernel of the specialization homomorphism is contained in the intersection of the kernels of the continuous homomorphisms from $\pi_1(\overline{X}_1)$ to finite groups of order prime to p (or equivalently, in the closed normal subgroup generated by a Sylow p-subgroup of the group of Galois type $\pi_1(\overline{X}_1)$); thus, if $\pi_1(\overline{X}_1)^{(p)}$ denotes the quotient group of $\pi_1(\overline{X}_1)$ by the preceding closed subgroup, and if $\pi_1(\overline{X}_0)^{(p)}$ is defined similarly, then the specialization homomorphism induces an* isomorphism

$$\pi_1(\overline{X}_1)^{(p)} \xrightarrow{\sim} \pi_1(\overline{X}_0)^{(p)}$$

Note that the proof of X.3.8 is purely algebraic. Proceeding as in X.2.6, we conclude *by transcendental methods:*

Corollary X.3.10. *Let X_0 be a proper, smooth, and connected curve of genus g over an algebraically closed field of characteristic p. With the notation introduced in X.3.9, the group $\pi_1(X_0)^{(p)}$ is isomorphic to $\Gamma^{(p)}$, where Γ is the group of Galois type generated by elements $s_i,\ t_i$ $(1 \leq i \leq g)$ subject to the relation*

$$(s_1 t_1 s_1^{-1} t_1^{-1}) \cdots (s_g t_g s_g^{-1} t_g^{-1}) = 1$$

284

Remarks X.3.11. When $\kappa(y_0)$ has characteristic zero, the result X.3.9 is well known by transcendental methods. Note that the proof of X.3.10 invokes the purity theorem in the mixed-characteristic case, but only for rings of dimension 2, where the proof of that theorem is easy and was recalled in the text.

Bibliography

[1] K. Kodaira, On compact analytic surfaces, Ann. of Math. **71** (1960), p. 111–152.

[2] S. Lang and J-P. Serre, Sur les revêtements non ramifiés des variétés algébriques, Amer. J. Math. **79** (1957), p. 319–330.

[3] M. Nagata, On the purity of branch loci in regular local rings, Illinois J. Math. **3** (1959), p. 328–333.

[4] J-P. Serre, Revêtements ramifiés du plan projectif (d'après S. Abhyankar), Séminaire Bourbaki, Mai 1960.

[5] O. Zariski, On the purity of the branch locus of algebraic functions, Proc. Nat. Acad. Sc. USA **44** (1958), p. 791–796.

[6] O. Zariski, Algebraic surfaces, Ergebnisse... 1948, Chelsea (New York).

CHAPTER XI

Examples and complements

XI.1. Projective spaces, unirational varieties 285

Proposition XI.1.1. *Let k be an algebraically closed field, and let $X = \mathbb{P}^r_k$ be projective space of dimension r over k. Then X is* simply connected, *i.e.* $\pi_1(X) = 0$.

For $r = 0$, this is trivial. If $r = 1$, we must show that if X' is a nonempty connected étale covering of $X = \mathbb{P}^1_k$, then $X' \xrightarrow{\sim} X$. The genus formula gives, in this case, if g and g' are the genera of X and X':

$$1 - g' = d(1 - g),$$

where d is the degree of X' over X. Since $g = 0$, we therefore have $1 - g' = d$, which requires $d = 1$ since $g' \geq 0$, and this proves $X' \xrightarrow{\sim} X$. When $r \geq 2$, we proceed by induction on r, assuming that $\mathbb{P}^{r'}$ is simply connected for $r' < r$. Applying this to a hyperplane in $\mathbb{P}^r$ and using (X X.2.9), it indeed follows that $\mathbb{P}^r$ is simply connected. Another proof: by (X X.1.6), we have $\pi_1(\mathbb{P}^1 \times \cdots \times \mathbb{P}^1) = \pi_1(\mathbb{P}^1) \times \cdots \times \pi_1(\mathbb{P}^1)$, and hence $(\mathbb{P}^1)^r$ is simply connected since $\mathbb{P}^1$ is, so $\mathbb{P}^r$ is simply connected by the birational invariance of the fundamental group (X X.3.4). This proof shows more generally:

Corollary XI.1.2. *Let X be a proper normal scheme over an algebraically closed field k; if X is a rational variety, i.e. it is integral and its function field is a purely transcendental extension of k, then X is simply connected.*

This result applies in particular to Grassmann varieties and, more generally, to 286
varieties G/H, where G is a connected linear group over k and H is an algebraic subgroup containing a Borel subgroup of G.

Recall that a *unirational variety over k* is a proper integral scheme over k whose function field K is contained in a purely transcendental extension K' of k that is finite over K (i.e. has the same transcendence degree over k as K), i.e. if there exists a dominant rational map $f\colon \mathbb{P}^r_k \to X$, with $r = \dim X$. If X is normal, it follows from the considerations preceding X X.3.4 that, for every connected étale covering X' of X, with function field L/K, the algebra $L \otimes_K K'$ over K' is unramified on the model $\mathbb{P}^r$, and is therefore completely split by X.1.1; this shows that L is K-isomorphic to a subextension of K'/K. Taking V V.8.2 into account, this therefore proves:

Corollary XI.1.3. *The fundamental group of a normal unirational variety over an algebraically closed field is finite.*

(N.B. Note that in the definition of "unirational", it was not necessary for K'/K to be finite).

Remarks XI.1.4. Of course, the results of this section are well known. On the other hand, J-P. Serre showed [**10**] that when X is a smooth projective unirational variety over an algebraically closed field of *characteristic zero,* X is simply connected. His proof is transcendental in that it uses the Hodge symmetry theorem.

[Added in 2003 (MR) Let us restrict ourselves to the case of an algebraically closed field k of characteristic $p \geq 0$. In characteristic $p > 0$, the definition of a unirational k-variety given above is that of a weakly unirational k-variety, as opposed to that of a strongly unirational k-variety, where one assumes in addition that K' is a separable extension of K.

In dimension 2, there exist smooth projective weakly unirational surfaces whose fundamental group (finite by 1.3) is nontrivial, and which are therefore not rational (T. Shioda, On unirationality of supersingular surfaces, Math. Ann. **225** (1977), p. 155–159.). On the other hand, a strongly unirational surface is rational: this follows from Castelnuovo's rationality criterion, extended to characteristic $p > 0$ by O. Zariski (cf. J-P. Serre, Sém. Bourbaki no. 146, 1957 and Œuvres–Collected papers, vol. 1, p. 467).

Over the field $\mathbb{C}$ of complex numbers, examples are known of smooth projective varieties of dimension ≥ 3 that are unirational but not rational (cf. P. Deligne, Variétés unirationnelles non rationnelles [d'après M. Artin et D. Mumford], Sém. Bourbaki no. 402, 1971-72, Lecture Notes vol. 317). This is the case, in particular, of a smooth cubic hypersurface in projective space $\mathbb{P}^4$. Some of these examples extend to characteristic > 0 to give strongly unirational nonrational varieties (cf. J.P. Murre, Reduction of the proof of the non-rationality of a non singular cubic threefold to a result of Mumford, Compositio Math. **27** (1973), p. 63–82).

Let V be a normal integral projective k-variety. We say that V is rationally connected if there exist an integral k-scheme T of finite type and a k-morphism $F : T \times \mathbb{P}^1 \to V$ such that the morphism

$$\begin{aligned} T \times \mathbb{P}^1 \times \mathbb{P}^1 &\to V \times V \\ (t, u, u') &\longmapsto (F(t, u), F(t, u')) \end{aligned} \tag{1}$$

is dominant. In particular, a rational curve passes through any two sufficiently general rational points of V. The terminology is justified by the fact that if V is rationally connected, two rational points can be joined by a finite chain of rational curves. If k has characteristic $p > 0$, we strengthen the preceding definition by requiring the map (1) to be generically smooth. This gives the notion of a separably rationally connected variety. Thus unirational varieties are rationally connected, and in characteristic $p > 0$, strongly unirational varieties are separably rationally connected. J. Kollár showed that separably rationally connected varieties have trivial algebraic fundamental group (cf. O. Debarre, Variétés rationnellement connexes [d'après T. Graber, J. Harris, J. Starr et A.J. de Jong], Sém. Bourbaki no. 906, 2001-2002).**]**

XI.2. Abelian varieties

Let k be an algebraically closed field, let A be an abelian variety over k, i.e. a group scheme over k that is proper over k, smooth over k, and connected, and finally let G be a commutative group scheme of finite type over k. Denote by $\mathrm{Ext}(A, G)$ the group of classes of commutative extensions of A by G, and by $\mathrm{H}^1(A, G)$ the group of classes of principal bundles over A with group G (compare no. XI.4 below), and consider the canonical homomorphism

$$\mathrm{Ext}(A, G) \to \mathrm{H}^1(A, G).$$

An argument of Serre [**5**, chap. VII, th. 5] shows that this is an injective homomorphism 287
whose image is the set of "primitive elements"of $\mathrm{H}^1(A, G)$, i.e. the elements ξ for which

$$\pi^*(\xi) = \mathrm{pr}_1^*(\xi) + \mathrm{pr}_2^*(\xi),$$

where the pr_i are the two projections from $A \times A$ onto A, and $\pi \colon A \times A \to A$ is the composition law of A (N.B. Serre states his theorem only for G linear and connected, and of course smooth over k, but, on simplifying the first part of his argument, one sees that these restrictions are unnecessary: it is enough to note that every morphism from A into a group scheme E of finite type over k which takes the identity to the identity is a group homomorphism, and to apply this to the sections over A of an extension E of A by G).

We shall apply this result when G is a finite separable group scheme over k, i.e. an ordinary finite group, assumed commutative. Using then $\pi_1(A \times A) \xrightarrow{\sim} \pi_1(A) \times \pi_1(A)$ (X X.1.6) and interpreting $\mathrm{H}^1(X, G)$ as $\mathrm{Hom}(\pi_1(X), G)$ for every algebraic scheme X, in particular for $X = A$ or $X = A \times A$, one sees that every class in $\mathrm{H}^1(A, G)$ is primitive, and hence there is an isomorphism

$$\mathrm{Ext}(A, G) \xrightarrow{\sim} \mathrm{H}^1(A, G),$$

in other words, *every principal covering of A with commutative structural group G, pointed above the origin of A, is endowed in a unique way with the structure of an algebraic group having the marked point as its origin and such that $A' \to A$ is a homomorphism of algebraic groups.* In particular, if A' is connected, it too is an abelian variety, isogenous to A.

On the other hand, since the functor $X \longmapsto \pi_1(X)$ from pointed algebraic schemes X to groups commutes with products (IX IX.1.7), it carries a group object in the first category to a group object in the category of groups, i.e. to a *commutative* group. Thus, *if A is an abelian variety, $\pi_1(A)$ is a commutative group.* Therefore, to determine $\pi_1(A)$, it 288
suffices to know the functor $G \longmapsto \mathrm{H}^1(A, G) = \mathrm{Hom}(\pi_1(A), G)$ as G ranges over the finite *commutative* groups. Finally, recall that for every integer $n > 0$, the multiplication-by-n homomorphism on A,

$$A \xrightarrow{n} A$$

is surjective and hence has finite kernel, i.e. it is an isogeny, and it follows that every isogeny $A' \to A$ is a quotient of an isogeny of the preceding type. From this and standard arguments (cf. for example [**6**]), one obtains:

Theorem XI.2.1 (Serre-Lang). *Let A be an abelian variety over an algebraically closed field k. For every integer $n > 0$, consider the ordinary finite group K_n underlying the kernel ${}_nA$ of multiplication by n on A, and, for every prime number ℓ, set*

$$T_\ell(A) = \varprojlim_r K_{\ell^r}$$

and

$$T_\cdot(A) = \prod_\ell T_\ell(A) = \varprojlim_n K_n$$

(where, if m is a multiple of n, $m = ns$, the map from K_m to K_n is multiplication by s). Then the group $\pi_1(A)$ is canonically isomorphic to $T_\cdot(A)$; consequently, for every prime number ℓ, the ℓ-primary component of $\pi_1(A)$ is canonically isomorphic to $T_\ell(A)$.

These isomorphisms are functorial as A varies. The module $T_\ell(A)$ is called the *ℓ-adic Tate module* of the abelian variety A. It is an additive functor in A; in particular, it gives rise to a representation of the ring $\mathrm{Hom}(A, A)$ of endomorphisms of A on $T_\ell(A)$, called the *ℓ-adic Weil representation*, which plays an important role in the theory of abelian varieties (cf. for example [**4**, chap VII]). Theorem XI.2.1 interprets it as the natural representation on the *ℓ-adic homology group* of A, $\mathrm{H}_1(A, \mathbb{Z}_\ell) = \pi_1(A)_\ell$, which is clearly more satisfactory 289
a priori, especially from the point of view of the Lefschetz formula, [**4**, Chap V]. Let us recall here Weil's results on the structure of $T_\ell(A)$:

a) If n is prime to $\mathrm{char}(k)$, then K_n is a free module of rank $2 \dim A$ over $\mathbb{Z}/n\mathbb{Z}$; hence, if ℓ is a prime number $\neq \mathrm{char}(k)$, $T_\ell(A)$ is a free module of rank $2 \dim A$ over the ring $\mathbb{Z}_\ell$ of ℓ-adic integers;
b) If n is a power of $\mathrm{char}(k) = p$, then K_n is a free module of rank $\nu \leq \dim A$ over $\mathbb{Z}/n\mathbb{Z}$, with ν independent of n; hence $T_p(A)$ is a free module of rank $\nu \leq \dim A$ over the ring $\mathbb{Z}_p$ of p-adic integers.

This shows that, in the theory of the fundamental group developed here, the fundamental group of a varying abelian variety does not vary regularly with the parameter: its p-primary component may drop abruptly at values of the parameter t corresponding to residual characteristic p. The best-known case of this phenomenon is that of elliptic curves. Note, however, that, whatever n may be (whether or not it is prime to the characteristic), the actual kernel ${}_nA$ in A of multiplication by n is a finite group scheme over k of degree n^{2g}, where $g = \dim A$, and it is nonseparable over k if n is a multiple of $p = \mathrm{char}(k)$. Moreover, when A varies in a family of abelian varieties, i.e. when there is an abelian scheme A over a base scheme S, one proves more generally that ${}_nA$ is a finite flat group scheme over S, of degree n^{2g} over S; that is, provided one takes the infinitesimal parts of the kernels ${}_nA$ into account, they behave regularly for every n. This suggests that the "true"fundamental group of an abelian variety A is the pro-algebraic group (the formal inverse limit of finite groups over k) $\varprojlim_n {}_nA$, where by the "true fundamental group"of an algebraic scheme X one means the pro-group that classifies the principal coverings of X whose structural group is an arbitrary finite group G over k (not necessarily separable over k). In this way, for example, the representations of $\mathrm{Hom}(A, A)$ on the p-primary component of the true fundamental group of A recover the Weil characteristic polynomial, defined by Weil using the primes $\ell \neq p$, more naturally than does Serre's construction [**8**].

XI.3. Projective cones, an example of Zariski 290

Continue to assume, for simplicity, that k is an algebraically closed field, and let V be a connected projective k-scheme, a closed subscheme of $\mathbb{P}^r_k$, which may, if desired, be assumed nonsingular. Let $Y = \hat{C}$ be the projective cone over V, let y_0 be its vertex, let $X = \hat{C}_V$ be the usual projective completion of the vector bundle $C_V = \mathbf{V}(\mathcal{O}_V(1))$ associated with $\mathcal{O}_V(1)$, and finally let

$$f\colon X \to Y$$

be the canonical morphism, contracting the zero section X_0 of C_V in X to a point (EGA II 8.6.4). Since X is a locally trivial bundle over V, with fibers $\mathbb{P}^1$ and hence simply connected fibers, the morphism $p\colon X \to V$ induces by (X X.1.4) an isomorphism:

$$\pi_1(X) \xrightarrow{\sim} \pi_1(V).$$

Since p induces an isomorphism $X_0 \to V$, we conclude that *an étale covering of X is completely split if and only if its restriction to X_0 is completely split.* Now, for every étale covering Y' of Y, the inverse image $X' = X \times_Y Y'$ is an étale covering of X that is completely split over the fiber X_0, and hence is trivial. Since the homomorphism $\pi_1(X) \to \pi_1(Y)$ is surjective (IX IX.3.4), we conclude that

$$\pi_1(Y) = (e)$$

in other words, *every projective cone is simply connected.* (N.B. in characteristic 0, the same result holds when Y is taken to be the affine cone).

Now suppose that V is regular, i.e. smooth over k; then X is regular, and for a suitable projective embedding of V, one obtains a *normal* projective cone Y. If V is not simply connected, and hence X is not simply connected, let X' be a nontrivial connected étale covering of X. Since the fibers of X over points $y \in Y$ distinct from y_0 are reduced to a point, we see that the restriction of X' to these fibers (in particular, to the generic fiber) is trivial; nevertheless, X is not obtained as the inverse image of an étale covering 291
of Y, since Y is simply connected and X' would be completely split. This shows that (X X.1.3 and X.1.4) become false if the hypothesis that f is separable is replaced by the weaker hypothesis that its fibers are separable algebraic schemes (or even smooth ones) over the $\kappa(s)$. Likewise, note that the fundamental groups of the geometric fibers $\overline{X}_y$ for $y \neq y_0$ are obviously reduced to (e), since these fibers are reduced to a point, whereas $\pi_1(X_0) \neq e$; thus the semicontinuity theorem (X X.2.4) likewise fails for f.

Finally, let us give the example, pointed out by Zariski, in which these same theorems fail when the hypothesis that f is separable is replaced by the hypothesis that f is flat.

Let $f\colon X \to Y$ be a morphism from a nonsingular projective surface to the rational line $Y = \mathbb{P}^1$, such that $K = k(X)$ is a "regular"extension of $k(f)$, i.e. primary and separable (i.e. the geometric generic fiber is connected and separable), and such that the divisor $(f) = X_0 - X_\infty$ is an n-fold multiple of a divisor (where n is an integer prime to the characteristic). It is possible to construct such examples in every characteristic. Let X' be the normalization of X in $K(f^{1/n})$, where $K = k(X)$ is the function field of X. The hypothesis on (f) implies that X' is étale over X. Let Y' be the normalization of Y in $k(t)(t^{1/n})$; it is ramified over Y exactly at the points $t = 0$ and $t = \infty$, and the restriction $X'|f^{-1}(U)$ is isomorphic to the inverse image of $Y'|U$. In particular, the restriction of X' to the *geometric* generic fiber of X over Y is completely split. Nevertheless, X' is not isomorphic to the inverse image of an étale covering of Y, since one sees at once that the latter would necessarily be Y', which is absurd because Y' is ramified over Y [1].

Here is (following Serre) a simple way to realize the conditions of this example, inspired by [**5**, no. 20]: take n to be a prime number ≥ 5, distinct from the characteristic, and let $G = \mathbb{Z}/n\mathbb{Z}$ act on $\underline{k}^{4\,*}$ by multiplying the coordinates by four distinct characters of G (which is possible since $n \geq 5$). Then G acts on projective space $\mathbb{P}^3_k$, and the only points fixed by G are the four points corresponding to the coordinate axes. The surface X' with equation $x^n + y^n + z^n + t^n = 0$ is smooth over k (Jacobian criterion), and contains none 292
of the fixed points; thus, since G has prime order, it acts on X "without fixed points", i.e. X is a principal covering of $X = X'/G$ with group G. Let $g = x/y$ in $k(X') = K'$; it is a Kummer generator of K' over $K = k(X)$ if the chosen characters were χ^i, $i = 0, 1, 2, 3$, with χ a primitive character. Let f be its n-th power, which is an element of K. We see at once that K' is a regular extension of $k(g)$; this follows from the fact that the plane curve with homogeneous equation in U, T, Z, $T^n + Z^n + (1 + g^n)U^n = 0$, is smooth over $k(g)$ (Jacobian criterion), and from the fact that every plane curve is known to be connected. On the other hand, $k(f) = K \cap k(g)$, since the right-hand side is an extension of $k(f)$ contained in $k(g)$, an extension of prime degree, and is different from $k(g)$ (since $g \notin K$). This implies that K is a regular extension of $k(f)$. Finally, the divisor of f on X is an n-fold multiple of a divisor, since its inverse image on X' is the divisor of g^n, and hence an n-fold multiple, and one can descend because X' is étale over X. We would be done if the rational map $f\colon X \to \mathbb{P}^1$ were a morphism, that is, if the divisors of the zeros and poles of f did not meet. In fact, one checks easily (again by looking on X') that the two divisors in question are the multiples by n of two smooth curves over k, meeting transversely at a point a. If we now replace X by the scheme obtained by blowing up a, the preceding conditions ($\operatorname{div}(f)$ divisible by n, and $k(X_1) = k(X)$ a regular extension of $k(X)$) remain satisfied; moreover, f is a *morphism* $X_1 \to \mathbb{P}^1$, so the required conditions hold.

XI.4. The cohomology exact sequence

Let S be a prescheme. The category $\mathbf{Sch}_{/S}$ of preschemes over S is then determined, as is the notion of a group in this category; such a group will also be called a *group prescheme over* S, or simply an *S-group*. To simplify the exposition and fix ideas, we shall most often restrict ourselves below to groups which are *affine* and *flat* over S [2]. This will suffice for the applications we have in mind, although of course many cases arise in which neither hypothesis holds.

Let G be such an S-group, and let P be a prescheme over S on which G acts on the 293

[1] One may observe, from the viewpoint of the "étale topology"(SGA 4 VII), that in this example $\mathrm{R}^1(f_{\text{ét}})_*(\mathbb{Z}/n\mathbb{Z})$ is "not separated"over S.

[*] [Added in 2003: $\underline{k}^4$ denotes affine space of dimension 4 over k.]

[2] In fact, for what follows, quasi-affine instead of affine would suffice; see footnote (5), p. 181.

right. In particular, there is a morphism

$$\pi\colon P \times_S G \longrightarrow P$$

in the category $\mathbf{Sch}_{/S}$, satisfying the familiar axioms. We say that P is *formally principal homogeneous under* G if the morphism

$$P \times_S G \longrightarrow P \times_S P$$

whose components are pr_1 and π is an isomorphism. Equivalently, for every object S' of $\mathbf{Sch}_{/S}$, the set

$$P(S') = \mathrm{Hom}_S(S', P),$$

regarded as a set with operator group

$$G(S') = \mathrm{Hom}_S(S', G),$$

is either empty or principal homogeneous, i.e. either empty or isomorphic to $G(S')$ with $G(S')$ acting by right translations.

We say that P is *trivial* if it is isomorphic to G, on which G acts by right translations, or equivalently if each of the operator sets $P(S')$ under $G(S')$ is trivial. One verifies, for example by the standard procedure of passing to the set-theoretic case, that *P is trivial if and only if it is formally principal homogeneous and has a section over S.* The latter condition may be stated categorically by saying that P has a section over the final object $e = S$ of $\mathbf{Sch}_{/S}$, i.e. that there is a morphism from e to P.

To define a principal homogeneous bundle P under G, a notion stronger than that of a formally principal homogeneous bundle, one must first specify in $\mathbf{Sch}_{/S}$ a class of morphisms to be used for descent and to play the role of localization morphisms that trivialize bundles. The most suitable choice varies with the context, and no one choice contains all the others[3]. Here it will be convenient to adopt the following definition:

Definition XI.4.1. *Let G be an S-group. A* principal homogeneous bundle *on the right under G is an S-prescheme P with a right action of the S-group G, such that there exist a covering of S by open subsets U_i, and for every i a faithfully flat and quasi-compact base-change morphism $S'_i \to U_i$, for which $P' = P \times_S S'$ is a trivial prescheme with operators under $G' = G \times_S S'$, where S' is the S-prescheme given by the disjoint sum of the S'_i.*

Observe that the base-change functor $X \longmapsto X' = X \times_S S'$, being left exact, sends groups to groups and objects with operator groups to objects with operator groups. Definition XI.4.1 is stable under base change. We also have: 294

Proposition XI.4.2. *Let G be an S-group, flat and quasi-compact over S, and let P be an S-prescheme on which G acts on the right. The following conditions are equivalent:*

(i) *P is a principal homogeneous bundle under G;*
(ii) *P is formally principal homogeneous under G, and the structural morphism $P \to S$ is faithfully flat and quasi-compact.*

If P is principal homogeneous under G, then, with the notation of XI.4.1, P' is faithfully flat and quasi-compact over S', since G' is so and P' is S'-isomorphic to G'. Hence P has the same properties over S. For surjectivity and quasi-compactness, see VIII VIII.3.1; flatness was omitted from the sorites of Exposé VIII. Conversely, if (ii) holds, take the base change $S' = P$, which is faithfully flat and quasi-compact over S. Then P' is formally principal homogeneous over S', since P is so over S and the base-change functor is left exact. Moreover, P' has a section over S', namely the diagonal section. It is therefore a trivial principal bundle, which completes the proof.

[3] See SGA 3, IV, especially §4.

Corollary XI.4.3. *If G is affine and flat over S, every principal homogeneous bundle P under G is affine and flat over S.*

Indeed, it becomes so after a faithfully flat and quasi-compact extension of the base, and VIII VIII.5.6 applies.

The usefulness of Definition XI.4.1 for S-groups *flat* and *affine* over S rests on VIII VIII.2.1: the morphisms $S' \to S$ considered in XI.4.1 are effective descent morphisms for the category of preschemes affine over other preschemes. Thanks to this fact, the verification of the statements outlined below is essentially categorical[4].

Let E be an S-prescheme on which the S-group G acts on the left, and let P be a principal homogeneous bundle on the right under G. We wish to define an associated bundle $E^{(P)}$, locally isomorphic to E. Let G act on the right on $P \times_S E$ by 295

$$(x, y) \longmapsto (xg, g^{-1}y),$$

the formula describing such actions in the set-theoretic context, and extended to categories by the standard procedure. Subject to its existence, set

$$E^{(P)} = (P \times_S E)/G.$$

Then $P \times_S E$ is a prescheme over $T = E^{(P)}$ with right operator group $G_T = G \times_S T$. To work comfortably, we would like $P \times_S E$ to be, in addition, a principal homogeneous bundle over T with group G_T.

To verify the existence of $E^{(P)}$ and this additional property, take the S' of Definition XI.4.1 and pass to the inverse image over S' of the original situation. Since P' is trivial, i.e. isomorphic to G'_d, one sees immediately that $E'^{(P')}$ exists and has the required exactness property. Indeed, $E' \times_{S'} P'$ is G'-isomorphic to $E' \times_{S'} G'$, so $E'^{(P')}$ is isomorphic to E'.

Moreover, formation of the associated bundle in the case of a trivial object with operators commutes with every base extension. Applying this to the various base extensions

$$S'' \rightrightarrows S' \qquad \text{and} \qquad S''' \substack{\longrightarrow\\[-1ex]\longrightarrow\\[-1ex]\longrightarrow} S',$$

where S'' and S''' are the double and triple fiber products of S' over S, shows that $E'^{(P')}$ *is endowed with a descent datum relative to $S' \to S$, and that $E^{(P)}$ exists with the required properties if and only if this descent datum is effective.* In that case $E^{(P)}$ is, of course, precisely the descended object. Here one uses the fact that $S' \to S$ is a descent morphism in the category of S-preschemes; see VIII VIII.5.2. It follows that *the associated bundle exists if E is affine over S.*

We shall apply this construction when an S-group homomorphism $G \to H$ is given, taking for E the S-prescheme H with the left action of G induced by the given morphism. The group H acts on itself on the right in a way that commutes with the action of G on H, and formation of the associated bundle, whenever it exists over S, commutes with base extension. It follows readily that H acts on the right on $P^{(H)}$, which is therefore a principal homogeneous bundle under H in the sense of XI.4.1, and is in fact trivialized by 296
the same morphism $S' \to S$ as P. In particular, *to every principal homogeneous bundle P under G and every homomorphism of S-groups $G \to H$, with H affine over S, there is functorially associated a principal homogeneous bundle with group H.* This construction is functorial in $(G \to H)$ and compatible with arbitrary base changes $T \to S$.

Definition XI.4.4. *Let G be an S-prescheme. We denote by* $\mathrm{H}^0(S, G)$ *the set of sections of G over S, regarded as a group when G is an S-group. In that case,* $\mathrm{H}^1(S, G)$

[4]Cf. loc. cit. in footnote (3), p. 179.

denotes the set of isomorphism classes of principal homogeneous bundles over S with group G, endowed with the distinguished point corresponding to trivial bundles[5].

Thus $\mathrm{H}^0(S, G)$ is a functor in the S-prescheme G, with values in the category of sets. This functor is left exact, and a fortiori commutes with finite products; it therefore sends groups to groups and commutative groups to commutative groups. Similarly, by formation of associated bundles, $\mathrm{H}^1(S, G)$ is a functor in the *affine* S-group G, with values in the category of pointed sets, and one readily verifies that it commutes with finite products. In particular, it sends groups in the category of affine S-groups, i.e. *commutative affine S-groups*, to groups in the category of pointed sets, and indeed to commutative groups, since the group objects in the first category are commutative. Thus, *if G is a commutative affine S-group, then $\mathrm{H}^1(S, G)$ is a commutative group*, and a homomorphism $G \to H$ of commutative affine S-groups gives rise to a group homomorphism

$$\mathrm{H}^1(S, G) \longrightarrow \mathrm{H}^1(S, H).$$

For simplicity, in what follows we restrict ourselves to *affine commutative* S-groups. Let

$$0 \longrightarrow G' \xrightarrow{u} G \xrightarrow{v} G'' \longrightarrow 0$$

be a sequence of morphisms of such groups. *We shall say that this sequence is exact if* $vu = 0$, which allows G to be regarded as a prescheme over G'' with right operator group $G'_{G''}$, *and if G is a principal homogeneous bundle over G'' with group* $G'_{G''} = G' \times_S G''$. 297
This implies in particular that $u\colon G' \to G$ is a kernel of v, and hence that the sequence

$$0 \longrightarrow \mathrm{H}^0(S, G') \longrightarrow \mathrm{H}^0(S, G) \longrightarrow \mathrm{H}^0(S, G'')$$

is exact. It also allows one to define a map

$$\partial\colon \mathrm{H}^0(S, G'') \longrightarrow \mathrm{H}^1(S, G')$$

as follows. To every section of G'' over S, i.e. every S-morphism $f\colon S \to G''$, associate the principal homogeneous bundle P_f over S, with group $G' \simeq f^*(G'_{G''})$, obtained by inverse image of the principal homogeneous bundle G over G''. As an S-prescheme, this is simply the inverse image under $v\colon G \to G''$ of the subprescheme which is the image of S under the immersion f; the action of G' on P_f is induced by the right action of G' on G.

We also leave to the reader the verification of the following proposition, which presents no difficulty apart from its formulation:

Proposition XI.4.5. *The map $\partial\colon \mathrm{H}^0(S, G'') \to \mathrm{H}^1(S, G')$ is a group homomorphism. The following sequence of homomorphisms is exact:*

$$0 \longrightarrow \mathrm{H}^0(S, G') \xrightarrow{u^0} \mathrm{H}^0(S, G) \xrightarrow{v^0} \mathrm{H}^0(S, G'') \xrightarrow{\partial} \mathrm{H}^1(S, G') \xrightarrow{u^1} \mathrm{H}^1(S, G) \xrightarrow{v^1} \mathrm{H}^1(S, G'').$$

All the homomorphisms other than ∂ arise from the functoriality of H^0*, respectively* H^1.

Remarks XI.4.6. The point of view presented here for the study of principal homogeneous bundles is visibly inspired by Serre [**7**], which the reader is strongly advised to consult. For a formalism that also applies to structural S-groups quasi-projective over S, so as in particular to include projective abelian schemes, it is preferable to modify XI.4.1 by requiring S' to be a disjoint sum of preschemes S'_i which are finite and locally free over open subsets S_i covering S. The preceding developments, including in particular XI.4.5, then remain valid after replacing the affine hypothesis everywhere by the quasi-projective hypothesis, and interpreting accordingly the definition given above of an exact sequence of S-groups. Indeed, one need only replace the reference to VIII VIII.2.1 by VIII VIII.7.7: 298

[5] This notation is compatible with the general cohomological notation (SGA 4, V) only when descent effectivity criteria are available. These are hardly assured unless G is affine, or merely quasi-affine; see VIII VIII.7.9.

the morphisms $S' \to S$ used here are effective descent morphisms for the fibered category of preschemes quasi-projective over other preschemes. One should note, however, that this second notion of principal homogeneous bundle is more restrictive than the first one in XI.4.1.

. An even more restrictive notion of principal homogeneous bundle is obtained by requiring S to be covered by open subsets S_i such that, for every i, $P|S_i$ is a trivial bundle with operators under $G|S_i$. We then say that P is a *locally trivial principal homogeneous bundle*. For fixed G, the classes of these bundles form a subset of $\mathrm{H}^1(S, G)$, in bijection with $\mathrm{H}^1(S, \mathcal{O}_S(G))$, where $\mathcal{O}_S(G)$ is the sheaf, in the ordinary sense, of sections of G over S; see [**2**]. For these H^1 sets still to give rise to the cohomology exact sequence XI.4.5, one must plainly suppose that

$$0 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 0$$

is exact in the appropriate sense for this new context: G must be a locally trivial bundle over G'' with group $G'_{G''}$. Equivalently, $u\colon G' \to G$ is a kernel of $v\colon G \to G''$, and G locally admits a section over G''.

. It is plainly very desirable to continue the exact sequence XI.4.5 by introducing the higher cohomology groups $\mathrm{H}^i(S, G)$. This is possible in the framework of *Weil cohomology*. Consider the category $\mathcal{B}$ of preschemes quasi-compact over S, equipped with the class $\mathcal{M}$ of faithfully flat and quasi-compact morphisms, called localizing morphisms. An abelian *Weil sheaf* on S, or more precisely on $(\mathcal{B}, \mathcal{M})$, is a contravariant functor $\mathcal{F}$ from $\mathcal{B}$ to the category of abelian groups, sending sums to products and every sequence

$$T'' = T' \times_T T' \overset{\mathrm{pr}_1, \mathrm{pr}_2}{\rightrightarrows} T' \xrightarrow{f} T\ , \qquad f \in \mathcal{M},$$

to an *exact* diagram of sets

$$\mathcal{F}(T) \longrightarrow \mathcal{F}(T') \rightrightarrows \mathcal{F}(T'').$$

Weil sheaves form an abelian category with exact direct limits and a generator, and hence with enough injective objects [**1**]. The right derived functors of

$$\Gamma(\mathcal{F}) = \mathcal{F}(S)$$

are denoted by $\mathrm{H}^i(S, \mathcal{F})$. Moreover, every commutative S-group plainly defines a Weil sheaf (VIII VIII.5.2), whose H^0 and H^1 are precisely $\mathrm{H}^0(S, G)$ and $\mathrm{H}^1(S, G)$. This gives a reasonable definition of the remaining groups $\mathrm{H}^i(S, G)$. One also proves that an exact 299
sequence of S-groups defines an exact sequence of Weil sheaves, recovering and extending the exact sequence XI.4.5[6].

. It would be appropriate to develop the noncommutative variants of XI.4.5, as in [**2**]. For a systematic development, in the proper framework, of the various cohomological notions outlined in this section, we refer to a work then in preparation by J. Giraud[7].

XI.5. Special cases of principal bundles

Suppose now that S is connected and endowed with a geometric point a. Its fundamental group $\pi_1(S, a)$ classifies the étale coverings of S: the category of étale coverings of S is equivalent to the category of finite sets on which π_1 acts continuously. It follows that a finite étale group scheme G over S is determined essentially by an ordinary finite group $\mathcal{G}$, on which π_1 acts continuously by group automorphisms. An étale covering P of S on which G acts on the right is determined essentially by a finite set $\mathcal{P}$ on which

[6]For a systematic study of this point of view, cf. SGA 4, I through IX.

[7]Cf. J. Giraud, *Cohomologie non abélienne*, Springer-Verlag, 1971.

π_1 acts continuously on the left, and on which $\mathcal{G}$ acts on the right compatibly with the action of π_1:

$$s(p \cdot g) = (sp) \cdot (sg) \qquad \text{for } s \in \pi_1,\ p \in \mathcal{P},\ g \in \mathcal{G}.$$

One verifies that P is a principal homogeneous bundle in the sense of XI.4.1 if and only if $\mathcal{P}$ is a principal homogeneous set under $\mathcal{G}$, for example by using the criterion XI.4.2. In other words, *the category of principal homogeneous bundles over S with group G is equivalent to the category of principal homogeneous bundles with group G in the category of finite sets on which π_1 acts continuously.* In particular, one obtains a canonical bijection, functorial in G:

$$(*) \qquad \mathrm{H}^1(S, G) \xrightarrow{\sim} \mathrm{H}^1(\pi_1, \mathcal{G}).$$

Here the right-hand side denotes the set of isomorphism classes of principal homogeneous bundles under $\mathcal{G}$ in the category of finite sets on which π_1 acts; there is no need to specify continuity. As is well known, this set is the quotient of the set $Z^1(\pi_1, \mathcal{G})$ of 1-cocycles $\varphi\colon \pi_1 \to \mathcal{G}$, satisfying 300

$$\varphi(1) = 1, \qquad \varphi(st) = \varphi(s)(s \cdot \varphi(t)),$$

by the natural action of $\mathcal{G}$.

An important case is that in which π_1 acts trivially on $\mathcal{G}$, i.e. when G is a completely split covering of S, isomorphic to the disjoint sum of $\mathcal{G}$ copies of S. One then also writes $\mathrm{H}^1(S, \mathcal{G})$ in place of $\mathrm{H}^1(S, G)$, and the bijection $(*)$ identifies this set with

$$\mathrm{H}^1(\pi_1, \mathcal{G}) = \mathrm{Hom}(\pi_1, \mathcal{G})/\{\text{inner automorphisms of } \mathcal{G}\}.$$

In this case, a principal homogeneous bundle over S with group G is nothing other than a *principal covering* of S with group $\mathcal{G}$ (V V.2.7). The preceding bijection is the one deduced from the correspondence between principal coverings of S with group $\mathcal{G}$, *pointed* above a, and continuous homomorphisms from $\pi_1(S, a)$ to $\mathcal{G}$; see the end of V, no. V.5.

The usefulness of connecting the theory of étale coverings with that of principal bundles, already implicit in A. Weil, *Généralisation des Fonctions Abéliennes*, and made explicit by S. Lang in his geometric class-field theory; see Serre [**5**], stems from the following fact. Every S-group finite and étale over S can be embedded into an S-group H, affine and smooth over S, with connected fibers, and commutative when G is commutative. The exact sequence XI.4.5, together where necessary with its noncommutative variants, then allows the discrete classification of principal coverings with group G to be studied through the continuous classification of principal bundles with group H, and conversely. For the idea of the general construction of the embedding $G \to H$, which appears to be used rather little in practice, see [**5**, VI 2.8]. In the next section we shall develop only two important classical special cases. We shall need the following auxiliary result:

Proposition XI.5.1. *Let S be a prescheme, and let G be an S-group isomorphic to $\mathbf{GL}(n)_S$, for example $\mathbf{G}_{m\,S}$, or to $\mathbf{G}_{a\,S}$. Then every principal homogeneous bundle under G is locally trivial.* 301

Here $\mathbf{GL}(n)_S$, with $n \geq 0$, denotes the S-group representing the contravariant functor

$$T \longmapsto \mathbf{GL}(n, \Gamma(T, \mathcal{O}_T))$$

in the S-prescheme T. In particular, the multiplicative group $\mathbf{G}_{m\,S}$ represents the contravariant functor

$$T \longmapsto \Gamma(T, \mathcal{O}_T^*),$$

and hence, as a prescheme over S, is isomorphic to

$$\mathrm{Spec}(\mathcal{O}_S[t, t^{-1}]),$$

where t is an indeterminate. Similarly, $\mathbf{G}_{a\,S}$ represents the contravariant functor

$$T \longmapsto \Gamma(T, \mathcal{O}_T),$$

and is therefore isomorphic, as an S-prescheme, to $\mathrm{Spec}(()\mathcal{O}_S[t])$. By dévissage, XI.5.1 recovers Rosenlicht's local-triviality result for the case in which G has a composition series whose successive factors are groups of the kind considered here. For a finer study of local triviality questions for principal homogeneous bundles, see [**7**] and [**3**].

The first assertion is proved by observing that

$$G(T) = \mathrm{Aut}(\mathcal{O}_T^n),$$

and that the morphisms $S' \to S$ occurring in XI.4.1, i.e. the faithfully flat and quasi-compact morphisms, are effective descent morphisms for the fibered category of modules locally isomorphic to $\mathcal{O}_T^n$, i.e. locally free of rank n (VIII VIII.1.12). The second assertion is proved similarly, noting that in this case

$$G(T) = \mathrm{Aut}(\mathcal{E}_T),$$

where $\mathcal{E}_T$ is the *trivial extension* of $\mathcal{O}_T$ by $\mathcal{O}_T$, and where the automorphisms must of course preserve the extension structure. The morphisms $S' \to S$ in XI.4.1 are effective descent morphisms for the fibered category of extensions of $\mathcal{O}_T$ by $\mathcal{O}_T$, as follows readily from VIII VIII.1.1, and such extensions are automatically locally trivial.

Remark XI.5.2. The same type of proof applies to the symplectic group $\mathbf{Sp}(2n)_S$, since a nondegenerate alternating form on a module locally isomorphic to $\mathcal{O}_S^{2n}$, i.e. a form defining an isomorphism of that module with its dual, is locally isomorphic to the standard form. The corresponding result for the orthogonal group is false, however, already when S is the spectrum of a field, since there may be quadratic forms over a field which are not isomorphic to the standard form. In fact, it is proved essentially in [**3**] that 302
the groups $\mathbf{GL}$, $\mathbf{Sp}$, $\mathbf{G}_a$, and those that can be dévisséd into such groups are, up to minor qualifications, the only ones for which a local-triviality result of the kind considered here holds.

Corollary XI.5.3. *There are canonical bijections*

$$\mathrm{H}^1(S, \mathbf{GL}(n)_S) \xleftarrow{\sim} \mathrm{H}^1(S, \mathbf{GL}(n, \mathcal{O}_S)),$$

in particular

$$\mathrm{H}^1(S, \mathbf{G}_{m\,S}) \xleftarrow{\sim} \mathrm{H}^1(S, \mathcal{O}_S^*),$$

and

$$\mathrm{H}^1(S, \mathbf{G}_{a\,S}) \xleftarrow{\sim} \mathrm{H}^1(S, \mathcal{O}_S),$$

where the right-hand sides denote the cohomology of the topological space S with coefficients in ordinary sheaves.

In particular, $\mathrm{H}^1(S, \mathbf{GL}(n)_S)$ is identified with the set of isomorphism classes of locally free modules of rank n on S, and $\mathrm{H}^1(S, \mathbf{G}_{a\,S})$ with the set of extension classes of the module $\mathcal{O}_S$ by itself.

XI.6. Application to principal coverings: Kummer and Artin–Schreier theories

Proposition XI.6.1. *Let S be a prescheme, let $n > 0$ be an integer, let*

$$u_n \colon \mathbf{G}_{m\,S} \longrightarrow \mathbf{G}_{m\,S}$$

be the n-th power homomorphism, and let $\boldsymbol{\mu}_{n\,S}$ be its kernel. Then $\boldsymbol{\mu}_{n\,S}$ is finite and locally free of rank n over S, and is étale over S if and only if the characteristic of every $s \in S$ is prime to n. The sequence of homomorphisms

$$0 \longrightarrow \boldsymbol{\mu}_{n\,S} \longrightarrow \mathbf{G}_{m\,S} \xrightarrow{u_n} \mathbf{G}_{m\,S} \longrightarrow 0$$

is exact in the sense of Section XI.4. It will be called the Kummer exact sequence *over S, relative to the integer n.*

We have 303

$$\mathbf{G}_m = \mathrm{Spec}()O_S[t, t^{-1}],$$

and u_n corresponds to the homomorphism on affine $\mathcal{O}_S$-algebras given by

$$u_n(t) = t^n.$$

The unit section of $\mathbf{G}_{m\,S}$, on the other hand, corresponds to the augmentation homomorphism of $\mathcal{O}_S$-algebras given by

$$\varepsilon(t) = 1,$$

whose kernel is the principal ideal $(t-1)$. Its image under u_n is therefore the principal ideal $(1-t^n)$, and hence

$$\boldsymbol{\mu}_{n\,S} = \mathrm{Spec}()(\mathcal{O}_S[t]/(1-t^n)).$$

This shows in particular that $\boldsymbol{\mu}_{n\,S}$ is finite over S and is defined by an algebra over S which is free of rank n, with basis t^i, $0 \le i \le n-1$.

For this group to be étale at $s \in S$, it is necessary and sufficient that the reduced algebra $k[t]/(1-t^n)$, where $k = \kappa(s)$, obtained by formally adjoining the n-th roots of unity to k, be separable over k. Equivalently, the roots of $1-t^n$ in an algebraic closure of k must all be distinct, which is the same as n being prime to the characteristic.

Finally, by the criterion XI.4.2, exactness of the sequence in XI.6.1 reduces to proving that u_n is faithfully flat. We may suppose S affine with ring A, so $\mathbf{G}_{m\,S}$ is affine with ring $B = A[t, t^{-1}]$. It is enough to check that u_n makes B a free module of rank n over B, or equivalently that u_n is injective and that

$$A[t, t^{-1}]$$

is a free module of rank n over $A[t^n, t^{-n}]$. The elements t^i, $0 \le i \le n-1$, plainly form a basis, completing the proof.

Definition XI.6.2. *The group* $\boldsymbol{\mu}_{n\,S}$ *is called the* Kummer group of rank n over S, *and a* Kummer principal covering of rank n over S *is a principal homogeneous bundle* 304
over S whose group is the Kummer group of rank n.

The set of these coverings is a group, denoted $\mathrm{H}^1(S, \boldsymbol{\mu}_{n\,S})$, or simply $\mathrm{H}^1(S, \boldsymbol{\mu}_n)$. Formation of the Kummer group of rank n over S is compatible with base extension, so $\boldsymbol{\mu}_{n\,S}$ comes by base extension from the *absolute Kummer group* $\boldsymbol{\mu}_n$ over $\mathrm{Spec}(()\mathbb{Z})$.

Write $(\mathbb{Z}/n\mathbb{Z})_S$ for the S-group defined by the ordinary finite group $\mathbb{Z}/n\mathbb{Z}$. If G is any S-group, the homomorphisms of S-groups

$$u\colon (\mathbb{Z}/n\mathbb{Z})_S \longrightarrow G$$

are in bijection, compatibly with base change, with the sections of G over S whose n-th power is the unit section. The correspondence sends u to the image under u of the section of $(\mathbb{Z}/n\mathbb{Z})_S$ over S defined by the generator 1 mod $n\mathbb{Z}$. Thus:

Corollary XI.6.3. *If $\boldsymbol{\mu}_{n\,S}$ is étale over S, there is a bijection between isomorphisms of S-groups*

$$(\mathbb{Z}/n\mathbb{Z})_S \xrightarrow{\sim} \boldsymbol{\mu}_{n\,S}$$

and sections of $\mathcal{O}_S^*$ having exact order n on every connected component of S. Such a section is called a primitive n-th root of unity over S. *Consequently, $\boldsymbol{\mu}_{n\,S}$ is isomorphic as an S-group to $(\mathbb{Z}/n\mathbb{Z})_S$ if and only if it is étale over S, i.e. the residue characteristics of S are prime to n, and there exists a primitive n-th root of unity over S.*

This explains the role in classical Kummer theory of the hypotheses that the base field, playing the role of S, have characteristic prime to n and contain the n-th roots of unity, together with the choice of a primitive n-th root of unity. Once the language of schemes is available, these hypotheses need no longer encumber the theory: one should reason directly with $\boldsymbol{\mu}_n$, rather than with $\mathbb{Z}/n\mathbb{Z}$. Combining XI.6.1, XI.4.5, and XI.5.3

gives the following general relation between the theory of Kummer principal coverings and that of Picard groups:

Proposition XI.6.4. *Let S be a prescheme. There is a canonical exact sequence* 305

$$0 \to \mathrm{H}^0(S, \boldsymbol{\mu}_n) \to \mathrm{H}^0(S, \mathcal{O}_S^*) \to \mathrm{H}^0(S, \mathcal{O}_S^*) \to \mathrm{H}^1(S, \boldsymbol{\mu}_n) \to \mathrm{H}^1(S, \mathcal{O}_S^*) \to \mathrm{H}^1(S, \mathcal{O}_S^*).$$

On setting $\mathrm{H}^1(S, \mathcal{O}_S^*) = \mathrm{Pic}(S)$, and, for every abelian group A, denoting the kernel and cokernel of multiplication by n on A by ${}_nA$ and A_n, respectively, one obtains the exact sequence

$$0 \longrightarrow \mathrm{H}^0(S, \mathcal{O}_S^*)_n \longrightarrow \mathrm{H}^1(S, \boldsymbol{\mu}_n) \longrightarrow {}_n\mathrm{Pic}(S) \longrightarrow 0.$$

We spell out two important cases in which one or the other outer term of this exact sequence vanishes:

Corollary XI.6.5. *Suppose that ${}_n\mathrm{Pic}(S) = 0$, for example if S is the spectrum of a local ring or of a factorial ring, and set*

$$A = \mathrm{H}^0(S, \mathcal{O}_S).$$

Then there is a canonical isomorphism

$$\mathrm{H}^1(S, \boldsymbol{\mu}_n) \xrightarrow{\sim} A^*/A^{*n}.$$

This is essentially the classical statement of Kummer theory when S is the spectrum of a field.

Corollary XI.6.6. *Suppose that every element of $\mathrm{H}^0(S, \mathcal{O}_S)$ is an n-th power, for example if $\mathrm{H}^0(S, \mathcal{O}_S)$ is a product of algebraically closed fields, or if S is reduced and proper over an algebraically closed field k. Then there is a canonical isomorphism*

$$\mathrm{H}^1(S, \boldsymbol{\mu}_n) \xrightarrow{\sim} {}_n\mathrm{Pic}(S).$$

In particular, when S is proper and connected over an algebraically closed field k, this relates the fundamental group of S to the finite-order points of the Picard scheme P of S over k. Thus, for n prime to the characteristic, one has an isomorphism

$$\mathrm{Hom}(\pi_1(S), \mathbb{Z}/n\mathbb{Z}) \simeq {}_nP(k),$$

which is often used in algebraic geometry. If the connected component P^0 of P is a complete group scheme of dimension g, the results recalled in Section XI.2, together with finiteness of the Néron–Severi torsion group, show that for every prime ℓ different from the characteristic, the ℓ-primary component of the abelianized fundamental group $\pi_1(S)$ is a finite-type module of rank $2g$ over the ring $\mathbb{Z}_\ell$ of ℓ-adic integers. It is moreover free for all but at most finitely many values of ℓ. 306

As Serre observed, this can be used under suitable hypotheses to prove that, when X is flat and projective over connected S, the Picard schemes of the fibers of X all have the same dimension, by applying the semicontinuity theorem X X.2.3. Serre's argument applies whenever the Picard scheme of X over S exists and the connected Picard schemes of the fibers of X over S are proper group schemes. This holds, for example, when the geometric fibers of X over S are normal, with X still flat and projective over S, and in particular when X is smooth and projective over S.

Now let p be a prime, and suppose that S is a prescheme of characteristic p, i.e. $p\mathcal{O}_S = 0$. The p-th power map on $\mathcal{O}_S$ is additive, and the corresponding morphism, obtained by replacing S by a variable T over S,

$$F \colon \mathbf{G}_{a\,S} \longrightarrow \mathbf{G}_{a\,S},$$

is therefore a homomorphism of S-groups, called the *Frobenius homomorphism.* Such a morphism is defined for every S-prescheme G obtained by base extension from a prescheme

G_0 over the prime field $\mathbb{Z}/p\mathbb{Z}$, and it is a group homomorphism if G_0 is a group prescheme. We set

$$\wp = \mathrm{id} - F\colon \mathbf{G}_{a\,S} \longrightarrow \mathbf{G}_{a\,S}.$$

Consider, on the other hand, the S-group $(\mathbb{Z}/p\mathbb{Z})_S$ defined by the ordinary finite group $\mathbb{Z}/p\mathbb{Z}$. We have said that, for every S-group G, the homomorphisms of S-groups from $(\mathbb{Z}/p\mathbb{Z})_S$ to G correspond bijectively to the sections of G over S whose p-th power is the identity section. When $G = \mathbf{G}_{a\,S}$, they therefore correspond to arbitrary sections 307
of G over S. Taking in particular the section of $\mathbf{G}_{a\,S}$ over S corresponding to the unit section of the sheaf of rings $\mathcal{O}_S$, we obtain a homomorphism of S-groups

$$i\colon (\mathbb{Z}/p\mathbb{Z})_S \to \mathbf{G}_{a\,S}$$

Proposition XI.6.7. *The sequence of homomorphisms of S-groups*

$$0 \to (\mathbb{Z}/p\mathbb{Z})_S \to \mathbf{G}_{a\,S} \to \mathbf{G}_{a\,S} \to 0$$

is exact (in the sense of no. XI.4). (It is called the Artin–Schreier exact sequence *over S.)*

It is enough to prove this over the prime field $k = \mathbb{Z}/p\mathbb{Z}$. It is enough to observe that the homomorphism $\wp^*\colon k[t] \to k[t]$ defined by $\wp^*(t) = t - t^p$ makes $k[t]$ a free module of rank p over $k[t]$: more precisely, $k[t]$ is a free module over $k[s]$, where $s = t - t^p$, with basis given by the t^i $(0 \leq i \leq p-1)$.

Using XI.4.5 and XI.5.3, we conclude:

Proposition XI.6.8. *There is a canonical exact sequence*

$$0 \to \mathrm{H}^0(S, \mathbb{Z}/p\mathbb{Z}) \to \mathrm{H}^0(S, \mathcal{O}_S) \to \mathrm{H}^0(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \\ \to \mathrm{H}^1(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathcal{O}_S),$$

and hence an exact sequence

$$0 \to \mathrm{H}^0(S, \mathcal{O}_S)/\wp\mathrm{H}^0(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \to \mathrm{H}^1(S, \mathcal{O}_S)^F \to 0,$$

(where the superscript F in the last term denotes the subgroup of invariants under the endomorphism F, equal to the kernel of $\wp = \mathrm{id} - F$).

Let us spell out two extreme cases:

Corollary XI.6.9. *Suppose that $\mathrm{H}^1(S, \mathcal{O}_S)^F = 0$, for example that S is an affine scheme. Then, putting $A = \mathrm{H}^0(S, \mathcal{O}_S)$, there is a canonical isomorphism*

$$\mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \xrightarrow{\sim} A/\wp A.$$

This is *Artin–Schreier theory* in its classical form, at least when S is the spectrum of a field.

Corollary XI.6.10. *Suppose that $\wp\mathrm{H}^0(S, \mathcal{O}_S) = \mathrm{H}^0(S, \mathcal{O}_S)$, for example that $\mathrm{H}^0(S, \mathcal{O}_S)$* 308
is a product of algebraically closed fields, or that S is proper over an algebraically closed field. Then there is a canonical isomorphism

$$\mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z}) \xrightarrow{\sim} \mathrm{H}^1(S, \mathcal{O}_S)^F.$$

Remarks XI.6.11. The last statement is due to J-P. Serre [**9**]. It is also possible to develop an analogous theory for the structure group $\mathbb{Z}/p^n\mathbb{Z}$, for arbitrary n, by using the Witt group scheme $\mathbf{W}_n$ in place of $\mathbf{G}_a$, cf. loc. cit. Notice that in characteristic $p > 0$, Kummer theory no longer gives information about principal coverings of order p, since $\boldsymbol{\mu}_p$ is then an *infinitesimal* group, i.e. radicial over the base, and hence has no direct connection with $\mathbb{Z}/p\mathbb{Z}$. Thus, at first sight, the theory of these coverings can no longer be handled (when, for definiteness, S is a proper scheme over an algebraically closed field) by the theory of the Picard scheme as in XI.6.6. Nevertheless, recalling that the Zariski tangent space at the origin of $\mathbf{Pic}_{S/K}$ [8] is identified with $\mathrm{H}^1(S, \mathcal{O}_S)$, we see that *knowledge of the group scheme* ${}_p\mathbf{Pic}_{S/k}$*, the kernel of multiplication by* p *in* $\mathbf{Pic}_{S/k}$*, determines* $\mathrm{H}^1(S, \mathbb{Z}/p\mathbb{Z})$ *as well as* $\mathrm{H}^1(S, \boldsymbol{\mu}_p)$*; note that it also determines* $\mathrm{H}^1(S, \boldsymbol{\alpha}_p)$. Here $\boldsymbol{\alpha}_p$ denotes the infinitesimal group scheme over the prime field that is the kernel of $F\colon \mathbf{G}_a \to \mathbf{G}_a$ (it may also be described as the spectrum of the restricted enveloping algebra of the trivial one-dimensional p-Lie algebra): indeed, the exact sequence XI.4.5 gives in this case

$$\mathrm{H}^1(S, \boldsymbol{\alpha}_p) \simeq \mathrm{Ker}(F\colon \mathrm{H}^1(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathcal{O}_S)),$$

and, more generally, if $\boldsymbol{\alpha}_{p^n}$ denotes the kernel in $\mathbf{G}_a$ of the n-th iterate of F, then

$$\mathrm{H}^1(S, \boldsymbol{\alpha}_{p^n}) \simeq \mathrm{Ker}(F^n\colon \mathrm{H}^1(S, \mathcal{O}_S) \to \mathrm{H}^1(S, \mathcal{O}_S)).$$

In fact, knowledge of ${}_p\mathbf{Pic}_{S/k}$ is equivalent to knowledge of $\mathrm{H}^1(S, G)$ for every finite commutative algebraic group G annihilated by p. More generally, knowledge of ${}_{p^n}\mathbf{Pic}_{S/k}$ 309
is equivalent to knowledge of $\mathrm{H}^1(S, G)$ for every finite commutative algebraic group G annihilated by p^n, by virtue of the following theorem, which in the situation under consideration encompasses both Kummer theory and Artin–Schreier theory:

Let G be a finite algebraic group over k, and let $D(G) = \mathbf{Hom}_{k\text{-groups}}(G, \mathbf{G}_m)$ be its *Cartier dual* (whose affine algebra is carried by the vector space dual to the affine algebra of G, i.e. by the hyperalgebra of G in the sense of Dieudonné–Cartier). Then there is a canonical isomorphism

$$(*) \qquad \mathrm{H}^1(S, G) \simeq \mathrm{Hom}_{k\text{-groups}}(D(G), \mathbf{Pic}_{S/k}).$$

(N.B. S is a proper scheme over the algebraically closed field k, with $\mathrm{H}^0(S, \mathcal{O}_S) = k$.) This formula can also be expressed by saying that the *true fundamental group* of S alluded to in no. XI.2, after abelianization, is isomorphic to the projective limit of the $D(P_i)$, where P_i ranges over the *finite* algebraic subgroups of $\mathbf{Pic}_{S/k}$. We denote it by $T^\bullet(\mathbf{Pic}_{X/k})$. When S is an abelian variety, we saw in XI.2.1 that this group is also isomorphic to the *true* Tate module $T_\bullet(S) = \varprojlim {}_nS$, and the isomorphism $(*)$ then takes the more striking form

$$\mathrm{Ext}^1(A, G) \simeq \mathrm{Hom}(D(G), B),$$

where A is an abelian variety, B its dual, and G a finite algebraic group over k. The results just indicated can moreover be generalized to the case where k is replaced by an arbitrary base prescheme, and to coefficient groups G other than finite groups.

[8] For the definition of $\mathbf{Pic}_{S/K}$, cf. A. Grothendieck, Sém. Bourbaki no. 232 (February 1962).

Bibliography

[1] A. Grothendieck, Sur quelques points d'algèbre homologique, Tôhoku Math. J. **9** (1957) p. 119–221. 310
[2] A. Grothendieck, A general theory of fibre spaces with structure sheaf, University of Kansas, 1955.
[3] A. Grothendieck, Torsion homologique et sections rationnelles, Séminaire Chevalley, June 16, 1958.
[4] S. Lang, Abelian varieties, Interscience tracts in pure and applied mathematics, no. 7, New York.
[5] J-P. Serre, Groupes algébriques et corps de classes, Actualités Scient. et Ind. no. 1264, Hermann, Paris, 1959.
[6] J-P. Serre, Groupes proalgébriques, Publications Mathématiques de l'IHES **7** (1960), p. 1–67.
[7] J-P. Serre, Espaces fibrés algébriques, Séminaire Chevalley, April 21, 1958.
[8] J-P. Serre, Quelques propriétés des variétés abéliennes en caractéristique p, Amer. J. Math. **80** (1958), p. 715–739.
[9] J-P. Serre, Sur la topologie des variétés algébriques en caractéristique p, Symposium Internacional de Topologia Algebraica, 1958.
[10] J-P. Serre, On the fundamental group of a unirational variety, J. London Math. Soc. **34** (1959), p. 481–484.

CHAPTER XII

Algebraic geometry and analytic geometry

311

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Proceeding as in [**10**], one associates with every scheme X locally of finite type over the field of complex numbers $\mathbb{C}$ an analytic space X^{an} whose underlying set is $X(\mathbb{C})$.

In no. XII.2 and XII.3 of this exposé, we give a "dictionary"between the usual properties of X and X^{an}, and between the properties of a morphism $f\colon X \to Y$ and those of the associated morphism $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$.

We then show that the comparison theorems for coherent sheaves on X and X^{an}, established in [**10**, no. 12] for a projective variety, remain valid when X is a proper scheme.

Finally, in no. XII.5 we prove the equivalence between the category of finite étale coverings of X and the category of finite étale coverings of X^{an}. As a bonus for the reader, we give a new proof of the Grauert-Remmert theorem [**6**], using resolution of singularities [**8**].

XII.1. Analytic space associated with a scheme

312

Let X be a scheme locally of finite type over $\mathbb{C}$. Let Φ be the functor from the category of analytic spaces [**4**, no. 9] to the category of sets which assigns to an analytic space $\mathcal{X}$ the set of morphisms of ringed spaces in $\mathbb{C}$-algebras $\mathrm{Hom}_{\mathbb{C}}(\mathcal{X}, X)$. We have the following theorem:

Theorem and definition XII.1.1. *The functor Φ is representable by an analytic space X^{an} and a morphism $\varphi\colon X^{\mathrm{an}} \to X$. We say that X^{an} is the analytic space associated with X.*

If $|X^{\mathrm{an}}|$ is the underlying set of X^{an}, φ induces a bijection from $|X^{\mathrm{an}}|$ onto the set $X(\mathbb{C})$ of points of X with values in $\mathbb{C}$. Moreover, for every point x of X^{an}, the morphism

$$\varphi_x\colon \mathcal{O}_{X,\varphi(x)} \to \mathcal{O}_{X^{\mathrm{an}},x},$$

which is necessarily local, yields upon passage to the completions an isomorphism

$$\widehat{\varphi}_x\colon \widehat{\mathcal{O}}_{X,\varphi(x)} \xrightarrow{\sim} \widehat{\mathcal{O}}_{X^{\mathrm{an}},x},$$

In particular, the morphism φ is flat.

Note that the fact that φ induces a bijection from X^{an} onto $X(\mathbb{C})$ follows from the universal property of X^{an}. We also have the following assertions:

a) If the theorem is true for a scheme Y, it is also true for every subscheme X of Y. Suppose first that X is an open subscheme of Y; if $\psi\colon Y^{\mathrm{an}} \to Y$ is the canonical morphism, $\psi^{-1}(X)$ is an open subset of Y^{an}, which we endow with 313
the analytic-space structure induced by that of Y^{an}. Since every morphism from an analytic space $\mathcal{X}$ to X factors through Y^{an} by the universal property of the latter, and therefore through X^{an}, which is the fiber product $Y^{\mathrm{an}} \times_Y X$, X^{an} is the analytic space associated with X. Finally, the assertion concerning the φ_x is clear.

[1]Based on unpublished notes by A. Grothendieck.

It remains only to consider the case where X is a closed subscheme of Y. Let I be the coherent $\mathcal{O}_Y$-ideal defining X; then $I \cdot \mathcal{O}_{Y^{\mathrm{an}}}$ is a coherent sheaf of ideals of $\mathcal{O}_{Y^{\mathrm{an}}}$ which defines a closed analytic subspace X^{an} of Y^{an}; as in the case of an open subscheme, one sees that X^{an} is the analytic space associated with X. Let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. For every point x of X^{an}, the morphism φ_x is none other than the morphism

$$\mathcal{O}_{Y,\psi(x)}/I_{\psi(x)} \to \mathcal{O}_{Y^{\mathrm{an}},x}/I_{\psi(x)} \cdot \mathcal{O}_{Y^{\mathrm{an}},x}$$

induced by ψ_x; its completion

$$\widehat{\varphi}_x\colon \widehat{\mathcal{O}}_{Y,\psi(x)}/I_{\psi(x)} \cdot \widehat{\mathcal{O}}_{Y,\psi(x)} \to \widehat{\mathcal{O}}_{Y^{\mathrm{an}},x}/I_{\psi(x)} \cdot \widehat{\mathcal{O}}_{Y^{\mathrm{an}},x}$$

is an isomorphism since $\widehat{\psi}_x$ is one, which proves a).

b) If we have two $\mathbb{C}$-schemes X_1, X_2 such that X_1^{an} and X_2^{an} exist, then $(X_1 \times X_2)^{\mathrm{an}}$ exists as well. Indeed, let $\varphi_1\colon X_1^{\mathrm{an}} \to X_1$, $\varphi_2\colon X_2^{\mathrm{an}} \to X_2$ be the canonical morphisms, and let p_1, p_2 be the two projections of $X_1^{\mathrm{an}} \times X_2^{\mathrm{an}}$. It follows formally from EGA I 1.8.1 that $X_1 \times X_2$ is the product of X_1 and X_2 in the category of locally ringed spaces; consequently, the morphisms $\varphi_1 \cdot p_1$ and $\varphi_2 \cdot p_2$ define a morphism $\varphi\colon X_1^{\mathrm{an}} \times X_2^{\mathrm{an}} \to X_1 \times X_2$, and the pair $(X_1^{\mathrm{an}} \times X_2^{\mathrm{an}}, \varphi)$ 314
represents the functor $\mathcal{X} \longmapsto \mathrm{Hom}_{\mathbb{C}}(\mathcal{X}, X_1 \times X_2)$.

c) If $\mathcal{E}^1$ denotes affine space of dimension 1, i.e. the topological space $\mathbb{C}$ endowed with the sheaf of holomorphic functions, the functor* $\mathcal{X} \longmapsto \mathrm{Hom}_{\mathbb{C}}(\mathcal{X}, E^1_{\mathbb{C}})$ is representable by $\mathcal{E}^1$, the canonical morphism $\varphi\colon \mathcal{E}^1 \to E^1_{\mathbb{C}}$ being the evident morphism. Indeed, to give a morphism from an analytic space $\mathcal{X}$ to $E^1_{\mathbb{C}}$ is equivalent to giving an element of $\Gamma(\mathcal{X}, \mathcal{O}_{\mathcal{X}})$, which is also equivalent to giving a morphism from $\mathcal{X}$ to $\mathcal{E}^1$. We plainly have a bijection $|\mathcal{E}^1| \simeq E^1(\mathbb{C})$, and, for every point $x \in \mathcal{E}^1$, the morphism $\widehat{\varphi}_x$ is none other than the identity morphism of a ring of formal power series in one variable over $\mathbb{C}$.

It follows from b) and c) that the theorem is true for affine space $E^n_{\mathbb{C}}$, $n \geq 0$. Using a), one sees that it is also true for every affine scheme X locally of finite type over $\mathbb{C}$. If X is no longer assumed affine and if (X_i) is a covering of X by affine open subsets, it follows from the universal property and from a) that the X_i^{an} glue together and thus define the analytic space X^{an} associated with X.

. Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type. If $\varphi\colon X^{\mathrm{an}} \to X$ and $\psi\colon Y^{\mathrm{an}} \to Y$ are the canonical morphisms, it follows from the universal property of Y^{an} that there exists a unique morphism $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$ such that the diagram

$$\begin{tikzcd} X^{\mathrm{an}} \arrow[r] \arrow[d, "f^{\mathrm{an}}"'] & X \arrow[d, "f"] \\ Y^{\mathrm{an}} \arrow[r] & Y \end{tikzcd}$$

is commutative. We have therefore defined a functor Φ from the category of $\mathbb{C}$-schemes 315
locally of finite type to the category of analytic spaces.

The functor Φ commutes with finite inverse limits. It suffices indeed to see that Φ commutes with fiber products. Now, if X, Y, Z are schemes locally of finite type over $\mathbb{C}$, it follows from the fact that $X \times_Z Y$ is the fiber product of X and Y over Z in the category of locally ringed spaces that $X^{\mathrm{an}} \times_{Z^{\mathrm{an}}} Y^{\mathrm{an}}$ satisfies the universal property characterizing $(X \times_Z Y)^{\mathrm{an}}$.

*[Added in 2003: $E^1_{\mathbb{C}}$ denotes the (algebraic) affine line over $\mathbb{C}$.]

. Let X be a $\mathbb{C}$-scheme locally of finite type, let X^{an} be the associated analytic space, and let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. If F is an $\mathcal{O}_X$-module, the inverse image $\varphi^* F = F^{\mathrm{an}}$ is a sheaf of modules over $\mathcal{O}_{X^{\mathrm{an}}}$. This defines a functor from the category of $\mathcal{O}_X$-modules to the category of modules on X^{an}. This functor commutes with inductive limits (EGA 0 4.3.2). Since the sheaf $\mathcal{O}_{X^{\mathrm{an}}}$ is coherent [**4**, no. 18 §2 th. 2], it sends coherent sheaves to coherent sheaves (EGA 0 5.3.11). Moreover:

Proposition XII.1.2. *The functor which assigns to an $\mathcal{O}_{X^{\mathrm{an}}}$-module F its inverse image F^{an} on X^{an} is exact, faithful, and conservative.*

Exactness follows from the fact that the morphism $\varphi\colon X^{\mathrm{an}} \to X$ is flat (XIII.1.1). Let us prove that the functor $F \longmapsto F^{\mathrm{an}}$ is faithful. In view of exactness, it suffices to show that if F^{an} is zero, then so is F. But, for every point x of X^{an}, we then have $F_{\varphi(x)} \otimes_{\mathcal{O}_{X,\varphi(x)}} \mathcal{O}_{X^{\mathrm{an}},x} = 0$. Since the morphism $\mathcal{O}_{X,\varphi(x)} \to \mathcal{O}_{X^{\mathrm{an}},x}$ is faithfully flat, we have $F_{\varphi(x)} = 0$ for every closed point $\varphi(x)$ of X, and, since X is Jacobson (EGA IV 10.4.8), this implies that F is zero.

The fact that the functor $F \longmapsto F^{\mathrm{an}}$ is conservative follows formally from exactness 316
and faithfulness.

XII.2. Comparison of the properties of a scheme and of its associated analytic space

Proposition XII.2.1. *Let X be a $\mathbb{C}$-scheme locally of finite type, let X^{an} be the associated analytic space, and let n be an integer. Consider the property P of being*

(i) *nonempty*
(i') *discrete*
(ii) *Cohen–Macaulay*
(iii) (S_n)
(iv) *regular*
(v) (R_n)
(vi) *normal*
(vii) *reduced*
(viii) *of dimension n.*

Then X has property P if and only if X^{an} does.

Let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. (i) follows from the equality $|X^{\mathrm{an}}| = X(\mathbb{C})$ (XIII.1.1) and from the fact that X is Jacobson (EGA IV 10.4.8). To say that X (respectivelyX^{an}) is discrete is equivalent to saying that $\dim X = 0$ (respectively$\dim X^{\mathrm{an}} = 0$, by [**4**, no. 19 §4 cor. 6]); (i') therefore follows from (viii).

Let P be one of the properties (ii) through (vii). The scheme X has property P if and only if P holds at every closed point of X; indeed, since X is excellent (EGA IV 7.8.6 (iii)), the set of points at which X has property P is open (loc. cit.), and if this open set 317
contains all the closed points, it is all of X. Thus, to say that X (respectivelyX^{an}) has property P is equivalent to saying that, for every point x of X^{an}, the local ring $\mathcal{O}_{X,\varphi(x)}$ (respectively$\mathcal{O}_{X^{\mathrm{an}},x}$) has property P. Since whether an excellent local ring has property P can be checked after passing to its completion, the proposition follows from the isomorphisms $\widehat{\mathcal{O}}_{X,\varphi(x)} \xrightarrow{\sim} \widehat{\mathcal{O}}_{X^{\mathrm{an}},x}$ in cases (ii) through (vii). The same is true in case (viii), in view of the equalities

$$\dim X = \sup_x \dim \mathcal{O}_{X,\varphi(x)} \qquad \dim X^{\mathrm{an}} = \sup_x \dim \mathcal{O}_{X^{\mathrm{an}},x}$$

where $x \in X^{\mathrm{an}}$. This completes the proof.

Proposition XII.2.2. *Let X be a $\mathbb{C}$-scheme locally of finite type, let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism, and let T be a locally constructible subset of X. Then*

$$\varphi^{-1}(\overline{T}) = \overline{\varphi^{-1}(T)}.$$

We may assume that T is a dense open subset of X. Let H be the reduced closed subscheme of X whose underlying space is $X - T$; the associated analytic space H^{an} is a closed analytic subspace of X^{an} whose underlying space is $X^{\mathrm{an}} - \varphi^{-1}(T)$. We must show that every point x of H^{an} belongs to $\overline{\varphi^{-1}(T)}$. At such a point x, the germ (X^{an}, x) contains the subgerm (H^{an}, x), and the latter is defined by a non-nilpotent ideal of $\mathcal{O}_{X^{\mathrm{an}},x}$. It then follows from the Nullstellensatz [**4**, no. 19 §4 cor. 3] that every open neighborhood of x contains points of X^{an} which do not belong to H^{an}, proving that $x \in \overline{\varphi^{-1}(T)}$.

Corollary XII.2.3. *Let X be a $\mathbb{C}$-scheme locally of finite type, let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism, and let T be a locally constructible subset of X. The subset T is open (respectivelyclosed, respectivelydense) if and only if $\varphi^{-1}(T)$ has the corresponding property.* 318

The corollary follows from XII.2.2 and from the fact that, since X is a Jacobson scheme (EGA IV 10.4.8), two locally constructible subsets of X having the same trace on the very dense subset $X(\mathbb{C})$ are equal.

Proposition XII.2.4. *Let X be a $\mathbb{C}$-scheme locally of finite type. Then X is connected (respectivelyirreducible) if and only if the same is true of X^{an}.*

Suppose that X^{an} is connected (respectivelyirreducible). The image $X(\mathbb{C})$ of X^{an} in X is then connected (respectivelyirreducible). It follows that X is connected (respectivelyirreducible), since the closed subsets of X and of $X(\mathbb{C})$ correspond bijectively (EGA IV 10.1.2).

Conversely, suppose that X is connected (respectivelyirreducible), and let us show that the same is true of X^{an}. We may reduce to the case where X is irreducible. Indeed, suppose that X is connected. Given a point x of X, the set of points $y \in X$ for which there exists a finite sequence of irreducible closed subschemes $X_1, \ldots, X_n$ of X, with $x \in X_1$, $y \in X_n$, $X_i \cap X_{i+1} \neq \emptyset$ for $1 \leq i \leq n-1$, is both open and closed, and hence is all of X. For a sequence $X_1, \ldots, X_n$ as above, we also have $X_i^{\mathrm{an}} \cap X_{i+1}^{\mathrm{an}} \neq \emptyset$ for $1 \leq i \leq n-1$; thus, once it is known that the X_i^{an} are connected, it follows that X^{an} is connected as well.

We now suppose that X is irreducible. We may further assume that X is affine. Indeed, if $(U_i)_{i \in I}$ is a covering of X by affine open subsets, any two of these open subsets have nonempty intersection, and the same property therefore holds for the covering $(U_i^{\mathrm{an}})_{i \in I}$ of X^{an}; once it is known that the U_i^{an} are irreducible, it follows that X^{an} is irreducible as well. 319

We may further assume that X is normal. Indeed, let $\tilde{X}$ be the normalization of X; since the morphism $\tilde{X} \to X$ is surjective, the same is true of $\tilde{X}^{\mathrm{an}} \to X^{\mathrm{an}}$. This shows that if $\tilde{X}^{\mathrm{an}}$ is irreducible, then X^{an} is irreducible as well.

We now suppose that X is affine and normal. Since the local rings of X^{an} are integral domains, saying that X^{an} is irreducible is equivalent to saying that it is connected. Indeed, if $\mathcal{F}$ is a closed analytic subset of X^{an}, the set of points x of X^{an} at which $\operatorname{codim}_x(\mathcal{F}, X^{\mathrm{an}}) = 0$ is a closed analytic subset of X^{an} [**4**, no. 20 A Cor. 1] that is also open; if X^{an} is connected, this shows that if $\mathcal{F} \neq X^{\mathrm{an}}$, then $\mathcal{F}$ is nowhere dense, and hence that X^{an} is irreducible. We are thus reduced to showing that X^{an} is connected.

Let

$$i\colon X \to P$$

be a compactification of X, where P is a normal projective $\mathbb{C}$-scheme and i is a dominant open immersion. It then follows from [**10**, no. 12 th. 1] that P^{an} is connected. Since X^{an} is obtained by removing from P^{an} a nowhere dense closed analytic subset, it follows from XII.2.5 below that X^{an} is connected as well.

Lemma XII.2.5. *Let $\mathcal{P}$ be a connected normal analytic space and $\mathcal{Y}$ a nowhere dense closed analytic subset. Then $\mathcal{X} = \mathcal{P} - \mathcal{Y}$ is connected.* 320

When $\mathcal{Y}$ has codimension ≥ 2, the proposition follows from [**11**, no. 3 prop. 4]. In the general case, after removing from $\mathcal{P}$ a closed analytic subset of codimension ≥ 2, we may assume that $\mathcal{P}$ and $\mathcal{Y}$ (regarded as a reduced analytic subspace of $\mathcal{P}$) are regular. By the implicit function theorem, every point y of $\mathcal{Y}$ has a neighborhood $\mathcal{U}$ isomorphic to a ball in an affine space $\mathcal{E}^n$, in such a way that $\mathcal{U} \cap \mathcal{Y}$ is defined by the vanishing of a certain number of coordinate functions. This shows that $\mathcal{U} - \mathcal{U} \cap \mathcal{Y}$ is connected, and therefore so is $\mathcal{X}$.

Corollary XII.2.6. *Let X be a $\mathbb{C}$-scheme locally of finite type; the morphism*

$$\pi_0(X^{\mathrm{an}}) \to \pi_0(X)$$

induced by the canonical morphism $X^{\mathrm{an}} \to X$ is bijective.

XII.3. Comparison of properties of morphisms

Proposition XII.3.1. *Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type, and let $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$ be the morphism induced by f on the associated analytic spaces. Let P be the property of being*

(i) *flat*
(ii) *net (i.e. unramified)*
(iii) *étale*
(iv) *smooth*
(v) *normal* 321
(vi) *reduced*
(vii) *injective*
(viii) *separated*
(ix) *an isomorphism*
(x) *a monomorphism*
(xi) *an open immersion.*

Then f has property P if and only if f^{an} does.

Denote by $\varphi\colon X^{\mathrm{an}} \to X$ and $\psi\colon Y^{\mathrm{an}} \to Y$ the canonical morphisms. Let x be a point of X^{an}, and put $y = f^{\mathrm{an}}(x)$. The morphisms $\mathcal{O}_{Y^{\mathrm{an}},y} \to \mathcal{O}_{X^{\mathrm{an}},x}$ and $\mathcal{O}_{Y,\psi(y)} \to \mathcal{O}_{X,\varphi(x)}$ induced by f and f^{an} yield the same morphism after passage to completions (XII.1.1). By [**2**, ch. 3 §5 prop. 4] (respectivelyEGA IV 17.4.4), saying that f^{an} satisfies property (i) (respectively(ii)) is therefore equivalent to saying that f satisfies (i) (respectively(ii)) at every closed point of X. Since the set of points of X at which (i) (respectively(ii)) holds is open (EGA IV 11.1.1 and I 3.3), this proves (i) and (ii), and hence also (iii).

Let P be property (iv) (respectively(v), respectively(vi)). In view of XII.2.1 ((v), (vi), (vii)), the geometric fibers of f^{an} at the various points y of Y^{an} are regular (respectively-normal, respectivelyreduced) if and only if the same holds for the geometric fibers of f at the various closed points $\psi(y)$ of Y. The cases (iv) (respectively(v), respectively(vi)) then follow from (i) and from the fact that the set of points of Y where the geometric fibers of f are regular is open (EGA IV 12.1.7).

(vii). If f is injective, then so is f^{an}. Conversely, suppose that f^{an} is injective, and let us show that f is as well. We may suppose that f is of finite type. Since f^{an} is injective, 322 the fibers of f over the closed points of Y are radicial; since the set of points of Y whose fiber is radicial is locally constructible (EGA IV 9.6.1), and since Y is a Jacobson scheme, all the fibers of f are radicial, and hence the morphism is injective.

(viii). Let $\Delta\colon X \to X \times_Y X$ and $\Delta^{\mathrm{an}}\colon X^{\mathrm{an}} \to X^{\mathrm{an}} \times_{Y^{\mathrm{an}}} X^{\mathrm{an}}$ be the diagonal immersions, and let $\Theta\colon X^{\mathrm{an}} \times_{Y^{\mathrm{an}}} X^{\mathrm{an}} \to X \times_Y X$ be the canonical morphism. By XII.2.3, saying that $\Delta(X)$ is closed in $X \times_Y X$ is equivalent to saying that $\Delta^{\mathrm{an}}(X^{\mathrm{an}})$ is closed in $X^{\mathrm{an}} \times_{Y^{\mathrm{an}}} X^{\mathrm{an}}$.

Since an open immersion is nothing but an étale injective morphism (EGA IV 17.9.1 and [**4**, no. 13 §1]), (xi) follows from (iii) and (vii). Since an isomorphism is the same thing as a surjective open immersion, (ix) follows from (xi) and (XII.3.2 (i)) below. To say that f is a monomorphism is equivalent to saying that the diagonal morphism $\Delta\colon X \to X \times_Y X$ is an isomorphism, so (x) follows from (ix).

PROPOSITION XII.3.2. *Let X and Y be two $\mathbb{C}$-schemes locally of finite type, let $f\colon X \to Y$ be a morphism of finite type, and let $f^{\mathrm{an}}\colon X^{\mathrm{an}} \to Y^{\mathrm{an}}$ be the morphism induced by f on the associated analytic spaces. Let P be the property of being*

(i) *surjective*
(ii) *dominant*
(iii) *a closed immersion*
(iv) *an immersion*
(v) *proper*[2] 323
(vi) *finite.*

Then f has property P if and only if f^{an} does.

Let $\varphi\colon X^{\mathrm{an}} \to X$ and $\psi\colon Y^{\mathrm{an}} \to Y$ be the canonical morphisms.

(i). If f is surjective, then for every point y of Y^{an}, $f^{-1}(\psi(y))$ is a nonempty closed subset of X; it therefore contains at least one closed point, which proves that f^{an} is surjective. Conversely, if f^{an} is surjective, $f(X)$ is a locally constructible subset of Y (EGA IV 1.8.4) containing all the closed points of Y; hence $f(X) = Y$.

(ii) follows from XII.2.2.

(iii). If f is a closed immersion, then so is f^{an} by XII.1.1 a). Conversely, if f^{an} is a closed immersion, then so is f by XII.3.1 (x) and XII.3.2 (v), since this is equivalent to saying that f is a proper monomorphism (EGA IV 8.11.5).

(iv). It is clear that if f is an immersion, then so is f^{an}. Conversely, suppose that f^{an} is an immersion, and let T be the image of X in Y, and $\overline{T}$ the scheme-theoretic closure of f. We have a factorization of f,

$$X \xrightarrow{i} \overline{T} \xrightarrow{j} J,$$

where j is a closed immersion, i is the canonical morphism, and from this we deduce the following factorization of f^{an}:

$$X^{\mathrm{an}} \xrightarrow{i^{\mathrm{an}}} \overline{T}^{\mathrm{an}} \xrightarrow{j^{\mathrm{an}}} Y^{\mathrm{an}} .$$

Since $T = f(X)$ is a locally constructible subset of Y (EGA IV 1.8.4), we have, by XII.2.2, 324
$\overline{T}^{\mathrm{an}} = \overline{f^{\mathrm{an}}(X^{\mathrm{an}})}$. It follows that $i^{\mathrm{an}}(X^{\mathrm{an}})$ is open in $\overline{T}^{\mathrm{an}}$, and hence that $i(X)$ is open in $\overline{T}$. Consider the canonical factorization of i:

$$X \xrightarrow{i_1} i(X) \xrightarrow{i_2} \overline{T}.$$

The morphism i_1^{an} is a proper monomorphism, and hence so is i_1 by XII.3.2 (v) and XII.3.1 (x); this proves that i_1, and hence also f, is an immersion.

(v). Suppose that f is proper, and let us show that f^{an} is as well. Since properness of f^{an} is local on Y^{an}, we may suppose that Y is affine. By Chow's lemma (EGA II 5.6.1), one can find a projective Y-scheme X' and a projective surjective morphism

$$g\colon X' \to X.$$

The morphism $(fg)^{\mathrm{an}} = f^{\mathrm{an}}g^{\mathrm{an}}$ is projective and hence proper, g^{an} is surjective, and it follows from [**1**, ch. 1 §10] that f^{an} is proper.

Conversely, suppose that f^{an} is proper, and let us show that f is as well. By XII.3.1 (viii), f is separated. It remains to prove that f is universally closed, and it even

[2]We shall say that a morphism of analytic spaces is proper if it is proper in the sense of [**1**, ch. 1 §10 no. 1] and if it is separated.

suffices to show that f is closed; indeed, for every Y-scheme Y' locally of finite type, the morphism

$$f_{(Y')} = h\colon X \times_Y Y' \to Y'$$

will also be closed, since h^{an} is proper. Let T be a closed subset of X; $f(T)$ is a locally constructible set, and we have

$$f^{\mathrm{an}}(\varphi^{-1}(T)) = \psi^{-1}(f(T)).$$

Since f^{an} is proper, $\psi^{-1}(f(T))$ is a closed subset of Y^{an}, and it therefore follows from XII.2.2 that 325

$$\psi^{-1}(\overline{f(T)}) = \psi^{-1}(f(T)).$$

This implies that $f(T) = \overline{f(T)}$, i.e. that f is closed and hence that f is proper.

(vi). To say that a morphism is finite is equivalent to saying that it is proper with finite fibers (EGA III 4.4.2 and [**4**, no. 19 §5]). Since the set of points where the fibers of f are finite is locally constructible (EGA IV 9.7.9), the fibers of f are finite if and only if the fibers of f^{an} are finite; (vi) therefore follows from (v).

Remark XII.3.3. a) Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type. The fact that f^{an} is a local isomorphism does not imply that the same is true of f. Indeed, if f is étale, then f^{an} is étale and hence a local isomorphism [**4**, no. 13 §1], but f need not be one.

b) The statement XII.3.2 is not true if f is not assumed to be of finite type. For example, f^{an} can be a closed immersion without f being one. Indeed, it suffices to take X to be the disjoint union of $\mathbb{Z}$ copies of $\mathrm{Spec}(\mathbb{C})$, Y to be the affine line, and f to be the morphism obtained by sending the points of X to distinct points of Y forming a discrete subset.

XII.4. Cohomological comparison and existence theorems

326

The purpose of this section is to reprove the results of [**3**, no. 2 th. 5 and th. 6]; the latter generalize the theorems established in [**10**, no. 12] for projective X to the case of a proper scheme, and extend them to the relative case. More general results concerning proper relative schemes over an analytic space are proved in [**7**, ch. VIII no. 3].

Recall that the Čech cohomology used in [**10**, no. 12] coincides with the usual cohomology in both the algebraic and the analytic cases (EGA III 1.4.1. and [**5**, II 5.10]).

. Let $f\colon X \to Y$ be a morphism of $\mathbb{C}$-schemes locally of finite type, and consider the commutative diagram

$$\begin{tikzcd} X^{\mathrm{an}} \arrow[r, "\varphi"] \arrow[d, "f^{\mathrm{an}}"'] & X \arrow[d, "f"] \\ Y^{\mathrm{an}} \arrow[r, "\psi"] & Y. \end{tikzcd}$$

If F is an $\mathcal{O}_X$-Module, then, for every integer $p \geq 0$, there are morphisms

$$\mathrm{R}^p f_* F \xrightarrow{i} \mathrm{R}^p f_*(\varphi_* F^{\mathrm{an}}) \xrightarrow{j} \mathrm{R}^p (f \cdot \varphi)_* F^{\mathrm{an}} \xrightarrow{k} \psi_*(\mathrm{R}^p f_*^{\mathrm{an}} F^{\mathrm{an}}),$$

where i is induced by the canonical morphism $F \to \varphi_* F^{\mathrm{an}}$, and j, k are "edge homomorphisms" of Leray spectral sequences. The composite $k.j.i$ gives rise to a canonical morphism

$$\theta_p\colon (\mathrm{R}^p f_* F)^{\mathrm{an}} \to \mathrm{R}^p f_*^{\mathrm{an}}(F^{\mathrm{an}}) \tag{4.1.1}$$

Theorem XII.4.1. *Let $f\colon X \to Y$ be a proper morphism of $\mathbb{C}$-schemes locally of finite type, and let F be a coherent $\mathcal{O}_X$-Module. Then, for every integer $p \geq 0$, the morphism* (4.1.1) 327

$$\theta_p\colon (\mathrm{R}^p f_* F)^{\mathrm{an}} \to \mathrm{R}^p f_*^{\mathrm{an}}(F^{\mathrm{an}})$$

is an isomorphism.

1) *The case where f is projective.* The proof is analogous to that of [**10**, no. 13]. Let us recall it briefly. One reduces to the case where X is a projective space of the form $\mathbb{P}^r_Y$ over Y. Let $\mathcal{Y} = Y^{\mathrm{an}}$, $\mathcal{P} = \mathbb{P}^r_{\mathcal{Y}}$; one first proves that

$$f^{\mathrm{an}}_* \mathcal{O}_{\mathcal{P}} = \mathcal{O}_{\mathcal{Y}}, \qquad \mathrm{R}^p f^{\mathrm{an}}_*(\mathcal{O}_{\mathcal{P}}) = 0 \qquad \text{for } p > 0$$

Indeed, to verify the preceding relations, one may reduce to the case where $\mathcal{Y}$ is a ball $\mathcal{B}$ in an affine space $\mathcal{E}^n$. Consider the "standard covering" $\{\mathcal{U}_i\}$ of $\mathcal{P}$ by $r+1$ open subsets isomorphic to $\mathcal{B} \times \mathcal{E}^r$. Since these open subsets are Stein, for every integer $p \geq 0$ there are isomorphisms

$$\mathrm{H}^p(\{\mathcal{U}_i\}, \mathcal{O}_{\mathcal{P}}) \xrightarrow{\sim} \mathrm{H}^p(\mathcal{P}, \mathcal{O}_{\mathcal{P}})$$

One can then express the sections of the structure sheaf $\mathcal{O}_{\mathcal{P}}$ over the open subsets $\mathcal{U}_i$ and over their intersections in terms of Laurent series; an easy computation shows that

$$\mathrm{H}^0(\mathcal{P}, \mathcal{O}_{\mathcal{P}}) \xrightarrow{\sim} \mathrm{H}^0(\mathcal{Y}, \mathcal{O}_{\mathcal{Y}}), \qquad \mathrm{H}^p(\mathcal{P}, \mathcal{O}_{\mathcal{P}}) = 0 \qquad \text{for } p > 0$$

The proof is then completed by repeating the argument of [**10**, no. 12 lemma 5], with the cohomology groups replaced by the cohomology sheaves.

2) *The case where f is proper.* We use EGA III 3.1.2 to reduce to the projective case. 328
Let $\mathcal{K}$ be the category of coherent $\mathcal{O}_X$-Modules for which θ_p is an isomorphism for every $p \geq 0$. It suffices to prove that, for every exact sequence $0 \to F' \to F \to F'' \to 0$ two of whose terms belong to $\mathcal{K}$, the same is true of the third; that a direct summand of an object of $\mathcal{K}$ belongs to $\mathcal{K}$; and that, for every point x of X, one can find an object F of $\mathcal{K}$ such that $F_x \neq 0$.

The first condition follows by applying the five lemma to the following commutative diagram, whose rows are exact:

$$\begin{tikzcd}
{} \arrow[r] & (\mathrm{R}^p f_* F')^{\mathrm{an}} \arrow[r] \arrow[d] & (\mathrm{R}^p f_* F)^{\mathrm{an}} \arrow[r] \arrow[d] & (\mathrm{R}^p f_* F'')^{\mathrm{an}} \arrow[r] \arrow[d] & (\mathrm{R}^{p+1} f_* F')^{\mathrm{an}} \arrow[r] \arrow[d] & {} \\
{} \arrow[r] & \mathrm{R}^p f^{\mathrm{an}}_* F'^{\mathrm{an}} \arrow[r] & \mathrm{R}^p f^{\mathrm{an}}_* F^{\mathrm{an}} \arrow[r] & \mathrm{R}^p f^{\mathrm{an}}_* F''^{\mathrm{an}} \arrow[r] & \mathrm{R}^{p+1} f^{\mathrm{an}}_* F'^{\mathrm{an}} \arrow[r] & {} ,
\end{tikzcd}$$

and the second condition is verified similarly.

To verify the third condition, one may restrict to the case where X is an irreducible scheme with generic point x. We could have assumed from the outset that Y is noetherian. By Chow's lemma (EGA II 5.6.1), one can find a projective Y-scheme X' and a projective surjective morphism $g \colon X' \to X$. Moreover, there is an integer n such that $\mathrm{R}^p g_*(\mathcal{O}_{X'}(n)) = 0$ for every $p > 0$ and the canonical morphism $g^* g_*(\mathcal{O}_{X'}(n)) \to \mathcal{O}_{X'}(n)$ is surjective (EGA III 2.2.1). If we put $F = g_*(\mathcal{O}_{X'}(n))$, then the sheaf F has the required property. Indeed, $F_x \neq 0$; moreover, since the Leray spectral sequence

$$\mathrm{R}^p f_*(\mathrm{R}^q g_*(\mathcal{O}_{X'}(n))) \Longrightarrow \mathrm{R}^{p+q}(f \cdot g)_*(\mathcal{O}_{X'}(n))$$

degenerates, there is an isomorphism

$$\mathrm{R}^p f_* F \xrightarrow{\sim} \mathrm{R}^p (f \cdot g)_*(\mathcal{O}_{X'}(n))$$

As in the algebraic case, there is a canonical isomorphism 329

$$\mathrm{R}^p f^{\mathrm{an}}_* F^{\mathrm{an}} \xrightarrow{\sim} \mathrm{R}^p (f \cdot g)^{\mathrm{an}}_*(\mathcal{O}_{X'}(n)^{\mathrm{an}}),$$

and the diagram

$$\begin{tikzcd}
(\mathrm{R}^p f_* F)^{\mathrm{an}} \arrow[r, "\sim"] \arrow[d, "\theta_p"'] & (\mathrm{R}^p (f \cdot g)_*(\mathcal{O}_{X'}(n)))^{\mathrm{an}} \arrow[d, "\psi_p"'] \\
\mathrm{R}^p f^{\mathrm{an}}_* F^{\mathrm{an}} \arrow[r, "\sim"] & \mathrm{R}^p (f \cdot g)^{\mathrm{an}}_*(\mathcal{O}_{X'}(n)^{\mathrm{an}})
\end{tikzcd}$$

is commutative. By 1), ψ_p is an isomorphism; hence so is θ_p, which completes the proof.

Corollary XII.4.2. *Let X be a proper $\mathbb{C}$-scheme, and let F be a coherent $\mathcal{O}_X$-Module. Then, for every integer $p \geq 0$, the canonical morphism*

$$\mathrm{H}^p(X, F) \to \mathrm{H}^p(X^{\mathrm{an}}, F^{\mathrm{an}})$$

is an isomorphism.

Theorem XII.4.3. *Let X be a proper $\mathbb{C}$-scheme. The functor which assigns to every coherent $\mathcal{O}_X$-Module F its inverse image F^{an} on X^{an} is an equivalence of categories.*

1) *The functor is fully faithful.* Indeed, let F and G be two coherent $\mathcal{O}_X$-Modules. The canonical morphism

$$\mathrm{Hom}_{\mathcal{O}_X}(F, G) \to \mathrm{Hom}_{\mathcal{O}_{X^{\mathrm{an}}}}(F^{\mathrm{an}}, G^{\mathrm{an}})$$

is identified with the canonical morphism

$$\mathrm{H}^0(X, \mathbf{Hom}_{\mathcal{O}_X}(F, G)) \to \mathrm{H}^0(X^{\mathrm{an}}, \mathbf{Hom}_{\mathcal{O}_X}(F, G))$$

(EGA 0_{I} 6.7.6). Since $\mathbf{Hom}_{\mathcal{O}_X}(F, G)$ is coherent, it follows from XII.4.2 that this morphism 330
is bijective.

2) *The functor is essentially surjective.* When X is projective, the assertion follows from [**10**, no. 12 th. 3]. The general case reduces to the preceding one by using Chow's lemma (EGA II 5.6.1). Indeed, let X' be a projective $\mathbb{C}$-scheme, let $f\colon X' \to X$ be a projective surjective morphism, and let U be a dense open subset of X such that f induces an isomorphism $f^{-1}(U) \simeq U$. We argue by noetherian induction on X; we may therefore suppose that, for every coherent sheaf $\mathcal{G}$ on X^{an} for which there exists a closed subset Y of X, distinct from X, satisfying $Y^{\mathrm{an}} \supset \mathrm{Supp}\,\mathcal{G}$, there is a coherent sheaf G on X such that $G^{\mathrm{an}} \simeq \mathcal{G}$.

Let $\mathcal{F}$ be a coherent sheaf of $\mathcal{O}_{X^{\mathrm{an}}}$-modules, and let $\mathcal{K}$ and $\mathcal{L}$ be the coherent sheaves defined by the condition that the sequence

$$0 \to \mathcal{K} \to \mathcal{F} \to f_*^{\mathrm{an}} f^{\mathrm{an}*}\mathcal{F} \to \mathcal{L} \to 0$$

be exact. Since X' is projective, there is a coherent $\mathcal{O}_{X'}$-Module F' such that $F'^{\mathrm{an}} \simeq f^{\mathrm{an}*}\mathcal{F}$; it then follows from XII.4.1 that there is an isomorphism $(f_*F')^{\mathrm{an}} \simeq f_*^{\mathrm{an}} f^{\mathrm{an}*}\mathcal{F}$. Since $\mathcal{K}|U^{\mathrm{an}}$ and $\mathcal{L}|U^{\mathrm{an}}$ vanish, there are coherent $\mathcal{O}_X$-Modules K and L such that $K^{\mathrm{an}} \simeq \mathcal{K}$ and $L^{\mathrm{an}} \simeq \mathcal{L}$. By 1), the morphism $f_*^{\mathrm{an}} f^{\mathrm{an}*}\mathcal{F} \to \mathcal{L}$ comes from a unique morphism $f_*F' \to L$; let $I = \mathrm{Ker}(f_*F' \to L)$. The sheaf $\mathcal{F}$ is then an extension of I^{an} by K^{an}, and
it remains to see that this extension is obtained by inverse image from an extension of I 331
by K. It therefore suffices to prove that the canonical morphism

$$(*) \qquad \mathrm{Ext}^q_{\mathcal{O}_X}(I, K)^{\mathrm{an}} \xrightarrow{\sim} \mathrm{Ext}^q_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}}) \quad q \neq 1$$

is bijective. But there are isomorphisms $\mathbf{Ext}^q_{\mathcal{O}_X}(I, K)^{\mathrm{an}} \xrightarrow{\sim} \mathbf{Ext}^q_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}})$ for every integer $q \geq 0$ (EGA 0_{III} 12.3.5), and a morphism of spectral sequences

$$\begin{array}{ccc} \mathrm{H}^p(X, \mathbf{Ext}^q_{\mathcal{O}_X}(I, K)) & \Longrightarrow & \mathrm{Ext}^{p+q}_{\mathcal{O}_X}(I, K) \\ \downarrow & & \downarrow \\ \mathrm{H}^p(X^{\mathrm{an}}, \mathbf{Ext}^q_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}})) & \Longrightarrow & \mathrm{Ext}^{p+q}_{\mathcal{O}_{X^{\mathrm{an}}}}(I^{\mathrm{an}}, K^{\mathrm{an}}). \end{array}$$

This morphism is an isomorphism because, by XII.4.2, it is an isomorphism on the terms E_2^{pq}, and this proves the bijectivity of $(*)$.

Corollary XII.4.4. *The functor which assigns to every proper $\mathbb{C}$-scheme X the analytic space X^{an} is fully faithful.*

We must show that, if X and Y are two proper $\mathbb{C}$-schemes, the canonical map

$$\mathrm{Hom}_{\mathbb{C}}(X, Y) \to \mathrm{Hom}(X^{\mathrm{an}}, Y^{\mathrm{an}})$$

is bijective. Now, to give a morphism from X to Y (respectivelyfrom X^{an} to Y^{an}) is equivalent to giving its graph, i.e. a closed subscheme Z of $X \times Y$ (respectivelya closed analytic subspace $\mathcal{Z}$ of $X^{\mathrm{an}} \times Y^{\mathrm{an}}$), such that the restriction of the first projection $X \times Y \to X$ to Z (respectivelyof $X^{\mathrm{an}} \times Y^{\mathrm{an}} \to X^{\mathrm{an}}$ to $\mathcal{Z}$) is an isomorphism. Since giving 332 a closed subscheme of $X \times Y$ (respectivelya closed analytic subspace of $X^{\mathrm{an}} \times Y^{\mathrm{an}}$) is equivalent to giving a coherent sheaf of ideals in $\mathcal{O}_{X\times Y}$ (respectivelyin $\mathcal{O}_{X^{\mathrm{an}}\times Y^{\mathrm{an}}}$), the corollary follows from XII.4.3.

Corollary XII.4.5. *Let X be a proper $\mathbb{C}$-scheme. The functor which assigns to every scheme X' finite (respectivelyfinite étale) over X the analytic space X'^{an} is an equivalence from the category of schemes finite (respectivelyfinite étale) over X to the category of analytic spaces finite (respectivelyfinite étale) over X^{an}.*

Indeed, to give a finite morphism $X' \to X$ (respectively$X'^{\mathrm{an}} \to X^{\mathrm{an}}$) is equivalent to giving a coherent sheaf of algebras over $\mathcal{O}_X$ (respectivelyover $\mathcal{O}_{X^{\mathrm{an}}}$) [**4**, no. 19 §5 th. 2]. The corollary therefore follows from XII.4.3 in the unparenthesized case, and the parenthesized case follows from it in view of XII.3.1 (iii).

XII.5. Comparison theorems for étale coverings

. Let us make precise the notion of a finite covering of an analytic space. If $\mathcal{X}$ is an analytic space, an analytic space $\mathcal{X}'$ finite over $\mathcal{X}$ is said to be a finite covering of $\mathcal{X}$ if every irreducible component of $\mathcal{X}'$ dominates an irreducible component of $\mathcal{X}$.

Theorem XII.5.1 ("Riemann existence theorem"). *Let X be a $\mathbb{C}$-scheme locally of finite type, and let X^{an} be the analytic space associated with X. The functor Ψ which assigns to every finite étale covering X' of X the space X'^{an} is an equivalence from the* 333 *category of finite étale coverings of X to the category of finite étale coverings of X^{an}.*

1) *The functor Ψ is fully faithful.* Let X' and X'' be two finite étale coverings of X, and let us prove that the canonical map

$$(*) \qquad \mathrm{Hom}_X(X', X'') \to \mathrm{Hom}_{X^{\mathrm{an}}}(X'^{\mathrm{an}}, X''^{\mathrm{an}})$$

is bijective. We may suppose that X' is connected. To give an X-morphism from X' to X'' is equivalent to giving a connected component X_i of $X' \times_X X''$ such that the morphism $X_i \to X'$ induced by the first projection is an isomorphism. Since the connected components of $X' \times_X X''$ correspond bijectively to the connected components of $X'^{\mathrm{an}} \times_{X^{\mathrm{an}}} X''^{\mathrm{an}}$ (XII.2.6), and since a morphism $X_i \to X'$ is an isomorphism if and only if $X_i^{\mathrm{an}} \to X'^{\mathrm{an}}$ is an isomorphism, this proves the bijectivity of $(*)$.

2) *The functor Ψ is essentially surjective.* Let $\mathcal{X}'$ be a finite étale covering of X^{an}, and let us prove that there exists an étale covering X' of X together with an isomorphism $X'^{\mathrm{an}} \xrightarrow{\sim} \mathcal{X}'$. In view of 1), the question is local on X, and we may therefore suppose that X is affine.

a) Reduction to the case where X is normal. We may suppose that X is reduced. Indeed, suppose that the theorem has been proved for X_{red}. The functor which assigns to a finite étale covering X' of X the finite étale covering $X'^{\mathrm{an}}_{\mathrm{red}}$ of $X^{\mathrm{an}}_{\mathrm{red}}$ is then an equivalence. Since it is obtained by composing Ψ with the functor Θ which assigns to a finite étale covering of X^{an} its inverse image over $X^{\mathrm{an}}_{\mathrm{red}}$, and since Θ is fully faithful, this shows that Ψ is an equivalence of categories.

We may suppose that X is normal. Indeed, let $\widetilde{X}$ be the normalization of X, and let 334 $p\colon \widetilde{X} \to X$ be the canonical morphism. Since p is finite, p is an effective descent morphism for the category of étale coverings (IX IX.4.7). Supposing that the theorem has been proved for $\widetilde{X}$, put $\widetilde{\mathfrak{X}'} = \mathfrak{X}' \times_{X^{\mathrm{an}}} \widetilde{X}^{\mathrm{an}}$. There then exist an étale covering $\widetilde{X}'$ of $\widetilde{X}$ and an isomorphism $\widetilde{X}'^{\mathrm{an}} \simeq \widetilde{\mathfrak{X}'}$. It follows from 1) that the natural descent datum on $\widetilde{\mathfrak{X}'}$ lifts to a descent datum on $\widetilde{X}'$ relative to $\widetilde{X} \to X$; this proves the existence of an étale covering X' of X together with an isomorphism $i\colon X'^{\mathrm{an}} \times_{X^{\mathrm{an}}} \widetilde{X}^{\mathrm{an}} \simeq \widetilde{\mathfrak{X}'}$ whose inverse images under the two projections from $\widetilde{X}^{\mathrm{an}} \times_{X^{\mathrm{an}}} \widetilde{X}^{\mathrm{an}}$ are the same. By IX IX.3.2, whose proof is valid in the analytic case, the morphism $\widetilde{X}^{\mathrm{an}} \to X^{\mathrm{an}}$ is a descent morphism for the category of étale coverings, and consequently i comes from an isomorphism $X'^{\mathrm{an}} \simeq \mathfrak{X}'$.

b) Reduction to the case where X is regular. Let U be the open subset of regular points of X, and let $i\colon U \to X$ and $i^{\mathrm{an}}\colon U^{\mathrm{an}} \to X^{\mathrm{an}}$ be the canonical morphisms; since X is normal, we have $\operatorname{codim}(X - U, X) \geq 2$. Suppose that there exists an étale covering U' of U such that $U'^{\mathrm{an}} \simeq \mathfrak{X}'|U^{\mathrm{an}}$, and let us show that U' then extends to an étale covering X' of X such that $X'^{\mathrm{an}} \simeq \mathfrak{X}'$. It suffices to see that U' extends to an étale covering X' of X; indeed, we will then have an isomorphism $X'^{\mathrm{an}}|U^{\mathrm{an}} \simeq \mathfrak{X}'|U^{\mathrm{an}}$. But if $\mathcal{F}$ and $\mathcal{G}$ are the coherent sheaves of algebras over $\mathcal{O}_{X^{\mathrm{an}}}$ defining $\mathfrak{X}'$ and X'^{an} respectively, the fact that X is normal and that $\operatorname{codim}(X - U, X) \geq 2$ implies that the canonical morphisms

$$\mathcal{F} \to i^{\mathrm{an}}_*(\mathcal{F}|U^{\mathrm{an}}) \qquad \mathcal{G} \to i^{\mathrm{an}}_*(\mathcal{G}|U^{\mathrm{an}})$$

are isomorphisms [**11**, no. 3 prop. 4]. It follows that $\mathcal{F}$ and $\mathcal{G}$, and hence also X'^{an} and 335
$\mathfrak{X}'$, are isomorphic.

Let $\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism. Since the problem of extending U' to X is local on X, it is enough to prove that, for every point y of $X^{\mathrm{an}} - U^{\mathrm{an}}$, the étale covering $U'_{\varphi(y)} = U' \times_X \operatorname{Spec}()O_{X,\varphi(y)}$ of $U_{\varphi(y)} = U \times_X \operatorname{Spec}()O_{X,\varphi(y)}$ extends to $\operatorname{Spec}()O_{X,\varphi(y)}$. Let H be the coherent $\mathcal{O}_U$-algebra defining U'. The canonical morphism

$$\alpha\colon (i_*H)^{\mathrm{an}} \to i^{\mathrm{an}}_*(H^{\mathrm{an}}) = \mathcal{F}$$

defines a morphism of sheaves of modules on $\operatorname{Spec}()O_{X^{\mathrm{an}},y}$:

$$\alpha_y\colon (i_*H)^{\mathrm{an}}_y \to \mathcal{F}_y,$$

whose restriction to $U_y = U_{\varphi(y)} \times_{\operatorname{Spec}()O_{X,\varphi(y)}} \operatorname{Spec}()O_{X^{\mathrm{an}},y}$ is an isomorphism. But this proves that $H|U_y$ is trivial, and hence that $U'_{\varphi(y)}$ extends to $\operatorname{Spec}()O_{X,\varphi(y)}$.

c) The case where X is affine and regular. Let

$$j\colon X \to P$$

be a compactification of X, where P is a projective $\mathbb{C}$-scheme and j is a dominant open immersion. By the resolution of singularities theorem [**8**], one can find a regular scheme R and a projective morphism $r\colon R \to P$ such that r induces an isomorphism $r^{-1}(X) \simeq X$ and $r^{-1}(X)$ is the complement in R of a divisor with normal crossings. Let 336

$$k\colon X \to R$$

be the canonical immersion. We shall show that there exists a finite normal covering (XII.5) $\mathcal{R}'$ of R^{an} extending the étale covering X'^{an}. By Proposition XII.5.3 below, such a covering is unique; the problem of extending X'^{an} is therefore local on R^{an} in a neighborhood of $R^{\mathrm{an}} - X^{\mathrm{an}}$. Now every point of $R^{\mathrm{an}} - X^{\mathrm{an}}$ has an open neighborhood $\mathcal{V}$ isomorphic to a ball in an affine space $\mathcal{E}^n$ such that $\mathcal{V} - \mathcal{V} \cap X^{\mathrm{an}}$ is defined by the vanishing of the first p coordinate functions $z_1, \dots, z_p$, where $0 \leq p \leq n$. The fundamental group of $\mathcal{U} = \mathcal{V} \cap X^{\mathrm{an}}$ is isomorphic to $\mathbb{Z}^p$, and every étale covering of $\mathcal{U}$ is a quotient of a covering of the form

$$\mathcal{U}'' = \mathcal{U}[T_1, \dots, T_p]/(T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p),$$

where the n_i are integers > 0, by a subgroup H of the Galois group $\mathbb{Z}/n_1\mathbb{Z} \times \cdots \times \mathbb{Z}/n_p\mathbb{Z}$ of $\mathcal{U}''$. Now $\mathcal{U}''$ extends to the regular covering

$$\mathcal{V}'' = \mathcal{V}[T_1, \dots, T_p]/(T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p),$$

of $\mathcal{V}$ on which H acts, and the quotient of $\mathcal{V}''$ by H is the desired extension.

The proof is then completed by XII.4.5. The covering $\mathcal{R}'$ comes from a finite covering R' of R; the restriction of R' to X is a covering X' of X such that $X'^{\mathrm{an}} \simeq \mathcal{X}'$, and by XII.3.1(iii), X' is an étale covering of X.

Corollary XII.5.2. *Let X be a connected $\mathbb{C}$-scheme locally of finite type, let* 337
$\varphi\colon X^{\mathrm{an}} \to X$ be the canonical morphism, and let x be a point of X^{an}. Let $\pi_1(X^{\mathrm{an}}, x)$ be the fundamental group of the topological space X^{an} at x, and let $\pi_1(X, \varphi(x))$ be the fundamental group of the scheme X at $\varphi(x)$ (V V.7). *Then $\pi_1(X, \varphi(x))$ is canonically isomorphic to the completion of $\pi_1(X^{\mathrm{an}}, x)$ with respect to the topology of subgroups of finite index.*

Indeed, let $\mathcal{C}$ be the category of finite étale coverings of X^{an}, let F be the functor from $\mathcal{C}$ to **Ens** which assigns to every finite étale covering $\mathcal{X}'$ of X^{an} the set of points of $\mathcal{X}'$ above x, and let $\widehat{\pi}_1(X^{\mathrm{an}}, x)$ be the profinite group associated with $\mathcal{C}$ and F as in V V.4. Since every finite étale covering of X^{an} is a quotient of the universal covering by a subgroup of finite index, $\widehat{\pi}_1(X^{\mathrm{an}}, x)$ is precisely the completion of $\pi_1(X^{\mathrm{an}}, x)$ for the topology of subgroups of finite index. The corollary therefore follows from XII.5.1 and V V.6.10.

Proposition XII.5.3. *Let $\mathcal{X}$ be a normal analytic space, and let $\mathcal{Y}$ be a closed analytic subset such that $\mathcal{U} = \mathcal{X} - \mathcal{Y}$ is dense in $\mathcal{X}$. Then the functor which assigns to every finite normal covering* (XII.5) *$\mathcal{X}'$ of $\mathcal{X}$ its restriction to $\mathcal{U}$ is fully faithful.*

Let $\mathcal{X}'$ and $\mathcal{X}''$ be two finite normal coverings of $\mathcal{X}$. We must show that the canonical map

$$\mathrm{Hom}_{\mathcal{X}}(\mathcal{X}', \mathcal{X}'') \to \mathrm{Hom}_{\mathcal{U}}(\mathcal{X}'|\mathcal{U}, \mathcal{X}''|\mathcal{U})$$

is bijective. Let u, v be two $\mathcal{X}$-morphisms from $\mathcal{X}'$ to $\mathcal{X}''$ whose restrictions to $\mathcal{U}$ are the same, and let us prove that $u = v$. The morphisms u and v coincide on the dense open 338
subset $\mathcal{U} \times_{\mathcal{X}} \mathcal{X}'$, and hence on the underlying topological spaces. By [**4**, no. 19 §4 cor. 5], this proves that $u = v$.

Now let u be a $\mathcal{U}$-morphism from $\mathcal{X}'|\mathcal{U}$ to $\mathcal{X}''|\mathcal{U}$, and let us show that it extends to all of $\mathcal{X}'$. We may suppose that $\mathcal{X}'$ is regular. Indeed, since $\mathcal{X}'$ is normal, one can find an open subset $\mathcal{V}$ of $\mathcal{X}$ whose complement is an analytic subset of codimension ≥ 2, such that $\mathcal{X}' \times_{\mathcal{X}} \mathcal{V} = \mathcal{V}'$ is regular. Let $\mathcal{V}'' = \mathcal{X}'' \times_{\mathcal{X}} \mathcal{V}$, and suppose the proposition proved for $\mathcal{V}$. Consider the commutative diagram

$$\begin{tikzcd}
\mathcal{V}' \arrow[rr, "i'"] \arrow[dr] \arrow[dd, "g'"'] & & \mathcal{X}' \arrow[dd, "f'"] & \\
& \mathcal{V}'' \arrow[rr, "i''"] \arrow[dl, "g''"'] & & \mathcal{X}'' \arrow[dl, "f''"'] \\
\mathcal{V} \arrow[rr, "i"] & & \mathcal{X} &
\end{tikzcd}$$

The morphism u determines a morphism of $\mathcal{O}_{\mathcal{V}}$-algebras $g''_*\mathcal{O}_{\mathcal{V}''} \to g'_*\mathcal{O}_{\mathcal{V}'}$, from which one deduces a morphism

$$i_*g''_*\mathcal{O}_{\mathcal{V}''} \to i_*g'_*\mathcal{O}_{\mathcal{V}'}.$$

In view of the isomorphisms $i'_*\mathcal{O}_{\mathcal{V}'} \simeq \mathcal{O}_{\mathcal{X}'}$, $i''_*\mathcal{O}_{\mathcal{V}''} \simeq \mathcal{O}_{\mathcal{X}''}$ [**11**, no. 3 prop. 4], one deduces a morphism of $\mathcal{O}_{\mathcal{X}}$-algebras

$$f''_*\mathcal{O}_{\mathcal{X}''} \to f'_*\mathcal{O}_{\mathcal{X}'},$$

and hence the desired morphism $\mathcal{X}' \to \mathcal{X}''$.

We henceforth suppose that $\mathcal{X}'$ is regular. Let $\mathcal{U}' = \mathcal{U} \times_{\mathcal{X}} \mathcal{X}'$, $\mathcal{Y}' = \mathcal{X}' - \mathcal{U}'$. We 339
regard $\mathcal{Y}'$ as a reduced analytic subspace of $\mathcal{X}'$; if $\mathcal{Y}'_1$ is the singular locus of $\mathcal{Y}'$, then $\dim \mathcal{Y}'_1 < \dim \mathcal{Y}'$ [**4**, no. 20 D. Th. 3]. Thus, by induction on the dimension of $\mathcal{Y}'$, we may suppose that $\mathcal{Y}'$ is smooth. Since it suffices to extend u to an open neighborhood of each

point of $\mathcal{Y}'$, the implicit function theorem allows us to suppose that $\mathcal{X}'$ is a ball in an affine space $\mathcal{E}^n$ and that $\mathcal{Y}'$ is the closed subset defined by the vanishing of the first p coordinate functions $z_1, \dots, z_p$, with $0 \leq p \leq n$.

The morphism u determines a section s of $p\colon \mathcal{X}' \times_{\mathcal{X}} \mathcal{X}'' \to \mathcal{X}'$ over $\mathcal{U}'$; after restricting $\mathcal{X}'$ if necessary, we may suppose that $p_*(\mathcal{O}_{\mathcal{X}' \times_{\mathcal{X}} \mathcal{X}''})$ is generated by elements $x_1, \dots, x_q$ of $\Gamma(\mathcal{X}', p_*\mathcal{O}_{\mathcal{X}' \times_{\mathcal{X}} \mathcal{X}''})$; let $u_1, \dots, u_q \in \Gamma(\mathcal{U}', \mathcal{O}_{\mathcal{X}'})$ be the images under s of $x_1|\mathcal{U}', \dots, x_q|\mathcal{U}'$. To say that s extends to $\mathcal{X}'$ is equivalent to saying that $u_1, \dots, u_q$ extend to sections of $\Gamma(\mathcal{X}', \mathcal{O}_{\mathcal{X}'})$. But since f is finite, each u_i is a Laurent series in $z_1, \dots, z_p$, with coefficients that are convergent power series in $z_{p+1}, \dots, z_n$, and satisfies integral-dependence relations. It follows that u_i is bounded, and hence is a convergent power series in $z_1, \dots, z_n$; consequently it extends to $\mathcal{X}'$.

One may ask whether the functor introduced in XII.5.3 is an equivalence of categories. The Grauert-Remmert theorem [**6**] answers this question; below we give a proof using resolution of singularities. One could also have used the Grauert-Remmert theorem to prove XII.5.1; this is what was done before [**8**] became available.

Theorem XII.5.4 (Grauert-Remmert theorem). *Let $\mathcal{X}$ be a normal analytic space, and let $\mathcal{Y}$ be a closed analytic subset such that $\mathcal{U} = \mathcal{X} - \mathcal{Y}$ is dense in $\mathcal{X}$. Let $\mathcal{U}'$ be a finite normal covering of $\mathcal{U}$; suppose that there exists a nowhere dense closed analytic subset $\mathcal{S}$ of $\mathcal{X}$ such that the restriction of $\mathcal{U}'$ to $\mathcal{U} - \mathcal{U} \cap \mathcal{S}$ is étale. Then there exists a finite normal covering $\mathcal{X}'$ of $\mathcal{X}$ which extends $\mathcal{U}'$, and $\mathcal{X}'$ is unique up to isomorphism.* 340

Uniqueness follows from XII.5.3. The problem of extending $\mathcal{U}'$ is therefore local on $\mathcal{X}$. We may suppose that $\mathcal{U}$ is regular and that $\mathcal{U}'$ is étale over $\mathcal{U}$. Indeed, the set of regular points of $\mathcal{U}$ is a dense open subset $\mathcal{V}$ of $\mathcal{X}$ whose complement is an analytic subset [**4**, no. 20 D th. 2], and it suffices to replace $\mathcal{U}$ by the open subset $\mathcal{V} - \mathcal{V} \cap \mathcal{S}$.

Let y be a point of $\mathcal{X} - \mathcal{U}$, and let us show that $\mathcal{U}'$ can be extended to a neighborhood of y. After restricting $\mathcal{X}$ to an open neighborhood of y, it follows from the resolution of singularities theorem [**8**] that one can find a regular analytic space $\mathcal{X}_1$ and a projective morphism $f\colon \mathcal{X}_1 \to \mathcal{X}$ which, upon restriction to $\mathcal{U}$, induces an isomorphism $\mathcal{U}_1 = f^{-1}(\mathcal{U}) \simeq \mathcal{U}$, such that $\mathcal{U}_1$ is the complement in $\mathcal{X}_1$ of a normal crossings divisor. Let us show that $\mathcal{U}'$ extends to a normal finite covering of $\mathcal{X}_1$. Since the question is local on $\mathcal{X}_1$, we may suppose that $\mathcal{X}_1$ is a ball in an affine space $\mathcal{E}^n$ and that $\mathcal{X}_1 - \mathcal{U}_1$ is defined by the vanishing of the first p coordinate functions $z_1, \dots, z_p$, with $0 \leq p \leq n$. The étale covering $\mathcal{U}'$ of $\mathcal{U}_1$ is a quotient of a covering of the form

$$\mathcal{U}_2 = \mathcal{U}_1[T_1, \dots, T_p] / \left(T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p\right)$$

by a subgroup H of the Galois group of $\mathcal{U}_2$. The covering $\mathcal{U}_2$ extends to the covering 341

$$\mathcal{X}_2 = \mathcal{X}_1[T_1, \dots, T_p] / \left(T_1^{n_1} - z_1, \dots, T_p^{n_p} - z_p\right)$$

of $\mathcal{X}_1$, on which H acts, and $\mathcal{X}_2/H$ extends $\mathcal{U}'$ to $\mathcal{X}_1$.

Let $\mathcal{X}_1'$ denote the finite normal covering of $\mathcal{X}_1$ which extends $\mathcal{U}'$, and let $\mathcal{F}_1$ denote the coherent $\mathcal{O}_{\mathcal{X}_1}$-algebra defined by $\mathcal{F}_1$. By the Grauert-Remmert finiteness theorem [**4**, no. 15 th. 1.1], $f_*\mathcal{F}_1$ is a coherent $\mathcal{O}_{\mathcal{X}}$-algebra. It therefore corresponds to a finite covering $\mathcal{X}'$ of $\mathcal{X}$, which is moreover normal since $\mathcal{X}_1'$ is normal, and $\mathcal{X}'$ is the desired extension of $\mathcal{U}'$.

Remark XII.5.5. In the statement XII.5.4, the hypothesis on the locus of points where the morphism $\mathcal{U}' \to \mathcal{U}$ is not étale cannot be omitted. For example, let $\mathcal{X}$ be the unit disk in the complex plane, let $\mathcal{U}$ be the complement of the origin in $\mathcal{X}$, and let $\mathcal{U}' = \mathcal{U}[T]/(T^2 - \sin 1/z)$, where z is the coordinate function on $\mathcal{X}$. Then $\mathcal{U}'$ is a normal finite covering of $\mathcal{U}$ which does not extend to $\mathcal{X}$. Indeed, suppose that $\mathcal{U}'$ extends to a finite covering $\mathcal{X}'$ of $\mathcal{X}$; the locus of points of $\mathcal{X}$ where the morphism $\mathcal{X}' \to \mathcal{X}$ is not étale would then be a closed analytic subset containing all points z such that $\sin 1/z = 0$, which is absurd.

The hypothesis on the singular locus of the morphism $\mathcal{U}' \to \mathcal{U}$ can, however, be omitted when $\operatorname{codim}(\mathcal{X} - \mathcal{U}, \mathcal{X}) \geq 2$. Indeed, we may suppose that $\mathcal{U}$ is regular. The locus of points of $\mathcal{U}$ where $\mathcal{U}' \to \mathcal{U}$ is not étale is a divisor of $\mathcal{U}$, and it follows from the Remmert-Stein theorem [**9**, th. 3] that it is the trace on $\mathcal{U}$ of a divisor of $\mathcal{X}$. Now, in 342
this case, if $\mathcal{A}$ is a coherent $\mathcal{O}_{\mathcal{U}}$-algebra such that $\mathcal{U}' = \operatorname{Spec}(\mathrm{an})(\mathcal{A})$, and if $i : \mathcal{U} \to \mathcal{X}$ is the canonical morphism, it suffices to take $\mathcal{X}' = \operatorname{Spec}(\mathrm{an})(i_*\mathcal{A})$; indeed, we know that $i_*\mathcal{A}$ is coherent [**11**, no. 1 th. 1].

Remark. **[**Added in 2003 (MR): There exist nontrivial finitely presented groups G which have no finite-index subgroups distinct from G, for example G. Higman's group (cf. J-P. Serre, Arbres et amalgames, Astérisque N°46, prop. 6, ch. I, §1). Consequently, if such a group is realized as the topological fundamental group of a scheme V over $\mathbb{C}$, say a smooth projective scheme, the topological space V_{top} underlying V is not simply connected, but V is algebraically simply connected. At present, no such V is known. Let us nevertheless mention that D. Toledo constructed smooth projective schemes V over $\mathbb{C}$ whose topological fundamental group is not Hausdorff for the topology of finite-index subgroups (the natural morphism $\pi_1(V_{\mathrm{top}}) \to \pi_1(V_{\mathrm{alg}})$ is not injective). [D. Toledo, Projective varieties with non-residually finite fundamental group, Publ. Math. IHES **77** (1993), p. 103–119]. **]**

Bibliography

[1] N. Bourbaki, Topologie Générale, Hermann, Paris (1960).
[2] N. Bourbaki, Algèbre Commutative, Hermann, Paris (1961).
[3] H. Cartan. Séminaire E.N.S., Paris (1956-57).
[4] H. Cartan. Séminaire E.N.S., Paris (1960-61).
[5] R. Godement, Théorie des Faisceaux, Hermann, Paris (1958).
[6] H. Grauert and R. Remmert, Komplexe Räume, Math. Ann. **136** (1958), p. 245–318.
[7] M. Hakim, Schémas relatifs, thesis, Paris (1967).
[8] H. Hironaka, Resolution of singularities of an algebraic variety over a field of characteristic zero, Ann. of Math. **39** (1964), p. 109–236.
[9] R. Remmert and K. Stein, Ueber die wesentlichen Singularitäten analytischer Mengen, Math. Ann. **126** (1953), p. 263–306.
[10] J-P. Serre, Géométrie algébrique et géométrie analytique, Ann. Inst. Fourier (Grenoble) **6** (1956), p. 1–42.
[11] J-P. Serre, Prolongement de faisceaux analytiques cohérents, Ann. Inst. Fourier (Grenoble) **16** (1966), p. 363–374.

CHAPTER XIII

Cohomological properness for sheaves of sets and sheaves of noncommutative groups

344

by Mme M. Raynaud[1]

The purpose of this exposé is to use étale cohomology to generalize certain results of IX and X. It also shows how the results of SGA 5 II that still make sense for sheaves of noncommutative groups can be extended to such sheaves. We assume familiarity with the notions of étale cohomology presented in SGA 4.

The main result (XIII.2.4) gives an important example of a nonproper morphism $f\colon U \to S$ that is *cohomologically proper in dimension* ≤ 1, that is, such that for certain sheaves of groups F on U (for the étale topology), the formation of f_*F and $\mathrm{R}^1 f_*F$ commutes with every base change $S' \to S$. This property is indeed satisfied by the open subset U of a scheme X proper over S that is the complement of a divisor D with normal crossings relative to S, at least if F is required to be finite constant, of order prime to the residue characteristics of S. If F is no longer assumed to have order prime to the residue characteristics of S, there is an analogous result with $\mathrm{R}^1 f_*F$ replaced by the subsheaf $\mathrm{R}^1_{\mathrm{t}} f_*F$ obtained by considering only torsors under F that are *tamely ramified on X relative to S*. In particular, this makes it possible to show that the tamely ramified fundamental group of a proper smooth algebraic curve over a separably closed field, with finitely many closed points removed, is topologically of finite type (XIII.2.11).

The no. XIII.4 is devoted to the homotopy exact sequence and the Künneth formula. 345

Finally, an appendix gives useful variants of Abhyankar's lemma proved in X.X.3.6.

XIII.0. Review of the theory of stacks

We shall use below the theory of stacks presented in [**1**] and [**2**]. We restrict ourselves to the case of the étale site of a scheme. Given a scheme X, denote by $X_{\text{ét}}$ the étale site of X. Recall that a stack F on X is a category fibered over $X_{\text{ét}}$ such that, for every scheme X' étale over X and every pair of objects x, y of the fiber $F_{X'}$, the presheaf $\mathbf{Hom}_{X'}(x, y)$ is a sheaf, and such that, for every surjective étale morphism $X'' \to X'$, every object of $F_{X''}$ equipped with descent data relative to $X'' \to X'$ is the inverse image of an object of $F_{X'}$.

We denote by $F(X')$ the category of cartesian sections of F/X'. More generally, if $\mathbf{Sch}_X$ is the category of schemes over X, endowed with the étale topology, the stack F can be extended to a stack $\mathcal{F}$ on $\mathbf{Sch}_X$, and for every morphism $f\colon X' \to X$, we again denote by $F(X')$ the category of cartesian sections of this stack $\mathcal{F}$ over X'.

A gerbe is a stack such that, for every scheme X' étale over X and every pair of objects x, y of $F_{X'}$, every morphism from x to y is an isomorphism, x and y are locally isomorphic, and the collection of objects X' of $X_{\text{ét}}$ for which $F_{X'}$ is nonempty forms a covering of $X_{\text{ét}}$. For example, the stack of torsors under a sheaf of groups is a gerbe that,

[1] Based on unpublished notes by A. Grothendieck.

moreover, has a cartesian section. Conversely, a gerbe that has a section, i.e. for which
there exists an object x of F_X, is equivalent to the stack of torsors under the sheaf of 346
groups $\mathbf{Aut}_X(x)$.

There is an evident notion of a subgerbe and of a maximal subgerbe of a stack F. Given a cartesian section x of $F(X)$, there is a unique maximal subgerbe G_x of F through which x factors. We call G_x the subgerbe generated by x; it is, by definition, a trivial gerbe. The presheaf SF defined by

$$SF(X') = \{\text{maximal subgerbes of } F|X'\}$$

is a sheaf, called the sheaf of maximal subgerbes of F. Let O be the presheaf defined by

$$O(X') = \{\text{classes of objects of } F_{X'} \text{ modulo isomorphism}\}.$$

By associating with every object x of $F_{X'}$ the maximal subgerbe of $F|X'$ generated by x, we obtain a morphism

$$O \to SF;$$

by [**2**, III 2.1.4], this morphism makes SF a sheaf associated with O.

A stack F is said to be *constructible* (*respectivelyind-$\mathbb{L}$-finite*, where $\mathbb{L}$ is a set of prime numbers) if, for every scheme X' étale over X and every object x of $F_{X'}$, the sheaf $\mathbf{Aut}_{X'}(x)$ has the corresponding property [**2**, VII 2.2.1]. A stack is said to be 1-constructible if it is constructible and its sheaf of maximal subgerbes is constructible.

XIII.1. Cohomological properness

. Let S be a scheme and $f\colon X \to Y$ a morphism of S-schemes. If S' is an S-scheme,
consider the following diagram, all of whose squares are cartesian: 347

(XIII.1.-1.1)
$$\begin{tikzcd} X & X' \arrow[l, "h"'] \arrow[d, "f'"] \\ Y \arrow[u, leftarrow, "f"] \arrow[d] & Y' \arrow[l, "g"'] \arrow[d] \\ S & S'. \arrow[l] \end{tikzcd}$$

If Y_1 is a scheme étale over Y, put $X_1 = X \times_Y Y_1$, $Y_1' = Y' \times_Y Y_1$, and consider the cartesian square

(XIII.1.-1.2)
$$\begin{tikzcd} X_1 \arrow[d, "f_1"'] & X_1' \arrow[l, "h_1"'] \arrow[d, "f_1'"] \\ Y_1 & Y_1'. \arrow[l, "g_1"] \end{tikzcd}$$

Definition XIII.1.1. *Let F be a stack on X. We say that (F, f) is* cohomologically proper relative to S in dimension $\leqslant -1$ *(respectivelyin dimension $\leqslant 0$, respectivelyin dimension $\leqslant 1$) if, for every S-scheme S', the canonical functor (defined in the evident way by the universal property of the inverse image of stacks)*

$$g^* f_* F \to f'_* h^* F \qquad \textit{(cf. 1.0.1)}$$

is faithful (respectivelyfully faithful, respectivelyan equivalence of categories).

If there is no possible ambiguity about S, in particular if $S = Y$, we say cohomologically proper instead of cohomologically proper relative to S.

. Let F be a sheaf of sets on X, and let Φ be the associated stack in discrete categories, i.e. the stack whose fiber over every scheme X_1 étale over X is the discrete category with object set $F(X_1)$. We say that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$) if (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$).

The canonical morphism 348

(XIII.1.-1.1)
$$g^*f_*F \to f'_*h^*F$$

induces, upon passing to the associated stacks in discrete categories, the canonical morphism

$$g^*f_*\Phi \to f'_*h^*\Phi.$$

Consequently, saying that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$) is equivalent to saying that, for every S-scheme S', the morphism (XIII.1.-1.1) is injective (respectivelybijective).

. Let F be a sheaf of groups on X, and let Φ be the stack of torsors on X with group F [**1**, II 2.3.2]. We say that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) if (Φ, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$). The cohomological properness condition can be made explicit as follows.

Proposition XIII.1.2. *Use the notation of* (XIII.1.-1.1) *and* (XIII.1.-1.2). *Let F be a sheaf of groups on X. Denote by F' (respectivelyF_1, respectivelyF'_1, etc.) the inverse image of F on X' (respectivelyon X_1, respectivelyon X'_1, etc.). Then the following conditions are equivalent:*

(i) *(F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$).*

(ii) *For every morphism $S' \to S$, every scheme Y_1 étale over Y, and every torsor P on X_1 with group F_1, if ${}^PF_1$ denotes the twist of F_1 by P* [**1**, II 4.1.2.3], *the canonical morphism*

$$a_0 \colon g_1^*(f_{1*}({}^PF_1)) \to f'_{1*}({}^{P'}F'_1)$$

is injective (respectivelya_0 is bijective and the canonical morphism

$$a_1 \colon g^*(\mathrm{R}^1f_*F) \to \mathrm{R}^1f'_*F'$$

is injective, respectivelya_0 and a_1 are bijective). 349

(ii bis) *For every morphism $S' \to S$, every scheme Y_1 étale over Y, every torsor P on X_1 with group F_1, and every torsor R under ${}^PF_1$, the canonical morphism*

$$\alpha_0 \colon g_1^*(f_{1*}R) \to f'_{1*}R'$$

is injective (respectivelyα_0 is bijective, respectivelythe morphisms α_0 and

$$\alpha_1 \colon g_1^*(\mathrm{R}^1f_{1*}({}^PF_1)) \to \mathrm{R}^1f'_{1*}({}^{P'}F'_1)$$

are bijective).

Proof. (i)$\Longrightarrow$(ii bis). By [**1**, II 4.2.5], every torsor R with group ${}^PF_1$ is of the form $R = Q \overset{F_1}{\wedge} P^\circ$, where Q is a torsor with group F_1 and P° is the opposite of P. We then have $R' \simeq Q' \overset{F'_1}{\wedge} P'^\circ$. Let Φ be the stack of torsors under F, and let x, y (respectivelyx', y') be the objects of the fiber category $(g^*f_*\Phi)_{Y'_1}$ (respectively$(f'_*\Phi')_{Y'_1}$) associated with P, Q (respectivelyP', Q'). We have

$$Q \overset{F_1}{\wedge} P^\circ \simeq \mathbf{Hom}_{F_1}(P, Q),$$

and it follows that there are canonical isomorphisms

$$\mathbf{Hom}_{Y_1'}(x,y) \simeq g_1^* f_{1*}(Q \overset{F_1}{\wedge} P^\circ), \quad \mathbf{Hom}_{Y_1'}(x',y') \simeq f_{1*}'(Q' \overset{F_1'}{\wedge} P'^\circ).$$

Consequently, the morphism α_0 is identified with the morphism

$$\mathbf{Hom}_{Y_1'}(x,y) \to \mathbf{Hom}_{Y_1'}(x',y'),$$

from which it follows that if (F,f) is cohomologically proper relative to S in dimension $\leqslant -1$, then α_0 is injective, and that if (F,f) is cohomologically proper in dimension $\leqslant 0$, then α_0 is bijective.

Now suppose that (F,f) is cohomologically proper relative to S in dimension $\leqslant 1$, i.e. that the canonical morphism 350

$$\varphi\colon g^* f_* \Phi \to f_*' \Phi'$$

is an equivalence. Let G be the sheaf of maximal subgerbes of the stack $f_*\Phi$ [**1**, III 2.1.8]; then there is an isomorphism $G \simeq \mathrm{R}^1 f_* F$. Since g^*G is the sheaf of maximal subgerbes of $g^* f_* \Phi$ [**2**, III 2.1.5.5], the morphism α_1 is obtained from $\varphi|Y_1'$ by taking the sheaves of maximal subgerbes, and is therefore an isomorphism.

(ii bis)$\Longrightarrow$(ii). It is enough to show that if the morphisms α_0 are bijective, then the morphisms a_1 are injective. Let Y_1' be a scheme étale over Y', and let s and t be two elements of $g^*(\mathrm{R}^1 f_* F)(Y_1')$ having the same image in $\mathrm{R}^1 f_*' F'(Y_1')$; we shall show that $s = t$. The assertion is local for the étale topology of Y_1', and, in view of the definition of the inverse image $g^*(\mathrm{R}^1 f_* F)$, we may suppose that Y_1' is the inverse image of a scheme Y_1 étale over Y, and that s and t come from torsors P and Q on X_1. The hypothesis on s and t then means that the inverse images P' and Q' of P and Q on X_1' are locally isomorphic for the étale topology of Y_1'. If we put $R = \mathbf{Hom}_{F_1}(P,Q)$, the fact that the morphism

$$g_1^* f_{1*} R \to f_{1*}' R'$$

is bijective proves that P and Q are locally isomorphic for the étale topology of Y_1, and hence that $s = t$.

(ii)$\Longrightarrow$(i). To prove that φ is faithful (respectivelyfully faithful), it is enough to show that if Y_1 is a scheme étale over Y, if P, Q are two torsors on X_1 with group F_1, and if x, y (respectivelyx', y') are the objects of $(g^* f_* \Phi)_{Y_1'}$ (respectively$(f_*' \Phi')_{Y_1'}$) associated with P, Q (respectivelyP', Q'), then the morphism

$$a\colon \operatorname{Hom}(x,y) \to \operatorname{Hom}(x',y')$$

is injective (respectivelybijective). But a is identified with the canonical morphism

$$\mathrm{H}^0(Y_1', g_1^* f_{1*}(Q \overset{F_1}{\wedge} P^\circ)) \to \mathrm{H}^0(Y_1', f_{1*}'(Q' \overset{F_1'}{\wedge} P'^\circ)).$$

If $\operatorname{Hom}(x,y) \neq \emptyset$, then $Q \overset{F_1}{\wedge} P^\circ$ is a torsor under ${}^P F_1$ that is locally trivial over Y_1. It 351
follows that $f_{1*}(Q \overset{F_1}{\wedge} P^\circ)$ is a torsor under $f_{1*}({}^P F_1)$, and that $g_1^* f_{1*}(Q \overset{F_1}{\wedge} P^\circ)$ is a trivial torsor. The morphism a is then identified with the canonical morphism

$$\mathrm{H}^0(Y_1, g_1^* f_{1*}({}^P F_1)) \to \mathrm{H}^0(Y_1', f_{1*}'({}^{P'} F_1')).$$

The same holds if $\operatorname{Hom}(x',y') \neq \emptyset$ and a_1 is injective, for then $Q' \overset{F_1'}{\wedge} P'^\circ$ is trivial, and the injectivity of a_1 implies that P and Q are locally isomorphic over Y_1. We conclude that if a_0 is injective (respectivelyif a_0 is bijective and a_1 injective), then φ is faithful (respectivelyfully faithful).

It remains to show that if a_0 and a_1 are bijective, the functor φ is essentially surjective. Let Y'' be a scheme étale over Y', put $X'' = X' \times_{Y'} Y''$, and let P'' be a torsor on X'' with group $F'' = F'|X''$. We shall show that there is an element x of $(g^* f_* \Phi)_{Y''}$ whose image in $(f_*' \Phi')_{Y''}$ is isomorphic to P''. Let p'' be the class of P''. Since a_1 is surjective, we can find a surjective étale morphism $Y_1'' \to Y''$, an étale morphism $Y_1 \to Y$ such that there is a morphism $Y_1'' \to Y_1'$, and a torsor P_1 on X_1 with group F_1, whose inverse image P_1'' on X_1'' is isomorphic to the inverse image of P''. Using the fact that φ is fully faithful, we see that the object x_1 of $(g^* f_* \Phi)_{Y_1''}$ corresponding to P_1'' is equipped with descent data relative to $Y_1'' \to Y''$, and hence comes from an element x of $(g^* f_* \Phi)_{Y''}$. Since the image of x in $(f'_* \Phi')_{Y''}$ is P'', this proves that φ is essentially surjective and completes the proof.

Example XIII.1.3. *Let $f\colon X \to Y$ be a proper morphism. It follows from* [**2**, VII 2.2.2] *that, for every ind-finite stack F on X, the pair (F, f) is cohomologically proper (relative to Y) in dimension $\leqslant 1$. In particular, for every sheaf of sets (respectivelyevery sheaf of groups, respectivelyevery ind-finite sheaf of groups) F on X, (F, f) is cohomologically proper in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 0$, respectivelyin dimension $\leqslant 1$).* 352

Remarks XIII.1.4. a) Let F be a sheaf of groups on X such that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$). If F is regarded as a sheaf of sets, (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$), but the converse is false.

For example, let Y be the spectrum of a strictly local discrete valuation ring, with closed point t and generic point s; let $f\colon X \to Y$ be a nonempty scheme over Y whose closed fiber is empty; let F be a nontrivial constant sheaf of groups on X; and let P be a torsor under F such that $\mathrm{H}^0(X_s, {}^P F|X_s) = 1$. Then $({}^P F, f)$ is cohomologically proper relative to Y in dimension $\leqslant -1$ when ${}^P F$ is regarded as a sheaf of sets. If ${}^P F$ is regarded as a sheaf of groups, there is an isomorphism ${}^{P^\circ}({}^P F) \simeq F$; since the canonical morphism

$$\mathrm{H}^0(X, F) \to \mathrm{H}^0(X_t, F|X_t) = 1$$

is not injective, this proves that $({}^P F, f)$ is not cohomologically proper relative to Y in dimension $\leqslant -1$.

b) Suppose that f is coherent (i.e. quasi-compact and quasi-separated). Let F be a stack on X. For every geometric point $\overline{y}$ of Y', denote by $\overline{Y}$ (respectively$\overline{Y'}$) the strict localization of Y (respectivelyY') at $\overline{y}$, and put $\overline{X} = X \times_Y \overline{Y}$, $\overline{X'} = X' \times_{Y'} \overline{Y'}$, etc. The pair (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) if and only if, for every S-scheme S' and every geometric point $\overline{y}$ of Y', the canonical functor

$$\overline{F}(\overline{X}) \to \overline{F'}(\overline{X'})$$

is faithful (respectivelyfully faithful, respectivelyan equivalence).

Indeed, if S' is an S-scheme, the functor 353

$$g^* f_* F \to f'_* F'$$

is faithful (respectivelyfully faithful, respectivelyan equivalence) if and only if the same is true of the functor induced on the fibers at the various geometric points $\overline{y}'$ of $\overline{Y'}$ [**2**, III 2.1.5.9]. The assertion therefore follows from the calculation of the geometric fibers of the direct image of a stack under a coherent morphism [**2**, VII 2.1.5].

c) Let F be a stack on X. The property that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) is local on Y for the étale topology.

Let S' be an S-scheme, and let F' be the inverse image of F on X' (cf. (XIII.1.-1.1)). If (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, then the same is true of (F', f'). But if (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$), the same need not be true of (F', f').

For example, let S' be a discrete valuation ring, let $f'\colon E_{S'} \to S'$ be affine space over S', let x be a closed point of $E_{S'}$ over the generic point of S', and let F' be the sheaf of sets on $E_{S'}$ whose restriction to $E_{S'} - \{x\}$ is the one-element constant sheaf and whose fiber at a geometric point over x has two elements. Then (F', f') is not cohomologically proper relative to S' in dimension $\leqslant -1$. Let $S = S'[Z]$, let $f\colon E_S \to S$ be affine space over S, and let T be a closed subset of $X = E_S$ that does not meet the closed subset $Z = 0$ and such that $f(T)$ contains the generic point of S. Let G be the inverse image of F' on X, and let F be the sheaf on X obtained by extending $G|X - T$ by the empty sheaf. Then (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$, but this is no longer true after the base change $S' \to S$ defined by $Z = 0$.

d) Let F be a stack on X such that (F, f) is cohomologically proper relative to Y in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$). Then, for every geometric point $\overline{y}$ of Y, the canonical functor 354

$$(f_*F)_{\overline{y}} \to F(X_{\overline{y}})$$

is faithful (respectivelyfully faithful, respectivelyan equivalence of categories).

Proposition XIII.1.5. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be two S-morphisms, and let Φ be a stack on X.*

1) *Suppose that (Φ, f) and $(f_*\Phi, g)$ are cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$). Then the same is true of (Φ, gf).*
2) *Suppose that (Φ, gf) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelythat (Φ, gf) is cohomologically proper relative to S in dimension $\leqslant 0$ and (Φ, f) is cohomologically proper relative to S in dimension $\leqslant -1$, respectivelythat (Φ, gf) is cohomologically proper relative to S in dimension $\leqslant 1$ and (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 0$). Then $(f_*\Phi, g)$ is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$, respectivelyin dimension $\leqslant 1$).*

For every S-scheme S', consider the following diagram, all of whose squares are cartesian:

(XIII.1.-1.1)
$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d, "h"'] & Y' \arrow[r, "g'"] \arrow[d, "k"'] & Z' \arrow[r] \arrow[d, "m"'] & S' \arrow[d] \\ X \arrow[r, "f"] & Y \arrow[r, "g"] & Z \arrow[r] & S. \end{tikzcd}$$

The canonical morphism

$$m^*(g_*f_*\Phi) \to g'_*f'_*(h^*\Phi)$$

is identified with the composite of the canonical morphisms 355

$$m^*(g_*f_*\Phi) \xrightarrow{i} g'_*(k^*f_*\Phi) \xrightarrow{j} g'_*f'_*(h^*\Phi).$$

1) The hypothesis implies that i and j are faithful (respectivelyfully faithful, respectivelyequivalences); hence the same is true of ji.
2) The hypothesis implies that ji is faithful (respectivelythat ji is fully faithful and j is faithful, respectivelythat ji is an equivalence and j is fully faithful); it follows that i is faithful (respectivelyfully faithful, respectivelyan equivalence).

Corollary XIII.1.6. *Let $f\colon X \to Y$ and $g\colon Y \to Z$ be two S-morphisms, and let F be a sheaf of groups on X. Suppose that (F, gf) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelythat (F, gf) is cohomologically proper relative to S in dimension $\leqslant 0$ and that (F, f) is cohomologically proper relative to S in dimension $\leqslant -1$). Then (f_*F, g) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$).*

Retain the notation of (XIII.1.-1.1), and, for every scheme Y_1 étale over Y, denote by f_1, F_1 the respective inverse images of f, F under the morphism $Y_1 \to Y$. Let Φ be the stack of torsors under F, and let Ψ be the stack of torsors under f_*F. There is a canonical functor

$$\varphi\colon \Psi \to f_*\Phi,$$

obtained by associating with every scheme Y_1 étale over Y and every torsor P on Y_1 with group $f_{1*}F_1$, the torsor $\widetilde{P}$ on X_1 obtained from f_1^*P by extension of structure group along $f_1^*f_{1*}F_1 \to F_1$. The functor φ is fully faithful. Indeed, if P and Q are two torsors on Y_1 with group $f_{1*}F_1$, there is a canonical morphism

$$\mathbf{Isom}_{f_{1*}F_1}(P,Q) \to f_{1*}(\mathbf{Isom}_{F_1}(\widetilde{P},\widetilde{Q}))$$

which is an isomorphism because this is so locally. It follows that the canonical morphism 356

$$\mathrm{Isom}_{f_{1*}F_1}(P,Q) \to \mathrm{Isom}_{F_1}(\widetilde{P},\widetilde{Q})$$

is an isomorphism, and hence that φ is fully faithful.

There is a commutative diagram

$$\begin{tikzcd} g'_*k^*\Psi \arrow[r, "i"] \arrow[d, "g'_*k^*(\varphi)"'] & m^*(g_*\Psi) \arrow[d, "m^*g_*(\varphi)"] \\ g'_*k^*(f_*\Phi) \arrow[r, "j"] & m^*(g_*f_*\Phi), \end{tikzcd}$$

where i and j are the base-change morphisms. It follows from XIII.1.5 2) that j is faithful (respectivelyfully faithful). Since $g'_*k^*(\varphi)$ and $m^*g_*(\varphi)$ are fully faithful, the diagram above shows that i is faithful (respectivelyfully faithful).

Corollary XIII.1.7. *Let $f\colon X \to Y$ be a coherent S-morphism, let $g\colon Y \to Z$ be a proper S-morphism, and let Φ be an ind-finite stack on X* [**2**, VII 2.2.1]*. Suppose that (Φ, f) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$); then the same is true of (Φ, gf).*

Since f is coherent, $f_*\Phi$ is an ind-finite stack (SGA 4 IX 1.6 (ii)). The corollary therefore follows from XIII.1.5 1) and XIII.1.3.

Corollary XIII.1.8. *Let $f\colon X \to Y$ be an integral S-morphism, and let $g\colon Y \to Z$ be an S-morphism. If F is a sheaf of sets on X, then (f_*F, g) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$) if and only if the same is true of (F, gf). If F is a sheaf of groups on X, then (f_*F, g) is cohomologically proper relative to S in dimension $\leqslant -1$ (respectively$\leqslant 0$, respectively$\leqslant 1$) if and only if the same is true of (F, gf).*

The assertion for a sheaf of sets follows from XIII.1.5 and from the fact that (F, f) 357
is cohomologically proper relative to S in dimension $\leqslant 0$. Let F be a sheaf of groups on X, and let Φ be the stack of torsors under F. By SGA 4 VIII 5.8, every torsor under F is locally trivial over Y. It follows that the stack $f_*\Phi$ is equivalent to the stack of torsors under f_*F, the equivalence being obtained by associating with every scheme Y_1 étale over Y and every torsor P on $X_1 = X \times_Y Y_1$ with group $F|X_1$, the torsor f_*P with group $f_*F|Y_1$. Since (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, the corollary therefore follows from XIII.1.5.

Definitions XIII.1.9.

. Let E be a category, and consider a diagram

$$\Phi \xrightarrow{p} \Phi_1 \underset{p_2}{\overset{p_1}{\rightrightarrows}} \Phi_2\ ,$$

where Φ, Φ_1, Φ_2 are categories fibered over E, the arrows are morphisms of fibered categories, and

$$a\colon p_1p \xrightarrow{\sim} p_2p$$

is an isomorphism of functors.

We say that the diagram above is exact if the following condition is satisfied:

a) For every pair of objects x, y of Φ and every morphism $f\colon p(x) \to p(y)$ such that $p_1(f) = p_2(f)$ (p_1p and p_2p being identified by means of a), there exists a unique morphism $g\colon x \to y$ such that $p(g) = f$.

. Consider the diagram

$$\Phi \xrightarrow{p} \Phi_1 \overset{p_1,p_2}{\rightrightarrows} \Phi_2 \overset{p_{12},p_{23},p_{31}}{\rightrightarrows\!\!\!\rightarrow} \Phi_3\ ,$$

where Φ, Φ_i, $1 \leqslant i \leqslant 3$, are categories fibered over E, and the arrows are morphisms of fibered categories. Suppose that we are given isomorphisms of functors 358

$$a\colon p_1p \xrightarrow{\sim} p_2p$$

$$a_1\colon p_{31}p_2 \xrightarrow{\sim} p_{12}p_1, \quad a_2\colon p_{12}p_2 \xrightarrow{\sim} p_{23}p_1, \quad a_3\colon p_{23}p_2 \xrightarrow{\sim} p_{31}p_1$$

such that the following diagram commutes:

$$\begin{tikzcd} p_{23}p_1p \arrow[r, "\mathrm{id}.a"] & p_{23}p_2p \arrow[r, "a_3.\mathrm{id}"] & p_{31}p_1p \arrow[d, "\mathrm{id}.a"] \\ p_{12}p_2p \arrow[u, "a_2.\mathrm{id}"] & p_{12}p_1p \arrow[l, "\mathrm{id}.a"'] & p_{31}p_2p \arrow[l, "a_1.\mathrm{id}"'] \end{tikzcd}\ .$$

We identify p_1p and p_2p, $p_{31}p_2$ and $p_{12}p_1$, and so on.

We say that the diagram above is exact if the following conditions are satisfied:

a) The analogue of condition a) of XIII.1.9.

b) For every object x_1 of Φ_1 and every isomorphism $u\colon p_1(x_1) \xrightarrow{\sim} p_2(x_1)$ such that

(XIII.1.-1.0.1) $$p_{23}(u)p_{31}(u) = p_{12}(u)^{-1},$$

there exists an object x of Φ and an isomorphism $i\colon p(x) \xrightarrow{\sim} x_1$ making the diagram

(XIII.1.-1.0.2) $$\begin{tikzcd} p_1p(x) \arrow[r, equal] \arrow[d, "p_1(i)"'] & p_2p(x) \arrow[d, "p_2(i)"] \\ p_1(x_1) \arrow[r, "\sim", "u"'] & p_2(x_1) \end{tikzcd}$$

commute.

. The notion of a morphism of exact diagrams of fibered categories over a category E is defined in the evident way.

. We shall use in particular the notion of an exact diagram in the case where E is a site and Φ, Φ_i, $1 \leqslant i \leqslant 3$, are stacks on E. 359

Let $f\colon E \to E'$ be a morphism of sites and let

(XIII.1.-1.0.1) $$\Phi \xrightarrow{p} \Phi_1 \overset{p_1,p_2}{\rightrightarrows} \Phi_2 \overset{p_{12},p_{23},p_{31}}{\rightrightarrows\!\!\!\rightarrow} \Phi_3$$

be an exact diagram of stacks on E. Taking direct images gives a diagram

$$f_*\Phi \longrightarrow f_*\Phi_1 \rightrightarrows f_*\Phi_2 \rightrightarrows\!\!\!\rightarrow f_*\Phi_3$$

which is evidently exact.

If $h\colon E'' \to E$ is a morphism of sites, one likewise has an exact diagram

$$h^*\Phi \xrightarrow{p''} h^*\Phi_1 \overset{p''_1,p''_2}{\rightrightarrows} h^*\Phi_2 \overset{p''_{12},p''_{23},p''_{31}}{\rightrightarrows\!\!\!\rightarrow} h^*\Phi_3$$

Let us first verify condition a) of XIII.1.9. Let F'' be an object of E'', let x'' and y'' be two objects of $(h^*\Phi)_{F''}$, let x''_1 and y''_1 be their respective images in $h^*\Phi_1$, and let x''_2, y''_2 be their images in $h^*\Phi_2$. Let $u''_1\colon x''_1 \to y''_1$ be a morphism such that $p''_1(u''_1) = p''_2(u''_1)$, and let us prove that u''_1 comes from a unique morphism $u''\colon x'' \to y''$. Since the question is local on F'', we may suppose that we have an object F_1 of E, a morphism from F'' to the inverse image F_1'' of F_1 under h, and objects x, y of Φ_{F_1} whose inverse images on F'' are x'' and y''. Let x_1, y_1 (respectivelyx_2, y_2) be the images of x, y in Φ_1 (respectively Φ_2). We may suppose that u_1'' comes from a morphism $u_1 \colon x_1 \to y_1$ such that $p_1(u_1) = p_2(u_1)$. By the exactness of (XIII.1.-1.0.1), we obtain a unique morphism $u \colon x \to y$ whose inverse image under h is the desired morphism u''.

The condition b) of XIII.1.9 is verified in the same way. Let x_1'' be an object of $(h^*\Phi_1)_{F''}$ 360
and let $u'' \colon p_1''(x_1'') \to p_2''(x_1'')$ be a morphism satisfying the relation

$$p_{23}''(u'')p_{31}''(u'') = p_{12}''(u'')^{-1},$$

and let us prove that there exists an object x'' of $(h^*\Phi)_{F''}$ and an isomorphism $i'' \colon p''(x'') \simeq x_1''$ making a diagram analogous to (XIII.1.-1.0.2) commute. Since the question is local on F'', we may suppose that we have an object F_1, a morphism $F'' \to F_1''$ as above, and an object x_1 of $(\Phi_1)_{F_1}$ whose inverse image in $(h^*\Phi_1)_{F''}$ is x_1''. Likewise, we may suppose that u'' comes from a morphism $u \colon p_1(x_1) \to p_2(x_1)$ satisfying (XIII.1.-1.0.2). The existence of an object x of Φ_{F_1} whose inverse image under h is an object x'' with the required property then follows from the exactness of (XIII.1.-1.0.1).

Examples XIII.1.10. 1) *Let $f \colon X_1 \to X$ be a* descent morphism *for the category of étale sheaves on variable schemes (for example, a universally submersive morphism (SGA 4 VIII 9.3)). Let $X_2 = X_1 \times_X X_1$, let $g \colon X_2 \to X$ be the canonical projection, and let F be a sheaf of sets on X. It then follows from loc. cit. that there is an exact sequence of sheaves of sets*

(XIII.1.-1.1) $$F \longrightarrow f_*f^*F \rightrightarrows g_*g^*F.$$

If Φ is the stack in discrete categories associated with F, and Φ_3 is the final stack on X, i.e. the stack all of whose fibers consist of a single object whose only morphism is the identity morphism, saying that the sequence (XIII.1.-1.1) *is exact is equivalent to saying that the diagram of stacks*

$$\Phi \longrightarrow f_*f^*\Phi \rightrightarrows g_*g^*\Phi \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} \Phi_3\ .$$

is exact.

2) *Let $f \colon X_1 \to X$ be an* effective descent morphism *for the category of étale sheaves* 361
on variable schemes (for example, a proper surjective morphism, an integral surjective morphism, or a faithfully flat morphism locally of finite presentation (SGA 4 VIII 9.4)). Let $X_2 = X_1 \times_X X_1$, let $g \colon X_2 \to X$ be the canonical projection, let $X_3 = X_1 \times_X X_1 \times_X X_1$, and let $h \colon X_3 \to X$ be the canonical morphism. Let Φ be a stack on X, let $\Phi_1 = f_*f^*\Phi$, $\Phi_2 = g_*g^*\Phi$, and $\Phi_3 = h_*h^*\Phi$. We then have an exact diagram

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} \Phi_3,$$

where the arrows are the canonical morphisms associated with the projections.

Indeed, regard Φ as a stack on the category $\mathbf{Sch}_X$ of schemes over X, endowed with the étale topology. Then, by [**2**, VII 2.2.8], *Φ is also a stack for the finest topology on $\mathbf{Sch}_X$ for which the covering morphisms are the effective descent morphisms for the category of étale sheaves. The exactness of the diagram above follows immediately.*

Proposition XIII.1.11. *Let S be a scheme, and let $f \colon X \to Y$ be an S-morphism.*

1) *Let*

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2$$

be an exact diagram of stacks on X. Suppose that (Φ_1, f) is cohomologically proper relative to S in dimension $\leqslant 0$ and that (Φ_2, f) is cohomologically proper *relative to S in dimension $\leqslant -1$. Then (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 0$.*

2) *Let*

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} \Phi_3$$

be an exact diagram of stacks on X. Suppose that (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 1$, that (Φ_2, f) is cohomologically proper relative to S in dimension $\leqslant 0$, and that (Φ_3, f) is cohomologically proper relative 362
to S in dimension $\leqslant -1$. Then (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 1$.

For every S-scheme S', consider the following commutative diagram, all of whose squares are cartesian:

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d, "h"'] & Y' \arrow[r] \arrow[d, "g"'] & S' \arrow[d] \\ X \arrow[r, "f"] & Y \arrow[r] & S \end{tikzcd}$$

Let us prove 2), the proof of 1) being analogous. Since the direct and inverse image functors take exact diagrams of stacks to exact diagrams (XIII.1), there is the following morphism of exact diagrams of stacks:

$$\begin{tikzcd} g^*f_*\Phi \arrow[r, "\pi"] \arrow[d, "\varphi"'] & g^*f_*\Phi_1 \arrow[r, shift left, "\pi_1"] \arrow[r, shift right, "\pi_2"'] \arrow[d, "\varphi_1"'] & g^*f_*\Phi_2 \arrow[r, shift left] \arrow[r] \arrow[r, shift right] \arrow[d, "\varphi_2"'] & g^*f_*\Phi_3 \arrow[d, "\varphi_3"'] \\ f'_*h^*\Phi \arrow[r] & f'_*h^*\Phi_1 \arrow[r, shift left] \arrow[r, shift right] & f'_*h^*\Phi_2 \arrow[r, shift left] \arrow[r] \arrow[r, shift right] & f'_*h^*\Phi_3 \end{tikzcd} .$$

By hypothesis, φ_1 is an equivalence of categories, φ_2 is fully faithful, and φ_3 is faithful. It therefore follows from the preceding diagram that φ is an equivalence.

Proposition XIII.1.12. *Let $f\colon X \to Y$ be an S-morphism.*

1) *Let*

$$F \longrightarrow G \rightrightarrows H$$

be an exact diagram of sheaves of sets on X. Suppose that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ and that (H, f) is cohomologically proper relative to S in dimension $\leqslant -1$. Then (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$.

2) *Let $F \to G$ be a monomorphism of sheaves of groups on X. If Y_1 is a scheme* 363
ètale over Y, put $X_1 = Y_1 \times_Y X$, and denote by f_1 (respectivelyF_1, respectivelyG_1) the inverse image of f (respectively$F$, respectively$G$) over Y_1 (cf. XIII.1.-1.2). Suppose that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$) and that, for every scheme Y_1 ètale over Y and every torsor Q under G_1, $(Q/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant -1$ (respectivelyin dimension $\leqslant 0$). Then (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$).

3) *Let $F \to G$ be a monomorphism of sheaves of groups on X. Suppose that (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$ and that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$. Then, for every torsor Q under G, $(Q/F, f)$ is cohomologically proper relative to S in dimension $\leqslant 0$.*

Proof. 1) Let Φ (respectivelyΦ_1, respectivelyΦ_2) be the stack in discrete categories associated with F (respectively$G$, respectively$H$), and let Φ_3 be the final stack on X. There is then an exact diagram

$$\Phi \longrightarrow \Phi_1 \rightrightarrows \Phi_2 \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} \Phi_3 .$$

By hypothesis, (Φ_1, f) is cohomologically proper relative to S in dimension $\leqslant 1$, and (Φ_2, f) is cohomologically proper relative to S in dimension $\leqslant 0$ (XIII.1). Since (Φ_3, f) is clearly cohomologically proper relative to S in dimension $\leqslant -1$, it follows from XIII.1.11 that (Φ, f) is cohomologically proper relative to S in dimension $\leqslant 1$, i.e. that (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$.

2) Let us first show that, if (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ and if the pairs $(Q/F_1, f_1)$ are cohomologically proper relative to S in dimension 364
$\leqslant -1$, then (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$. By XIII.1.2, it suffices to prove that, for every scheme Y_1 ètale over Y and every torsor P on X_1 with group F_1, $({}^PF_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant 0$ when ${}^PF_1$ is regarded as a sheaf of sets, and that the canonical morphism

$$d\colon g^*(\mathrm{R}^1 f_* F) \to \mathrm{R}^1 f'_* F'$$

is injective. The first assertion follows at once from 1), for if Q denotes the torsor obtained from P by extension of structure group along $F_1 \to G_1$, there is an isomorphism ${}^QG_1/{}^PF_1 \xrightarrow{\sim} Q/F_1$.

Let us show that d is injective. It suffices to prove that, if Y_1 is a scheme ètale over Y, and if P and $\tilde{P}$ are two torsors under F_1 whose inverse images P' and $\tilde{P}'$ on X'_1 are isomorphic, then, after replacing Y_1 by a surjective ètale extension, P and $\tilde{P}$ become isomorphic. Choose an isomorphism $p'\colon P' \xrightarrow{\sim} \tilde{P}'$. Let Q (respectively $\tilde{Q}$) be the torsor obtained from P (respectively$\tilde{P}$) by extension of structure group along $F_1 \to G_1$. The inverse images Q' (respectively$\tilde{Q}'$) of Q (respectively$\tilde{Q}$) on X'_1 are obtained from P' (respectively$\tilde{P}'$) by the extension of the structure group $F'_1 \to G'_1$; let $q'\colon Q' \xrightarrow{\sim} \tilde{Q}'$ be the isomorphism obtained in the same way from p'. Since (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$, after replacing Y_1 by a surjective ètale extension, we may suppose that q' is the inverse image of an isomorphism $q\colon Q \xrightarrow{\sim} \tilde{Q}$. Associated with the torsor P (respectively $\tilde{P}$) is a section x of Q/F_1 (respectivelya section $\tilde{x}$ of $\tilde{Q}/F_1$), and, for P and $\tilde{P}$ to be isomorphic, it is necessary and sufficient that there be an isomorphism $Q \xrightarrow{\sim} \tilde{Q}$ such that the induced isomorphism

$$e\colon \mathrm{H}^0(X_1, Q/F_1) \to \mathrm{H}^0(X_1, \tilde{Q}/F_1)$$

carries x to $\tilde{x}$. Take the isomorphism q. The sections $e(x)$ and $\tilde{x}$ of $\mathrm{H}^0(X_1, \tilde{Q}/F_1)$ have the same image in $\mathrm{H}^0(X'_1, \tilde{Q}'/F'_1)$. Since $(\tilde{Q}/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant -1$, after replacing Y_1 by a surjective ètale extension, we indeed have 365
$e(x) = \tilde{x}$, which proves the injectivity of d.

To complete the proof, it remains to show that, if (G, f) is cohomologically proper relative to S in dimension $\leqslant 1$, and if, for every scheme Y_1 ètale over Y and every torsor Q on X_1 with group F_1, $(Q/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant 0$, then the morphism d is surjective. Let P' be a torsor on X'_1 with group F'_1, and let Q' be the torsor under G'_1 obtained from P' by extension of structure group. The data of P' are equivalent to those of Q' together with a section x' of $\mathrm{H}^0(X'_1, Q'/F'_1)$. It follows from the surjectivity of the morphism

$$g^*(\mathrm{R}^1 f_* G) \to \mathrm{R}^1 f'_* G'$$

that, after replacing Y_1 by a surjective ètale extension, there is a torsor Q under G_1 whose inverse image on X'_1 is isomorphic to Q'. Using the fact that $(Q/F_1, f_1)$ is cohomologically proper relative to S in dimension $\leqslant 0$, we may likewise suppose that there is an element x of $\mathrm{H}^0(X_1, Q/F_1)$ whose image in $\mathrm{H}^0(X'_1, Q'/F'_1)$ is x'. Together, Q and x determine a torsor P under F_1 whose inverse image on X_1 is isomorphic to P', which proves the surjectivity of d.

3) Let us show that $(Q/F, f)$ is cohomologically proper relative to S in dimension $\leqslant -1$, i.e. that, for every S-scheme S', and every scheme Y_1 ètale over Y, if x, $\tilde{x}$ are two

elements of $\mathrm{H}^0(X_1, Q_1/F_1)$ whose images x', $\tilde{x}'$ in $\mathrm{H}^0(X'_1, Q'_1/F'_1)$ are equal, then, after a surjective extension of Y_1, we have $x = \tilde{x}$. Associated with x (respectively $\tilde{x}$) is a torsor P (respectively $\tilde{P}$) under F_1 such that Q_1 is obtained from P (respectively $\tilde{P}$) by extension of structure group along $F_1 \to G_1$. It follows from the relation $x' = \tilde{x}'$ that there is an isomorphism $u' \colon P' \xrightarrow{\sim} \tilde{P}'$ such that the isomorphism induced on Q'_1 by u' is the identity. Since (F, f) is cohomologically proper relative to S in dimension $\leqslant 0$, it follows that, after 366
a surjective ètale extension of Y_1, there is an isomorphism $u \colon P \to \tilde{P}$ lifting u'; the fact that (G, f) is cohomologically proper relative to S in dimension $\leqslant -1$ then implies that $x = \tilde{x}$.

Let us show that $(Q/F, f)$ is cohomologically proper relative to S in dimension $\leqslant 0$. Let Y'' be a scheme ètale over Y', and let x'' be an element of $\mathrm{H}^0(X'', Q''/F'')$. Associated with x'' is a torsor P'' on X'' with group F''. Since (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, one can find surjective ètale morphisms $Y''_1 \to Y''$ and $Y_1 \to Y$ such that there is a morphism $Y''_1 \to Y'_1$, and a torsor P on X_1 with group F_1 whose inverse image on X''_1 is isomorphic to the inverse image of P''. It then follows from the fact that (G, f) is cohomologically proper relative to S in dimension $\leqslant 0$ that one can even choose Y''_1 and Y_1 so that the torsor obtained from P by extension of structure group along $F_1 \to G_1$ is isomorphic to Q_1; the torsor P corresponds to an element x of $\mathrm{H}^0(X_1, Q_1/F_1)$ whose image in $\mathrm{H}^0(X''_1, Q''_1/F''_1)$ is isomorphic to the inverse image of x'', which completes the proof.

Proposition XIII.1.13. *Let $f \colon X \to S$ be the structure morphism of an S-scheme, and let F be a sheaf of sets or groups on X (respectivelya sheaf of ind-$\mathbb{L}$-groups, where $\mathbb{L}$ is a set of prime numbers). Suppose that F is locally constant, that (F, f) is cohomologically proper in dimension $\leqslant 0$ (respectivelyin dimension $\leqslant 1$), and that f is locally 0-acyclic (respectivelylocally 1-aspherical for $\mathbb{L}$)* (SGA 4 XV 1.11). *Then, for every specialization $\overline{s}_1 \to \overline{s}_2$ of geometric points of S, the specialization morphism* (SGA 4 VIII 7.1)

$$a_0 \colon (f_*F)_{\overline{s}_2} \to (f_*F)_{\overline{s}_1}$$

is an isomorphism, and, if F is a sheaf of groups, the morphism

$$a_1 \colon (\mathrm{R}^1 f_*F)_{\overline{s}_2} \to (\mathrm{R}^1 f_*F)_{\overline{s}_1}$$

is injective (respectivelythe morphisms a_0 and a_1 are isomorphisms).

The proof is obtained by reproducing verbatim that of SGA 4 XVI 2.3, with the 367
expression “proper”replaced by the expression “cohomologically proper”.

Corollary XIII.1.14. *Let $f \colon X \to S$ be a morphism, let Φ be a stack on X, and let $\mathbb{L}$ be a set of prime numbers. Suppose that, for every scheme X_1 étale over X and every pair of objects x, y of Φ_{X_1}, the sheaf $\mathbf{Hom}_{X_1}(x, y)$ is locally constant, the sheaf $\mathbf{Aut}_{X_1}(x)$ is a locally constant ind-$\mathbb{L}$-group, and the sheaf $S\Phi$ of maximal subgerbes of Φ* [**1**, III 2.1.7] *is locally constant. Suppose that (Φ, f) is cohomologically proper in dimension $\leqslant 1$ and that f is locally 1-aspherical for $\mathbb{L}$. Then, for every specialization $\overline{s}_1 \to \overline{s}_2$ of geometric points of S, the specialization morphism*

$$a \colon (f_*\Phi)_{\overline{s}_2} \to (f_*\Phi)_{\overline{s}_1}$$

is an equivalence of categories.

Let $\overline{S}_1$ (respectively$\overline{S}_2$) be the strict localization of S at $\overline{s}_1$ (respectivelythe strict localization of S at $\overline{s}_2$), and let $\overline{X}_2$, $\overline{\Phi}_2$ (respectively$\overline{X}_1$, $\overline{\Phi}_1$) be the inverse images of X_2, Φ_2 on $\overline{S}_2$ (respectivelyof X_1, Φ_1 on $\overline{S}_1$), and consider the cartesian square

$$\begin{tikzcd} \overline{X}_1 \arrow[r, "h"] \arrow[d, "\overline{f}_1"'] & \overline{X}_2 \arrow[d, "\overline{f}_2"] \\ \overline{S}_1 \arrow[r, "g"] & \overline{S}_2 \end{tikzcd} .$$

We must show that the functor

$$\varphi\colon \overline{\Phi}_2(\overline{X}_2) \to \overline{\Phi}_1(\overline{X}_1)$$

is an equivalence. The functor φ is fully faithful; in fact, let x, y be two objects of $(\overline{\Phi}_2)_{\overline{X}_2}$; 368
the canonical morphism

$$\mathrm{Hom}_{\overline{X}_2}(x,y) \to \mathrm{Hom}_{\overline{X}_1}(\varphi(x),\varphi(y))$$

is identified with the canonical morphism

$$\mathrm{H}^0(\overline{X}_2, \mathbf{Hom}_{\overline{X}_2}(x,y)) \to \mathrm{H}^0(\overline{X}_1, h^*(\mathbf{Hom}_{\overline{X}_2}(x,y)).$$

This morphism is an isomorphism by XIII.1.13.

Let us show that φ is an equivalence. Let x_1 be an object of $\overline{\Phi}_1(\overline{X}_1)$, and let G_1 be the maximal subgerbe of $\overline{\Phi}_1$ generated by x_1. The morphism

$$\mathrm{H}^0(\overline{X}_2, S\overline{\Phi}_2) \to \mathrm{H}^0(\overline{X}_1, h^*(S\overline{\Phi}_2)) = \mathrm{H}^0(\overline{X}_1, S\overline{\Phi}_1)$$

is bijective, and hence there exists a maximal subgerbe G_2 of $\overline{\Phi}_2$ such that h^*G_2 is isomorphic to G_1. It then suffices to prove that the functor

$$G_2 \to h_*h^*G_2$$

is an equivalence. In this form, however, the question is local for the étale topology on $\overline{X}_2$. We may therefore suppose that G_2 is a gerbe of torsors under the automorphism group of an object of G_2, in which case the assertion follows from XIII.1.13.

Corollary XIII.1.15. *The hypotheses are those of XIII.1.13. Suppose in addition that F is a sheaf of sets (respectivelyof ind-$\mathbb{L}$-groups) and that f_*F (respectivelyR^1f_*F) is constructible. Then f_*F (respectivelyR^1f_*F) is locally constant.*

The corollary follows from XIII.1.13 by SGA 4 IX 2.11.

Remark XIII.1.16. Recall that the condition that f be locally 0-acyclic is satisfied if f is flat with separable fibers and X and Y are locally Noetherian (SGA 4 XV 4.1), and that the condition that f be locally 1-aspherical for $\mathbb{L}$ is satisfied if f is smooth, with $\mathbb{L}$ 369
the set of prime numbers distinct from the residual characteristics of S (SGA 4 XV 2.1).

XIII.2. A special case of cohomological properness: relative normal crossings divisors

. Let R be a discrete valuation ring with fraction field K, and let L be an étale K-algebra. Then L is the direct product of finitely many fields L_i, where L_i is an étale extension of K. If L'_i denotes the Galois extension generated by L_i in an algebraic closure of L_i, we say that L is *tamely ramified* over R if the L'_i are tamely ramified extensions in the sense of X X.3, i.e. if an inertia group I_i of $L'_i|K$ has order prime to the residue characteristic p of R.

It is known that I_i is in any case an extension of a cyclic group of order prime to p by a p-group. (This follows from [**5**, ch. IV prop. 7 cor. 4] when the residue extension of R is separable. The proof given in loc. cit. extends to the general case as follows. Retain the hypotheses and notation of loc. cit. , but without assuming the residue extension separable. Let H_i be the subgroup of the inertia group G_0 consisting of the elements s of G_0 such that $s\pi/\pi \in U^i$ for every uniformizer π of A_L. One then checks that G_0/H_1 is a group of order prime to p and that, for $i \geqslant 1$, the H_i/H_{i+1} are p-groups, which gives the asserted result.)

. When R is strictly local, there is the following simple characterization: the K-algebra L is tamely ramified over R if and only if the $[L_i : K]$ are prime to p. Moreover, if L is tamely ramified over R, the L_i are cyclic extensions of K. Indeed, when R is strictly
local, I_i is equal to the Galois group of L'_i over K. As just recalled, I_i is an extension of 370
a cyclic group of order prime to p by a p-group. If L'_i is assumed tamely ramified over R, then I_i is a cyclic group of order prime to p. It follows that $[L_i : K]$ is prime to p and that $L_i = L'_i$. Conversely, if $[L_i : K]$ is prime to p, then I_i cannot contain a nontrivial normal p-subgroup; I_i is therefore a cyclic group of order prime to p, which proves that L_i is tamely ramified over R.

. Let R be a discrete valuation ring with fraction field K, let L be an étale K-algebra, let $\overline{R}$ be a strict localization of R, let $\overline{K}$ be its fraction field, and put $\overline{L} = L \otimes_K \overline{K}$. Then L is tamely ramified over R if and only if $\overline{L}$ is tamely ramified over $\overline{R}$. We may indeed reduce to the case where L is a field. Write $\overline{L} = \prod_i \overline{L}_i$, where the $\overline{L}_i$ are field extensions of $\overline{K}$. If L' is the Galois extension generated by L, and if $\overline{L'} = L' \otimes_K \overline{K}$, then $\overline{L'}$ likewise decomposes as a product of fields, $\overline{L'} = \prod_j \overline{L'_j}$ and every $\overline{L}_i$ is a subextension of at least one $\overline{L'_j}$. Since L' is a Galois extension of K, the $\overline{L'_j}$ are Galois extensions of $\overline{K}$. Suppose that L is tamely ramified over R. Since the Galois group of $\overline{L'_j}|\overline{K}$ is isomorphic to the inertia group of $L'|K$, the $\overline{L'_j}$ are also tamely ramified over $\overline{R}$, and hence so are the $\overline{L}_i$. Conversely, suppose that $\overline{L}$ is tamely ramified over $\overline{R}$. For each j, let v_j be the discrete valuation of $\overline{L'_j}$ extending the valuation of $\overline{K}$, and also denote by v_j the induced valuation on L'. As j varies, v_j ranges over the valuations of L' extending the valuation of K. Let $G = \mathrm{Gal}(L'|K)$, $H = \mathrm{Gal}(L'|L)$, and let I_j be the inertia group of $L'|K$ at v_j, and J_j the inertia group of $L'|L$ at v_j. The group I_j is an extension of a cyclic group of order prime to p by a p-group P_j. Since the $\overline{L}_i$ are tamely ramified over $\overline{R}$, I_j/J_j has order prime to p, and hence $P_j \subset J_j$. Consequently, the group H contains every P_j, and therefore
also the group P generated by the P_j as j varies. But P is invariant in G, since an 371
inner automorphism of G permutes the I_j, and hence also the P_j. It follows that P is a subgroup of H normal in G, and therefore, since L' is the Galois extension generated by L, that $P = 1$. This proves that L is tamely ramified over R.

More generally, let $R \to R'$ be a morphism of discrete valuation rings such that the image of a uniformizer π of R is a uniformizer π' of R', and such that the residue extension $\kappa(R')|\kappa(R)$ is separable. Let K be the fraction field of R, K' the fraction field of R', L an étale K-algebra, and $L' = L \otimes_K K'$. Then L is tamely ramified over R if and only if L' is tamely ramified over R'. We may indeed suppose that R and R' are strictly local. By XIII.2, it is enough to prove that when L is a field, so is L'. Let $\widetilde{R}$ be the normalization of R in L, let $\widetilde{\pi}$ be a uniformizer of $\widetilde{R}$, and put $\widetilde{R}' = \widetilde{R} \otimes_R R'$. Since the extension $\kappa(\widetilde{R})|\kappa(R)$ is radicial and the extension $\kappa(R')|\kappa(R)$ is étale, $\kappa(\widetilde{R}) \otimes_{\kappa(R)} \kappa(R')$ is a field [EGA IV 4.3.2 and 4.3.5]. This proves that $\widetilde{R}'$ is a local ring, and, since π maps to π' in R', we have $\kappa(\widetilde{R}') = \widetilde{R}'/(\widetilde{\pi})$. Consequently, $\widetilde{R}'$ is a discrete valuation ring [**5**, ch. I §2 prop. 2], and hence L' is a field.

. By reducing to the strictly local case, we see that a subalgebra of a tamely ramified algebra is tamely ramified; that the tensor product of two tamely ramified algebras is tamely ramified; that a tamely ramified algebra remains so after extending the discrete valuation ring; and that an algebra which becomes tamely ramified after a tamely ramified extension is itself tamely ramified.

. Let X be an S-scheme, and let D be a divisor ≥ 0 on X. Recall (SGA 5 II 4.2) 372
that D is said to have *strict normal crossings relative to S* if there exists a finite family $(f_i)_{i \in I}$ of elements of $\Gamma(X, \mathcal{O}_X)$ such that $D = \sum_{i \in I} \mathrm{div}(f_i)$ and the following condition is satisfied:

. For every point x of $\operatorname{Supp} D$, X is smooth over S at x, and, if $I(x)$ denotes the set of $i \in I$ such that $f_i(x) = 0$, the subscheme $V((f_i)_{i\in I(x)})$ is smooth over S of codimension $\operatorname{card} I(x)$ in X.

The divisor D is said to have *normal crossings relative to* S if, locally on X for the étale topology, it has strict normal crossings.

Let D be a divisor with normal crossings relative to S. Put $Y = \operatorname{Supp} D$, $U = X - Y$, and denote by $i\colon U \to X$ the canonical immersion. For every geometric point $\overline{s}$ of S and every maximal point y of the geometric fiber $Y_{\overline{s}}$, the ring $R = \mathcal{O}_{X_{\overline{s}},y}$ is a discrete valuation ring.

In the remainder of this number, we shall use the following technical definition:

Definition XIII.2.1. *Let F be a sheaf of sets on U. We say that F is* tamely ramified on X *(*along D*)* relative to S *if, for every geometric point $\overline{s}$ of S, the following condition is satisfied:*

For every maximal point y of $Y_{\overline{s}}$, the restriction of F to the fraction field K of $\mathcal{O}_{X_{\overline{s}}}$ is representable by the spectrum of an étale K-algebra L which is tamely ramified over $\mathcal{O}_{X_{\overline{s}},y}$.

Most often, when no confusion can result, we shall omit mention of D from the terminology.

Definition XIII.2.2. *If F is a sheaf of groups on U, tamely ramified on X relative* 373
to S, we denote by

$$\mathrm{H}^1_{\mathrm{t}}(U, F)$$

the subset of $\mathrm{H}^1(U, F)$ consisting of the classes of torsors under F which are tamely ramified on X relative to S.

Let

$$\begin{array}{ccc} U & \xrightarrow{i} & X \\ {\scriptstyle g}\downarrow & \swarrow{\scriptstyle f} & \\ T & & \end{array}$$

be a commutative diagram of S-schemes, with i as in XIII.2; we denote by

$$\mathrm{R}^1_{\mathrm{t}} g_* F$$

the sheaf on T associated with the presheaf $T' \longmapsto \mathrm{H}^1_{\mathrm{t}}(U', F)$, where T' ranges over the schemes étale over T and where $U' = U \times_T T'$; $\mathrm{R}^1_{\mathrm{t}} g_* F$ is a subsheaf of $\mathrm{R}^1 g_* F$.

Note that, if g is coherent, if $\overline{t}$ is a geometric point of T, if $\overline{T}$ is the strict localization of T at $\overline{t}$, and if $\overline{U} = U \times_T \overline{T}$, then there is an isomorphism

(XIII.2.-1.-1.1) $$(\mathrm{R}^1_{\mathrm{t}} g_* F)_{\overline{t}} \simeq \mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F})$$

. Let $C_t((U, X)/S)$, or simply C_t, be the category of étale coverings of U that are tamely ramified on X relative to S. Suppose that U is connected, and let a be a geometric point of U. Let Γ_t be the functor which assigns to an étale covering U' of U tamely ramified on X relative to S the set of geometric points of U' above a. It follows from XIII.2 that the pair (C_t, Γ_t) satisfies axioms (G_1) through (G_6) of V V.4. Consequently, Γ_t is representable by a pro-object called the *tamely ramified universal covering of* (U, X) 374
relative to S pointed at a. The opposite of the group of U-automorphisms of the tamely ramified universal covering is called the *tamely ramified fundamental group* and is denoted by

$$\pi_1^{\mathrm{t}}((U, X)/S, a) \text{ or simply } \pi_1^{\mathrm{t}}(U, a) \text{ or even } \pi_1^{\mathrm{t}}(U).$$

It is clearly a quotient of the fundamental group $\pi_1(U, a)$ (V V.6.9).

. Let F be a sheaf of groups on U, let P be a right torsor with group F, and let Q be a left torsor with group F. Suppose that P and Q are tamely ramified on X relative to S. Then the same is true of $P \overset{F}{\wedge} Q$. Indeed, we reduce to showing that, if R is a discrete valuation ring with fraction field K, if F is a finite étale group scheme over K, and if P and Q are two torsors under F tamely ramified over R, then the same is true of $P \overset{F}{\wedge} Q$. Now $T = P \overset{F}{\wedge} Q$ is a quotient of $P \times_K Q$. If L, M, N denote the K-algebras representing T, P, Q, respectively, then L is a subalgebra of $M \otimes_K N$, and it follows from XIII.2 that L is tamely ramified over R.

It follows from the preceding that, if F is a sheaf of groups on U and if there exists a torsor with group F tamely ramified on X relative to S, then F is tamely ramified on X relative to S. Indeed, the opposite torsor P° of P is tamely ramified on X relative to S, since it is isomorphic to P as a sheaf of sets. If ${}^P F$ is the group obtained by twisting F by P, there is an isomorphism

$$F \simeq P^\circ \overset{{}^P F}{\wedge} P$$

and consequently F is tamely ramified on X relative to S.

As before, we see that, if $F \to F'$ is a morphism of sheaves of groups on U tamely ramified on X relative to S, and if P is a torsor under F tamely ramified on X relative to S, then the torsor P' deduced from P by extension of the structure group $F \to F'$ is 375
tamely ramified on X relative to S.

In particular, the canonical morphism

$$\mathrm{H}^1(U, F) \to \mathrm{H}^1(U, F')$$

induces, by restriction to $\mathrm{H}^1_{\mathrm{t}}(U, F)$, a canonical morphism

$$\mathrm{H}^1_{\mathrm{t}}(U, F) \to \mathrm{H}^1_{\mathrm{t}}(U, F')$$

. Let $S' \to S$ be a morphism, and denote by U' (respectivelyX', etc.) the inverse image of U (respectively$X$, etc.) on S'. If F is a sheaf of sets on U tamely ramified on X relative to S, it follows from Definition XIII.2.1 and from XIII.2 that F' is tamely ramified on X' relative to S'.

If now F is a sheaf of groups on U, the inverse image on S' of a torsor under F tamely ramified on X relative to S is a torsor under F' tamely ramified on X' relative to S'. In particular, there is a canonical functor

(XIII.2.-1.-1.1) $$C_t((U, X)/S) \to C_t((U', X')/S').$$

Suppose that U and U' are connected, and let a be a geometric point of U and a' a geometric point of U' above a; the preceding gives a canonical morphism

(XIII.2.-1.-1.2) $$\pi_1^{\mathrm{t}}(U', a') \to \pi_1^{\mathrm{t}}(U, a).$$

If $S' \to S$ is a morphism and $h\colon T' \to T$ is the canonical projection, the morphism

$$h^*(\mathrm{R}^1 g_* F) \to \mathrm{R}^1 g'_* F'$$

restricts to a canonical morphism

(XIII.2.-1.-1.3) $$h^*(\mathrm{R}^1_{\mathrm{t}} g_* F) \to \mathrm{R}^1_{\mathrm{t}} g'_* F'$$

. Let F be a sheaf of groups on U, tamely ramified on X relative to S. With the 376
notation of XIII.2.2, there are canonical exact sequences:

(XIII.2.-1.-1.1) $$\begin{array}{ccccccc} 1 & \longrightarrow & \mathrm{H}^1(X, i_*F) & \longrightarrow & \mathrm{H}^1_{\mathrm{t}}(U, F) & \longrightarrow & \mathrm{H}^0(X, \mathrm{R}^1_{\mathrm{t}} i_* F) \\ 1 & \longrightarrow & \mathrm{R}^1 f_*(i_*F) & \longrightarrow & \mathrm{R}^1_{\mathrm{t}} g_* F & \longrightarrow & f_*(\mathrm{R}^1_{\mathrm{t}} i_* F). \end{array}$$

The first is obtained from the exact sequence (SGA 4 III 3.2):

$$1 \to \mathrm{H}^1(X, i_*F) \to \mathrm{H}^1(U, F) \to \mathrm{H}^0(X, \mathrm{R}^1 i_* F).$$

Indeed, it suffices to show that the image of $\mathrm{H}^1(X, i_*F)$ in $\mathrm{H}^1(U, F)$ is in fact contained in $\mathrm{H}^1_{\mathrm{t}}(U, F)$ and that the image of $\mathrm{H}^1_{\mathrm{t}}(U, F)$ in $\mathrm{H}^0(X, \mathrm{R}^1 i_*F)$ is in fact contained in $\mathrm{H}^0(X, \mathrm{R}^1_{\mathrm{t}} i_*F)$. Now the inverse image on U of a torsor under i_*F is a torsor under i^*i_*F which is clearly tamely ramified on X relative to S; the same is therefore true after extension of the structure group $i^*i_*F \to F$, which proves the existence of the arrow $\mathrm{H}^1(X, i_*F) \to \mathrm{H}^1_{\mathrm{t}}(U, F)$. The fact that the image of $\mathrm{H}^1_{\mathrm{t}}(U, F)$ in $\mathrm{H}^0(X, \mathrm{R}^1 i_*F)$ is contained in $\mathrm{H}^0(X, \mathrm{R}^1_{\mathrm{t}} i_*F)$ follows at once from the definition of $\mathrm{R}^1_{\mathrm{t}} i_*F$. This proves the existence of the first exact sequence, and the second follows from it by localization.

. Retain the notation of XIII.2. We shall define a notion of a tamely ramified object of a stack Φ on U when the latter is given, locally *on* X and on S for the étale topology, as the *inverse image of a stack* Ψ *on* S.

First let G be a gerbe on U, and suppose that we are given a surjective étale morphism $S_1 \to S$, a surjective étale morphism $X_2 \to X \times_S S_1$, a trivial gerbe H on S_1, and an isomorphism

$$G|U_2 \to H|U_2,$$

where $U_1 = U \times_X X_1$, $U_2 = U \times_X X_2$. Upon choosing a trivialization of $H|X_2$, the 377
isomorphism above identifies $G|U_2$ with the stack of torsors under a sheaf of groups F. An object x of G_U is said to be tamely ramified on X relative to S if the restriction of x to U_2 is a torsor tamely ramified on X relative to S. By XIII.2, this notion does not depend on the chosen trivialization of $H|X_2$.

Now let Φ be a stack on U, and suppose that we are given a surjective étale morphism $S_1 \to S$, a surjective étale morphism $X_2 \to X \times_S S_1$, a stack Ψ on S_1, and an isomorphism

$$i\colon \Phi|U_2 \to \Psi|U_2.$$

Let x be an object of Φ_U, let G_x be the maximal subgerbe of Φ generated by x [**1**, III 2.1.7], and let $S\Phi$ be the sheaf of maximal subgerbes of Φ. The isomorphism i induces an isomorphism

$$S\Phi|U_2 \to S\Psi|U_2.$$

It follows from XIII.5.8 that, after replacing S_1 by a surjective étale extension if necessary, there is a unique maximal subgerbe H of Ψ, which may be assumed trivial, such that i defines an isomorphism

$$G_x|U_2 \to H|U_2.$$

The object x is said to be tamely ramified on X relative to S if it is so as an object of G_x equipped with the isomorphism above.

. Let Φ be a stack on U given, locally on X and S, as the inverse image of a stack on S, and let

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[d, "g"'] & X \arrow[dl, "f"] \\ T & \end{tikzcd}$$

be a diagram as in XIII.2.2. For every scheme T' étale over T, if $U' = U \times_T T'$, consider the subset $(g^{\mathrm{t}}_*\Phi)_{T'}$ of $(g_*\Phi)_{T'} = \Phi_{U'}$, consisting of the objects of $\Phi_{U'}$ that are tamely 378
ramified on X relative to S. The *tamely ramified direct image* of Φ under g, denoted

$$g^{\mathrm{t}}_*\Phi,$$

is the full subcategory of $g_*\Phi$ whose objects over a scheme T' étale over T are the elements of $(g^{\mathrm{t}}_*\Phi)_{T'}$. It is clear that $g^{\mathrm{t}}_*\Phi$ is a substack of $g_*\Phi$.

. By reducing to the case of a stack of torsors, we see that if Φ is a stack on U which, locally for the étale topology of S and X, is the inverse image of a stack on S, then the canonical morphism

$$h^*(g_*\Phi) \to g'_*\Phi'$$

restricts to a canonical morphism

$$h^*(g_*^{\mathrm{t}}\Phi) \to g'^{\mathrm{t}}_*\Phi'.$$

Remarks XIII.2.3. a) If F is a constructible locally constant sheaf of sets on U, then, for F to be tamely ramified on X relative to S, it is enough that the condition of XIII.2.1 hold at the geometric points of S lying over the maximal points of S. To see this, we may suppose that D is a strict normal crossings divisor. The sheaf F is represented by an étale covering V of U. If $\overline{s}$ is a geometric point of S and y a maximal point of $Y_{\overline{s}}$, denote by $\overline{S}$ the strict localization of S at $\overline{s}$, by $\overline{X}$ the strict localization of X at $\overline{y}$, and put $\overline{U} = U \times_X \overline{X}$, $\overline{V} = V \times_X \overline{X}$. If the condition of XIII.2.1 holds at geometric points over the maximal points of S, it follows from XIII.5.5 below that $\overline{V}$ is a covering of $\overline{U}$ tamely ramified on $\overline{X}$ relative to $\overline{S}$. Consequently, V is an étale covering of U tamely ramified on X relative to S.

b) Let F be a sheaf of groups on U tamely ramified on X relative to S. If $\overline{s}$ is a 379
geometric point of S and y a maximal point of $Y_{\overline{s}}$, denote by K the fraction field of $\mathcal{O}_{X_{\overline{s}},y}$. Suppose that, for every point $\overline{s}$ and every point y, the K-algebra L whose spectrum represents $F|K$ has rank prime to the residue characteristic p of $\mathcal{O}_{X_{\overline{s}}}$. By an abuse of language, we shall sometimes say that F is prime to the residue characteristics of S. When this is so, every torsor P under F is tamely ramified on X relative to S.

Indeed, let $\overline{R}$ be the strict localization of $\mathcal{O}_{X_{\overline{s}},y}$ at $\overline{y}$, let $\overline{K}$ be its fraction field, and let $\overline{F}$ be the inverse image of F on $\overline{K}$. We show that F may be assumed constant. Since F is tamely ramified on X relative to S, $\overline{F}$ is represented by the spectrum of a $\overline{K}$-algebra $L = \prod L_i$, where the L_i are extensions of $\overline{K}$ of degree prime to p. We can therefore find an extension K' of $\overline{K}$ of degree prime to p such that $\overline{F}|K'$ is a constant sheaf. By XIII.2, to prove that $P|\overline{K}$ is tamely ramified over $\overline{R}$, it is enough to see that $P|K'$ is tamely ramified over the integral closure of $\overline{R}$ in K'; this reduces us to the case where $\overline{F}$ is constant. Henceforth suppose that $\overline{F}$ is constant. The $\overline{K}$-algebra H representing $P|\overline{K}$ is then a product of mutually isomorphic extensions H_i of $\overline{K}$. Since the rank of H is prime to p, the same is true of $[H_1 : \overline{K}]$, which proves that H is tamely ramified on X relative to S.

c) Let X be a regular scheme, let D be a normal crossings divisor of X (SGA 5 I 3.1.5), let $U = X - \operatorname{Supp} D$, and let F be a sheaf of sets on U. If y is a maximal point of $\operatorname{Supp} D$, denote by K the fraction field of $\mathcal{O}_{X,y}$. We say that F is *tamely ramified relative to D* if, for every maximal point y of $\operatorname{Supp} D$, $F|K$ is represented by a K-algebra tamely ramified over $\mathcal{O}_{X,y}$.

Theorem XIII.2.4. *Let $f\colon X \to S$ be an S-scheme, let D be a divisor on X with* 380
normal crossings relative to S (XIII.2)*, put $Y = \operatorname{Supp} D$ and $U = X - Y$, and let $i\colon U \to X$ be the canonical immersion. Let F be a sheaf of sets (respectivelygroups) on U, satisfying one of the following conditions:*

a) *Locally for the étale topology* on *X and on S, F is the inverse image of a sheaf of sets (respectivelya constructible sheaf of groups) on S.*
b) *F is constructible locally constant on U and is tamely ramified on X relative to S.*

Then the following conclusions hold:

1) *(F, i) is cohomologically proper relative to S in dimension ≤ 0 (respectivelyfor every morphism $h\colon S' \to S$, if $i'\colon U' \to X'$ is the inverse image of i on S', if*

$F' = F|U'$, and if $k = h_{(X)}$, the canonical morphism

$$\Psi\colon k^*(\mathrm{R}^1_{\mathrm{t}} i_* F) \to \mathrm{R}^1_{\mathrm{t}} i'_* F'$$

is an isomorphism).

If F is a sheaf of groups prime to the residue characteristics of S (XIII.2.3.b)) (and tamely ramified on X relative to S), then (F, i) is cohomologically proper relative to S in dimension ≤ 1.

2) *If F is a constructible sheaf of sets (respectivelygroups), then i_*F (respectively$\mathrm{R}^1_{\mathrm{t}} i_* F$) is constructible.*

Proof. For every S-scheme S', consider the following diagram, all of whose squares are cartesian:

$$\begin{tikzcd}
U \arrow[dr, "i"] \arrow[dd, "g"'] & & U' \arrow[ll] \arrow[dr, "i'"] \arrow[dd, "g'"] & \\
& X \arrow[dl, "f"] & & X' \arrow[ll, "k"'] \arrow[dl, "f'"] \\
S & & S' \arrow[ll, "h"] &
\end{tikzcd}$$

Since the question is local on X for the étale topology, we may suppose that D is a 381
strict normal crossings divisor relative to S (XIII.2); moreover, after restricting X to a neighborhood of Y if necessary, we may suppose that X is smooth over S.

Proof of XIII.2.4 1).

. *Case of a sheaf of sets satisfying* a). We may suppose that $F = g^*G$, where G is a sheaf on S. It then follows from (SGA 4 XVI 3.2) that the canonical morphism

(XIII.2.-1.-1.1) $$f^*G \to i_*F$$

is an isomorphism. For every S-scheme $h\colon S' \to S$, there is similarly an isomorphism $f'^*G' \to i'_*F'$; consequently, the canonical morphism

$$\varphi\colon k^*(i_*F) \to i'_*F'$$

is identified with the natural isomorphism

$$k^*f^*G \simeq f'^*G'$$

. *Case of a sheaf of sets satisfying* b). We must show that φ is an isomorphism, and it is enough to check this at every geometric point $\overline{x}'$ of X'. Let $\overline{S}$ (respectively$\overline{X}$, respectively$\overline{S'}$, respectively$\overline{X'}$) be the strict localization of S (respectively X, respectively S', respectivelyX') at the corresponding image of $\overline{x}'$, and put $\overline{U} = U_{(\overline{X})}$, $\overline{U'} = U_{(\overline{X'})}$, etc. The morphism $\varphi_{\overline{x}}$ is identified with the canonical morphism

$$\overline{\varphi}\colon \mathrm{H}^0(\overline{U}, \overline{F}) \to \mathrm{H}^0(\overline{U'}, \overline{F'})$$

One can find a principal covering V of $\overline{U}$, of the type appearing in XIII.5.4, such that the inverse image of $\overline{F}$ on V is a constant sheaf with value C. If Π is the Galois group of V over $\overline{U}$, then Π acts on $\overline{F}|V$, and there is an isomorphism

(XIII.2.-1.-1.1) $$\mathrm{H}^0(\overline{U}, \overline{F})) \simeq \mathrm{H}^0(V, C_V)^{\Pi}$$

where the right-hand side denotes the set of elements of $\mathrm{H}^0(V, C_V)$ invariant under Π. 382
Since $V' = V \times_{\overline{U}} \overline{U'}$ is a principal covering of $\overline{U'}$ with Galois group $\Pi' \simeq \Pi$, one sees that the morphism $\overline{\varphi}$ is obtained, by taking the invariants under Π, from the canonical morphism

$$\mathrm{H}^0(V, C_V) \to \mathrm{H}^0(V', C_{V'}).$$

Since V and V' are connected (XIII.5.4), this morphism, and hence also $\overline{\varphi}$, is an isomorphism.

Note that if, in addition, F is a sheaf of groups and P is a torsor on $\overline{U}$ with group $\overline{F}$, tamely ramified relative to D, it follows from the preceding proof and from XIII.2 that $({}^{P}\overline{F}, \bar{i})$ is cohomologically proper relative to S in dimension ≤ 0.

. *Case of a sheaf of groups.* To show that Ψ is an isomorphism, it suffices to prove that, for every geometric point $\overline{y}'$ of Y', the morphism

$$\Psi_{\overline{y}'} \colon (k^*(\mathrm{R}^1_{\mathrm{t}} i_* F))_{\overline{y}'} \to (\mathrm{R}^1_{\mathrm{t}} i'_* F')_{\overline{y}'}$$

is an isomorphism. Now, by XIII.2.2, $\Psi_{\overline{y}'}$ is identified with the canonical morphism

$$\overline{\Psi} \colon \mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \to \mathrm{H}^1_{\mathrm{t}}(\overline{U'}, \overline{F'}).$$

Let $\widetilde{U}$ be the tamely ramified universal covering of $\overline{U}$ (XIII.2), and let $\widetilde{F}$ be the inverse image of $\overline{F}$ on $\widetilde{U}$. It follows from XIII.5.8 in case a) and from XIII.5.5 in case b) that $\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F})$ is identified with the subset of $\mathrm{H}^1(\overline{U}, \overline{F})$ consisting of the classes of $\overline{F}$-torsors whose inverse image on $\widetilde{U}$ is trivial. On the other hand, a standard (cf. IX IX.5, p. 183) argument shows that the set of elements of $\mathrm{H}^1(\overline{U}, \overline{F})$ whose inverse image on $\widetilde{U}$ is trivial is identified with $\mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F}))$. Thus one obtains a canonical isomorphism

$$\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \xrightarrow{\sim} \mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F})). \tag{XIII.2.-1.-1.1}$$

Consequently, the morphism $\overline{\Psi}$ is identified with the canonical morphism

$$\mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F})) \to \mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U'}), \mathrm{H}^0(\widetilde{U}', \widetilde{F}')).$$

Let us show that this morphism is an isomorphism. The morphism $\pi_1^{\mathrm{t}}(\overline{U'}) \to \pi_1^{\mathrm{t}}(\overline{U})$ is an 383
isomorphism by XIII.5.6, and the same is true of the morphism $\mathrm{H}^0(\widetilde{U}, \widetilde{F}) \to \mathrm{H}^0(\widetilde{U}', \widetilde{F}')$. Indeed, this is clear in case b), since $\widetilde{F}$ is constant and $\widetilde{U}$ and $\widetilde{U}'$ are connected. In case a), let $\overline{G}$ be a constructible sheaf of groups on $\overline{S}$ such that $\overline{F} = \overline{g}^* \overline{G}$; since the morphisms $\widetilde{U} \to \overline{S}$ and $\widetilde{U}' \to \overline{S'}$ are 0-acyclic (XIII.5.8), there are isomorphisms

$$\mathrm{H}^0(\widetilde{U}, \widetilde{F}) \simeq \mathrm{H}^0(\overline{S}, \overline{G}) \simeq \mathrm{H}^0(\overline{S'}, \overline{G'}) \simeq \mathrm{H}^0(\widetilde{U}', \widetilde{F}'),$$

which implies that Ψ is an isomorphism. The last assertion of XIII.2.4 1) follows from the foregoing, in view of XIII.2.3 b).

Proof of XIII.2.4 2). The case of a constructible sheaf of sets satisfying a) follows immediately from (XIII.2.-1.-1.1). Let $F = g^* G$ be a sheaf of groups satisfying a), where G is a constructible sheaf; we may suppose that S is affine. Let $(S_j)_{j \in J}$ be a finite family of reduced closed subschemes of S whose union covers S, such that the inverse image of G on S_j is a locally constant sheaf. In view of XIII.2.4 1), in order to prove that $\mathrm{R}^1_{\mathrm{t}} i_* F$ is constructible, it suffices to prove this after the base change $S_j \to S$, for every $j \in J$. We are therefore reduced to case b), where F is locally constant.

From now on, suppose that F is a sheaf of sets or of groups satisfying b). Since the question is local for the étale topology on X, we may suppose that X is of finite presentation over S, and, by passage to the limit, we may suppose that X and S are noetherian.

Let $D = \sum_{1 \leq i \leq r} \operatorname{div} f_i$, where, for every point x of $\operatorname{Supp} D$, if $I(x)$ is the set of i such that $f_i(x) = 0$, the closed subscheme $V((f_i)_{i \in I(x)})$ is smooth over S of codimension $\operatorname{card} I(x)$ in X. Let $\mathcal{P}$ be the set of subsets of $[1, r]$, and, for every $I \in \mathcal{P}$, put

$$X_I = \Big(\bigcap\nolimits_{i \in I} V(f_i) \Big) \cap \Big(\bigcap\nolimits_{i \notin I} X_{f_i} \Big).$$

Let z be a point of X_I. After first passing to an étale neighborhood of z, one can find an 384
open subset W of X containing z and a principal covering V of $U \cap W$, tamely ramified on W relative to S, of the type considered in XIII.5.7, such that the inverse image of F on V is a constant sheaf with value C. Let π be the Galois group of the covering V. For every geometric point $\overline{x}$ of X_I, one then has, by (XIII.2.-1.-1.1),

$$\mathrm{H}^0(\overline{U}, \overline{F}) \simeq \mathrm{H}^0(V, C_V)^{\pi}.$$

It follows that $i_*F|X_I \cap W$ is locally constant (SGA 4 IX 2.13), and hence i_*F is constructible.

Finally, let us show that if F is a locally constant sheaf of groups, then $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X_I}$ is constructible. If $\overline{x}$ is a geometric point of X, we have obtained in (XIII.2.-1.-1.1) the expression

$$\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \xrightarrow{\sim} \mathrm{H}^1(\pi_1^{\mathrm{t}}(\overline{U}), \mathrm{H}^0(\widetilde{U}, \widetilde{F})).$$

If p is the residue characteristic of $\overline{X}$, then by XIII.5.6, $\pi_1^{\mathrm{t}}(\overline{U}) = \prod_{l \neq p} \mathbb{Z}_l(1)^{\operatorname{card} I}$. Let $\mathbb{L}$ be the set of prime numbers that divide the order of the finite group $\mathrm{H}^0(\widetilde{U}, \widetilde{F})$, and let $K = \prod_{l \in \mathbb{L} - \{p\} \cap \mathbb{L}} \mathbb{Z}_l(1)^{\operatorname{card} I}$. It follows from [**4**, I §5 ex.2] that there is an isomorphism

$$\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F}) \simeq \mathrm{H}^1(K, \mathrm{H}^0(\widetilde{U}, \widetilde{F})).$$

Since K is topologically finitely generated and $\mathrm{H}^0(\widetilde{U}, \widetilde{F})$ is finite, we first deduce that the fibers of the sheaf $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X_I}$ are finite. Moreover, the set $\mathbb{L}$ does not depend on the point $\overline{x}$. For every $q \in \mathbb{L}$, let $X_{I,q}$ be the closed subset of X_I defined by the equation $q = 0$, and let $X_{I'}$ be the open subset of X_I complementary to the union of the $X_{I,q}$. Then $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X_{I,q}}$ and $\mathrm{R}^1_{\mathrm{t}} i_*F|_{X_I'}$ are locally constant; indeed, a specialization morphism between geometric points of $X_{I,q}$ (respectivelyof $X_{I'}$) induces an isomorphism on the groups K (XIII.5.7), hence also on the sets $\mathrm{H}^1_{\mathrm{t}}(\overline{U}, \overline{F})$, and one may apply SGA 4 IX 2.13.

Corollary XIII.2.5. *Let $f\colon X \to S$ be a morphism, let D be a divisor on X with normal crossings relative to S* (XIII.2)*, let $Y = \operatorname{Supp} D$, $U = X - Y$, and let $i\colon U \to X$ be the canonical immersion. Let Φ be a stack on U, and suppose that we are given surjective étale morphisms $S_1 \to S$ and $X_2 \to X \times_S S_1$, a stack Ψ on S_1, and an isomorphism $\Phi|U_2 \simeq \Psi|U_2$ (cf. XIII.2).* 385

Then, for every morphism $h\colon S' \to S$, if $k = h_{(X)}$, the canonical functor

$$\varphi\colon k^* i_* \Phi \to i'_* \Phi'$$

is fully faithful. If Ψ is constructible, the canonical functor

$$\psi\colon k^* i^{\mathrm{t}}_* \Phi \to i'^{\mathrm{t}}_* \Phi'$$

is an equivalence of categories.

Moreover, if the stack Ψ is constructible (respectivelyif Ψ is 1*-constructible* (XIII.0) *and S is locally noetherian), $i_*\Phi$ is constructible (respectivelyi$^t_*\Phi$ is 1-constructible).*

Let us show that φ is fully faithful. It is enough to see that, for every geometric point $\overline{x}'$ of X', the same is true of the functor

$$\overline{\varphi}\colon \Phi(\overline{U}) \to \Phi'(\overline{U'})$$

(we again use the notation of XIII.2). Let a, b be two objects of $\Phi(\overline{U})$, and let a', b' be their images under $\overline{\varphi}$. Since the morphism $\overline{U} \to \overline{S}$ is locally 0-acyclic (XIII.5.8), we have an isomorphism

$$\mathrm{H}^0(\overline{U}, \overline{S\Phi}) \xrightarrow{\sim} \mathrm{H}^0(S, \overline{S\Psi}).$$

Consequently, a and b are inverse images of objects of Ψ, and the same is therefore true of $F = \mathbf{Hom}_{\overline{U}}(a, b)$. Since the sheaf $F' = \mathbf{Hom}_{\overline{U'}}(a', b')$ is the inverse image on $\overline{U'}$ of F, it follows from XIII.2 that the canonical morphism

$$\mathrm{H}^0(\overline{U}, F) \to \mathrm{H}^0(\overline{U'}, F')$$

is an isomorphism, which proves that $\overline{\varphi}$ is fully faithful.

Let us show that, for every geometric point $\overline{x}'$ of X', the functor 386

$$\overline{\psi}\colon i^{\mathrm{t}}_* \Phi(\overline{X'}) \to i'^{\mathrm{t}}_* \Phi'(\overline{X'})$$

is an equivalence. By what precedes, $\overline{\psi}$ is fully faithful. Let us show that $\overline{\psi}$ is essentially surjective. Let a' be an object of $\Phi'(\overline{U'})$ tamely ramified on X' relative to S' (XIII.2), and let us show that it is the image of a tamely ramified object of $\Phi(\overline{U})$. It follows from XIII.2.4 1) that the canonical morphism

$$\mathrm{H}^0(\overline{U}, \overline{S\Phi}) \to \mathrm{H}^0(\overline{U'}, \overline{S\Phi}')$$

is an isomorphism. Let G' be the maximal subgerbe of Φ' generated by a'; there is then a maximal subgerbe G of $\overline{\Phi}$, the inverse image of a gerbe on $\overline{S}$, such that

$$\overline{m}^* G \simeq G',$$

where m is the morphism $\overline{U'} \to \overline{U}$. The canonical functor

$$(*) \qquad \overline{k}^* \overline{i}^{\mathrm{t}}_* G \to \overline{i}'^{\mathrm{t}}_* G'$$

is an equivalence, because G identifies with a gerbe of torsors under a constructible sheaf of groups coming from $\overline{S}$, and we can apply XIII.2.4 1). It then follows from $(*)$ that there exists an object a of $G(\overline{U})$, tamely ramified on X relative to S, whose inverse image on $\overline{U'}$ is a'. This proves that ψ is an equivalence.

If Ψ is constructible, then so is $i_*\Phi$; indeed, an object x of $i_*\Phi$ is, locally for the étale topology on S, the inverse image of an object y of Ψ; it therefore follows from XIII.2.-1.-1.1 that $\mathbf{Aut}(x)$ is the inverse image of $\mathbf{Aut}(y)$, and hence is constructible. Finally, if Ψ is 1-constructible, then so is $i^{\mathrm{t}}_*\Phi$ by XIII.6.5 below.

Corollary XIII.2.6. *The notation is that of XIII.2.4. Suppose that S has characteristic zero at every point s such that $Y_s \neq \emptyset$. Then, if $\mathcal{F}$ is a locally constant constructible sheaf of groups on U (respectivelya constructible stack on U which, locally on X and S, is the inverse image of a constructible stack on S), $(\mathcal{F}, i)$ is cohomologically proper relative to S in dimension ≤ 1.* 387

Since every constructible sheaf of sets on U is tamely ramified on X relative to S, the corollary follows from XIII.2.4 (respectivelyXIII.2.5).

Corollary XIII.2.7. *The notation is that of XIII.2.4, but in addition we are given an S-scheme T, a proper morphism $p\colon X \to T$, and we suppose that X and T are of finite presentation over S; let $q = pi$. Let $\mathcal{F}$ be a constructible sheaf of sets on U satisfying one of conditions* a) *or* b) *of XIII.2.4 (respectivelya sheaf of groups satisfying one of conditions* a), b) *of XIII.2.4, respectivelya stack on U which, locally on X and S, is the inverse image of a constructible stack G on S). Then the following conclusions hold:*

1) *$(\mathcal{F}, q)$ is cohomologically proper relative to S in dimension ≤ 0 (respectivelyfor every morphism $h\colon S' \to S$, if $m = h_{(T)}$, the canonical morphism:*

$$\Theta\colon m^*(\mathrm{R}^1_{\mathrm{t}} q_* \mathcal{F}) \to \mathrm{R}^1_{\mathrm{t}} q'_* \mathcal{F}'$$

is an isomorphism, respectivelyfor every morphism $S' \to S$, the canonical morphism

$$\xi\colon m^*(q^{\mathrm{t}}_* \mathcal{F}) \to q'^{\mathrm{t}}_* \mathcal{F}'$$

is an equivalence).

2) *The sheaf $q_*\mathcal{F}$ (respectivelythe sheaf $\mathrm{R}^1_{\mathrm{t}} q_*\mathcal{F}$, respectivelythe stack $q^{\mathrm{t}}_*\mathcal{F}$) is constructible. In the last case, if S is assumed locally noetherian and G is 1-constructible, then so is $q^{\mathrm{t}}_*\mathcal{F}$.*

The first part follows immediately from XIII.2.4, XIII.2.5, and the proof of XIII.1.7. 388
Let us prove 2). If $\mathcal{F}$ is a constructible sheaf of sets on U satisfying XIII.2.4 a) or XIII.2.4 b), it follows from XIII.2.4 2) that $i_*\mathcal{F}$ is constructible; hence so is $q_*\mathcal{F} = p_*(i_*\mathcal{F})$ (SGA 4 XIV 1.1).

Let $\mathcal{F}$ be a constructible sheaf of groups on U satisfying XIII.2.4 a) or XIII.2.4 b), and let us prove that $\mathrm{R}^1_{\mathrm{t}} q_*\mathcal{F}$ is constructible. By passage to the limit (EGA IV 8.10.5 and 17.7.8) and using 1), we may suppose that S is noetherian. Let Φ then be the stack on X

whose fiber over every scheme X' étale over X consists of the torsors on $U' = U \times_X X'$, with group $\mathcal{F}|U$, which are tamely ramified on X relative to S; thus

$$S(i_*^{\mathrm{t}}\Phi) \simeq \mathrm{R}_{\mathrm{t}}^1 i_*\mathcal{F}$$

and this sheaf is constructible by XIII.2.4 2). It follows from XIII.6.5 below that $S(p_*\Phi)$ is constructible, i.e. that $\mathrm{R}_{\mathrm{t}}^1 q_*\mathcal{F}$ is constructible.

Finally, if $\mathcal{F}$ is a stack on U which, locally on X and S, is the inverse image of a constructible stack on S, $i_*^{\mathrm{t}}\mathcal{F}$ is constructible, and hence so is $q_*^{\mathrm{t}}\mathcal{F} = p_*(i_*^{\mathrm{t}}\mathcal{F})$. If in addition S is locally noetherian and SG is constructible, $S(i_*^{\mathrm{t}}\mathcal{F})$ is constructible by 6.3; hence the same is true of $S(q_*i_*^{\mathrm{t}}\mathcal{F})$, i.e. of $S(q_*^{\mathrm{t}}\mathcal{F})$ XIII.6.4.

Corollary XIII.2.8. *Let*

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[d, "g"'] & X \arrow[dl, "f"] \\ S & \end{tikzcd}$$

be a commutative diagram of schemes in which U is the open complement in X of a divisor with normal crossings relative to S, and f is a proper *morphism of finite presentation. Let $\mathbb{L}$ be a set of prime numbers. Suppose that g is locally 0-acyclic (respectivelylocally 1-aspherical for $\mathbb{L}$). Then, if $\mathcal{F}$ is a sheaf of sets on U (respectivelya sheaf of $\mathbb{L}$-groups) that is locally constant and constructible on U, and tamely ramified on X relative to S, $f_*\mathcal{F}$ (respectively$\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F}$) is locally constant constructible, and $(\mathcal{F}, f)$ is cohomologically proper relative to S in dimension ≤ 0 (respectivelythe formation of $\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F}$ commutes with every base change $S' \to S$). In the non-respective case, if $\mathcal{F}$ is a sheaf of groups, then for every specialization $\overline{s}_1 \to \overline{s}_2$ of geometric points of S, the specialization morphism* 389

$$(\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F})_{\overline{s}_2} \to (\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F})_{\overline{s}_1}$$

is injective.

The corollary follows immediately from XIII.2.6 and from XIII.1.13 (respectivelyfrom the analogue of XIII.1.13 for $\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F}$, which is proved as in loc. cit.).

Corollary XIII.2.9. *Let*

$$\begin{tikzcd} U \arrow[r, "i"] \arrow[d, "g"'] & X \arrow[dl, "f"] \\ S & \end{tikzcd}$$

be a commutative diagram of schemes in which U is the open complement in X of a divisor with normal crossings relative to S, and f is a proper smooth morphism of finite presentation. Let $\mathbb{L}$ be the set of prime numbers distinct from the residue characteristics of S. Let $\mathcal{F}$ be a locally constant constructible sheaf of $\mathbb{L}$-groups on U, tamely ramified on X relative to S. Then $\mathrm{R}^1 f_*\mathcal{F}$ is locally constant constructible, and $(\mathcal{F}, f)$ is cohomologically proper relative to S in dimension ≤ 1. 390

The corollary follows from XIII.2.8 and from the fact that $\mathrm{R}_{\mathrm{t}}^1 f_*\mathcal{F} = \mathrm{R}^1 f_*\mathcal{F}$ (XIII.2.3 b)).

. If U is a connected scheme, a is a geometric point of U, and $\mathbb{L}$ is a set of prime numbers, we denote by

$$\pi_1^{\mathbb{L}}(U, a) \tag{2.10.0}$$

the inverse limit of the finite quotients of $\pi_1(U, a)$ whose orders have all their prime factors in $\mathbb{L}$.

We shall define specialization homomorphisms for the fundamental group, generalizing X.X.2.

Let $g\colon U \to S$ be a coherent morphism with geometrically connected fibers (respectivelya morphism of the form $g = fi$, where $f\colon X \to S$ is a proper morphism of finite presentation and $i\colon U \to X$ is an open immersion such that U is the complement in X of a divisor with normal crossings relative to S (cf. XIII.2.8)). Let $\mathbb{L}$ be a set of prime numbers and suppose, in the first case, that for every finite constant $\mathbb{L}$-group C, (C_U, g) is cohomologically proper relative to S in dimension ≤ 1. Let $\overline{s}_1 \to \overline{s}_2$ be a specialization morphism of geometric points of S, let $\overline{S}$ be the strict localization of S at $\overline{s}_2$, and put $\overline{U} = U \times_S \overline{S}$. There is a commutative diagram

$$\begin{tikzcd}
U_{\bar{s}_1} \arrow[r, "h_1"] \arrow[d] & \bar{U} \arrow[d, "\bar{g}"] & U_{\bar{s}_1} \arrow[l, "h_2"'] \arrow[d] \\
\bar{s}_1 \arrow[r] & \bar{S} & \bar{s}_2 . \arrow[l]
\end{tikzcd}$$

Let a_1 be a geometric point of $U_{\bar{s}_1}$, a_2 a geometric point of $U_{\bar{s}_2}$; the morphisms h_1 and h_2 define canonical homomorphisms 391

$$\pi_1\colon \pi_1^{\mathbb{L}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathbb{L}}(\bar{U}, a_1) \qquad \pi_2\colon \pi_1^{\mathbb{L}}(U_{\bar{s}_2}, a_2) \to \pi_1^{\mathbb{L}}(\bar{U}, a_2)$$
$$(\text{respectively } \pi_1\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathrm{t}}(\bar{U}, a_1) \qquad \pi_2\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_2}, a_2) \to \pi_1^{\mathrm{t}}(\bar{U}, a_2))$$

(V V.7 and XIII.2.-1.-1.2). The hypotheses of cohomological properness (respectivelyXIII.2.8) show that π_2 is an isomorphism. If we choose a path class from a_1 to a_2, we obtain an isomorphism

$$\pi_{12}\colon \pi_1^{\mathbb{L}}(\bar{U}, a_1) \xrightarrow{\sim} \pi_1^{\mathbb{L}}(\bar{U}, a_2) \quad (\text{respectively } \pi_{12}\colon \pi_1^{\mathrm{t}}(\bar{U}, a_1) \xrightarrow{\sim} \pi_1^{\mathrm{t}}(\bar{U}, a_2))$$

and hence a homomorphism $\pi = \pi_2^{-1}\pi_{12}\pi_1$

$$\pi\colon \pi_1^{\mathbb{L}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathbb{L}}(U_{\bar{s}_2}, a_2) \quad (\text{respectively } \pi\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_1}, a_1) \to \pi_1^{\mathrm{t}}(U_{\bar{s}_2}, a_2)).$$

Changing the path class from a_1 to a_2 amounts to modifying π by an inner automorphism of $\pi_1^{\mathbb{L}}(X_{\bar{s}_2}, a_2)$ (respectivelyof $\pi_1^{\mathrm{t}}(X_{\bar{s}_2}, a_2)$). One calls any one of the homomorphisms defined above the *specialization homomorphism for the fundamental group* associated with the morphism $\bar{s}_1 \to \bar{s}_2$, and writes simply

$$\pi\colon \pi_1^{\mathbb{L}}(X_{\bar{s}_1}) \to \pi_1^{\mathbb{L}}(X_{\bar{s}_2}) \quad (\text{respectively } \pi\colon \pi_1^{\mathrm{t}}(X_{\bar{s}_1}) \to \pi_1^{\mathrm{t}}(X_{\bar{s}_2}))$$

for one of the homomorphisms defined above.

Lemma XIII.2.10. *Let $f\colon X \to S$ be a proper morphism of finite presentation, let D be a divisor on X with normal crossings relative to S, put $Y = \operatorname{Supp} D$, $U = X - Y$, and let $i\colon U \to X$ be the canonical morphism. Let $\bar{s}_1 \to \bar{s}_2$ be a specialization morphism of geometric points of S, let y_1 be a geometric point of $Y_{\bar{s}_1}$, and let y_2 be a geometric point of $Y_{\bar{s}_2}$ such that the projection z_1 of y_1 on X is a generization of the projection z_2 of y_2. Let I_{y_1} be an inertia subgroup of $\pi_1^{\mathrm{t}}(\bar{U}_{\bar{s}_1})$ at y_1. Then the image of I_{y_1} under the specialization homomorphism*

$$\pi\colon \pi_1^{\mathrm{t}}(U_{\bar{s}_1}) \to \pi_1^{\mathrm{t}}(U_{\bar{s}_2})$$

is an inertia subgroup of $\pi_1^{\mathrm{t}}(U_{\bar{s}_2})$ at y_2. 392

Indeed, let $\overline{X}$ (respectively$\widetilde{X}$) be the strict localization of X at y_2 (respectivelyat y_1), and put $\overline{U} = U \times_X \overline{X}$ (respectively$\widetilde{U} = U \times_X \widetilde{X}$). There is a canonical morphism $\widetilde{U} \to \overline{U}$, and it follows from XIII.1.9 that there is a commutative diagram

$$\begin{tikzcd}
\pi_1^{\mathrm{t}}\left(\widetilde{U}_{\bar{s}_1}\right) \arrow[r, "\pi'"] \arrow[d] & \pi_1^{\mathrm{t}}\left(\overline{U}_{\bar{s}_2}\right) \arrow[d] \\
\pi_1^{\mathrm{t}}\left(U_{\bar{s}_1}\right) \arrow[r, "\pi"] & \pi_1^{\mathrm{t}}\left(U_{\bar{s}_2}\right)
\end{tikzcd}$$

where π' is the composite of the canonical homomorphism $\pi_1^{\mathrm{t}}(\widetilde{U}_{\overline{s}_1}) \to \pi_1^{\mathrm{t}}(\overline{U}_{\overline{s}_1})$ and the specialization homomorphism. Since $\pi_1^{\mathrm{t}}(\widetilde{U}_{\overline{s}_1})$ (respectively$\pi_1^{\mathrm{t}}(\overline{U}_{\overline{s}_1})$) is an inertia group of $\pi_1^{\mathrm{t}}(U_{\overline{s}_1})$ at y_1 (respectivelyof $\pi_1^{\mathrm{t}}(U_{\overline{s}_2})$ at y_2), it suffices to prove that π' is surjective. This follows from the expression obtained in XIII.5.6.

Corollary XIII.2.11. *Let X be a connected proper smooth curve of genus g over a separably closed field k of characteristic $p \geq 0$. Let U be the open subset obtained by removing from X the n distinct closed points $a_1, \dots, a_n$. Then the tamely ramified fundamental group $\pi_1^{\mathrm{t}}(U)$* (XIII.2) *can be generated by $2g + n$ elements x_i, y_i, σ_j, with $1 \leq i \leq g$, $1 \leq j \leq n$, such that σ_j is a generator of an inertia group corresponding to a_j, and the following relation holds:*

$$(*) \qquad \prod_{1\leq i\leq g} (x_i y_i x_i^{-1} y_i^{-1}) \cdot \prod_{1\leq j\leq n} \sigma_j = 1.$$

For every finite group G of order prime to p, *generated by elements $\overline{x}_i, \overline{y}_i, \overline{\sigma}_j$ satisfying the relation $(*)$, there exists an étale covering of U with group G, corresponding to a homomorphism $\pi_1^{\mathrm{t}}(U) \to G$ which sends x_i, y_i, σ_j to $\overline{x}_i, \overline{y}_i, \overline{\sigma}_j$, respectively. In other words,* 393
if p' denotes the set of prime numbers distinct from p, then $\pi_1^{p'}(U)$ is the pro-p'-group generated by x_i, y_i, σ_j subject to the single relation $(*)$.

Proof. We may suppose that k is algebraically closed. Suppose first that k has characteristic zero. There then exists an algebraically closed subfield k' of k, of finite transcendence degree over $\mathbb{Q}$, such that X is obtained by extension of scalars from a proper smooth curve X' defined over k', and we may suppose that the points $a_1, \dots, a_n$ come from rational points $a'_1, \dots, a'_n$ of X'. Since k' has finite transcendence degree over $\mathbb{Q}$, there is an embedding of k' into the field of complex numbers $\mathbb{C}$; put $\widetilde{U} = U' \times_{k'} \mathbb{C}$. Let k'' be an algebraically closed extension of k' such that there are k'-morphisms from k and from $\mathbb{C}$ to k''. If $g' \colon U' \to k'$ is the structural morphism, and if $\mathcal{F}$ is a finite constant sheaf of groups on U'', it follows from XIII.2.9 that the specialization morphisms

$$(\mathrm{R}^1 g'_* \mathcal{F}')_{\mathbb{C}} \to (\mathrm{R}^1 g'_* \mathcal{F}')_{k''} \longleftarrow (\mathrm{R}^1 g'_* \mathcal{F}')_k$$

are isomorphisms. In terms of fundamental groups, this shows that there is an isomorphism, defined up to inner automorphism,

$$\pi_1(U) \to \pi_1(\widetilde{U}),$$

and it is clear that this isomorphism sends an inertia group relative to a point of $X' - U'$ to an inertia group relative to the same point. We may therefore suppose that $k = \mathbb{C}$. In this case, the Riemann existence theorem (XII XII.5.2) shows that the fundamental group $\pi_1(U)$ is precisely the completion, for the topology of subgroups of finite index, of the fundamental group of the analytic space associated with U. The latter can be computed by transcendental methods [**3**, ch. 7 §47]; it can be generated by $2g+n$ elements x_i, y_i, σ_j 394
such that σ_j is the image of a generator of the local fundamental group $\pi_1(D_j)$ of a small disk centered at a_j, i.e. a generator of an inertia group corresponding to the point a_j, and these elements satisfy the single relation $(*)$.

Now suppose that k has characteristic $p > 0$. There exists a complete discrete valuation ring A with residue field k and fraction field K of characteristic zero, and a connected scheme X_1, proper and smooth over $S = \mathrm{Spec}(A)$, such that $X_1 \times_S \mathrm{Spec}(k) \simeq X$ (III III.7.4). The points a_j then lift to sections s_j of X_1 over S. Let Y_{1j} be the reduced closed subscheme with underlying space $s_j(S)$, let Y_1 be the union of the Y_{1j}, put $U_1 = X_1 - Y_1$, and let $g_1 \colon U_1 \to S$ be the structural morphism. Let $\overline{K}$ be an algebraically closed extension of K, and put $\overline{U} = U_1 \times_S \overline{K}$. If C is a finite constant group, it follows from XIII.2.8 that the specialization morphism

$$(\mathrm{R}^1 g_{1*} C_{U_1})_k \to (\mathrm{R}^1 g_{1*} C_{U_1})_{\overline{K}}$$

is injective, and even bijective if C has order prime to p. In terms of fundamental groups, this means that the specialization homomorphism (XIII.1.9)

$$\pi\colon \pi_1(\overline{U}) \to \pi_1^{\mathrm{t}}(U)$$

is surjective, and that the specialization homomorphism

$$\pi_1^{p'}(\overline{U}) \to \pi_1^{p'}(U)$$

is bijective. Finally, if x_i, y_i, σ_j are generators of $\pi_1(\overline{U})$ such that σ_j is a generator of an inertia group corresponding to the point $b_j = Y_{1j}(\overline{K})$ of $\overline{X}$, then, by XIII.1.10, $\pi(\sigma_j)$ is a generator of an inertia group corresponding to a_j. This completes the proof.

Remark. **[**Added in 2003 (MR): Let k be an algebraically closed field of characteristic $p > 0$, let X be a connected proper smooth algebraic curve over k of genus g, and let U be an affine open subset of X, the complement of $r \geq 1$ rational points of X. We have the fundamental group $\pi_1(U)$, its quotient $\pi_1^{\mathrm{t}}(U)$, which classifies the finite étale coverings of U tamely ramified at the points of $X - U$, and the quotient $\pi_1^{p'}(U)$ of $\pi_1^{\mathrm{t}}(U)$, which classifies the Galois étale coverings of U whose Galois group has order prime to p.

In (Coverings of algebraic curves, Amer. J. Math. **79** (1957), p. 825–856), S. Abhyankar formulated a number of conjectures on the structure of the finite groups that occur as Galois groups of finite étale coverings of U.

The conjectures concerning Galois groups of connected étale coverings of U whose order is prime to p are proved and made precise in Corollary XIII.2.11. Before turning to the finite quotients of $\pi_1(U)$, let us first give some indications about the "size"of this group. Let $X = \mathrm{Spec}(()A)$. We know that $\mathrm{Hom}_{\mathrm{cont}}(\pi_1(U), \mathbb{Z}/p\mathbb{Z})$ is described by Artin–Schreier theory (Corollary. XI XI.6.9). This group is isomorphic to $A/\wp(A)$, where $\wp$ is the map from A to A that sends a to $a^p - a$. Suppose first that U is the affine line $\mathbf{A}^1$, with ring $k[T]$. Let E be the set of elements of $k[T]$ of the form $\sum_i a_i T^i$, with $a_i = 0$ whenever p divides i. It is immediate that the composite map $E \to k[T] \to k[T]/\wp k[T]$ is bijective. Hence the coefficients a_i, $(i, p) = 1$, behave like coordinates on a space parametrizing the cyclic coverings of degree p of the affine line. In particular, it follows that $\pi_1(\mathbf{A}^1)$ is not topologically finitely generated and that, if one makes an extension $k \to k'$ of algebraically closed fields, the natural homomorphism $\pi_1(\mathbf{A}^1 \times_k k') \to \pi_1(\mathbf{A}^1)$, which is surjective, is not bijective, unlike in the proper case. In general, the curve U can be realized as a finite scheme over the affine line, and we conclude that the same phenomena occur for $\pi_1(U)$ (and indeed, more generally, for $\pi_1(V)$ for every connected affine k-scheme V of finite type and dimension ≥ 1).

With this said, if G is a finite group, let $G^{(p')}$ denote the largest quotient group of G whose order is prime to p. For a finite group G to be a topological quotient of $\pi_1(U)$, it is necessary that $G^{(p')}$ be a topological quotient of $\pi_1^{p'}(U)$, a condition that can in principle be decided using Corollary XIII.2.11. In the article cited above, Abhyankar conjectured that this necessary condition is also sufficient. This conjecture was proved by M. Raynaud in the case of the affine line and by D. Harbater in the general case (Invent. Math. **116** (1994), p. 425–462 and **117**, p. 1–25). For example, for the affine line $\pi_1^{p'}(\mathbf{A}^1) = 1$, and it follows that a finite group G is the Galois group of a connected étale covering of $\mathbf{A}^1$ if and only if $G^{(p')} = 1$, that is, if and only if G is generated by its Sylow p-subgroups. Thus every finite simple group whose order is a multiple of p has the required property. **]**

XIII.3. Cohomological properness and generic local acyclicity 395

Theorem XIII.3.1. *Let S be an irreducible scheme with generic point s, let X and Y be two S-schemes of finite presentation, and let $f\colon X \to Y$ be an S-morphism. For every S-scheme S', denote by Y', X', etc. the inverse images of Y, X, etc. under the morphism $S' \to S$. The following properties hold:*

1) a) *One can find a nonempty open subset S' of S such that, for every finite constant sheaf of sets F' on X', f'_*F' is constructible, and (F', f') is cohomologically proper relative to S' in dimension ≤ 0.*

b) *Let F be a constructible sheaf of sets on X. Then one can find a nonempty open subset S' of S (depending on F) such that f'_*F' is constructible and (F', f') is cohomologically proper relative to S' in dimension ≤ 0.*

2) *Suppose that the schemes of finite type of dimension $\leq \dim X_s$ over an algebraic closure $\overline{k}$ of $\kappa(s)$ are strongly desingularizable* (SGA 5 I 3.1.5). *Then the following additional properties hold:*

a) *One can find a nonempty open subset S' of S such that, for every finite constant sheaf of groups F' on X' whose order is prime to the residual characteristics of S, if Φ' is the stack of torsors under F', then $f'_*\Phi'$ is 1-constructible, and (F', f') is cohomologically proper relative to S' in dimension ≤ 1.*

b) *Let $\mathbb{L}$ be the set of prime numbers distinct from the residual characteristics of S, and let Φ be an ind-$\mathbb{L}$-stack that is 1-constructible on X* (XIII.0) *such that, for every scheme X_1 étale over X and every pair of objects x, x_1 of Φ_{X_1}, the sheaf $\mathbf{Hom}_{X_1}(x, x_1)$ is constructible. Suppose moreover that S is locally noetherian. Then one can find a nonempty open subset S' of S such that $f'_*\Phi'$ is 1-constructible, such that, for every pair of objects y, y_1 of a fiber $(f'_*\Phi')_{Y_1}$, $\mathbf{Hom}_{Y_1}(y, y_1)$ is constructible, and such that (Φ', f') is cohomologically proper* 396
relative to S' in dimension ≤ 1.

Proof. We may suppose that S is affine; by SGA 4 VIII 1.1, we may suppose that S is integral; finally, by passage to the limit, we may suppose that S is the spectrum of an algebra of finite type over $\mathbb{Z}$; in particular, S is then noetherian. Since the question is local on Y, we may suppose that Y is affine. Moreover, in order to prove the theorem, it suffices to prove it after a finite extension $S' \to S$, where S' is an integral scheme and where $S' \to S$ is a composite of étale morphisms and finite surjective radicial morphisms.

1) Case of constant sheaves of sets. 1) 1. Reduction to the case where X is normal over S. Let $X_{1\overline{s}}$ be the normalization of $(X_{\overline{s}})_{\mathrm{red}}$; after restricting S to a nonempty open subset and making a radicial extension of S, we may suppose that $X_{1\overline{s}}$ comes from a scheme X_1 normal over S, and that the morphism $X_{1\overline{s}} \to X_{\overline{s}}$ comes from a finite surjective morphism $p\colon X_1 \to X$ (EGA IV 8.8.2 and 9.6.1). Suppose that the theorem has been proved for fp. After restricting S to an open subset, we may suppose that, for every constant sheaf of sets F on X, (p^*F, fp) is cohomologically proper relative to S in dimension ≤ 0 and that $f_*p_*(p^*F)$ is constructible. By XIII.1.8, (p_*p^*F, f) is then cohomologically proper relative to S in dimension ≤ 0. The morphism

$$F \to p_*p^*F = G$$

is a monomorphism. Since f_*F is a subsheaf of f_*G, it already follows that f_*F is constructible (SGA 4 IX 2.9 (ii)) and that (F, f) is cohomologically proper relative to S in dimension ≤ -1.

Let $X_2 = X_1 \times_X X_1$, and let $q\colon X_2 \to X$ be the canonical morphism. By XIII.1.10 1), there is an exact sequence 397

$$F \longrightarrow G \rightrightarrows q_*q^*F \ .$$

By what we have just proved, applied to fq in place of f, after restricting S to a nonempty open subset, we may suppose that, for every constant sheaf of sets F on X, (q^*F, fq) is cohomologically proper relative to S in dimension ≤ -1, and hence that (q_*q^*F, f) is cohomologically proper relative to S in dimension ≤ -1. It then follows from XIII.1.12 1) that (F, f) is cohomologically proper relative to S in dimension ≤ 0.

1) 2. Reduction to the case where X is normal affine over S.

Let U_s be an affine open subset of X_s dense in X_s. After restricting S to a nonempty open subset, we may suppose that $U_s \to X_s$ lifts to an open immersion $i\colon U \to X$, schematically dominant relative to S (EGA IV 8.9.1). Since the morphism $X \to S$ is normal, by SGA 2 XIV 1.18 we have

$$\operatorname{prof} \text{ét}_{S-U}(X) \geq 2;$$

consequently, for every constant sheaf F on X, the canonical morphism

$$F \to i_* i^* F$$

is an isomorphism. It follows that, if the theorem is assumed proved for fi and i, then, after restricting S to a nonempty open subset, (i^*F, i) and (i^*F, fi) are cohomologically proper relative to S in dimension ≤ 0. The same is therefore true of (F, f) (XIII.1.5 2)). Moreover, $f_*F = (fi)_*(i^*F)$ is constructible, which completes the reduction.

1) 3. End of the proof.

We may suppose that S is normal (EGA IV 7.8.3). One can find a compactification of X_s: 398

$$\begin{tikzcd} X_s \arrow[r, "j_s"] \arrow[d, "f_s"'] & P_s \arrow[dl, "g_s"] \\ Y_s & \end{tikzcd}$$

where j_s is a dominant open immersion and g_s is a proper morphism; after making a radicial extension of $\kappa(s)$ and replacing P_s by its normalization, which does not change X_s, we may suppose that P_s is geometrically normal. After restricting S to a nonempty open subset and making a surjective radicial extension, we may suppose that the diagram above comes from a diagram

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[d, "f"'] & P \arrow[dl, "g"] \\ Y & \end{tikzcd}$$

where P is a scheme normal over S, j is a schematically dominant open immersion relative to S, and g is a proper morphism (EGA IV 6.9.1, 9.9.4 and 9.6.1). For every finite constant sheaf of sets F on X with value C, j_*F is the constant sheaf with value C (SGA 4 2.14.1), and the same is true after every base change $S' \to S$; it follows that (F, j) is cohomologically proper relative to S in dimension ≤ 0. The same is therefore true of (F, f), since g is proper (XIII.1.7). Since g is proper, $f_*F = g_*C_P$ is constructible, which completes the proof of 1) a).

2) Case of a constructible sheaf of sets. Let F be a constructible sheaf of sets on X. By SGA 4 IX 2.14 (ii), one can find a finite family of morphisms $p_i\colon Z_i \to X$ and, on each Z_i, a finite constant sheaf of sets C_i, such that there is a monomorphism 399

$$j\colon F \to \prod_i p_{i*} C_i = G.$$

By 1) a), after restricting S to a nonempty open subset, we may suppose that the (C_i, fp_i) are cohomologically proper relative to S in dimension ≤ 0, and that the $f_* p_{i*} C_i$ are constructible. We already conclude that (G, f) is cohomologically proper relative to S in dimension ≤ 0 (XIII.1.8), hence that (F, f) is cohomologically proper relative to S in dimension ≤ -1, and that f_*F is constructible. Let K be the amalgamated sum $K = G \coprod_F G$; since F and G are constructible, so is K. We therefore conclude from the preceding that, after restricting S to a nonempty open subset, we may suppose that (K, f) is cohomologically proper relative to S in dimension ≤ -1. It then follows from XIII.1.12 1) that (F, f) is cohomologically proper relative to S in dimension ≤ 0.

3) Case of constant sheaves of groups. If F is a constant sheaf of groups on X, we denote by Φ the stack of torsors under F.

3) 1. Let us first show that, after restricting S to a nonempty open subset, for every constant sheaf of groups F on X whose order is prime to the residue characteristics of S, (F, f) is cohomologically proper relative to S in dimension ≤ 0, and $f_*\Phi$ is constructible.

For this, we reduce to the case where X is smooth over S. After making a finite extension of $\kappa(s)$, which is permissible because it can be regarded as the composite of an étale extension and a radicial extension, one can find a proper surjective morphism $p_s\colon X_{1s} \to X_s$, where X_{1s} is a scheme smooth over S of the same dimension as X_s, and, after restricting S to a nonempty open subset, we may suppose that p_s comes from a proper surjective morphism $p\colon X_1 \to X$, where X_1 is a scheme smooth over S (EGA IV 400
9.6.1 and 12.1.6). Let $X_2 = X_1 \times_X X_1$, and let $q\colon X_2 \to X$ be the canonical morphism. There is an exact diagram of stacks on X

$$\Phi \longrightarrow p_*p^*\Phi \rightrightarrows q_*q^*\Phi$$

(XIII.1.10 2)). Assuming that the theorem has been proved in the smooth case, we first see that we may suppose $f_*p_*p^*\Phi$ constructible; hence the same is true of $f_*\Phi$ (XIII.3.2 below). Moreover, by XIII.1.5 2), we may suppose $(p_*p^*\Phi, f)$ cohomologically proper relative to S in dimension ≤ 0; it follows that (Φ, f) is cohomologically proper relative to S in dimension ≤ -1. We may therefore suppose that $(q_*q^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ -1, and this implies that (Φ, f) is cohomologically proper relative to S in dimension ≤ 0 (XIII.1.11 1)).

We then reduce, as in 1) 2, to the case where X is smooth and affine over S. Let

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[d, "f"'] & P \arrow[dl, "q"] \\ Y & \end{tikzcd}$$

be a compactification of X, where i is a dominant open immersion and q is a proper morphism. Since $\dim P_s = \dim X_s$, we may apply the resolution of singularities hypothesis to $P_{\overline{s}}$. After making an étale extension and a radicial extension of S, one can find a proper morphism $r\colon Z \to P$, where Z is smooth over S, $r^{-1}(X) \simeq X$, and $r^{-1}(X)$ is the complement in Z of a divisor with normal crossings relative to S. Every torsor under F is then tamely ramified on Z relative to S (XIII.2.3 b)). It therefore follows from XIII.2.7 that (F, f) is cohomologically proper relative to S in dimension ≤ 0, which proves our assertion.

3) 2. Reduction to the case where X is smooth over S. 401

After making a finite extension of $\kappa(s)$, one can find a proper surjective morphism $p_s\colon X_{1s} \to X$, where X_{1s} is a scheme smooth over s, and we may suppose that p_s comes from a proper surjective morphism $p\colon X_1 \to X$, where X_1 is smooth over S. Suppose that the theorem has been proved for fp, and let us prove it for f. Let F be a finite constant sheaf of groups on X, of order prime to the residue characteristics of S, and let Φ be the stack of torsors under F. Let $X_2 = X_1 \times_X X_1$, $X_3 = X_1 \times_X X_1 \times_X X_1$, and let $q\colon X_2 \to X$, $r\colon X_3 \to X$ be the canonical morphisms. By XIII.1.10 2), there is an exact diagram of stacks

$$\Phi \longrightarrow p_*p^*\Phi \rightrightarrows q_*q^*\Phi \mathrel{\substack{\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow\\[-0.6ex]\textstyle\rightarrow}} r_*r^*\Phi\ .$$

To prove that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1, it is enough to show that the same is true of $(p_*p^*\Phi, f)$, that $(q_*q^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0, and that $(r_*r^*\Phi, f)$ is cohomologically proper relative to S in dimension ≤ -1 (XIII.1.11 2)). By 3) 1 above, we may suppose that, for every finite constant sheaf of groups F, $(q^*\Phi, fq)$, $(q^*\Phi, q)$, $(r^*\Phi, fr)$, $(r^*\Phi, r)$ are cohomologically proper relative to S in dimension ≤ 0. It then follows from XIII.1.5 2) that $(q_*q^*\Phi, f)$ and $(r_*r^*\Phi, f)$ are cohomologically proper relative to S in dimension ≤ 0. Since the theorem is assumed proved in the smooth case, $(p^*\Phi, fp)$ and $(p^*\Phi, p)$, and hence also $(p_*p^*\Phi, f)$ (XIII.1.5 2)), are cohomologically proper relative to S in dimension ≤ 1. This indeed shows that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1.

Moreover, $f_*p_*p^*\Phi$ is 1-constructible by hypothesis; by XIII.3.1, we may suppose that $f_*q_*q^*\Phi$ is constructible; it therefore follows from XIII.3.2 below that $f_*\Phi$ is 1-constructible.

3) 3. Reduction to the case where X is smooth and affine over S. 402

By 3) 2, we may suppose that X is smooth over S. Let $(U_i)_{i\in I}$ be a finite covering of X by affine open subsets, let X_1 be the disjoint union of the U_i, and let $p\colon X_1 \to X$ be the canonical morphism. Since p is an effective descent morphism for the category of étale sheaves of finite type on variable schemes, we see as in 3) 2 that, if the theorem is assumed proved for the U_i, i.e. for X_1, it is also true for X.

3) 4. The case where X is smooth and affine over S.

As in 3)1, after restricting S to a nonempty open subset and making a surjective étale extension and a surjective radicial extension of S, one can find a commutative diagram

$$\begin{tikzcd} X \arrow[r, "i"] \arrow[d, "f"'] & P \arrow[dl, "q"] \\ Y & \end{tikzcd}$$

where P is a scheme smooth over S, X is the complement in P of a divisor with normal crossings relative to S, and g is a proper morphism. If F is a constant sheaf of order prime to the residue characteristics of S, every torsor under F is tamely ramified on P relative to S (XIII.2.3 b)). The fact that (F, f) is cohomologically proper relative to S in dimension ≤ 1 and that $f_*\Phi$ is constructible then follows from XIII.2.7.

4) Proof of 2) b). 4) 1. The case where Φ is a gerbe.

One can find a surjective étale morphism of finite type $p\colon X_1 \to X$ such that $p^*\Phi$ is a trivial gerbe. By descent, as in 3) 2, we see that it is enough to prove the theorem for X_1, $X_1 \times_X X_1$, and $X_1 \times_X X_1 \times_X X_1$. We are therefore reduced to the case where Φ is the 403
gerbe of torsors under a constructible sheaf of groups F whose fibers have order prime to the residue characteristics of S.

By SGA 4 IX 2.14, one can find a finite family of finite morphisms $p_i\colon Z_i \to X$ and, for each i, a finite constant sheaf of groups C_i, of order prime to the residue characteristics of S, such that there is a monomorphism

$$j\colon F \to \prod_i p_{i*}C_i = G.$$

Let Φ_i be the stack of torsors under C_i, and let Ψ be the stack of torsors under G. It follows from 2) a) that, after restricting S to a nonempty open subset, we may suppose that the (C_i, fp_i) are cohomologically proper relative to S in dimension ≤ 1, and that the stacks $f_*p_{i*}\Phi_i$ are 1-constructible. It then follows from XIII.1.8 that the $(p_{i*}C_i, f)$ are cohomologically proper relative to S in dimension ≤ 1; hence the same is true of (G, f). Moreover, since $p_{i*}\Phi_i$ is equivalent to the stack of torsors under the group $p_{i*}C_i$ (SGA 4 VIII 5.8), we see that $f_*\Psi$ is 1-constructible.

Since $\mathrm{R}^1 f_*G$ is constructible, one can find a sheaf representable by an étale Y-scheme of finite type T and an epimorphism

$$a\colon T \to \mathrm{R}^1 f_*G$$

(SGA 4 IX 2.7); moreover, we may suppose that the image of the identity section of $T(T)$ is defined by a torsor Q on $X \times_Y T = X_T$, with group $G|X_T$. Let $f_T\colon X_T \to T$ be the canonical morphism, and put $F_T = F|X_T$, etc. By 1) b), after restricting S to a nonempty open subset, we may suppose that $(Q/F_T, f_T)$ is cohomologically proper relative to S in dimension ≤ 0 and that $f_{T*}(Q/F_T)$ is constructible. It then follows from XIII.3.3 that $f_*\Psi$ is constructible.

Let us show that (F, f) is cohomologically proper relative to S in dimension ≤ 1. By XIII.1.12 2), it is enough to prove that, for every scheme Y_1 étale over Y and every torsor Q_1 on $X_1 = X \times_Y Y_1$, if $f_1\colon X_1 \to Y_1$ is the canonical morphism, then $(Q_1/F_1, f_1)$ is cohomologically proper relative to S in dimension ≤ 0. But, by definition of T, Q_1 is, locally for the étale topology on Y_1, the inverse image of Q, which proves our reduction. 404

4) 2. The general case.

Using Lemma XIII.6.2, 4) 1, and 1) a), we see that, after restricting S to a nonempty open subset, we may suppose that $S(f_*\Phi)$ is constructible and $(S\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0. We can then find a sheaf representable by an étale Y-scheme of finite type T and an epimorphism

$$a\colon T \to S(f_*\Phi).$$

We again use the notation of 4) 1, and in addition put $Z = T \times_Y T$, $X_Z = X \times_Y Z$, and denote by $f_Z\colon X_Z \to Z$ the canonical morphism. We may suppose that T is chosen so that the image q of the identity section of $T(T)$ under a is defined by an object p of $(f_*\Phi)_T = \Phi_{X_T}$. Let p_1 and p_2 (respectively q_1 and q_2) be the inverse images of p (respectively q) under the two projections from Z to T. After restricting S to a nonempty open subset, we may suppose that $(\mathbf{Aut}_{X_T}(p), f_T)$ is cohomologically proper relative to S in dimension ≤ 1, that $(\mathbf{Hom}_{X_Z}(p_1, p_2), f_Z)$ is cohomologically proper relative to S in dimension ≤ 0, and that $f_{T*}(\mathbf{Aut}_{X_T}(p))$, $f_{Z*}(\mathbf{Hom}_{X_Z}(p_1, p_2))$ are constructible (a) 1) and 4) 1)).

It follows first that, for every scheme Y_1 étale over Y and every pair of objects y, y_1 of $(f_*\Phi)_{Y_1}$, the sheaf $\mathbf{Aut}_{Y_1}(y)$ (respectively$\mathbf{Hom}_{Y_1}(y, y_1)$) is constructible: locally for the étale topology of Y_1, such a sheaf is the inverse image of $f_{T*}(\mathbf{Aut}_{X_T}(p))$ (respectivelyof $f_{Z*}(\mathbf{Hom}_{X_Z}(p_1, p_2))$).

It remains to prove that (Φ, f) is cohomologically proper relative to S in dimension ≤ 1. For this, it is enough to show that, for every S-scheme S' and every geometric point $\overline{y}'$ of Y', if $\overline{Y}$ denotes the strict localization of Y at y', and $\overline{X} = X \times_Y \overline{Y}$, etc., then the canonical functor 405

$$\overline{\varphi}\colon \Phi(\overline{X}) \to \Phi'(\overline{X'})$$

is an equivalence of categories.

Let us show that $\overline{\varphi}$ is fully faithful. Let $x, y \in \Phi(\overline{X})$, and let x', y' be their images in $\Phi'(\overline{X'})$. We must show that the canonical morphism

$$(*) \qquad \mathrm{Hom}_{\overline{X}}(x, y) \to \mathrm{Hom}_{\overline{X'}}(x', y')$$

is bijective. By the definition of T, there are two morphisms from $\overline{Y}$ to T such that x and y are the inverse images of p under these two morphisms. This amounts to giving a morphism $\overline{Y} \to Z$, and hence a morphism $h\colon \overline{X} \to X_Z$ such that

$$h^*(p_1) = x \qquad h^*(p_2) = y.$$

Consequently, there is a canonical isomorphism

$$\mathbf{Hom}_{\overline{X}}(x, y) = h^*(\mathbf{Hom}_{X_Z}(p_1, p_2)).$$

But, since $(\mathbf{Hom}_{X_Z}(p_1, p_2), f_Z)$ is cohomologically proper relative to S in dimension ≤ 0, the same is true of $(\mathbf{Hom}_{\overline{X}}(x, y), \overline{f})$, which proves that the morphism $(*)$ is bijective.

Let us show that $\overline{\varphi}$ is essentially surjective. Let $x' \in \Phi'(\overline{X'})$. Since $(S\Phi, f)$ is cohomologically proper relative to S in dimension ≤ 0, the canonical morphism

$$\mathrm{H}^0(\overline{X}, \overline{S\Phi}) \to \mathrm{H}^0(\overline{X'}, \overline{S\Phi}')$$

is bijective. Let G' be the maximal subgerbe of Φ' generated by x'. There is then a maximal subgerbe G of $\overline{\Phi}$ whose inverse image on $\overline{X'}$ is G'. By 4) 1, $(G, \overline{f})$ is cohomologically proper relative to S in dimension ≤ 1. Consequently, the canonical functor

$$G(\overline{X}) \to G'(\overline{X'})$$

is an equivalence of categories. This proves the existence of an object x of $\Phi(\overline{X})$ whose 406
image in $\Phi'(X')$ is isomorphic to x', and completes the proof of the theorem.

Lemma XIII.3.2. *Let S be a locally Noetherian scheme, and let*

$$\Phi \xrightarrow{p} \Phi_1 \overset{p_1, p_2}{\rightrightarrows} \Phi_2$$

be an exact diagram of stacks on S (XIII.1.9). *If Φ_1 is constructible, then so is Φ. If, for every scheme S' étale over S and every pair of objects x_1, y_1 of $(\Phi_1)_{S'}$, the sheaf $\mathbf{Hom}_{S'}(x_1, y_1)$ is constructible, then for every pair of objects x, y of $\Phi_{S'}$, the same is true of $\mathbf{Hom}_{S'}(x, y)$. Suppose that Φ_1 is 1-constructible (XIII.0) and that Φ_2 is constructible; then Φ is 1-constructible.*

For every scheme S' étale over S and every object x of $\Phi_{S'}$, there is a monomorphism

$$\mathbf{Aut}_{S'}(x) \to \mathbf{Aut}_{S'}(p(x)).$$

It follows from SGA 4 IX 2.9 that if Φ_1 is constructible, then so is Φ, and the second assertion of the lemma is proved in the same way.

Now suppose that Φ_1 is 1-constructible and that Φ_2 is constructible. The morphism p induces a morphism on the sheaves of maximal subgerbes,

$$\varphi\colon S\Phi \to S\Phi_1.$$

Let G be the image of $S\Phi$ under φ. By SGA 4 IX 2.9, G is a constructible sheaf. We can therefore find a sheaf represented by an étale S-scheme T of finite type, and an epimorphism

$$a\colon T \to G$$

(SGA 4 IX 2.7). Moreover, T can be chosen so that the image y of the identity section 407
of $T(T)$ under a is defined by an object x_1 of $(\Phi_1)_T$ of the form $x_1 = p(x)$, where $x \in \Phi_T$.

It is enough to show that, for every point s of S, there is a nonempty open subset U of $\overline{\{s\}}$ such that $S\Phi|U$ is constructible locally constant. Let $s \in S$, let $\overline{s}$ be a geometric point over s, and let $\overline{y}_1, \dots, \overline{y}_n$ be the elements of $G_{\overline{s}}$. By the definition of T, there are morphisms $h_i\colon \overline{s} \to T$ such that $h_i^*(y) = \overline{y}_i$. Let S' be the fiber product over S

of n schemes isomorphic to T, let $h\colon \overline{s} \to S'$ be the fiber product of the h_i, and let y_i (respectivelyx_i) be the inverse image of y (respectively$x$) under the i-th projection from S' to T. Let F_i be the subsheaf of $S\Phi|S'$ that is the inverse image of y_i, and let us show that the F_i are constructible. The sheaf F_i is a quotient of the sheaf F'_i such that, for every scheme S'' étale over S',

$$F'_i(S'') = \{\text{classes modulo isomorphism of objects } z \text{ of } \Phi_{S''} \text{ equipped} \\ \text{with an isomorphism } i\colon p(z) \simeq p(x_i|S'')\}.$$

It is enough to show that the F'_i are constructible. Put $z_i = p_1p(x_i)$, $z'_i = p_2p(x_i)$. There is a monomorphism

$$\Psi_i\colon F'_i \to \mathbf{Isom}_{S'}(z_i, z'_i),$$

obtained by associating with every scheme S'' étale over S' and every object z of $\Phi_{S''}$ equipped with an isomorphism $i\colon p(z) \simeq p(x_i|S'')$, the isomorphism from z_i to z'_i defined by requiring the diagram

$$\begin{tikzcd} p_1p(z) \arrow[r, "p_1(i)"] \arrow[d, "j"'] & z_i \arrow[d] \\ p_2p(z) \arrow[r, "p_2(i)"] & z'_i \end{tikzcd}$$

to be commutative (j is the canonical morphism associated with the exact diagram). 408

The morphism Ψ_i is injective: to say that two objects z, z', equipped with isomorphisms $i\colon p(z) \xrightarrow{\sim} p(x_i|S'')$ and $i'\colon p(z') \xrightarrow{\sim} p(x_i|S'')$, define the same element of $\mathbf{Isom}_{S''}(z_i, z'_i)$ amounts to saying that $p_1(i'^{-1}i) = p_2(i'^{-1}i)$, i.e. that $i'^{-1}i$ comes from an isomorphism $z \to z'$. Since F'_i is a subsheaf of $\mathbf{Isom}_{S'}(z_i, z'_i)$, it is constructible.

Since there is a nonempty open subset U of $\overline{\{s\}}$ such that $y_1|U, \dots, y_n|U$ generate G, it follows from Lemma XIII.6.3 below that $S\Phi|U$ is constructible. Thus, after restricting U if necessary, $S\Phi|U$ is constructible locally constant.

Lemma XIII.3.3. *Let S be a locally Noetherian scheme, let $f\colon X \to S$ be a morphism, let $F \to G$ be a monomorphism of sheaves of groups on X, let Ψ (respectivelyΨ_1) be the stack of torsors under F (respectivelyunder G), and put $\Phi = f_*\Psi$, $\Phi_1 = f_*\Psi_1$. Suppose that we are given a sheaf on S represented by an étale S-scheme T of finite type and a surjective morphism*

$$a\colon T \to S\Phi_1 \simeq \mathrm{R}^1 f_* G,$$

such that there is a torsor Q on $X_T = X \times_S T$ with group $G|X_T$, defining in $\mathrm{R}^1 f_*G(T)$ the image under a of the identity section of $T(T)$. Let $f_T\colon X_T \to T$ be the canonical morphism, and put $F_T = F|X_T$. Suppose that Φ_1 is 1-constructible and that $f_{T*}(Q/F_T)$ is constructible. Then Φ is 1-constructible.

Indeed, it is enough to repeat the proof of XIII.3.2, with the existence of the morphisms Ψ_i replaced by the isomorphisms

$$F'_i \xrightarrow{\sim} f_*(Q/F_T)|S'.$$

Remark XIII.3.4. *Suppose that $k = \kappa(s)$ has characteristic zero. Then schemes of* 409
finite type over $\overline{k}$, of dimension $\leq \dim X$, admit strong resolution of singularities, and the proof of XIII.3.1 gives the following results:

a) *There is a nonempty open subset S_1 of S such that, for every scheme S' over S_1 whose maximal points have characteristic zero and every constructible locally constant sheaf of sets F on $X' = X \times_S S'$, (F, f') is cohomologically proper relative to S' in dimension ≤ 0.*

b) *If all the residue characteristics of S are zero, there is a nonempty open subset S_1 of S such that, for every scheme S' over S_1 and every constructible locally constant sheaf of groups F on X', (F, f') is cohomologically proper relative to S' in dimension ≤ 1.*

Indeed, it is enough to repeat the proof of XIII.3.1.2) a). Proposition XIII.2.7, used in 3) 4, applies to a locally constant sheaf F, because, with the notation of 3) 4, every torsor under F is tamely ramified on P relative to S, since all the residue characteristics of S are zero.

Corollary XIII.3.5. *Let k be a field of characteristic $p \geq 0$, let p' be the set of prime numbers distinct from p, and let $f\colon X \to k$ be a coherent morphism.*

1) *For every sheaf of sets F, (F, f) is cohomologically proper in dimension ≤ 0.*
2) *Suppose that one of the following two conditions holds:*
   a) *f is of finite type, and schemes of finite type of dimension $\leq \dim X$ over an algebraic closure of k admit strong resolution of singularities.*
   b) *Schemes of finite type over an algebraic closure of k admit strong resolution of singularities.* 410

   Then, for every sheaf of ind-p'-groups F, (F, f) is cohomologically proper in dimension ≤ 1.

Let F be a sheaf of sets (respectivelyof ind-p'-groups). By SGA 4 IX 2.7.2, F can be written as a filtered inductive limit

$$F = \varinjlim F_i,$$

where the F_i are constructible sheaves of sets (respectivelyof ind-p'-groups). Since f is coherent, f_* (respectively$\mathrm{R}^1 f_*$) commutes with inductive limits (SGA 4 VII 3.3). If the (F_i, f) are known to be cohomologically proper in dimension ≤ 0 (respectivelyif every sheaf of sets is cohomologically proper in dimension ≤ 0 and the (F_i, f) are cohomologically proper in dimension ≤ 1), then the same is true of (F, f). We may therefore suppose that F is constructible.

If f is assumed to be of finite type (respectivelyto satisfy a)), the proposition follows from XIII.3.1 1) b) (respectivelyfrom XIII.3.1 2) b)). We now prove XIII.3.5 without assuming that f is of finite type. For every scheme S' over k and every geometric point $\overline{s}$ of S', denote by $\overline{k}$ (respectively$\overline{S'}$) the strict localization of k at $\overline{s}$ (respectivelyof S' at $\overline{s}$), let $\overline{X}$ be the inverse image of X on $\overline{k}$, and consider the cartesian square

$$\begin{tikzcd} \overline{X} \arrow[d, "\overline{f}"'] & \overline{X'} \arrow[l, "g"'] \arrow[d, "\overline{f}'"] \\ \overline{k} & \overline{S'} \arrow[l] \end{tikzcd}$$

It is enough to prove that, for every S' and every $\overline{s}$, the canonical morphism

$$\mathrm{H}^0(\overline{X}, \overline{F}) \to \mathrm{H}^0(\overline{X'}, \overline{F'}) \qquad (\text{respectively} \mathrm{H}^1(\overline{X}, \overline{F}) \to \mathrm{H}^1(\overline{X'}, \overline{F'}))$$

is an isomorphism. It is enough to show the relations 411

$$(*) \qquad \overline{F} \simeq g_* g^* \overline{F} \qquad (\text{respectively} \mathrm{R}^1 g_*(g^* \overline{F}) = 0).$$

In this form, the question is local on X for the étale topology. We may therefore suppose that X is affine; by passage to the limit, we may suppose that X is of finite type over k. It is then known that (F, f) is cohomologically proper in dimension ≤ 0 (respectively≤ 1), and that the same is true when X is replaced by a scheme étale and of finite type over X. This proves $(*)$.

Theorem XIII.3.6. *Let S be an irreducible scheme with generic point s, and let $f\colon X \to S$ be a morphism of finite presentation. Suppose that schemes of finite type of* *dimension* $\leqslant \dim X_s$ *over an algebraic closure* $\bar{k}$ *of* k *admit resolution of singularities* (EGA IV 7.9.1). *If* $\mathbb{L}$ *denotes the set of prime numbers distinct from the residue characteristics of* S*, then there is a nonempty open subset* S_1 *of* S *such that the morphism* $f|S_1$ *is universally locally* 1*-aspherical for* $\mathbb{L}$*.*

We may suppose that S is integral and X is reduced (SGA 4 VIII 1.1). By passage to the limit, we may suppose that S is Noetherian. Moreover, to prove the theorem, it is enough to prove it after a finite extension $S_1 \to S$, where S_1 is an integral scheme and $S_1 \to S$ is a composite of étale extensions and surjective radicial extensions.

First let us show that, after restricting S to a nonempty open subset if necessary, f is universally locally 0-acyclic. After a radicial extension of $\kappa(s)$, we may suppose that $(X_s)_{\mathrm{red}} \to s$ is separable. Thus, after restricting S to a nonempty open subset and making a surjective radicial extension of S, we may suppose that f is flat with geometrically separable fibers (EGA IV 12.1.1), which implies that f is universally 0-acyclic (SGA 4 XV 4.1).

Let us show that, after restricting S to a nonempty open subset if necessary, f is
universally locally 1-aspherical for $\mathbb{L}$. After making a finite extension of $\kappa(s)$, which is 412
allowed because it may be regarded as a composite of an étale extension and a radicial extension, we can find a proper surjective morphism $p_s\colon Y_s \to X_s$, where Y_s is a scheme smooth over s and has the same dimension as X_s. We may suppose that p_s comes from a proper surjective morphism $p\colon Y \to X$, where Y is smooth over S (EGA IV 9.6.1 and 12.1.6). It is enough to show that, after restricting S to a nonempty open subset if necessary, for every diagram with cartesian squares

$$\begin{tikzcd}
S'' \arrow[d, "i"'] & X'' \arrow[l, "f''"'] \arrow[d, "j"] \\
S' \arrow[d] & X' \arrow[l, "f'"'] \arrow[d] \\
S & X, \arrow[l, "f"']
\end{tikzcd}$$

where i is étale of finite presentation, and every sheaf of ind-$\mathbb{L}$-groups F on S'', if Φ is the stack of torsors under F, then the canonical morphism

$$f'^{*} i_{*}\Phi \to j_{*} f''^{*}\Phi$$

is an equivalence. Put $Z = Y \times_X Y$ and $T = Y \times_X Y \times_X Y$. There is a natural commutative diagram

$$\begin{tikzcd}
S'' \arrow[d, "i"'] & X'' \arrow[l, "f''"'] \arrow[d, "j"'] & Y'' \arrow[l, "p''"'] \arrow[d] & Z'' \arrow[l, shift left] \arrow[l, shift right] \arrow[d] & T'' \arrow[l, shift left=2] \arrow[l] \arrow[l, shift right=2] \arrow[d] \\
S' \arrow[d] & X' \arrow[l, "f'"'] \arrow[d] & Y' \arrow[l, "p'"'] \arrow[d] & Z' \arrow[l, shift left] \arrow[l, shift right] \arrow[d] & T' \arrow[l, shift left=2] \arrow[l] \arrow[l, shift right=2] \arrow[d] \\
S & X \arrow[l, "f"'] & Y \arrow[l, "p"'] & Z' \arrow[l, shift left] \arrow[l, shift right] & T' \arrow[l, shift left=2] \arrow[l] \arrow[l, shift right=2]
\end{tikzcd}$$

Let $q\colon Z \to X$ and $r\colon T \to X$ be the canonical morphisms, with q', r', q'', r'' having the 413
evident meanings. By XIII.1.10 2), there is the following essentially commutative diagram, whose rows are exact:

$$\begin{tikzcd}
f'^{*} i_{*}\Phi \arrow[r] \arrow[d, "a"'] & p'_{*}p'^{*}(f'^{*} i_{*}\Phi) \arrow[r, shift left] \arrow[r, shift right] \arrow[d, "b"'] & q'_{*}q'^{*}(f'^{*} i_{*}\Phi) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] \arrow[d, "c"'] & r'_{*}r'^{*}(f'^{*} i_{*}\Phi) \arrow[d, "d"'] \\
j_{*} f''^{*}\Phi \arrow[r] & j_{*}p''_{*}p''^{*}(f''^{*}\Phi) \arrow[r, shift left] \arrow[r, shift right] & j_{*}q''_{*}q''^{*}(f''^{*}\Phi) \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & j_{*}r''_{*}r''^{*}(f''^{*}\Phi)
\end{tikzcd}$$

Since Y is smooth over S, the morphism fp is universally locally 1-aspherical for $\mathbb{L}$ (SGA XV 2.1), and it follows from [**2**, VII 2.1.7] that b is an equivalence of categories. On the other hand, after restricting S to a nonempty open subset if necessary, we may suppose that the morphisms $Z \to S$ and $T \to S$ are universally locally 0-acyclic. It follows that the functors c and d are fully faithful, and the diagram above then shows that a is an equivalence, completing the proof.

Corollary XIII.3.7. *Let k be a field of characteristic $p \geqslant 0$, let p' be the set of prime numbers distinct from p, and let $f\colon X \to k$ be a coherent morphism. Suppose that one of the following two conditions holds:*

a) *f is of finite type, and schemes of finite type of dimension $\leqslant \dim X$ over an algebraic closure of k admit resolution of singularities.*
b) *Schemes of finite type over an algebraic closure of k admit resolution of singularities.*

Then f is universally locally 1-aspherical for p'.

Case a) follows from XIII.3.6. In case b), since the question is local on X, we may suppose that X is affine; passage to the limit (SGA 4 XV 1.3) reduces us to the case where X is of finite type over k.

Corollary XIII.3.8. *Let S be an irreducible scheme with generic point s, and let* 414
$f\colon X \to S$ be a morphism of finite presentation. Suppose that schemes of finite type of dimension $\leqslant \dim X_s$ over an algebraic closure $\overline{k}$ of $\kappa(s)$ admit strong resolution of singularities (SGA 5 I 3.1.5)*. If $\mathbb{L}$ denotes the set of prime numbers distinct from the residue characteristics of S, then there is a nonempty open subset S_1 of S such that, for every specialization $\bar{s}_1 \to \bar{s}_2$ of geometric points of S_1, the specialization morphism* (XIII.2)

$$\pi_1^{\mathbb{L}}(X_{\bar{s}_1}) \to \pi_1^{\mathbb{L}}(X_{\bar{s}_2})$$

is bijective.

By XIII.3.1 and XIII.3.6, after restricting S to a nonempty open subset if necessary, we may suppose that f is locally 1-aspherical for $\mathbb{L}$, and that, for every finite constant sheaf of $\mathbb{L}$-groups F on X, (F, f) is cohomologically proper in dimension $\leqslant 1$. It then follows from XIII.1.13 that, for every specialization $\bar{s}_1 \to \bar{s}_2$ of geometric points of S_1, the specialization morphism

$$\left(\mathrm{R}^1 f_* F\right)_{\bar{s}_2} \to \left(\mathrm{R}^1 f_* F\right)_{\bar{s}_1}$$

is bijective. The corollary is precisely the translation of the preceding statement into the language of fundamental groups.

XIII.4. Exact homotopy sequences

. Let X and S be two connected schemes, let $f\colon X \to S$ be a morphism, let a be a geometric point of X, and let $\mathbb{L}$ be a set of prime numbers. Let K be the kernel of the canonical homomorphism $\pi_1(X, a) \to \pi_1(S, a)$, and let N be the smallest normal pro-subgroup of K such that K/N is the pro-$\mathbb{L}$-group $K^{\mathbb{L}}$. Then N is normal in $\pi_1(X, a)$, and we denote by

$$\pi_1'(X, a)$$

the quotient of $\pi_1(X, a)$ by N. If a is a geometric point of a geometric fiber $X_{\bar{s}}$, the 415
canonical morphisms

$$\pi_1(X_{\bar{s}}, a) \to \pi_1(X, a) \to \pi_1(S, a)$$

define canonical morphisms

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi_1'(X, a) \xrightarrow{v} \pi_1(S, a)$$

We have $vu = 0$.

Proposition XIII.4.1. *Let S be a connected scheme, and let $f\colon X \to S$ be a locally* 0*-acyclic morphism* (SGA 4 XV 1.11)*; suppose moreover that f is* 0*-acyclic (which, when f is coherent, amounts to saying that the geometric fibers of f are connected* (SGA 4 XV 1.16)*). Let $\mathbb{L}$ be a set of prime numbers. If S' is a scheme étale over S, denote by X' and f' the inverse images of X and f over S'. Suppose that, for every étale covering S' of S and every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, (E, f') is cohomologically proper relative to S' in dimension $\leqslant 0$ and f'_*E is constructible. Then, if $\bar{s}$ is a geometric point of S and a is a geometric point of the fiber $X_{\bar{s}}$, the sequence of group homomorphisms*

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi'_1(X, a) \xrightarrow{v} \pi_1(S, a) \to 1 \tag{XIII.4.-1.1}$$

is exact.

This statement generalizes X.X.1.4, whose proof we shall repeat.

Let us first show that v is surjective. It suffices to show that, for every connected étale covering S' of S, X' is also connected (V V.6.9). Let C be a set having at least two elements. Since f is 0-acyclic, the canonical morphism

$$\mathrm{H}^0(S', C_{S'}) \to \mathrm{H}^0(X', C_{X'})$$

is bijective, and hence X' is connected, which proves the surjectivity of v.

By definition of $K^{\mathbb{L}}$ (XIII.4), there is an exact sequence 416

$$1 \to K^{\mathbb{L}} \to \pi'_1(X, a) \to \pi_1(S, a) \to 1$$

Let $\widetilde{S}$ be the universal covering of S, and put $\widetilde{X} = \widetilde{S} \times_S X$; the group $K^{\mathbb{L}}$ classifies the Galois coverings P whose group is an $\mathbb{L}$-group and for which there exist an étale covering S' of S and a Galois covering Q of $X' = X \times_S S'$ such that there is an isomorphism $P \simeq Q \times_{X'} \widetilde{X}$. For the sequence (XIII.4.-1.1) to be exact, it is necessary and sufficient that the canonical morphism

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \to K^{\mathbb{L}}$$

be surjective. By the interpretation of $K^{\mathbb{L}}$, this amounts to saying that, for every étale covering S' of S and every Galois covering Q of X' whose group is an $\mathbb{L}$-group, such that $P = Q \times_{X'} \widetilde{X}$ is connected, $Q|X_{\bar{s}}$ is connected. Let us show that this last condition is satisfied. Indeed, let S' be an étale covering of S, and let Q be a Galois covering of X' whose group is an $\mathbb{L}$-group F, such that $Q|X_{\bar{s}}$ is disconnected; let us show that, after possibly replacing S' by an étale covering, Q becomes disconnected. There is a subgroup G of F distinct from F and a torsor R under $G|X_{\bar{s}}$ such that $Q|X_{\bar{s}}$ is obtained by extension of structure group along $G \to F$ from R. The étale covering $E = Q/G$ of X' is such that $E|X_{\bar{s}}$ has a section. By XIII.1.15, f_*E is locally constant constructible and, after possibly replacing S' by an étale covering, we may even suppose that f_*E is constant. Since (E, f') is cohomologically proper relative to S' in dimension $\leqslant 0$ and $\mathrm{H}^0(X_{\bar{s}}, E|X_{\bar{s}})$ is nonempty, one sees that E has a section. But this proves that Q is disconnected, which completes the proof.

From XIII.4.1, we deduce the following lemma, which will be used in XIII.4.6.

Lemma XIII.4.2. *Let S be a connected scheme, let $f\colon X \to S$ be a locally* 0*-acyclic and* 0*-acyclic morphism, and let $\mathbb{L}$ be a set of prime numbers. Suppose that, for every* 417 *finite constant sheaf of $\mathbb{L}$-groups F on X, (F, f) is cohomologically proper relative to S in dimension $\leqslant 1$, and that, for every étale covering S' of S and every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, f'_*E is constructible. Then, if $\bar{s}$ is a geometric point of S and a is a geometric point of the fiber $X_{\bar{s}}$, the sequence of group homomorphisms*

$$\pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \to \pi'_1(X, a) \to \pi_1(S, a) \to 1$$

is exact.

The hypotheses of XIII.4.1 are satisfied. Indeed, it follows from XIII.1.12 3) that, for every scheme S' étale over S and every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, (E, f') is cohomologically proper relative to S' in dimension $\leqslant 0$.

PROPOSITION XIII.4.3. *Let S be a connected scheme, let $\mathbb{L}$ be a set of prime numbers, let $f\colon X \to S$ be a* 0*-acyclic morphism, locally* 1*-aspherical for* $\mathbb{L}$ (SGA 4 XV 1.11), *and let $g\colon S \to X$ be a section of f. Let $\bar{s}$ be a geometric point of S, and let a be a geometric point of the fiber $X_{\bar{s}}$. Suppose that, for every constant sheaf of $\mathbb{L}$-groups F, (F, f) is cohomologically proper in dimension $\leqslant 1$, that the direct image under f of the stack of torsors under F is a* 1*-constructible stack* (XIII.0)*, and that, for every étale covering E of X' that is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, f'_*E is constructible. Then the sequence of group homomorphisms*

$$1 \to \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \xrightarrow{u} \pi'_1(X, a) \xrightarrow{v} \pi_1(S, a) \to 1 \tag{XIII.4.-1.1}$$

is exact.

In view of XIII.4.2, it suffices to prove the injectivity of the morphism u, i.e. to prove that, for every principal covering $\overline{Z}$ of $X_{\bar{s}}$ whose group is an $\mathbb{L}$-group C, there exist 418
an étale covering Z of X and a morphism from a connected component of $Z|X_{\bar{s}}$ to $\overline{Z}$ (V V.6.8). Thus let $\overline{Z}$ be a principal covering of $X_{\bar{s}}$ whose group is an $\mathbb{L}$-group C, and let $\bar{z}$ be its class in $\mathrm{H}^1(X_{\bar{s}}, C_{X_{\bar{s}}})$. By XIII.1.4 d), there is a canonical isomorphism

$$\left(\mathrm{R}^1 f_* C_X\right)_{\bar{s}} \xrightarrow{\sim} \mathrm{H}^1(X_{\bar{s}}, C_{X_{\bar{s}}}),$$

and by XIII.1.15, $\mathrm{R}^1 f_* C_X$ is a locally constant constructible sheaf. Thus one can find an étale covering S' of S such that $\mathrm{R}^1 f_* C_X|S'$ is constant. If $\bar{s} \to S'$ is a geometric point above the geometric point $\bar{s} \to S$, there is an element z of $\mathrm{H}^0(S', \mathrm{R}^1 f_* C_X)$ whose image in $\mathrm{H}^1(X_{\bar{s}}, C_{X_{\bar{s}}})$ is $\bar{z}$. By the lemma XIII.4.4 below, one can find an étale covering S'_1 of S' and a torsor P on X'_1 with group C whose image in $\mathrm{H}^0(S'_1, \mathrm{R}^1 f_* C)$ is equal to the restriction of z. The torsor P is represented by an étale covering Z of $X'_1 = X \times_S S'_1$ such that $Z \times_{X'_1} X_{\bar{s}}$ is isomorphic to $\overline{Z}$. If Z is regarded as an étale covering of X, there is then a morphism from $Z \times_X X_{\bar{s}}$ to $\overline{Z}$, which completes the proof.

LEMMA XIII.4.4. *Let $f\colon X \to S$ be a* 0*-acyclic and locally* 0*-acyclic morphism, and let g be a section of f. Let C be a finite constant group such that (C_X, f) is cohomologically proper in dimension $\leqslant 0$ and the direct image under f of the stack of torsors under C_X is constructible. Then, for every section z of $\mathrm{H}^0(S, \mathrm{R}^1 f_* C_X)$, one can find an étale covering S_1 of S and, if $X_1 = X \times_S S_1$, an element of $\mathrm{H}^1(X_1, C_{X_1})$ whose image under the canonical morphism*

$$\mathrm{H}^1(X_1, C_{X_1}) \to \mathrm{H}^0(S_1, \mathrm{R}^1 f_* C_{X_1})$$

is equal to the restriction of z to $\mathrm{H}^0(S_1, \mathrm{R}^1 f_* C_{X_1})$.

For every scheme S' étale over S, put $X' = X \times_S S'$, and denote by g' (respectivelyF', etc.) the inverse image of g (respectively$F$, etc.) under the morphism $S' \to S$. The 419
presheaf G on S defined by

$$G(S') = \begin{Bmatrix} \text{isomorphism classes of torsors } P \text{ on } X' \text{ with group } C_{X'}, \text{ endowed with} \\ \text{an isomorphism } g'^* P \xrightarrow{\sim} C_{S'} \end{Bmatrix}$$

is then a sheaf. Indeed, this follows by descent from the fact that an automorphism of a torsor P on X' is completely determined by its restriction to $g'(S')$. Moreover, there is a surjective morphism

$$G \to \mathrm{R}^1 f_* F.$$

Let z be an element of $\mathrm{H}^0(S, \mathrm{R}^1 f_* C_X)$, and let H be the subsheaf of G that is the inverse image of z. It suffices to show that H is a locally constant constructible sheaf. Since this property is local on S, we may suppose that z comes from an element of $\mathrm{H}^1(X, C_X)$ represented by a torsor P such that g^*P is isomorphic to C_S. To give an isomorphism $i\colon g^*P \xrightarrow{\sim} C_S$ amounts to giving a global section of $\mathbf{Aut}_{C_S}(g^*P)$, and two isomorphisms i and i' define the same element of $G(X)$ if and only if ii'^{-1} is the image of an element of $\mathrm{Aut}_{C_X}(P)$. Consider the canonical injection

$$f_* \mathbf{Aut}_{C_X}(P) \to \mathbf{Aut}_{C_S}(g^*P) \simeq C,$$

so H is identified with the quotient of $\mathbf{Aut}_{C_S}(g^*P)$ by $f_* \mathbf{Aut}_{C_X}(P)$. By XIII.1.15, $f_* \mathbf{Aut}_{C_X}(P)$ is locally constant; hence the same is true of H, which completes the proof.

Examples XIII.4.5. *Note that, if S is a connected scheme, the hypotheses of XIII.4.1 are satisfied when the morphism f is proper, flat, and of finite presentation, with separable, connected geometric fibers, and $\mathbb{L}$ is arbitrary (cf. X X.1.3). The hypotheses of XIII.4.3 are satisfied if, in addition, f is smooth and has a section, with $\mathbb{L}$ denoting the set of prime numbers distinct from the residual characteristics of S (SGA 4 XV 2.1 and XVI 5.2).*

The hypotheses of XIII.4.1 are also satisfied if S is connected, and if there is a scheme Z that is proper, flat, and of finite presentation over S, with separable, connected geometric fibers, such that X is the complement in Z of a normal crossings divisor relative to S, with $\mathbb{L}$ the set of prime numbers distinct from the residual characteristics of S (XIII.2.9). *The hypotheses of XIII.4.3 are satisfied if, in addition, f is smooth and has a section.* 420

. We retain the notation and hypotheses of XIII.4.3. If $\bar{s}$ is a geometric point of S and $a = g(\bar{s})$, the section g defines a morphism

$$w\colon \pi_1(S, a) \to \pi_1'(X, a),$$

such that $\pi_1'(X, a)$ is identified with the semidirect product of $\pi_1(S, a)$ by $\pi_1(X_{\bar{s}}, a)$. The profinite group $\pi_1(S, a)$ therefore acts on $\pi_1(X_{\bar{s}}, a)$. Since $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is a strict inverse limit of groups invariant under the action of $\pi_1(S, a)$, specifying $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ together with this action is equivalent to specifying a strict inverse system of finite étale group schemes over S, which we denote by

$$\pi_1^{\mathbb{L}}(X/S, g, \bar{s}) \quad \text{or simply} \quad \pi_1^{\mathbb{L}}(X/S, g).$$

The following properties then hold:

. For every finite étale group scheme G over S whose fibers are $\mathbb{L}$-groups, the set E of classes of torsors P under the pullback G_X of G to X, equipped with an isomorphism $g^*P \xrightarrow{\sim} G$, is canonically isomorphic to the set

$$\mathrm{Hom}_S(\pi_1^{\mathbb{L}}(X/S, g, \bar{s}), G) \quad \text{modulo inner automorphisms of } G.$$

. For every finite étale group scheme G over S whose fibers are $\mathbb{L}$-groups, the sheaf $\mathrm{R}^1 f_* G_X$ is canonically isomorphic to the sheaf associated with the presheaf

$$S' \longmapsto \mathrm{Hom}_{S'}(\pi_1^{\mathbb{L}}(X/S, g, \bar{s}), G) \text{ modulo inner automorphisms of } G.$$

(S' denotes a scheme étale over S).

. Let S' be a connected S-scheme , let $\bar{s}$ be a geometric point of S', and let X', g' be the respective pullbacks of X, g to S'. Then $\pi_1^{\mathbb{L}}(X'/S', g', \bar{s})$ is canonically isomorphic to the pullback of $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})$ to S'. For every geometric point ξ of S, the fiber $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})_\xi$ is isomorphic to $\pi_1^{\mathbb{L}}(X_\xi)$. 421

Indeed, specifying G is equivalent to specifying an abstract $\mathbb{L}$-group $\mathbf{G}$ on which $\pi_1(S, a)$ acts, thereby inducing an action of $\pi_1'(X, a)$ on $\mathbf{G}$. The isomorphism defined in XIII.4 is then obtained, by restriction to the subset E, from the canonical morphism

$$\mathrm{H}^1(\pi_1'(X, a), \mathbf{G}) \to \mathrm{H}^1(\pi_1^{\mathbb{L}}(X_{\bar{s}}, a), \mathbf{G}) = \mathrm{Hom}(\pi_1^{\mathbb{L}}(X_{\bar{s}}, a), \mathbf{G})/\text{inner aut. } G,$$

with the set E mapping bijectively onto the subset of morphisms from $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ to $\mathbf{G}$ that are compatible with the action of $\pi_1(S, a)$. Assertion XIII.4 then follows from the definition of $\pi_1^{\mathbb{L}}(X/S, g, \bar{s})$, taking into account the exact homotopy sequence (XIII.4.-1.1), and XIII.4 follows from XIII.4 and XIII.4.

Proposition XIII.4.6 (Künneth formula). *Let k be a separably closed field of characteristic $p \geq 0$, let X and Y be two connected k-schemes, let a be a geometric point of X, let b be a geometric point of Y, and let c be a geometric point of $X \times_k Y$ lying over a and b. Suppose that one of the following two conditions holds:*

a) *X is of finite type over k, and schemes of finite type over an algebraic closure $\bar{k}$ of k of dimension $\leq \dim X$ are strongly desingularizable.* (SGA 5 I 3.1.5).
b) *X is quasi-compact and quasi-separated, and every scheme of finite type over $\bar{k}$ is strongly desingularizable.*

Then, if p' is the set of prime numbers distinct from p, the morphism

(XIII.4.-1.0) $$\pi_1^{p'}(X \times_k Y, c) \to \pi_1^{p'}(X, a) \times \pi_1^{p'}(Y, b),$$

induced by the homomorphisms on fundamental groups associated with the projections 422

$$X \times_k Y \to X \quad \textit{and} \quad X \times_k Y \to Y,$$

is an isomorphism.

We may assume that k is algebraically closed and that X is reduced (SGA 4 VIII 1.1). Let $Z = X \times_k Y$, and let $g \colon X \to k$ and $f \colon Z \to Y$ be the canonical morphisms. The morphism g, and hence also f, is universally locally 1-aspherical for p' by XIII.3.8. Since X is connected, f is 0-acyclic (SGA 4 XV 1.16). Moreover, it follows from XIII.3.5 that, for every finite p'-group C, (C_X, g) is cohomologically proper relative to k in dimension ≤ 1. It follows that (C_Z, f) is cohomologically proper relative to Y in dimension ≤ 1 (XIII.1.4 c)) and that f_*C_Z and $\mathrm{R}^1 f_* C_Z$ are constant sheaves. Consequently, g satisfies all the hypotheses of XIII.4.2. We therefore have the exact sequence

$$\pi_1^{p'}(X_b, c) \to \pi_1^{p'}(Z, c) \to \pi_1^{p'}(Y, b) \to 1.$$

Moreover, the composite morphism

$$\pi_1^{p'}(X_b, c) \to \pi_1^{p'}(Z, c) \to \pi_1^{p'}(Y, b)$$

is an isomorphism, and we therefore have the exact sequence

$$1 \to \pi_1^{p'}(X, a) \to \pi_1^{p'}(Z, c) \to \pi_1^{p'}(Y, b) \to 1.$$

On the other hand, the morphism (XIII.4.-1.0) defines a morphism from this exact sequence to the exact sequence

$$1 \to \pi_1^{p'}(X, a) \to \pi_1^{p'}(X, a) \times \pi_1^{p'}(Y, b) \times \pi_1^{p'}(Y, b) \to 1,$$

and it follows that the morphism (XIII.4.-1.0) is an isomorphism.

. Let

$$\begin{tikzcd} X & X_U \arrow[l] & X_{\bar{s}} \arrow[l] \\ S \arrow[u, leftarrow, "f"'] & U \arrow[l] \arrow[u, leftarrow, "f_U"'] & \bar{s} \arrow[l] \arrow[u, leftarrow] \end{tikzcd}$$

be a diagram whose squares are cartesian, where S is an arcwise connected scheme (SGA 4 IX 2.12), U is a connected open subset of S, and $\bar{s}$ is a geometric point of U. Let a be a geometric point of $X_{\bar{s}}$, and let $\mathbb{L}$ be a set of prime numbers. Let g be a section of f, and suppose that the following conditions are satisfied: 423

a) The morphism f is 0-acyclic and locally 0-acyclic, and, for every étale covering S' of S and every étale covering E of $X \times_S S'$ which is a quotient of a Galois covering whose group is an $\mathbb{L}$-group, $(E, f_{(S')})$ is cohomologically proper relative to S' in dimension ≤ 0.
b) The morphism f_U is locally 1-aspherical for $\mathbb{L}$, and, for every finite constant sheaf of $\mathbb{L}$-groups F on X_U, (F, f_U) is cohomologically proper in dimension ≤ 1, and the fibers of $\mathrm{R}^1 f_{U*}F$ are finite.

We then deduce from XIII.4.1 and XIII.4.3 the following commutative diagram, whose rows are exact:

(XIII.4.-1.0)
$$\begin{tikzcd} 1 \arrow[r] & \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \arrow[r] \arrow[d, equal] & \pi_1'(X_U, a) \arrow[r] \arrow[d] & \pi_1(U, a) \arrow[r] \arrow[d] & 1 \\ & \pi_1^{\mathbb{L}}(X_{\bar{s}}, a) \arrow[r] & \pi_1'(X, a) \arrow[r] & \pi_1(S, a) \arrow[r] & 1 \end{tikzcd}$$

Thanks to the section g, we have morphisms

$$\pi_1(U, a) \to \pi_1'(X_U, a), \quad \pi_1(S, a) \to \pi_1'(X, a)$$

from these we deduce a morphism from the amalgamated sum of $\pi_1(S, a)$ and $\pi_1'(X_U, a)$ over $\pi_1(U, a)$ into $\pi_1'(X, a)$:

(XIII.4.-1.1)
$$\varphi\colon \pi = \pi_1(S, a) \coprod_{\pi_1(U,a)} \pi_1'(X_U, a) \to \pi_1'(X, a).$$

Suppose that the following condition is satisfied:

c) If $T = S - U$, then $\operatorname{prof}\text{ét}_T(S) \geq 2$ (SGA 2 XIV 1.1).

Then the functor which assigns to an étale covering of S its restriction to U is fully faithful (SGA 2 XVI 1.4). It follows that the morphism $\pi_1(U, a) \to \pi_1(S, a)$ is surjective (V V.6.9), and we deduce from the diagram (XIII.4.-1.0) that the same is true of the 424
morphism $\pi_1'(X_U, a) \to \pi_1'(X, a)$; a fortiori, φ is an epimorphism. Let

(XIII.4.-1.2)
$$K = \operatorname{Ker}(\pi_1(U, a) \to \pi_1(S, a)).$$

The group π of (XIII.4.-1.1) is identified with the quotient of $\pi_1'(X_U, a)$ by the closed normal subgroup generated by the image L of K in $\pi_1'(X_U, a)$. Regard $\pi_1'(X_U, a)$ as the semidirect product of $\pi_1(U, a)$ by $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$. The group K then acts by inner automorphisms on $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$, and the quotient $\pi = \pi_1'(X_U, a)/L$ is identified with the semidirect product of $\pi_1(U, a)/K = \pi_1(S, a)$ by the group $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K$ of coinvariants of $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ under K. Finally, we have an epimorphism

(XIII.4.-1.3)
$$\varphi\colon \pi = \pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K \cdot \pi_1(S, a) \to \pi_1'(X, a).$$

The following proposition gives conditions under which the morphism φ is an isomorphism.

Proposition XIII.4.7. *The notation is that of XIII.4. In addition to conditions* a), b), c)*, suppose that the following conditions are satisfied:*

d) *For every point t of $T = S - U$, the morphism f is locally 1-aspherical for $\mathbb{L}$ at $g(t)$.*
e) *For every point $t \in T$, every irreducible component of the fiber X_t contains $g(t)$ and, for every point x of $X_t - \{g(t)\}$ which is not maximal, one has*

$$\operatorname{prof}\operatorname{hop}_x(X) \geq 3 \quad \text{(SGA 2 XIV 1.2)}$$

and the ring $\mathcal{O}_{X,x}$ is noetherian.

Then the morphism (XIII.4.-1.3) *is an isomorphism.*

As stated above, the group π is identified with the quotient of $\pi_1'(X_U, a)$ by the closed normal subgroup L generated by the image of K (XIII.4.-1.2) in $\pi_1'(X_U, a)$. This amounts to saying that π classifies the principal coverings Z of X_U such that $g_U^{-1}(Z)$ extends to 425
an étale covering of S, and which induce on $X_{\bar{s}}$ a covering obtained by extension of the structure group from a principal covering whose group is an $\mathbb{L}$-group. To prove that φ is an isomorphism, it is enough to show that such a covering Z extends to all of X.

Let us first show that Z extends to an étale covering of an open subset containing X_U and $g(S)$. Let W be a scheme étale over X whose image contains X_U and $g(S)$, and put $W_U = W \times_S U$. Since the morphism $W \to S$ is 0-acyclic and $\operatorname{prof.\acute{e}t}_T S \geq 2$, we have

$$\operatorname{prof.\acute{e}t}_{(W-W_U)} W \geq 2$$

(SGA 2 XIV 1.13); consequently, if $Z|W_U$ extends to an étale covering of W, this extension is unique up to unique isomorphism. It follows that the problem of extending Z to a neighborhood of $g(S) \cup X_U$ is local for the étale topology near the points of $g(T)$. If t is a point of T, put $x = g(t)$, and denote by $\bar{X}$ (respectively$\bar{S}$) the strict localization of X at $\bar{x}$ (respectivelyof S at $\bar{t}$), put $\bar{U} = U \times_S \bar{S}$, $\bar{X}_U = \bar{X} \times_X X_U$, and let $\bar{g}\colon \bar{S} \to \bar{X}$ be the morphism induced by g. It is enough to show that, for every point t of T, the inverse image $\bar{Z}$ of Z on $\bar{X}_U$ extends to $\bar{X}$ or, equivalently, is trivial. But, by definition of Z, the inverse image of $\bar{Z}$ on $\bar{U}$ is trivial. To prove that $\bar{Z}$ is trivial, it is enough to show that it is of the form $\bar{f}_U^* E$, where E is a principal covering of $\bar{U}$; indeed, we will then have $E \simeq \bar{g}_U^* \bar{f}_U^* E \simeq \bar{g}_U^* \bar{Z}$, whence the result, since $\bar{g}_U^* \bar{Z}$ is trivial. Since the morphism $\bar{f}_U$ is 0-acyclic and locally 0-acyclic, to prove that $\bar{Z}$ comes from $\bar{U}$, it is enough to show that, for every algebraic geometric point over a point of $\bar{U}$, which we may suppose to be the point $\bar{s}$, $\bar{Z}|X_{\bar{s}}$ is trivial (SGA 4 XV 1.15). But $\bar{Z}|X_{\bar{s}}$, being obtained by extension of the structure group from a principal covering whose group is an $\mathbb{L}$-group, is trivial because the morphism $\bar{f}$ is 1-aspherical for $\mathbb{L}$.

We have therefore shown that there exists an open neighborhood V of $g(S) \cup X_U$ such 426
that Z extends to an étale covering Z_V of V. Let us show that Z_V extends to all of X. It is enough to see that, for every point x of $X - V$, one has

$$\operatorname{prof\,hop}_x X \geq 3.$$

But this follows from hypothesis e) and from the fact that a point x of $X - V$ cannot be maximal in its fiber X_t: since every irreducible component of X_t contains $g(t)$, every maximal point of X_t belongs to V.

Corollary XIII.4.8. *The hypotheses are those of XIII.4.7, but suppose in addition that $\pi_1(S, a) = 1$. Then there is an isomorphism*

$$(*) \qquad \pi_1^{\mathbb{L}}(X, a) \xrightarrow{\sim} \pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K.$$

In particular, if $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is topologically finitely presented and if K acts on $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ through a finitely generated group, then $\pi_1^{\mathbb{L}}(X, a)$ is topologically finitely presented.

The isomorphism $(*)$ was proved in XIII.4.7. Suppose that $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)$ is the quotient of the free pro-$\mathbb{L}$-group on n generators $L(x_1, \ldots, x_n)$ by the closed normal subgroup generated by the elements $y_1, \ldots, y_p$ of $L(x_1, \ldots, x_n)$, and suppose that K acts through a group generated by elements $k_1, \ldots, k_q$. If, for every $i \in [1, n]$, $j \in [1, q]$, z_{ij} denotes an element of $L(x_1, \ldots, x_n)$ lifting the element $(k_j \cdot x_i)x_i^{-1}$, then $\pi_1^{\mathbb{L}}(X_{\bar{s}}, a)_K$ is the quotient of $L(x_1, \ldots, x_n)$ by the closed normal subgroup generated by the elements $(y_i)_{i\in[1,p]}$, $(z_{ij})_{i\in[1,n], j\in[1,q]}$.

Remarks XIII.4.9. a) Conditions a) through e) of XIII.4 are satisfied when S is a connected normal scheme, U is a dense retrocompact open subset of S, and f is a proper morphism of finite presentation, with geometrically connected and irreducible fibers at every point t of T, f being moreover separable, smooth at the points of $X_U \cup g(T)$, $\mathbb{L}$ 427 being the set of prime numbers distinct from the residue characteristics of S, and X being regular at every point of X_t. Indeed, condition a) follows from SGA 4 XV 4.1 and 1.4; conditions b) and d) follow from XIII.1.3 and SGA 4 XV 2.1 and XVI 5.2. Finally, e) follows from SGA XIV 1.11.

b) Corollary XIII.4 applies to compute the fundamental group $\pi_1^{p'}(X)$ of a proper smooth surface X over a separably closed field k of characteristic p (p' denoting the set of prime numbers distinct from p). The method was communicated to us by J.P. MURRE; it consists in reducing, by blowing up X, to the case where there is a fibration $X \to \mathbb{P}^1_k$ and an open subset U of $\mathbb{P}^1_k$ satisfying the hypotheses of XIII.4 (see SGA 7 for more details). The same method may be used more generally (loc. cit.) to prove that, if X is a connected k-scheme of finite type and if the schemes of finite type of dimension $\leq \dim X$ over an algebraic closure of k are strongly desingularizable (SGA 5 I 3.1.5), then $\pi_1^{p'}(X)$ is topologically finitely presented.

XIII.5. Appendix I: Variations on Abhyankar's lemma

This appendix contains several variants of Abhyankar's lemma.

PROPOSITION XIII.5.1. *Let* $X = \operatorname{Spec}(A)$ *be a regular local scheme, and let* $D = \sum_{1\leq i\leq r} \operatorname{div} f_i$ *be a normal crossings divisor, where the* f_i *are elements of the maximal ideal of* A *belonging to a regular system of parameters. Let* n_i, $1 \leq i \leq r$, *be integers* ≥ 0, *and put*

$$X' = X[T_1, \dots, T_r]/(T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

$U' = U \times_X X'$. *Then* X' *is regular, and* U' *is the complement in* X' *of the normal crossings divisor* $\sum_{1\leq i\leq r} \operatorname{div} T_i$. *If the integers* n_i *are prime to the residue characteristic* p *of* X, 428
then U' *is a connected étale covering of* U, *tamely ramified relative to* D (XIII.2.3 c)).

Indeed, X' is the spectrum of a local ring A' whose maximal ideal is generated by $T_1, \dots, T_r$. (We may suppose that $r = \dim(A)$.) Since A' is finite and flat over A, and hence has dimension r, A' is regular (EGA 0_{IV} 17.1.1), and the T_i form a regular system of parameters of A'. Suppose that the n_i are prime to p. Since all the f_i are invertible on U, the fact that U' is étale over U follows from I I.7.4. Moreover, U' is tamely ramified relative to D. Indeed, let x_i be the generic point of $V(f_i)$, let $\bar{R}$ be the strict localization of $R = \mathcal{O}_{X,x_i}$, and let $\bar{K}$ be the fraction field of $\bar{R}$. Then the $\bar{K}$-algebra representing $U'|\bar{K}$ is obtained from the field $\bar{K}[T_i]/(T_i^{n_i} - f_i)$ by an unramified extension; it is therefore tamely ramified over $\bar{R}$.

PROPOSITION XIII.5.2 (Absolute Abhyankar lemma). *Let* X *be a regular local scheme, let*

$$D = \sum_{1\leq i\leq r} \operatorname{div} f_i$$

be a normal crossings divisor as in XIII.5.1, put $Y = \operatorname{Supp} D$, $U = X - Y$, *and let* V *be an étale covering of* U *tamely ramified relative to* D. *If* x_i *is the generic point of the closed subset* $V(f_i)$, *then* $\mathcal{O}_{X,x_i}$ *is a discrete valuation ring with fraction field* K_i, *and* $V|K_i = \operatorname{Spec}(() \prod_{j\in J_i} L_j)$, *where the* L_j *are finite separable extensions of* K_i. *Let* n_j *be the order of the inertia group of a Galois extension generated by* L_j, *and let* n_i *be the least common multiple of the* n_j *as* j *ranges over* J_i. *If*

$$X' = X[T_1, \dots, T_r]/(T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

$U' = U_{(X')}$, $V' = V_{(X')}$, *etc., then the étale covering* V' *of* U' *extends, uniquely up to unique isomorphism, to an étale covering of* X', *and the* n_i *are prime to the residue characteristic* p *of* X.

Uniqueness follows from the fact that X' is normal (XIII.5.1). Indeed, an étale 429
covering of X' extending V' is isomorphic to the normalization of X' in the fiber of V' at the generic point of X' (I.10.2). If $\bar{x}'$ is a geometric point of Y', denote by $\bar{X}'$ the strict localization of X' at $\bar{x}'$, and put $\bar{V}' = V'_{(\bar{X}')}$, etc. By descent, in view of uniqueness, it is enough to show that for every geometric point $\bar{x}'$ of Y', the étale covering $\bar{V}'$ of $\bar{U}'$ extends to $\bar{X}'$. Since an étale covering of an open subset of the regular scheme X' containing all points x' such that $\dim \mathcal{O}_{X',x'} \leq 1$ extends to all of X' (SGA 2 XIV 1.11), we may even restrict ourselves to the points $\bar{x}'$ lying over a maximal point of Y'. At such a point $\bar{x}'$, the fact that $\bar{V}'$ extends to an étale covering of $\bar{X}'$ follows from X.3.6.

Let us show that the n_i are prime to p. Otherwise, for example, we would have $p|n_1$. After replacing X by

$$X[T_1, \dots, T_r]/(T_1^{n_1/p} - f_1, T_2^{n_2} - f_2, \dots, T_r^{n_r} - f_r),$$

we reduce to the case $X' = X[T_1]/(T_1^p - f_1)$. It is enough to show that V extends to an étale covering of X, since we would then have $n_1 = 1$, contrary to the hypothesis. For this, we may suppose that X is strictly local. Let Z be the closed subscheme of X defined by $p = 0$, and let $Z_1 = Z \cap X_{f_1}$; Z_1 is a nonempty open subset of Z. By what precedes, the étale covering V' of U' extends to an étale covering W' of X'. Let W''_1 and W''_2 be the inverse images of W' under the two projections $X'' = X' \times_X X' \rightrightarrows X'$. We show that the descent isomorphism $u\colon W''_1|U'' \to W''_2|U''$ extends to an X''-morphism $W''_1 \to W''_2$, which will necessarily be descent data on W' relative to $X' \to X$. Let Z'' (respectivelyZ''_1) be the inverse image of Z (respectivelyZ_1) in X''. Since the morphism $Z'' \to Z$ is radicial, there is an isomorphism $v\colon W''_1|Z'' \to W''_2|Z''$ extending the isomorphism $u|Z''_1$. But since X is Henselian, there is a bijection

$$\operatorname{Hom}_{X''}(W''_1, W''_2) \simeq \operatorname{Hom}_{Z''}(W''_1|Z'', W''_2|Z''),$$

and hence a morphism $w\colon W''_1 \to W''_2$ lifting v. The open-and-closed subscheme of X'' 430
over which u and w coincide contains Z''_1, and therefore equals X''. Thus V extends to X.

. Retain the hypotheses and notation of XIII.5.2, and suppose in addition that X is strictly local. It follows from loc. cit. that every connected étale covering of U tamely ramified relative to D is a quotient of a tamely ramified covering of the form

$$U' = U[T_1, \dots, T_r]/(T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

where the n_i are integers prime to p. Let $\boldsymbol{\mu}_n$ be the group of n-th roots of unity of U. The group of U-automorphisms of U' is precisely $\boldsymbol{\mu}_{n_1} \times \dots \times \boldsymbol{\mu}_{n_r}$, an n_i-th root of unity ξ_i acting on U' by sending T_i to $\xi_i T_i$. We therefore have the following statement:

Corollary XIII.5.3. *Let X be a regular strictly local scheme of residue characteristic $p \geq 0$, let $D = \sum_{1 \leq i \leq n} \operatorname{div} f_i$ be a normal crossings divisor on X, and put $U = X - \operatorname{Supp} D$. Let*

$$\widetilde{U} = \varprojlim_{(n_i)} U[T_1, \dots, T_r]/(T_1^{n_1} - f_1, \dots, T_r^{n_r} - f_r),$$

where the projective limit is taken over the filtered ordered set (under divisibility) of families of integers $n_i > 0$ prime to p. Then $\widetilde{U}$ is a universal tamely ramified covering of U. Consequently, the tamely ramified fundamental group of U is

$$\pi_1^{\mathrm{t}}(U) \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell[1]^r \qquad \textit{(canonical isomorphism)}$$

where $\mathbb{Z}_\ell[1] = \varprojlim_{n>0} \boldsymbol{\mu}_{\ell^n}$. The group $\pi_1^{\mathrm{t}}(U)$ is noncanonically isomorphic to $\prod_{\ell \neq p} \mathbb{Z}_\ell^r$.

Proposition XIII.5.4. *Let $f\colon X \to S$ be a morphism of schemes, and let $D =$
$\sum_{1 \leq i \leq r} \operatorname{div} f_i$ be a divisor on X with normal crossings relative to S* (XIII.2)*, where, for* 431
every point x of $Y = \operatorname{Supp} D$, if $I(x) \subset [1, r]$ is the set of i such that $f_i(x) = 0$, the subscheme $V((f_i)_{i \in I(x)})$ is smooth over S, of codimension $\operatorname{card} I(x)$ *in X. Let $U = X - Y$.*

Let x be a point of Y, put $X_1 = \operatorname{Spec}()O_{X,x}$, $U_1 = U \times_X X_1$, let n_i, $i \in I(x)$, be integers, and put

$$X' = X[T_i]_{i \in I(x)}/(T_i^{n_i} - f_i).$$

Then, if x' is the point of X' over x, X' is smooth over S at x'. If the integers n_i are prime to the characteristic p of $\kappa(x)$, then $U_1' = U_1 \times_X X'$ is a connected étale covering of U_1, tamely ramified on X_1 relative to S_1 (XIII.2.1).

If $s = f(x)$, the geometric fiber $X'_{\overline{s}}$ is regular at the point x' (XIII.5.1). Since X' is flat over S in a neighborhood of Y, this proves that X' is smooth over S at x' (EGA IV 12.1.6). If the integers n_i are prime to p, U_1' is an étale covering of U_1 (I.7.4); it is tamely ramified on X_1 relative to S because this is so on the geometric fibers at every point of S (XIII.5.1). Finally, the connectedness of U_1' follows from SGA 4 XVI 3.2.

Proposition XIII.5.5 (Relative Abhyankar lemma). *Let X be an S-scheme and let D be a normal crossings divisor relative to S, as in* (XIII.5.4). *Let $Y = X - \operatorname{Supp} D$, $U = X - Y$, let x be a point of Y, let X_1 be the strict localization of X at a geometric point above x, let $U_1 = U \times_X X_1$, and let V_1 be an étale covering of U_1. Suppose that, for every maximal point s of S, $V_{1\overline{s}}$ is tamely ramified on $X_{1\overline{s}}$ relative to $\overline{s}$. Then there exist integers n_i prime to the characteristic p of $\kappa(x)$, with $i \in I(x)$, such that, upon putting*

$$X_1' = X_1[T_i]_{i \in I(x)}/(T_i^{n_i} - f_i),$$

$U_1' = U_1 \times_{X_1} X_1'$, etc., the étale covering V_1' of U_1' extends uniquely, up to unique isomorphism, to an étale covering of X_1'. In particular, V_1 is tamely ramified on X_1 432
relative to S.

We may assume that S is local noetherian with closed point $f(x)$. For every maximal point s of S and every $i \in I(x)$, let x_i be the generic point of the closed subset $V(f_i)$ of the fiber $X_{1\overline{s}}$. The local ring $(\mathcal{O}_{X_1,x_i})_{\mathrm{red}}$ is a discrete valuation ring with fraction field K_i, and we have $V_1|K_i = \operatorname{Spec}(()\prod_{j \in I(x_i)} L_j)$, where L_j is a finite separable extension of K; let n_j denote the order of the inertia group of a Galois extension generated by L_j, and let n_i be the l.c.m. of the n_j as s ranges over the maximal points of S and $j \in I(x_i)$.

With the n_i so chosen, we shall show that V_1' extends uniquely to an étale covering of X_1'. Uniqueness follows from the fact that, since X' is smooth over S at the points of Y, we have $\operatorname{prof} \operatorname{\acute{e}t}_{Y_1'}(X_1') \geqslant 2$ (SGA 4 XVI 3.2 or SGA 2 XIV 1.19). Let x_1' be a point of Y_1', let $\overline{x}_1'$ be a geometric point above x_1', and denote by $\overline{X_1'}$ the strict localization of X_1' at $\overline{x}_1'$, $\overline{U_1'} = U'_{1(\overline{X_1'})}$, etc. By descent, in view of uniqueness, it suffices to show that $\overline{V}_1'$ extends to $\overline{X_1'}$. Moreover, it is enough to take for x_1' only the maximal points of Y_1'; indeed, we shall then have an extension of V_1' over an open subset W_1' of X_1' containing the maximal points of Y_1'. Now, if $Z_1' = X_1' - W_1'$, then $\operatorname{codim}(Z'_{1s}, X'_{1s}) \geqslant 2$ if s is a maximal point of S, and $\operatorname{codim}(Z'_{1s}, X'_{1s}) \geqslant 1$, $\operatorname{prof} \operatorname{\acute{e}t}_s(S) \geqslant 1$ if s is a point of S that is not maximal; the fact that V_1' extends to all of X_1' then follows from (SGA 2 XIV 1.20). But at a geometric point $\overline{x}_1'$ above a maximal point of Y_1', $X'_{1\mathrm{red}}$ is the spectrum of a discrete valuation ring, and the fact that $\overline{V_1'}$ extends to $\overline{X_1'}$ then follows from X X.3.6.

Let us show that the n_i are prime to p. Indeed, if this were not so, there would be an index $i_0 \in I(x)$ such that p divides n_{i_0}. After replacing X by $X_1[T_{i_0}, T_i]_{i \in I(x)}/(T_i^{n_{i_0}/p} - f_{i_0}, T_i^{n_i} - f_i)$, we are reduced to the case $X_1' = X_1[T]/T^p - f_{i_0}$. By what precedes, the étale covering V_1' of U_1' extends to an étale covering E_1' of X_1'. Let η be the closed point 433
of S; since the morphism $X'_{1\eta} \to X_{1\eta}$ is radicial, $V_{1\eta}$ extends to an étale covering $E_{1\eta}$ of $X_{1\eta}$. It then follows, as in XIII.5.2, that E_1' is endowed with descent data relative to the morphism $X_1' \to X_1$ extending the natural descent data on $E_1'|U_1'$. It follows that V_1 extends to X_1; but this implies $n_{i_0} = 1$, contrary to the hypothesis $n_{i_0} = p$.

Corollary XIII.5.6. *Let X be an S-scheme, and let $D = \sum_{1 \leqslant i \leqslant r} \operatorname{div} f_i$ be a normal crossings divisor relative to S, as in XIII.5.4. Let $\overline{x}$ be a geometric point of X, let $\overline{X}$ be the strict localization of X at $\overline{x}$, let $\overline{Y} = Y_{(\overline{X})}$, $\overline{U} = \overline{X} - \overline{Y}$, and let*

$$\widetilde{U} = \varprojlim_{(n_i)} \overline{U}[T_i]_{i \in I(x)}/(T_i^{n_i} - f_i),$$

where the projective limit is taken over the filtered set of families of integers $n_i > 0$ prime to the characteristic p of $\kappa(x)$. Then $\widetilde{U}$ is a universal tamely ramified covering of $\overline{U}$ relative to S. Consequently, the tamely ramified fundamental group of $\overline{U}$ is

$$\pi_1^{\mathrm{t}}(\overline{U}) \simeq \prod_{\ell \neq p} \mathbb{Z}_\ell[1]^{I(x)} \qquad \textit{(canonical isomorphism).}$$

The group $\pi_1^{\mathrm{t}}(\overline{U})$ is noncanonically isomorphic to $\prod_{\ell \neq p} \mathbb{Z}_\ell^{I(x)}$.

Remark XIII.5.7. *Let X be an S-scheme, let $D = \sum_{1 \leqslant i \leqslant r} \operatorname{div} f_i$ be a normal crossings divisor relative to S, as in XIII.5.4, and let $U = X - \operatorname{Supp} D$. For every subset $I \subset [1, r]$, let*

$$X_I = \Big(\bigcap_{i \in I} V(f_i)\Big) \cap \Big(\bigcap_{i \in \complement I} X_{f_i}\Big).$$

Let p be a prime integer or zero, and let Z be a subset of X_I all of whose points have characteristic p. Let

$$\widetilde{U}_I = \varprojlim_{(n_i)} U[T_i]_{i \in I}/(T_i^{n_i} - f_i),$$

where the projective limit is taken over the filtered set of families of integers $n_i > 0$ prime 434
to p. Then, for every geometric point $\overline{x}$ of Z, the pullback of $\widetilde{U}_I$ to $\overline{U}$ identifies with the universal tamely ramified covering of $\overline{U}$.

Corollary XIII.5.8. *Use the notation of XIII.5.6. Let $\overline{S}$ be the strict localization of S at $\overline{x}$, and let*

$$\overline{g} \colon \overline{U} \to \overline{S} \quad \textit{and} \quad \tilde{g} \colon \widetilde{U} \to \widetilde{S}$$

be the canonical morphisms. Then the morphisms $\overline{g}$ and $\tilde{g}$ are 0-acyclic (SGA 4 XV 1.3). *Let G be a constructible sheaf of groups on $\overline{S}$, let $F = \overline{g}^* G$, and let P be a torsor under F. Then P is tamely ramified on $\overline{X}$ relative to $\overline{S}$ if and only if its pullback $\widetilde{P}$ to $\widetilde{U}$ is trivial.*

Indeed, for every scheme $\overline{X'} = \overline{X}[T_i]_{i \in I(x)}/(T_i^{n_i} - f_i)$, where the n_i are integers > 0 prime to p, the morphism $\overline{f}' \colon \overline{X'} \to \overline{S}$ is 0-acyclic. The geometric fibers of $\overline{f}'$ at the various points of $\overline{S}$ are therefore connected, and indeed irreducible. The same is consequently true of the geometric fibers of the morphisms $\overline{g}' \colon \overline{U'} \to \overline{S}$, which proves that the $\overline{g}'$, and hence also $\tilde{g}$, are 0-acyclic (SGA 4 XV 1.16).

It is clear that a torsor P on $\overline{U}$ under F whose pullback to $\widetilde{U}$ is trivial is tamely ramified on $\overline{X}$ relative to $\overline{S}$. Conversely, suppose that P is tamely ramified on $\overline{X}$ relative to $\overline{S}$; we shall show that its pullback to $\widetilde{U}$ is trivial.

It follows from (SGA 4 IX 2.14 (ii)) that there exist a finite morphism $n \colon S_1 \to \overline{S}$, a constant sheaf of groups C on S_1, and a monomorphism $G \to n_* C$. Consider the following commutative diagram formed by Cartesian squares:

$$\begin{tikzcd}
\widetilde{U}_1 \arrow[r] \arrow[d, "r"'] & U_1 \arrow[r, "g_1"] \arrow[d, "q"'] & S_1 \arrow[d, "n"'] \\
\widetilde{U} \arrow[r] & \overline{U} \arrow[r, "\overline{g}"] & \overline{S}.
\end{tikzcd}$$

Let C_1 (respectively$\widetilde{C}_1$) be the pullback of C to U_1 (respectivelyto $\widetilde{U}_1$). We have a 435 commutative diagram in which i and j are isomorphisms (SGA 4 VIII 5.8):

$$(*)\qquad \begin{tikzcd} \mathrm{H}^1(\overline{U}, q_*C_1) \arrow[r, "i"] \arrow[d] & \mathrm{H}^1(U_1, C_1) \arrow[d] \\ \mathrm{H}^1(\widetilde{U}, r_*\widetilde{C}_1) \arrow[r, "j"] & \mathrm{H}^1(\widetilde{U}_1, \widetilde{C}_1). \end{tikzcd}$$

Let Q be the torsor under q_*C_1 obtained from P by extension of structure group along $F \to q_*C_1$. By XIII.2, Q is tamely ramified on $\overline{X}$ relative to $\overline{S}$. The torsor Q corresponds, via i, to a torsor Q_1 under C_1, and it is clear that Q_1 is tamely ramified on $X_1 = \overline{X} \times_{\overline{S}} S_1$ relative to S_1. It therefore follows from XIII.5.6 that the pullback $\widetilde{Q}_1$ of Q_1 to $\widetilde{U}_1$ is trivial, and diagram $(*)$ then shows that the pullback $\widetilde{Q}$ of Q to $\widetilde{U}$ is trivial.

Consider the following commutative diagram, whose second row is exact (SGA 4 XII 3.1):

$$(**)\qquad \begin{tikzcd} \mathrm{H}^0(\overline{S}, n_*C/G) \arrow[r] \arrow[d, "k"'] & \mathrm{H}^1(\overline{S}, G) = 1 \arrow[d] & \\ \mathrm{H}^0(\widetilde{U}, r_*\widetilde{C}_1/\widetilde{F}) \arrow[r] & \mathrm{H}^1(\widetilde{U}, \widetilde{F}) \arrow[r] & \mathrm{H}^1(\widetilde{U}, r_*\widetilde{C}_1). \end{tikzcd}$$

Since the morphism $\widetilde{U} \to \overline{S}$ is 0-acyclic, k is an isomorphism. It then follows from $(**)$ that $\widetilde{P}$ is trivial.

XIII.6. Appendix II: finiteness theorem for direct images of stacks

Proposition XIII.6.1. *Let S be a locally noetherian scheme and let $f\colon X \to S$ be a morphism. If S' is an S-scheme, we denote by X' (respectivelyff', etc.) the pullback of X (respectivelyf$f$, etc.) by the morphism $S' \to S$. Suppose that, for every scheme S'* 436
étale over S and every constructible sheaf of sets F on X', f'_*F is constructible, and that, for every constructible sheaf of groups F on X', $\mathrm{R}^1 f'_*F$ is constructible. Let Φ be a 1-constructible stack on X (XIII.0). *Then $f_*\Phi$ is 1-constructible.*

For every scheme S' étale over S and every object x of $(f_*\Phi)_{S'}$, there is an isomorphism

$$\mathbf{Aut}_{S'}(x) \simeq f'_*\,\mathbf{Aut}_{X'}(x),$$

where, on the right-hand side, x is regarded as an object of $\Phi_{X'}$. The hypotheses therefore imply that $f_*\Phi$ is constructible. Let $S\Phi$ be the sheaf of maximal subgerbes of Φ [**1**, III 2.1.7]. Since $f_*(S\Phi)$ is constructible, we may apply (SGA 4 IX 2.7) to it, and the fact that $f_*\Phi$ is 1-constructible then follows from the sublemma below.

Lemma XIII.6.2. *Let S be a locally noetherian scheme, let $f\colon X \to S$ be a morphism, and let Φ be a stack on X. Suppose we are given a sheaf on S, representable by an étale S-scheme of finite type T, a surjective morphism*

$$a\colon T \to f_*(S\Phi)$$

and an object p of the fiber Φ_{X_T} (where $X_T = X \times_S T$), defining in $f_*(S\Phi)(T) = S\Phi(X_T)$ an element equal to the image q under a of the identity section of $T(T)$. Let $f_T\colon X_T \to T$ be the canonical morphism, and suppose that the sheaf $\mathrm{R}^1 f_{T*}(\mathbf{Aut}_{X_T}(p))$ is constructible; then the same holds for $S(f_*\Phi)$.

The canonical morphism $f^*f_*\Phi \to \Phi$ gives a morphism

$$S(f^*f_*\Phi) \simeq f^*(S(f_*\Phi)) \to S\Phi,$$

and hence a canonical morphism

$$\varphi\colon S(f_*\Phi) \to f_*(S\Phi).$$

Let $F = S(f_*\Phi)$, and let G be the image of F under φ; by (SGA 4 IX, 2.9), G is a 437
constructible sheaf.

It is enough to show that, for every point s of S, there exists a nonempty open neighborhood U of s such that $F|U$ is locally constant constructible. Let $s \in S$, let $\overline{s}$ be a geometric point lying over s, and let $\overline{q}_1, \ldots, \overline{q}_n$ be the elements of $G_{\overline{s}}$. By the definition of T, there exist S-morphisms $h_i \colon \overline{s} \to S'$ such that $\overline{q}_i = h_i^*(q)$. Let S' be the fiber product over S of n schemes isomorphic to T, let $\overline{s} \to S'$ be the fiber product of the h_i, let $X' = X \times_S S'$, and let q_i (respectivelyp_i) be the pullback of q (respectively$p$) by the i-th projection of S' onto T. If Ψ_i is the maximal subgerbe of $\Phi|X'$ generated by p_i, the sheaf $F_i = \mathrm{R}^1 f'_*(\mathbf{Aut}_{X'}(p_i))$ is precisely the sheaf $S(f'_*\Psi_i)$ of maximal subgerbes of $f'_*\Psi_i$. In particular, the canonical injection $\Psi_i \to \Phi|X'$ gives a morphism

$$\alpha_i \colon F_i \to F|S'.$$

We shall show that α_i is a bijection from F_i onto the inverse image of q_i in $F|S'$. For every scheme S'' étale over S', every section y of $F_i(S'')$ has image $q_i|S''$ in $F(S'')$, because, locally for the étale topology on S'', y is defined by an object x of $\Phi_{X''}$ that is isomorphic to $p_i|X''$. Conversely, if $y \in F(S'')$ has image $q_i|S''$ in $F(S'')$, then, locally for the étale topology on S'', y is defined by an object x of $\Phi_{X''}$ that is isomorphic to p_i; consequently, x is an object of $\Psi_{iX''}$ and therefore $y \in F_i(S'')$.

The proof is completed using XIII.6.3 below. Indeed, one can find an open neighborhood U' of s such that $q_1|U', \ldots, q_n|U'$ are sections of $G(U')$ and generate this sheaf. Since the $F_i|U'$ and $G|U'$ are constructible, so is $F|U'$ by XIII.6.3; after replacing U by a smaller open subset if necessary, $F|U$ is locally constant, which completes the proof.

Lemma XIII.6.3. *Let S be a locally noetherian scheme, and let $F \to G$ be a surjective* 438
morphism of sheaves of groups on S. Let q_i be a finite family of sections of G on X that generate G, and, for each i, let F_i be the subsheaf of F that is the inverse image of q_i. Then, if G and the F_i are constructible, so is F.

To prove that F is constructible, it is enough to show that, for every point s of S, there exists an open neighborhood U of s such that $F|U$ is locally constant constructible. Let s be a point of S. Since the sheaves F_i and G are constructible, one can find an open neighborhood U of s such that $F_i|U$ and $G|U$ are locally constant. We now show that $F|U$ is locally constant. By (SGA 4 IX 2.13 (i)), it is enough to see that, if $\overline{s}$ is a geometric point lying over s, $\tilde{s}$ is a geometric point of U, and $\overline{s} \to \tilde{s}$ is a specialization morphism, the canonical morphism

$$F_{\tilde{s}} \to F_{\overline{s}}$$

is bijective.

Consider the commutative diagrams

$$\begin{tikzcd}
(F_i)_{\tilde{s}} \arrow[r, "\tilde{a}_i"] \arrow[d, "\wr"] & F_{\tilde{s}} \arrow[r, "\tilde{a}"] \arrow[d, "b"'] & G_{\tilde{s}} \arrow[d, "\wr"] \\
(F_i)_{\overline{s}} \arrow[r, "\overline{a}_i"] & F_{\overline{s}} \arrow[r, "\overline{a}"] & G_{\overline{s}}
\end{tikzcd}$$

Let $\overline{q}_i$ (respectively$\tilde{q}_i$) be the pullback of q_i in $G_{\overline{s}}$ (respectively$G_{\tilde{s}}$); the morphisms $\overline{a}$ and $\tilde{a}$ are surjective, and the morphism $\overline{a}_i$ (respectively$\tilde{a}_i$) induces a bijection from $(F_i)_{\overline{s}}$ (respectively$(F_i)_{\tilde{s}}$) onto $\overline{a}^{-1}(q_i)$ (respectivelyonto $\tilde{a}^{-1}(q_i)$). It therefore follows from the diagram above that b is an isomorphism.

Corollary XIII.6.4. *Let S be a locally noetherian scheme and let $f \colon X \to S$ be* 439
a proper morphism. If Φ is a 1-constructible stack on X, then $f_*\Phi$ is a 1-constructible stack.

The proof of XIII.6.1 also proves the following result, in view of XIII.2.4 2).

COROLLARY XIII.6.5. *Let S be a locally noetherian scheme, let $f\colon X \to S$ be a morphism, let D be a divisor on X with normal crossings relative to S* (XIII.2), *let $Y = \operatorname{Supp} D$, let $U = X - Y$, and let $i\colon U \to X$ be the canonical immersion. Let Φ be a stack on U given, locally for the étale topology on X and S, as the pullback of a stack Ψ that is 1-constructible on S. Then the stack $i^{\mathrm{t}}_{*}\Phi$ is 1-constructible.*

Bibliography

[1] J. GIRAUD, Cohomologie non abélienne de degré 2, thesis, Paris (1966).
[2] J. GIRAUD, Cohomologie non abélienne, Springer, Berlin-New York (1971).
[3] H. SEIFERT, W. THRELFALL, Lehrbuch der Topologie, Chelsea, New York (1934).
[4] J-P. SERRE, Cohomologie galoisienne, Lecture Notes no. 5, Springer, Berlin (1964).
[5] J-P. SERRE, Corps locaux, Hermann, Paris (1962).

Index of Notation